Principle of Finantial Computing

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October 25, 2026

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1. Introduction

Lecture 1
September 10, 2026

Slides pp. 2–3

You must go into finance, Amory.
— F. Scott Fitzgerald (1896–1940),
This Side of Paradise (1920)

The two most dangerous words in Wall Street vocabulary are "financial engineering."
— Wilbur Ross (2007)

The two epigraphs above already frame the tension this course lives in. Finance is where quantitative modelling pays off most visibly, and it is also where a misplaced faith in a model does the most damage. Our job is to learn the models *and* to keep track of what they assume.

§1.1. Class Information

Slides pp. 4–6

The course follows the instructor's own textbook, and everything administrative lives on the course web page rather than in these notes.

- **Textbook.** Yuh-Dauh Lyuu, *Financial Engineering & Computation: Principles, Mathematics, Algorithms*. Cambridge University Press, 2002.
- **Official web page.**

  www.csie.ntu.edu.tw/~lyuu/finance1.html

  - Lecture notes will be uploaded *before* class.
  - Homeworks and teaching assistants will also be announced there.
  - Do not mistake last year's homeworks for this year's!
- **Software.** Some of the software used in the course is collected at

  www.csie.ntu.edu.tw/~lyuu/capitals.html
- **Questions.** Please ask many questions in class. In a large class this is the best way for the instructor to remember you.[1]

§1.2. Grading

Slides p. 7

Assessment is by **programming assignments** plus **one test**. Every homework is to be treated as an examination, so the academic-integrity rules are strict:

- You are expected to write your own codes and to turn in your source code.
- Do not copy or collaborate with fellow students.

[1] "[A] science concentrator [...] said that in his eighth semester of [Harvard] college, there was not a single science professor who could identify him by name." (*New York Times*, September 3, 2003.)

- Never ask your friends to write programs for you.
- Do not ask AI to code for you.
- Never give your codes to other students or publish your codes.

§1.3. Scope of the Course

Slides pp. 8–9

§1.3.i. What This Course Is About

The course runs from financial theory down to running code, in roughly this order:

- Financial theories in pricing.
- Mathematical backgrounds.
- Derivative securities.
- Pricing models.
- Efficient algorithms in pricing financial instruments.
- Research problems.
- Help in finding your thesis directions.

The last two items are worth emphasising: pricing problems that are merely "computable" are usually not good enough, and the gap between a correct algorithm and a *fast* one is exactly where the open research questions sit.

§1.3.ii. What This Course Is *Not* About

Equally important is what will *not* be covered.

- **How to program.**[2] Programming ability is assumed, not taught — but it matters. A software bug cost Knight Capital Group, Inc. US$457.6 million on August 1, 2012.[3]
- **Basic calculus, probability, combinatorics, and algebra.**
- **Details of the financial markets.**
- **How to be rich.**
- **How the markets will perform tomorrow.**
- **Professional behavior.**

Pricing theory tells you what a security is worth *given* a model of the market. It does not tell you what the market will do next.

§1.4. Useful Journals

Slides pp. 10–13

The literature of the field is spread across finance, mathematics, operations research and computer science journals. The following list is the one given in class.

[2] See http://www.csie.ntu.edu.tw/train/.
[3] Kirilenko & Lo (2013).

- *Applied Mathematical Finance.*
- *Applied Mathematics and Computations.*
- *Communications on Pure and Applied Mathematics.*
- *Computers & Mathematics with Applications.*
- *European Journal of Finance.*
- *European Journal of Operational Research.*
- *Finance and Stochastics.*
- *Finance Research Letters.*
- *Financial Analysts Journal.*
- *Financial Management.*
- *Insurance: Mathematics and Economics.*
- *International Journal of Finance & Economics.*
- *International Journal of Theoretical and Applied Finance.*
- *Journal of Banking & Finance.*
- *Journal of Computational Finance.*
- *Journal of Derivatives.*
- *Journal of Economic Dynamics & Control.*
- *Journal of Empirical Finance.*
- *Journal of Finance.*
- *Journal of Financial Economics.*
- *Journal of Financial Markets.*
- *Journal of Fixed Income.*
- *Journal of Futures Markets.*
- *Journal of Financial and Quantitative Analysis.*
- *Journal of Portfolio Management.*
- *Journal of Real Estate Finance and Economics.*
- *Journal of Risk and Uncertainty.*
- *Management Science.*
- *Mathematical Finance.*
- *Quantitative Finance.*
- *Review of Financial Studies.*
- *Review of Derivatives Research.*
- *Review of Finance.*
- *Risk Magazine.*
- *SIAM Journal on Financial Mathematics.*
- *Stochastics and Stochastics Reports.*

§1.5. The Modelers' Hippocratic Oath

Slides p. 14

Before writing down a single model, it is worth reading the oath proposed by Emanuel Derman and Paul Wilmott on January 7, 2009, in the wake of the financial crisis.

- I will remember that I didn't make the world, and it doesn't satisfy my equations.
- Though I will use models boldly to estimate value, I will not be overly impressed by mathematics.
- I will never sacrifice reality for elegance without explaining why I have done so.
- Nor will I give the people who use my model false comfort about its accuracy. Instead, I will make explicit its assumptions and oversights.
- I understand that my work may have enormous effects on society and the economy, many of them beyond my comprehension.

Intuition . A model is a deliberate simplification. The discipline is not in believing it, but in knowing precisely *where* it is wrong and *how much* that costs.

§1.6. A First Look at the Markets and the Data

Slides p. 15

Two facts motivate everything that follows. First, the markets we want to price in are enormous, so even small pricing errors translate into large amounts of money. Second, the

price series they produce have a very definite statistical shape, and any pricing model has to be compatible with it.

§1.6.i. Outstanding U.S. Debts

The fixed-income market alone is measured in tens of trillions of dollars, and over the period below it roughly quadrupled. Note in particular how the **mortgage-related** and **asset-backed** columns grow — these are precisely the securities whose pricing requires the algorithmic machinery of this course.

Year 年份	Municipal 市政債	Treasury 美國公債	Mortgage-related 房貸相關	U.S. corporate 公司債	Fed agencies 聯邦機構債	Money market 貨幣市場	Asset-backed 資產擔保	Total 總計
85	859.5	1,437.7	372.1	776.5	293.9	847.0	0.9	4,587.6
86	920.4	1,619.0	534.4	959.6	307.4	877.0	7.2	5,225.0
87	1,010.4	1,724.7	672.1	1,074.9	341.4	979.8	12.9	5,816.2
88	1,082.3	1,821.3	772.4	1,195.7	381.5	1,108.5	29.3	6,391.0
89	1,135.2	1,945.4	971.5	1,292.5	411.8	1,192.3	51.3	7,000.0
90	1,184.4	2,195.8	1,333.4	1,350.4	434.7	1,156.8	89.9	7,745.4
91	1,272.2	2,471.6	1,636.9	1,454.7	442.8	1,054.3	129.9	8,462.4
92	1,302.8	2,754.1	1,937.0	1,557.0	484.0	994.2	163.7	9,192.8
93	1,377.5	2,989.5	2,144.7	1,674.7	570.7	971.8	199.9	9,928.8
94	1,341.7	3,126.0	2,251.6	1,755.6	738.9	1,034.7	257.3	10,505.8
95	1,293.5	3,307.2	2,352.1	1,937.5	844.6	1,177.3	316.3	11,228.5
96	1,296.0	3,459.7	2,486.1	2,122.2	925.8	1,393.9	404.4	12,088.1
97	1,367.5	3,456.8	2,680.2	2,346.3	1,022.6	1,692.8	535.8	13,102.0
98	1,464.3	3,355.5	2,955.2	2,666.2	1,296.5	1,978.0	731.5	14,447.2
99	1,532.5	3,281.0	3,334.2	3,022.9	1,616.5	2,338.2	900.8	16,026.4
00	1,567.8	2,966.9	3,564.7	3,372.0	1,851.9	2,661.0	1,071.8	17,056.1
01	1,688.4	2,967.5	4,125.5	3,817.4	2,143.0	2,542.4	1,281.1	18,565.3
02	1,783.8	3,204.9	4,704.9	3,997.2	2,358.5	2,577.5	1,543.3	20,170.1

Table 1.1.: Outstanding U.S. debts, in billions of U.S. dollars.

§1.6.ii. The Global OTC Derivatives Market

Slides p. 16

Over-the-counter derivatives grew even faster, by close to an order of magnitude between 2000 and the 2008 crisis, after which the notional outstanding plateaued and then declined as trades were compressed and cleared centrally.

Remark 1.6.1. **衍生性商品**（derivative）是價值取決於某個標的（利率、匯率、股價、信用等）的合約。**場外交易**（over-the-counter, OTC）指買賣雙方私下議定、不經交易所撮合的交易，與合約標準化、由結算所集中結算的**交易所交易**相對。OTC 合約可客製化（金額、到期日、標的皆可議），代價是雙方直接承擔**對手方風險**；市場由少數大型銀行（dealer）造市，未平倉部位以利率交換（interest rate swap）為大宗。

Figure 1.1 的縱軸為**名目本金**（notional amount）——僅是計算利息與收益的參考金額，並非真正易手的資金，故市場實際市值遠小於此。2008 年金融危機後，美國 Dodd–Frank 法與歐盟 EMIR 要求標準化的 OTC 商品（尤其信用違約交換 CDS 與利率交換）改採**中央結算**（central clearing），並藉**壓縮**（compression）沖銷互相抵銷的部位，名目餘額因而在 2013 年前後見頂並回落。

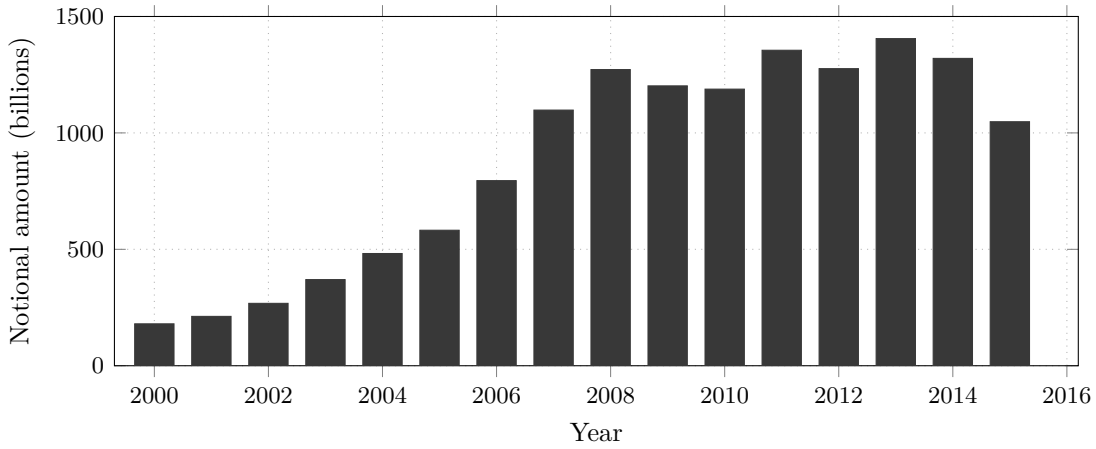


Figure 1.1.: The global OTC derivatives market. Bar heights are read off the chart shown in class.

§1.6.iii. The S&P 500 Index and Its Returns

Slides pp. 17–19

Robert Shiller[4] maintains a monthly series of the S&P 500 index reaching back to 1871. Plotted in levels (Figure 1.2), it is dominated by the post-1980 growth; the 1929 crash is barely visible on this scale.

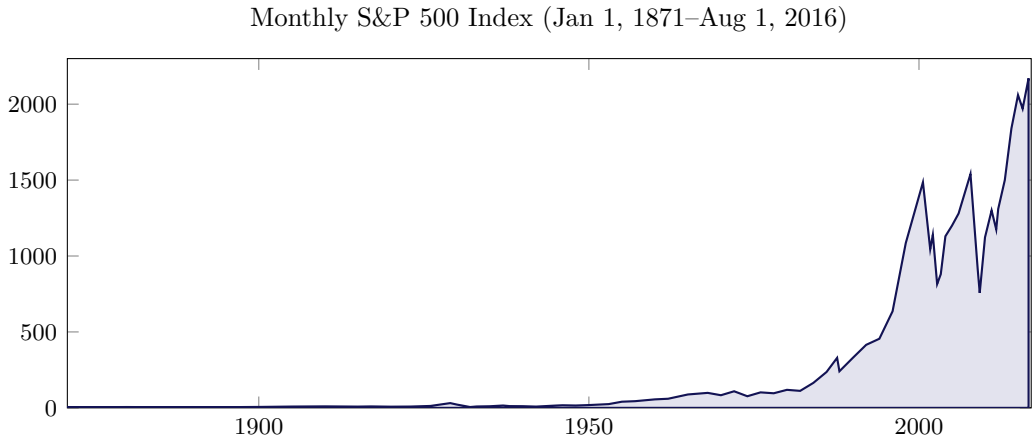


Figure 1.2.: The S&P 500 index in levels, reconstructed from the Shiller series.

Levels are not what we model, however. Prices are non-stationary and their scale is arbitrary, so we work with the **log return** over one period,

$$r_t \;=\; \ln \frac{S_t}{S_{t-1}},$$

where S_t is the index level at time t. Figure 1.3 plots the monthly log returns of the same series.

Remark 1.6.2. 為什麼不直接對價位建模？**價位**（levels）是**非定態**的——統計性質隨時間漂移，1900 年與 2000 年的「正常水準」天差地遠，無法用一組固定參數描述；而且指數從幾點起編是人為選擇，尺度本身沒有經濟意義。改用**對數報酬** $r_t = \ln(S_t/S_{t-1})$ 一次補齊三件事：

- S_t/S_{t-1} 是本期的成長倍數（漲 2% 即 1.02）。變化小時 $\ln(1+x) \approx x$，所以 r_t 近似百分比變化，且與絕對點數無關。

[4]Co-winner of the 2013 Nobel Prize in Economic Sciences.

- **可加性**：跨期報酬 $\ln(S_t/S_{t-k}) = r_t + r_{t-1} + \cdots + r_{t-k+1}$，相加即可，不必連乘。
- 報酬率序列大致**定態**——繞著 0 波動、沒有趨勢，正適合建模。

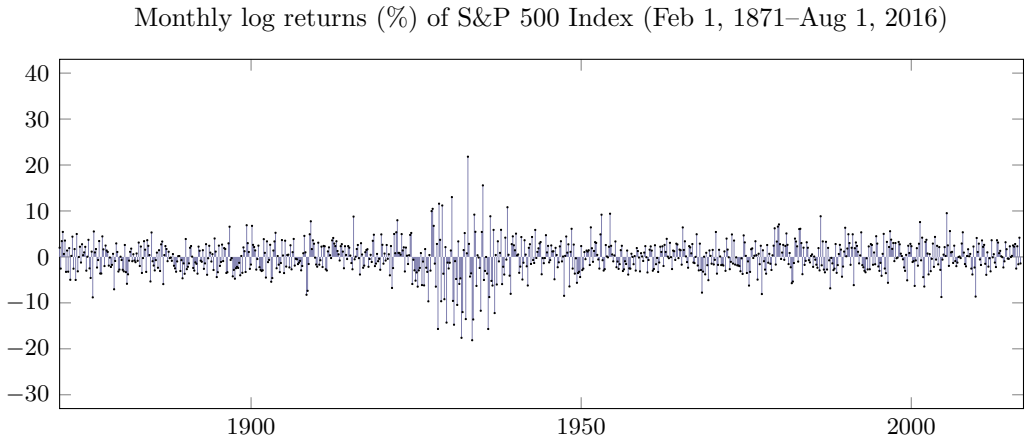


Figure 1.3.: Monthly log returns of the S&P 500 index. The series is a schematic reproduction of the plot shown in class: it reproduces the mean, the typical amplitude, the fat tails and the volatility cluster of the early 1930s, not the individual monthly observations.

Three features stand out, and they will constrain every model we build:

- The returns hover around zero and show no obvious trend, in sharp contrast with the levels.
- Their amplitude is *not* constant: quiet stretches alternate with violent ones (**volatility clustering**), the early 1930s being the extreme case.
- Large moves are far more frequent than a normal distribution would allow.

The last point is easiest to see from the empirical density (Figure 1.4): the body is a sharp peak near zero, but the tails stay noticeably above zero out to ±30%.

Remark 1.6.3. 兩個關鍵術語：**波動聚集**（volatility clustering）指抖動的**幅度**會成群出現——大波動旁邊通常還是大波動，平靜期也連成一片（1930 年代初最極端），也就是說波動率本身不是常數。**厚尾**（fat tails）指極端漲跌遠比常態分佈預測的頻繁。看Figure 1.4：曲線畫的是每月報酬率的出現頻率，中央為 0 附近又高又尖的峰（多數月份幾乎不動），但兩側尾巴延伸到 ±30% 仍明顯高於零，而常態分佈在那裡的機率早已可忽略。這種「尖峰厚尾」的形狀，正是接下來那句結論的依據。

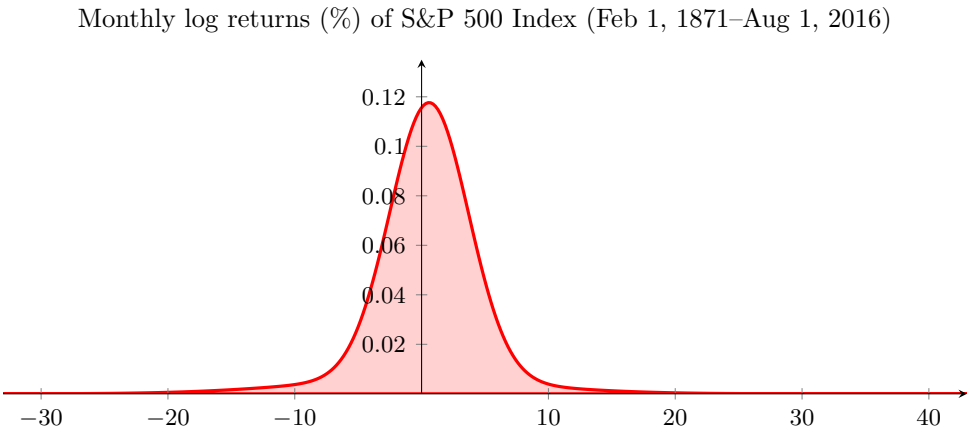


Figure 1.4.: Distribution of the monthly log returns of the S&P 500 index: a sharp central peak with tails much heavier than the normal.

Prices are not normally distributed, and their volatility is not constant. Any model that assumes otherwise is buying tractability with realism.
價格並非常態分佈，波動率也非定值；凡是假設相反的模型，都是以寫實性為代價，換取數學上的易處理性。

§1.7. Analysis of Algorithms

Slides pp. 20–21

I can calculate the motions
of the heavenly bodies,
but not the madness of people.
— Isaac Newton (1642–1727)

It is unworthy of excellent men
to lose hours like slaves
in the labor of computation.
— Gottfried Wilhelm Leibniz (1646–1716)

Having seen how large the markets are and how awkward their data is, we now fix the vocabulary for the other half of the course: what it means for a pricing problem to be solvable, and at what cost.

§1.7.i. Computability and Algorithms

Slides p. 22

Definition 1.7.1. An **algorithm** is a precise procedure that can be turned into a computer program.

Not every question admits one. **Uncomputable** problems are those for which no algorithm exists at all, no matter how much time we are willing to spend. Two standard examples — both, uncomfortably, about programs themselves:

- Does this program have infinite loops?
- Is this program bug free?

Everything else is **computable**, and computable problems split again according to how expensive the algorithm is: **intractable** problems admit algorithms in principle but not ones we can afford to run, while **tractable** problems admit algorithms that are practical at the sizes we care about. Figure 1.5 summarises the picture.

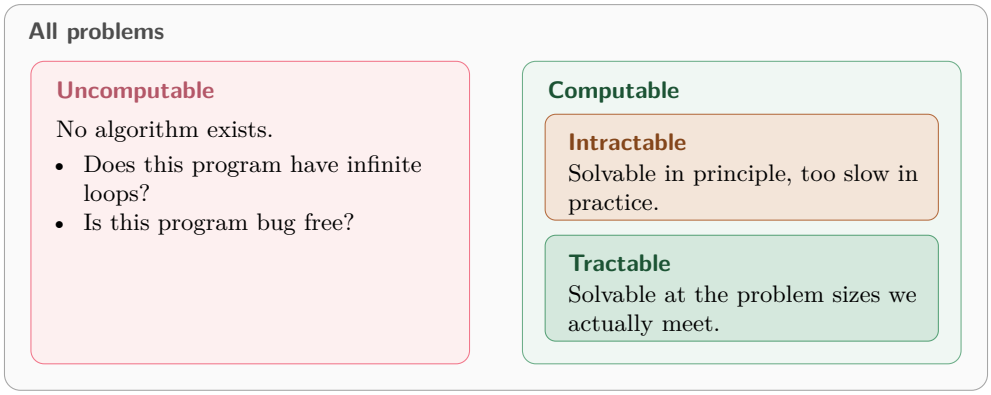


Figure 1.5.: The classification of problems. This course lives almost entirely in the bottom-right box — and much of the research effort consists of moving pricing problems into it.

Remark 1.7.2. The conventional formal criterion is how running time scales with the input size n. A problem is counted *tractable* when it admits an algorithm of *polynomial* time, $O(n^k)$ for some fixed k, and *intractable* when it admits none — every algorithm needs superpolynomial time, in most cases exponential. The rationale is practical: a polynomial bound keeps the cost manageable as n grows, whereas anything superpolynomial blows up.

§1.7.ii. Complexity

Slides p. 23

To measure cost without committing to a machine or a language, we count operations rather than seconds.

Definition 1.7.3. A set of basic operations — $+$, $-$, $\times$, $/$, log, x^y, e^x, ... — are assumed to take one unit of time each. The total number of such operations performed is the total work done by an algorithm, its **computational complexity**. The amount of memory space it uses is its **space complexity**.

Remark 1.7.4. Two conventions are worth making explicit.

- We concentrate on the *abstract* complexity of an algorithm instead of its detailed implementation. Constant factors from the language, compiler and hardware are deliberately ignored.
- Despite that, complexity is a good guide to an algorithm's *actual* running time. It is the coarse measure that survives a change of machine.

§1.7.iii. Common (Asymptotic) Complexities

Slides p. 24

Let n stand for the "size" of the problem — the number of elements, the number of cash flows, the number of time periods, and so on. Complexity is then expressed as a function of n, and the standard names are as follows.

Name	Complexity	Remark
Linear	$O(n)$	
Quadratic	$O(n^2)$	
Cubic	$O(n^3)$	
Superpolynomial	e.g. $2^{O(\sqrt{n})}$	higher than any polynomial[5]
Exponential time	$2^{O(n)}$	the usual worst case

Table 1.2.: Common asymptotic complexities, in increasing order of growth.

Note 1.7.5. The superpolynomial entry is not a curiosity. Several pricing problems that were only known to admit exponential-time algorithms have been pushed down to $2^{O(\sqrt{n})}$ — an improvement large enough to turn an unusable algorithm into a usable one, and precisely the kind of result this course aims at.

[5] E.g., T. Dai (`B82506025`, `R86526008`, `D8852600`) & Lyuu (2007); Lyuu & C. Wang (`F95922018`) (2011); H. Chiu (`R98723059`) (2012).

2. Basic Financial Mathematics

Slides pp. 25–26

In the fifteenth century
mathematics was mainly concerned with
questions of commercial arithmetic and
the problems of the architect.
— Joseph Alois Schumpeter (1883–1950)

I'm more concerned about
the return of *my money than*
the return on *my money.*
— Will Rogers (1879–1935)

Everything in this chapter follows from one idea: a dollar today and a dollar next year are different goods, and the interest rate is the exchange rate between them. Once that exchange rate is pinned down, valuing a stream of payments is arithmetic — and, as we will see, arithmetic whose cost we can still argue about.

§2.1. The Time Line

Slides p. 27

Before any formula, fix the convention for talking about time. Time is measured in **periods**; time 0 is now, and period k is the interval running from time $k-1$ to time k. A cash flow "at time k" arrives at the *end* of period k.

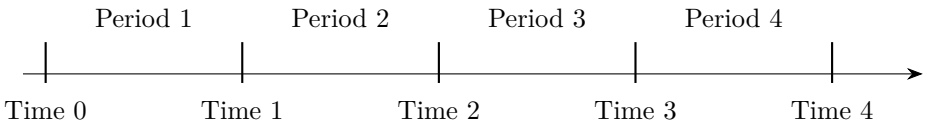


Figure 2.1.: The time line. Period k spans time $k-1$ to time k.

§2.2. Time Value of Money

Slides p. 28

Definition 2.2.1. Let r be the **interest rate** per period. The **future value** (FV) and **present value** (PV) of a single amount over n periods are related by

$$\begin{aligned} \mathrm{FV} &= \mathrm{PV}(1+r)^n, \\ \mathrm{PV} &= \mathrm{FV} \times (1+r)^{-n}. \end{aligned} \tag{2.1}$$

Intuition . $(1+r)^n$ carries money forward in time and $(1+r)^{-n}$ carries it backward. *Discounting* — the second operation — is the one we use constantly, because a security's price is quoted today while its payoffs arrive later.
$(1+r)^n$ 把錢往未來搬，$(1+r)^{-n}$ 把錢往現在搬。我們不斷用到的是後者——**折現**（discounting）——因為證券的價格報在今天，收益卻在日後才到。

The idea is old: it goes back to Fibonacci (1170–1240), and was put on a modern footing by Irving Fisher (1867–1947).

§2.3. Compounding

Slides pp. 29–30

Equation (2.1) is stated per *period*, but rates are quoted per *year*. Reconciling the two is what compounding conventions are for — and getting the convention wrong is a classic source of pricing errors.

§2.3.i. Periodic Compounding

Suppose the *annual* interest rate r is compounded m times per annum. Each compounding period earns r/m, so one dollar grows along

$$1 \to \left(1+\frac{r}{m}\right) \to \left(1+\frac{r}{m}\right)^2 \to \left(1+\frac{r}{m}\right)^3 \to \cdots$$

Hence, after n years,

$$\mathrm{FV} = \mathrm{PV}\left(1+\frac{r}{m}\right)^{nm}. \tag{2.2}$$

§2.3.ii. Common Compounding Methods

Method	m		
Annual	1	Weekly	52
Semiannual	2	Daily	365
Quarterly	4	Continuous	∞
Monthly	12		

§2.3.iii. Translating between Rate and Frequency

Slides pp. 31–32

An annual interest rate of r compounded m times a year is "equivalent to" an interest rate of r/m per $1/m$ year. Read in reverse: if a loan asks for a return of 1% *per month*, the *annual* interest rate will be 12% with monthly compounding.

Example 2.3.1. The annual interest rate is 10% compounded twice per annum. Each dollar will grow to be

$$\left[1+(0.1/2)\right]^2 = 1.1025$$

one year from now. The rate is therefore "equivalent to" an interest rate of 10.25% compounded once *per annum*, since

$$1 + 0.1025 = 1.1025.$$

The 10.25% in the example is the **effective annual rate**: the single annually-compounded rate that produces the same money after one year. The quoted 10% is only a **nominal rate**, and by itself it does not determine anything.

An interest rate is meaningless without the compounding frequency.

§2.3.iv. The Rule of 72

Slides pp. 33–35

Let the annual interest rate be r with annual compounding. How many years T will it take for your money to double? The identity to solve is

$$(1+r)^T = 2, \qquad \text{so} \qquad T = \frac{\ln 2}{\ln(1+r)}.$$

Is there an easier way? The **rule of 72** is a heuristic that estimates T without a logarithm:

$$T \approx \frac{72}{r\ (\%)}. \tag{2.3}$$

So it takes about $72/12 = 6$ years to double the GDP if the annual growth rate is 12%.

Reason. Expanding $\ln(1+r) = r - r^2/2 + O(r^3)$ gives, for small r,

$$\frac{1}{\ln(1+r)} \approx \frac{1}{r\,(1-r/2)} \approx \frac{1}{r} + \frac{1}{2},$$

and therefore

$$\frac{\ln 2}{\ln(1+r)} \approx \frac{0.693}{r} + 0.3466. \qquad \blacksquare$$

Remark 2.3.2. The leading constant is 0.693, i.e. 69.3 when r is measured in percent — so a "rule of 69.3" is the asymptotically correct one. The extra +0.3466 is a positive offset, so inflating the numerator to 72 compensates for it over the range of rates one actually meets. That 72 is also divisible by $2, 3, 4, 6, 8, 9, 12$ is a large part of why it, rather than 70, became the folk rule.

Figure 2.2 plots the error of both variants: the true interest rate minus the approximation, in percent, against the number of years to double.

Remark 2.3.3. 這張圖把 Eq. (2.3) **反過來用**：平常是「給利率 r，估翻倍要幾年 T」，這裡改成「給橫軸的 T（想要幾年翻倍），估需要多少利率」——把公式解成 $r \approx 72/T$。縱軸畫的是兩個**利率**估計值的差：真正在 T 年翻倍所需的 $r_{\text{真}} = 2^{1/T} - 1$，減去 Rule of 72 反解出的 $r_{\text{估}} = 72/T$（兩者都乘 100 化成百分比）。所以縱軸單位是「利率的百分點」，不是「年」；橫軸的 T 在這裡是已知條件，不是被估計的量。

代入數字比較有感覺：$T = 1$ 年時，真正利率 100% 對 Rule of 72 的 72%，差 28 個百分點，極不準；$T = 8$ 年時，真正利率約 9.05% 對 9%，差只有 0.05，正是 caption 說的「超過約 8 年後極準」；$T = 20$ 年時差變成負的（約 -0.07），代表 Rule of 72 在長期會略為高估所需利率（"overshoot"）。

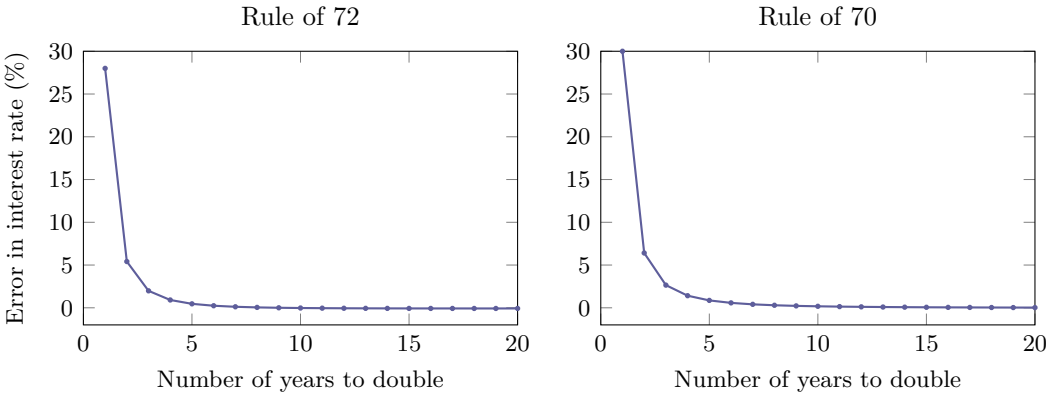


Figure 2.2.: How good is the rule of 72? Plotted is the true interest rate minus the approximation (in %). Both are badly off for very short doubling times and excellent beyond about eight years; the rule of 72 slightly *over*shoots there, the rule of 70 slightly undershoots.

§2.3.v. Continuous Compounding

Slides pp. 36–37

Let $m \to \infty$ in Eq. (2.2), so that

$$\left(1 + \frac{r}{m}\right)^m \longrightarrow e^r.$$

Then

$$\mathrm{FV} = \mathrm{PV} \times e^{rn}, \tag{2.4}$$

where $e = 2.71828\ldots$ [1]

Continuous compounding is easier to work with, because the exponents simply add. Suppose the annual interest rate is r_1 for n_1 years and r_2 for the following n_2 years. Then the FV of one dollar will be

$$e^{r_1 n_1 + r_2 n_2}$$

after $n_1 + n_2$ years — no nested products, just a sum in the exponent.

§2.3.vi. Conversion between Compounding Methods

Slides pp. 38–40

The word "equivalent" has already been used twice in quotation marks. Here is what it means.

Definition 2.3.4 (Equivalent rates)**.** The **accumulation factor** $A(n)$ of a quoting convention is the amount one dollar grows to under it after n years:

$$A(n) = \begin{cases} e^{rn}, & r \text{ compounded continuously,} \\ \left(1 + \dfrac{r}{m}\right)^{mn}, & r \text{ compounded } m \text{ times per annum.} \end{cases}$$

Two quoted annual rates are **equivalent** if they turn one dollar into the same amount of money at every horizon — that is, if their accumulation factors agree as functions of n.

[1] Jacob Bernoulli (1654–1705) in 1685.

Note 2.3.5. Checking one year is enough. Each accumulation factor above is a pure power of its own one-year factor, $A(n) = A(1)^n$, so $A_1(1) = A_2(1)$ already forces $A_1(n) = A_2(n)$ for every $n \geq 0$. Equivalently: two rates are equivalent exactly when they share the same **effective annual rate** $A(1) - 1$ of Section 2.3.iii.

Let r_1 be the annual rate with continuous compounding and r_2 the *equivalent* rate compounded m times per annum. How are they related?

Proposition 2.3.6

The principle is that both interest rates must produce the same amount of money after one year, that is,

$$\left(1 + \frac{r_2}{m}\right)^m = e^{r_1}.$$

Therefore[a]

$$r_1 = m \ln\left(1 + \frac{r_2}{m}\right),$$
$$r_2 = m\left(e^{r_1/m} - 1\right).$$

[a] Are they really equivalent? In what sense are they equivalent? Contributed by Mr. Chen, Tung-Li (D09922014) on February 24, 2023.

The same principle handles two periodic conventions. Suppose r_1 is the annual rate compounded m_1 times per annum and r_2 is the *equivalent* rate compounded m_2 times per annum. Then

$$\left(1 + \frac{r_1}{m_1}\right)^{m_1} = \left(1 + \frac{r_2}{m_2}\right)^{m_2}. \tag{2.5}$$

Observation . Every conversion in this subsection is the same one-line demand: *equal value after one year*. There is nothing to memorise beyond that.

§2.4. Present Value

Slides p. 41

§2.4.i. The PV Formula

Definition 2.4.1. The PV of the cash flow $C_1, C_2, \ldots, C_n$ at times $1, 2, \ldots, n$ is

$$\mathrm{PV} = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \cdots + \frac{C_n}{(1+y)^n}, \tag{2.6}$$

where y is the discount rate per period.

Remark 2.4.2. 我們想要衡量一筆包含未來的錢的現金流折算到現在價值多少，這條公式就是把單筆折現 $\mathrm{PV} = \mathrm{FV}(1+y)^{-n}$ **對每一筆現金流各做一次，再加起來**：在 time k 收到的 C_k 要往回搬 k 期，所以除以 $(1+y)^k$——越晚到的錢折得越兇——加總便是整串現金流今天的總值。y 是每期折現率，字母取自 *yield*，角色與前一章的 r 相同。

This formula and its variations are the engine behind most of financial calculations.[2] The difficulty is never the algebra; it is deciding what goes into it:

[2] Cochrane (2005), "Asset pricing theory all stems from one simple concept [...]: price equals expected discounted payoff."

- What is y?
- What are the C_i?
- What is n?

Note 2.4.3. Equation (2.6) is taken on faith for now. It will be justified later in the course from "sound" principles rather than assumed — see Theorem 7.3.1, where the only principle used is that there is no free lunch.

§2.4.ii. Evaluating the PV Formula

Slides pp. 42–44

Even a formula this simple raises the question the previous chapter set up: how much work does it actually take? The direct transcription is Algorithm 2.1.

Algorithm 2.1: Evaluating the PV in Eq. (2.6) directly

```
1 x := 0;
2 for i = 1, 2, ..., n do
3 |   x := x + C_i/(1 + y)^i;
4 return x;
```

Computing $(1+y)^i$ by repeated multiplication costs i operations, so the algorithm takes time proportional to $\sum_{i=1}^{n} i = O(n^2)$ if only $+, -, \times$ and $/$ are allowed. It can be improved to $O(n)$ by applying $a^b = e^{b \ln a}$ in step 3, since we agreed to count x^y as taking one unit of time.

But there is a better route, which needs no exponentiation at all (Algorithm 2.2).

Algorithm 2.2: Evaluating the PV by Horner's rule

```
1 x := 0;
2 d := 1 + y;
3 for i = n, n − 1, ..., 1 do
4 |   x := (x + C_i)/d;
5 return x;
```

Proposition 2.4.4 (Horner's rule)

The underlying identity is

$$\left(\cdots\left(\left(\frac{C_n}{1+y}+C_{n-1}\right)\frac{1}{1+y}+C_{n-2}\right)\frac{1}{1+y}+\cdots\right)\frac{1}{1+y},$$

due to Horner (1786–1837) in 1819.

The algorithm takes $O(n)$ time, and it is the most efficient possible in terms of the absolute number of arithmetic operations.[3] It can also be computed in parallel in $O(\log n)$ time using prefix sums.[4]

Two algorithms for the same formula, $O(n^2)$ against $O(n)$ — and the fast one is also the numerically simpler one. This is the pattern the whole course chases.

[3] A. Borodin & Munro (1975).

[4] Contributed by Mr. Chen, Yi-Feng (R08922077) on February 24, 2021.

§2.5. Annuities

Slides pp. 45–47

§2.5.i. Annuities Certain

Definition 2.5.1. An **annuity** pays out the same C dollars at the end of each year for n years.

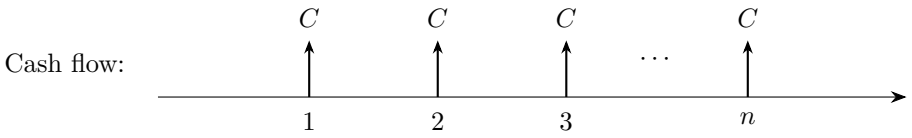


Figure 2.3.: The cash flow of an n-year annuity certain.

With a rate of r, specialising Eq. (2.6) to $C_i \equiv C$ and summing the geometric series gives

$$\mathrm{PV} = \sum_{i=1}^{n} C(1+r)^{-i} = C\,\frac{1-(1+r)^{-n}}{r},$$

while the FV at the end of the nth year is

$$\mathrm{FV} = \sum_{i=0}^{n-1} C(1+r)^{i} = C\,\frac{(1+r)^{n}-1}{r}.$$

The study of these sums goes back to Jan de Witt (1625–1672) in 1671 and Nicholas Bernoulli (1687–1759) in 1709.

§2.5.ii. General Annuities

Suppose instead that m payments of C dollars each are received per year (the **general annuity**), and let r be compounded m times per annum.[5] Then the same geometric sum with period rate r/m and nm periods gives

$$\begin{aligned}\mathrm{PV} &= \sum_{i=1}^{nm} C\Big(1+\frac{r}{m}\Big)^{-i} = C\,\frac{1-\big(1+\frac{r}{m}\big)^{-nm}}{\frac{r}{m}}, \\ \mathrm{FV} &= \sum_{i=0}^{nm-1} C\Big(1+\frac{r}{m}\Big)^{i} = C\,\frac{\big(1+\frac{r}{m}\big)^{nm}-1}{\frac{r}{m}}.\end{aligned} \tag{2.7}$$

Remark 2.5.2. Annuity is paid at the end of every year, so note the indices of above formulas.

§2.6. Amortization

Slides pp. 48–49

[5] If your r has a compounding frequency different from m, convert it with Eq. (2.5) first.

§2.6.i. The Amortization Method

Definition 2.6.1. **Amortization** is a method of repaying a loan through regular payments of interest *and* principal.

The mechanics are worth stating carefully, because every line of the schedule below follows from them:

- The size of the loan (the original balance) is reduced by the *principal* part of each payment.
- The *interest* part of each payment pays the interest incurred on the *remaining* principal balance.
- As the principal gets paid down over the term of the loan, the interest part of the payment diminishes.

An amortized loan is exactly a general annuity seen from the lender's side: the loan amount is the PV of the payment stream. Inverting Eq. (2.7) therefore gives the regular payment,

$$C = \text{loan amount} \times \frac{\frac{r}{m}}{1-\left(1+\frac{r}{m}\right)^{-nm}}. \tag{2.8}$$

For monthly payments, take $m = 12$.

Remark 2.6.2. 每期還款總金額一樣，只是把每期還款拆賬成上期剩餘本金造成的利息跟一部分本金。

Remark 2.6.3. 初看之下兩者的計息方式並不一樣：**攤還**（amortization）貸款的利息是對剩餘本金計算的，而**年金**（annuity）公式 (2.7) 裡根本沒有出現「餘額」，只是把每期的 C 折現後相加。兩者一致的理由是：折現本身就是「利息只算到該筆錢實際在外流通為止」的數學編碼。以下三點說的是同一件事。

(a) 這是恆等式，不是類比。記 $i \triangleq r/m$、$N \triangleq nm$，並令 B_k 為第 k 期付款後的剩餘本金。「利息按剩餘本金計、付款先付息再沖本」這句話的全部內容就是遞迴

$$B_k = B_{k-1}(1+i) - C, \qquad B_0 = \text{loan amount}.$$

移項得「往回推一期」的式子

$$B_{k-1} = \frac{C}{1+i} + \frac{B_k}{1+i},$$

意思是：上一期的餘額 = 本期付款的折現 + 本期餘額的折現。所謂「反覆代入自己」，就是把這條式子不斷套用在右邊剩下的那個 B 上。從 $k=1$ 開始：

$$B_0 = \frac{C}{1+i} + \frac{B_1}{1+i}.$$

右邊出現了 B_1，而同一條式子取 $k=2$ 給出 $B_1 = \frac{C}{1+i} + \frac{B_2}{1+i}$，代進去：

$$B_0 = \frac{C}{1+i} + \frac{1}{1+i}\left(\frac{C}{1+i} + \frac{B_2}{1+i}\right) = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \frac{B_2}{(1+i)^2}.$$

現在右邊剩下 B_2，再取 $k=3$ 把 $B_2 = \frac{C}{1+i} + \frac{B_3}{1+i}$ 代進去：

$$B_0 = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \frac{B_3}{(1+i)^3}.$$

規律很清楚：每代入一次，就多出一項付款的折現 $C/(1+i)^k$，而殘留的餘額項往後推一期、多除一個 $(1+i)$。一路代到 $k=N$ 為止（可用歸納法嚴格證明），得

$$B_0 = \sum_{k=1}^{N} \frac{C}{(1+i)^k} + \frac{B_N}{(1+i)^N}.$$

而 *fully amortized* 的意思正是 $B_N = 0$，末項消失，於是貸款額 = 付款流的**現值**（present value），即 (2.7)。年金並不是用來近似攤還貸款的模型，它就是那條餘額遞迴式的封閉解。

(b) 折現如何「知道」餘額在變。把 B_0 拆成 N 筆互不相干的零息小貸款，第 k 筆的金額恰為 $C(1+i)^{-k}$。它獨自滾 k 期複利後成為 $C(1+i)^{-k}(1+i)^k = C$，剛好被第 k 期的付款一次結清，此後即離開帳本、不再生息。每一筆都只在自己尚未被償還的期間生息，合起來就是「利息只對剩餘本金計算」。關鍵在於折現因子的指數是該筆付款自己的期數 k 而非全期 N——已經還掉的錢不再出現在後面各項裡。

(c) 兩個假設要分開。 (a) 的恆等式對任意付款序列 $C_1, \ldots, C_N$ 都成立，此時 $B_0 = \sum_k C_k(1+i)^{-k}$；「餘額計息」本身只保證這件事，與付款是否等額無關。真正額外用到的假設是 等額付款（$C_k \equiv C$），它使上式成為等比級數、能求和成封閉解 (2.7)，我們也才能反解出 (2.8)。

§2.6.ii. Example: Home Mortgage

Slides pp. 50–54

By paying down the principal consistently, the risk to the lender is lowered, and when the borrower sells the house, only the remaining principal is due the lender. Consider the equal-payment case, i.e., fixed-rate, level-payment, fully amortized mortgages — these are called traditional mortgages in the U.S.[6]

Remark 2.6.4. 上段中譯：藉由持續償還本金，放款人承擔的風險逐步降低；當借款人賣掉房子時，只需清償當時的剩餘本金即可。以下只考慮等額付款的情形，也就是 **固定利率**（fixed-rate）、**等額付款**（level-payment）、**完全攤還**（fully amortized）的房貸——這類房貸在美國被稱為 **傳統房貸**（traditional mortgage）。

註腳中譯：《經濟學人》(2016)：「多數國家的銀行以短期或浮動利率房貸來壓低自身風險；美國的借款人則享有較好的條件——便宜的 30 年期固定利率房貸，而且可以免費提前清償（即無提前清償違約金）。」

Example 2.6.5. Consider a 15-year, \$250,000 loan at 8.0% interest rate. Solve Eq. (2.7) with PV = 250000, $n = 15$, $m = 12$ and $r = 0.08$. That is,

$$250000 = C\,\frac{1-\left(1+\frac{0.08}{12}\right)^{-15\times 12}}{\frac{0.08}{12}},$$

which gives a monthly payment of $C = 2389.13$.

Month	Payment	Interest	Principal	Remaining principal
				250,000.000
1	2,389.13	1,666.667	722.464	249,277.536
2	2,389.13	1,661.850	727.280	248,550.256
3	2,389.13	1,657.002	732.129	247,818.128
		...		
178	2,389.13	47.153	2,341.980	4,730.899
179	2,389.13	31.539	2,357.591	2,373.308
180	2,389.13	15.822	2,373.308	0.000
Total	430,043.438	180,043.438	250,000.000	

Table 2.1.: The amortization schedule of the 15-year, \$250,000 loan at 8%.

[6] *The Economist* (2016), "In most countries banks minimize their risk by offering short-term or floating-rate mortgages. American borrowers get a better deal: cheap 30-year fixed-rate mortgages that can be repaid early free."

In every month:

- The principal and interest parts add up to \$2,389.13.
- The remaining principal is reduced by the amount indicated under the Principal heading.[7] The Principal column forms a geometric sequence.[8]
- The interest is computed by multiplying the remaining balance of the previous month by 0.08/12.

Note also that:

- The Principal column adds up to \$250,000 — it must, since the loan is fully amortized.
- The Payment column adds up to \$430,043.438! The difference, \$180,043.438, is the total interest paid over 15 years.
- If the borrower plans to pay off the mortgage right *after* the 178th month's monthly payment, he needs to pay another \$4,730.899.[9]

Figure 2.4 shows the third bullet of Table 2.1 graphically: the payment is level, but its composition rotates from almost all interest to almost all principal.

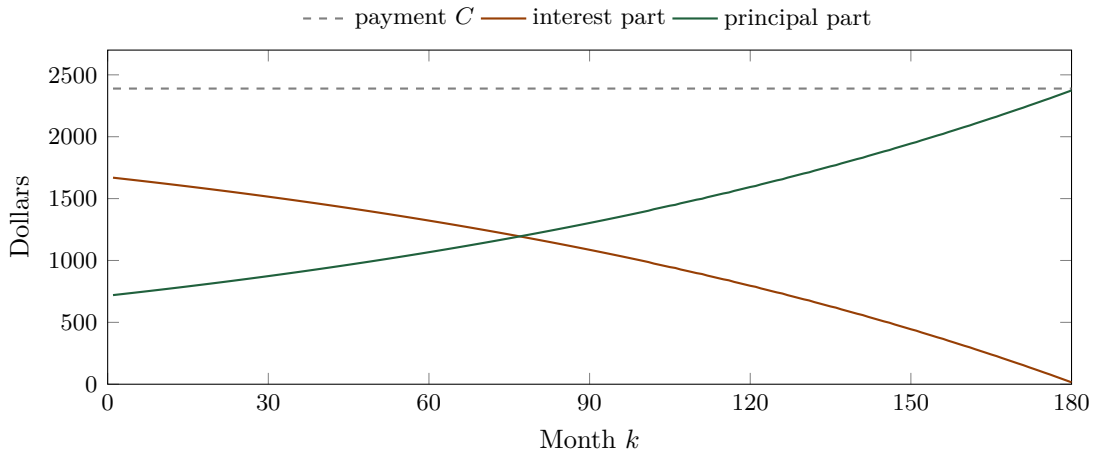


Figure 2.4.: Composition of the level monthly payment over the life of the loan, computed from the example above. The two parts always sum to C, and the principal part is a geometric sequence with ratio $1 + r/m$.

§2.6.iii. Calculating the Remaining Principal

Slides pp. 55–57

Two ways to answer "how much is still owed after k payments?"

Method 1: run the schedule

- A month's principal payment = monthly payment − (previous month's remaining principal) × (monthly interest rate).
- A month's remaining principal = previous month's remaining principal − principal payment calculated above.
- Generate the amortization schedule until the particular month you are interested in.

This method is relatively slow — it costs $O(k)$ work — but it is universal in its applicability. It can, for example, accommodate prepayments and floating interest rates, neither of which has a closed form.

[7] This column varies with r. Thanks to a lively class discussion on Feb 24, 2010. In fact, every column varies with r. Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.
[8] Contributed by Mr. Sun, Ao (R05922147) on February 22, 2017.
[9] Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.

Method 2: discount the remaining cash flows

Right after the kth payment, the remaining principal is the PV of the future $nm-k$ cash flows because we have

$$B_{k-1}(1+i)-C=B_k$$

and we can solve this to obtain

$$B_k=\sum_{i=1}^{nm-k}C\Big(1+\frac{r}{m}\Big)^{-i}=C\,\frac{1-\big(1+\frac{r}{m}\big)^{-nm+k}}{\frac{r}{m}}. \tag{2.9}$$

This method is much faster — $O(1)$ instead of $O(k)$ — but it is limited in applications because it makes more assumptions: a level payment, a fixed rate, and no prepayment.

Exercise 2.6.6. Use Eq. (2.9) together with Eq. (2.8) to prove that the Principal column of Table 2.1 forms a geometric sequence.

The same trade-off as in Algorithm 2.2: the closed form is faster, the loop is more general. Which one you may use is decided by the contract, not by the mathematics.

§2.7. Yields and the Internal Rate of Return

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So far the rate was given and we computed a value. The rest of the chapter runs the other way: the price is observed in the market, and we solve for the rate it implies.

§2.7.i. Bond Equivalent Yield and Mortgage Equivalent Yield

The term **yield** denotes the return of investment. Two widely used yields are the **bond equivalent yield** (BEY) and the **mortgage equivalent yield** (MEY). Recall Eq. (2.2), $\mathrm{FV}=\mathrm{PV}\big(1+\frac{r}{m}\big)^{nm}$:

- BEY corresponds to the r above that equates PV with FV when $m=2$.
- MEY corresponds to the r above that equates PV with FV when $m=12$.

Observation . BEY and MEY may differ numerically, but they refer to the same yield — they are the same economic fact quoted under two different compounding conventions, exactly as in Equation 2.5.

Remark 2.7.1. **殖利率**（yield）、**債券等值殖利率**（bond equivalent yield, BEY）、**房貸等值殖利率**（mortgage equivalent yield, MEY）。

上面兩點的意思是：市場給的事實只有「今天付 PV、n 年後拿回 FV」，而年化利率並非事實而是報價——必須先選定一年複利幾次，才解得出

$$r=m\left[\left(\frac{\mathrm{FV}}{\mathrm{PV}}\right)^{1/(nm)}-1\right],$$

其值隨 m 而異。取 $m=2$ 者稱 BEY，取 $m=12$ 者稱 MEY。例如今天付 100、一年後拿回 108.16，同一筆現金流可報價為 8.16%（$m=1$，即**有效年利率**（effective annual rate, EAR））、8.00%（BEY）或 7.87%（MEY）：m 愈大，達成同樣成長所需的**名目年利率**（nominal rate）愈小。

兩個名字來自報價慣例——美國公債每半年付息（$m = 2$），房貸每月繳款（$m = 12$）。因此跨市場比較前務必先換算，否則會誤判：房貸的 8% MEY 相當於債券的 8.13% BEY，兩個 8% 並不一樣。

§2.7.ii. The Internal Rate of Return

Definition 2.7.2. The **internal rate of return** (IRR) is the yield y which equates an investment's PV with its price P,

$$P = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \cdots + \frac{C_n}{(1+y)^n}.$$

Remark 2.7.3. **內部報酬率**（internal rate of return, IRR）；用在債券上時通常稱 **到期殖利率**（yield to maturity, YTM）。

上述定義口語一點的說法是：期初投入 P 元，此後在各個時點分別回收 $C_1, \ldots, C_n$ 元；IRR 就是讓這些回收恰好抵掉當初那筆投入的利率。說得再具體些，令 B_k 表示第 k 期回收之後尚未收回的投資餘額：期初分文未收，故 $B_0 = P$；此後餘額每期先以利率 y 增值，再扣掉當期收到的 C_k。IRR 就是使餘額在最後一期恰好歸零的那個 y：

$$B_0 = P, \qquad B_k = B_{k-1}(1+y) - C_k, \qquad B_n = 0.$$

把這條遞迴展開即回到上面的定義式。之所以稱為「內部」，是因為算它只用到交易本身的現金流，不必從外部指定折現率。

Table 2.1 正是這條遞迴的逐期紀錄：餘額每月以 0.08/12 滾息、再減去 2,389.13 元，到第 180 個月剛好歸零。所以 8% 就是那筆房貸對放款人而言的 IRR——我們當初是先給定利率去求月付款，而 IRR 是同一條式子反過來問：給定價格與現金流，利率是多少？

IRR 的好處是自動把時間價值算進去。例如今天付 1,000、往後三年各收回 400，解 $1000 = \sum_{k=1}^{3} 400(1+y)^{-k}$ 得 $y = 9.70\%$；若同樣的 1,200 元要到第三年才一次收到，則只有 $1.2^{1/3} - 1 = 6.27\%$。總金額相同，但早收到的錢報酬率高得多。

惟下述再投資假設不可輕忽：承上例，若中途收回的 400 只能以 3% 再投資，實際年化報酬僅 7.33%，而非宣稱的 9.70%。

最後，y 在定義式中同時出現在 n 個不同的次方裡，求解 y 等同於解一個 n 次多項式的根，一般沒有公式解，只能用 Section 2.8 的數值方法逼近。

Remark 2.7.4. The IRR assumes all cash flows are reinvested at the *same* rate as the internal rate of return, because

$$\mathrm{FV} = C_1(1+y)^{n-1} + C_2(1+y)^{n-2} + \cdots + C_n.$$

So it must be used with caution. The issue disappears when there are no intermediate cash flows — which is one reason zero-coupon bonds (Section 2.9.i) are so convenient.

§2.8. Numerical Methods for IRRs

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§2.8.i. The Root-Finding Problem

Define

$$f(y) \triangleq \sum_{t=1}^{n} \frac{C_t}{(1+y)^t} - P,$$

where P is the market price, and solve $f(y) = 0$ for a real $y \geq -1$.[10]

Proposition 2.8.1

If $C_t > 0$ for all t, then $f(y)$ is monotonically decreasing in y, so a unique real-number solution exists.

There is no closed form for y in general, so we need iterative methods. The three below trade off robustness against speed.

Remark 2.8.2. *Trade off A against B* 意為在 A 與 B 之間權衡取捨：想多要一點 B，就得犧牲一些 A。此處的兩個軸是**穩健性**（robustness，是否保證收斂、對起始值是否敏感）與**速度**（收斂階數），三者的對照見 Table 2.2。

二分法最穩健：只要初始區間滿足 $f(a)f(b) < 0$ 便保證收斂，但每步僅將區間減半，故最慢。牛頓法與割線法則都不保證收斂。例如 $f(x) = x^{1/3}$ 的牛頓疊代為 $x_{k+1} = x_k - 3x_k = -2x_k$，自任何非零起點出發都翻倍發散；而 $f(x) = x^3 - 2x + 2$ 自 $x_0 = 0$ 出發則在 $0 \leftrightarrow 1$ 之間永遠循環，一步都不前進。此外 $f'(x_k) \approx 0$ 時步長會爆炸，多根時亦可能收斂到非預期的那一個；割線法還多一個分母 $f(x_k) - f(x_{k-1})$ 趨零的風險。理論上能保證的只有局部收斂：起點充分接近單根時保證二次收斂，但定理並未告訴你「充分接近」究竟是多近。

不過就 IRR 這個問題而言，Proposition 2.8.1 其實已把風險排除了。當所有 $C_t > 0$ 時 f 不僅單調遞減，還滿足 $f''(y) = \sum_t t(t+1)C_t(1+y)^{-t-2} > 0$ 而為凸函數。凸函數的切線恆在曲線下方，故 $f(x_{k+1}) \geq 0$；又 f 遞減，這表示第一步之後每個迭代點都落在根的左側。此後 $f(x_k) \geq 0$ 而 $f'(x_k) < 0$，步長 $-f/f'$ 恆非負，迭代遂單調遞增且以根為上界，必定收斂。所以該命題不只是在說「根唯一」，更是在替牛頓法背書：對付息全為正的 IRR，牛頓法自任何 $y_0 > -1$ 出發都保證收斂。

實務上的求根程式（如 Brent's method）多半混用：先以二分法將根框住，再切換至割線或牛頓法加速，一旦迭代點跳出括號便退回二分法——保證來自二分法，速度來自另外兩者。

§2.8.ii. The Bisection Method

Start with a and b where $a < b$ and $f(a)\,f(b) < 0$. Then $f(\xi)$ must be zero for some $\xi \in [a, b]$. If we evaluate f at the midpoint $c \triangleq (a+b)/2$, either

(1) $f(c) = 0$,
(2) $f(a)\,f(c) < 0$, or
(3) $f(c)\,f(b) < 0$.

In the first case we are done; in the second we continue the process with the new bracket $[a, c]$; and in the third we continue with $[c, b]$. The bracket is halved in the latter two cases, so after n steps ξ lies within a bracket of length $(b-a)/2^n$.

[10] Negative interest rates became a reality for German and Swiss bonds in 2015. In 2016, Sweden, Denmark, and Japan imposed negative interest rates on excess reserves. As many as 355 corporate bonds were issued with negative yields as of June of 2016. Even so, -100% should be a natural lower bound because why would anyone or financial institutions want to have every cent confiscated?

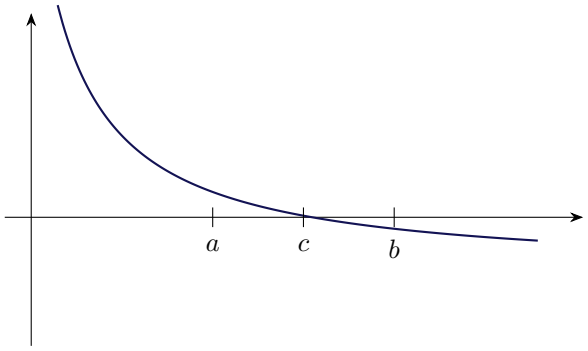


Figure 2.5.: The bisection method. f is decreasing with $f(a) > 0 > f(b)$, so the root is bracketed; evaluating at c halves the bracket.

§2.8.iii. The Newton–Raphson Method

It converges faster than the bisection method. Start with a first approximation x_0 to a root of $f(x) = 0$. Then

$$x_{k+1} \triangleq x_k - \frac{f(x_k)}{f'(x_k)}.$$

When computing yields,

$$f'(x) = -\sum_{t=1}^{n} \frac{t\, C_t}{(1+x)^{t+1}}. \tag{2.10}$$

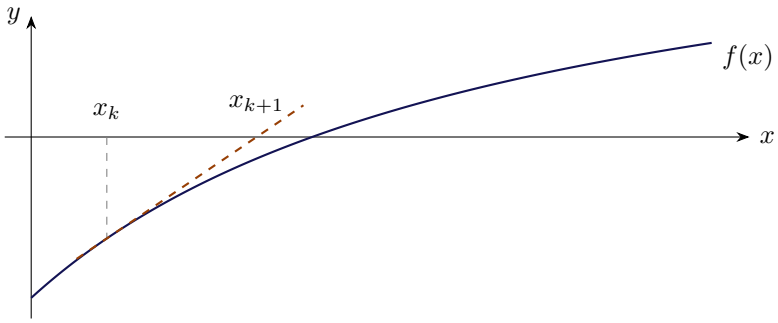


Figure 2.6.: The Newton–Raphson method. The tangent at x_k is followed to the axis to give x_{k+1}.

§2.8.iv. The Secant Method

A variant of the Newton–Raphson method: replace differentiation with difference. Start with two approximations x_0 and x_1, then compute the $(k+1)$st approximation with

$$x_{k+1} = x_k - \frac{f(x_k)\,(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})},$$

so $[\,f(x_k) - f(x_{k-1})\,]/(x_k - x_{k-1})$ replaces $f'(x_k)$.

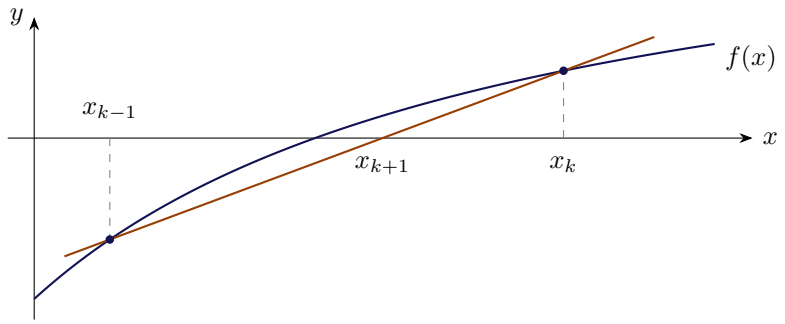


Figure 2.7.: The secant method. The chord through the two latest iterates is followed to the axis to give x_{k+1}; compare Figure 2.6, which follows the tangent at a single iterate instead.

Its convergence rate is 1.618, slightly worse than the Newton–Raphson method's 2. But the secant method does not evaluate $f'(x_k)$, which saves about 50% in computation efforts per iteration. For reference, the convergence rate of the bisection method is 1.

Method	Convergence rate	Needs f'?	Convergence guaranteed?
Bisection	1	no	yes, once bracketed
Secant	1.618	no	no
Newton–Raphson	2	yes	no (see Proposition 2.8.1)

Table 2.2.: The three root-finding methods. Higher rate means fewer iterations, but each Newton–Raphson iteration costs roughly twice a secant one — and the last column is what that speed is bought with. Bisection cannot fail once $f(a)f(b) < 0$; the other two may diverge, cycle, or land on the wrong root.

Remark 2.8.3. **割線法**（secant method）的幾何意義見 Figure 2.7：牛頓法在一個點上作切線、追到 x 軸得下一個近似值，割線法則改取最近兩個點連成的割線，追到 x 軸得下一個近似值——名稱即由此而來。公式也是這樣來的：通過 $(x_{k-1}, f(x_{k-1}))$ 與 $(x_k, f(x_k))$ 的直線斜率為差商 $[f(x_k) - f(x_{k-1})]/(x_k - x_{k-1})$，令該直線的 y 座標為零解出交點，就得到上式。換言之，割線法就是把牛頓法裡的 $f'(x_k)$ 換成差商——用兩點的實際落差去估斜率，而不去要求函數可微或另外寫一份導數程式。

所謂省下約 50% 的計算量，關鍵在於 $f(x_{k-1})$ 在上一輪就已經算過、可直接沿用，故每輪只需一次新的函數求值；牛頓法每輪則要各算一次 f 與 f'，而由 (2.10) 可見算 f' 的成本與算 f 相當。若改以每次函數求值（而非每輪疊代）來衡量效率，牛頓法為 $2^{1/2} \approx 1.414$、割線法為 1.618：割線法反而略勝一籌。

至於 1.618 這個怪數字，來自誤差遞迴 $e_{k+1} \approx C\, e_k e_{k-1}$：設 $e_{k+1} \sim A\, e_k^p$ 代入可得 $p^2 = p + 1$，故 $p = (1+\sqrt{5})/2$，恰為黃金比例。

最後可與二分法對照：兩者都持有兩個點，但二分法只看 f 的正負號並取中點，割線法則利用 f 的數值大小外推。多用了資訊所以更快，代價是失去保證——割線法不保證收斂，且當 $f(x_k) \approx f(x_{k-1})$ 時分母趨近於零，數值上會不穩定。

§2.8.v. Systems of Nonlinear Equations

It is not easy to extend the bisection method to higher dimensions — there is no natural notion of "bracketing" a root in the plane. But the Newton–Raphson method extends directly.

Let (x_k, y_k) be the kth approximation to the solution of the two simultaneous equations

$$f(x, y) = 0, \qquad g(x, y) = 0.$$

The $(k+1)$st approximation (x_{k+1}, y_{k+1}) satisfies the following *linear* equations,

$$\begin{bmatrix} \dfrac{\partial f(x_k,y_k)}{\partial x} & \dfrac{\partial f(x_k,y_k)}{\partial y} \\ \dfrac{\partial g(x_k,y_k)}{\partial x} & \dfrac{\partial g(x_k,y_k)}{\partial y} \end{bmatrix} \begin{bmatrix} \Delta x_{k+1} \\ \Delta y_{k+1} \end{bmatrix} = - \begin{bmatrix} f(x_k,y_k) \\ g(x_k,y_k) \end{bmatrix},$$

with unknowns

$$\Delta x_{k+1} \triangleq x_{k+1} - x_k, \qquad \Delta y_{k+1} \triangleq y_{k+1} - y_k.$$

The above has a unique solution for $(\Delta x_{k+1}, \Delta y_{k+1})$ when the 2×2 matrix — the Jacobian — is invertible. Finally, set

$$x_{k+1} = x_k + \Delta x_{k+1}, \qquad y_{k+1} = y_k + \Delta y_{k+1}.$$

Intuition . Each step replaces the nonlinear system by its linearisation at the current point and solves that exactly. In one dimension the linearisation is the tangent line of Figure 2.6; here it is the tangent plane.

Remark 2.8.4. 這節處理的是「n 條方程式、n 個未知數，且方程式非線性」的情形，前面求 IRR 不過是 $n = 1$ 的特例。金融上最常見的來源是**校準**（calibration）：以市場上觀測到的 n 個價格，反解出模型的 n 個未知參數——例如用兩檔債券的市價反推某個兩參數利率模型的兩個參數。

二分法之所以推不到高維，是因為它的保證來自中間值定理：一維上 f 一變號就把根夾住了。但在平面上 $f(x,y) = 0$ 的解集通常是一條曲線而非一個點，我們要的是它與 $g(x,y) = 0$ 那條曲線的交點；此時並沒有「把區域對半切、且能判斷根落在哪一半」的辦法，**框住根**（bracketing）這個概念本身就不存在。

至於那條矩陣方程式並非憑空而來，它就是一階泰勒展開後令其為零。在 (x_k, y_k) 附近，

$$f(x_{k+1}, y_{k+1}) \approx f(x_k, y_k) + \frac{\partial f}{\partial x}\,\Delta x_{k+1} + \frac{\partial f}{\partial y}\,\Delta y_{k+1},$$

g 亦然。我們希望新的一點恰好是解，於是令兩式右側皆為零、把常數項移到等號另一邊，寫成矩陣形式就是上式。

因此這整套做法與一維的牛頓法其實是同一件事。把該矩陣方程記為 $J\Delta = -F$，則 $\Delta = -J^{-1}F$，亦即 $\mathbf{x}_{k+1} = \mathbf{x}_k - J^{-1}F(\mathbf{x}_k)$：一維的「除以導數 $f'(x_k)$」在多維成了「左乘 Jacobian 的反矩陣」，而一維要求 $f'(x_k) \neq 0$ 也正對應到此處要求 J 可逆。繞這一圈的用意則在於，非線性方程組難解，線性方程組卻可直接以高斯消去法解出。

§2.9. Bonds

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§2.9.i. Zero-Coupon Bonds

Definition 2.9.1. A **zero-coupon bond** (or **pure discount bond**) makes a single payment at maturity. By Eq. (2.1), the price of one that pays F dollars in n periods is

$$F/(1+r)^n, \tag{2.11}$$

where r is the interest rate per period.

Zero-coupon bonds can be used to meet future obligations as there is no reinvestment risk — the caveat attached to the IRR above simply does not arise.

Remark 2.9.2. **零息債券**（zero-coupon bond），亦稱**純折價債券**（pure discount bond）；F 為**面值**（par value，即票面金額），n 為到期期數。口語地說，它就是一張「中途完全不付息、到期一次付你一筆固定金額」的債券——正因為中途沒有任何**票息**（coupon），才叫「零息」。它的報酬全部來自買價與面值之間的價差：由 (2.11) 可知，只要 $r > 0$ 價格便必然低於面值，即必定折價發行，故又稱純折價債券。

上段所謂沒有**再投資風險**（reinvestment risk），指的正是先前 IRR 那條但書在此不成立：中途沒有現金流，就沒有錢需要再投資，也就無從假設再投資利率。因此零息債券的 IRR 就是你實際拿到的報酬，一分不差；這也是為何它適合用來支應未來某個時點的確定支出。

例中 20.83 這個數字採債券市場的報價慣例——以面額 *100* 為基準，故 20.83 意為面額 100 元的債券今天只賣 20.83 元。另可留意 $45.64\% \times 45.64\% \approx 20.83\%$：20 年期的價格恰為 10 年期價格的平方，因為折現是連乘而非相加。

Example 2.9.3. The interest rate is 8% compounded semiannually. A zero-coupon bond that pays the par value 20 years from now will be priced at $1/(1.04)^{40}$, that is, 20.83% of its par value.[11] It will be quoted as 20.83. If the bond matures in 10 years instead of 20, its price would be 45.64.

§2.9.ii. Level-Coupon Bonds

A **level-coupon bond** is described by its **coupon rate** and its **par value**, paid at maturity. Write F for the par value and C for the coupon.

Remark 2.9.4. **固定票息債券**（level-coupon bond，亦稱附息債券）；c 為**票面利率**（coupon rate），F 為**面值**（par value），C 為每期的**票息**（coupon）。口語地說，它就是「每期付你固定的 C 元，到期那一期除了 C 之外再一次還你 F 元」的債券，現金流見 Figure 2.8。留意原文的 *paid at maturity* 修飾的是 par value 而非 coupon rate：票息是每期都付，面值則是到期才一次付清。所謂 *level*（等額）是指每期票息固定不變，以與浮動利率債券相區別。

票面利率與票息的換算為下文的 $C = Fc/m$：c 是年利率，乘上面值得一年的總票息，再除以一年付息次數 m 即為每期票息。例如面值 100、票面利率 8%、每半年付息一次，則 $m = 2$、每期票息 $C = 100 \times 0.08/2 = 4$ 元。

最要緊的是分清 c 與 r 這兩個利率：c 由合約寫死，決定你每期收到多少錢，發行後永不改變；r 則是市場折現率，隨市場浮動，決定這串現金流今天值多少錢。兩者的大小關係恰好決定債券售價高於、等於或低於面值，即**溢價**（premium）、**平價**（par）與**折價**（discount）三種情形：當 $c = r$ 時 $C = Fr/m$，代入 (2.12) 後前項化為 $F\big[1-(1+\frac{r}{m})^{-n}\big]$，與後項 $F(1+\frac{r}{m})^{-n}$ 相加恰得 $P = F$；又 P 對 r 遞減，故

$$c = r \Rightarrow P = F \text{ (平價)}, \qquad c < r \Rightarrow P < F \text{ (折價)}, \qquad c > r \Rightarrow P > F \text{ (溢價)}.$$

而 $c = 0$ 即退化為前一小節的零息債券，價格恰為 (2.11)——零息債券只是本式的特例。

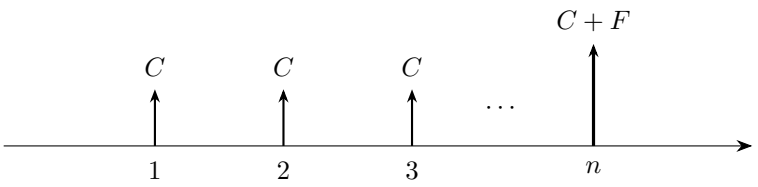


Figure 2.8.: The cash flow of a level-coupon bond.

[11] One fifth!

Remark 2.9.5. Figure 2.8 隱含一項理想化的假設：今天恰好站在付息日上，故各筆現金流剛好落在 $1, 2, \ldots, n$ 這些整數期上。真實的交割日多半落在兩個付息日之間，屆時每一筆現金流都要往前挪，指數由 i 變成 $\omega + i$——這正是 (2.15) 所處理的情形。

不過要留意，最後一期同時收到 C 與 F 並非理想化，真實債券的最後一次付息本就與本金同日支付。這一點在 (2.15) 中依然成立：末筆票息（$i = n-1$）與 F 的折現指數同為 $\omega + n - 1$，可見放寬假設後兩者仍同時發生。

Coupon bonds can be thought of as a matching package of zero-coupon bonds, at least theoretically.[a]

[a] "You see, Daddy didn't bake the cake, and Daddy isn't the one who gets to eat it. But he gets to slice the cake and hand it out. And when he does, little golden crumbs fall off the cake. And Daddy gets to eat those," wrote Tom Wolfe (1931–2018) in *Bonfire of the Vanities* (1987).

Remark 2.9.6. 這條 moral 的具體意思是：一張 n 期的附息債券，可視為 n 張面額 C、到期日分別為 $1, 2, \ldots, n$ 的零息債券，再加上一張面額 F、到期日為 n 的零息債券，所組成的投資組合。下方的定價公式 (2.12) 正是把這個拆解逐項寫出來：第一項是 n 張票息零息債券的現值之和，第二項則是面值那一張的現值。

所謂 *at least theoretically*，是指這種拆解未必真能在市場上執行；但在美國公債市場它確實可行。自 1985 年起的 **STRIPS** 制度（Separate Trading of Registered Interest and Principal of Securities）允許交易商將一張公債的每一期票息與本金拆開、各自登記為獨立的零息債券單獨買賣，亦允許反向重組回原債券。雙向皆可執行，套利便會迫使「整張債券的價格」等於「各零息碎片價格之和」——這條 moral 因而不只是心算技巧，而是由無套利條件所保證。

這個觀點日後還有一個更要緊的用處：拆開之後每一張零息債券各有自己的到期日，理應各用該期限的即期利率折現。(2.12) 對所有期別一律套用單一的 r，其實是一種簡化，正確的做法是改用一整條利率期限結構。

至於註腳的典故，出自 Tom Wolfe 的小說 *Bonfire of the Vanities*（1987）：書中債券業務員 Sherman McCoy 的妻子向女兒解釋爸爸的工作——蛋糕不是爸爸烤的，也不是爸爸吃的，但由爸爸負責切開、分送給大家；切的時候會掉下金色的碎屑，那些碎屑就歸爸爸。放在此處是一語雙關：交易商正是把債券「切片」（strip）後分銷，自己既不創造現金流也不領取現金流，賺的就是買賣價差那些「碎屑」。

Splitting the cash flow of Figure 2.8 into an annuity of coupons plus a zero-coupon bond for the par value, and applying Eq. (2.7) and Eq. (2.11), gives the pricing formula

$$\begin{aligned} P &= \sum_{i=1}^{n} \frac{C}{\left(1 + \frac{r}{m}\right)^{i}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{n}} \\ &= C\,\frac{1 - \left(1 + \frac{r}{m}\right)^{-n}}{\frac{r}{m}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{n}}. \end{aligned} \tag{2.12}$$

Here n is the number of cash flows, m the number of payments per year, and r the annual rate compounded m times per annum. Note $C = Fc/m$ when c is the annual coupon rate.

Note 2.9.7. Because the sum collapses to a closed form, the price P can be computed in $O(1)$ time — no loop over the n coupons is needed.

§2.9.iii. Yields to Maturity

Definition 2.9.8. The **yield to maturity** (YTM) is the r that satisfies Eq. (2.12) with P being the bond price.

Example 2.9.9. For a 15% BEY, a 10-year bond with a coupon rate of 10% paid semiannually sells for

$$5 \times \frac{1-[1+(0.15/2)]^{-2\times 10}}{0.15/2} + \frac{100}{[1+(0.15/2)]^{2\times 10}} = 74.5138$$

percent of par. So 15% is the yield to maturity if the bond sells for 74.5138.[12]

Remark 2.9.10. **到期殖利率**（yield to maturity, YTM）的名稱可拆成兩半來理解：*yield* 指報酬率，*to maturity* 則指這個報酬率是在「一路持有到到期日」的前提下算出來的。換言之它回答的是：若今天以價格 P 買進、此後每期領取票息、到期領回面值且中途不賣出，這串現金流相當於年利率多少？

之所以要把持有期間寫進名字裡，是因為債券還有別的「殖利率」，各自假設不同的持有期間。**當期收益率**（current yield）$= Cm/P$ 只比較票息收入與價格，完全不管到期時的資本利得或損失——以上例而言其值為 $10/74.5138 = 13.42\%$，低於 YTM 的 15%，差額正是價格將由 74.51 回升至面值 100 的那段資本利得。**贖回殖利率**（yield to call）則假設持有至發行人提前贖回之日而非到期日；若中途賣出，實際得到的又是另一個持有期報酬率。既然數字取決於你假設持有多久，名字裡自然得講明是哪一種。

至於它與前面 IRR 的關係：YTM 就是把 (2.12) 反解 r，亦即債券這筆投資的 IRR，只是債券市場習慣改稱 YTM。因此 IRR 的但書一併繼承：YTM 要成為你實際拿到的報酬，除了持有到期、發行人不違約之外，還須能把每一期收到的票息都以 r 再投資出去。零息債券沒有中途票息、無從再投資，其 YTM 恰為實際報酬——這正是前面說它「沒有再投資風險」的意思。

儘管有這些但書，YTM 仍是債券市場的標準報價，因為它把價格、票息與到期日壓縮成單一數字，使票面利率與年期各異的債券得以直接比較。

§2.9.iv. Price Behavior

Slides pp. 75–79

(1) Direction

Bond prices fall when interest rates rise, and vice versa. This is immediate from Eq. (2.12) — every term is decreasing in r — and yet "only 24 percent answered the question correctly."[13]

Remark 2.9.11. 價格與市場利率反向的道理在於：票息 C 與面值 F 已由合約寫死，市場利率變動改變的不是你會收到多少錢，而是那些錢今天值多少。利率上升時，新發行的債券給得更多，你手上這張給得少的自然得降價才賣得掉；利率下跌則反之。註腳所引 2001 年 CNN 的調查中僅 24% 的人答對，正是因為多數人誤把「利率上升」聯想成「債券更值錢」——真正在變動的是折現率，而非票息。

(2) Premium, par, and discount

A level-coupon bond sells[14]

- at a **premium** (above its par value) when its coupon rate c is above the market interest rate r;
- at **par** (at its par value) when its coupon rate is equal to the market interest rate;
- at a **discount** (below its par value) when its coupon rate is below the market interest rate.

Bonds selling at par, at a premium and at a discount are called **par bonds**, **premium bonds** and **discount bonds** respectively.

[12] Note that the yield 15% exceeds the coupon rate 10%.
[13] CNN, December 21, 2001.
[14] Consult the text for proofs.

Yield (%)	Price (% of par)
7.5	113.37
8.0	108.65
8.5	104.19
9.0	**100.00**
9.5	96.04
10.0	92.31
10.5	88.79

Table 2.3.: A 9% coupon bond. The bond trades at par exactly when the yield equals the coupon rate (bold row), at a premium above, at a discount below.

Remark 2.9.12. **溢價**（premium）、**平價**（par）與**折價**（discount）；依售價分類者分別稱為**平價債券**（par bonds）、**溢價債券**（premium bonds）與**折價債券**（discount bonds）。三者的定義與判準已見 Remark 2.9.4：$c = r$ 時 $P = F$，且 P 對 r 遞減，故 $c > r$ 時售價高於面值（溢價）、$c < r$ 時低於面值（折價）。

以下補上溢價與折價的金額。記 $a \triangleq \left[1-(1+\frac{r}{m})^{-n}\right]/\frac{r}{m}$ 為年金現值因子，則代入 $C = Fc/m$ 後，(2.12) 可寫成 $P = \frac{Fc}{m}\,a + F(1+\frac{r}{m})^{-n}$。在同一式中把 c 換成 r，右側恰化簡為 F，於是有恆等式 $F = \frac{Fr}{m}\,a + F(1+\frac{r}{m})^{-n}$——這正是平價的情形。兩式相減，末項相消，故

$$P - F = \frac{F(c-r)}{m}\,a = \sum_{i=1}^{n} \frac{\frac{F(c-r)}{m}}{\left(1+\frac{r}{m}\right)^{i}},$$

末式是把 $a = \sum_{i=1}^{n}(1+\frac{r}{m})^{-i}$ 展開的結果，意思一目了然：第 i 期你比市場行情多領（或少領）$F(c-r)/m$ 元，把這 n 筆差額各自折現回今天再加總，就是溢價（或折價）的金額。

以 Table 2.3 的 9% 債券驗證：殖利率 8% 時每期多領 0.5 元，$a = 17.292$，故 $P - F = 0.5 \times 17.292 = 8.65$，恰為表中 108.65 與面值 100 之差。另可留意，無論溢價或折價，到期時價格必然回到面值，故溢價債券的價格會隨時間逐步下滑至 F——那是溢價的攤銷，並非虧損。

Remark 2.9.13. 在此說明如何讀 Table 2.3。表中描述的是一張 15 年期、票面利率 $c = 9\%$、每半年付息一次的債券，即 $n = 30$、$m = 2$、每期票息 $C = Fc/m = 4.5$ 元；整張表自始至終都是這同一張債券，票面利率固定為 9% 不變。變動的只有左欄的市場殖利率 r，右欄則是把該 r 代入 (2.12) 算得的價格，並依債券市場慣例以面額 100 為基準表示（故 100.00 即等於面值）。例如第一列 $r = 7.5\%$：$a = \left[1-(1.0375)^{-30}\right]/0.0375 = 17.829$，$P = 4.5 \times 17.829 + 100 \times (1.0375)^{-30} = 113.37$，與表中相符。粗體列即 $r = c = 9\%$ 的平價情形；其上方各列殖利率較低故溢價，下方各列殖利率較高故折價——留意表格說明裡的 *above* ／ *below* 指的是粗體列的上下方，而非殖利率的高低。這七列也就是 Figure 2.9 曲線上標出的七個點。

(3) Convexity

The price–yield relationship is not merely decreasing; it is *convex*. Concretely, take any yield change of size $\Delta > 0$. The price *gain* from a fall of Δ is larger than the price *loss* from a rise of that same Δ,

$$P(r-\Delta) - P(r) \;>\; P(r) - P(r+\Delta),$$

and the gap widens as Δ grows.

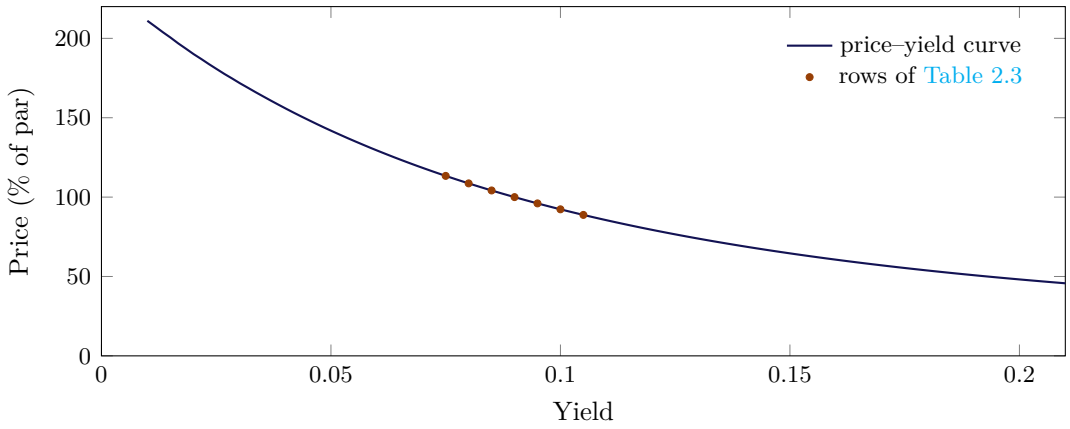


Figure 2.9.: Convexity of the price–yield curve, for the 15-year, 9% coupon bond paid semiannually that generates Table 2.3. The seven tabulated prices are marked; they lie on the curve.

Remark 2.9.14. 凸性即 $\partial^2 P/\partial r^2 > 0$：(2.12) 的每一項皆形如 $A\big(1+\frac{r}{m}\big)^{-i}$，其二階導數 $\frac{i(i+1)A}{m^2}\big(1+\frac{r}{m}\big)^{-i-2} > 0$，逐項為凸，總和自然為凸。其實際意涵是漲跌不對稱：由 Table 2.3，殖利率自 9% 降至 8% 價格漲 8.65，升至 10% 卻只跌 7.69，且幅度愈大這個不對稱愈明顯。換言之利率無論往哪個方向變動同樣幅度，持有人賺的總是多於賠的，故凸性本身即具價值，市場會為凸性較高的債券付出較高的價格——這是日後存續期間與凸性避險的基礎。

§2.10. Day Count Conventions

Slides pp. 80–83

Every formula above assumed cash flows land on exact period boundaries. Real bonds settle on arbitrary calendar dates, so we need a rule converting two dates into a number of days. The rule is a *convention*, not a fact, and different markets use different ones.

§2.10.i. Actual/Actual

The first "actual" refers to the actual number of days in a month; the second refers to the actual number of days in a year.

Example 2.10.1. The number of days between June 17, 1992, and October 1, 1992, is 106: 13 days in June, 31 days in July, 31 days in August, 30 days in September, and 1 day in October.

Remark 2.10.2. 此處的「幾天」指的是兩個日期之間經過的天數，即「結束日減起始日」，而非兩端所涵蓋的日曆日數。以上例而言，6 月只算 13 天（6/18 至 6/30，不含起始日 6/17），10 月則算 1 天（含結束日 10/1），故為「不算頭、算尾」。

不過要留意，改採「算頭不算尾」會得到同一個答案（6 月 14 天、10 月 0 天，合計仍是 106），因為兩者算的都是半開區間的長度。真正會算錯的是兩端都算（得 107）與兩端都不算（得 105）。

之所以採此慣例，是為了讓天數具有**可加性**（additivity）：任取中間日期 B，恆有「A 到 B」的天數加「B 到 C」的天數等於「A 到 C」的天數。若兩端都算，銜接點 B 便會被重複計入，串接多段期間時就會逐段多算一天。下一小節的 (2.13) 也正是「結束日減起始日」的形式，可見兩種慣例在這一點上完全一致——差別只在每月、每年各算幾天，不在端點怎麼取。

§2.10.ii. 30/360

Each month has 30 days and each year 360 days. In general, the number of days from date (y_1, m_1, d_1) to date (y_2, m_2, d_2) is

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1). \tag{2.13}$$

Example 2.10.3. The number of days between June 17, 1992, and October 1, 1992, is now 104: 13 days in June, 30 days in July, 30 days in August, 30 days in September, and 1 day in October.

Remark 2.10.4. (2.13) 看似繁瑣，其實只是「把每個日期換算成一個序號，再相減」。既然每月一律 30 天、每年一律 360 天，日期 (y, m, d) 在這套虛擬曆法上的序號就是 $360y + 30m + d$；兩個日期的序號相減即得該式，三項分別是「差幾年 $\times 360$」、「再差幾個月 $\times 30$」與「最後差幾號」。

關鍵在於後兩項允許為負，不必手動借位。以上例的 6/17 至 10/1 為例，$30 \times (10 - 6) + (1 - 17) = 120 - 16 = 104$；那個 -16 看似奇怪，但代數已自動處理了跨月的借位。

如此的好處是完全不必查曆法——不管某月是 28、29、30 還是 31 天，任兩日期的間隔都能純以算術求得，且每個「月」所孳生的利息一律相等。代價則是這套曆法裡沒有 31 號，故 d_1 或 d_2 為 31 時須先改成 30，即下文的調整。

If d_1 or d_2 is 31, we must change it to 30 *before* applying formula (2.13).[15] Hence:

- There are 3 days between February 28 and March 1.
- There are 2 days between February 29 and March 1.
- There are 29 days between March 1 and March 31.

Remark 2.10.5. An equivalent formula to (2.13) without any adjustment is (check it)

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1 - 1) + \max(30 - d_1,\, 0) + \min(d_2,\, 30).$$

There are many variations on the "30/360" convention regarding 31, February 28, and February 29.[16]

Remark 2.10.6. 上式的四項各有明確意義，合起來正是「起始月的殘餘 + 中間的整月 + 結束月已過的部分」：

- $\max(30 - d_1,\, 0)$：起始月從 d_1 號到月底還剩幾天。若 $d_1 = 31$ 則為 0，等於把 31 號視同月底。
- $30 \times (m_2 - m_1 - 1)$：夾在起訖兩月之間的完整月份，每月 30 天。
- $\min(d_2,\, 30)$：結束月從月初到 d_2 號已過幾天。若 $d_2 = 31$ 則取 30。
- $360 \times (y_2 - y_1)$：整年的部分。

以 6/17 至 10/1 驗證：$\max(30 - 17, 0) = 13$（6 月）、$30 \times (10 - 6 - 1) = 90$（7、8、9 三個整月）、$\min(1, 30) = 1$（10 月），合計 104——恰好就是前例「13 天 + 30 + 30 + 30 + 1 天」那句話的代數寫法。

所謂 *without any adjustment*，是指 31 號的處理已被 max 與 min 吸收，不必如前文那樣先改數字再套公式，因此可直接寫成程式。至於 *check it*：令 $d' \triangleq \min(d,\, 30)$ 表示調整後的日，並注意恆等式 $\max(30 - d_1,\, 0) = 30 - d_1'$，則

$$\underbrace{30(m_2 - m_1 - 1)}_{\text{少算了一個月}} + \underbrace{(30 - d_1')}_{\max(30-d_1,0)} + \underbrace{d_2'}_{\min(d_2,30)} = 30(m_2 - m_1) + (d_2' - d_1'),$$

右側加上 $360(y_2 - y_1)$ 後正是調整後的 (2.13)，故兩式等價。

末句提醒的變體差異不可輕忽：本節採用的是歐式的 30E/360，對 d_1 與 d_2 一律截至 30，規則對稱；而美式的 30/360（Bond Basis）在 $d_2 = 31$ 但 $d_1 < 30$ 時並不把 d_2 改為 30，對二月底的處理也另有規定，算出的天數因而可能不同。故實務上合約必須載明採用哪一種慣例，否則利息金額會對不起來。

[15] This is the simplest of all the "30/360" variations (called the "30E/360" convention), used mainly in the Eurobond market (Kosowski & Neftci, 2015).

[16] Kosowski & Neftci (2015).

§2.11. Full Price, Accrued Interest, and the Clean Price

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§2.11.i. The Full Price

In reality, the settlement date may fall on any day between two coupon payment dates. Let

$$\omega \triangleq \frac{\text{no. of days between the settlement and the next coupon payment date}}{\text{no. of days in the coupon period}}. \tag{2.14}$$

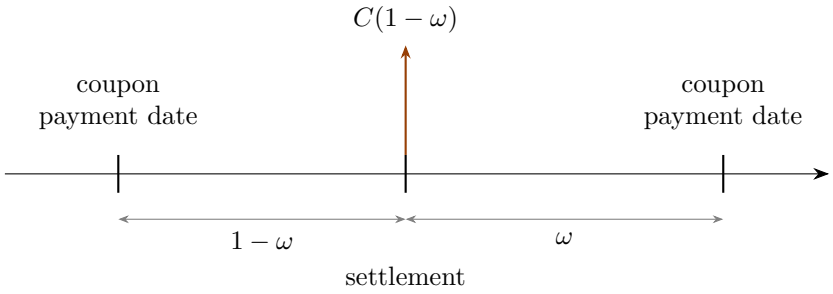


Figure 2.10.: A settlement date falling between two coupon payment dates. ω is the fraction of the coupon period still to run.

Discounting each remaining cash flow over its true (fractional) number of periods, the price is now calculated by

$$\begin{aligned} \text{PV} &= \frac{C}{\left(1+\frac{r}{m}\right)^{\omega}} + \frac{C}{\left(1+\frac{r}{m}\right)^{\omega+1}} + \cdots \\ &= \sum_{i=0}^{n-1} \frac{C}{\left(1+\frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1+\frac{r}{m}\right)^{\omega+n-1}}. \end{aligned} \tag{2.15}$$

This is the **full price**, also called the **dirty price** or **invoice price**.

Remark 2.11.1. **交割日**（settlement date）是債券易手之日：賣方交出債券、買方付錢，此後所有尚未發生的現金流都歸買方。於是問題成了「這些剩餘現金流在交割日這一刻值多少」，而 (2.15) 算的正是這個數，稱為**全價**（full price），亦稱**含息價**（dirty price）或**發票價格**（invoice price）。

麻煩在於交割日多半不落在**付息日**（coupon payment date）上，各筆現金流因而不再位於整數期。(2.14) 的 ω 正是用來度量這個零頭：分子是交割日到下一個付息日的天數，分母是整個付息期間的天數，故 ω 表示「本期還剩多少比例沒走完」，$1-\omega$ 則是已走完的部分（見 Figure 2.10）。

有了 ω，(2.15) 做的事便與 (2.12) 相同——把每筆剩餘現金流各自折現回交割日再加總——只是每筆的期數都多帶一個 ω。$i=0$ 是最近的那筆票息，只需再等 ω 期；$i=1,\dots,n-1$ 是其後各筆，每筆再晚一整期；末筆票息與 F 同日支付，故指數同為 $\omega+n-1$。取 $\omega=1$（交割日恰在付息日之後）即化回 (2.12)，可見後者只是本式的特例。

以 Example 2.11.4 的數字（$C=5$、$F=100$、$n=4$、$\frac{r}{m}=0.015$、$\omega=1/3$）逐項展開：

i	距今期數 $\omega+i$	現金流	折現值
0	1/3	5	4.9752
1	4/3	5	4.9017
2	7/3	5	4.8293
3	10/3	5	4.7579
–	10/3	100	95.1583
		合計	**114.6225**

而淨價加應計利息為 $111.2891 + 3.3333 = 114.6224$，兩者相符（尾差來自四捨五入）——這就是該例所謂「可用 (2.15) 驗證殖利率為 3%」的意思。

§2.11.ii. Accrued Interest and the Clean Price

The quoted price in the U.S./U.K. does *not* include the accrued interest; it is called the **clean price** or **flat price**. The buyer pays the invoice price: the quoted price *plus* the **accrued interest** (AI). AI equals

$$C \times \frac{\text{no. of days from the last coupon payment to the settlement date}}{\text{no. of days in the coupon period}} = C \times (1-\omega).$$

Hence the yield to maturity is the r satisfying Eq. (2.15) when PV is the invoice price:

$$\text{clean price} + \text{AI} = \sum_{i=0}^{n-1} \frac{C}{\left(1+\frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1+\frac{r}{m}\right)^{\omega+n-1}}.$$

Intuition . The clean price is quoted precisely so that it does *not* saw-tooth: the full price jumps down by C on every coupon date, and stripping out the accrued interest removes that artefact, leaving a number that moves only with the yield.

Remark 2.11.2. invoice price 是整個 bond 剩餘的價值，包含賣方還沒領到的那一份 coupon，所以 coupon 應該照比例 $(1-\omega)$ 分給賣方，也就是說 full price 扣掉 AI (分給賣方的利息) 才是 clean price (對買方來說 bond 對他真正有的價值) 。

Remark 2.11.3. 先說清楚**全價**（full price）是什麼：它就是 (2.15) 算出的那個數，也就是買方在交割日實際要掏出來的錢。它的毛病在於，縱使殖利率與債券本身都毫無變化，這個數字仍會隨付息週期上下起伏。以 $F=100$、$c=r=10\%$、$m=2$（故 $C=5$）為例，全價恰為 $100\times 1.05^{1-\omega}$：剛付完息時（$\omega=1$）為 100，其後隨下一筆票息逼近而一路爬升，至付息前夕（$\omega\to 0$）達 105；票息一經發放，該筆現金流即自剩餘現金流中消失，全價隨之驟降回 100。如此週而復始，全價的時間序列遂呈鋸齒狀。於是看到「甲債券報 105、乙債券報 100」時，你分不出甲究竟真的比較貴，還是只是離付息日比較近而已。

市場的對策是把全價拆成兩塊：隨持有天數線性累積的那部分稱為**應計利息**（accrued interest, AI），扣掉它之後的餘額稱為**淨價**（clean price），亦稱 flat price——注意此處的 flat 與先前的平價（par）毫無關係，切勿混淆。承上例，淨價 $= 100\times 1.05^{1-\omega} - 5(1-\omega)$，整個付息週期內都停在 100 附近（$\omega=1/2$ 時為 99.97），鋸齒已被濾掉。美、英市場報的就是淨價，它只隨殖利率與信用狀況變動。

AI 之所以等於 $C\times(1-\omega)$，是因為票息按持有天數比例分攤：賣方持有了本期的 $1-\omega$，就該分得這一期票息的 $1-\omega$。但整筆 C 會在下一個付息日全額付給買方，故買方須在交割時先把屬於賣方的那一份補上——這便是買方實付**發票價格**（invoice price）= 報價 + AI 的由來，而它正好等於全價。

於是最容易出錯的一點是：反解殖利率時，(2.15) 左邊的 PV 必須填發票價格而非報價，這正是本小節末那條 “clean price + AI $=\cdots$” 的用意。

§2.11.iii. Example ("30/360")

Example 2.11.4. A bond with a 10% coupon rate and paying interest semiannually, with clean price 111.2891. The settlement date is July 1, 1993, and the maturity date is March 1, 1995.

There are 60 days between July 1, 1993, and the next coupon date, September 1, 1993, so $\omega = 60/180$. The accrued interest is

$$(10/2) \times \left(1 - \tfrac{60}{180}\right) = 3.3333$$

per \$100 of par value.

The yield to maturity is 3%. This can be verified by Eq. (2.15) with $\omega = 60/180$, $n = 4$, $m = 2$, $F = 100$, $C = 5$, PV $= 111.2891 + 3.3333$ and $r = 0.03$.

Remark 2.11.5. 疑問：例中驗證 YTM 時，不由分說直接取 $F = 100$；但若不知道這張債券的面額究竟是多少，YTM 不是就算不出來了嗎？

先說 $F = 100$ 是怎麼來的。它並非假設這張債券的面額恰好是 100 元，而是因為債券價格一律以面額的百分比報價：所謂 clean price 111.2891 意即「面額的 111.2891%」，而例中也明言 AI 是 *per \$100 of par value*。本章三處都採此慣例——零息債券例的 "quoted as 20.83"、Table 2.3 的欄位 "% of par"，以及本例。取 $F = 100$ 只是挑一個讓報價直接等於數值的單位罷了。

更要緊的是，YTM 根本不依賴面額。把 (2.15) 兩邊同除以 F 並代入 $C = Fc/m$，得

$$\frac{\text{PV}}{F} = \sum_{i=0}^{n-1} \frac{c/m}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{1}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}},$$

式中 F 已完全消失，只剩「價格佔面額的比例」、票面利率 c、ω 與 n。面額加倍時票息與價格同步加倍，解出的 r 不變；故你不必知道面額究竟是 100、1,000 還是 1,000,000，只要知道價格相對於面額是多少——而那正是報價所提供的資訊。

真正需要用到實際面額的只有結算金額：買方應匯出 $\frac{\text{報價} + \text{AI}}{100} \times$ 實際面額。以本例而言，若買的是 1,000,000 面額，實付 $1.146224 \times 1{,}000{,}000 = 1{,}146{,}224$ 元，但殖利率仍是 3%。

§2.11.iv. Price Behavior (2) Revisited

Previously we said a bond sells at par if the yield to maturity equals the coupon rate. But that assumed the settlement date is *on* a coupon payment date.

Suppose the settlement date for a bond selling at par[17] falls between two coupon payment dates. Then its yield to maturity is *less* than the coupon rate.[18] The reason: exponential growth *to* C is replaced by linear growth, hence overpaying the accrued interest.

Remark 2.11.6. 疑問：前面才說「殖利率等於票面利率 $\Leftrightarrow$ 賣在平價」，但這只在交割日是在付息日才會是對的，為何交割日一落在兩個付息日之間，平價的債券殖利率反而低於票面利率？原文只丟下一句「指數成長被線性成長取代」與「見課本習題」。

情境：關鍵在「賣在平價」講的是哪個價——註腳已註明是報價（即淨價）等於面值，但買方實付的是淨價加 AI。於是真正該問的是：這筆實付金額，與「殖利率真的等於 c 時本應支付的金額」相比，究竟是多還是少？

脈絡：先講清楚前提——「賣在平價」是假設而非推導結果，註腳已言明它指報價（淨價）等於面值。於是買方實付

$$\underbrace{F}_{\text{淨價（依假設等於面額）}} + \underbrace{C(1-\omega)}_{\text{AI}} = F + F\tfrac{c}{m}(1-\omega) = F\left[1 + \tfrac{c}{m}(1-\omega)\right],$$

[17] The *quoted price* equals the par value.
[18] See Exercise 3.5.6 of the textbook for proof.

此處用到票息 $C = Fc/m$ 由合約決定、與殖利率無關。（留意 F 在此身兼二職：它既是 (2.15) 裡的面額，又因假設而等於報價。）

接著問：若殖利率真的等於票面利率 c，本應付多少？把 (2.15) 提出 $(1+\frac{c}{m})^{1-\omega}$，得

$$\mathrm{PV} = \left(1+\tfrac{c}{m}\right)^{1-\omega}\left[\sum_{k=1}^{n}\frac{C}{\left(1+\frac{c}{m}\right)^{k}} + \frac{F}{\left(1+\frac{c}{m}\right)^{n}}\right],$$

中括號恰為 (2.12) 在殖利率等於 c 時的值，即 F；故本應支付 $F(1+\frac{c}{m})^{1-\omega}$。

令 $x \triangleq 1-\omega$ 表示本期已走完的比例，兩者遂可並排寫成

$$\underbrace{F\left(1+\tfrac{c}{m}\right)^{x}}_{\text{本應支付}} \qquad \text{與} \qquad \underbrace{F\left[1+\tfrac{c}{m}\,x\right]}_{\text{慣例實付}}.$$

後者正是前者在 $x=0$ 與 $x=1$ 兩端之間的直線內插——兩式在 $x=0$ 時同為 F（剛付完息，AI $=0$），在 $x=1$ 時同為 $F(1+\frac{c}{m})$（付息前夕，AI $=C$），兩端完全吻合，差別只在中間。而 $(1+\frac{c}{m})^{x}$ 為凸函數，凸函數的圖形在兩端點之間必位於該連線之下（見 Figure 2.11），故對 $0<x<1$ 恆有

$$\left(1+\tfrac{c}{m}\right)^{x} \;<\; 1+\tfrac{c}{m}\,x.$$

亦即實付金額高於「殖利率等於 c 時應付的金額」；又價格對殖利率遞減，買方付得愈多代表殖利率愈低，故實際殖利率必然低於票面利率。

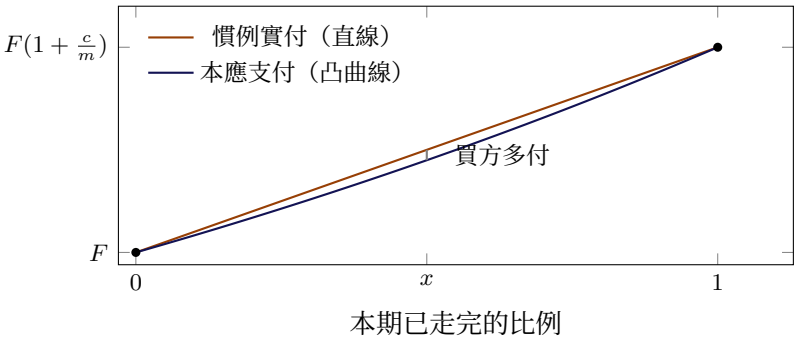


Figure 2.11.: 慣例的線性累積與正確的指數累積。兩者在期初與期末完全一致，中間則因直線位於凸曲線之上而使買方略為多付。（為看得清楚，圖中的 c/m 遠大於實際。）

代個數字看規模——$F=100$、$c=10\%$、$m=2$、$\omega=1/2$ 時，本應付 $100\times 1.05^{0.5} = 102.4695$，慣例卻要付 $100+2.5=102.5$，多付約 0.03；偏差在期中最大，兩端則歸零。

歸根結柢，這是 AI 採「按天數線性分攤」所付出的代價。該約定之所以線性，是為了讓買賣雙方不必對任何利率有共識就能算出同一個數字（公式裡根本沒有 r）；換來的偏差就記在這裡。這也正是下方 moral 的意思：平價、溢價、折價那些漂亮的敘述，都是「交割日恰在付息日上」這個乾淨模型底下的結論。

The tidy statements about premium, par and discount are statements about a model with clean period boundaries. Day count conventions are where that model meets the calendar — and the tidy statements need amending.

3. Bond Price Volatility

Lecture 2
September 17, 2026

Slides pp. 92–93

Well, Beethoven, what is this?[1]
— Attributed to Prince Anton Esterházy

You mean to tell me that
the success of my program and my re-election
hinges on the Federal Reserve
and a bunch of f–king bond traders?
— Bill Clinton to Robert Rubin, January 7, 1993

The previous chapter answered the question "what is a bond worth?". Given a yield y we can compute a price P, and given a price we can solve back for the yield. This chapter answers the next question, which is the one a risk manager actually loses sleep over: *if the yield moves, how much does the price move?*

The whole chapter is an exercise in Taylor expansion. Write the price as a function $P(y)$ of the yield; then

$$\Delta P \;=\; \frac{\partial P}{\partial y}\,\Delta y \;+\; \frac{1}{2}\frac{\partial^2 P}{\partial y^2}\,(\Delta y)^2 \;+\; \cdots$$

The first-order term, suitably normalised, is called **duration**; the second-order term, suitably normalised, is called **convexity**. Everything else — Macaulay duration, modified duration, effective duration, dollar duration, $D_{\%}$, effective convexity — is one of these two numbers in a different set of units, or computed by a different method. Keeping that map in mind is the best defence against the terminology.

Duration is the first derivative of price with respect to yield; convexity is the second. Everything in this chapter is bookkeeping around those two facts.
存續期間（duration）就是價格對殖利率的一階導數，凸性（convexity）就是二階導數；本章其餘的一切，都只是圍繞這兩件事的記帳工作。

§3.1. Price Volatility

Slides pp. 94–95

Volatility here measures how bond prices respond to interest rate changes. It is the key input to the risk management of interest rate-sensitive securities: a desk that holds $1 billion of bonds needs to know, in dollars, what a one-basis-point move costs it.

The question to make precise is this: *what is the sensitivity of the percentage price change to changes in interest rates?* Note the two deliberate choices hidden in that sentence. We measure the response in *percentage* terms, so that bonds of different sizes can be compared, and we measure it against a change in the *yield*, which is the single number we have agreed to summarise the whole discount curve by.

[1] Mass in C major, Op. 86 (1808).

Definition 3.1.1 (Price volatility)**.** The **price volatility** of a security with price P and yield y is

$$-\frac{\dfrac{\partial P}{\partial y}}{P}. \tag{3.1}$$

Intuitively, for a small yield move Δy,

$$\frac{\Delta P}{P} \;\approx\; -\,\text{price volatility} \times \Delta y. \tag{3.2}$$

Remark 3.1.2. (3.1) 的兩個設計都有理由。

為何要除以 P？ $\partial P/\partial y$ 的單位是「元 per 單位殖利率」，面額 100 的債券和面額 1,000 的同一檔債券會得到差十倍的數字。除以 P 之後得到的是百分比變化對殖利率變化的敏感度，與部位大小無關，才能拿不同債券互相比較。

為何要加負號？價格對殖利率遞減（$\partial P/\partial y < 0$），加上負號後，**價格波動性**（price volatility）對一般債券是正數，講「這檔債券的波動性是 8」比講「是 -8」自然。代價是 (3.2) 裡必須把負號再寫回去。

Note 3.1.3. (3.1) is a *local* statement: it is exact only in the limit $\Delta y \to 0$. How badly it fails for a finite Δy is precisely what convexity, in Section 3.10, will measure.

§3.2. Price Volatility of a Level-Coupon Bond

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For a level-coupon bond the derivative can be computed in closed form. Recall from the previous chapter that with n periods to maturity, a coupon C per period, par value F, and a *periodic* yield y,

$$P \;=\; \frac{C}{y}\Big[\,1-(1+y)^{-n}\,\Big] \;+\; F(1+y)^{-n}. \tag{3.3}$$

Proposition 3.2.1

The price volatility of a level-coupon bond is

$$-\,\frac{(C/y)\,n-\big(C/y^2\big)\big[(1+y)^{n+1}-(1+y)\big]-nF}{(C/y)\big[(1+y)^{n+1}-(1+y)\big]+F(1+y)}\,. \tag{3.4}$$

Here F is the par value and C is the coupon payment per period.

Proof. Differentiate (3.3) term by term:

$$\frac{\partial P}{\partial y} = -\frac{C}{y^2} + \frac{C}{y^2}(1+y)^{-n} + \frac{C}{y}\,n(1+y)^{-n-1} - nF(1+y)^{-n-1}.$$

Multiplying through by $(1+y)^{n+1}$ clears every negative power:

$$(1+y)^{n+1}\frac{\partial P}{\partial y} = \frac{C}{y}\,n - \frac{C}{y^2}\Big[(1+y)^{n+1}-(1+y)\Big] - nF,$$

which is the numerator of (3.4). Multiplying (3.3) by the same factor,

$$(1+y)^{n+1}P = \frac{C}{y}\Big[(1+y)^{n+1}-(1+y)\Big] + F(1+y),$$

which is the denominator. The quotient is therefore $\big(\partial P/\partial y\big)/P$, and (3.4) is its negative, as (3.1) requires. ∎

Two observations about the formula.

- **It simplifies with $C = Fc/m$.** If the bond pays an annual coupon rate c with m payments per year, then $C = Fc/m$, and F divides out of numerator and denominator alike. What is left depends only on c/m, y and n — as it must, since price volatility is a percentage quantity and cannot depend on how big the bond is.
- **It is positive.** For the bond above,

$$-\frac{\partial P/\partial y}{P} > 0,$$

because P is a sum of positive cash flows discounted by $(1+y)^{-i}$, every one of which falls as y rises. Higher yield, lower price: the sign is not an accident but a restatement of the discounting formula.

§3.3. Macaulay Duration

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Definition 3.3.1 (Macaulay duration)**.** The **Macaulay duration**[2] (MD) is a weighted average of the times to an asset's cash flows. The weights are the cash flows' PVs divided by the asset's price. Formally, for cash flows $C_1, \dots, C_n$ at times $1, \dots, n$,

$$\text{MD} \triangleq \frac{1}{P}\sum_{i=1}^{n} \frac{C_i}{(1+y)^i}\, i. \tag{3.5}$$

The definition really is an average. Put

$$w_i \triangleq \frac{1}{P}\frac{C_i}{(1+y)^i},$$

the fraction of today's price that is accounted for by the ith cash flow. Since $P = \sum_i C_i(1+y)^{-i}$ we have $\sum_i w_i = 1$, and (3.5) reads $\text{MD} = \sum_i w_i\, i$. So MD is the *centre of mass* of the cash flows when each is given the weight of its present value.

Remark 3.3.2. **麥考利存續期間**（Macaulay duration, MD）在問一句話：這檔債券的錢，平均要等幾期才拿得到？

i 是第 i 筆現金流距今幾期（不是到期日，到期日是 n）。直接把 $1, \dots, n$ 平均，等於不管每期收到的錢多寡，通通當同一回事；但債券到期那期通常連本金帶票息一起付，金額遠大於中途單純的票息，不該跟其他期一樣輕。所以要用每筆現金流的現值佔總價格的比重 w_i 來加權——金額愈大、時間愈近，權重愈大，MD 就愈被拉向那一期。這就是 (3.5) 是**加權平均**（weighted average）、w_i 叫**現值權重**（PV weight）的原因。

下面例子裡 $w_5 = 0.744$，於是 MD= 4.34：比單純平均的 3 大得多（因為多數價值壓在最後一期），又比到期日 5 略小（因為中途先收到票息）。把每期現值想成掛在時間軸上的砝碼，MD 就是這根槓桿的平衡點，即**重心**（centre of mass），見 Figure 3.1。

這個數字有用，是因為錢等得愈久、折現折得愈兇，價格對殖利率也就愈敏感——所以「平均要等幾期」同時也是「殖利率一動，價格會動多少」的度量，這正是下一小節 (3.6) 要證明的事。

單位是「期」，換算成年要除以每年付息次數 k（Section 3.5）。

[2] Frederick Macaulay (1882–1970) in 1938.

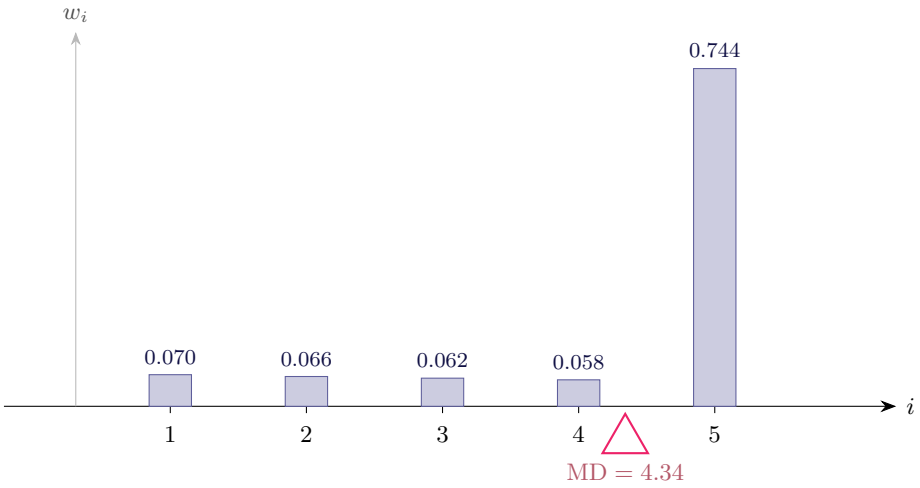


Figure 3.1.: Macaulay duration as a centre of mass. The bars are the PV weights w_i of a five-period bond with $F = 100$, period coupon rate 8% and yield $y = 6\%$; the balance point sits at MD $= 4.34$ periods, short of the maturity $n = 5$ because the four coupons pull it left.

Example 3.3.3 (Not on the slides)**.** A five-period bond with $F = 100$, coupon $C = 8$ per period and periodic yield $y = 6\%$ has

$$P = \sum_{i=1}^{4} \frac{8}{1.06^i} + \frac{108}{1.06^5} = 108.4247,$$

with PV weights $w_1, \ldots, w_5 = 0.0696,\ 0.0657,\ 0.0620,\ 0.0584,\ 0.7443$. Hence

$$\text{MD} = 1(0.0696) + 2(0.0657) + 3(0.0620) + 4(0.0584) + 5(0.7443) = 4.3422 \text{ periods}.$$

Almost three quarters of the price is the final principal repayment, which is why MD sits so close to $n = 5$.

Question 3.3.4. What if $C_i = (1 + c)^i$ for some constant c, letting $n \to \infty$ and assuming $c > y$?[a]

[a] Contributed by Mr. Chen, Yu-Hsing (B06901048, R11922045) on March 3, 2023.

Answer. Write $q \triangleq (1 + c)/(1 + y) > 1$, so that $C_i(1 + y)^{-i} = q^i$. Then

$$P = \sum_{i=1}^{n} q^i = \frac{q(q^n - 1)}{q - 1} \xrightarrow[n\to\infty]{} \infty,$$

and the unnormalised numerator $\sum_{i=1}^{n} i\, q^i$ diverges faster still. Their ratio behaves like n for large n, so MD $\to \infty$ as well.

The example is not a curiosity but a warning. The weighted-average reading of (3.5) presumes that the weights w_i sum to a finite number; when the coupons grow faster than the discount factor shrinks, the price itself is infinite and there is nothing left to average. Any security whose cash flows grow at a rate exceeding its own discount rate has no finite price and no finite duration. ■

§3.3.i. MD as a Measure of Price Volatility

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Here is the fact that makes MD worth defining: it is the bridge between the "average waiting time" picture of (3.5) and the "price sensitivity" picture of (3.1).

Theorem 3.3.5

The Macaulay duration, in periods, equals

$$\text{MD} \;=\; -(1+y)\,\frac{\partial P}{\partial y}\,\frac{1}{P}. \tag{3.6}$$

Proof. From $P = \sum_{i=1}^{n} C_i(1+y)^{-i}$,

$$\frac{\partial P}{\partial y} = -\sum_{i=1}^{n} i\,C_i(1+y)^{-i-1} = -\frac{1}{1+y}\sum_{i=1}^{n} i\,C_i(1+y)^{-i}.$$

Multiply both sides by $-(1+y)/P$:

$$-(1+y)\frac{\partial P}{\partial y}\frac{1}{P} = \frac{1}{P}\sum_{i=1}^{n}\frac{C_i}{(1+y)^i}\,i = \text{MD}. \qquad \blacksquare$$

Intuition . (3.6) says MD $= (1+y) \times$ price volatility. Two quantities defined for entirely different reasons — one a statement about *time*, the other a statement about *sensitivity* — turn out to differ only by the factor $1+y$.

The reason is visible in the proof: differentiating $(1+y)^{-i}$ brings down exactly the factor i, which is the time index. Long cash flows are discounted more steeply, so a security's sensitivity to y is governed by how far out its money sits.

§3.3.ii. MD of a Level-Coupon Bond

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Proposition 3.3.6

The MD of a level-coupon bond is

$$\text{MD} \;=\; \frac{1}{P}\left[\sum_{i=1}^{n}\frac{iC}{(1+y)^i} \;+\; \frac{nF}{(1+y)^n}\right]. \tag{3.7}$$

It can be simplified to

$$\text{MD} \;=\; \frac{c(1+y)\big[(1+y)^n-1\big]+ny(y-c)}{cy\big[(1+y)^n-1\big]+y^2}, \tag{3.8}$$

where c is the *period* coupon rate.

(3.7) is just (3.5) with the cash flows of a level-coupon bond written out: a coupon C in each of periods $1,\dots,n$, plus the par value F at period n.

For (3.8), write $C = Fc$ and let $A(n) \triangleq \sum_{i=1}^{n}(1+y)^{-i} = \big[1-(1+y)^{-n}\big]/y$ be the ordinary annuity factor, so (3.3) reads $P = F\big[\,c\,A(n)+(1+y)^{-n}\,\big]$. The bracket in (3.7) factors the same way:

$$\sum_{i=1}^{n}\frac{iC}{(1+y)^{i}}+\frac{nF}{(1+y)^{n}} \;=\; F\left[c\sum_{i=1}^{n}\frac{i}{(1+y)^{i}}+\frac{n}{(1+y)^{n}}\right].$$

So the numerator and the denominator P of (3.7) carry the *same* factor F, and it cancels — this is why F disappears. C disappears too, but only because it was never independent: it enters solely through $C = Fc$, and once F is gone all that is left of it is the rate c. Summing the remaining arithmetic–geometric series and simplifying gives (3.8), which — exactly as with (3.4) — depends only on c, y and n, never on the bond's size.

Two consequences are worth stating separately.

- **A coupon bond has** MD $< n$**.** If $c > 0$ then some of the weight w_i sits at times $i < n$, so the centre of mass in Figure 3.1 is pulled to the left of n.
- **A zero-coupon bond has** MD $= n$**.** With $c = 0$ there is a single cash flow, $w_n = 1$, and the average of one number is that number. Its duration *is* its term to maturity.

§3.4. What MD Does and Does Not Mean

§3.4.i. When the Formulas Apply

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Note 3.4.1. Formulas (3.6) and (3.7) hold only if the coupon C, the par value F, and the maturity n are all independent of the yield y — that is, only if the cash flow is independent of yields.

This is a far stronger assumption than it looks, and the slides' argument for why deserves unpacking.

Suppose the market yield declines. Mechanically, (3.8) then reports a *longer* MD: lower y discounts the distant cash flows less severely, so the weights w_i shift towards large i and the centre of mass moves right. For a plain bond that is the end of the story, and a longer MD does mean greater price volatility.

But now suppose the security's maturity *itself* shortens when yields fall. Then the actual price volatility, as originally defined in (3.1), may *decrease* at the very moment the formula says the duration lengthened. Formula and reality point in opposite directions.

Remark 3.4.2. 白話翻譯一下上面這段在講什麼。

MD 的公式（(3.6)、(3.7)）都偷偷假設了一件事：C、F、n 這些現金流的內容不會因為 y 變動而跟著變——也就是「不管殖利率變成怎樣，這張債券該付多少錢、哪一期付，永遠不變」。一般公司債、公債確實如此，但**可贖回債券**（callable bond）跟房貸抵押證券不是：殖利率一降，發行人可能提前贖回、屋主可能**提前還款**（prepayment）再融資，現金流本身就跟著 y 一起變了。

現在把「公式算出來的」跟「實際上發生的」分開看：

- **公式算出來的**：套用 (3.8)，殖利率 y 下降 → 折現折得比較輕 → 較晚那幾期的權重 w_i 變大 → MD（加權平均的期數）變長。這一步純粹是代公式，跟債券實際上會不會被贖回無關。
- **實際上發生的**：如果這張債券在殖利率下降時被提前贖回或提前還款，它的壽命反而縮短了。壽命一短，價格對利率的敏感度（(3.1) 那個真正的定義）通常也跟著下降。

為什麼壽命短，敏感度就低？最乾淨的例子是零息債：$P = F(1+y)^{-n}$，算一下 (3.1) 就得到價格波動性 $= n/(1+y)$——敏感度就是到期期數 n 除以 $1+y$，n 愈小，敏感度愈低，一目了然。直覺上也對：n 愈小，這筆錢愈快回到你手上，y 的變動只需要「折現少數幾期」，累積效果自然小；n 愈大，同一個 y 的變動要連乘（或連加，在指數上）更多次，效果就被放大。一般附息債也是同樣道理：Section 3.3.i 已經證明 MD 就是 $(1+y)\times$ 價格波動性，而 MD 本身是現金流到期期數的加權平均，到期日一旦提前，這個加權平均自然跟著縮小，價格波動性也跟著等比例縮小。

§3.4.ii. How Not To Think about MD

Slides p. 101

The MD has its origin in measuring the length of time a bond investment is outstanding, and the name still carries that connotation. But it should be seen mainly as measuring *price volatility*, for the simple reason that the temporal reading breaks down in cases the volatility reading handles without complaint.

- **Duration can be longer than maturity.** A leveraged inverse floater — a note whose coupon is, say, $20\% - 2\times$ the reference rate — behaves like a long position in a fixed-rate bond financed by a short position in a floater. Its price sensitivity can exceed that of *any* fixed-rate bond of the same maturity.
- **Duration can be negative.** An interest-only (IO) strip of a mortgage pool loses value when rates fall, because prepayments cut short the very interest payments it owns. Its price *rises* with yields, so $\partial P/\partial y > 0$ and its duration is negative. Figure 3.2 on p. 49 shows exactly such a price curve.

Neither of these makes sense under the maturity interpretation: no security is outstanding for longer than its maturity, and none is outstanding for a negative length of time. Many, if not most, duration-related terms can only be comprehended as measuring volatility.

Remark 3.4.3. 簡單說：MD 本質上是「利率敏感度」(sensitivity)，「加權平均到期時間」只是現金流固定不變時剛好等於敏感度的副產品；現金流一旦隨 y 變化，MD 就可能跑出時間該有的範圍。

- **槓桿反向浮動利率債券（leveraged inverse floater）**：票息 $= 20\% - 2\times$ 參考利率。等同於「多一張固定利率債（fixed-rate bond）、用借來的錢放空一張浮動利率債（floater）」——浮動債每期重設票息，久期（duration）近乎 0，固定債久期則正常。用槓桿多買固定債、空浮動債對沖，等於把固定債的利率敏感度用槓桿倍數放大，MD 自然能超過同期限的一般定息債。
- **IO strip（interest-only strip，純利息證券）**：只分到房貸池的利息，不分本金。y（連帶房貸利率）一下降，屋主**提前還款**（prepayment，也叫 refinance 轉貸）的比例上升；本金一旦提前還完，後面本該收的利息就直接消失，不只是折現折得少。這個「現金流消失」的損失蓋過折現效果，導致價格跟 y 反向變動，$\partial P/\partial y > 0$，MD 變負。

Read "duration" as "price sensitivity measured in units of time", never as "time".
把 duration 讀成「以時間為單位表示的價格敏感度」，而不是「時間」。

§3.5. Conversion to Annual Units

Slides p. 102

So far everything has been measured in *periods*, with y the periodic yield. Market conventions quote years and annual yields, so both formulas need converting. Let k be the compounding

frequency per annum and let y now denote the *annual* yield, so that the periodic yield is y/k and cash flow i arrives at time i/k years.

For the MD to be year-based, modify (3.7) to

$$\frac{1}{P}\left[\sum_{i=1}^{n}\frac{i}{k}\frac{C}{\left(1+\frac{y}{k}\right)^{i}}+\frac{n}{k}\frac{F}{\left(1+\frac{y}{k}\right)^{n}}\right]. \tag{3.9}$$

Nothing has changed except that each time index i has been re-expressed as i/k years. (n itself is still a count of *periods*, not years; only the time label in the numerator has been converted to i/k and n/k years, while the discounting exponent stays in periods.) Formula (3.6) likewise becomes

$$\text{MD} \;=\; -\Big(1+\frac{y}{k}\Big)\frac{\partial P}{\partial y}\,\frac{1}{P}. \tag{3.10}$$

By definition, then,

$$\text{MD (in years)} \;=\; \frac{\text{MD (in periods)}}{k}.$$

Remark 3.5.1. 注意 MD(in periods) 是指每期利率 $\frac{y}{k}$ 算出來的 md，然後 MD(in year) 只是把時間的單位從期換到年，利率還是一樣。三個式子講的是同一件事，只是單位不同，但實務上最常出錯的就是單位。

(3.9) 這一步很直覺：i/k、n/k 只是把「第 i 期」重新標成「第 i/k 年」，k 是常數可以直接從加總裡拿出來，所以整個式子自然就是 MD(periods)$/k$，沒有懸念。

(3.10) 看起來像是另外冒出來的一個公式，但它不是重新定義出來湊答案的，而是可以從 (3.6) 用連鎖律（chain rule）硬算出來，結果剛好也等於 MD(periods)$/k$。推導如下。

把每期的殖利率記成 $y_p = y/k$（也就是 (3.6) 裡原本那個 y，這裡改名是為了不跟年殖利率 y 搞混）。因為 P 其實是 y_p 的函數，y 只是透過 $y_p = y/k$ 間接影響 P，連鎖律給出

$$\frac{\partial P}{\partial y} \;=\; \frac{\partial P}{\partial y_p}\cdot\frac{\partial y_p}{\partial y} \;=\; \frac{1}{k}\,\frac{\partial P}{\partial y_p}.$$

直覺上：y 動 1 個單位，真正拿去折現的 $y_p = y/k$ 只被拖著動 $1/k$ 個單位，所以價格對「年殖利率 y」的敏感度，天生就比對「期殖利率 y_p」的敏感度小了 k 倍——這就是那個 $1/k$ 的來源，跟前面的係數完全無關。

把這個關係代進 (3.10)：

$$-\Big(1+\frac{y}{k}\Big)\frac{\partial P}{\partial y}\,\frac{1}{P} \;=\; -(1+y_p)\cdot\frac{1}{k}\,\frac{\partial P}{\partial y_p}\cdot\frac{1}{P} \;=\; \frac{1}{k}\underbrace{\left[-(1+y_p)\frac{\partial P}{\partial y_p}\,\frac{1}{P}\right]}_{=\ \text{MD(periods)，即 (3.6)}} \;=\; \frac{\text{MD(periods)}}{k}.$$

注意係數 $\big(1+y/k\big)$ 只是原本的 $(1+y_p)$ 換了個寫法而已，並沒有跟著 k 做任何調整；真正帶出 $1/k$ 的，是求導變數從 y_p 換成 y 時連鎖律多乘出來的那個因子。兩者是獨立的兩件事，只是恰好都跟 k 有關，才會讓最後結果乾淨地等於 MD(periods)$/k$——跟直接把加總式除以 k（(3.9) 的做法）殊途同歸，這正是三個式子彼此一致、講的是同一件事的原因。

§3.6. Modified Duration

Slides p. 103

The factor $1+y$ in (3.6) is a nuisance if what one actually wants is the sensitivity itself. Dividing it out gives the quantity traders quote.

Definition 3.6.1 (Modified duration)**.**

$$\text{modified duration} \;\triangleq\; -\frac{\partial P}{\partial y}\,\frac{1}{P} \;=\; \frac{\text{MD}}{1+y}. \tag{3.11}$$

So modified duration is exactly the price volatility (3.1) of the first section, under a new name. By the Taylor expansion,

$$\text{percent price change} \approx -\text{modified duration} \times \text{yield change}. \tag{3.12}$$

Note 3.6.2. Taylor-expanding P to first order, $P(y+\Delta y) \approx P(y)+\frac{\partial P}{\partial y}\Delta y$; dividing by $P(y)$ and substituting (3.11) gives exactly (3.12) — the same computation that produced (3.2). "price volatility" and "modified duration" are two names for that one term.

Note 3.6.3. Modified duration equals MD under continuous compounding. With continuous compounding $P = \sum_i C_i e^{-yt_i}$, so $\partial P/\partial y = -\sum_i t_i C_i e^{-yt_i}$ and the ratio $-\big(\partial P/\partial y\big)/P$ is the PV-weighted average of the t_i with no correction factor at all. The $1+y$ in (3.11) is an artifact of discrete compounding; it is the discrete analogue of the fact that $\frac{d}{dy}e^{-yt} = -t\,e^{-yt}$ exactly, whereas $\frac{d}{dy}(1+y)^{-i} = -i(1+y)^{-i-1}$ picks up one extra factor of $(1+y)^{-1}$.

§3.6.i. Example

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Example 3.6.4. Consider a bond whose modified duration is 11.54 with a yield of 10%. If the yield increases instantaneously from 10% to 10.1%, the approximate percentage price change is

$$-11.54 \times 0.001 = -0.01154 = -1.154\%.$$

Remark 3.6.5. 注意單位：殖利率從 10% 升到 10.1%，變化量是 0.1 個百分點，也就是 $\Delta y = 0.001$，即十個**基點**（basis point, bp），不是 0.1。

§3.6.ii. Modified Duration of a Portfolio

Slides p. 105

Proposition 3.6.6

By calculus, the modified duration of a portfolio equals

$$\sum_i \omega_i D_i, \tag{3.13}$$

where D_i is the modified duration of the ith asset and ω_i is the market value of that asset expressed as a percentage of the market value of the portfolio.

Proof. Let the portfolio value be $V = \sum_i P_i$ and $\omega_i = P_i/V$. Then

$$-\frac{1}{V}\frac{\partial V}{\partial y} = -\frac{1}{V}\sum_i \frac{\partial P_i}{\partial y} = \sum_i \frac{P_i}{V}\left(-\frac{1}{P_i}\frac{\partial P_i}{\partial y}\right) = \sum_i \omega_i D_i. \qquad \blacksquare$$

§3.7. Effective Duration

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Yield changes may alter the cash flow, or the cash flow may be too complex for simple formulas — exactly the callable bonds and mortgage securities that broke (3.7) in Section 3.4. We need a general numerical formula for volatility, one that asks the pricing model rather than a closed form.

Definition 3.7.1 (Effective duration)**.**

$$\text{effective duration} \triangleq \frac{P_- - P_+}{P_0\,(y_+ - y_-)}, \tag{3.14}$$

where P_- is the price if the yield is decreased by Δy, P_+ is the price if the yield is increased by Δy, P_0 is the initial price, y is the initial yield, and Δy is small.

So $y_\pm = y \pm \Delta y$ and $y_+ - y_- = 2\Delta y$: (3.14) is the **central difference** approximation to $-(\partial P/\partial y)/P$. The prices P_-, P_0, P_+ are obtained by re-running the pricing model three times — with the cash flows allowed to change in response to the yield, which is the whole point.

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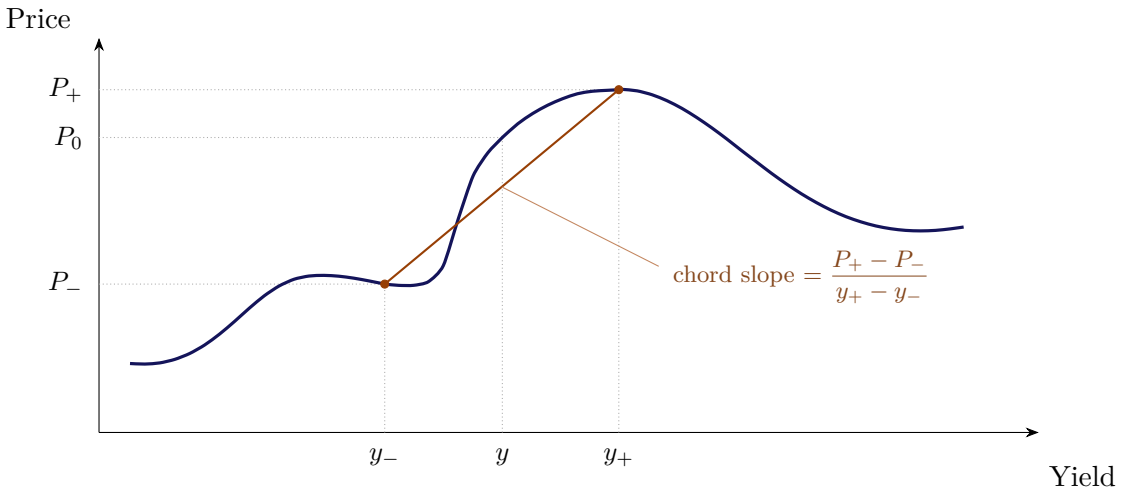


Figure 3.2.: Effective duration reads the slope of the chord through (y_-, P_-) and (y_+, P_+), not of the tangent at y. The curve is traced from the plot shown in class. It is deliberately badly behaved—a plateau on the left, a steep rise through the bracket, and a peak just past y_+—and it is *increasing* in the yield over $[y_-, y_+]$, so this security's duration there is negative. No closed-form duration formula can handle such a price curve.

Remark 3.7.2. 為什麼要定義 effective duration？ (3.7) 的算法是先把價格寫成 y 的封閉公式，再對 y 解析微分。一旦現金流本身會隨 y 改變——可贖回債被贖回、房貸提前還款——這樣的封閉公式根本寫不出來，(3.7) 也就沒有意義（正是 Section 3.4 點出的情形）。effective duration 的辦法是繞開微分：把定價模型當黑箱，讓 y 分別偏移 $\pm\Delta y$、現金流也跟著一起變動，重新定價三次得到 P_-, P_0, P_+，再用連接 (y_-, P_-) 與 (y_+, P_+) 的割線斜率去近似 (y, P_0) 處切線斜率 $-\partial P/\partial y$。也就是說，(3.14) 不管定價公式長什麼樣、甚至有沒有公式，都能算——這正是「一個泛用的數值公式」的意思。(順帶一提：一般中央差分寫成 $(P_+ - P_-)/(y_+ - y_-)$ 去近似 $\partial P/\partial y$；(3.14) 的分子把順序反過來寫成 $P_- - P_+$，分母卻維持 $y_+ - y_-$ 不變，這不是筆誤，而是把 duration 定義裡本來就有的那個負號直接吃進分子，等價於在標準中央差分前面乘上 -1。)

這張圖在講什麼？橫軸是殖利率 y、縱軸是價格 P；虛線標出 (y_-, P_-)、(y, P_0)、(y_+, P_+) 三個點。曲線刻意畫得不規則——左邊一段平台、中間陡然爬升、過了 y_+ 後反而往下掉——用來代表 MBS、**純息分割債券**（interest-only strip, IO）這類現金流會隨 y 改變的證券，這種形狀一般封閉公式完全處理不了。橙色直線是連接 (y_-, P_-) 與 (y_+, P_+) 的割線，它的斜率（取負號、除以 P_0）正是 (3.14) 在算的量。在圖上區間 $[y_-, y_+]$ 裡曲線是遞增的，即 $P_- < P_+$，於是 $P_- - P_+ < 0$，代入 (3.14) 得到 effective duration < 0——這就是 Section 3.4「duration 可以是負的」那個 IO strip 例子的具體樣子。

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One can compute the effective duration of just about any financial instrument: all it takes is a pricing routine that can be called three times. An alternative is to use the one-sided difference

$$\frac{P_0 - P_+}{P_0\,\Delta y}, \tag{3.15}$$

which is more economical — two model runs instead of three, and P_0 is usually already known — but theoretically less accurate.

Note 3.7.3. The accuracy claim is the standard one from numerical differentiation. Expanding $P(y \pm \Delta y)$ about y,

$$\frac{P_- - P_+}{2\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{6}\frac{\partial^3 P}{\partial y^3}(\Delta y)^2 + \cdots, \qquad \frac{P_0 - P_+}{\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{2}\frac{\partial^2 P}{\partial y^2}\,\Delta y + \cdots.$$

The central difference loses the first-order error term to symmetry: its leading error is $O\big((\Delta y)^2\big)$ against $O(\Delta y)$ for the one-sided version. Halving Δy buys a factor of four in one case and only a factor of two in the other.

Remark 3.7.4. P_0 是現在的價格，通常本來就知道；中央差分 (3.14) 分子不含 P_0，真正要多算的是 P_- 與 P_+ 兩次；單邊差分 (3.15) 直接拿現成的 P_0 當基準，只需多算一次 P_+，所以少一次定價。

不準在哪裡：把 y 對稱地往兩邊推，二階（曲率）誤差在 P_- 與 P_+ 裡符號相反、相減即抵消，只剩三階項，誤差 $O((\Delta y)^2)$；只往一個方向推就沒有這種抵消，誤差停在 $O(\Delta y)$。把 Δy 減半，中央差分誤差縮到 1/4、單邊只縮到 1/2——這正是「省一次定價」換來的代價。

§3.8. The Practices: Duration in Percentage Terms

Slides p. 109

Duration is usually expressed in percentage terms — call it $D_{\%}$ — for quick mental calculation.[3] The percentage price change expressed in percentage terms is then approximated by

$$-D_{\%} \times \Delta r \tag{3.16}$$

when the yield increases instantaneously by $\Delta r\%$.

Remark 3.8.1. 先說清楚一個容易誤會的地方：$D_{\%}$ 不是把 D 換算成百分比形式後得到的另一個數字。它只是 (3.16) 這條式子裡的係數；式子裡真正換了單位的是 Δr（百分點）跟價格變化（百分比），不是係數本身。下面會證明 $D_{\%}$ 其實跟原來的 duration 數值完全相同——證完之後的說明會解釋為什麼這件事一點都不奇怪。

Example 3.8.2. Suppose $D_{\%} = 10$ and $\Delta r = 2$. Price will drop by 20%, as $10 \times 2 = 20$.

Exercise 3.8.3. $D_{\%}$ in fact equals the original duration. Prove it!

Proof. Let D be the modified duration, so that $\Delta P/P \approx -D\,\Delta y$ with Δy in decimals. Writing the yield change in percentage points, $\Delta r = 100\,\Delta y$, and the price change in percent, $100\,\Delta P/P$, gives

$$100\,\frac{\Delta P}{P} \;\approx\; -D\,(100\,\Delta y) \;=\; -D\,\Delta r.$$

Comparing with (3.16) yields $D_{\%} = D$. The two factors of 100 — one on each side — cancel exactly, which is why the convention costs nothing. ∎

[3] Neftci (2008): "Market professionals do not like to use decimal points."

Remark 3.8.4. $D_{\%}$ **是哪一種 duration？**不是新的第五種量。前面算出來的那個數字 D——這裡是 modified duration (3.11)，若現金流會隨 y 變就是 effective duration (3.14)——$D_{\%}$ 就是同一個 D，數值完全不變，只是換一套單位讀它：把 Δy（小數）換成以百分點記的 $\Delta r = 100\,\Delta y$，把 $\Delta P/P$（小數）換成以百分比記的 $100\,\Delta P/P$。

§3.9. Hedging

Slides p. 110

Hedging offsets the price fluctuations of the position to be hedged by the hedging instrument in the opposite direction, leaving the total wealth unchanged.

Remark 3.9.1. **部位**（position）指你手上持有的一筆資產或負債；要避險的部位（the position to be hedged）就是你原本就有、想保護價值的那筆持倉，例如一個債券投資組合。避險工具（hedging instrument）是額外配上去的另一筆部位，專門用來抵銷前者的風險，通常是另一種對 y 敏感度相反的證券。

「往相反方向抵銷」的意思是：y 一變動，原部位賺（賠）多少，避險工具那邊就賠（賺）差不多多少；兩筆加在一起，總財富幾乎不受 y 變動影響——這就是 leaving the total wealth unchanged。「差不多多少」要怎麼配、配多少，就是本節接下來要算的東西。

Percentages are the wrong unit for that job: two positions of different size with equal and opposite *modified* durations do not cancel. What must cancel is the change in *dollars*.

Definition 3.9.2 (Dollar duration)**.**

$$\text{dollar duration} \triangleq \text{modified duration} \times \text{price} = -\frac{\partial P}{\partial y}. \tag{3.17}$$

The approximate dollar price change is then

$$\text{price change} \approx -\,\text{dollar duration} \times \text{yield change}. \tag{3.18}$$

One can hedge a bond portfolio with a dollar duration D by bonds with a dollar duration $-D$: the two first-order terms cancel and the combined position is, to first order, insensitive to y.

> **Intuition .** Modified duration is a *rate*; dollar duration is a *quantity*. Hedging is an accounting identity about quantities, so the hedge ratio must be computed in dollar durations.

Note 3.9.3. The hedge is only first-order and only instantaneous. It fails for large moves — that is the second-order term, i.e. convexity — and it fails when the two legs' yields do not move together, which no single-y framework can express.

Remark 3.9.4. 「first-order」「second-order」呼應本章最開頭的泰勒展開 $\Delta P = (\partial P/\partial y)\Delta y + \frac{1}{2}(\partial^2 P/\partial y^2)(\Delta y)^2 + \cdots$：一階項只跟 Δy 成正比，也就是 duration 那一項；二階項跟 $(\Delta y)^2$ 成正比，也就是 convexity 那一項。

怎麼看出避險只做到一階：(3.18) 這條式子本身就只有一階項，完全沒寫 $(\Delta y)^2$ 那一塊。配 D 與 $-D$ 兩筆 dollar duration，只保證兩邊代入 (3.18) 算出來的一階項互相抵銷；至於兩邊真正的二階項（convexity），公式裡從沒要求它們相等，通常也真的不相等，所以加總後還留著一段沒被抵銷的 $(\Delta y)^2$ 尾巴。Δy 很小時這尾巴可以忽略，避險看起來完美；Δy 一大，這段二階尾巴就顯現出來，避險失靈——這就是「fails for large moves」。

至於「兩腳殖利率沒有一起動」：**腳**（leg）就是避險這筆交易的其中一邊——被避險的部位是一腳，拿來避險的工具是另一腳，兩腳各自有自己的價格、自己的殖利率。這是另一個完全不同的限制，跟 order 高低無關：整套 D、$-D$ 的算法都假設這兩腳共用同一個 y（利率平行移動），y 一變兩腳一起變、變一樣多。如果兩腳的殖利率各自變動、不同步（例如一邊是公司債、一邊是公債，信用利差自己在變），就沒有單一個 y 可以同時代入兩腳的一階分析了。

§3.10. Convexity

Slides p. 111

Duration is the first derivative. The second derivative gets its own name.

Definition 3.10.1 (Convexity)**.**

$$\text{convexity (in periods)} \triangleq \frac{\partial^2 P}{\partial y^2}\,\frac{1}{P}. \tag{3.19}$$

Exercise 3.10.2. The convexity of a level-coupon bond is positive. Prove it!

For options, the analogues of duration and convexity are delta and gamma (Section 9.4).

Proof. Write the coupons and the par value as separate terms,

$$P \;=\; \sum_{i=1}^{n} C_i(1+y)^{-i} \;+\; F(1+y)^{-n},$$

where $C_i = C$ is the coupon paid at period i and F is the par value returned at maturity. Differentiating term by term,

$$\frac{\partial P}{\partial y} = -\sum_{i=1}^{n} i\,C_i\,(1+y)^{-i-1} \;-\; nF(1+y)^{-n-1},$$

and once more,

$$\frac{\partial^2 P}{\partial y^2} = \sum_{i=1}^{n} i(i+1)\,C_i\,(1+y)^{-i-2} \;+\; n(n+1)\,F(1+y)^{-n-2} \;>\; 0,$$

since $C_i \geq 0$, $F \geq 0$, not all of them are zero, and $1+y > 0$. Dividing by $P > 0$ leaves the sign unchanged. The argument uses nothing about level coupons: *any* security with non-negative, yield-independent cash flows has positive convexity. ■

For a bond with positive convexity, the price rises more for a rate decline than it falls for a rate increase of equal magnitude. （也就是說，凸性為正的債券，殖利率下降時價格漲的幅度，會大於殖利率上升同樣幅度時價格跌的幅度。）That asymmetry is the content of Figure 3.3: the true price curve bends away from its own tangent, and it bends *upwards*, so the tangent understates the price on both sides.

Slides p. 112

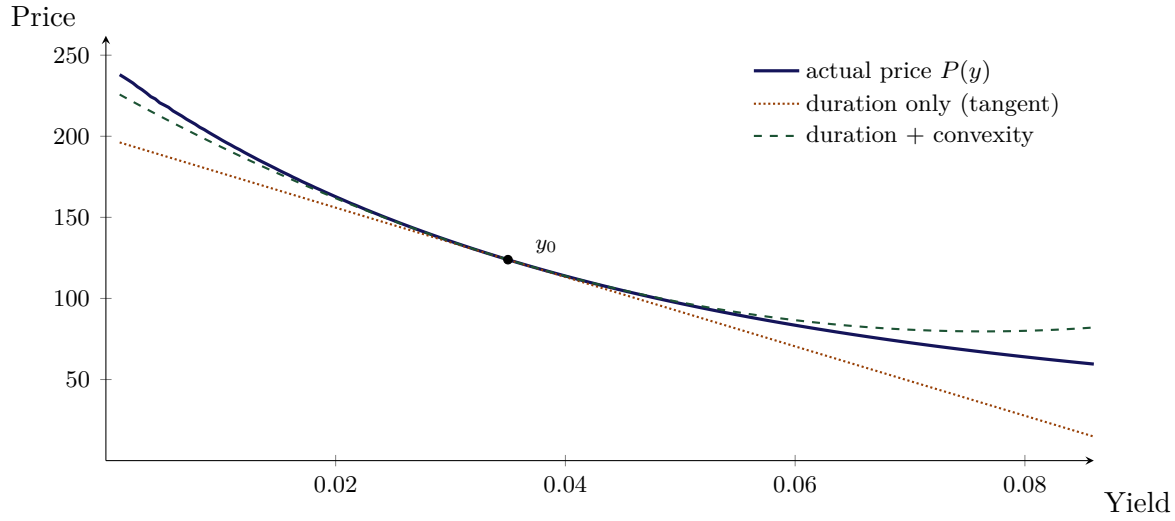


Figure 3.3.: Price versus yield for a level-coupon bond ($n = 30$, $C = 4.8$, $F = 100$), expanded about $y_0 = 3.5\%$. The tangent — the duration-only approximation — lies below the true curve everywhere, understating the gain from a rate fall and overstating the loss from a rate rise. Adding the convexity term recovers almost all of the discrepancy over the plotted range.

Remark 3.10.3. 把這張圖讀成三句話：

- **實線**是真正的價格曲線，對殖利率遞減且向上凸。
- **點線**是只用 duration 的線性近似，也就是曲線在 y_0 的切線。由於曲線是凸的，切線恆在曲線下方：殖利率跌時低估漲幅，殖利率漲時高估跌幅——兩邊都對持有人有利，這就是「正凸性是好事」的圖解。
- **虛線**加進了凸性的二次項，在相當寬的範圍內幾乎與實線重合。

注意切線與真實曲線的落差隨 $|\Delta y|$ 呈平方成長：$\Delta y = 1\%$ 時幾乎看不出來，$\Delta y = 4\%$ 時已經差了幾十塊。這就是為什麼小幅度時可以只用 duration，大幅度時非補上凸性不可。

§3.10.i. Consequences and Units

Slides p. 113

So between two bonds with the same price and duration, the one with a higher convexity is more valuable.[4]

Suppose there are k periods per annum. Convexity measured in periods and convexity measured in years are related by

$$\text{convexity (in years)} \;=\; \frac{\text{convexity (in periods)}}{k^2}. \tag{3.20}$$

Note 3.10.4. This is the chain rule, applied twice. Let $u = y/k$ be the periodic yield, so the price is $P(u)$, and let $\widetilde{P}(y) \triangleq P(y/k)$ be the same price viewed as a function of the annual yield y. Since $u = y/k$ is linear in y, $du/dy = 1/k$ is constant, and

$$\frac{\partial \widetilde{P}}{\partial y} \;=\; \frac{\partial P}{\partial u}\cdot\frac{du}{dy} \;=\; \frac{1}{k}\,\frac{\partial P}{\partial u},$$

[4] Do you spot a problem here (Christensen & Sørensen, 1994)?

which is MD(years) = MD(periods)/k up to the $(1+u)$-vs.-1 normalising factor in (3.6)–(3.10). Differentiating once more, and noting that $\partial P/\partial u$ is again a function of $u = y/k$, the chain rule applies a *second* time:

$$\frac{\partial^2 \widetilde{P}}{\partial y^2} \;=\; \frac{\partial}{\partial y}\left(\frac{1}{k}\,\frac{\partial P}{\partial u}\right) \;=\; \frac{1}{k}\cdot\frac{\partial^2 P}{\partial u^2}\cdot\frac{du}{dy} \;=\; \frac{1}{k^2}\,\frac{\partial^2 P}{\partial u^2}.$$

Dividing both sides by the same P gives exactly (3.20): the k^2 is two factors of k^{-1}, one from each application of the chain rule, not an independent convention.

More generally, this is an induction on the order r of the derivative. The base case $r = 1$ is the MD computation above. For the inductive step, suppose $\partial^r \widetilde{P}/\partial y^r = k^{-r}\,\partial^r P/\partial u^r$. The right-hand side is a function of $u = y/k$ composed with $y \mapsto y/k$, so differentiating both sides once more in y picks up one further factor $du/dy = 1/k$:

$$\frac{\partial^{r+1} \widetilde{P}}{\partial y^{r+1}} \;=\; k^{-r}\cdot\frac{\partial^{r+1} P}{\partial u^{r+1}}\cdot\frac{du}{dy} \;=\; k^{-(r+1)}\,\frac{\partial^{r+1} P}{\partial u^{r+1}}.$$

Hence any quantity that is an rth derivative with respect to the yield converts between periods and years by a factor of k^{-r}: k^{-1} for duration ($r = 1$), k^{-2} for convexity ($r = 2$).

Remark 3.10.5. 那個腳註的「問題」在哪裡？

表面上的推論是：兩檔債券今天價格相同、duration 相同，但一檔凸性較高。殖利率不論漲跌，凸性高的那檔表現都不會比較差（漲得多、跌得少）。既然如此，它應該比較值錢——但它們今天價格明明一樣。這聽起來像是白吃的午餐，也就是**套利**（arbitrage）機會。

Christensen 與 Sørensen（1994）點出的正是這個矛盾：在**無套利**（no-arbitrage）的**期限結構**（term structure）模型裡，這種情形不可能持續存在。真正的機制是凸性要付錢買——市場會把凸性高的債券的殖利率壓低（或說價格推高），高凸性帶來的好處恰好被較低的收益率抵銷。

更根本的問題出在推論的隱含前提：整條殖利率曲線平行移動。這個假設本身就與無套利相容不了（若曲線只能平行移動，就能建構出套利部位）。一旦改用真正的期限結構模型，「同價同 duration 但凸性較高」的兩檔債券根本不會同時存在於市場上。

§3.11. Use of Convexity

Slides p. 114

The approximation $\Delta P/P \approx -\,\text{duration} \times \text{yield change}$ works for small yield changes. For larger yield changes, keep the next term of the Taylor expansion:

$$\begin{aligned}\frac{\Delta P}{P} \;&\approx\; \frac{\partial P}{\partial y}\,\frac{1}{P}\,\Delta y \;+\; \frac{1}{2}\,\frac{\partial^2 P}{\partial y^2}\,\frac{1}{P}\,(\Delta y)^2 \\ &=\; -\,\text{duration} \times \Delta y \;+\; \frac{1}{2} \times \text{convexity} \times (\Delta y)^2.\end{aligned} \tag{3.21}$$

Recall Figure 3.3: the dashed curve *is* (3.21), and the gap between it and the dotted tangent is the convexity term doing its work.

Note 3.11.1. Because convexity is positive for an ordinary bond, the second term is positive whichever way y moves. The correction always *raises* the predicted price relative to the duration-only estimate — which is exactly the asymmetry stated on p. 111 of the slides.

§3.12. The Practices: Convexity in Percentage Terms

Slides p. 115

Convexity is likewise usually expressed in percentage terms — call it $C_{\%}$ — for quick mental calculation. The percentage price change expressed in percentage terms is approximated by

$$-D_{\%} \times \Delta r + C_{\%} \times (\Delta r)^2/2 \tag{3.22}$$

when the yield increases instantaneously by $\Delta r\%$.

Example 3.12.1. Price will drop by 17% if $D_{\%} = 10$, $C_{\%} = 1.5$ and $\Delta r = 2$, because

$$-10 \times 2 + \frac{1}{2} \times 1.5 \times 2^2 = -20 + 3 = -17.$$

Exercise 3.12.2. $C_{\%}$ equals convexity divided by 100. Prove it!

Proof. Start from (3.21) with Δy in decimals and substitute $\Delta y = \Delta r/100$:

$$100\,\frac{\Delta P}{P} \;\approx\; -D\,(100\,\Delta y) + \frac{1}{2}\,\mathrm{conv}\,\frac{(100\,\Delta y)^2}{100} \;=\; -D\,\Delta r + \frac{1}{2}\,\frac{\mathrm{conv}}{100}\,(\Delta r)^2.$$

Comparing with (3.22) gives $D_{\%} = D$ as before and $C_{\%} = \mathrm{conv}/100$. ∎

Remark 3.12.3. $D_{\%} = D$ **這件事哪裡容易搞錯？**直覺上很容易以為「掛了 %」代表這個數字本身被換成了百分比形式，所以 $D_{\%}$ 應該跟 D 長得不一樣（比方乘或除以 100）。但 $D_{\%}$、$C_{\%}$ 從來不是把 D、conv 本身換算單位，它們只是 (3.16)、(3.22) 這兩條式子裡的係數；式子裡真正換單位的是變量：輸入端 $\Delta y \to \Delta r = 100\Delta y$，輸出端 $\Delta P/P \to 100\,\Delta P/P$。係數本身要變成多少，要看它前面到底黏著幾個 Δy。

§3.13. Effective Convexity

Slides p. 116

Just as effective duration replaced the closed-form first derivative by a difference quotient, effective convexity replaces the second.

Definition 3.13.1 (Effective convexity)**.**

$$\text{effective convexity} \;\triangleq\; \frac{P_+ + P_- - 2P_0}{P_0\big(0.5 \times (y_+ - y_-)\big)^2}, \tag{3.23}$$

where P_- is the price if the yield is decreased by Δy, P_+ the price if it is increased by Δy, P_0 the initial price, y the initial yield, and Δy small.

Since $0.5 \times (y_+ - y_-) = \Delta y$, the denominator is just $P_0(\Delta y)^2$, and (3.23) is the standard central second difference $\big[P_+ + P_- - 2P_0\big]/(\Delta y)^2$ normalised by P_0. Effective convexity is most relevant when a bond's cash flow is interest rate sensitive — the callable bonds and mortgage securities for which (3.19) cannot even be written down, and for which the convexity often turns out *negative*.

Remark 3.13.2. 這條式子在算什麼？沿用本章最開頭的泰勒展開 $\Delta P = P'\Delta y + \frac{1}{2}P''(\Delta y)^2 + \frac{1}{6}P'''(\Delta y)^3 + \cdots$（這裡 $'$ 表示對 y 微分），在 y 兩側各展開一次：

$$P_+ = P_0 + P'\Delta y + \tfrac{1}{2}P''(\Delta y)^2 + \tfrac{1}{6}P'''(\Delta y)^3 + \cdots, \qquad P_- = P_0 - P'\Delta y + \tfrac{1}{2}P''(\Delta y)^2 - \tfrac{1}{6}P'''(\Delta y)^3 + \cdots.$$

effective duration 是把這兩式相減，讓一階項 $P'\Delta y$ 留下來、其他項互相抵銷或變高階小量。effective convexity 反過來，把兩式相加：一階、三階這些奇數階的項符號相反、正好消掉，只留下偶數階：

$$P_+ + P_- = 2P_0 + P''(\Delta y)^2 + O\big((\Delta y)^4\big) \implies \frac{P_+ + P_- - 2P_0}{(\Delta y)^2} = P'' + O\big((\Delta y)^2\big).$$

再除以 P_0，換成跟 (3.19) 一樣「除以 P」的正規化，就得到 (3.23)。所以這條公式跟 effective duration 是同一招的延伸：effective duration 用兩點的差逼近一階導數，effective convexity 用三點的加權和逼近二階導數。

幾何意義。 y 是 y_-、y_+ 的中點，所以連接 (y_-, P_-) 與 (y_+, P_+) 的割線，在 y 這一點的高度正好是兩端點的平均 $(P_- + P_+)/2$。因此

$$P_+ + P_- - 2P_0 \;=\; 2\left[\frac{P_- + P_+}{2} - P_0\right]$$

就是「割線在 y 處的高度」與「曲線實際在 y 處的高度 P_0」之間差距的兩倍，見 Figure 3.4。曲線向上凸（跟 Figure 3.3 一樣）時，曲線落在割線下方，這個差距是正的，effective convexity 隨之為正；曲線若在這段反而向下凹，差距變負，算出來的 effective convexity 就是負的——這正是 MBS、可贖回債券的凸性常常是負的原因。

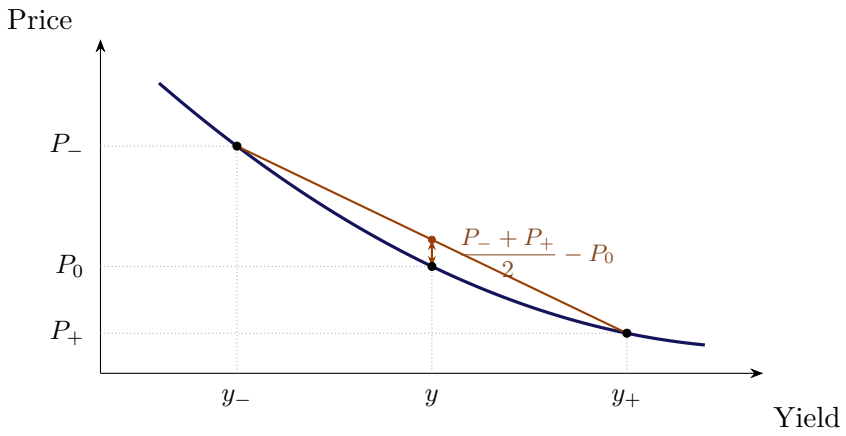


Figure 3.4.: The chord through (y_-, P_-) and (y_+, P_+) sits above the curve at the midpoint y whenever the curve is convex there. Effective convexity is, up to the factor $2/\big[(\Delta y)^2 P_0\big]$, exactly that vertical gap $(P_- + P_+)/2 - P_0$.

How to choose the right Δy is a delicate matter.

§3.14. A Numerical Caveat: Second Differences in Floating Point

Slides p. 117

The last sentence is not a throwaway remark, and the slides make the point with a deliberately trivial example. Take $f(x) = x^2$, whose second derivative at $x = 1$ is exactly 2, and form the central second difference[5]

$$\frac{(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2}{(\Delta x)^2}.$$

[5]The slide writes the target as $d^2 f(x)^2/dx^2$; the quantity actually approximated is $d^2 f/dx^2 = 2$.

In exact arithmetic the numerator is $2(\Delta x)^2$ and the quotient is 2 for *every* Δx — there is no truncation error at all, since f is a quadratic. In floating point the answer is a disaster.

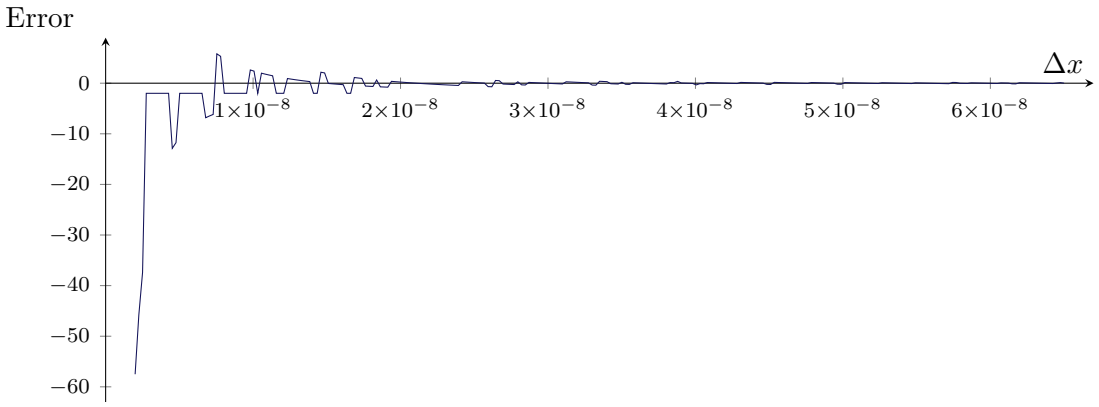


Figure 3.5.: The difference between $\big[(1+\Delta x)^2+(1-\Delta x)^2-2\big]/(\Delta x)^2$ and its exact value 2, computed in IEEE double precision. The flat stretches at -2 are the cases where the numerator underflows to exactly zero; the spikes are the cases where it survives as one or two units in the last place.

Remark 3.14.1. 這個現象叫做**災難性抵銷**（catastrophic cancellation），值得一步一步看清楚。

取 $\Delta x \approx 10^{-8}$。在雙精度浮點數中，1 附近的**機器精度**（machine epsilon）是 $\varepsilon = 2^{-52} \approx 2.22\times10^{-16}$，也就是說 1 附近能分辨的最小間隔就是這個數量級。

而 $(1+\Delta x)^2+(1-\Delta x)^2-2$ 的真值是 $2(\Delta x)^2 \approx 2\times10^{-16}$——恰好落在 ε 的同一個數量級。換句話說，我們要算的整個答案都藏在最後一個位元裡。多數情況下捨入直接把它抹成 0，商數變成 0，誤差就是 $0-2=-2$（圖中那些平坦段）；偶爾殘留一兩個 ulp，除以 $(\Delta x)^2\approx10^{-16}$ 之後就被放大成 ±10 甚至 -57 的巨大誤差（圖中的尖峰）。

推論：Δy 不是愈小愈好。把**截斷誤差**（truncation error）與**捨入誤差**（round-off error）寫在一起，設 u 為相對捨入誤差的量級：

$$\underbrace{\tfrac{1}{12}\big|P^{(4)}\big|(\Delta y)^2}_{\text{截斷誤差，隨}\Delta y\text{ 遞減}} \;+\; \underbrace{\frac{4u\,|P|}{(\Delta y)^2}}_{\text{捨入誤差，隨}\Delta y\text{ 遞增}}\,.$$

兩者相權，二階差分的最佳步長約為 $\Delta y \sim \varepsilon^{1/4} \approx 1.2\times10^{-4}$；一階（中央）差分則是 $\Delta y \sim \varepsilon^{1/3} \approx 6\times10^{-6}$。所以 (3.23) 的 Δy 取 1 個 basis point（10^{-4}）左右通常是合理的，取 10^{-8} 則會得到一堆雜訊。

This numerical issue is common in financial engineering but does not have general solutions yet.[6] Section 9.7.v meets it again when computing an option's gamma.

§3.15. Interest Rates and Bond Prices: Which Determines Which?

Slides p. 118

Question 3.15.1. Do interest rates determine bond prices, or do bond prices determine interest

[6] See pp. 872ff of the slides.

rates?[a]

[a]Contributed by Mr. Wang, Cheng (R01741064) on March 5, 2014.

If you have one, you have the other. Given a price, the yield is the root of (3.3); given a yield, the price is its value. So they are just two names given to the same thing: the **cost of fund (資金成本)**.

The difference is one of habit rather than of substance:

- Traders most likely work with prices.
- Banks most likely work with interest rates.

Price and yield are one quantity in two coordinate systems. Duration and convexity are the derivatives that translate between them.
價格與殖利率是同一個量的兩套座標；duration 與 convexity 就是在兩者之間換算的導數。

4. Term Structure of Interest Rates

Slides pp. 119–120

Why is it that the interest of money is lower,
when money is plentiful?
— Samuel Johnson (1709–1784)

If you have money, don't lend it at interest.
Rather, give [it] to someone
from whom you won't get it back.
— Thomas Gospel 95

Every formula so far has carried a single interest rate. One y discounted all of a bond's cash flows in (2.6); one y was perturbed to get duration and convexity in the last chapter. That was a convenient fiction, and this chapter retires it.

Money borrowed for three months and money borrowed for thirty years do not cost the same, and the pattern of rates across maturities — the **term structure** — is itself the object to be measured. The plan is:

(1) say what the term structure is and what its standard shapes are;
(2) replace the single y by a whole schedule of **spot rates** $S(1), S(2), \ldots$, one per maturity, and show why yield-to-maturity discounting was inconsistent all along;
(3) extract those spot rates from observed bond prices (**bootstrapping**), and see where that procedure runs into the data;
(4) use the curve to measure credit risk (**spreads**) and relate it back to the yield curve we started from.

Remark 4.0.1. **信用風險**（credit risk）：發行人到期還不出錢（違約）的風險。公司債可能違約，殖利率因此比公債高，這多出來的部分就是**利差**（spread），細節見 Section 4.8。

§4.1. What the Term Structure Is

Slides p. 121

The term structure is concerned with how interest rates change with maturity. Concretely, the set of yields to maturity for bonds form the term structure — subject to two conditions that do all the work:

- The bonds must be of equal quality.
- They differ solely in their terms to maturity.

The term structure is fundamental to the valuation of fixed-income securities.

Remark 4.1.1. **期限結構**（term structure）就是「殖利率隨到期日怎麼變化」畫出來的一條曲線——橫軸是到期日，縱軸是殖利率。

具體做法是拿一群債券各自的到期殖利率當縱軸上的點，但只有兩個條件都成立時，畫出來的才算是期限結構：品質相同、只差到期日。這樣才能保證點跟點之間的殖利率差異純粹來自到期日不同；若拿信用品質不同的債券比，測到的其實是信用風險（Section 4.8），畫出來的就不是期限結構。

最後一句「fundamental to valuation」是說：債券價格是把每期現金流分別折現再加總，不同到期日理當用不同的折現率，而期限結構正是提供這些折現率的來源——這也是本章稍後要用一整組 spot rate $S(1), S(2), \ldots$ 取代單一 y 的原因。

Slides p. 122

Three names get used loosely and are worth separating once and for all:

- The term "**term structure**" often refers exclusively to the yields of *zero-coupon* bonds. This is the convention adopted here, and the reason will become clear in Section 4.4: only a zero has a single, unambiguous date attached to it.
- A **yield curve** plots the yields to maturity of *coupon* bonds against maturity.
- A **par yield curve** is constructed from bonds trading near par.

Note 4.1.2. 這三條曲線是用同一個市場的資料，算出來的三種不同東西。

一檔 coupon bond 的 yield 為什麼會「混雜」不同到期日的利率：它的 YTM 是用同一個折現率 y 去折現所有期數的現金流、再反解出來的單一數字；coupon 越高，越早期的現金流在價格裡佔比越重，YTM 就越偏向那些早期的折現率。換句話說，同樣 n 年到期、但 coupon 不同的兩檔債券，YTM 一般並不相同——同一個到期日底下，YTM 其實不只一個值，會隨挑到哪個 coupon 而變。這就是「yield curve 是 term structure 被混雜後的版本」的意思。

par yield curve 就是從這些不同的 coupon 裡挑出一個標準答案：要求債券恰好 trade at par，即 $P = F$。由價格公式 (3.3)，

$$P = \frac{C}{y}\left[1 - (1+y)^{-n}\right] + F(1+y)^{-n},$$

令 $P = F$，兩邊減去 $F(1+y)^{-n}$、再除以共同的 $\left[1-(1+y)^{-n}\right]$，直接解出 $F = C/y$，也就是 coupon rate C/F 剛好等於 yield y。所以「trade at par」不是巧合，而是代數上唯一決定了 coupon 該取多少：對每個到期日，par yield curve 只採用「coupon 恰好等於自己 yield」的那一檔債券，不再讓 coupon 隨便挑——這就是把「混雜」固定下來的意思。實務上很少有債券剛好以 par 交易，所以會取 trading near par 的債券近似，此時 $C/F \approx y$ 只是近似相等，不是恰好相等。

市面上大多數所謂的「yield curve」，包括 Figure 4.1 裡的美國公債利率曲線，其實都是 par yield curve。

§4.1.i. The Yield Curve of U.S. Treasuries

Slides p. 123

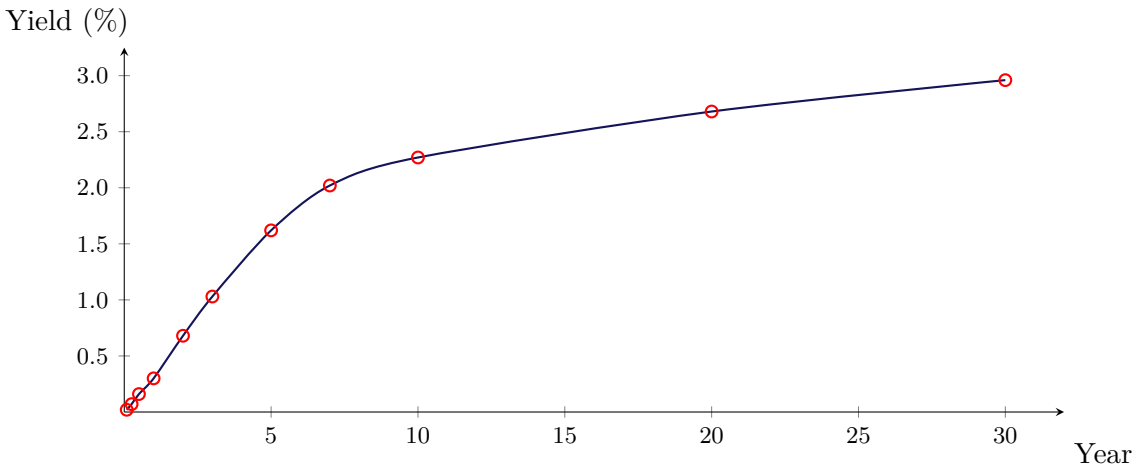


Figure 4.1.: Yield curve of U.S. Treasuries as of July 24, 2015, as shown in class. The circles are the eleven quoted maturities — 1 and 3 and 6 months, then 1, 2, 3, 5, 7, 10, 20 and 30 years — and the line is an interpolation through them. Note how unevenly the quotes are spaced: eight of the eleven sit inside the first ten years.

Remark 4.1.3. 這張圖有兩個訊息，一個明顯、一個要留意。

明顯的是形狀：短端幾乎貼地（1 個月只有 0.02%），長端接近 3%，是標準的**正常**（normal）曲線。這是 2015 年的美國——金融海嘯後聯準會把短率壓在零附近已近七年。

要留意的是橫軸上的點有多稀疏。市場只在十一個到期日上報價，其中八個擠在前十年；10 年到 20 年之間整整十年沒有任何資料點。

§4.1.ii. Four Typical Shapes

Slides p. 124

- A **normal** yield curve is upward sloping.
- An **inverted** yield curve is downward sloping.
- A **flat** yield curve is flat.
- A **humped** yield curve is upward sloping at first but then turns downward sloping.

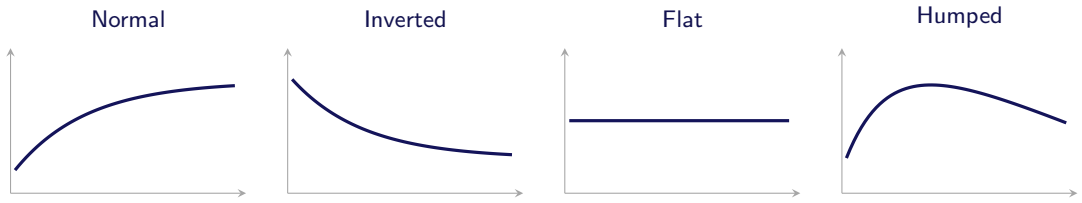


Figure 4.2.: The four shapes named in class. The horizontal axis is maturity and the vertical axis is yield in every panel.

Remark 4.1.4. 這四種形狀不只是分類，市場普遍把它們讀成對未來的訊息：

- **正常**（normal）：最常見。長天期資金較貴，通常解讀為經濟擴張、或投資人對長期持有要求額外補償。
- **倒掛**（inverted）：**短率**（short rate）高於長率。歷史上多次出現在景氣衰退之前，因此被當作預警訊號——市場預期央行日後必須降息。
- **平坦**（flat）：常是正常與倒掛之間的過渡狀態。
- **駝峰**（humped）：中期利率最高，兩端較低。

倒掛那一條的推理值得補一句。長天期利率大致是「今天的短率」與「市場預期未來各期短率」的平均：把錢鎖十年，跟每期滾動短天期投資十年，預期報酬應該相當，否則資金會往報酬高的那邊跑到兩者拉平為止（這正是之後預期理論的核心）。既然長率是預期短率的平均，長率低於今天的短率就只有一種解釋——市場預期未來的短率會比現在低，也就是預期央行日後會降息。而央行通常是在景氣轉弱時才降息，倒掛因此被當成衰退的預警訊號。

要強調的是：這些都是經驗上的關聯，不是本課程要證明的定理。曲線形狀為何如此，是 Section 4.7 之後幾節（預期理論等）才會處理的問題；此處只需認得這四個名字。

§4.2. Spot Rates

Slides p. 125

Definition 4.2.1 (Spot rate)**.** The i-period **spot rate** $S(i)$ is the yield to maturity of an i-period zero-coupon bond. The PV of one dollar i periods from now is by definition

$$\big[1 + S(i)\big]^{-i}. \tag{4.1}$$

Three immediate remarks, all of them from the slides:

- (4.1) *is* the price of an i-period zero-coupon bond, by (2.11). There is nothing to prove: a zero has one cash flow, so its yield and its discount rate are the same number by construction.

- The one-period spot rate $S(1)$ is called the **short rate**.
- The **spot rate curve** — that is, the term structure, per our convention — is the plot of spot rates against maturity:
$$S(1),\ S(2),\ \ldots,\ S(n).$$

Intuition . A spot rate is a *price*, not a forecast. $S(5) = 4\%$ says that one dollar delivered five periods from now trades today at 1.04^{-5} dollars; it says nothing about what the short rate will be in five periods' time.
即期利率（spot rate）是價格，不是預測。$S(5) = 4\%$ 的意思是「五期後的一塊錢今天值 1.04^{-5} 元」，而不是「五期後的短率會是 4%」。

§4.3. What Is Wrong with Yield-to-Maturity Discounting

Slides p. 126

In the bond price formula (2.6),

$$\sum_{i=1}^{n} \frac{C}{(1+y)^i} + \frac{F}{(1+y)^n},$$

every cash flow is discounted at the same yield y. Now consider two **riskless bonds**[1] with *different* yields to maturity, the difference arising purely because their cash flows differ:

$$\mathrm{PV}_1 = \sum_{i=1}^{n_1} \frac{C}{(1+y_1)^i} + \frac{F}{(1+y_1)^{n_1}}, \tag{4.2}$$

$$\mathrm{PV}_2 = \sum_{i=1}^{n_2} \frac{C}{(1+y_2)^i} + \frac{F}{(1+y_2)^{n_2}}. \tag{4.3}$$

Note 4.3.1. **無風險債券**（riskless bond，亦稱 default-free bond）指的是承諾的現金流必定如期足額收到，發行人不會違約；實務上以公債（如美國國庫券）為代表。
這個假設在這裡不是裝飾，而是整個論證的關鍵：

- 兩檔債券都無風險，才能說它們在第 i 期付的那一塊錢是同一件商品——同一天、同樣金額、同樣確定。既然是同一件商品，就不該有兩個價格。
- 若兩者信用不同，$y_1 \neq y_2$ 反而理所當然：差額就是**信用價差**（credit spread），是對違約風險的補償，談不上矛盾。把風險排除後，殖利率的差異就只剩下現金流的時間分佈可以解釋，這正是下一頁要指出的問題所在。

要注意「無風險」只針對違約而言，不表示沒有**利率風險**：市場利率變動時，債券價格照樣會漲跌（見 Table 2.3），只是持有到期必定拿得到 C 與 F。

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The yield-to-maturity methodology discounts their *contemporaneous* cash flows with different rates. But shouldn't they be discounted at the same rate?

[1] **Riskless** (or **default-free**) means the promised cash flows are certain to arrive in full and on time — the issuer cannot default. Government bonds such as U.S. Treasuries are the standard proxy.

Remark 4.3.2. 這一頁的投影片只丟出一個問句就結束了，但這是本章的樞紐，值得補完。

把 (4.2) 與 (4.3) 的第 i 期（i 同時小於 n_1 與 n_2）拿出來對照：兩檔債券都在同一天付出同樣的 C 元，兩者都無風險，可是前者用 $(1+y_1)^{-i}$ 折現、後者用 $(1+y_2)^{-i}$ 折現。既然 $y_1 \neq y_2$，同一天、同樣金額、同樣無風險的一塊錢，被賦予了兩個不同的現值。

這違反**一價法則**（law of one price）：同一件商品不能有兩個價格。而且這不只是美感問題——真有兩個價格就有**套利**（arbitrage）空間：從便宜的那邊買進未來的一塊錢、往貴的那邊賣出，鎖住無風險利潤。

問題出在哪裡？ y 是「把一整串現金流壓成一個數字」的內部報酬率，它同時被當成第 1 期到第 n 期的折現率使用。但這串現金流的時間分佈（票息高低、到期長短）會影響這個平均值，於是不同的債券得到不同的 y。換句話說，y 是債券的屬性，而折現率應該是日期的屬性。

解方就在下一節：每個日期給它自己的折現率 $S(i)$，與是哪一檔債券付這筆錢無關。**期限結構**（term structure）平坦時（$S(1)=S(2)=\cdots$）兩種作法才會一致；這也解釋了為什麼 y 一直「差不多堪用」——它是對真實折現曲線的一個加權平均近似。

§4.4. Spot Rate Discount Methodology

Slides p. 128

The repair is to stop thinking of a bond as one instrument with one yield. A cash flow $C_1, C_2, \ldots, C_n$ is equivalent to a *package of zero-coupon bonds*, with the ith bond paying C_i dollars at time i.

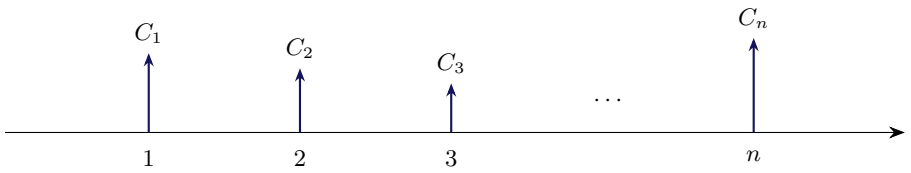


Figure 4.3.: A cash flow seen as a package of zero-coupon bonds. Each arrow is a separate security, to be priced by its own spot rate.

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Pricing each arrow at the spot rate matching *its own* date and adding up, a level-coupon bond has the price

$$P = \sum_{i=1}^{n} \frac{C}{\big[1+S(i)\big]^{i}} + \frac{F}{\big[1+S(n)\big]^{n}}. \tag{4.4}$$

- This pricing method incorporates information from the term structure.
- It discounts each cash flow at the matching spot rate.

The discount rate belongs to the *date*, not to the *bond*.
折現率屬於日期，不屬於債券。

Note 4.4.1. Compare (4.4) with (2.6). They have the same shape; the only change is that the single y has been replaced by a schedule $S(1), \ldots, S(n)$. Every complaint of the previous section evaporates: two riskless bonds now discount their common time-i dollar by the same factor, because that factor no longer knows which bond it came from.

§4.5. Discount Factors

Slides p. 130

It is worth giving those common factors a name, because they, rather than the rates, are what the pricing formula actually uses.

Definition 4.5.1 (Discount factors)**.** In general, any riskless security having a cash flow $C_1, C_2, \ldots, C_n$ should have a market price of

$$P = \sum_{i=1}^{n} C_i \, d(i), \tag{4.5}$$

where

$$d(i) \triangleq \big[1 + S(i)\big]^{-i}, \qquad i = 1, 2, \ldots, n,$$

are called the **discount factors**.

- $d(i)$ is the PV of one dollar i periods from now.
- The above formula will be justified later in the course, rather than assumed; the argument is Theorem 7.3.1.
- The discount factors are often interpolated to form a continuous function called the **discount function**.

Note 4.5.2. 最後一點的意思是：$d(1), d(2), \ldots, d(n)$ 只在整數期上有定義，但現實中要折現的日期不見得落在這些格點上（下一期票息可能是 2.5 期後才付）。把這串離散的點連成一條曲線 $d(t)$（t 為任意實數），就得到**折現函數**（discount function）。

舉個例子。設市場給出

$$S(1) = 3\%, \quad S(2) = 3.5\%, \quad S(3) = 4\%,$$

亦即

$$d(1) = 1.03^{-1} = 0.97087, \quad d(2) = 1.035^{-2} = 0.93351, \quad d(3) = 1.04^{-3} = 0.88900.$$

現在要問：2.5 期後的 1,000 元今天值多少？表上沒有 $d(2.5)$，只能內插。最簡單的線性內插是

$$d(2.5) \approx \frac{d(2) + d(3)}{2} = 0.91125,$$

於是 PV $\approx 1{,}000 \times 0.91125 = 911.25$ 元。

順帶一提，內插的對象會影響答案：若改成先內插利率 $S(2.5) \approx 3.75\%$ 再換算，得

$$d(2.5) = 1.0375^{-2.5} = 0.91207,$$

PV 變成 912.07 元，兩者差了約 0.82 元。差距雖小，但在名目金額動輒上億的部位上並非可以忽略；實務上因此不用直線，而是對 d（或 $\ln d$）配適三次樣條等較平滑的曲線。

Remark 4.5.3. 為什麼**折現因子**（discount factor）要多取一個名字？因為 (4.5) 對 $d(i)$ 是線性的，對 $S(i)$ 卻不是。

$d(i)$ 是價格（今天買一張「i 期後付一元」的憑證要花多少錢），$S(i)$ 只是把同一件事換成年化報酬率的說法。凡是關於「一籃子現金流值多少」的敘述，用 d 寫出來都是加權求和，線性代數就能處理；換成 S 就會冒出 $[1+S(i)]^{-i}$ 這種非線性的東西。後面的 bootstrapping、避險、以及由債券價格反推曲線的迴歸，全都是在 d 的座標下進行的。

另外，用 d 寫，**無套利**（no-arbitrage）的條件也特別乾淨。所謂套利，是指一個不必掏錢、卻穩賺不賠的部位（正式定義見 Section 7.1）；此處可交易的就是那 n 張零息債券，一組買賣就是一組 $d(i)$ 的加權和。於是無套利等價於一句話：$d(i) > 0$——不會有人把未來的一塊錢白送給你。

至於常見的 $1 > d(1) > d(2) > \cdots$（愈晚的錢愈不值錢），那是額外假設「現金可以零成本保存」之後才有的結論，不是無套利本身的內容。保管現金其實要成本，這也正是負利率（2015 年起的德國、瑞士公債）能夠存在而不被套利掉的原因。

§4.6. Extracting Spot Rates from the Yield Curve

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The market does not quote $S(1), S(2), \ldots$ directly; beyond the very short end it quotes coupon bonds. Spot rates must therefore be *extracted*, and the extraction is recursive.

Remark 4.6.1. 「beyond the very short end」指的是殖利率曲線最左端以外的部分。最短的那一段（美國約一年以內的 Treasury bills）本身就是零息債券，價格與 $d(1)$ 是同一件事，所以 $S(1)$ 可以直接讀出來；但兩年、五年、十年、三十年這些期限，市場上掛牌的都是每期付息的**附息債券**（coupon bond），沒有純粹「n 期後付一元」的憑證在交易。

這就是必須反推的原因：一張 n 期附息債券的價格是 $d(1), \ldots, d(n)$ 的加權和，單一個價格裡混著 n 個未知數。但若 $S(1), \ldots, S(n-1)$ 已經由更短的債券定出來，那麼這個加權和裡只剩 $d(n)$ 未知，於是一式一未知、可以解出來。

- Start with the short rate $S(1)$. Note that short-term Treasuries are zero-coupon bonds, so $S(1)$ can simply be read off.
- Compute $S(2)$ from the two-period coupon bond price P by solving

$$P \;=\; \frac{C}{1+S(1)} \;+\; \frac{C+100}{\big[1+S(2)\big]^2}. \tag{4.6}$$

The point is that (4.6) contains exactly one unknown: $S(1)$ is already known, so the first term is a number, and $S(2)$ can be solved for in closed form.

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Inductively, we are given the market price P of the n-period coupon bond and

$$S(1),\; S(2),\; \ldots,\; S(n-1).$$

Then $S(n)$ can be computed from (4.4), repeated here:

$$P \;=\; \sum_{i=1}^{n} \frac{C}{\big[1+S(i)\big]^i} \;+\; \frac{F}{\big[1+S(n)\big]^n}.$$

The running time can be made to be $O(n)$. The procedure is called **bootstrapping**.

§4.6.i. Why $O(n)$

The naive reading of the induction costs $O(n^2)$: at step n one re-sums $n-1$ discounted coupons. The saving comes from noticing that the sum only ever grows by one term — provided we accumulate *discount factors* rather than discounted coupons, since each bond has its own coupon.

Write $A(k) \triangleq \sum_{i=1}^{k} d(i)$ for the running annuity factor. The n-period bond with coupon C_n and par F satisfies

$$P_n \;=\; C_n\, A(n-1) \;+\; (C_n + F)\, d(n),$$

so that

$$d(n) \;=\; \frac{P_n - C_n A(n-1)}{C_n + F}, \qquad S(n) \;=\; d(n)^{-1/n} - 1, \qquad A(n) \;=\; A(n-1) + d(n). \tag{4.7}$$

Each step is $O(1)$ work, so the whole curve costs $O(n)$.

Algorithm 4.1: Bootstrapping the spot rate curve

Input: prices $P_1, \dots, P_n$ and coupons $C_1, \dots, C_n$ of bonds maturing in $1, \dots, n$ periods; par value F

Output: spot rates $S(1), \dots, S(n)$

```
A := 0;
for k = 1, 2, ..., n do
    d := (P_k − C_k A)/(C_k + F);
    S(k) := d^{−1/k} − 1;
    A := A + d;
return S(1), ..., S(n);
```

Note 4.6.2. The kth root costs one unit of time under the convention of Section 1.7.ii, where x^y was declared a basic operation. Without that convention the count would carry an extra factor for the exponentiation — a good illustration of how much a complexity claim depends on the cost model agreed in advance.

§4.6.ii. A Worked Example

Example 4.6.3 (Not on the slides)**.** Suppose annual periods, $F = 100$, and that the market quotes *par* bonds — bonds priced at 100, so that each one's coupon rate equals its own yield — at

$$y_1 = 3.5\%, \qquad y_2 = 4.0\%, \qquad y_3 = 4.3\%, \qquad y_4 = 4.5\%.$$

For a par bond, $P_k = F$ and $C_k = F y_k$, so (4.7) simplifies to $d(k) = \big[\,1 - y_k A(k-1)\,\big]/(1+y_k)$. Running Algorithm 4.1:

k	y_k	$A(k-1)$	$d(k)$	$S(k)$
1	3.5%	0	0.966184	3.5000%
2	4.0%	0.966184	0.924378	4.0100%
3	4.3%	1.890561	0.880830	4.3204%
4	4.5%	2.771391	0.837596	4.5301%

Checking the second row against (4.6) directly: $100 - 4/1.035 = 96.135266$, and $\big(104/96.135266\big)^{1/2} - 1 = 4.0100\%$ as claimed.

Remark 4.6.4. 這張表有兩件事值得盯著看。

第一，$S(k)$ 一律高於同期的 y_k（$k = 1$ 時相等），而且差距隨到期日拉長而擴大：1、10、20、30 個基點。這不是巧合，而是 Section 4.9 要證明的定理——曲線正常時必然如此。

第二，$d(k)$ 嚴格遞減（$0.966 > 0.924 > 0.881 > 0.838$），符合前一節說的無套利要求。若某組報價算出來的 d 不遞減，那不是曲線奇特，而是資料有問題。

§4.7. Some Problems

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Algorithm 4.1 is clean; the data are not.

- Treasuries of the same maturity might be selling at different yields (the **multiple cash flow problem**).
- Some maturities might be missing from the data points (the **incompleteness problem**).
- Treasuries might not be of the same quality.

- Interpolation and fitting techniques are needed in practice to create a smooth spot rate curve.[2]

Remark 4.7.1. 逐條說明：

（一）多重現金流問題（multiple cash flow problem）。同一到期日可能有好幾檔公債，票息不同。依上一節的分析，票息不同的債券本來就會有不同的 y——這正是我們一開始要拋棄 y 的理由。麻煩在於它們解出來的 $S(n)$ 也會互相矛盾，於是必須決定採信哪一檔，或做某種平均。

（二）不完整問題（incompleteness problem）。Algorithm 4.1 要求每一期 $1, 2, \ldots, n$ 都有一檔債券，現實中不可能。Figure 4.1 裡 10 年到 20 年之間沒有任何報價，半年期的格點更是大半空著。缺的格子只能猜。

（三）品質不齊。新發行券（on-the-run）流動性遠優於**舊券**（off-the-run），殖利率因而偏低；稅負待遇、能否作為附買回擔保品也都造成差異。這些價差與期限無關，卻全被算進曲線裡。

（四）插補（interpolation）。前三項的結果是：實務上得到的不是一條曲線，而是一團帶雜訊且有缺口的點。要變成可用的 $d(\cdot)$，必須挑一個插補或擬合方法——而那個腳註的力道很重：通常沒有經濟學上的理由。選哪條曲線，往往只是選哪一種數學上的平滑性。

Note 4.7.2. 第（三）條的三種價差各有來歷。

流動性。價格與殖利率反向，「殖利率低」即「大家願意出更高的價錢」。隨時能脫手、買賣價差極窄，本身就是有價值的方便，投資人願意為它多付錢；流動性差的券則必須便宜一點賣，多給的那一截就是**流動性溢酬**（liquidity premium）。交易集中在最新標售的那一檔，它一旦被下一次標售取代就沒人碰了。

稅負。殖利率是稅前的，投資人在意的是稅後。同一到期日、票息不同的兩檔債券，現金流的稅務性質不同——高票息每年課一次利息所得，折價券的報酬大半以到期價差實現、稅率與時點都不一樣——於是稅前價格被推到不同水準。

附買回。**附買回**（repo）是「賣出並約定買回」，實質上是拿債券當擔保品借現金，價差即利息。若某檔券特別多人想借（例如放空者必須交割它），它的 repo 利率會低於一般擔保品，持有人得以用低於市場的成本取得資金。持有它既然多賺這一點，大家就願意接受它債券本身的殖利率低一些；不被接受當擔保品的券則反過來要賣便宜。

§4.7.i. Which One?

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The incompleteness problem is worth seeing rather than reading about. Here are the same eleven observations of Figure 4.1, joined three different ways.

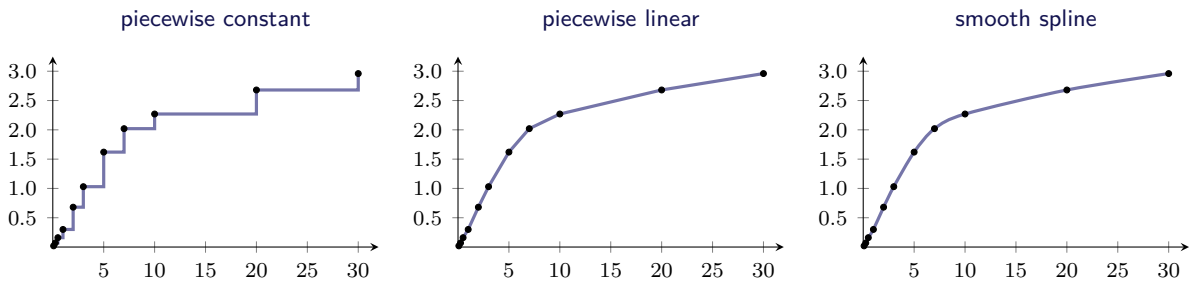


Figure 4.4.: Which one? Three curves through the identical eleven quotes of Figure 4.1; the horizontal axis is the maturity in years and the vertical axis the yield in percent, as before. Every one of them reprices all observed bonds exactly; they disagree about everything in between.

[2] Often without economic justifications.

The data fix the curve only where the data are. Everywhere else the curve is an assumption — and any security priced off that stretch inherits it.
資料只能決定有資料的地方；其餘部分是假設。凡是用那段曲線定價的商品，都繼承了這個假設。

§4.8. Yield Spread and Static Spread

Everything so far assumed riskless cash flows. A corporate bond promises the same shape of cash flow but may not deliver, and the market charges for that. The two measures below quantify the charge; they differ in whether they use the term structure or ignore it. Merton's model (Proposition 10.1.8) later derives the charge from the firm's balance sheet.

§4.8.i. Yield Spread

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Consider a risky bond with the cash flow $C_1, C_2, \ldots, C_n$ and selling for P.

- Calculate the IRR of the risky bond.
- Calculate the IRR of a riskless bond with comparable maturity.
- The **yield spread (殖利率價差/利差)** is their difference.

§4.8.ii. Static Spread

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Were the risky bond riskless, it would fetch

$$P^* = \sum_{t=1}^{n} \frac{C_t}{\big[1 + S(t)\big]^t}. \tag{4.8}$$

But as risk must be compensated, in reality $P < P^*$.

Definition 4.8.1 (Static spread)**.** The **static spread** is the amount s by which the spot rate curve has to shift in parallel to price the risky bond:

$$P = \sum_{t=1}^{n} \frac{C_t}{\big[1 + s + S(t)\big]^t}. \tag{4.9}$$

Unlike the yield spread, the static spread explicitly incorporates information from the term structure.

Remark 4.8.2. 兩者差在哪裡，用一句話講：**殖利率價差**（yield spread）**拿風險債券的 IRR 去減另一檔債券的 IRR；靜態價差**（static spread）**則是把整條無風險曲線平移到剛好能解釋這個價格**。

殖利率價差的毛病承襲自 Section 4.3：兩個 IRR 各自是各自現金流的加權平均，票息與期限只要不同，比較就不乾淨。靜態價差把每一期的現金流都對到它自己的 $S(t)$，再問「所有期別同時加上多少，價格才對得上」，因此拿掉了現金流形狀造成的干擾。

先確認 (4.9) 這個 s 真的存在、而且只有一個。把右邊看成 s 的函數 $\varphi(s)$：票息 C_t 皆為正，每一項 $C_t[1+s+S(t)]^{-t}$ 都隨 s 遞減，故 φ 連續且嚴格遞減。又 $\varphi(0) = P^* > P$（不加價差會定出太高的價格），而 $s \to \infty$ 時 $\varphi(s) \to 0 < P$。由中間值定理，恰有一個 $s > 0$ 滿足 $\varphi(s) = P$：價格與靜態價差是一一對應的，報其中一個等於報另一個。期限結構平坦時（$S(t) \equiv S$），(4.9) 退化成 $P = \sum C_t/(1+S+s)^t$，此時靜態價差恰好等於殖利率價差——兩者的差異完全來自曲線的斜率。

「靜態」二字則是留給後面的伏筆：它假設現金流固定不變。一旦債券附有贖回權或提前還款權（Section 3.13 那些商品），現金流會隨利率改變，就需要**選擇權調整價差**（option-adjusted spread, OAS）。

Example 4.8.3 (Not on the slides)**.** Take the spot curve bootstrapped above, $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, and a three-period risky bond paying $6, 6, 106$. Then (4.8) gives $P^* = 104.7114$. Suppose it actually trades at $P = 95$.

- Its IRR solves $95 = \sum_t C_t(1+y)^{-t}$, giving $y = 7.9380\%$. Against the comparable riskless three-period par yield of 4.3%, the *yield spread* is 363.8 basis points.
- Solving (4.9) gives a *static spread* of $s = 364.7$ basis points.

The two differ by about a basis point here because the curve is only mildly sloped over three periods — by Remark 4.8.2 a flat curve would make them agree exactly; on a steep curve, or a long bond, the gap is far larger.

§4.9. Of Spot Rate Curve and Yield Curve

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We began with the yield curve and replaced it with the spot rate curve. How do the two relate? Write y_i for the yield to maturity of the i-period coupon bond.

- $S(k) \geq y_k$ if $y_1 < y_2 < \cdots$ (yield curve is normal).
- $S(k) \leq y_k$ if $y_1 > y_2 > \cdots$ (yield curve is inverted).
- $S(k) \geq y_k$ if $S(1) < S(2) < \cdots$ (spot rate curve is normal).
- $S(k) \leq y_k$ if $S(1) > S(2) > \cdots$ (spot rate curve is inverted).
- If the yield curve is flat, the spot rate curve coincides with the yield curve.

Throughout, y_k is the **par yield**: the coupon rate at which the k-period bond trades at par. In particular $y_1 = S(1)$, the one-period bond being zero-coupon.

The last three bullets follow from one small lemma, which is worth isolating because it is the precise sense in which "a yield is an average of spot rates"; the first two need one more step.

Lemma 4.9.1

Let a security with non-negative cash flows $C_1, \ldots, C_k$, not all zero, be priced by (4.5), and let y_k be its yield to maturity. Then

$$\min_{1\leq i\leq k} S(i) \;\leq\; y_k \;\leq\; \max_{1\leq i\leq k} S(i).$$

Proof. Put $f(y) \triangleq \sum_{i=1}^{k} C_i(1+y)^{-i}$, which is strictly decreasing for $y > -1$, and let $\overline{S} \triangleq \max_i S(i)$. Since $(1+\overline{S})^{-i} \leq [1+S(i)]^{-i}$ for every i,

$$f(\overline{S}) \;\leq\; \sum_{i=1}^{k} C_i\big[1+S(i)\big]^{-i} \;=\; P \;=\; f(y_k),$$

and f decreasing gives $y_k \leq \overline{S}$. The other inequality is symmetric. ∎

The lemma sandwiches y_k between the smallest and the largest spot rate the bond touches. Whenever the hypothesis names the *spot* curve, that sandwich already identifies which end $S(k)$ sits at, and the third, fourth and fifth bullets drop out:

- If the spot rate curve is normal then $\max_{i\leq k} S(i) = S(k)$, so $y_k \leq S(k)$.
- If it is inverted then $\min_{i\leq k} S(i) = S(k)$, so $y_k \geq S(k)$.

- If the curve is flat, min = max and y_k is squeezed to the common value; spot and yield curves coincide.

The first two bullets assume instead that the *yield* curve is monotone, and the lemma says nothing about where $S(k)$ then lies: a normal yield curve does not force a normal spot curve.[3] So they have to be proved on their own, by induction along the curve.

Proposition 4.9.2

若 $y_1 < y_2 < \cdots$（yield curve 為 normal），則每個 k 皆有 $S(k) \geq y_k$；若 $y_1 > y_2 > \cdots$（inverted），則每個 k 皆有 $S(k) \leq y_k$。

Proof. 以下只證 normal 的情形；inverted 只需把每一個不等號反向。以下 rate 皆設為正。

記號。除了 (4.7) 的 annuity factor $A(k) = \sum_{i \leq k} d(i)$——以 spot rate 逐期折現的那一個——再記

$$a_k(y) \triangleq \sum_{i=1}^{k} (1+y)^{-i} = \frac{1-(1+y)^{-k}}{y}$$

為同一筆 annuity 改用單一 flat rate y 折現的因子，亦即 Remark 2.9.12 的 annuity present value factor。折現率愈高、現值愈低，故 a_k 對 y 嚴格遞減；這是後面唯一用到的性質。

策略。直接比較 $S(k)$ 與 y_k 無從下手，因為 y_k 是前 k 期整段曲線的綜合結果，不是第 k 期一個人的事。辦法是把待證的「$S(k) \geq y_k$」改寫成一個只涉及前 $k-1$ 期的等價條件（第二步），如此它便能由前一個到期日的結論推出，於是可以沿到期日歸納（第三步）。第零步與第一步只是把要用的兩條等式擺好。

第零步：y_k 是怎麼來的。曲線 $d(1), \ldots, d(n)$ 已由 bootstrapping 定出，於是任何現金流的價格都算得出來。考慮 k 期、par value 為 1、每期付 coupon c 的 bond，它的價格是 c 的一次函數

$$P(c) = c\,A(k) + d(k).$$

$A(k) > 0$ 故 P 嚴格遞增，又 $P(0) = d(k) < 1$ 而 $c \to \infty$ 時 $P \to \infty$，因此方程 $P(c) = 1$ 有唯一解

$$y_k = \frac{1-d(k)}{A(k)} > 0.$$

這個唯一的 coupon rate 就是 par yield y_k：它完全由曲線決定，沒有挑選的餘地，而那張 bond 是據此算出來的合成物，不必真的在市場上交易。反過來說，「它的價格是 1」並非觀察或假設，而是我們解 c 時所要求的條件。

（把 coupon 固定成使 bond 以 par 成交的那一個，是為了讓 y_k 有定義。同一到期日、coupon 不同的 bond，YTM 本來就不同——即 Section 4.7 的 multiple cash flow problem——若不指定 coupon，「$y_1 < y_2 < \cdots$」根本不是一個意義明確的條件。）

第一步：同一張 bond 的兩條定價式。把 y_k 代回，第一條等式就只是 $P(y_k) = 1$ 的展開：

$$1 = y_k A(k) + d(k) = y_k A(k-1) + (1+y_k)\,d(k). \tag{4.10}$$

第二個等號把第 k 期的最後一張 coupon 從 $A(k)$ 拆出來、與同時到達的 principal 併在一起，好讓「前 $k-1$ 期」與「第 k 期」各據一項。

第二條等式來自 YTM 的定義：YTM 是能還原同一個價格的那個單一 flat rate。這張 bond 以 par 成交，而依 Remark 2.9.12，par 等價於 coupon rate 等於 YTM，故它的 YTM 恰為 y_k，於是

$$1 = y_k\, a_{k-1}(y_k) + (1+y_k)(1+y_k)^{-k}. \tag{4.11}$$

[3] $S(1), \ldots, S(4) = 0.5\%, 1\%, 16\%, 15.5\%$ gives par yields $0.5000\%, 0.9975\%, 13.7364\%, 13.7853\%$, which increase although $S(4) < S(3)$.

兩式並未把 spot rate 與 YTM 混用：(4.10) 全程以曲線折現，y_k 在其中只是 coupon 的金額；(4.11) 全程以 flat rate 折現，y_k 在其中才兼任 *discount rate*。兩式共用的僅是左邊那個價格 1——也正因如此才能相減。

第二步：相減，換成只涉及前 $k-1$ 期的條件。兩式左邊同為 1，相減得

$$0 \;=\; y_k\big[\,A(k-1) - a_{k-1}(y_k)\,\big] \;+\; (1+y_k)\big[\,d(k) - (1+y_k)^{-k}\,\big]. \tag{4.12}$$

(手法與 Remark 2.9.12 導出 premium 金額時相同：兩條同值的定價式相減，差異自然浮現。)兩個中括號因此不同號。而待證的 $S(k) \geq y_k$ 即 $d(k) \leq (1+y_k)^{-k}$，說的正是第二個中括號 ≤ 0，故

$$S(k) \geq y_k \qquad \Longleftrightarrow \qquad A(k-1) \;\geq\; a_{k-1}(y_k). \tag{4.13}$$

右邊只牽涉前 $k-1$ 期，這正是要的形式。它的意思也很直白：總價已被釘在 1，曲線若對前 $k-1$ 期的 coupon 折現得不比 flat rate y_k 吝嗇，那麼前面讓掉的就必須在第 k 期補回來，$S(k)$ 只能高於 y_k。

第三步：歸納。$k=1$ 時一期 bond 沒有中間 coupon，$y_1 = S(1)$，結論以等號成立。

設 $S(k-1) \geq y_{k-1}$，即 $d(k-1) \leq [1+y_{k-1}]^{-(k-1)}$。第零步用在到期日 $k-1$ 上給出 $1 = y_{k-1}A(k-1) + d(k-1)$，解出 annuity factor 後

$$A(k-1) \;=\; \frac{1-d(k-1)}{y_{k-1}} \;\geq\; \frac{1-[1+y_{k-1}]^{-(k-1)}}{y_{k-1}} \;=\; a_{k-1}(y_{k-1}) \;\geq\; a_{k-1}(y_k),$$

其中第一個不等號用歸納假設，第二個用 $y_{k-1} < y_k$ 與 a_{k-1} 遞減。這恰是 (4.13) 的右邊，故 $S(k) \geq y_k$。 ■

整段論證只在一個地方用到 normal 的假設：第三步把 $A(k-1)$ 的下界從 y_{k-1} 換成 y_k。這也說明為什麼假設非得是關於整條曲線不可——單獨一個 k 的資訊推不出任何結論。

§4.9.i. Shapes

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- The spot rate curve often has the same shape as the yield curve.
  - If the spot rate curve is inverted (normal, resp.), then the yield curve is inverted (normal, resp.).
- But this is only a trend, not a mathematical truth.[4]

Note 4.9.3. The asymmetry is worth stating plainly, because the slides put theorem and trend in the same list.

The lemma above is a *theorem*: given the shape of the spot curve, the position of y_k relative to $S(k)$ is settled at every k individually. The claim on this page is different — it is about the *shape* of the whole yield curve, i.e. about how y_k compares with y_{k+1}, and that comparison involves two different bonds with two different sets of weights. Nothing forces the weighting to behave monotonically, and the textbook exhibits a spot curve that is monotone while the induced yield curve is not.

The practical reading: use the spot curve to price, and treat the yield curve's shape as a summary for conversation, not as an input.

§4.10. Forward Rates

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The yield curve contains information regarding future interest rates currently "expected" by the market. To extract it, compare two ways of holding money from time 0 to time j.

[4] See a counterexample in the text.

- **The maturity strategy.** Invest \$1 for j periods to end up with $\big[1+S(j)\big]^j$ dollars at time j.
- **The rollover strategy.** Invest \$1 in bonds for i periods and at time i invest the proceeds in bonds for another $j-i$ periods, where $j>i$. This will have

$$\big[1+S(i)\big]^i\big[1+S(i,j)\big]^{j-i}$$

dollars at time j, where $S(i,j)$ denotes the $(j-i)$-period spot rate i periods from now.

The first number is known today. The second is not: $S(i,j)$ is a rate that will only be quoted at time i. The question the market implicitly answers is what value of $S(i,j)$ would make the two strategies tie.

Remark 4.10.1. rollover strategy 是這樣一件事：兩種策略都是今天投入一元、都要在時點 j 把錢拿回來，差別只在中途要不要換手。

maturity strategy 一次買一張 j 期的 zero-coupon bond，錢鎖到 j 期，結果 $[1+S(j)]^j$ 今天就已確定。rollover strategy 則先買一張較短的 i 期 bond（$i<j$），到第 i 期拿回本利和 $[1+S(i)]^i$，再把這筆錢全數投入一張剩餘 $j-i$ 期的 bond，持到 j 期。所謂 rollover（展期）就是指到期後把錢原封不動轉進下一張。

記號 $S(i,j)$ 要留意：兩個參數是起點與終點，不是起點與期限。它指「站在第 i 期時，期限 $j-i$ 期的 spot rate」，因此今天的 $S(j)$ 在這個記號下就是 $S(0,j)$。

關鍵的不對稱在於：$S(i)$ 與 $S(j)$ 都是今天報價表上的數字，$S(i,j)$ 卻要等到第 i 期才會被報出來。所以 maturity strategy 的終值今天是已知數，rollover strategy 的終值今天還是未知數——兩者唯一的差別就在這裡。於是可以反問：$S(i,j)$ 要是多少，兩條路才會打成平手？那個數就是下面的 forward rate。

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Definition 4.10.2 (Forward rate)**.** When $S(i,j)$ equals

$$f(i,j) \triangleq \left[\frac{\big(1+S(j)\big)^j}{\big(1+S(i)\big)^i}\right]^{1/(j-i)} - 1, \tag{4.14}$$

we will end up with the same $\big[1+S(j)\big]^j$ dollars. The $f(i,j)$ are the **(implied) forward (interest) rates** — more precisely, the $(j-i)$-period forward rate i periods from now.

(4.14) is just the tie condition solved for the unknown:

$$\big[1+S(i)\big]^i\big[1+f(i,j)\big]^{j-i} = \big[1+S(j)\big]^j \quad\Longleftrightarrow\quad \big[1+f(i,j)\big]^{j-i} = \frac{\big(1+S(j)\big)^j}{\big(1+S(i)\big)^i}.$$

As expected, $f(0,j)=S(j)$: with $i=0$ the rollover strategy *is* the maturity strategy.

§4.10.i. The Time Line

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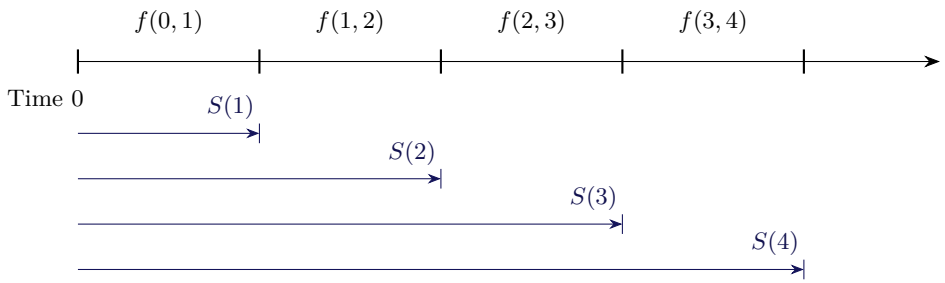


Figure 4.5.: The time line. The spot rates below run from today out to each maturity; the one-period forward rates above cover the individual legs between consecutive dates.

Remark 4.10.3. 這張圖是整節的骨架，值得盯著看清楚兩者的分工。

下半部的 $S(k)$ 都從 time 0 出發——每一條都是「從今天走到第 k 期」的平均利率，所以彼此重疊。**上半部的 $f(k,k+1)$ 則彼此不重疊**，各自只負責一小段。

用速度打比方：$S(k)$ 是走完前 k 期的平均速度，$f(k,k+1)$ 是第 $k+1$ 期的瞬時速度。平均速度由瞬時速度累積而成——這正是 (4.15) 要說的事；而平均速度上升，必然是因為當下的瞬時速度高於既有的平均，這正是 Section 4.11 的不等式。

§4.10.ii. Forward Rates and Future Spot Rates

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Note 4.10.4. We did not assume any *a priori* relation between $f(i,j)$ and the future spot rate $S(i,j)$ — that is the subject of the term structure theories in Section 4.15. We merely looked for the future spot rate that, *if realized*, will equate the two investment strategies.

The $f(i,i+1)$ are the **instantaneous forward rates** or **one-period forward rates**.

Remark 4.10.5. $f(i,j)$ 與 $S(i,j)$ 是兩個不同的東西，這是本章最常被誤解的地方。

遠期利率（forward rate）$f(i,j)$ 是今天的數字：由 (4.14) 從今天查得到的 $S(i)$ 與 $S(j)$ 算出來，今天就已完全確定。$S(i,j)$ 是未來的數字：第 i 期市場真正報出來的那個 $(j-i)$ 期 spot rate，今天還只是隨機變數。

兩者容易混為一談，是因為 forward rate 的定義寫成「當 $S(i,j)$ 等於 $f(i,j)$ 時兩策略打成平手」。那是一個條件句：$f(i,j)$ 回答的是「若要打平，$S(i,j)$ 該是多少」，並非斷言 $S(i,j)$ 真的會等於它。到了第 i 期，實現的 $S(i,j)$ 通常不等於 $f(i,j)$，而兩種策略孰優孰劣正是由這個落差決定。

這也是為什麼前面「market expects」那句話加了引號：f 是市場定價隱含的數字，至於 $f(i,j)$ 是否等於市場對 $S(i,j)$ 的期望值，是一個需要額外假設的理論命題（Section 4.15 的無偏預期理論就是這樣一個假設），而不是推導出來的結果。

Section 4.13 會更進一步：$f(i,j)$ 不只是算出來的數字，而是今天就能用現有 bond 鎖定的真實利率——這與「預測未來」仍是兩回事。

§4.11. Spot Rates and Forward Rates

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Proposition 4.11.1

When the spot rate curve is normal, the forward rate dominates the spot rates,

$$f(i,j) \;>\; S(j) \;>\; \cdots \;>\; S(i).$$

When the spot rate curve is inverted, the forward rate is dominated by the spot rates,

$$f(i,j) \;<\; S(j) \;<\; \cdots \;<\; S(i).$$

Proof. Suppose the curve is normal, so $S(i) < S(j)$ for $i < j$. From (4.14),

$$\big[1+f(i,j)\big]^{j-i} = \frac{\big(1+S(j)\big)^{j}}{\big(1+S(i)\big)^{i}} \;>\; \frac{\big(1+S(j)\big)^{j}}{\big(1+S(j)\big)^{i}} = \big[1+S(j)\big]^{j-i},$$

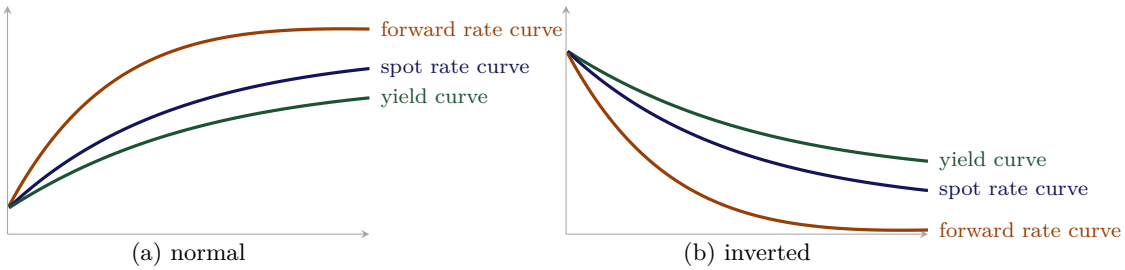
where the inequality replaces $S(i)$ in the denominator by the larger $S(j)$. Taking $(j-i)$th roots gives $f(i,j) > S(j)$. The inverted case reverses every inequality. ∎

Example 4.11.2 (Not on the slides)**.** Take the curve bootstrapped in Section 4.6, namely $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, $S(4) = 4.5301\%$. The one-period forward rates are

$$f(0,1) = 3.5000\%, \quad f(1,2) = 4.5225\%, \quad f(2,3) = 4.9440\%, \quad f(3,4) = 5.1617\%.$$

Each sits above the spot rate for the same terminal date, and the forward curve rises faster than the spot curve — the marginal rate outruns the average it is pulling up.

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(a) normal (b) inverted

Figure 4.6.: The three curves never cross: forward above spot above yield when the *spot rate* curve is normal, and exactly the reverse when it is inverted. Horizontal axis maturity, vertical axis rate.

Remark 4.11.3. 先說三條曲線為何能畫在同一張圖上。$f(i,j)$ 有兩個變數，圖上畫的是它的對角線切片 $k \mapsto f(k-1,k)$，即固定 $j-i=1$ 的 one-period forward rate。這沒有損失資訊，因為 (4.14) 串接起來會 telescoping：

$$\big[1+f(i,j)\big]^{j-i} \;=\; \frac{\big(1+S(j)\big)^{j}}{\big(1+S(i)\big)^{i}} \;=\; \prod_{k=i}^{j-1}\big[1+f(k,k+1)\big],$$

任何 $f(i,j)$ 都是那條曲線上各點的幾何平均。另兩條同樣是切片：$S(k)$ 就是 $S(0,k)$（固定今天出發），y_k 則是把 coupon 固定成 par 後唯一的那一條。三者各被固定掉一個自由度，剩下的參數都讀成到期期限，因此共用橫軸。

再說順序，關鍵是**每一條都是它上面那一條的平均**，而平均的窗口一路回溯到今天。

其一，(4.15) 在 $i=0$ 時說

$$\big[1+S(k)\big]^k \;=\; \prod_{m=1}^{k}\big[1+f(m-1,m)\big],$$

即 $1+S(k)$ 是前 k 個 one-period forward 之 gross factor 的幾何平均（連續複利下 (4.19) 即字面上的算術平均）。滑動平均要上升，新進的那一項就必須高於平均本身，故 spot curve 為 normal 時 $f(k-1,k)>S(k)$——這正是 Proposition 4.11.1。

其二，y_k 是單一折現率卻要處理 bond 的所有現金流，依 Lemma 4.9.1 它夾在 $\min_{i\le k} S(i)$ 與 $\max_{i\le k} S(i)$ 之間，實為 $S(1),\dots,S(k)$ 以各期現金流現值為權重的加權平均；coupon 落在較早、較低的日期，故 normal 時 $y_k \le S(k)$，zero-coupon bond 取等號。

兩者疊起來即

$$\underbrace{f(k-1,k)}_{\text{只第}k\text{ 期}} \;>\; \underbrace{S(k)}_{\text{平均}1\sim k\text{ 期}} \;\ge\; \underbrace{y_k}_{\text{再按 coupon 加權}}\quad.$$

窗口愈寬就算進愈多較早、較低的 rate，值被拉得愈低；inverted 時較早的 rate 較高，同一機制把每個平均往上拉，順序於是整串反轉。兩條不等式對每個 k 逐點成立，故三條曲線從不交叉。

最後，前提必須掛在 *spot* curve 的單調性上：上述兩條不等式是定理，而「yield curve 的形狀跟著 spot curve 一樣單調」只是傾向（見 Section 4.9 那則 note）。若只知道 yield curve 為 normal，spot curve 仍可能非單調，圖上的順序便未必成立。

§4.12. Forward Rates ≡ Spot Rates ≡ Yield Curve

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The three curves are not merely ordered; they carry exactly the same information. The FV of \$1 at time n can be derived in two ways.

- Buy n-period zero-coupon bonds and receive
$$\big[1+S(n)\big]^n.$$
- Buy one-period zero-coupon bonds today and a series of such bonds at the forward rates as they mature. The FV is
$$\big[1+S(1)\big]\big[1+f(1,2)\big]\cdots\big[1+f(n-1,n)\big].$$

Remark 4.12.1. 為什麼兩種算法的 FV 一定相同？可以從代數與經濟兩面看。

代數面：telescoping。先注意 $1+S(1)=1+f(0,1)$，再把 (4.14) 取 $j=i+1$ 代進每一個因子：

$$\prod_{k=1}^{n}\big[1+f(k-1,k)\big] \;=\; \frac{\big(1+S(1)\big)^1}{\big(1+S(0)\big)^0}\cdot\frac{\big(1+S(2)\big)^2}{\big(1+S(1)\big)^1}\cdots\frac{\big(1+S(n)\big)^n}{\big(1+S(n-1)\big)^{n-1}} \;=\; \big[1+S(n)\big]^n.$$

相鄰分子分母兩兩相消，只剩頭尾。換句話說，forward rate 本來就是定義成讓 rollover 與 maturity 打平的那個數（見 (4.14) 下方的 tie condition）；這裡只是把 n 段「每段各自打平」串起來，整條路自然也打平。

經濟面：無套利。第二種做法用的是今天就鎖定的 $f(k-1,k)$，而不是屆時才報價的未來 spot rate $S(k-1,k)$。Section 4.13 會看到，每一段 forward loan 都能用今天交易的零息債券複製出來，所以這條 rollover 路徑的 FV 在今天就已完全確定，和第一種一樣是無風險的。兩個無風險策略、同樣今天投入 \$1、同樣在時點 n 收錢，終值若不同，就買進終值高的、放空終值低的，今天淨投入為零、時點 n 穩賺差額——這是**套利**（arbitrage）。因此兩者必須相等。

要小心的是：若第二種做法改成「到期後按當時的市場利率再投資」，FV 就變成隨機的 $\big[1+S(1)\big]\big[1+S(1,2)\big]\cdots\big[1+S(n-1,n)\big]$，一般不會等於 $\big[1+S(n)\big]^n$。相等是 forward rate 的性質，不是未來 spot rate 的性質。

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Theorem 4.12.2

Since the two are identical,

$$S(n) \;=\; \Big\{\big[1+S(1)\big]\big[1+f(1,2)\big]\cdots\big[1+f(n-1,n)\big]\Big\}^{1/n} - 1. \tag{4.15}$$

Hence the forward rates — specifically the one-period forward rates — determine the spot rate curve.

Other equivalencies can be derived similarly, such as

$$f(T,T+1) \;=\; \frac{d(T)}{d(T+1)} - 1. \tag{4.16}$$

Proof of (4.16). By definition $d(k) = \big[1+S(k)\big]^{-k}$, so

$$\frac{d(T)}{d(T+1)} = \frac{\big(1+S(T+1)\big)^{T+1}}{\big(1+S(T)\big)^{T}} = 1+f(T,T+1)$$

by (4.14) with $i=T$, $j=T+1$. ■

Spot rates, forward rates, discount factors and the yield curve are four coordinate systems for one object. Any one of them can be computed from any other; none contains extra information.
即期利率、遠期利率、折現因子、殖利率曲線，是同一個東西的四套座標。任一套都能算出其餘三套，沒有誰多帶了資訊。

§4.13. Locking in the Forward Rate

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So far $f(n,m)$ is an arithmetic construct. The following trade turns it into a rate one can actually contract at today, using nothing but zero-coupon bonds that trade now.

- Buy one n-period zero-coupon bond for $1/(1+S(n))^n$ dollars.
- Sell $(1+S(m))^m/(1+S(n))^n$ m-period zero-coupon bonds.
- No net initial investment, because the cash inflow equals the cash outflow, both being $1/(1+S(n))^n$.
- At time n there will be a cash inflow of \$1.
- At time m there will be a cash outflow of $(1+S(m))^m/(1+S(n))^n$ dollars.

Why the initial cost is zero. Each m-period zero sells for $(1+S(m))^{-m}$, so selling $(1+S(m))^m/(1+S(n))^n$ of them raises

$$\frac{(1+S(m))^m}{(1+S(n))^n} \times \frac{1}{(1+S(m))^m} = \frac{1}{(1+S(n))^n},$$

which is exactly what the n-period zero costs. ■

Remark 4.13.1. 這一頁要說的是：$f(n,m)$ 不只是從 spot curve 算出來的數字，今天就能用市場上現成的零息債券做出一筆利率為 $f(n,m)$ 的借款。這組交易（買 n 期 zero、同時放空適當數量的 m 期 zero）之後會被稱為 **locking trade**：lock 指「鎖定」，因為它讓人在今天就把未來 n 到 m 這段期間的利率鎖死在 $f(n,m)$，不必等到時點 n 再看市場報價 $S(n,m)$。

兩條腿各做什麼。買一張 n 期零息債券，是今天把錢借出去、第 n 期拿回 \$1；賣出（放空）m 期零息債券，則是今天借錢進來，第 m 期每張還 \$1。單看任何一條腿，今天都有現金進出。

數量怎麼挑。賣出的張數 $(1+S(m))^m/(1+S(n))^n$ 是刻意選的：它讓放空拿到的錢剛好付得起那張 n 期債券（上面的 proof）。於是今天兩邊互相抵銷，一毛錢都不用出。

剩下的現金流。把時點 0 抵銷掉之後，只剩第 n 期收到 \$1、第 m 期付出 $(1+S(m))^m/(1+S(n))^n$。這就是「第 n 期借進 \$1、第 m 期還本帶息」的借款，只是今天就已經簽好。還款金額除以 \$1 就是 $m-n$ 期的 gross factor，由 (4.14) 它正好等於 $\big(1+f(n,m)\big)^{m-n}$，所以這筆借款的利率就是 $f(n,m)$（下一頁 Figure 4.7）。

這為什麼重要。未來的 spot rate $S(n,m)$ 今天還不知道，但 $f(n,m)$ 今天就能鎖定，而且所有條件都已定死，沒有任何不確定性。這也是上一節「兩種算法 FV 相同」的無套利論證所依靠的事實：rollover 路徑每一段用的都是這樣鎖定的利率，所以它和直接買 n 期債券一樣是無風險的。反過來，如果市場上另有一筆 forward loan 的利率不等於 $f(n,m)$，就可以一邊用它、一邊用這組零息債券做反向操作，賺取無風險利潤。

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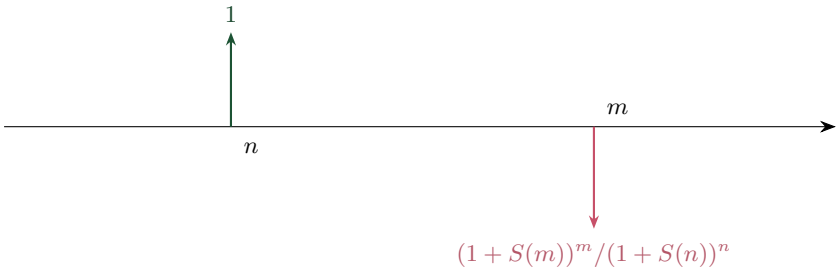


Figure 4.7.: The cash flow locked in today. Nothing changes hands at time 0; one dollar arrives at time n and is repaid at time m.

This implies the interest rate between times n and m equals $f(n,m)$, because by (4.14)

$$\frac{(1+S(m))^m}{(1+S(n))^n} = \big(1+f(n,m)\big)^{m-n}.$$

§4.13.i. Forward Loans

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We had generated the cash flow of a type of forward contract called the **forward loan**. Agreed upon today, it enables one to

- borrow money at time n in the future, and
- repay the loan at time $m > n$ with an interest rate equal to the known forward rate $f(n,m)$.

Question 4.13.2. Can the spot rate curve be arbitrarily drawn?[a]

[a] Contributed by Mr. Dai, Tian-Shyr (B82506025, R86526008, D88526006) in 1998.

Answer. 不行。每一筆 forward loan 都能用市場上交易的 zero-coupon bond 複製出來，所以曲線隱含的 forward rate 是真的能成交的利率——它們本身也必須不能被套利。

假設某個 n 使得 $f(n,n+1) < 0$，由 (4.16) 這等價於 $d(n) < d(n+1)$，也就是較晚到期的 zero 反而比較貴。這時賣出一張 $(n+1)$ 期 zero、買進一張 n 期 zero：今天淨收入 $d(n+1) - d(n) > 0$，時點 n 收到 \$1，時點 $n+1$ 要付出 \$1。把時點 n 收到的那 \$1 以現金形式保管到時點 $n+1$，剛好用來支付後面的義務。結果是今天白拿一筆正的錢，將來沒有任何負債——這就是 arbitrage。

因此曲線必須滿足

$$1 > d(1) \geq d(2) \geq \cdots, \qquad \text{等價地} \qquad \big(1+S(n)\big)^n \text{ 對} n \text{ 非遞減.}$$

(取 $n=0$、$d(0)=1$ 用同一個論證就得到 $d(1)<1$。) 特別地，inverted curve 不能下降得任意快；Section 4.14 會給出同一個界的連續時間版本。 ■

Remark 4.13.3. 先說為什麼會冒出這個問題。到目前為止，spot rate curve 一直被當成輸入：給任何一組 $S(1), S(2), \ldots$，都能用 (4.14) 算出一組 $f(i,j)$，代數上從來不會卡住。如果 f 只是算出來的數字，那 spot curve 畫成什麼形狀似乎都可以。

但上一節剛證明了 $f(n,m)$ 不只是數字：用 zero-coupon bond 就能在今天把它鎖定成一筆真的 forward loan。一旦 forward rate 是能成交的價格，它就得接受市場的檢驗——例如「n 到 $n+1$ 之間借錢還能拿到利息補貼」這種 forward rate 若出現，就會被套利。所以問題其實是在問：forward loan 的可交易性反過來對 spot curve 加了什麼限制？答案就是上面的單調性條件：$\big(1+S(n)\big)^n$ 不能隨 n 變小。這也呼應前面「曲線是 normal 還是 inverted」的討論：inverted 是允許的，但只能緩慢下降，不能降到讓某段 forward rate 變負。

再說上面那段論證偷偷用了一個假設：現金可以無成本地保管。正因為存著鈔票的報酬率是 0，任何低於 0 的遠期利率才會被套利掉。

現實中這個假設並不完全成立——大額現金的保管、保險與運送都要錢，而且央行可以對準備金課負利率。2014 年之後的歐元區與日本公債確實出現過負的即期與遠期利率，而且持續多年。所以正確的說法是：曲線不能任意畫，但下界不是 0，而是 −(持有現金的成本)。

§4.13.ii. Synthetic Bonds

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Read the locking trade as an identity between portfolios. We had seen that

$$\text{forward loan} \;=\; n\text{-period zero} \;-\; \big[1+f(n,m)\big]^{m-n} \times m\text{-period zero.}$$

Thus

$$n\text{-period zero} \;=\; \text{forward loan} \;+\; \big[1+f(n,m)\big]^{m-n} \times m\text{-period zero.}$$

We have created a **synthetic** zero-coupon bond with forward loans and other zero-coupon bonds. This is useful if the n-period zero is unavailable or illiquid.

Remark 4.13.4. 這些式子在說什麼。把每個金融商品看成「各時點收付多少錢」的一張清單，式子就是在加減這些清單：$+$ 是持有 (long)，$-$ 是放空 (short)，係數是張數，$=$ 表示兩邊每個時點的現金流都一樣。令 $K = \big[1+f(n,m)\big]^{m-n}$，逐期列出來：

	時點n	時點m
n-period zero	$+1$	0
$K \times m$-period zero	0	$+K$
forward loan	$+1$	$-K$

第一式：第一列減第二列，得到第三列——這就是上一節的做法 (買 n 期 zero、放空 K 張 m 期 zero，今天淨成本 0)。第二式：把同一件事移項，第三列加第二列，得回第一列。

搞這個幹嘛。第二式告訴你：就算市場上買不到（或很少成交）n 期 zero，只要能交易 m 期 zero 和 forward loan，就能自己**合成**（synthesize）一張現金流一模一樣的 n 期 zero。現金流一樣，價格就必須一樣，否則可以套利；驗算也對得上：forward loan 今天不用錢，組合成本 $K\,d(m) = d(n)$。

Note 4.13.5. This is the first appearance of a move that will run through the rest of the course: an identity between cash flows is an identity between *prices*, so any instrument that can be replicated must cost what its replicating portfolio costs. Here it fills a hole in the maturity spectrum — exactly the incompleteness problem of Section 4.7, attacked by construction rather than by interpolation.

§4.14. Spot and Forward Rates under Continuous Compounding

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Every formula above simplifies under continuous compounding, because products of gross rates become sums of rates. Recall (2.4): one dollar grows to e^{rn} rather than $(1+r)^n$.

- The pricing formula:

$$P \;=\; \sum_{i=1}^{n} C e^{-iS(i)} \;+\; F e^{-nS(n)}. \tag{4.17}$$

- The market discount function:

$$d(n) \;=\; e^{-nS(n)}. \tag{4.18}$$

- The spot rate is an *arithmetic average* of forward rates:

$$S(n) \;=\; \frac{f(0,1) + f(1,2) + \cdots + f(n-1,n)}{n}. \tag{4.19}$$

推導(4.19). 與 Section 4.12 同一個論證：\$1 放到時點 n，有兩種方式。直接買 n 期 zero，連續複利下得到 $e^{nS(n)}$；逐期 rollover、每期用今天鎖定的 one-period forward rate，得到

$$e^{f(0,1)}\, e^{f(1,2)} \cdots e^{f(n-1,n)} \;=\; e^{f(0,1)+f(1,2)+\cdots+f(n-1,n)}.$$

兩者都是今天就確定的無風險終值，無套利下必須相等：

$$e^{nS(n)} \;=\; e^{f(0,1)+f(1,2)+\cdots+f(n-1,n)}.$$

兩邊取 ln 得 $nS(n) = f(0,1) + \cdots + f(n-1,n)$，再除以 n 即得。

也可以逐項看：one-period 的 tie condition $e^{(k-1)S(k-1)} e^{f(k-1,k)} = e^{kS(k)}$ 取 ln 得 $f(k-1,k) = kS(k) - (k-1)S(k-1)$，對 $k = 1, \ldots, n$ 相加時中間項兩兩相消（telescoping），只剩 $nS(n) - 0 \cdot S(0) = nS(n)$。 ■

Remark 4.14.1. 把 (4.19) 與離散版的 (4.15) 並排看，就知道連續複利的好處在哪裡。

離散版是**幾何平均**（geometric mean）：$1 + S(n)$ 等於 n 個 $1+f$ 相乘再開 n 次方。連續版是**算術平均**（arithmetic mean）：$S(n)$ 就是 n 個 f 直接相加除以 n。

原因只有一個：$e^a e^b = e^{a+b}$。指數把乘法換成加法，於是「利率的累積」變成單純的求和，微分與積分也才用得上——這就是為什麼從這裡之後的理論工作幾乎一律採用連續複利。

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- The formula for the forward rate:

$$f(i,j) \;=\; \frac{jS(j) - iS(i)}{j-i}. \tag{4.20}$$

- The one-period forward rate:

$$f(j, j+1) \;=\; -\ln \frac{d(j+1)}{d(j)}. \tag{4.21}$$

推導(4.20). 與離散版 (4.14) 同一個 tie condition：\$1 放到時點 j，maturity strategy 得 $e^{jS(j)}$；rollover strategy 先放 i 期得 $e^{iS(i)}$，再以鎖定的 $f(i,j)$ 放 $j-i$ 期，得 $e^{iS(i)}e^{(j-i)f(i,j)}$。兩者相等：

$$e^{iS(i)+(j-i)f(i,j)} \;=\; e^{jS(j)}.$$

取 ln 得 $iS(i)+(j-i)f(i,j) = jS(j)$，移項後除以 $j-i$ 即得。 ■

推導(4.21). 由 (4.18)，$d(k) = e^{-kS(k)}$，所以

$$\frac{d(j+1)}{d(j)} \;=\; \frac{e^{-(j+1)S(j+1)}}{e^{-jS(j)}} \;=\; e^{-[(j+1)S(j+1)-jS(j)]} \;=\; e^{-f(j,j+1)},$$

最後一步是 (4.20) 取 $i = j$、$j \to j+1$（分母 $j-i=1$）。兩邊取 ln 再加負號即得。 ■

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Definition 4.14.2 (Instantaneous forward rate curve)**.**

$$f(T) \;\triangleq\; \lim_{\Delta T \to 0} f(T, T+\Delta T) \;=\; S(T) + T\,\frac{\partial S}{\partial T}. \tag{4.22}$$

Proof. By (4.20) with $i = T$ and $j = T+\Delta T$,

$$f(T, T+\Delta T) = \frac{(T+\Delta T)S(T+\Delta T) - T\,S(T)}{\Delta T},$$

which is the difference quotient of the function $T \mapsto T\,S(T)$. Its limit is $\dfrac{d}{dT}\big[T\,S(T)\big] = S(T) + T\,S'(T)$. ■

Two consequences are recorded on the slide:

- $f(T) > S(T)$ if and only if $\partial S/\partial T > 0$, i.e. the spot rate curve is normal. This is the continuous-time restatement of Section 4.11, and (4.22) makes it immediate: $f - S = T\,\partial S/\partial T$.
- If $S(T) < -T\,(\partial S/\partial T)$, then $f(T) < 0$.

Remark 4.14.3. 第二條正是前面「曲線不能任意畫」的連續版本。

離散版是 Section 4.13 裡那題 question 的答案：forward rate 不能是負的，否則借長賣短、中間持有現金就能套利。連續版把「一期 forward rate」換成「瞬間 forward rate $f(T)$」，其餘一字不改。兩邊逐行對照：

	離散	連續
條件	$f(n, n+1) \ge 0$	$f(T) \ge 0$
用d 寫	$d(n) \ge d(n+1)$	$d(T) = e^{-TS(T)}$ 遞減
用S 寫	$(1+S(n))^n$ 遞增	$T\,S(T)$ 遞增

每一欄的三行互相等價：離散欄由 (4.16)，$1 + f(n, n+1) = d(n)/d(n+1)$，而 $d(n) = (1+S(n))^{-n}$；連續欄由 (4.22)，$f(T) = \dfrac{d}{dT}\big[T\,S(T)\big]$，而 $d(T) = e^{-TS(T)}$。最後一行的對應也很直接：$(1+S(n))^n$ 是 \$1 放 n 期的終值，$e^{TS(T)}$ 是連續版的終值，$T\,S(T)$ 就是它的指數。所以兩邊說的是同一件事——*\$1* 放得愈久，終值不能變少。

§4.14.i. An Example

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Example 4.14.4. Let the interest rates be continuously compounded, and suppose the spot rate curve is[5]

$$S(T) \triangleq 0.08 - 0.05\, e^{-0.18T}.$$

Then by (4.22) the forward rate curve is

$$\begin{aligned} f(T) &= S(T) + T\, S'(T) \\ &= 0.08 - 0.05\, e^{-0.18T} + 0.009\, T\, e^{-0.18T}, \end{aligned}$$

since $S'(T) = 0.009\, e^{-0.18T}$.

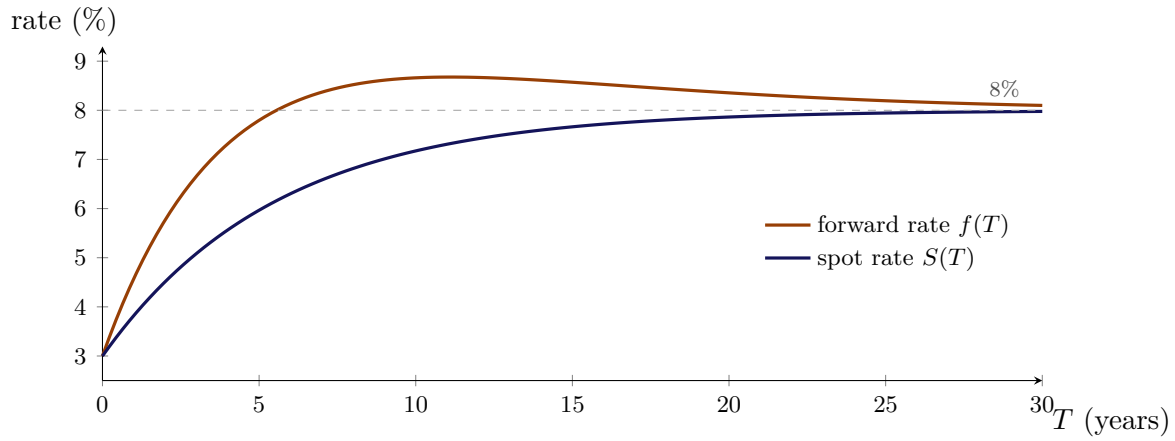


Figure 4.8.: The Hull–White spot curve and the forward curve it implies. The spot curve rises monotonically to its asymptote of 8%; the forward curve overshoots it, peaking at 8.68% near $T = 11.1$ before falling back. Both start at 3%, since $f(0) = S(0)$.

Remark 4.14.5. 遠期曲線衝過長期水準再折返，這件事值得解釋——它不是畫錯了。

$S(T)$ 是 $[0, T]$ 上所有瞬時利率的平均。既然平均要從 3% 一路爬到 8%，而起點被壓在 3%，那麼中段的瞬時利率就必須高於 8%，才拉得動整條平均線。等到平均值接近 8%、不再需要被拉動時，f 才降回 8%。

這正是「邊際領先平均」的標準圖像，也是實務上常說的：遠期曲線是即期曲線的放大版——即期曲線的任何斜率，在遠期曲線上都被乘上 T 放大了（見 (4.22)）。

§4.15. Term Structure Theories

Everything up to here was arithmetic: given prices, the forward rates follow by identity. The remaining question is economic — what, if anything, does $f(i, j)$ say about the *future* spot rate $S(i, j)$? Answering it requires an assumption, and the candidate assumptions are the term structure theories.

§4.15.i. Unbiased Expectations Theory

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[5] J. Hull & White (1994).

Theorem 4.15.1 (Unbiased expectations theory)

The forward rate equals the average future spot rate,

$$f(a,b) \;=\; E\big[\,S(a,b)\,\big]. \tag{4.23}$$

- It does *not* imply that the forward rate is an accurate predictor for the future spot rate.
- It implies the maturity strategy and the rollover strategy produce the same result at the horizon "on average".

Remark 4.15.2. 先釐清定理本身：$f(a,b)$ 是今天就算得出來的數字，$S(a,b)$ 則要到時點 a 才會在市場上報出來，今天看它是隨機變數。這個理論假設前者等於後者的期望值。它是一個關於市場的假設，不是前面那種由無套利推出來的定理。

第一點：「無偏」不等於「準」。 $f(a,b) = E[S(a,b)]$ 只是說 forward rate 沒有系統性地偏高或偏低：把很多次的預測誤差 $S(a,b) - f(a,b)$ 平均起來是 0。但任何一次實際報出的 $S(a,b)$ 都可能離 $f(a,b)$ 很遠，例如今天算出 $f(1,2) = 5\%$，一年後的實際利率是 3% 或 7% 都跟理論不衝突。

第二點：兩種策略「平均」打平。 回想 Section 4.10 的兩種策略。maturity strategy 的終值今天已經確定；rollover strategy 的終值要用到未來的 $S(a,b)$，所以是隨機的。forward rate 的定義是：若 rollover 後半段的利率剛好等於 $f(a,b)$，兩者就打平。這個理論再假設 $f(a,b)$ 是 $S(a,b)$ 的期望值，於是 rollover 的終值平均而言等於 maturity 的終值；實際走出來的某一條路徑則可能高也可能低。「on average」加了引號，是因為這句話只在粗略意義下成立：終值裡的 $S(a,b)$ 是出現在 $(1+S(a,b))^{b-a}$ 裡，而一般來說 $E\big[(1+S)^k\big] \neq (1+E[S])^k$。下一小節的「bad」expectations theory 就是在這種細節上出問題。

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The theory ties the shape of today's curve to the market's expectations.

Proposition 4.15.3

Under (4.23), a normal spot rate curve is due to the fact that the market expects the future spot rate to rise. Conversely, the spot rate is expected to fall if and only if the spot rate curve is inverted.

Proof. First, $f(j,j+1) > S(j+1)$ if and only if $S(j+1) > S(j)$. Indeed, by (4.14),

$$1 + f(j,j+1) = \frac{\big(1+S(j+1)\big)^{j+1}}{\big(1+S(j)\big)^{j}},$$

so $f(j,j+1) > S(j+1)$ is equivalent to $\big(1+S(j+1)\big)^{j} > \big(1+S(j)\big)^{j}$, i.e. to $S(j+1) > S(j)$. Substituting (4.23) gives

$$E\big[\,S(j,j+1)\,\big] > S(j+1) > \cdots > S(1) \qquad \text{if and only if} \qquad S(j+1) > \cdots > S(1). \qquad \blacksquare$$

Remark 4.15.4. 第一個小結論不需要任何理論——它純粹是 (4.14) 的代數重寫，也就是 Section 4.11 的特例。理論的作用只在最後一步：把算出來的 $f(j,j+1)$ 換成期望值 $E[S(j,j+1)]$。

那串不等式為什麼等於「預期利率上升」？ 關鍵是把每個符號讀成「哪個時點的一期利率」：

- $S(1) = S(0,1)$ 是今天的一期利率；
- $S(j,j+1)$ 是時點 j 的一期利率，今天未知，$E[S(j,j+1)]$ 就是市場對它的預期。

「預期未來利率上升」的意思就是：預期時點 j 的一期利率高於今天的一期利率，即 $E[S(j,j+1)] > S(1)$。那串不等式的頭尾正是這件事：

$$\underbrace{E\big[S(j,j+1)\big]}_{\text{預期的未來一期利率}} \;>\; S(j+1) \;>\; \cdots \;>\; \underbrace{S(1)}_{\text{今天的一期利率}},$$

中間的 $S(j+1) > \cdots > S(1)$ 就是「spot curve 是 normal」。所以整句讀成：曲線 normal $\iff$ 預期的未來一期利率高過今天的一期利率。

直覺上：由 (4.15)，$S(j+1)$ 是 $S(1), f(1,2), \ldots, f(j,j+1)$ 的（幾何）平均；在此理論下這些 f 就是各期的預期一期利率。所以長天期 spot rate 就是「今天的利率加上未來各期預期利率」的平均。平均值隨期限上升（normal），唯一的原因就是後面加進來的預期利率比較高——市場預期利率會漲。

所以這條命題的內容其實是：**在無偏預期理論（unbiased expectations theory）之下**，「殖利率曲線正常」與「市場預期短率將上升」是同一件事。這也是新聞上「倒掛的殖利率曲線預示降息」那句話的理論根據——但別忘了它建立在假設之上，而下一小節正是要說明這一類假設有多脆弱。

§4.15.ii. A "Bad" Expectations Theory

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Consider the natural-sounding strengthening: the expected returns[6] on all possible riskless bond strategies are equal for all holding periods. It turns out to be internally inconsistent.

Over a *two*-period horizon, buying a two-period bond and rolling over one-period bonds must tie:

$$\big(1+S(2)\big)^2 \;=\; \big(1+S(1)\big)\,E\big[\,1+S(1,2)\,\big]. \tag{4.24}$$

After rearrangement,

$$\frac{1}{E\big[\,1+S(1,2)\,\big]} \;=\; \frac{1+S(1)}{\big(1+S(2)\big)^2}. \tag{4.25}$$

Remark 4.15.5. 這個理論在說什麼。先固定一個**持有期間**（holding period）H：今天投入 \$1，看時點 H 手上有多少。在這 H 期內，用無違約風險的債券可以有很多玩法：買 H 期 zero 持有到期、買短的一路 rollover、買更長的 zero 然後在時點 H 提前賣掉……這個理論主張：只要持有期間相同，所有這些策略的預期 one-plus return 都相等，而且這對每一個 H 都成立。其中 **one-plus return** = 期末價值 / 期初投入，算的是整個持有期間：例如兩期 zero 持有兩期是 $(1+S(2))^2$，不是 $1+S(2)$。

背後的想法是：若投資人只看期望報酬、不在乎風險，那麼哪個策略的期望報酬比較高，大家就會去搶，價格被推到直到打平為止。注意「無風險」只是指不會違約；rollover 或提前賣出時，未來的利率與價格今天仍未知，所以報酬還是隨機的，才需要取期望值。

所以什麼時候「要相等」？只比同一個持有期間的策略：

- $H=2$（這一頁）：兩期 zero 持有到期 vs. 一期 zero rollover 兩次；
- $H=1$（p. 158–159）：一期 zero 持有到期（報酬確定是 $1+S(1)$）vs. 兩期 zero 持有一期後賣掉（賣價未知）。

p. 159 說「the theory says the returns are equal」，就是因為那兩個策略的持有期間都是 1，理論要求它們預期報酬相同。

跟 unbiased expectations theory 的關係。不是先相信 unbiased expectations theory 再去推廣它。單看 $H=2$ 那條式子，它其實就是 unbiased expectations theory 用在 $f(1,2)$ 上（$E[S(1,2)] = f(1,2)$）；單看 $H=1$ 那條式子，它就是下一小節的 local expectations theory。兩條各自都可以成立，這個「bad」theory 的問題是把「投資人不在乎風險」這個直覺同時套到所有持有期間上。接下來三頁要證明：兩條式子不可能同時成立，除非未來利率毫無不確定性。所以結論不是「unbiased expectations theory 錯了」，而是「expectations theory 只能挑一個持有期間來要求，不能全都要」。

證明的策略。先用 $H=2$ 得到一條等式（這一頁），再用 $H=1$ 得到另一條（下一頁），兩條合起來會跟 Jensen's inequality 衝突。

兩期的持有期間。今天投入 \$1，看時點 2 的結果：

- 買兩期 zero 持有到期：終值 $(1+S(2))^2$，今天就確定；

[6] More precisely, the one-plus returns.

- 買一期 zero，時點 1 再以當時的利率 $S(1,2)$ 滾一期：終值 $(1+S(1))(1+S(1,2))$，因為 $S(1,2)$ 今天未知，只能取期望值。

理論說兩者預期相等，就是 (4.24)。到這裡還只是 unbiased expectations theory 的兩期版本（$E[1+S(1,2)] = 1 + E[S(1,2)]$），沒有任何問題。

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Now consider two *one*-period strategies.

- Strategy one buys a two-period bond for $(1+S(2))^{-2}$ dollars and sells it after one period. Its price then will be $(1+S(1,2))^{-1}$, so the expected return is

$$E\big[\,(1+S(1,2))^{-1}\,\big]/(1+S(2))^{-2}.$$

- Strategy two buys a one-period bond with a return of $1+S(1)$.

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The theory says the returns are equal:

$$\frac{1+S(1)}{\big(1+S(2)\big)^2} \;=\; E\left[\frac{1}{1+S(1,2)}\right].$$

Combining this with (4.25) yields

$$E\left[\frac{1}{1+S(1,2)}\right] \;=\; \frac{1}{E\big[\,1+S(1,2)\,\big]}. \tag{4.26}$$

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Proposition 4.15.6

(4.26) is impossible save for a certain economy — that is, unless $S(1,2)$ is known today.（除非是一個確定性的經濟——也就是未來利率 $S(1,2)$ 今天就已知、沒有任何隨機性——否則 (4.26) 不可能成立。這裡的 save for 是「除了」。）

Proof. Jensen's inequality states that $E[\,g(X)\,] > g\big(E[X]\big)$ for any nondegenerate random variable X and strictly convex function g (i.e. $g''(x) > 0$). Use

$$g(x) \;\triangleq\; (1+x)^{-1},$$

for which $g''(x) = 2(1+x)^{-3} > 0$ whenever $x > -1$. With $X = S(1,2)$,

$$E\left[\frac{1}{1+S(1,2)}\right] \;>\; \frac{1}{1+E[\,S(1,2)\,]} \;=\; \frac{1}{E[\,1+S(1,2)\,]},$$

which contradicts (4.26) unless $S(1,2)$ is degenerate, i.e. deterministic. ∎

Remark 4.15.7. 這個結果的教訓比它的算式重要，值得講清楚。

錯在哪裡？不在於某一個式子算錯，而在於這個理論要求太多：它同時宣稱所有期間長度、所有策略的期望報酬都相等。兩期視角給出 (4.24)，一期視角給出另一條式子，兩者合起來就逼出 (4.26)——一個只有在未來利率毫無不確定性時才成立的等式。

根源是「期望值」與「取倒數」不可交換。債券價格是利率的凸函數（$1/(1+x)$ 凸），所以「先取期望再折現」與「先折現再取期望」必然不同，差額就是 **Jensen 不等式**（Jensen's inequality）留下的缺口。這與 Section 3.10 的凸性其實是同一件事的兩個面貌：那裡凸性讓持有人「漲賺得比跌賠得多」，這裡凸性讓 $E[1/(1+X)] > 1/E[1+X]$。

出路是只挑一種視角來要求，別全都要——這就是下一小節的**局部預期理論**（local expectations theory）。

§4.15.iii. Local Expectations Theory

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Theorem 4.15.8 (Local expectations theory)

The expected rate of return of any bond over *a single period* equals the prevailing one-period spot rate:

$$\frac{E\big[\,(1+S(1,n))^{-(n-1)}\,\big]}{(1+S(n))^{-n}} \;=\; 1+S(1) \qquad \text{for } \textit{all } n>1. \tag{4.27}$$

Remark 4.15.9. (4.27) 就是「今天買一張 n 期 zero、持有一期後賣掉」的預期 one-plus return（賣價 / 買價）：

- 分母 $(1+S(n))^{-n}$：這張 n 期 zero 今天的價格（買價）；
- 分子：到了時點 1，它變成剩 $n-1$ 期的 zero，當時的價格是 $(1+S(1,n))^{-(n-1)}$，其中 $S(1,n)$ 是時點 1 才報出的利率，今天未知，所以取期望值（預期賣價）；
- 右邊 $1+S(1)$：直接買一期 zero 持有到期的報酬，今天就確定。

兩邊比的是比值，與買多少無關：左邊用面額 \$1 的 zero 算，右邊若也用面額 \$1 的一期 zero，買價 $(1+S(1))^{-1}$、到期拿 \$1，比值同樣是 $1+S(1)$，跟「投入 \$1 拿回 $1+S(1)$」一樣。所以整條式子說：持有一期的話，買哪一張 *zero* 預期報酬都一樣，都等於一期利率。$n=2$ 就是 p. 159 那條式子；這裡只要求持有期間 $H=1$，不再同時要求 $H=2$，因此不會碰到上一小節的 Jensen 矛盾。

等價的寫法是把分母移過去：

$$\underbrace{(1+S(n))^{-n}}_{\text{今天價格}} \;=\; \frac{E\big[\,\text{時點1 的價格}\,\big]}{1+S(1)},$$

即今天的價格 = 下一期的預期價格用一期利率折現回來。(today's price of every bond is the expectation of its price next period, discounted at the known one-period rate. That is precisely the recursive structure a tree or lattice model needs, which is why the theory reappears later in the course as the *risk-neutral* pricing convention.)

§4.16. Duration, in Practice

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We had assumed parallel shifts in the spot rate curve — the caveat flagged back in Section 3.6.ii. Now that the curve is a genuine object with n separate coordinates, that assumption can be relaxed.

To handle more general shifts, define a vector $[c_1, c_2, \ldots, c_n]$ that characterizes the shift:

- Parallel shift: $[1, 1, \ldots, 1]$.
- Twist: $[1, 1, \ldots, 1, -1, \ldots, -1]$, $[1.8, 1.6, 1.4, 1, 0, -1, -1.4, \ldots]$, etc.
- ...

At least one c_i should be 1 as the reference point.

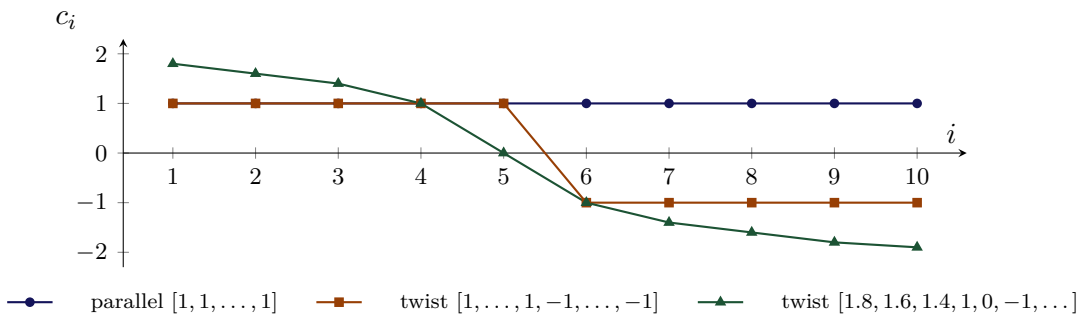


Figure 4.9.: Shift vectors. Each describes how much the spot rate at maturity i moves per unit of the common factor y; a parallel shift is the special case in which every maturity moves alike.

Remark 4.16.1. 問題在哪。之前算 duration 時說「利率上升 Δy，價格掉多少」，暗中假設曲線上每一期的 spot rate 都一起上升 Δy，也就是整條曲線平移（parallel shift）。但實際上曲線常常不是這樣動的：可能短天期漲、長天期跌（曲線變平或倒掛），也可能中間動得多、兩端動得少。

這一頁的做法。把「曲線怎麼動」拆成兩部分：

$$S(i) \;\longrightarrow\; S(i) + y\,c_i,$$

- y：一個數字，代表動多大；
- $[c_1,\ldots,c_n]$：一個向量，代表動的形狀——第 i 期跟著 y 動幾倍。

以 $y = 0.1\%$ 為例：

- $[1,1,\ldots,1]$：每一期都 $+0.1\%$，就是原本的平移；
- $[1,\ldots,1,-1,\ldots,-1]$：前幾期 $+0.1\%$、後幾期 -0.1%，短端漲長端跌，曲線被「扭」了（twist）；
- $[1.8,1.6,1.4,1,0,-1,\ldots]$：第 1 期 $+0.18\%$、第 4 期 $+0.1\%$、第 5 期不動、再往後轉為下跌，是比較平滑的扭轉。

上圖就是把這三個向量畫出來：橫軸是期別 i，縱軸是 c_i。

為什麼要有一個 $c_i = 1$。 y 和 c_i 只以乘積出現，$[c_i]$ 全部乘 2、y 除以 2 是同一種變動，所以要固定尺度。規定某一期的 $c_i = 1$，y 就有了明確意義：那一期的 spot rate 動了多少。

下一頁再用這個 y 定義 duration：y 動一點點時，價格變動的百分比。它是原本 duration 的推廣：若形狀選平移 $[1,\ldots,1]$、且曲線本來就是平的（$S(i)$ 全等於某個 r），價格變成 $P(y) = \sum_i C_i/(1+r+y)^i$，這正是把殖利率從 r 移到 $r+y$ 的價格函數，於是 $-\dfrac{1}{P}\dfrac{\partial P}{\partial y}\Big|_{y=0} = \dfrac{1}{P}\sum_i \dfrac{i\,C_i}{(1+r)^{i+1}}$，恰好就是 (3.11) 的 modified duration。

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Definition 4.16.2 (Duration under a general shift)**.** Let

$$P(y) \;\triangleq\; \sum_i \frac{C_i}{\big(1 + S(i) + y\,c_i\big)^i} \tag{4.28}$$

be the price associated with the cash flow $C_1, C_2, \ldots$, and define duration as

$$-\left.\frac{\partial P(y)/P(0)}{\partial y}\right|_{y=0} \qquad \text{or} \qquad -\frac{P(\Delta y) - P(-\Delta y)}{2P(0)\,\Delta y}.$$

Modified duration equals the above when

$$[\,c_1, c_2, \ldots, c_n\,] = [\,1,1,\ldots,1\,], \qquad S(1) = S(2) = \cdots = S(n).$$

Remark 4.16.3. 這一節把第三章的 duration 放回正確的位置，值得把三件事對齊。

（一）y 換了角色。 (4.28) 裡的 y 不再是殖利率，而是一個抽象的衝擊大小；真正決定各期利率怎麼動的是向量 $[c_i]$。y 動一單位，第 i 期的即期利率就動 c_i 單位。

（二）為什麼要有「至少一個 $c_i = 1$」。 y 與 $[c_i]$ 只以乘積 yc_i 出現，把 $[c_i]$ 全部乘以 2、y 全部除以 2，模型完全一樣。固定某個 $c_i = 1$ 就是挑一個參考期別來定尺度，讓算出來的 duration 有明確的單位：「當那一期的利率動 1 時，價格動幾個百分比」。

（三）第二個公式就是中央差分。它與 (3.14) 的 effective duration 是同一個數值技巧，只是這裡擾動的是整條曲線而非單一殖利率——連同 Section 3.13 之後那段關於 Δy 取多大的討論，也照樣適用。

實務上把不同的 $[c_i]$ 各算一個 duration，就得到**關鍵利率存續期間**（key rate duration）：取 $[c_i]$ 為只在某一期為 1、其餘為 0 的向量，便可量出部位對曲線上各個點的個別暴險，而不是只有一個總數。一個 duration 為零的部位，仍可能對**扭轉**（twist）有巨大暴險。twist 指曲線繞著某個期別「轉動」：一端上升、另一端下降，改變的是曲線的斜率而不是整體高低（例如短端 -0.1%、長端 $+0.1\%$，曲線變陡）。例如持有長天期債券、同時放空短天期債券，把數量配到平移下的 duration 為零；平移時兩邊損益互相抵銷，但遇到上述變陡的 twist，長債因長端利率上升而跌價、放空的短債因短端利率下降而漲價，兩邊同時虧錢。

§4.17. Some Loose Ends on Dates

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Every formula in this chapter has indexed time by integers $1, 2, \ldots, n$. Converting those to actual dates raises a list of unglamorous questions that nonetheless have to be answered before any of it can be coded:

- Holidays.
- Weekends.
- Business days ($T + 2$, etc.).
- Shall we treat a year as 1 year whether it has 365 or 366 days?

Note 4.17.1. These are the same concerns as the day count conventions of Section 2.10, now compounded by the settlement calendar. None of them is deep, and all of them are mandatory: a discount factor $d(i)$ is only as well defined as the date i names, and two desks using different holiday calendars will bootstrap two different curves from the same quotes.

（翻譯：這些問題和 Section 2.10 的 day count convention 是同一類，只是現在還多疊了一層**交割日曆**（settlement calendar）的問題。它們都不深奧，卻每一個都非處理不可：折現因子 $d(i)$ 的定義只能跟「第 i 期」指的是哪一天一樣明確；兩個交易櫃檯若用不同的假日日曆，拿同一組報價 bootstrap 出來的曲線就會不同。其中 $T + 2$ 指成交後第 2 個營業日才交割（錢券互換），所以現金流的實際日期還要跳過週末與假日來算。）

5. Fundamental Statistical Concepts

Slides pp. 165–166

There are three kinds of lies:
lies, damn lies, and statistics.
— Misattributed to Benjamin Disraeli (1804–1881)

If 50 million people believe a foolish thing,
it's still a foolish thing.
— George Bernard Shaw (1856–1950)

One death is a tragedy,
but a million deaths are a statistic.
— Josef Stalin (1879–1953)

Everything so far has been deterministic. A bond's cash flows were known, the curve was observed, and the only uncertainty admitted — in the term structure theories of Section 4.15 — was smuggled in through an expectation operator that was never defined.

This chapter supplies the vocabulary that the rest of the course runs on. Nothing in it is deep; what matters is that every later model, from the binomial tree to Black–Scholes to Monte Carlo pricing, is assembled from these pieces. Three of them will do most of the work:

- **Conditional expectation**, which is how a model says "given what we know today".
- **The normal distribution**, which is what we will assume *log* returns follow.
- **The lognormal distribution**, which is what prices then follow — and the reason (5.29) below turns into the Black–Scholes formula later on.

The chapter is built in layers, and each section uses only what came before it. Section 5.1–Section 5.3 develop means, variances and covariances from the linearity of expectation, including the Cauchy–Schwarz inequality, Chebyshev's inequality and the law of large numbers. Section 5.4 defines conditional expectation and proves the tower law. Section 5.5–Section 5.7 introduce densities, the normal distribution and its moment generating function, and the multivariate normal distribution. Section 5.8 and Section 5.9 turn this theory into algorithms for simulation, and prove the central limit theorem along the way. Finally, Section 5.10 combines everything into the lognormal model for prices, ending with the expected payoff of a call.

Note 5.0.1. The epigraphs are a warning worth keeping. Figure 1.4 already showed that real returns are *not* normal: the tails are far too heavy. We use the normal distribution anyway, because it is the only one that stays tractable under the operations we need. The discipline recommended by the Modelers' Hippocratic Oath in Section 1.5 applies exactly here — know precisely where the assumption is wrong.

§5.1. Moments

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The expectation $E[X]$ tells us where a random variable is centred, but nothing about how widely it is spread or how it moves together with other random variables. Moments fill in that picture. Throughout, we take for granted the two basic properties of expectation: it is *linear*, $E[aX + bY] = a\,E[X] + b\,E[Y]$, and *monotone*, $X \le Y$ implies $E[X] \le E[Y]$. Apart from one limiting argument in Lemma 5.1.4, everything in this section is derived from these two facts.

§5.1.i. Variance and Covariance

For $n \geq 1$, the **nth moment** of X is $E[X^n]$, and its **nth central moment** is $E\big[(X-\mu)^n\big]$, where $\mu = E[X]$. The mean is the first moment. The second central moment measures spread and gets its own name.

Definition 5.1.1. The **variance** of a random variable X is defined as

$$\operatorname{Var}[X] \triangleq E\big[(X - E[X])^2\big]. \tag{5.1}$$

The **covariance** between random variables X and Y is

$$\operatorname{Cov}[X,Y] \triangleq E\big[(X-\mu_X)(Y-\mu_Y)\big], \tag{5.2}$$

where μ_X and μ_Y are the means of X and Y, respectively. Random variables X and Y are **uncorrelated** if $\operatorname{Cov}[X,Y] = 0$.

The covariance is positive when X and Y tend to be above their means at the same time, and negative when one tends to be above its mean while the other is below. Taking $Y = X$ in (5.2) recovers $\operatorname{Cov}[X,X] = \operatorname{Var}[X]$: variance is covariance with oneself. So most facts about variance are special cases of facts about covariance.

Proposition 5.1.2

For random variables X, Y, Z with finite variances and constants a, b, c:

1. (Computational form) $\operatorname{Cov}[X,Y] = E[XY] - E[X]\,E[Y]$; in particular $\operatorname{Var}[X] = E[X^2] - \big(E[X]\big)^2$.
2. (Symmetry) $\operatorname{Cov}[X,Y] = \operatorname{Cov}[Y,X]$.
3. (Bilinearity) $\operatorname{Cov}[aX+bY,\, Z] = a\,\operatorname{Cov}[X,Z] + b\,\operatorname{Cov}[Y,Z]$, and likewise in the second argument.
4. (Constants do not matter) $\operatorname{Cov}[X+c,\, Y] = \operatorname{Cov}[X,Y]$ and $\operatorname{Cov}[c,\, Y] = 0$.
5. (Scaling of variance) $\operatorname{Var}[a+bX] = b^2\,\operatorname{Var}[X]$.

Proof. **1.** Expand the product and use linearity, remembering that μ_X and μ_Y are constants:

$$E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y = E[XY] - \mu_X\mu_Y.$$

2. The definition is symmetric in X and Y.

3. The mean of $aX+bY$ is $a\mu_X + b\mu_Y$, so $(aX+bY) - E[aX+bY] = a(X-\mu_X) + b(Y-\mu_Y)$. Multiply by $Z-\mu_Z$ and take expectations. The second argument follows by symmetry.

4. $X+c$ has mean $\mu_X + c$, so $(X+c) - E[X+c] = X - \mu_X$ and nothing changes. A constant equals its own mean, so $c - E[c] = 0$.

5. By items 3 and 4, $\operatorname{Var}[a+bX] = \operatorname{Cov}[a+bX,\, a+bX] = \operatorname{Cov}[bX,\, bX] = b^2\,\operatorname{Cov}[X,X]$. ∎

Items 2 and 3 say that covariance behaves like a dot product of vectors, with the variance playing the role of squared length. This analogy is worth keeping in mind: the Cauchy–Schwarz inequality in Section 5.2 and the formula for the variance of a sum in Section 5.3 are both the random-variable versions of familiar facts about dot products. Item 5 says that shifting a random variable does not change its spread, while scaling it by b scales the standard deviation by $|b|$.

§5.1.ii. What Variance Controls

Variance is defined as an average squared distance from the mean. The next results make precise the idea that a small variance forces X to stay close to its mean.

Lemma 5.1.3 (Markov's inequality)
If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \;\leq\; \frac{E[X]}{a}.$$

Proof. Let $\mathbf{1}_{\{X \geq a\}}$ be the random variable that equals 1 when $X \geq a$ and 0 otherwise. Then $a\,\mathbf{1}_{\{X \geq a\}} \leq X$: when $X \geq a$ the left side is $a \leq X$, and otherwise it is $0 \leq X$. Taking expectations and using monotonicity gives $a\,P(X \geq a) \leq E[X]$. ■

Lemma 5.1.4
If $X \geq 0$ and $E[X] = 0$, then $X = 0$ with probability one.

Proof. By Lemma 5.1.3, $P(X \geq 1/n) \leq n\,E[X] = 0$ for every $n \geq 1$. The event $\{X > 0\}$ is the union of the increasing events $\{X \geq 1/n\}$, so $P(X > 0) = \lim_{n \to \infty} P(X \geq 1/n) = 0$. ■

Proposition 5.1.5 1. $\mathrm{Var}[X] \geq 0$, and $\mathrm{Var}[X] = 0$ if and only if X is a constant with probability one.
2. (Chebyshev's inequality) If $\sigma = \sqrt{\mathrm{Var}[X]} > 0$ and $k > 0$, then

$$P\big(\,|X - \mu| \geq k\sigma\,\big) \;\leq\; \frac{1}{k^2}.$$

Proof. **1.** $\mathrm{Var}[X]$ is the expectation of the non-negative random variable $(X - \mu)^2$, so it is non-negative. If it is zero, Lemma 5.1.4 gives $(X - \mu)^2 = 0$, that is, $X = \mu$, with probability one. Conversely, a constant has zero variance by item 5 of Proposition 5.1.2 with $b = 0$.
2. Apply Lemma 5.1.3 to $(X-\mu)^2 \geq 0$ with $a = k^2\sigma^2$: $P\big((X-\mu)^2 \geq k^2\sigma^2\big) \leq \sigma^2/(k^2\sigma^2)$. ■

Chebyshev's inequality holds for *every* distribution with finite variance, which is why it is weak: it only says that at most $1/4$ of the probability lies more than two standard deviations from the mean. For a normal distribution the true figure is about 4.6%. The gap between the two is the room that heavy tails can occupy.

The third and fourth central moments, scaled by σ, describe the shape of the distribution beyond its spread:

$$\text{skewness} \;=\; \frac{E\big[(X-\mu)^3\big]}{\sigma^3}, \qquad \text{kurtosis} \;=\; \frac{E\big[(X-\mu)^4\big]}{\sigma^4}.$$

Skewness measures asymmetry, and kurtosis measures how much of the variance comes from rare, large deviations. Both are unchanged by the scaling $X \mapsto a + bX$ with $b > 0$, for the same reason as item 5 of Proposition 5.1.2. Section 5.6 shows that the normal distribution has kurtosis 3.

§5.1.iii. Independence and Uncorrelatedness

Uncorrelatedness is often confused with a much stronger notion.

Definition 5.1.6. Random variables X and Y are **independent** if, for all sets A and B,

$$P(X \in A,\; Y \in B) = P(X \in A)\,P(Y \in B).$$

In plain words: knowing the value of X tells you nothing at all about Y. The following form of the definition is the one used in calculations.

Proposition 5.1.7

X and Y are independent if and only if

$$E\big[f(X)\,g(Y)\big] = E\big[f(X)\big]\,E\big[g(Y)\big]$$

for all functions f and g for which the expectations exist.

Proof. Suppose X and Y are independent and discrete. Independence with $A = \{x\}$ and $B = \{y\}$ gives $P(X = x, Y = y) = P(X = x)\,P(Y = y)$, so

$$E\big[f(X)\,g(Y)\big] = \sum_{x,y} f(x)\,g(y)\,P(X = x)\,P(Y = y) = \Big(\sum_x f(x)\,P(X = x)\Big)\Big(\sum_y g(y)\,P(Y = y)\Big),$$

which is $E[f(X)]\,E[g(Y)]$. With densities, independence means the joint density factors as $f_X(x)f_Y(y)$, and the same computation works with integrals in place of sums.

Conversely, take $f = \mathbf{1}_A$ and $g = \mathbf{1}_B$, the functions that equal 1 on A (resp. B) and 0 elsewhere. Then $E[\mathbf{1}_A(X)\,\mathbf{1}_B(Y)] = P(X \in A, Y \in B)$ and $E[\mathbf{1}_A(X)] = P(X \in A)$, so the identity is exactly the definition of independence. ∎

Corollary 5.1.8

Independent random variables with finite means are uncorrelated.

Proof. First, $E[XY]$ is finite: applying Proposition 5.1.7 to the non-negative functions $|x|$ and $|y|$ gives $E\big[|XY|\big] = E\big[|X|\big]\,E\big[|Y|\big] < \infty$.[1] Now take $f(x) = x$ and $g(y) = y$: $E[XY] = E[X]\,E[Y]$, so $\mathrm{Cov}[X, Y] = E[XY] - E[X]\,E[Y] = 0$ by the computation in item 1 of Proposition 5.1.2, which uses only that $E[X]$, $E[Y]$ and $E[XY]$ are finite. ∎

No second moments are needed here: for dependent random variables, finite variances are what guarantee that $\mathrm{Cov}[X, Y]$ is finite (by the Cauchy–Schwarz inequality of Section 5.2), but independence already makes $E[XY]$ finite.

The proof shows exactly how much weaker uncorrelatedness is. $\mathrm{Cov}[X, Y] = 0$ is the single identity $E[XY] = E[X]E[Y]$, which is Proposition 5.1.7 for *one* choice of f and g. Independence demands it for all f and g, including nonlinear ones such as x^2 or e^x. So the converse of the corollary fails.

Example 5.1.9. Let $X \sim N(0, 1)$ and $Y = X^2$. With $f(x) = x$ and $g(y) = y$, we get $E[XY] = E[X^3] = 0 = E[X]\,E[X^2]$, so X and Y are uncorrelated. (The odd moments of X vanish because its density is symmetric about 0.) But with $f(x) = x^2$ and $g(y) = y$, we get $E[X^2Y] = E[X^4] = 3$ while $E[X^2]\,E[Y] = 1$, so X and Y are not independent. (The value $E[X^4] = 3$ is computed in Section 5.6.) Of course they are not: Y is computed directly from X.

Section 5.2 explains what uncorrelatedness *does* say: that the best straight line through the data is flat.

[1] For non-negative f and g the identity in Proposition 5.1.7 holds even when the expectations are infinite; the proof is the same.

§5.2. Correlation

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Covariance mixes two things: how strongly two variables move together, and how large they are. Measuring a price in cents instead of dollars multiplies every covariance with it by 100, although nothing about the co-movement has changed. Correlation removes the size.

Definition 5.2.1. The **standard deviation** of X is the square root of the variance,

$$\sigma_X \triangleq \sqrt{\mathrm{Var}[X]}.$$

The **correlation** (or **correlation coefficient**) between X and Y is

$$\rho_{X,Y} \triangleq \frac{\mathrm{Cov}[X,Y]}{\sigma_X \sigma_Y}, \tag{5.3}$$

provided both have nonzero standard deviations.[2]

§5.2.i. Invariance under Rescaling

Proposition 5.2.2

For constants a, c and nonzero b, d,

$$\rho_{a+bX,\, c+dY} = \mathrm{sign}(bd)\, \rho_{X,Y}.$$

In particular, correlation is unchanged by changing the units or the origin of either variable.

Proof. By Proposition 5.1.2, $\mathrm{Cov}[a+bX, c+dY] = bd\,\mathrm{Cov}[X,Y]$, while $\sigma_{a+bX} = |b|\,\sigma_X$ and $\sigma_{c+dY} = |d|\,\sigma_Y$. Dividing, the ratio is $bd/(|b||d|) = \mathrm{sign}(bd)$ times $\rho_{X,Y}$. ∎

§5.2.ii. The Range of Correlation

The fact that ρ lies between -1 and 1 comes from the following inequality, the random-variable version of $|\boldsymbol{u}\cdot\boldsymbol{v}| \le \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$.

Lemma 5.2.3 (Cauchy–Schwarz inequality)

For random variables U and V with $E[U^2]$ and $E[V^2]$ finite,

$$\big(E[UV]\big)^2 \le E[U^2]\,E[V^2].$$

If $E[U^2] > 0$, equality holds if and only if $V = t_0 U$ with probability one for some constant t_0.

Proof. First, $E[UV]$ is finite, because $|UV| \le \frac{1}{2}(U^2+V^2)$. If $E[U^2] = 0$, then $U = 0$ with probability one by Lemma 5.1.4, and both sides are zero. Otherwise, for every real number t, the random variable $(V-tU)^2$ is non-negative, so

$$0 \le q(t) \triangleq E\big[(V-tU)^2\big] = E[V^2] - 2t\,E[UV] + t^2 E[U^2].$$

This is a quadratic in t that opens upward and never goes below zero, so it has at most one real root and its discriminant is not positive: $4\big(E[UV]\big)^2 - 4E[U^2]\,E[V^2] \le 0$.

[2] Wilmott (2009): "the correlations between financial quantities are notoriously unstable." It may even break down "at high-frequency time intervals" (Budish, Cramton, & Shim, 2015).

If equality holds, the discriminant is zero and q has a double root $t_0 = E[UV]/E[U^2]$. Then $E\big[(V - t_0U)^2\big] = 0$, so $V = t_0U$ with probability one by Lemma 5.1.4. Conversely, if $V = t_0U$, both sides equal $t_0^2\big(E[U^2]\big)^2$. ■

Theorem 5.2.4

$-1 \le \rho_{X,Y} \le 1$. Moreover $|\rho_{X,Y}| = 1$ if and only if $Y = a + bX$ with probability one for some constants a and $b \neq 0$, that is, Y is an **affine** function of X;[a] in that case $\rho_{X,Y}$ has the sign of b.

[a] An *affine* function（仿射函數）is one of the form $f(x) = a + bx$: a linear function plus a constant. Its graph is any straight line, not necessarily one through the origin. Strictly speaking, a *linear* function also requires $a = 0$, but $\rho = \pm 1$ does not; for example, $Y = 3 + 2X$ has $\rho = 1$.

Proof. Apply Lemma 5.2.3 to $U = X - \mu_X$ and $V = Y - \mu_Y$. Then $E[UV] = \mathrm{Cov}[X, Y]$, $E[U^2] = \sigma_X^2 > 0$ and $E[V^2] = \sigma_Y^2$, so $\mathrm{Cov}[X, Y]^2 \le \sigma_X^2\sigma_Y^2$. Dividing by $\sigma_X^2\sigma_Y^2$ gives $\rho_{X,Y}^2 \le 1$.

Equality holds if and only if $Y - \mu_Y = t_0(X - \mu_X)$, that is, $Y = a + bX$ with $b = t_0$ and $a = \mu_Y - t_0\mu_X$. Here $b \neq 0$, since otherwise Y would be constant and $\sigma_Y = 0$. Finally, Proposition 5.2.2 with X in place of Y gives $\rho_{X,\,a+bX} = \mathrm{sign}(b)\,\rho_{X,X} = \mathrm{sign}(b)$. ■

So $\rho = \pm 1$ describes a perfect straight-line relationship, and every other value lies strictly in between. The next subsection says what the values in between mean.

§5.2.iii. Correlation and the Best Straight Line

Suppose we want to predict Y by a straight-line function $a + bX$ of X, and we measure the quality of the prediction by the mean squared error $E\big[(Y - a - bX)^2\big]$.

Proposition 5.2.5

Assume $\sigma_X > 0$. The mean squared error $E\big[(Y - a - bX)^2\big]$ is minimised by

$$b^* = \frac{\mathrm{Cov}[X, Y]}{\mathrm{Var}[X]} = \rho_{X,Y}\,\frac{\sigma_Y}{\sigma_X}, \qquad a^* = \mu_Y - b^*\mu_X,$$

and the minimum value is

$$E\big[(Y - a^* - b^*X)^2\big] \;=\; \sigma_Y^2\,\big(1 - \rho_{X,Y}^2\big). \tag{5.4}$$

Proof. For fixed b, write $W = Y - bX$. Then $E\big[(W - a)^2\big] = \mathrm{Var}[W] + \big(E[W] - a\big)^2$, by item 1 of Proposition 5.1.2 applied to $W - a$, together with $\mathrm{Var}[W - a] = \mathrm{Var}[W]$ from item 5. This is smallest when $a = E[W] = \mu_Y - b\mu_X$, and the minimum is $\mathrm{Var}[W]$. By bilinearity,

$$\mathrm{Var}[\,Y - bX\,] = \mathrm{Var}[Y] - 2b\;\mathrm{Cov}[X, Y] + b^2\;\mathrm{Var}[X],$$

a quadratic in b that opens upward, with its minimum at $b^* = \mathrm{Cov}[X, Y]/\,\mathrm{Var}[X]$. Substituting b^* gives

$$\mathrm{Var}[Y] - \frac{\mathrm{Cov}[X, Y]^2}{\mathrm{Var}[X]} = \sigma_Y^2\left(1 - \frac{\mathrm{Cov}[X, Y]^2}{\sigma_X^2\sigma_Y^2}\right) = \sigma_Y^2\big(1 - \rho_{X,Y}^2\big).$$ ■

Equation (5.4) gives the cleanest reading of ρ. Before seeing X, the best constant prediction of Y is μ_Y, with mean squared error σ_Y^2. The best straight line in X reduces this error by the factor $1 - \rho^2$. So ρ^2 is the *fraction of the variance of Y explained by a straight line in X*. When

$|\rho| = 1$ the line explains everything, in agreement with Theorem 5.2.4. When $\rho = 0$ the best slope is $b^* = 0$: the line is flat and explains nothing.

This is the precise sense in which uncorrelated means "no *linear* relationship". It does not rule out other relationships. In Example 5.1.9 the points (X, Y) lie on the parabola $y = x^2$, which is symmetric about $x = 0$, so the best straight line through it is flat, even though Y is completely determined by X. Section 5.4 shows that dropping the restriction to straight lines leads to the conditional expectation, which predicts $Y = X^2$ perfectly.

Remark 5.2.6. 開頭那個腳註不是客套話，值得展開。

（一）相關係數不穩定。 ρ 是用歷史樣本估出來的，而它本身會隨時間漂移——尤其在市場承壓時，資產之間的相關係數往往一起往 1 靠攏。用平時估出來的 ρ 去算危機時的投資組合風險，會系統性地低估損失；2008 年的結構型商品定價就是栽在這裡。

（二）高頻下會崩解。把觀測間隔縮短到毫秒等級，兩個本來高度相關的資產（例如 S&P 500 期貨與其 ETF）算出來的相關係數會急遽下降，這就是所謂的 **Epps 效應**（Epps effect）。原因是兩邊的成交並非同時發生，取樣間隔一短，非同步造成的雜訊就蓋過真正的共同波動。Budish、Cramton 與 Shim（2015）指出，這個在極短時間尺度上反覆出現的價差，正是高頻交易「軍備競賽」的獵物。

（三）相關係數只看得到直線關係。做風險管理時，常用「兩個資產相關係數很低」來說明 diversification（分散投資）有效。但由 (5.4)，ρ 量的只是直線能解釋多少；真正出事的時候，資產之間的關係往往是非線性的——平常看起來各走各的，一遇到危機卻一起大跌。這種關係，就像 Example 5.1.9 的拋物線一樣，相關係數看不出來。

結論：ρ 是一個取樣間隔與樣本期間的函數，報告時必須說清楚是用什麼頻率、什麼期間算的；而且即使 ρ 很小，也不代表兩個資產沒有關係。

§5.3. Variance of a Sum

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A portfolio is a weighted sum of asset returns, so the variance of a weighted sum is the first formula of risk management. Because covariance is bilinear (Proposition 5.1.2), it is also a direct computation.

Proposition 5.3.1

The variance of a weighted sum of random variables equals

$$\operatorname{Var}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \operatorname{Cov}[X_i, X_j]. \tag{5.5}$$

It becomes

$$\sum_{i=1}^{n} a_i^2 \operatorname{Var}[X_i] \tag{5.6}$$

when the X_i are uncorrelated.[a]

[a] Bienaymé (1853).

Proof. Since variance is covariance with oneself, apply bilinearity in each argument:

$$\operatorname{Var}\Big[\sum_i a_i X_i\Big] = \operatorname{Cov}\Big[\sum_i a_i X_i, \sum_j a_j X_j\Big] = \sum_i \sum_j a_i a_j \operatorname{Cov}[X_i, X_j].$$

If the X_i are uncorrelated, every off-diagonal term vanishes and only the $i = j$ terms, $a_i^2 \operatorname{Var}[X_i]$, survive. ∎

The same argument with two different weight vectors gives the covariance of two weighted sums,

$$\operatorname{Cov}\Big[\sum_i a_i X_i,\ \sum_j b_j X_j\Big] = \sum_i \sum_j a_i b_j\ \operatorname{Cov}[\,X_i, X_j\,]. \tag{5.7}$$

§5.3.i. The Covariance Matrix

The double sums are easier to handle in matrix notation. Collect the random variables into a column vector $\boldsymbol{X} = (X_1, \dots, X_n)^{\mathsf{T}}$ and define its **covariance matrix**

$$C \triangleq \big[\operatorname{Cov}[\,X_i, X_j\,]\big]_{1 \le i,j \le n},$$

whose diagonal holds the variances. With weights $\boldsymbol{a} = (a_1, \dots, a_n)^{\mathsf{T}}$, (5.5) and (5.7) read

$$\operatorname{Var}[\,\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X}\,] = \boldsymbol{a}^{\mathsf{T}} C\, \boldsymbol{a}, \qquad \operatorname{Cov}[\,\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X},\, \boldsymbol{b}^{\mathsf{T}}\boldsymbol{X}\,] = \boldsymbol{a}^{\mathsf{T}} C\, \boldsymbol{b}.$$

Proposition 5.3.2 1. C is symmetric and **positive semidefinite**: $\boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a} \ge 0$ for every $\boldsymbol{a} \in \mathbb{R}^n$.
2. C is **positive definite** ($\boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a} > 0$ for every $\boldsymbol{a} \neq \boldsymbol{0}$) if and only if no nontrivial combination $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X}$ is constant with probability one.
3. If A is an $m \times n$ matrix and $\boldsymbol{U} = A\boldsymbol{X}$, then the covariance matrix of $\boldsymbol{U}$ is ACA^{T}.

Proof. **1.** Symmetry is item 2 of Proposition 5.1.2, and $\boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a} = \operatorname{Var}[\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X}] \ge 0$.
2. By Proposition 5.1.5, $\boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a} = \operatorname{Var}[\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X}] = 0$ exactly when $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{X}$ is constant with probability one.
3. $U_k = \sum_i A_{ki} X_i$ is a weighted sum with weights given by the kth row of A. By (5.7), $\operatorname{Cov}[U_k, U_l] = \sum_{i,j} A_{ki}\, C_{ij}\, A_{lj}$, which is the (k,l) entry of ACA^{T}. ∎

Item 2 has a concrete meaning: C fails to be positive definite exactly when one of the variables is redundant, in the sense that some combination of them carries no risk at all. Item 3 is what Section 5.9 uses to manufacture correlated random variables from independent ones.

§5.3.ii. Averages and the Law of Large Numbers

The uncorrelated case (5.6) has a consequence that the whole of Monte Carlo simulation rests on.

Theorem 5.3.3 (Weak law of large numbers)
Let $X_1, X_2, \dots$ be uncorrelated with common mean μ and common variance σ^2, and let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then $E[\overline{X}_n] = \mu$, $\operatorname{Var}[\overline{X}_n] = \sigma^2/n$, and for every $\varepsilon > 0$,

$$P\big(\,|\overline{X}_n - \mu| \ge \varepsilon\,\big) \ \le\ \frac{\sigma^2}{n\varepsilon^2} \ \longrightarrow\ 0 \qquad (n \to \infty).$$

Proof. The mean follows from linearity. By (5.6) with $a_i = 1/n$, $\operatorname{Var}[\overline{X}_n] = n \cdot \sigma^2/n^2 = \sigma^2/n$. Chebyshev's inequality (Proposition 5.1.5) with $k\sigma/\sqrt{n} = \varepsilon$ gives the bound. ∎

So the average of many independent draws settles down to the mean, and the standard error shrinks like $\sigma/\sqrt{n}$: to gain one more decimal digit of accuracy, one needs 100 times as many draws. This is exactly how Monte Carlo pricing estimates an expectation, and why it is slow.

The same formula also shows the limits of diversification. If an equally weighted portfolio holds n assets, each with variance σ^2 and every pair with correlation ρ, then (5.5) gives

$$\operatorname{Var}\Big[\frac{1}{n}\sum_{i=1}^{n} X_i\Big] = \frac{1}{n^2}\Big(n\sigma^2 + n(n-1)\rho\sigma^2\Big) = \frac{\sigma^2}{n} + \Big(1 - \frac{1}{n}\Big)\rho\sigma^2 \;\longrightarrow\; \rho\sigma^2.$$

The uncorrelated part of the risk can be diversified away; the common part $\rho\sigma^2$ cannot. (The formula also shows that such a ρ cannot be negative when n is large, since the variance must stay non-negative: a common correlation ρ among n variables requires $\rho \geq -1/(n-1)$.)

§5.4. Conditional Expectation

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Definition 5.4.1. "$X \mid I$" denotes X conditional on the information set I; the information set can be another random variable's value, or the past values of X, say. The **conditional expectation**

$$E[\,X \mid I\,]$$

is the expected value of X conditional on I. It is itself a *random variable*.

The definition above is informal: it does not say how to compute $E[X \mid I]$, nor in what sense it is "the" expected value. Before proving anything we make it precise, first in the simplest case and then in general. Throughout, X is assumed to have a finite mean, $E\big[|X|\big] < \infty$.

§5.4.i. The Elementary Case

By the **elementary case** we mean the following situation: the information set I consists of the value of a single random variable Y, and Y is *discrete*, taking finitely or countably many values y, each with $P(Y = y) > 0$. In this case $E[X \mid I]$, which we then write $E[X \mid Y]$, can be computed by an explicit formula. Once we learn that $Y = y$, the natural forecast of X is its average under the conditional distribution,

$$g(y) \;\triangleq\; E[\,X \mid Y = y\,] \;=\; \sum_x x\, P(X = x \mid Y = y) \;=\; \sum_x x\, \frac{P(X = x,\, Y = y)}{P(Y = y)},$$

defined for every y with $P(Y = y) > 0$. This is an ordinary number for each fixed y. The conditional expectation is what we get by plugging in the random value of Y:

$$E[\,X \mid Y\,] \;\triangleq\; g(Y).$$

Before Y is observed we do not know which y will occur, so $g(Y)$ is random; this is the sense in which $E[X \mid I]$ is a random variable. When X and Y have a joint density $f(x, y)$, the analogous formula $g(y) = \int x\, f(x, y)\, dx \big/ \int f(x, y)\, dx$ is the natural guess. This is *not* the elementary case, since $P(Y = y) = 0$ for every y and the division by $P(Y = y)$ above no longer makes sense; justifying this formula is one of the reasons for the general definition below.

Example 5.4.2. Roll a fair die and let X be the result. Suppose we are only told whether X is even, and let $Y = 1$ if it is and $Y = 0$ otherwise. Then

$$E[\,X \mid Y = 1\,] = \frac{2 + 4 + 6}{3} = 4, \qquad E[\,X \mid Y = 0\,] = \frac{1 + 3 + 5}{3} = 3.$$

So $E[X \mid Y]$ is the random variable that equals 4 with probability $\frac{1}{2}$ and 3 with probability $\frac{1}{2}$. It is coarser than X: it only resolves as much as Y does.

§5.4.ii. The General Definition

The formula for g breaks down when $P(Y = y) = 0$ for every y, and it says nothing when the information is something like "all prices up to time t". What survives in every case is the following property, which we first check in the elementary case.

Lemma 5.4.3

In the elementary case, with $E[X \mid Y] = g(Y)$ as defined above, for every bounded function h,

$$E\big[\,E[\,X \mid Y\,]\,h(Y)\,\big] \;=\; E\big[\,X\,h(Y)\,\big]. \tag{5.8}$$

Proof. Group the outcomes by the value of Y:

$$E\big[\,g(Y)\,h(Y)\,\big] = \sum_y g(y)\,h(y)\,P(Y=y) = \sum_y h(y) \sum_x x\,P(X=x,\,Y=y) = \sum_{x,y} x\,h(y)\,P(X=x,\,Y=y),$$

and the last sum is $E[X\,h(Y)]$. ∎

Read (5.8) as a statement about bets. Any bet whose payoff $W = h(Y)$ can be settled using only the information in Y pays the same on average whether it is scaled by the true outcome X or by the forecast $E[X \mid Y]$. In other words, as far as anyone who only knows Y can tell, $E[X \mid Y]$ and X are the same. This becomes the definition.

Say that a random variable W is **determined by** I if its value is known once the information in I is revealed.[3]

Definition 5.4.4. $E[X \mid I]$ is a random variable M such that

1. M is determined by I, and
2. $E[\,M\,W\,] = E[\,X\,W\,]$ for every bounded W determined by I.

Such an M always exists; this is a theorem of measure theory (the Radon–Nikodym theorem), which we take for granted. It is also unique, which is what entitles us to say "*the* conditional expectation".

Proposition 5.4.5 (Uniqueness)

If M_1 and M_2 both satisfy Definition 5.4.4, then $M_1 = M_2$ with probability one.

Proof. Let $D = M_1 - M_2$. It is determined by I, and subtracting the two versions of condition 2 gives $E[DW] = 0$ for every bounded W determined by I. Take $W = \mathbf{1}_{\{D>0\}}$, which equals 1 when $D > 0$ and 0 otherwise; it is determined by I because D is. Then $E\big[D\,\mathbf{1}_{\{D>0\}}\big] = 0$. But $D\,\mathbf{1}_{\{D>0\}} \geq 0$, so by Lemma 5.1.4 it is zero with probability one, that is, $P(D > 0) = 0$. The same argument with $W = \mathbf{1}_{\{D<0\}}$ gives $P(D < 0) = 0$. ∎

The payoff of the abstract definition is that every property below is proved the same way: write down a candidate, and check the two conditions of Definition 5.4.4. Uniqueness then says the candidate *is* the conditional expectation.

§5.4.iii. Basic Properties

[3] In measure-theoretic probability, I is a σ-field $\mathcal{G}$, and "determined by I" means "$\mathcal{G}$-measurable". When I is the value of Y, the random variables determined by I are exactly the functions $h(Y)$.

Proposition 5.4.6

Let X and Z have finite means, and let a, b be constants.

1. (Linearity) $E[\,aX + bZ \mid I\,] = a\,E[\,X \mid I\,] + b\,E[\,Z \mid I\,]$.
2. (Known quantities come out) If W is bounded and determined by I, then $E[WX \mid I\,] = W\,E[\,X \mid I\,]$. In particular $E[\,W \mid I\,] = W$.
3. (Irrelevant information is ignored) If X is independent of I, then $E[\,X \mid I\,] = E[\,X\,]$.
4. (Monotonicity) If $X \geq 0$, then $E[\,X \mid I\,] \geq 0$. Consequently $X \leq Z$ implies $E[X \mid I] \leq E[Z \mid I]$.

Proof. Write $M_X = E[X \mid I]$ and $M_Z = E[Z \mid I]$, and let V denote an arbitrary bounded random variable determined by I.

1. The candidate $aM_X + bM_Z$ is determined by I, and $E\big[(aM_X + bM_Z)V\big] = a\,E[M_XV] + b\,E[M_ZV] = a\,E[XV] + b\,E[ZV] = E\big[(aX + bZ)V\big]$.

2. The candidate WM_X is determined by I, since both factors are. The product WV is bounded and determined by I, so it is a legitimate test variable for M_X: $E\big[(WM_X)V\big] = E\big[M_X(WV)\big] = E\big[X(WV)\big] = E\big[(WX)V\big]$. For the special case, take $X = 1$, whose conditional expectation is clearly 1.

3. The candidate is the constant $E[X]$, which is determined by any I. Since V is determined by I and X is independent of I, X and V are independent, so $E[XV] = E[X]\,E[V] = E\big[E[X]\,V\big]$.

4. Take $V = \mathbf{1}_{\{M_X<0\}}$. Then $M_XV \leq 0$, while $E[M_XV] = E[XV] \geq 0$ because $X \geq 0$. So $E[M_XV] = 0$ with $M_XV \leq 0$, forcing $M_XV = 0$, that is, $P(M_X < 0) = 0$. For the consequence, apply this to $Z - X \geq 0$ and use linearity. ∎

Items 2 and 3 are the two extremes. If you already know W, conditioning on I cannot tell you anything more about it, so it behaves like a constant. If I carries no information about X, the best you can do is the unconditional mean. Everything else lies between the two.

§5.4.iv. The Tower Law

Theorem 5.4.7 (Law of iterated conditional expectations, or the tower law)

$$E[\,X\,] \;=\; E\big[\,E[\,X \mid I\,]\big]. \tag{5.9}$$

Proof. The constant $W = 1$ is bounded and determined by any I, so condition 2 of Definition 5.4.4 gives $E\big[E[X \mid I] \cdot 1\big] = E[X \cdot 1]$. ∎

In the elementary case, (5.9) reads

$$E[\,X\,] = \sum_y E[\,X \mid Y = y\,]\,P(Y = y),$$

which is the **law of total expectation**: the overall average is the weighted average of the group averages, weighted by the size of each group. In the die example, $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 3 = 3.5 = E[X]$.

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Theorem 5.4.8

If I_2 contains at least as much information as I_1, then

$$E[\,X \mid I_1\,] \;=\; E\big[\,E[\,X \mid I_2\,] \mid I_1\,\big]. \tag{5.10}$$

Proof. Let $M_2 = E[X \mid I_2]$. We check that the candidate $M_1 = E[X \mid I_1]$ satisfies the two conditions for being $E[M_2 \mid I_1]$. It is determined by I_1 by definition. Now let V be bounded and determined by I_1. Because I_2 contains everything in I_1, V is also determined by I_2, so it is a legitimate test variable for both M_1 and M_2:

$$E[\,M_1 V\,] = E[\,XV\,] = E[\,M_2 V\,].$$

The outer equalities are condition 2 for M_1 and for M_2, respectively. By uniqueness, $M_1 = E[M_2 \mid I_1]$. ∎

(5.9) is the special case where I_1 is the trivial information set (knowing nothing), since conditioning on nothing is the ordinary expectation. The other order of conditioning is even simpler: $E\big[E[X \mid I_1] \mid I_2\big] = E[X \mid I_1]$, by item 2 of Proposition 5.4.6, because $E[X \mid I_1]$ is determined by I_1 and hence by I_2. Either way, when two conditionings are stacked, *the smaller information set wins.*

Here I_1 contains price information up to time t_1, and I_2 contains price information up to a later time $t_2 > t_1$. In general,

$$I_1 \subseteq I_2 \subseteq \cdots$$

means the players never forget past data, so the information sets are increasing over time.[4]

§5.4.v. Conditional Expectation as the Best Forecast

Section 5.2.iii showed that the best *straight line* $a+bX$ for predicting Y has slope $\mathrm{Cov}[X,Y]/\,\mathrm{Var}[X]$. Conditional expectation answers the same question without the restriction to straight lines.

Theorem 5.4.9

Suppose $E[X^2] < \infty$. Among all random variables Z determined by I with $E[Z^2] < \infty$, the mean squared error $E\big[(X-Z)^2\big]$ is smallest for $Z = E[X \mid I]$. More precisely,

$$E\big[(X-Z)^2\big] \;=\; E\big[(X-M)^2\big] + E\big[(M-Z)^2\big], \qquad M = E[\,X \mid I\,]. \tag{5.11}$$

Proof. Expand $X - Z = (X - M) + (M - Z)$:

$$E\big[(X-Z)^2\big] = E\big[(X-M)^2\big] + 2\,E\big[(X-M)(M-Z)\big] + E\big[(M-Z)^2\big].$$

The random variable $V = M - Z$ is determined by I, so condition 2 of Definition 5.4.4 gives $E[XV] = E[MV]$, that is, $E\big[(X-M)V\big] = 0$, and the cross term vanishes.[5] The last term is non-negative and equals zero when $Z = M$. ∎

Equation (5.11) is Pythagoras' theorem. The forecast error $X - M$ is orthogonal to everything determined by I, so M is the "foot of the perpendicular" from X onto the space of forecasts that use only I. The regression line of Section 5.2.iii is the foot of the perpendicular onto the smaller space of straight lines $a + bX$, which is why it can do worse. For $X \sim N(0,1)$ and $Y = X^2$, the best line is flat and explains nothing, whereas $E[Y \mid X] = X^2$ predicts Y perfectly.

[4] Hirsa & Neftci (2014). This idea is formalised by sigma fields and filtrations in probability theory.

[5] Condition 2 was stated for bounded V. It extends to V with $E[V^2] < \infty$ by truncating V at $\pm n$ and letting $n \to \infty$; we skip the details.

§5.4.vi. Conditional Variance

Define the **conditional variance** by analogy with (5.1),

$$\operatorname{Var}[X \mid I] \triangleq E\big[(X - E[X \mid I])^2 \mid I\big] = E[X^2 \mid I] - \big(E[X \mid I]\big)^2.$$

The second form follows by expanding the square and using item 2 of Proposition 5.4.6 to pull out $E[X \mid I]$, just as in the unconditional case.

Proposition 5.4.10 (Law of total variance)
If $E[X^2] < \infty$, then

$$\operatorname{Var}[X] = E\big[\operatorname{Var}[X \mid I]\big] + \operatorname{Var}\big[E[X \mid I]\big]. \tag{5.12}$$

Proof. Let $M = E[X \mid I]$; by the tower law $E[M] = E[X]$. Taking expectations of the second form of $\operatorname{Var}[X \mid I]$ and using the tower law on $E[X^2 \mid I]$,

$$E\big[\operatorname{Var}[X \mid I]\big] = E[X^2] - E[M^2], \qquad \operatorname{Var}[M] = E[M^2] - \big(E[X]\big)^2.$$

Adding the two gives $E[X^2] - (E[X])^2 = \operatorname{Var}[X]$. ∎

The total uncertainty in X splits into two parts: the uncertainty that remains *after* learning I, on average, and the uncertainty in the forecast $E[X \mid I]$ itself, which comes from not yet knowing what I will say. Since both parts are non-negative, $\operatorname{Var}\big[E[X \mid I]\big] \le \operatorname{Var}[X]$: forecasts are always less volatile than the thing they forecast. In the die example, $\operatorname{Var}[X] = \frac{35}{12}$, the forecast takes the values 3 and 4 with equal probability so $\operatorname{Var}\big[E[X \mid Y]\big] = \frac{1}{4}$, and within each group the variance is $\frac{8}{3}$; indeed $\frac{8}{3} + \frac{1}{4} = \frac{35}{12}$.

Remark 5.4.11. 這一節補了很多數學，這裡把它們串起來，看它們在後面會怎麼用到。

（一）$E[X \mid I]$ 為什麼是隨機變數。從骰子的例子就看得出來：在還不知道是奇數還是偶數之前，「條件期望值」可能是 3 也可能是 4，所以它本身是個隨機變數。一般的定義（Definition 5.4.4）則是反過來刻畫它：只要是能用 I 結算的賭注，用 $E[X \mid I]$ 算出來的平均報酬都跟用 X 算的一樣。後面所有性質都是從這個定義，加上唯一性推出來的。

（二）(5.10) 的意思。用白話講：今天對「明天的預測」做預測，得到的就是今天的預測。如果你今天就能確定明天會把預測往上調，那今天就應該直接調上去。之後會定義**鞅**（martingale）：一個過程 M_t 若對所有 $t_1 \le t_2$ 都滿足 $E[M_{t_2} \mid I_{t_1}] = M_{t_1}$，就叫做鞅。由 (5.10) 馬上可以得到：對任何 X，$M_t = E[X \mid I_t]$ 都是一個鞅。無套利定價說的正是「折現後的價格是某個機率下的條件期望值」，所以折現後的價格是鞅。

（三）條件期望值是最好的預測。 Theorem 5.4.9 說 $E[X \mid I]$ 是只用 I 能做到的最小均方誤差預測。Section 5.2.iii 的回歸直線只在直線裡面找最好的，條件期望值則是在所有函數裡面找。這兩者的差距，正是前面說「相關係數看不到非線性關係」的原因。

（四）為什麼要寫出 $I_1 \subseteq I_2 \subseteq \cdots$。它把「資訊只會增加、不會忘記」這個假設明確寫出來。這也是 (5.10) 能用的前提：證明裡用到「由 I_1 決定的東西，也由 I_2 決定」。數學上這一串資訊集叫做**過濾**（filtration），是隨機過程裡描述「隨時間知道什麼」的方式。

§5.5. The Normal Distribution

The random variables so far could be discrete or continuous; nothing depended on the difference. From here on we work mostly with continuous distributions, so we first collect the tools for handling densities.

§5.5.i. Distribution Functions and Densities

The **distribution function** of a random variable X is $F_X(x) \triangleq P(X \le x)$. It increases from 0 to 1 as x runs from $-\infty$ to ∞. X has a **density** f_X if

$$P(a < X \le b) = \int_a^b f_X(x)\,dx \qquad \text{for all } a < b,$$

in which case $f_X = F_X'$ wherever f_X is continuous. Expectations are then computed by integrals instead of sums, $E[g(X)] = \int_{-\infty}^{\infty} g(x)\,f_X(x)\,dx$, and every result proved so far holds with the same proof.

We will often need the density of a transformed variable $Y = g(X)$. The rule is to go through the distribution function.

Proposition 5.5.1 (Change of variables)

Let X have density f_X, and let g be differentiable and strictly monotone with inverse $h = g^{-1}$. Then $Y = g(X)$ has density

$$f_Y(y) = f_X\big(h(y)\big)\,\big|h'(y)\big|.$$

Proof. If g is increasing, then $Y \le y$ exactly when $X \le h(y)$, so $F_Y(y) = F_X(h(y))$. Differentiating by the chain rule gives $f_Y(y) = f_X(h(y))\,h'(y)$, and $h' > 0$. If g is decreasing, then $Y \le y$ exactly when $X \ge h(y)$, so $F_Y(y) = 1 - F_X(h(y))$ and $f_Y(y) = -f_X(h(y))\,h'(y)$, with $h' < 0$. ∎

The factor $|h'(y)|$ accounts for stretching: if g spreads a small interval of x-values over a longer interval of y-values, the same probability is spread more thinly, and the density drops accordingly.

§5.5.ii. Definition and Standardisation

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Definition 5.5.2. A random variable X has the **normal distribution** with mean μ and variance σ^2 if its probability density function is

$$\frac{1}{\sigma\sqrt{2\pi}}\,e^{-(x-\mu)^2/(2\sigma^2)}. \tag{5.13}$$

This is expressed by $X \sim N(\mu, \sigma^2)$. The **standard normal distribution** has zero mean, unit variance, and the following distribution function:

$$\text{Prob}[\,X \le z\,] \;\triangleq\; N(z) \;=\; \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{z} e^{-x^2/2}\,dx. \tag{5.14}$$

We write $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$ for the standard normal density, so that $N' = \phi$. The definition calls μ and σ^2 the mean and the variance, and it calls (5.13) a density; all three claims need checking. The first step reduces everything to the standard case.

Proposition 5.5.3 (Standardisation)

$X \sim N(\mu, \sigma^2)$ if and only if $Z = (X - \mu)/\sigma \sim N(0, 1)$. Consequently $X = \mu + \sigma Z$ and

$$P(X \le x) = N\left(\frac{x - \mu}{\sigma}\right).$$

Proof. Apply Proposition 5.5.1 to $g(x) = (x - \mu)/\sigma$, whose inverse $h(z) = \mu + \sigma z$ has $h' = \sigma$: the density of Z is $\sigma \cdot e^{-(\sigma z)^2/(2\sigma^2)}/(\sigma\sqrt{2\pi}) = \phi(z)$. The converse is the same computation with g and h exchanged. ■

So a single function $N(\cdot)$ serves every normal distribution; this is why it appears in every option pricing formula.

Proposition 5.5.4 1. $\int_{-\infty}^{\infty} \phi(z)\, dz = 1$, so (5.13) is a density.
2. If $Z \sim N(0, 1)$, then $E[Z] = 0$ and $\mathrm{Var}[Z] = 1$. Hence $X \sim N(\mu, \sigma^2)$ has mean μ and variance σ^2.
3. (Symmetry) $N(-z) = 1 - N(z)$ for every z.

Proof. **1.** Let $I = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$. Then I^2 is a double integral over the plane, which we evaluate in polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, $dx\, dy = r\, dr\, d\theta$:

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\, dx\, dy = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\, r\, dr\, d\theta = 2\pi\Big[-e^{-r^2/2}\Big]_0^{\infty} = 2\pi.$$

So $I = \sqrt{2\pi}$ and $\int \phi = 1$.

2. $z\,\phi(z)$ is an odd function, so $E[Z] = \int z\,\phi(z)\, dz = 0$. For the variance, note that $\phi'(z) = -z\,\phi(z)$ and integrate by parts:

$$E[Z^2] = \int_{-\infty}^{\infty} z \cdot z\,\phi(z)\, dz = \Big[-z\,\phi(z)\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z)\, dz = 0 + 1.$$

For general $X = \mu + \sigma Z$, use linearity and item 5 of Proposition 5.1.2.

3. ϕ is even, so $P(Z \le -z) = P(Z \ge z) = 1 - N(z)$. ■

Together, standardisation and symmetry answer every question about normal probabilities with a table of $N(z)$ for $z \ge 0$. For instance, $P(|X - \mu| \le k\sigma) = N(k) - N(-k) = 2N(k) - 1$, which is about 68.3%, 95.4% and 99.7% for $k = 1, 2, 3$. Compare this with Chebyshev's guarantee in Proposition 5.1.5, which for $k = 2$ promises only 75%.

Note 5.5.5. The integral in (5.14) has no closed form in elementary functions, so $N(\cdot)$ is always evaluated numerically. Every option pricing formula in this course will end up calling it, which makes a fast and accurate $N(\cdot)$ a genuine implementation concern rather than a detail.

§5.6. Moment Generating Functions

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Definition 5.6.1. The **moment generating function** of a random variable X is defined as

$$\theta_X(t) \triangleq E\big[e^{tX}\big].$$

Since $e^{tX} > 0$, the expectation always makes sense, but it may be $+\infty$. We always have $\theta_X(0) = 1$. We say that the moment generating function *exists* if $\theta_X(t) < \infty$ for every t in some interval $(-h, h)$ with $h > 0$. All the results below assume this.

§5.6.i. Moments from Derivatives

Proposition 5.6.2

If θ_X exists, then every moment $E[X^n]$ is finite,

$$\theta_X(t) \;=\; \sum_{n=0}^{\infty} \frac{E[\,X^n\,]}{n!}\, t^n \qquad \text{for } |t| < h, \tag{5.15}$$

and consequently $\theta_X^{(n)}(0) = E[X^n]$ for every $n \geq 0$.

Proof. Expand the exponential: $e^{tX} = \sum_{n\geq 0} t^n X^n / n!$. Taking the expectation term by term gives (5.15).[6] The right-hand side of (5.15) is a power series in t, and the coefficient of t^n in the Taylor series of θ_X at 0 is $\theta_X^{(n)}(0)/n!$. Comparing coefficients gives $\theta_X^{(n)}(0) = E[X^n]$. ■

This is where the name comes from: θ_X packs all the moments of X into one function, and differentiating n times at 0 pulls out the nth moment. In particular,

$$E[\,X\,] = \theta_X'(0), \qquad \mathrm{Var}[\,X\,] = \theta_X''(0) - \big(\theta_X'(0)\big)^2.$$

It is often easier to differentiate $\ln \theta_X$ instead. Using $\theta_X(0) = 1$,

$$\big(\ln \theta_X\big)'(0) = \theta_X'(0) = E[\,X\,], \qquad \big(\ln \theta_X\big)''(0) = \theta_X''(0) - \big(\theta_X'(0)\big)^2 = \mathrm{Var}[\,X\,]. \tag{5.16}$$

§5.6.ii. Transformations and Sums

Proposition 5.6.3 1. (Affine maps) For constants a and b, $\theta_{a+bX}(t) = e^{at}\,\theta_X(bt)$.
2. (Independent sums) If X and Y are independent, then $\theta_{X+Y}(t) = \theta_X(t)\,\theta_Y(t)$.

Proof. **1.** $E\big[e^{t(a+bX)}\big] = e^{at}\,E\big[e^{(bt)X}\big]$.
2. Apply Proposition 5.1.7 with $f(x) = e^{tx}$ and $g(y) = e^{ty}$: $E\big[e^{t(X+Y)}\big] = E\big[e^{tX}e^{tY}\big] = E\big[e^{tX}\big]\,E\big[e^{tY}\big]$. ■

Item 2 is the main reason moment generating functions are useful. The distribution of $X+Y$ is given by a convolution integral, which is usually hard to compute. The moment generating function turns that convolution into a plain multiplication.

To go back from θ_{X+Y} to the distribution of $X+Y$ we need one more fact, which we quote without proof.

Theorem 5.6.4 (Uniqueness)

If $\theta_X(t) = \theta_Y(t) < \infty$ for all t in some interval $(-h, h)$ with $h > 0$, then X and Y have the same distribution.

So a moment generating function that exists identifies the distribution. To show that some random variable has a certain distribution, it is enough to compute its moment generating function and recognise it.

§5.6.iii. The Normal Distribution

[6] Swapping E and $\sum$ needs justification. For $|t| < h$, every partial sum is bounded in absolute value by $e^{|tX|} \leq e^{tX} + e^{-tX}$, whose expectation $\theta_X(t) + \theta_X(-t)$ is finite. The dominated convergence theorem then allows the swap. The same bound shows $E\big[|X|^n\big] < \infty$, because $|tX|^n/n! \leq e^{|tX|}$.

Theorem 5.6.5

The moment generating function of $X \sim N(\mu, \sigma^2)$ is

$$\theta_X(t) \;=\; \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right). \tag{5.17}$$

Proof. First let $Z \sim N(0,1)$. Complete the square in the exponent, $tz - \frac{1}{2}z^2 = \frac{1}{2}t^2 - \frac{1}{2}(z-t)^2$:

$$\theta_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz - z^2/2}\, dz = e^{t^2/2}\, \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2}\, dz = e^{t^2/2}.$$

The last integral equals 1 because $e^{-(z-t)^2/2}/\sqrt{2\pi}$ is the $N(t,1)$ density. For general $X \sim N(\mu, \sigma^2)$, Proposition 5.5.3 gives $X = \mu + \sigma Z$. By item 1 of Proposition 5.6.3,

$$\theta_X(t) = e^{\mu t}\, \theta_Z(\sigma t) = e^{\mu t}\, e^{\sigma^2 t^2/2}.$$

■

As a check, $\ln \theta_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$, so (5.16) gives mean μ and variance σ^2, as it should. The normal distribution is exactly the distribution whose $\ln \theta$ is a polynomial of degree two; all its higher-order structure is fixed by the mean and the variance.

Corollary 5.6.6 1. If $X \sim N(\mu, \sigma^2)$ and $b \neq 0$, then $a + bX \sim N(a + b\mu,\, b^2\sigma^2)$.
2. If $X_1, \ldots, X_n$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, then $\sum_i X_i \sim N\big(\sum_i \mu_i,\, \sum_i \sigma_i^2\big)$.

Proof. **1.** By Proposition 5.6.3, $\theta_{a+bX}(t) = e^{at} \exp\big(\mu b t + \frac{1}{2}\sigma^2 b^2 t^2\big) = \exp\big((a + b\mu)t + \frac{1}{2}b^2\sigma^2 t^2\big)$, which is the moment generating function of $N(a + b\mu,\, b^2\sigma^2)$. Apply Theorem 5.6.4.

2. By independence, $\theta_{\sum X_i}(t) = \prod_i \exp\big(\mu_i t + \frac{1}{2}\sigma_i^2 t^2\big) = \exp\big((\sum_i \mu_i)t + \frac{1}{2}(\sum_i \sigma_i^2)t^2\big)$. Apply Theorem 5.6.4 again. ■

Corollary 5.6.7

If $Z \sim N(0,1)$, then $E[Z^n] = 0$ for odd n, and

$$E\big[\, Z^{2k} \,\big] = \frac{(2k)!}{2^k\, k!} = 1 \cdot 3 \cdot 5 \cdots (2k-1).$$

In particular $E[Z^2] = 1$, $E[Z^4] = 3$ and $E[Z^6] = 15$.

Proof. Expand $\theta_Z(t) = e^{t^2/2}$ as a power series:

$$e^{t^2/2} = \sum_{k=0}^{\infty} \frac{(t^2/2)^k}{k!} = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k\, k!}.$$

Only even powers appear, so by (5.15) all odd moments vanish. Matching the coefficient of t^{2k} with $E[Z^{2k}]/(2k)!$ gives $E[Z^{2k}] = (2k)!/(2^k k!)$. Finally, $(2k)! = \big(2 \cdot 4 \cdots 2k\big)\big(1 \cdot 3 \cdots (2k-1)\big) = 2^k k! \cdot \big(1 \cdot 3 \cdots (2k-1)\big)$. ■

The value $E[Z^4] = 3$ is the benchmark behind the word "kurtosis": a distribution with $E[(X-\mu)^4]/\sigma^4 > 3$ has heavier tails than the normal, which is what Figure 1.4 showed for real returns.

Note 5.6.8. The moment generating function does not always exist. A lognormal $Y = e^X$ with X normal has every moment finite, yet $E[e^{tY}] = \infty$ for every $t > 0$, because e^{te^x} grows far faster than the normal density $e^{-x^2/2}$ decays. So Proposition 5.6.2 cannot be run in reverse: finite moments do not imply an existing moment generating function. The usual remedy is the **characteristic function** $E[e^{itX}]$, with $i = \sqrt{-1}$, which always exists because $|e^{itX}| = 1$, and which also determines the distribution.

Remark 5.6.9. 這一節的重點可以濃縮成三句話，後面會一再用到。

（一）動差生成函數把所有動差包在一起。把 e^{tX} 展開再取期望值，第 n 項的係數就是 $E[X^n]/n!$，所以在 0 微分 n 次就得到第 n 階動差（Proposition 5.6.2）。實際在算的時候，對 $\ln \theta_X$ 微分通常比較方便：一次微分是平均數，二次微分是變異數。常態分佈的 $\ln \theta_X$ 剛好是 t 的二次式，所以它完全由平均數和變異數決定；$E[Z^4] = 3$ 這種高階動差，也是從同一個式子展開出來的。

（二）獨立相加，在動差生成函數上就是相乘。這配合唯一性定理，就能直接證明「獨立常態相加還是常態」（Corollary 5.6.6）。下一節的多元常態分佈，以及後面把每一期報酬加起來得到整段報酬的論證，都是建立在這件事上。

（三）在 $t = n$ 取值就得到對數常態的動差。由 (5.17)，$E[e^{nX}] = e^{n\mu + n^2\sigma^2/2}$。若 $Y = e^X$，這就是 $E[Y^n]$，所以 Section 5.10 的平均數與變異數 (5.25) 只是把 $n = 1, 2$ 代進去而已。但要注意，Y 自己的動差生成函數並不存在（上面的 note），所以處理對數常態時，永遠是透過 $\ln Y$ 這個常態變數來算，而不是直接算 Y 的動差生成函數。

§5.7. The Multivariate Normal Distribution

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A portfolio involves several assets at once, so a single normal variable is not enough: we need a joint distribution for several of them, with a prescribed covariance matrix. The independent case was settled in Section 5.6. If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then

$$\sum_i X_i \;\sim\; N\left(\sum_i \mu_i,\; \sum_i \sigma_i^2\right).$$

This was proved in Corollary 5.6.6.

Dropping independence requires a definition rather than a theorem. Let $X_i \sim N(\mu_i, \sigma_i^2)$, which may not be independent, and suppose

$$\sum_{i=1}^n t_i X_i \;\sim\; N\left(\sum_{i=1}^n t_i \mu_i,\; \sum_{i=1}^n \sum_{j=1}^n t_i t_j \operatorname{Cov}[X_i, X_j]\right) \tag{5.18}$$

for every linear combination $\sum_{i=1}^n t_i X_i$ with $\sum_{i=1}^n \sum_{j=1}^n t_i t_j \operatorname{Cov}[X_i, X_j] \neq 0$.

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Then the X_i are said to have a **multivariate normal distribution**, or to be **jointly normal**.[7] With $M \equiv C^{-1}$ and the (i,j)th entry of the matrix M being $M_{i,j}$, the probability density function for the X_i is

$$\frac{1}{\sqrt{(2\pi)^n \det(C)}} \exp\left[-\frac{1}{2}\sum_{i=1}^n \sum_{j=1}^n (X_i - \mu_i)\, M_{ij}\, (X_j - \mu_j)\right], \tag{5.19}$$

with a positive definite covariance matrix

$$C \;\triangleq\; \left[\operatorname{Cov}[X_i, X_j]\right]_{1 \le i,j \le n}.$$

[7] Corrected by Mr. Huang, Guo-Hua (R98922107) on March 10, 2010.

The mean and the variance in (5.18) are not an extra assumption: they are forced by linearity and by (5.5). The real content of the definition is that every linear combination is *normal*. The variance condition only excludes combinations that are constant, which are "normal with variance 0" in a degenerate sense. The rest of this section explains where such random variables come from, where the density (5.19) comes from, and what is special about them.

§5.7.i. Construction from Independent Normals

Proposition 5.7.1

Let $Z_1, \dots, Z_k$ be independent standard normal variables, $\boldsymbol{Z} = (Z_1, \dots, Z_k)^{\mathsf{T}}$, and let A be an $n \times k$ matrix and $\boldsymbol{\mu} \in \mathbb{R}^n$. Then $\boldsymbol{X} = \boldsymbol{\mu} + A\boldsymbol{Z}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance matrix $C = AA^{\mathsf{T}}$.

Proof. The covariance matrix of $\boldsymbol{Z}$ is the identity, so that of $\boldsymbol{X}$ is AA^{T} by item 3 of Proposition 5.3.2. For a weight vector $\boldsymbol{t}$, set $\boldsymbol{c} = A^{\mathsf{T}}\boldsymbol{t}$; then

$$\boldsymbol{t}^{\mathsf{T}}\boldsymbol{X} = \boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu} + \sum_{j=1}^{k} c_j Z_j.$$

Each $c_j Z_j$ with $c_j \neq 0$ is $N(0, c_j^2)$, and they are independent, so by Corollary 5.6.6 the sum is normal with variance $\sum_j c_j^2 = \boldsymbol{t}^{\mathsf{T}} AA^{\mathsf{T}} \boldsymbol{t}$, provided this is nonzero. Its mean is $\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu}$. This is (5.18). ∎

Conversely, the distribution of a multivariate normal vector is determined by $\boldsymbol{\mu}$ and C alone. Indeed, for every $\boldsymbol{t}$ the variable $\boldsymbol{t}^{\mathsf{T}}\boldsymbol{X}$ is normal, so by (5.17) at $t = 1$,

$$E\big[e^{\boldsymbol{t}^{\mathsf{T}}\boldsymbol{X}}\big] = \exp\Big(\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu} + \tfrac{1}{2}\,\boldsymbol{t}^{\mathsf{T}} C \boldsymbol{t}\Big),$$

and the multivariate version of Theorem 5.6.4, which we quote, says that this function of $\boldsymbol{t}$ determines the joint distribution. Hence every multivariate normal vector with mean $\boldsymbol{\mu}$ and covariance matrix C has the same distribution as $\boldsymbol{\mu} + A\boldsymbol{Z}$, for any matrix A with $AA^{\mathsf{T}} = C$. Finding such an A is the subject of Section 5.9.

§5.7.ii. The Density

When C is positive definite, we can choose A square and invertible with $AA^{\mathsf{T}} = C$ (Section 5.9 shows how). The density (5.19) is then the result of a change of variables. In n dimensions, Proposition 5.5.1 becomes: if $\boldsymbol{X} = \boldsymbol{g}(\boldsymbol{Z})$ with $\boldsymbol{g}$ invertible and $\boldsymbol{h} = \boldsymbol{g}^{-1}$, then

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = f_{\boldsymbol{Z}}\big(\boldsymbol{h}(\boldsymbol{x})\big)\,\big|\det D\boldsymbol{h}(\boldsymbol{x})\big|,$$

where $D\boldsymbol{h}$ is the Jacobian matrix of partial derivatives. The absolute determinant measures how $\boldsymbol{h}$ scales volumes, just as $|h'|$ measured how h scales lengths.

Proposition 5.7.2

If C is positive definite, the multivariate normal vector $\boldsymbol{X}$ has the density (5.19).

Proof. The Z_i are independent, so the density of $\boldsymbol{Z}$ is the product $\prod_i \phi(z_i) = (2\pi)^{-n/2} \exp\big(-\tfrac{1}{2}\boldsymbol{z}^{\mathsf{T}}\boldsymbol{z}\big)$. Since $\boldsymbol{X}$ has the same distribution as $\boldsymbol{\mu} + A\boldsymbol{Z}$, we may take $\boldsymbol{X} = \boldsymbol{\mu} + A\boldsymbol{Z}$, so $\boldsymbol{h}(\boldsymbol{x}) = A^{-1}(\boldsymbol{x} - \boldsymbol{\mu})$ and $D\boldsymbol{h} = A^{-1}$. Then

$$\boldsymbol{h}(\boldsymbol{x})^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}) = (\boldsymbol{x} - \boldsymbol{\mu})^{\mathsf{T}}(A^{-1})^{\mathsf{T}} A^{-1}(\boldsymbol{x} - \boldsymbol{\mu}) = (\boldsymbol{x} - \boldsymbol{\mu})^{\mathsf{T}} C^{-1} (\boldsymbol{x} - \boldsymbol{\mu}),$$

because $(A^{-1})^{\mathsf{T}}A^{-1} = (AA^{\mathsf{T}})^{-1} = C^{-1}$. Also $|\det A^{-1}| = 1/|\det A| = 1/\sqrt{\det C}$, since $\det C = \det A \det A^{\mathsf{T}} = (\det A)^2$. Substituting gives (5.19), whose double sum is $(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}M(\boldsymbol{x}-\boldsymbol{\mu})$. ■

When C is only positive semidefinite, some combination $\boldsymbol{t}^{\mathsf{T}}\boldsymbol{X}$ is constant (Proposition 5.3.2), so $\boldsymbol{X}$ lives on a lower-dimensional subspace and has no density; the definition (5.18) still applies.

§5.7.iii. Uncorrelated Jointly Normal Variables Are Independent

Section 5.1.iii showed that uncorrelated random variables need not be independent. For jointly normal variables they must be.

Proposition 5.7.3

If $X_1, \dots, X_n$ are jointly normal and pairwise uncorrelated, then they are independent.

Proof. Discard any X_i with zero variance; a constant is independent of everything. The remaining covariance matrix is diagonal with positive entries σ_i^2, hence positive definite, with $C^{-1} = \operatorname{diag}(1/\sigma_i^2)$ and $\det C = \prod_i \sigma_i^2$. The density (5.19) becomes

$$\prod_{i=1}^{n} \frac{1}{\sigma_i\sqrt{2\pi}}\, e^{-(x_i-\mu_i)^2/(2\sigma_i^2)},$$

a product of one-variable densities, which is exactly independence. ■

The hypothesis is *joint* normality; normal marginals are not enough.

Example 5.7.4. Let $X \sim N(0,1)$, and let S be independent of X with $P(S=1) = P(S=-1) = \frac{1}{2}$. Put $Y = SX$. By the symmetry of ϕ, $Y \sim N(0,1)$. Also $\operatorname{Cov}[X,Y] = E[SX^2] = E[S]\,E[X^2] = 0$. But $|Y| = |X|$, so X and Y are far from independent. Proposition 5.7.3 is not contradicted, because (X,Y) is not jointly normal: $X+Y$ equals 0 with probability $\frac{1}{2}$, which no normal variable with positive variance does.

§5.7.iv. Conditional Expectation for Normal Variables

Proposition 5.7.3 has a striking consequence that ties together Section 5.2.iii and Section 5.4. In general the best predictor $E[Y \mid X]$ can be any function of X, and the best straight line can do much worse. For jointly normal variables the two coincide.

Proposition 5.7.5

Let (X,Y) be bivariate normal with $\sigma_X > 0$, and let $b^* = \operatorname{Cov}[X,Y]/\operatorname{Var}[X]$. Then

$$E[\,Y \mid X\,] = \mu_Y + b^*(X-\mu_X), \qquad \operatorname{Var}[\,Y \mid X\,] = \sigma_Y^2\big(1-\rho_{X,Y}^2\big).$$

In fact, given X, the variable Y is normal with this mean and this variance.

Proof. Let $W = Y - b^*X$. Any combination $aX + cW = (a - cb^*)X + cY$ is a combination of X and Y, so (X,W) is again bivariate normal. By bilinearity, $\operatorname{Cov}[X,W] = \operatorname{Cov}[X,Y] - b^*\operatorname{Var}[X] = 0$, so X and W are independent by Proposition 5.7.3. Now $Y = b^*X + W$. Given X, the first term is known and the second has the same distribution as without conditioning, namely normal with mean $E[W] = \mu_Y - b^*\mu_X$ and variance $\operatorname{Var}[W]$. In the language of Proposition 5.4.6, items 2 and 3 give $E[Y \mid X] = b^*X + E[W]$. Finally $\operatorname{Var}[W] = \operatorname{Var}[Y - b^*X] = \sigma_Y^2(1-\rho_{X,Y}^2)$ was computed in the proof of Proposition 5.2.5. ■

So for jointly normal variables, the regression line *is* the conditional expectation, and the residual uncertainty $\sigma_Y^2(1-\rho^2)$ does not even depend on the observed value of X. All of the dependence between X and Y is captured by the single number ρ.

Remark 5.7.6. 這一節把前面幾節串在一起，重點有三個。

（一）為什麼用「所有線性組合都是常態」來定義。這個定義比直接寫密度函數更廣：當 C 不可逆時（例如 $X_3 = X_1 + X_2$），密度不存在，但定義一樣適用。而且金融裡最常碰到的就是線性組合，因為投資組合就是資產的加權和；這個定義直接告訴你任何投資組合的分佈。要注意每個邊際分佈都是常態，並不代表聯合起來是多元常態，Example 5.7.4 就是反例。

（二）多元常態都可以由獨立標準常態做出來。 Proposition 5.7.1 說 $\boldsymbol{\mu} + A\boldsymbol{Z}$ 是多元常態，共變異數矩陣是 AA^{T}；反過來，任何多元常態都長這樣。所以要模擬多元常態，只要產生獨立的標準常態（Section 5.8），再找一個 A 使 $AA^{\mathsf{T}} = C$（Section 5.9）就好。

（三）在常態的世界裡，相關係數就是一切。 Section 5.2 說過，相關係數只量得到直線關係，而且不相關不代表獨立。但如果變數是聯合常態，不相關就等於獨立（Proposition 5.7.3），而且最好的預測剛好就是回歸直線（Proposition 5.7.5），所以 ρ 已經描述了全部的相依關係。這也是聯合常態假設危險的地方：真實市場在危機時一起崩跌的那種非線性相依，在這個模型裡根本不存在。

§5.8. Generation of Univariate Normal Distributions

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Monte Carlo pricing needs normal variates by the million, and a computer supplies only uniform ones. This section explains how to turn the latter into the former.

§5.8.i. Uniform Variables and the Inverse Transform

A random variable ξ is **uniformly distributed** over $(0,1)$ if it has density 1 on $(0,1)$ and 0 elsewhere, that is, $P(\xi \le x) = x$ for $0 < x \le 1$. Its mean and variance are

$$E[\xi] = \int_0^1 x\,dx = \frac{1}{2}, \qquad \operatorname{Var}[\xi] = \int_0^1 x^2\,dx - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

The most direct way to turn a uniform variable into any other distribution is the following.

Proposition 5.8.1 (Inverse transform)

Let F be a continuous, strictly increasing distribution function, and let ξ be uniform over $(0,1)$. Then $F^{-1}(\xi)$ has distribution function F.

Proof. Since F is increasing, $F^{-1}(\xi) \le x$ exactly when $\xi \le F(x)$, which has probability $F(x)$. ∎

So $N^{-1}(\xi)$ is a standard normal variable. The catch is that N^{-1}, like N itself, has no closed form and must be approximated numerically. The inverse transform works well for distributions whose F^{-1} is explicit. The one we need below is: if ξ is uniform over $(0,1)$, then

$$P\big(-2\ln\xi > s\big) = P\big(\xi < e^{-s/2}\big) = e^{-s/2}, \qquad s \ge 0, \tag{5.20}$$

so $-2\ln\xi$ has density $\frac{1}{2}e^{-s/2}$ on $(0,\infty)$, the **exponential distribution** with mean 2.

§5.8.ii. The Box–Muller Transform

The trick for avoiding N^{-1} is to generate two normal variables at a time. A pair of independent standard normals has a rotationally symmetric joint density, so it is natural to describe it in polar coordinates.

Theorem 5.8.2 (Box–Muller)

Let S and Θ be independent, with S exponential with mean 2 and Θ uniform over $[0, 2\pi)$. Then $\sqrt{S}\cos\Theta$ and $\sqrt{S}\sin\Theta$ are independent standard normal variables. In particular, if ω is uniform over $(0, 1)$ and independent of Θ, one may take $S = -2\ln\omega$.

Proof. Start from the other end. Let Z_1, Z_2 be independent standard normal, with joint density $\frac{1}{2\pi}e^{-(z_1^2+z_2^2)/2}$, and write them in polar coordinates, $Z_1 = R\cos\Theta'$ and $Z_2 = R\sin\Theta'$. For any region B of the (s, θ) plane, with $s = r^2$ and $dz_1\,dz_2 = r\,dr\,d\theta = \frac{1}{2}\,ds\,d\theta$,

$$P\big((R^2, \Theta') \in B\big) = \iint_B \frac{1}{2\pi}\,e^{-s/2}\cdot\frac{1}{2}\,ds\,d\theta = \iint_B \Big(\frac{1}{2}e^{-s/2}\Big)\Big(\frac{1}{2\pi}\Big)\,ds\,d\theta.$$

The joint density of (R^2, Θ') factors into an exponential density with mean 2 in s times a uniform density on $[0, 2\pi)$ in θ. So R^2 and Θ' are independent with exactly the distributions of S and Θ. Since $(s, \theta) \mapsto (\sqrt{s}\cos\theta, \sqrt{s}\sin\theta)$ maps both pairs the same way, $(\sqrt{S}\cos\Theta, \sqrt{S}\sin\Theta)$ has the same joint distribution as (Z_1, Z_2). The last claim is (5.20). ■

§5.8.iii. The Polar Method

The Box–Muller transform still needs cos and sin. The **polar method** obtains the angle from a random point instead.

- Let X be uniformly distributed over $(0, 1]$ so that

$$\text{Prob}[\,X \le x\,] = x, \qquad 0 < x \le 1.$$

- Repeatedly draw two samples x_1 and x_2 from X until

$$\omega \triangleq (2x_1 - 1)^2 + (2x_2 - 1)^2 < 1.$$

- Then $c(2x_1 - 1)$ and $c(2x_2 - 1)$ are independent standard normal variables, where[8]

$$c \triangleq \sqrt{-2(\ln\omega)/\omega}.$$

[8] As they are normally distributed, to prove independence it suffices to prove that they are uncorrelated, which is easy. Thanks to a lively class discussion on March 5, 2014. Strictly speaking, this shortcut needs them to be *jointly* normal (Proposition 5.7.3); normal marginals alone are not enough (Example 5.7.4). The proof below establishes independence directly.

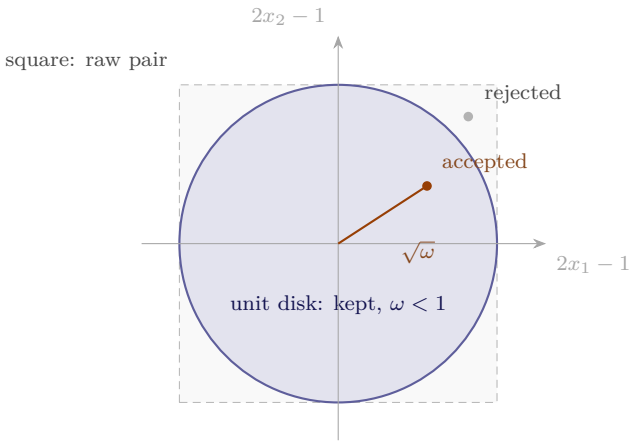


Figure 5.1.: The polar method. A raw pair lands uniformly in the square; the draw is retried unless it falls inside the unit disk. Accepted points are uniform on the disk, which makes the squared radius ω uniform on $(0, 1)$ and independent of the angle.

The acceptance step is **rejection sampling**. Its correctness rests on two facts.

Lemma 5.8.3 1. An accepted pair $(V_1, V_2) = (2x_1 - 1,\ 2x_2 - 1)$ is uniformly distributed on the unit disk, and the expected number of attempts per accepted pair is $4/\pi \approx 1.27$.
2. If (V_1, V_2) is uniform on the unit disk, then its squared radius $\omega = V_1^2 + V_2^2$ is uniform over $(0, 1)$, its angle Θ is uniform over $[0, 2\pi)$, and ω and Θ are independent.

Proof. **1.** A raw pair is uniform on the square $(-1, 1]^2$ of area 4, so it lands in a region A with probability $\text{area}(A)/4$. For A inside the disk,

$$P(\text{pair} \in A \mid \text{accepted}) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

which is the uniform distribution on the disk. Each attempt succeeds independently with probability $p = \pi/4$, and the number of attempts until the first success has mean $1/p$.

2. The event $\{\omega \le s,\ \Theta \le \alpha\}$ is a sector of radius $\sqrt{s}$ and angle α, with area $\frac{1}{2}\alpha s$. Therefore

$$P(\omega \le s,\ \Theta \le \alpha) = \frac{\alpha s/2}{\pi} = s \cdot \frac{\alpha}{2\pi}, \qquad 0 \le s \le 1,\ 0 \le \alpha < 2\pi.$$

This is the product of the uniform distribution function on $(0, 1)$ and the uniform distribution function on $[0, 2\pi)$, which gives both the marginals and independence. ∎

Theorem 5.8.4

The two numbers $c(2x_1 - 1)$ and $c(2x_2 - 1)$ produced by the polar method are independent standard normal variables.

Proof. By Lemma 5.8.3, the accepted pair is $(V_1, V_2) = (\sqrt{\omega}\cos\Theta, \sqrt{\omega}\sin\Theta)$ with ω uniform over $(0, 1)$ and independent of the uniform angle Θ. Hence

$$c\,V_1 = \sqrt{\frac{-2\ln\omega}{\omega}}\,\sqrt{\omega}\cos\Theta = \sqrt{-2\ln\omega}\,\cos\Theta, \qquad c\,V_2 = \sqrt{-2\ln\omega}\,\sin\Theta,$$

which are independent standard normals by Theorem 5.8.2. ∎

Remark 5.8.5. 這個方法看起來很神祕，其實就是把 Box–Muller 裡「產生一個均勻角度再取 cos、sin」這一步，換成「在圓裡隨機丟一個點」。點的方向直接給出 $\cos\Theta = V_1/\sqrt{\omega}$ 和 $\sin\Theta = V_2/\sqrt{\omega}$，而點到圓心距離的平方 ω 剛好又是一個獨立的均勻變數，可以拿來當 Box–Muller 裡的 ω，一個點用了兩次。代價是大約 21% 的點會被丟掉，但整個過程不需要計算三角函數；在早期的硬體上，三角函數比開根號和取對數昂貴得多。而且每次都產生兩個獨立的常態變數。

§5.8.iv. A Dirty Trick and a Right Attitude

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Let ξ_i be independent and uniformly distributed over $(0,1)$. A simple method to generate the standard normal variable is to calculate[9]

$$\left(\sum_{i=1}^{12} \xi_i\right) - 6. \tag{5.21}$$

But why use 12? Recall the mean and variance of ξ_i are $1/2$ and $1/12$, respectively.

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The general formula is

$$\frac{\left(\sum_{i=1}^{n} \xi_i\right) - (n/2)}{\sqrt{n/12}}, \tag{5.22}$$

which is the central limit theorem applied to n uniforms, standardised. By linearity and (5.6), the sum $\sum_{i=1}^{n} \xi_i$ has mean $n/2$ and variance $n/12$, so (5.22) always has mean 0 and variance 1; the theorem below says that its distribution approaches $N(0,1)$ as n grows. Choosing $n = 12$ makes the denominator $\sqrt{12/12} = 1$ and the numerator's correction 6, yielding a formula without the need of division and square-root operations.[10]

Theorem 5.8.6 (Central limit theorem)

Let $X_1, X_2, \ldots$ be independent and identically distributed with mean μ and variance $\sigma^2 > 0$. Then for every z,

$$P\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \le z\right) \longrightarrow N(z) \qquad (n \to \infty).$$

Proof sketch. We assume in addition that the moment generating function of X_i exists; this covers the uniform case. Let $Y_i = (X_i - \mu)/\sigma$, which has mean 0 and variance 1, and let $T_n = \sum_{i=1}^{n} Y_i/\sqrt{n}$ be the standardised sum. By (5.15), $\theta_Y(t) = 1 + \frac{1}{2}t^2 + \varepsilon(t)\,t^2$ with $\varepsilon(t) \to 0$ as $t \to 0$. By Proposition 5.6.3,

$$\theta_{T_n}(t) = \left(\theta_Y\left(t/\sqrt{n}\right)\right)^n = \left(1 + \frac{t^2/2 + \varepsilon(t/\sqrt{n})\,t^2}{n}\right)^n \longrightarrow e^{t^2/2},$$

using $(1 + a_n/n)^n \to e^a$ whenever $a_n \to a$. The limit is the moment generating function of $N(0,1)$. The *continuity theorem*, a strengthening of Theorem 5.6.4 that we quote, says that convergence of moment generating functions near 0 implies convergence of the distribution functions. ∎

[9] Jäckel (2002): "this is not a highly accurate approximation and should only be used to establish ballpark estimates."

[10] Contributed by Mr. Chen, Shih-Hang (R02723031) on March 5, 2014.

The theorem says nothing about how fast the convergence is. In relative terms the approximation is worst far out in the tails, where the probabilities are tiny to begin with, so a small absolute error can still be a large relative one. Figure 5.2 shows what this means for $n = 12$.

Always blame your random number generator last.[a] Instead, check your programs first.

[a] "The fault, dear Brutus, lies not in the stars but in ourselves that we are underlings." William Shakespeare (1564–1616), *Julius Caesar*.

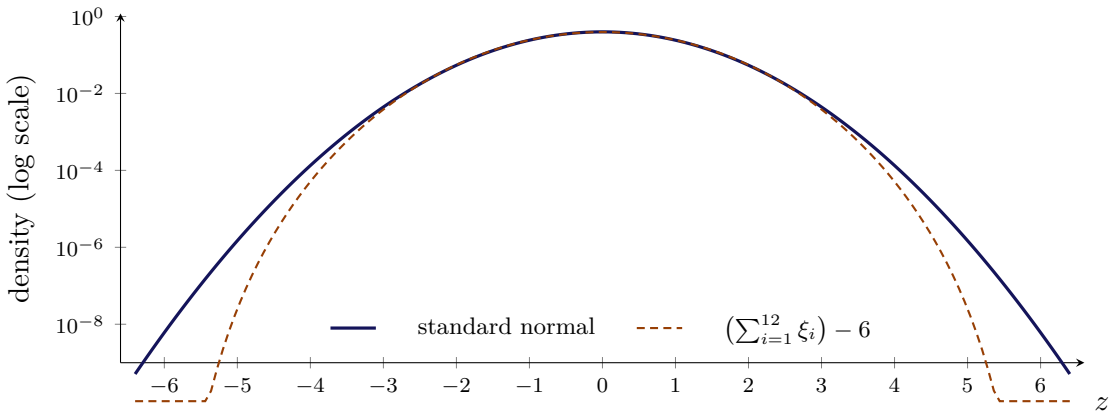


Figure 5.2.: Why (5.21) is only a ballpark approximation. On a logarithmic scale the two densities agree to the eye out to about $|z| = 2$ and then part company: at $z = 4$ the trick delivers 38% of the correct density, at $z = 5$ only 1.7%, and beyond $|z| = 6$ exactly zero, since a sum of twelve uniforms cannot leave $[-6, 6]$.

§5.9. Generation of Bivariate Normal Distributions

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Pairs of normally distributed variables with correlation ρ can be generated as follows. Let X_1 and X_2 be independent standard normal variables, and set

$$\begin{aligned} U &\triangleq aX_1, \\ V &\triangleq a\rho X_1 + a\sqrt{1-\rho^2}\, X_2. \end{aligned} \tag{5.23}$$

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Proposition 5.9.1

U and V are the desired random variables, with

$$\operatorname{Var}[U] = \operatorname{Var}[V] = a^2, \qquad \operatorname{Cov}[U, V] = \rho a^2.$$

Proof. $\operatorname{Var}[U] = a^2 \operatorname{Var}[X_1] = a^2$. Since X_1 and X_2 are uncorrelated, (5.6) gives

$$\operatorname{Var}[V] = a^2\rho^2 \operatorname{Var}[X_1] + a^2(1-\rho^2) \operatorname{Var}[X_2] = a^2\rho^2 + a^2(1-\rho^2) = a^2.$$

Finally, using $\operatorname{Cov}[X_1, X_2] = 0$ and $\operatorname{Var}[X_1] = 1$,

$$\operatorname{Cov}[U, V] = a^2\rho \operatorname{Var}[X_1] + a^2\sqrt{1-\rho^2} \operatorname{Cov}[X_1, X_2] = \rho a^2. \qquad \blacksquare$$

Note that the mapping from (X_1, X_2) to (U, V) is a one-to-one correspondence for $a \neq 0$ and $|\rho| < 1$; when $\rho = \pm 1$, $V = \pm U$ and the pair has no density. By (5.3), the correlation of U and V is $\rho a^2/(a \cdot a) = \rho$, as intended.

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The mapping in matrix form is

$$\begin{bmatrix} U \\ V \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}. \tag{5.24}$$

The proposition only checks variances and covariances, but more is true. This has the form $A\boldsymbol{X}$ with $\boldsymbol{X} = (X_1, X_2)^\mathsf{T}$, so by Proposition 5.7.1 the pair (U, V) is bivariate normal. Since a multivariate normal distribution is determined by its mean and covariance matrix, (U, V) has *exactly* the bivariate normal distribution with variances a^2 and correlation ρ, not merely the right second moments.

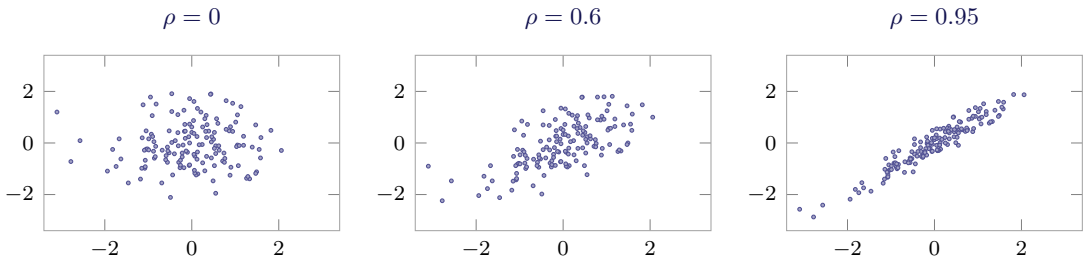


Figure 5.3.: Bivariate normal samples generated by (5.24) with $a = 1$, from the same underlying (X_1, X_2) draws. Raising ρ shears the cloud towards the diagonal without changing either marginal.

§5.9.i. The General Case: Cholesky Decomposition

The matrix in (5.24) was not pulled out of thin air. With $a = 1$ it is the lower-triangular matrix

$$L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \qquad \text{with} \qquad LL^\mathsf{T} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

the correlation matrix we wanted. By Proposition 5.7.1, to generate an n-dimensional normal vector with mean $\boldsymbol{\mu}$ and covariance matrix C we need, in general, a matrix L with $LL^\mathsf{T} = C$; then $\boldsymbol{\mu} + L\boldsymbol{Z}$, with $\boldsymbol{Z}$ a vector of independent standard normals, does the job. Such an L can always be found, and can be chosen lower triangular.

Theorem 5.9.2 (Cholesky decomposition)

If C is a symmetric positive definite $n \times n$ matrix, there is a lower-triangular matrix L with positive diagonal entries such that $C = LL^\mathsf{T}$.

Proof. Induction on n. For $n = 1$, $C = [c]$ with $c > 0$, and $L = [\sqrt{c}]$. For $n > 1$, write

$$C = \begin{bmatrix} c_{11} & \boldsymbol{c}^\mathsf{T} \\ \boldsymbol{c} & C' \end{bmatrix}, \qquad L = \begin{bmatrix} \ell_{11} & \boldsymbol{0}^\mathsf{T} \\ \boldsymbol{\ell} & L' \end{bmatrix},$$

where c_{11} is a number, $\boldsymbol{c}$ is a column of length $n-1$ and C' is $(n-1) \times (n-1)$. Multiplying out, $LL^\mathsf{T} = C$ amounts to

$$\ell_{11}^2 = c_{11}, \qquad \ell_{11}\boldsymbol{\ell} = \boldsymbol{c}, \qquad L'L'^\mathsf{T} = C' - \boldsymbol{\ell}\boldsymbol{\ell}^\mathsf{T} = C' - \frac{\boldsymbol{c}\boldsymbol{c}^\mathsf{T}}{c_{11}}.$$

Taking $\boldsymbol{a} = (1, 0, \ldots, 0)^{\mathsf{T}}$ in the definition of positive definiteness gives $c_{11} > 0$, so the first two equations are solved by $\ell_{11} = \sqrt{c_{11}}$ and $\boldsymbol{\ell} = \boldsymbol{c}/\sqrt{c_{11}}$. For the third, it suffices by induction that $C'' \triangleq C' - \boldsymbol{c}\boldsymbol{c}^{\mathsf{T}}/c_{11}$ be symmetric positive definite. Symmetry is clear. For $\boldsymbol{y} \neq \boldsymbol{0}$ of length $n-1$, put $\boldsymbol{a} = \left(-\boldsymbol{c}^{\mathsf{T}}\boldsymbol{y}/c_{11},\ \boldsymbol{y}\right)^{\mathsf{T}} \neq \boldsymbol{0}$; expanding,

$$0 < \boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a} = \frac{(\boldsymbol{c}^{\mathsf{T}}\boldsymbol{y})^2}{c_{11}} - 2\,\frac{(\boldsymbol{c}^{\mathsf{T}}\boldsymbol{y})^2}{c_{11}} + \boldsymbol{y}^{\mathsf{T}} C' \boldsymbol{y} = \boldsymbol{y}^{\mathsf{T}} C'' \boldsymbol{y}. \qquad \blacksquare$$

The proof is also the algorithm: fill in L one column at a time, each step costing a square root and some divisions. It shows why positive definiteness is needed: the square roots taken along the way are square roots of variances of the form $\boldsymbol{a}^{\mathsf{T}} C \boldsymbol{a}$, and they break down as soon as some combination has zero or negative "variance". If C is only positive semidefinite, a lower-triangular L with $LL^{\mathsf{T}} = C$ still exists but some diagonal entries are 0; one of the variables is then an exact combination of the others (Proposition 5.3.2).

Remark 5.9.3. 這一節的做法，是 Proposition 5.7.1 的具體實現：先產生獨立的標準常態，再乘上一個 L，使 LL^{T} 等於想要的共變異數矩陣。Figure 5.3 可以直接看出這個線性變換在做什麼：ρ 變大時，點雲被往對角線方向「剪」過去，但每個座標自己的分佈都沒有改變。

實務上最常出問題的是正定性。由市場資料估出來的相關係數矩陣，理論上一定是半正定的（Proposition 5.3.2），但如果每一對資產是用不同的樣本期間分別估計的，或者手動調整過某些相關係數，拼起來的矩陣就可能不再是半正定。這時 Cholesky 分解會在某一步需要對負數開根號，模擬根本無法開始，必須先把矩陣修正成最接近的正定矩陣。這也呼應 Section 5.2 的提醒：相關係數本身就不太穩定，一整個矩陣的相關係數更需要小心處理。

§5.10. The Lognormal Distribution

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Definition 5.10.1. A random variable Y is said to have a **lognormal distribution** if $\ln Y$ has a normal distribution.

This is the distribution we will use for prices: if the log return $\ln(S_T/S_0)$ is normal, then S_T/S_0 is lognormal. Every question about a lognormal Y is answered by translating it into a question about the normal variable $\ln Y$, using the tools of Section 5.5 and Section 5.6.

§5.10.i. Density, Median and Moments

Let $X \sim N(\mu, \sigma^2)$ and $Y \triangleq e^X$.

Proposition 5.10.2 1. Y takes values in $(0, \infty)$ and has density

$$f(y) = \frac{1}{y\,\sigma\sqrt{2\pi}}\, e^{-(\ln y - \mu)^2/(2\sigma^2)}, \qquad y > 0.$$

2. The median of Y is e^{μ}: $P(Y \le e^{\mu}) = \frac{1}{2}$.
3. $E[Y^n] = e^{n\mu + n^2\sigma^2/2}$ for every real n.

Proof. **1.** Apply Proposition 5.5.1 with $g = \exp$, whose inverse $h = \ln$ has $h'(y) = 1/y$.

2. exp is increasing, so $P(Y \le e^{\mu}) = P(X \le \mu) = N(0) = \frac{1}{2}$ by Proposition 5.5.3 and the symmetry of ϕ.

3. $E[Y^n] = E[e^{nX}] = \theta_X(n)$, which is (5.17) at $t = n$. $\blacksquare$

The mean and variance of Y are

$$\begin{aligned} \mu_Y &= e^{\mu+\sigma^2/2}, \\ \sigma_Y^2 &= e^{2\mu+\sigma^2}\big(e^{\sigma^2}-1\big), \end{aligned} \tag{5.25}$$

respectively. They follow from $E[Y^n] = e^{n\mu+n^2\sigma^2/2}$ — which is just (5.17) evaluated at $t = n$. For the variance, $E[Y^2] - \mu_Y^2 = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2}$.

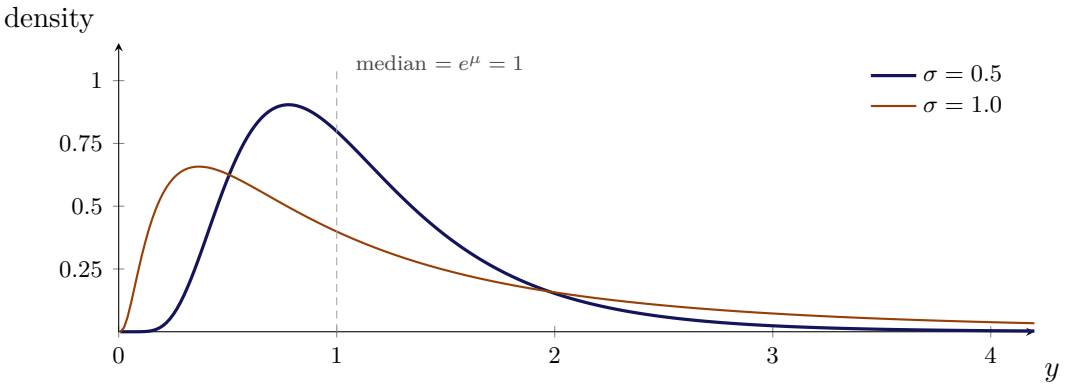


Figure 5.4.: Lognormal densities with $\mu = 0$. The support is $(0, \infty)$, the shape is right-skewed, and the right tail thickens rapidly with σ. The median sits at e^{μ} regardless of σ, while the mean $e^{\mu+\sigma^2/2}$ drifts to the right of it.

§5.10.ii. Jensen's Inequality

The mean $e^{\mu+\sigma^2/2}$ is strictly larger than the median e^{μ}, and the gap grows with σ. This is an instance of a general inequality, already used in Section 4.15 and needed again for options.

A function g is **convex** if through every point $(m, g(m))$ of its graph there is a straight line, a **supporting line**, that lies entirely below the graph. For differentiable g the supporting line is the tangent line, and for twice differentiable g convexity means $g'' \geq 0$. A convex function may have kinks, such as the call payoff $\max(x-a, 0)$ at $x = a$; there any line through the kink with slope between the left and right slopes supports the graph. The function is **strictly convex** if each supporting line touches the graph only at its own point, which holds when $g'' > 0$.

Theorem 5.10.3 (Jensen's inequality)

If g is convex and X and $g(X)$ have finite means, then

$$E\big[g(X)\big] \;\geq\; g\big(E[X]\big).$$

If g is strictly convex, equality holds only when X is constant with probability one.

Proof. Let $m = E[X]$ and let $\ell(x) = g(m) + k(x-m)$ be a supporting line of g at m (for differentiable g, $k = g'(m)$). Convexity says $g(x) \geq \ell(x)$ for all x, so by monotonicity and linearity

$$E\big[g(X)\big] \;\geq\; E\big[\ell(X)\big] = g(m) + k\big(E[X]-m\big) = g(m).$$

If equality holds, the non-negative random variable $g(X) - \ell(X)$ has mean 0, so it is 0 with probability one by Lemma 5.1.4. For strictly convex g that happens only at $X = m$. ∎

The picture is simple: averaging the points of a convex curve lands *above* the curve. With $g(x) = e^x$ and $Y = e^X$,

$$E[Y] = E\big[e^X\big] \;>\; e^{E[X]} = e^{\mu},$$

and (5.25) says that the ratio is exactly $e^{\sigma^2/2}$. Rare large values of Y pull the mean up, while small values of Y are bounded below by 0 and cannot pull it down as much.

§5.10.iii. Products of Lognormal Variables

The next result is why the lognormal distribution fits prices over many periods.

Proposition 5.10.4

If $Y_1, \dots, Y_n$ are independent and lognormal with $\ln Y_i \sim N(\mu_i, \sigma_i^2)$, then $\prod_i Y_i$ is lognormal with $\ln \prod_i Y_i \sim N\big(\sum_i \mu_i, \sum_i \sigma_i^2\big)$. In particular, if the log returns $r_t = \ln(S_t/S_{t-1})$, $t = 1, \dots, T$, are independent and normal, then $S_T = S_0 \exp\big(\sum_t r_t\big)$ is lognormal.

Proof. $\ln \prod_i Y_i = \sum_i \ln Y_i$ is a sum of independent normal variables, which is normal by Corollary 5.6.6. For the second claim, $S_T/S_0 = \prod_t S_t/S_{t-1}$, and multiplying by the constant S_0 only shifts the mean of the logarithm by $\ln S_0$. ■

Even without normality, Theorem 5.8.6 applied to the log returns suggests that $\ln(S_T/S_0)$ is approximately normal when T is large; this is the usual informal justification for the lognormal model.

Remark 5.10.5. 為什麼金融上用**對數常態分佈**（lognormal distribution）描述價格，用常態描述對數報酬？前面證明過的三件事就是理由。

（一）價格不能為負。常態分佈的值域是整條實線，但股價不會是負的。$Y = e^X$ 自動把值域限制在 $(0, \infty)$（Proposition 5.10.2）。

（二）報酬用加的，價格用乘的。 Section 1.6.iii 的對數報酬 $r_t = \ln(S_t/S_{t-1})$ 有一個好處：跨期報酬就是逐期報酬的和。獨立常態相加還是常態，所以價格一定是對數常態（Proposition 5.10.4）；就算每期報酬不是常態，中央極限定理也讓對數常態成為合理的近似。

（三）平均數比中位數大。由 Jensen 不等式，$E[Y] = e^{\mu+\sigma^2/2}$ 嚴格大於中位數 e^{μ}，多出來的 $\sigma^2/2$ 完全來自波動。這就是之後會一再出現的「$-\sigma^2/2$ 修正項」的來源：想讓價格的期望成長率等於 r，指數裡就要寫 $r - \sigma^2/2$（見 Lemma 8.14.15）。

§5.10.iv. Converting between the Two Parametrisations

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Proposition 5.10.6

Suppose Y is lognormally distributed with mean μ and variance σ^2. Then $\ln Y$ has a normal distribution with

$$\begin{aligned} E[\ln Y] &= \ln\Big[\mu/\sqrt{1+(\sigma/\mu)^2}\Big], \\ \operatorname{Var}[\ln Y] &= \ln\big[1+(\sigma/\mu)^2\big]. \end{aligned} \tag{5.26}$$

Proof. Write $m = E[\ln Y]$ and $s^2 = \operatorname{Var}[\ln Y]$. By (5.25), $\mu = e^{m+s^2/2}$ and $\sigma^2 = \mu^2\big(e^{s^2} - 1\big)$. The second equation gives $e^{s^2} = 1+(\sigma/\mu)^2$, which is the formula for s^2. Then $m = \ln\mu - s^2/2 = \ln\mu - \ln\sqrt{1+(\sigma/\mu)^2}$, which is the formula for m. ■

Note 5.10.7. Beware the notation in Proposition 5.10.6: μ and σ^2 now denote the mean and variance of Y *itself*, not of $\ln Y$ as in (5.25). (5.26) is the inverse of (5.25), and the two must not be mixed.

We say X and Y are **joint-lognormally distributed** if $(\ln X, \ln Y)$ is bivariate normal.

Proposition 5.10.8

If X and Y are joint-lognormally distributed, then

$$\begin{aligned} E[\,XY\,] &= E[\,X\,]E[\,Y\,]\,e^{\mathrm{Cov}[\,\ln X,\ln Y\,]}, \\ \mathrm{Cov}[\,X,Y\,] &= E[\,X\,]E[\,Y\,]\left(e^{\mathrm{Cov}[\,\ln X,\ln Y\,]}-1\right). \end{aligned} \tag{5.27}$$

Proof. Write $\ln X \sim N(m_X, s_X^2)$, $\ln Y \sim N(m_Y, s_Y^2)$ and $c = \mathrm{Cov}[\ln X, \ln Y]$. Since $(\ln X, \ln Y)$ is bivariate normal, $\ln X + \ln Y$ is normal with mean $m_X + m_Y$ and, by (5.5), variance $s_X^2 + s_Y^2 + 2c$. So $XY = e^{\ln X + \ln Y}$ is lognormal, and by Proposition 5.10.2

$$E[XY] = e^{m_X+m_Y+(s_X^2+s_Y^2+2c)/2} = e^{m_X+s_X^2/2}\,e^{m_Y+s_Y^2/2}\,e^{c} = E[X]\,E[Y]\,e^{c}.$$

The covariance is $E[XY] - E[X]E[Y]$. ∎

In particular, X and Y are uncorrelated exactly when $\ln X$ and $\ln Y$ are, and the correlation of the prices always has the same sign as that of the log returns.

§5.10.v. Partial Expectations

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Option payoffs are not smooth: a call pays only when the price ends above the strike. Its expected payoff therefore involves the expectation of Y restricted to an event $\{Y > a\}$, and the probability of that event.

Theorem 5.10.9

Let Y be lognormally distributed such that $\ln Y \sim N(\mu, \sigma^2)$, with density f. Then

$$\int_a^\infty y\,f(y)\,dy \;=\; e^{\mu+\sigma^2/2}\,N\!\left(\frac{\mu-\ln a}{\sigma}+\sigma\right). \tag{5.28}$$

Proof. Substitute $y = e^x$, so that $\ln Y = X \sim N(\mu, \sigma^2)$ and the integral becomes

$$\int_{\ln a}^\infty e^x\,\frac{1}{\sigma\sqrt{2\pi}}\,e^{-(x-\mu)^2/(2\sigma^2)}\,dx.$$

Completing the square in the exponent,

$$x - \frac{(x-\mu)^2}{2\sigma^2} = \mu + \frac{\sigma^2}{2} - \frac{\left(x-\mu-\sigma^2\right)^2}{2\sigma^2},$$

so the integral equals

$$e^{\mu+\sigma^2/2}\int_{\ln a}^\infty \frac{1}{\sigma\sqrt{2\pi}}\,e^{-(x-\mu-\sigma^2)^2/(2\sigma^2)}\,dx = e^{\mu+\sigma^2/2}\,\mathrm{Prob}\big[\,Z > \tfrac{\ln a-\mu-\sigma^2}{\sigma}\,\big],$$

with Z standard normal. By the symmetry $1-N(z) = N(-z)$ this is $e^{\mu+\sigma^2/2}N\big((\mu+\sigma^2-\ln a)/\sigma\big)$, which is (5.28). ∎

The other ingredient is the probability of the event itself.

Proposition 5.10.10

With Y as above and $a > 0$,

$$\operatorname{Prob}[\,Y > a\,] = N\!\left(\frac{\mu - \ln a}{\sigma}\right).$$

Proof. $Y > a$ exactly when $\ln Y > \ln a$. By Proposition 5.5.3 this has probability $1 - N\big((\ln a - \mu)/\sigma\big)$, which equals $N\big((\mu - \ln a)/\sigma\big)$ by symmetry (Proposition 5.5.4). ∎

Corollary 5.10.11

With Y as above and $a > 0$, let

$$d_2 = \frac{\mu - \ln a}{\sigma}, \qquad d_1 = d_2 + \sigma.$$

Then

$$E\big[\max(Y - a, 0)\big] = e^{\mu + \sigma^2/2}\, N(d_1) - a\, N(d_2). \tag{5.29}$$

Proof. $\max(Y - a, 0) = (Y - a)\,\mathbf{1}_{\{Y > a\}}$, so its expectation is $\int_a^\infty y\, f(y)\, dy - a \operatorname{Prob}[Y > a]$. Apply (5.28) to the first term and Proposition 5.10.10 to the second. ∎

(5.28) is the partial expectation of a lognormal variable — the expected value of Y restricted to the event $Y > a$.
It is not a technical curiosity: it is the entire content of the first term of the Black–Scholes formula. A call option pays $Y - a$ exactly when $Y > a$, so its expected payoff splits as in Corollary 5.10.11, the first piece supplying $S_0 N(d_1)$ and the second $a\,N(d_2)$; Theorem 8.14.19 carries this out. Every remaining chapter of this course is, in one way or another, an elaboration of that split.
換句話說，(5.29) 就是 Black–Scholes 公式的骨架：買權在 $Y > a$ 時付 $Y - a$，期望報酬拆成兩塊，前者給出 $S_0 N(d_1)$，後者給出 $a\,N(d_2)$。之後要做的，只是決定該用哪一個機率分佈來取期望值，以及怎麼折現。

6. Option Basics

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Lecture 3
September 24, 2026

The shift toward options as
the center of gravity of finance [. . .]
— Merton H. Miller[1] (1923–2000)

Too many potential physicists and engineers
spend their careers shifting money around
in the financial sector,
instead of applying their talents to
innovating in the real economy.
— Barack Obama (2016)

Every instrument so far has had a cash flow that is *fixed* the moment the contract is signed. A bond pays its coupons whether the world turns out well or badly; the only uncertainty is the rate at which we discount them, and all of Chapters 3 and 4 was about that rate. An option is the first instrument in this course whose cash flow at expiration is **contingent**: it depends on what the underlying asset happens to be worth on that day.

Remark 6.0.1. **或有的**（contingent）：現金流「視情況而定」。債券的票息與本金在簽約時就寫死，唯一的不確定是折現率；選擇權到期時能拿到多少錢，卻要看標的資產當天的價格——這正是它比債券難定價的根本原因。

That single change is what makes the pricing problem hard, and it is the problem the rest of the course solves. This chapter, however, prices nothing. Its job is to fix

(i) the vocabulary — call, put, strike, exercise, American, European;

(ii) the *payoff functions*, which are the terminal conditions every later pricing model must reproduce;

(iii) the market conventions that decide what the contract actually promises — dividends, splits, short selling; and

(iv) the standard portfolios built out of options: hedges, spreads, and combinations.

The last item is worth more than it looks. Two of the spreads below, pushed to a limit, deliver model-*independent* prices for two exotic securities, and — pushed one step further — read the market's own probability distribution for S_T straight off the quoted call prices. All of that comes for free, before writing down any stochastic process at all.

§6.1. Calls and Puts

Slides pp. 187–189

Definition 6.1.1. A **call** gives its holder the *right*, but not the obligation, to *buy* one unit of the underlying asset by paying a **strike price** (also called the exercise price).

[1] Co-winner of the 1990 Nobel Prize in Economic Sciences.

The holder pays for that right up front. The price paid is the **option premium**, and it is the only cash flow that is certain: the cash flow at expiration is contingent, because the holder will simply walk away if buying at the strike is a bad deal. Figure 6.1 draws the two dates.

Remark 6.1.2.
- **買權**（call）：持有人有權利（但沒有義務）買進一單位標的資產。
- **履約價**（strike price，又稱 exercise price）：履約時買進（或賣出）標的所用的約定價格。
- **權利金**（option premium）：買方今天為取得這個權利付給賣方的價格。它是整筆交易中唯一確定會發生的現金流；到期時的交割則不一定發生，因為若按履約價買進不划算，持有人會直接放棄。

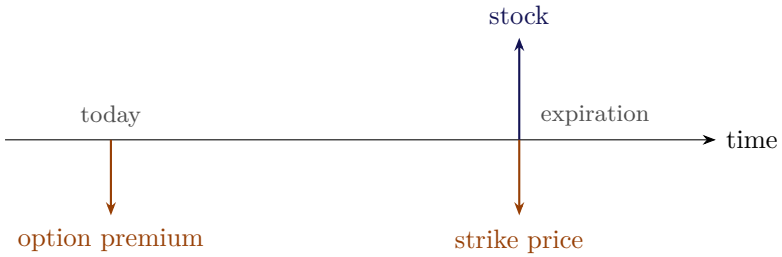


Figure 6.1.: Cash flows of a long call. Arrows below the axis are payments made by the holder, arrows above it are receipts. The premium leaves today with certainty; the exchange of the strike price for the stock happens at expiration *only if* the holder chooses to exercise.

Definition 6.1.3. A **put** gives its holder the right to *sell* one unit of the underlying asset for the strike price.

Remark 6.1.4. **賣權**（put）：持有人有權利以履約價賣出一單位標的資產。與買權恰好相反——買權是「付履約價、拿股票」，賣權是「交出股票、收履約價」；權利金一樣是今天先付。

Remark 6.1.5. **多頭／買方部位**（long）與**空頭／賣方部位**（short）：描述你站在合約的哪一邊。
- **Long**：買進並持有。long a call ／ long a put 就是買進買權／賣權、付出權利金、拿到權利；long a stock 就是持有股票。
- **Short**：賣出。short a call ／ short a put 就是賣出選擇權、收下權利金，並承擔對方行使權利時必須配合的義務；short a stock 則是賣出股票，連自己沒有的股票也能賣（做法留待後面 Short Selling 一節說明）。

同一份合約，long 方賺的就是 short 方賠的，兩邊損益正好互為相反數（零和）。注意：long ／ short 說的是「買或賣這份合約」，不是「看漲或看跌」——long a put 是買進賣權，押的其實是股價下跌。

The put reverses the two contingent arrows: the holder delivers the stock and receives the strike price (Figure 6.2). The premium still leaves today.

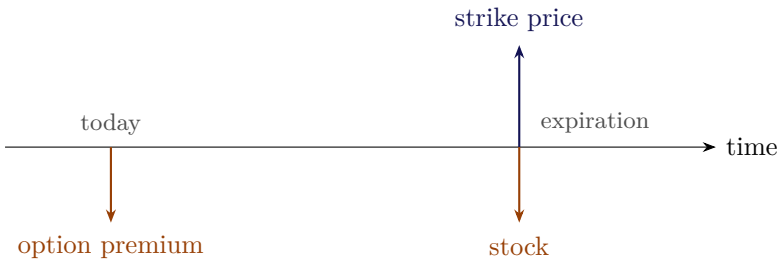


Figure 6.2.: Cash flows of a long put — Figure 6.1 with the roles of stock and strike price interchanged.

> **Notation .** Options sold on their own, as the two figures above assume, are called *standalone*. An **embedded option** is one that cannot be separated from its host: it has to be traded along with the underlying asset. The callable bond of Chapter 3 is the standing example — the issuer's right to redeem early is an option, but nobody trades it apart from the bond.

Remark 6.1.6. **內嵌選擇權**（embedded option）：附在其他證券上、無法單獨拆出來交易的選擇權；相對的是單獨買賣的獨立（standalone）選擇權。例如第三章的可贖回債券：發行人「可提前贖回」的權利本質上是一個選擇權，但它只能跟著債券一起買賣。

Remark 6.1.7. How to price options? That is the question the next several chapters answer, and it is old: it can be traced to Aristotle's (384 B.C.–322 B.C.) *Politics*, if not earlier.[2]

§6.2. Exercise

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To **exercise** an option is to invoke the right it grants, and the mechanics are exactly the two contingent arrows drawn above:

- When a *call* is exercised, the holder pays the strike price in exchange for the stock.
- When a *put* is exercised, the holder receives the strike price from the writer in exchange for the stock.

The counterparty — the **writer**, who sold the option and pocketed the premium — has no choice in the matter: the writer must take the other side whenever the holder exercises. This asymmetry is the whole content of the word "option".

Remark 6.2.1.
- **履約**（exercise）：持有人行使選擇權賦予的權利——買權履約是付履約價換股票，賣權履約是交出股票換履約價。
- **賣方**（writer，又稱立權人）：賣出選擇權、收下權利金的一方。賣方沒有選擇：只要持有人要履約，就必須配合交割。「權利在買方、義務在賣方」這個不對稱，就是「選擇權」這個名字的意思。

Some options may be exercised *before* the expiration date. This is called **early exercise**, and deciding whether to do it is an optimization problem in its own right, not a detail of the contract. Section 7.7 settles the question completely for calls; for puts it stays open for the rest of the course.

Remark 6.2.2. **提前履約**（early exercise）：在到期日之前就履約。這不是合約的小細節，而是一個最佳化問題——提前履約就放棄了「繼續等、看股價會不會更有利」的機會，何時划算需要計算。後面會證明：在一定條件下買權永遠不該提前履約；賣權則沒有這麼簡單的答案。

§6.3. American and European

Slides p. 191

Definition 6.3.1. An **American option** can be exercised at any time up to the expiration date. A **European option** can only be exercised at expiration.

[2] Aristotle reports that Thales, foreseeing a large olive harvest, paid deposits in winter for the use of every olive press in Miletus and Chios — the deposits are option premiums, and the presses are the underlying asset.

Remark 6.3.2. **美式選擇權**（American option）到期日前任何時刻都能履約；**歐式選擇權**（European option）只能在到期日當天履約。名稱與地理無關，差別只在「什麼時候可以履約」。美式多了提前履約的權利，所以價值至少不低於歐式。

Proposition 6.3.3

An American option is worth at least as much as an otherwise identical European option.

Proof. The American contract contains every right the European contract contains, plus the right to exercise early. The holder of the American option may simply choose never to exercise early, thereby reproducing the European payoff exactly. A strictly larger set of available strategies cannot be worth less. ■

Note 6.3.4. The names are geographical only by accident; both types trade on both continents. Note also that Proposition 6.3.3 is an inequality, not an equality — and the gap is genuinely hard to compute. European options on a non-dividend-paying stock admit a closed-form price (Black–Scholes); American puts do not, which is why so much of the numerical machinery later in the course exists — Section 7.7 proves the gap is *zero* for calls on such a stock, and Section 8.3.i shows where it reopens for puts.

§6.4. Convenient Conventions

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Notation . The following symbols are used without further comment for the rest of the course.

Symbol	Meaning
C	call value
P	put value
X	strike price
S	stock price
D	dividend

We always assume $S \geq 0$.[a] When the time matters we write S_t for the stock price at time t, and S_T for its value at expiration.

[a] Contributed by Mr. Tang, Bert (B08902102) on March 10, 2021.

Remark 6.4.1. C（call value）與 P（put value）指的是一份買權、一份賣權這張合約本身值多少錢，而不是履約後拿到多少。

- **到期前**：就是選擇權的市場價格，也就是現在買進它要付的權利金。例如 $C = 3$ 代表現在花 \$3 可以買到一份買權。
- **到期時**：已經沒有未來可等，合約的價值就只剩「現在履約能拿到多少錢」——下一節會把它寫成 S 和 X 的式子。

所以同一個符號 C、P，在到期前代表價格，到期時才剛好等於履約所得。

§6.5. Payoff, Mathematically Speaking

Slides pp. 193–194

At expiration the holder's decision is trivial, because there is no future left to speculate about: exercise if and only if doing so produces cash.

Definition 6.5.1. The **payoff** of a call at expiration is

$$C = \max(0, S - X), \tag{6.1}$$

and the payoff of a put at expiration is

$$P = \max(0, X - S). \tag{6.2}$$

The $\max(0, \cdot)$ is the contract's optionality made arithmetic. A call will be exercised only if the stock price is *higher* than the strike price — otherwise the stock can be bought more cheaply in the market, and the holder lets the option lapse. Symmetrically, a put will be exercised only if the stock price is *less* than the strike price.

Remark 6.5.2. **到期報酬**（payoff）：選擇權在到期日當天的價值（尚未扣掉權利金，扣掉後才是利潤 profit）。到期時沒有未來可以等，決策很單純：有錢賺就履約，否則放棄。所以買權拿到 $\max(0, S - X)$、賣權拿到 $\max(0, X - S)$，外面那層 $\max(0, \cdot)$ 就是「可以選擇不履約」寫成算式。

Two facts about (6.1) and (6.2) drive everything that follows.
They are *non-negative*: an option can never impose a loss on its holder beyond the premium already paid. Hence $C \geq 0$ and $P \geq 0$ at *all* times, not just at expiration.
They are *convex* and *kinked* at $S = X$. Convexity is why options are worth more than their intrinsic value before expiration (Jensen's inequality again), and the kink is the single point where every numerical method in this course has to work hardest.

Figure 6.3 draws all four elementary positions with $X = 50$. The writer's payoff is the holder's payoff reflected through the horizontal axis — options are a zero-sum contract between the two parties, so the two pictures in each column must sum to zero pointwise.

§6.5.i. Intrinsic Value and Moneyness

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Definition 6.5.3. At any time t *before* the expiration date, the **intrinsic value** of a call is

$$\max(0, S_t - X),$$

and the intrinsic value of a put is

$$\max(0, X - S_t).$$

The intrinsic value is what the option would pay if expiration were today. It is a *lower* bound on an American option's value (exercising is always available) and, as Lemma 7.6.1 will show, a lower bound on a European *call*'s value as well — though not, it turns out, on a European put's. Whatever the option is worth above its intrinsic value is its **time value** — the market's charge for the remaining chance that the stock moves further in the holder's favour.

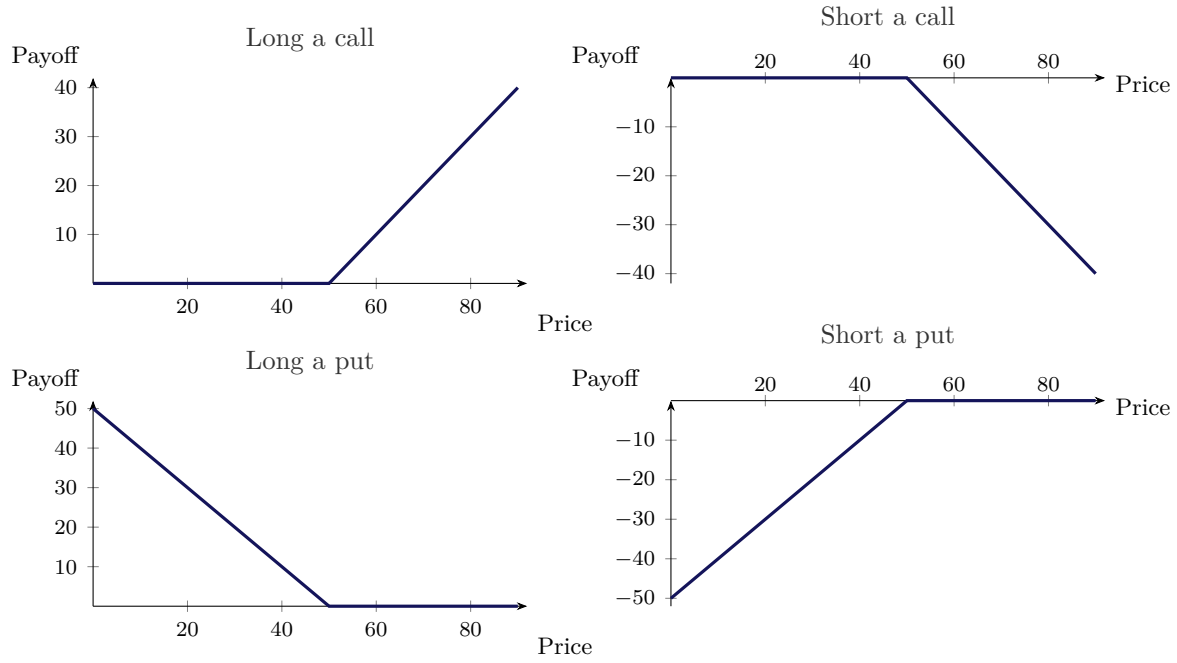


Figure 6.3.: Payoffs at expiration of the four elementary option positions, with strike $X = 50$. These are payoffs, not profits: the premium has not been subtracted. The kink always sits at $S = X$. Note the asymmetry in the writer's risk — a short call has unbounded loss as $S \to \infty$, whereas a short put loses at most X, since the stock cannot fall below zero.

Remark 6.5.4.
- **內含價值**（intrinsic value）：假如現在就是到期日，選擇權能拿到的錢——買權 $\max(0, S_t - X)$、賣權 $\max(0, X - S_t)$。
- **時間價值**（time value）：選擇權價格超過內含價值的部分，是市場對「剩下的時間裡股價還可能往有利方向走」所收的費用。到期時歸零。

所以「選擇權價格 = 內含價值 + 時間價值」。

Definition 6.5.5. A call is **in the money** if $S > X$, **at the money** if $S = X$, and **out of the money** if $S < X$. A put is in the money if $S < X$, at the money if $S = X$, and out of the money if $S > X$.

Remark 6.5.6.
- **價內**（in the money）：現在履約有正的內含價值（買權 $S > X$、賣權 $S < X$）。
- **價平**（at the money）：$S = X$，現在履約不賺不賠。
- **價外**（out of the money）：現在履約沒有好處，內含價值為 0（買權 $S < X$、賣權 $S > X$）。

Note 6.5.7. The direction reverses between calls and puts, so the words describe *whether the option currently has positive intrinsic value*, not whether the stock is high or low.

Options that are in the money at expiration should be exercised — the payoff is positive and free. Astonishingly, they are not always.[3]

Finding an option's value at *any* time before expiration — that is, splitting it into intrinsic value plus time value with an actual number attached — is a major intellectual breakthrough. Chapter 8 makes it, and the rest of the course elaborates on it.

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Figure 6.4 shows what the answer looks like ahead of time. The dashed line in each panel is

[3] About 11% of option holders let in-the-money options expire worthless.

the intrinsic value, the payoff already drawn in Figure 6.3; the solid curve is the option's value one month before expiration. Three features are visible and will be proved later:

- The solid curve is *smooth*: the kink at $S = X$ has been rounded off, because with time left the market does not yet know which side of the strike the stock will land on.
- The call's curve lies *above* its intrinsic value everywhere — that gap is time value — and is monotone and convex, sandwiched between the intrinsic value and the stock price itself.
- The put's curve does *not*. Look at the left edge of the right-hand panel: for small S the solid curve dips *below* the dashed one, so a European put can be worth less than its intrinsic value. This is not a drawing error, and Lemma 7.6.4 explains it.

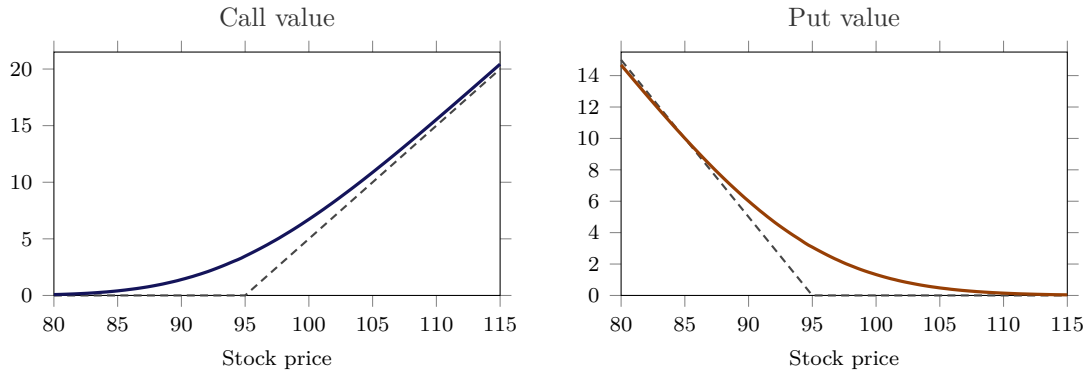


Figure 6.4.: Value of a European call and put with strike $X = 95$ one month before expiration (solid), against their intrinsic values (dashed). Computed from Black–Scholes with $r = 5\%$ and $\sigma = 30\%$; the formula itself is derived in Theorem 8.14.19, and only the *shape* matters here. The vertical gap between the two lines is the time value, largest at the money and vanishing deep in or out of the money.

Remark 6.5.8. 怎麼讀這張圖。兩張圖都是「離到期還有一個月」這個時間點的快照。橫軸是當下的股價 S；實線是在這個股價下選擇權現在值多少錢（C 或 P）；虛線是在這個股價下的內含價值（買權 $\max(0, S-95)$、賣權 $\max(0, 95-S)$）。同一個 S 上，實線減虛線就是時間價值。

時間價值可以是負的嗎？可以，右圖左端就是例子；但買權不會。持有買權而不馬上履約，有兩個好處：

(i) 履約價 X 晚一個月才付，這段期間的錢還能生利息；

(ii) 到期時若股價跌到 X 以下，可以直接不履約，不必承擔損失。

兩者都只會讓價值變高，不會變低，所以買權的實線永遠在虛線之上（Lemma 7.6.1 會正式證明）。

賣權為什麼在股價很低時跑到虛線下面？賣權的好處 (ii) 還在，但 (i) 反過來了：賣權履約是收到 X，而歐式賣權要等到期才能收，晚一個月拿到的 X 比今天拿到的 X 少了一個月利息，對持有人不利。股價很低時，這個賣權幾乎一定會被履約，(ii) 的保護用不太到；而且股價最低只到 0，再往下跌的空間也有限。剩下的主要就是 (i) 的利息損失，於是價值大約是「一個月後收 X 的現值 $-S$」，比內含價值 $X - S$ 還小，時間價值變成負的。美式賣權則可以現在就履約拿 $X - S$，所以不會掉到虛線以下。

§6.6. Corporate Actions: Dividends and Splits

The payoff formulas above refer to "the stock", but a share is not a fixed object: companies pay dividends and split their shares, and each event moves S for reasons that have nothing to

do with the market's opinion of the company. Whether the option contract is *adjusted* for such an event therefore changes what the option is worth. The convention differs by event type.

§6.6.i. Cash Dividends

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Exchange-traded stock options are *not* cash dividend-protected (or simply **protected**): the option contract is not adjusted for cash dividends.

Why this matters: on the **ex-dividend** date the stock price falls by an amount roughly equal to the cash dividend, because a buyer from that day on no longer receives it. The drop is mechanical, but the unprotected option absorbs it in full.

Remark 6.6.1.
- **受保護**（protected）：合約會因應現金股利調整條款（例如調降履約價），就稱為受股利保護。交易所掛牌的股票選擇權不受保護。
- **除息**（ex-dividend）：從除息日起買進股票的人拿不到這次股利，所以股價在除息日會下跌約一個股利的金額。

兩者合起來：除息造成的股價下跌全數反映在不受保護的選擇權上，所以買權變便宜、賣權變貴。

Consequently

cash dividends are detrimental for calls, and the opposite is true for puts.

> **Intuition .** The option holder owns exposure to the share price but not the share, and therefore collects no dividend. A dividend transfers value out of the share price and into the pockets of shareholders — over whom the call holder has no claim. So a call on a dividend-paying stock is worth less than a call on an otherwise identical stock that retains its cash, and a put is worth correspondingly more. This is also the reason early exercise of an American call can be optimal: exercising just before the ex-dividend date is the only way to capture the dividend. Theorem 7.7.3 makes that precise — those dates are the *only* ones at which it can pay to exercise early.

Remark 6.6.2. 舉例：股價 100、每股配現金股利 3 元。除息日之後買的人領不到這 3 元，只願意付約 97，所以股價掉到約 97。
- **股東**不吃虧：股價少 3 元，但口袋多了 3 元現金，總財富不變。
- **選擇權持有人**手上沒有股票，領不到這 3 元；合約又不受保護、履約價不調整，所以他只看到股價少了 3 元。

買權是股價愈高愈值錢，股價平白少 3 元就是損失；賣權是股價愈低愈值錢，反而得利。這就是上面那句話的意思：股利把價值從「股價」搬進「股東口袋」，而選擇權只連結到股價那一邊。

這也解釋了美式買權為什麼可能提前履約：在除息前先履約、把股票拿到手，就能領到那 3 元股利，而不是眼睜睜看著股價下跌。

§6.6.ii. Stock Splits and Stock Dividends

Slides pp. 199–200

Options *are* adjusted for stock splits. After an n-**for-**m **stock split**, m shares become n shares. Accordingly, the option is rewritten so that
- the strike price becomes m/n times its previous value, and
- the number of shares covered by one option becomes n/m times its previous value.

Exchange-traded stock options are adjusted for **stock dividends** in the same way. The point of the adjustment is that the total strike outlay $X \times$ (shares covered) is unchanged, so the split — which alters nothing real about the company — alters nothing about the option.

Remark 6.6.3.
- **n 換 m 股票分割**（n-for-m stock split）：原本 m 股變成 n 股，每股價格約變成原來的 m/n 倍，公司總價值不變。
- **股票股利**（stock dividends，台灣俗稱配股）：以新股代替現金發放股利，效果與股票分割相同。

這兩種事件選擇權會調整：履約價乘 m/n、每口涵蓋股數乘 n/m，使履約總金額不變——公司實質沒變，選擇權的價值也就不該變。

Example 6.6.4. Consider an option to buy 100 shares of a company for \$50 per share. A 2-for-1 split changes the terms to a strike price of \$25 per share for 200 shares. The outlay on exercise is $100 \times 50 = 200 \times 25 = \$5{,}000$ either way.

Note 6.6.5. Unless stated otherwise, **we assume options are unprotected** from here on. Cash dividends will therefore show up explicitly as a term in every pricing formula, while splits will never be mentioned again.

§6.7. Short Selling

Slides pp. 201–203

Several of the portfolios below, and essentially every no-arbitrage argument later in the course, require taking a *negative* position in the stock. The mechanism that makes this possible is short selling.

Definition 6.7.1. **Short selling** (or **shorting**)[4] means selling an asset that is not owned, with the intention of buying it back later.

Remark 6.7.2. **放空**（short selling / shorting）：賣出自己沒有的資產，打算之後再買回來歸還。實際做法是向券商借股票賣掉、日後買回還給對方，所以股價下跌就賺錢。放空讓我們能持有「負的」股票部位，下面的投資組合以及後面章節的定價論證幾乎都要用到。

Concretely, if you short 1,000 XYZ shares:

1. The broker borrows the 1,000 shares from another client and sells them in the market.
2. This action generates proceeds for the investor — cash now, and an obligation to return the shares later.
3. The investor closes out the short position by buying 1,000 XYZ shares and returning them.

Clearly, the investor profits if the stock price falls: the repurchase costs less than the sale brought in. The payoff of a short position is the payoff of a long position with the sign flipped (Figure 6.5), which is exactly the property that makes shorting the algebraic tool it is.

[4] Invented by Le Maire (1558–1624) in 1608.

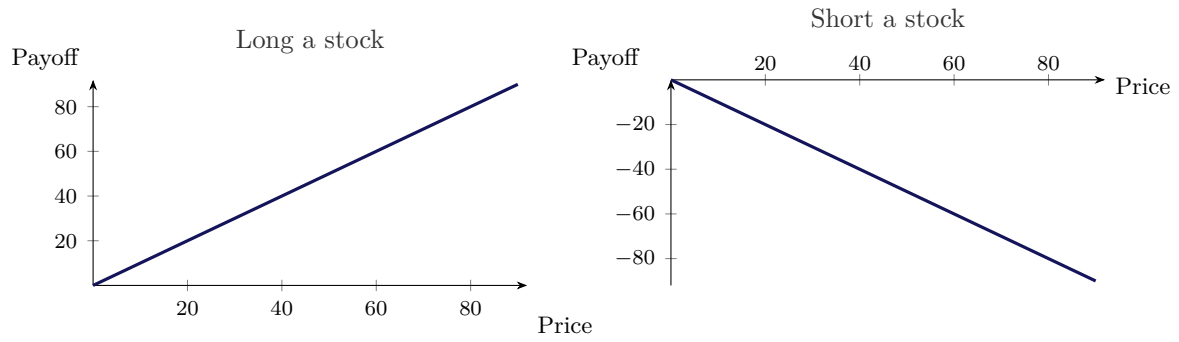


Figure 6.5.: Payoff of a stock position. Unlike the option payoffs of Figure 6.3 these are straight lines through the origin — the stock has no kink, and hence no optionality.

Note 6.7.3. Two warnings, both of which the models we build will quietly ignore.

Not all assets can be shorted — there must be a lender, and lendable inventory is neither free nor guaranteed.

In reality, short selling is not simply the opposite of going long.[5] The short seller pays a borrowing fee, can be forced to close the position when the lender recalls the shares, faces an unbounded loss, and may be barred from shorting altogether by regulators in a falling market.

Remark 6.7.4. 上面這段提醒：課程裡的模型會把放空當成「持有 -1 股」，損益剛好是持有股票的相反數；但現實中放空沒有這麼理想。

(i) **不是什麼都能放空**。放空要先借到股票，得有人願意出借、市場上也要有足夠可借的量，這兩件事都不一定成立，借券也不是免費的。

(ii) **放空不只是「買進的反面」**。放空者還要面對幾個買進者沒有的問題：

- 要付借券費；
- 出借人可以隨時要回股票，放空者就得被迫買回平倉，不一定挑得到好時機；
- 損失沒有上限：買股票最多賠光本金（股價跌到 0），但放空時股價可以一直漲，買回的成本也跟著無限增加；
- 股市大跌時，主管機關可能直接禁止放空。

之後的定價論證會把這些摩擦全部忽略，假設想放空多少就能放空多少、而且沒有成本。

§6.8. Covered Positions: Hedges

Slides pp. 204–205

A **covered position** is an option combined with other securities. Three families are worth naming: *hedges*, which combine an option with its own underlying stock; *spreads* (Section 6.9), which combine options of the *same* type; and *combinations* (Section 6.10), which mix calls with puts.

Definition 6.8.1. A **hedge** combines an option with its underlying stock in such a way that one protects the other against loss.

Remark 6.8.2.

- **有掩護部位**（covered position）：選擇權與其他證券組合成的部位，分三類——避險（搭配標的股票）、價差（同類型選擇權）、組合（買權搭賣權）。
- **避險**（hedge）：選擇權搭配其標的股票，讓其中一方為另一方的損失提供保護。

Two hedges account for almost all practical use:

[5] Kosowski & Neftci (2015). See https://tw.news.appledaily.com/headline/daily/20180307/37950481/ for an example in Taiwan on February 6, 2018.

- **Covered call**: a long position in the stock together with a *short* call.[6] It is called "covered" because the stock on hand can be delivered to the buyer of the call if the call is exercised — the writer is never forced to buy at an arbitrarily high market price.
- **Protective put**: a long position in the stock together with a *long* put. The put is an insurance policy on the shares: whatever happens, they can be sold for at least X.

Remark 6.8.3.
- **掩護性買權**（covered call）：持有股票，同時賣出買權。被履約時手上已有股票可交，所以叫「有掩護」；代價是股價漲過 X 的部分讓給買方，報酬被封頂在 X。
- **保護性賣權**（protective put）：持有股票，同時買進賣權。賣權就像股票的保險，股價再怎麼跌都能以 X 賣出，報酬有地板 X。

Writing S_T for the stock price at expiration, the two payoffs are

$$\begin{aligned} \text{covered call:} \quad & S_T - \max(0, S_T - X) \;=\; \min(S_T,\, X), \\ \text{protective put:} \quad & S_T + \max(0, X - S_T) \;=\; \max(S_T,\, X). \end{aligned} \tag{6.3}$$

So the covered call *caps* the outcome at X and the protective put *floors* it at X. A firm's bonds are a covered call on the firm's total value (Section 10.1).

Note 6.8.4. Payoff 是什麼？到期時手上這整個部位值多少錢，不扣掉期初建立部位花的錢（買股票的錢、付或收的權利金）。扣掉期初成本之後才是利潤（profit），也就是 Figure 6.6 裡的虛線。兩種避險都持有一股股票，而股價不會小於 0，所以 payoff 一定 ≥ 0；但利潤可以是負的。

左邊怎麼來的？把部位裡每樣東西到期時的價值加起來：
- **掩護性買權** = 持有一股 + 賣出一個買權。股票到期值 S_T；賣出的買權如果被履約，你要付給對方 $\max(0, S_T - X)$（(6.1)，站在 short 方所以是減號）。合起來是 $S_T - \max(0, S_T - X)$。
- **保護性賣權** = 持有一股 + 買進一個賣權。股票到期值 S_T；買來的賣權到期值 $\max(0, X - S_T)$（(6.2)，long 方所以是加號）。合起來是 $S_T + \max(0, X - S_T)$。

分 $S_T \geq X$ 與 $S_T < X$ 代入，就得到右邊的 $\min(S_T, X)$ 和 $\max(S_T, X)$。

Note 6.8.5. Both strategies break even only if the stock price rises above a certain level, so **both are bullish**. This is easy to miss, because a covered call is often described as "defensive". The premium received merely offsets a fixed amount of downside; the position still loses roughly dollar-for-dollar with the stock once that cushion is gone, as the left-hand tail in Figure 6.6 shows.

[6] A short position has a payoff opposite in sign to that of a long position, as in Figure 6.3. Some ETFs package this payoff directly, such as the Global X Nasdaq 100 Covered Call ETF (QYLD).

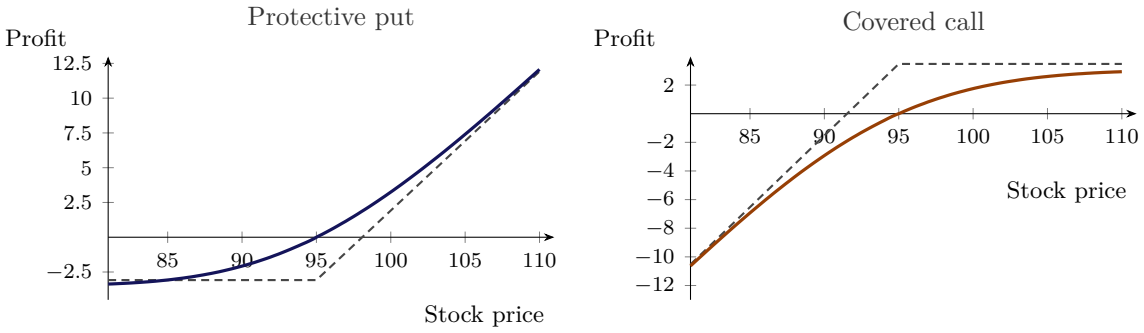


Figure 6.6.: Profit of the two hedges, with $X = 95$ and the portfolio set up when $S = 95$ one month before maturity. Dashed lines are profits at expiration, given by (6.3) less the setup cost. Solid lines are profits at the moment the portfolio is set up, as a function of where the stock price goes; they are smooth because the options still carry time value. Both pictures slope upward on the left, which is what "bullish" means.

Remark 6.8.6. 圖在畫什麼。兩張圖都是在 $S_0 = 95$、離到期一個月時建立部位，履約價 $X = 95$。當時的賣權價格 $P_0 \approx 3.08$、買權價格 $C_0 \approx 3.48$（和 Figure 6.4 一樣由 Black–Scholes 公式算出，公式後面才推導，這裡當成市場報價即可）。縱軸是利潤 = 部位價值 − 期初成本，其中保護性賣權的成本是 $S_0 + P_0$，掩護性買權是 $S_0 - C_0$（賣出買權會先收錢）。

- **虛線**：到期時的利潤，橫軸是到期股價 S_T。
- **實線**：建倉當下的利潤——若股價馬上變成橫軸上的 S（離到期仍有一個月），部位值多少、扣掉成本賺賠多少。

虛線為什麼是折線。把 (6.3) 減掉成本：

$$\text{保護性賣權：}\ \max(S_T, X) - (S_0 + P_0), \qquad \text{掩護性買權：}\ \min(S_T, X) - (S_0 - C_0).$$

兩者都只在 $S_T = X$ 折一次，每段斜率不是 0 就是 1。保護性賣權在 X 以下是高度 $X - S_0 - P_0 = -3.08$ 的水平線（賣權托底，最多賠權利金），X 以上斜率 1，要漲過 $S_0 + P_0 = 98.08$ 才賺錢。掩護性買權反過來：X 以上是高度 $X - S_0 + C_0 = +3.48$ 的水平線（被封頂，最多賺權利金），X 以下斜率 1，跌破 $S_0 - C_0 = 91.52$ 就開始賠。

實線為什麼是平滑曲線。建倉當下部位值 $S + P(S)$ 或 $S - C(S)$。把選擇權價格拆成內含價值加時間價值 TV：

$$\text{實線} = \text{虛線} + \mathrm{TV}_{\mathrm{put}}(S), \qquad \text{實線} = \text{虛線} - \mathrm{TV}_{\mathrm{call}}(S).$$

所以兩線的差距就是時間價值。時間價值在價平附近最大，於是實線在折點附近離虛線最遠，折角被磨圓。保護性賣權是買賣權，中間實線在虛線上方；最左端歐式賣權的時間價值為負（Figure 6.4），實線略低於虛線。掩護性買權是賣買權，減掉 $\mathrm{TV}_{\mathrm{call}} \geq 0$，實線處處在虛線下方。

回到 Note 6.8.5。那個 note 的主張是：保護性賣權和掩護性買權都是看漲（bullish）的部位；並提醒掩護性買權雖然常被說成「防守型」，其實擋不住大跌。現在可以從圖上直接看出來：

- **都看漲**：兩張圖的虛線與實線都往右上升（斜率 ≥ 0），股價愈高利潤愈高、股價跌就不賺甚至賠，方向和單純持有股票一樣。
- **掩護性買權不防守**：它比單純持股多出來的，只有期初收的權利金 $C_0 = 3.48$，這就是 Note 6.8.5 說的「只抵銷固定金額的下跌」。股價跌破 $S_0 - C_0 = 91.52$ 之後，虛線左段斜率 1，股價每跌 1 元就賠 1 元，跟持股完全一樣——也就是 Note 6.8.5 說的「墊子用完之後，幾乎一塊錢對一塊錢地跟著股票虧」。

那兩者各防什麼？和單純持股（利潤 $S_T - S_0$）比：

- **保護性賣權防大跌**：股價跌得再深，最多賠 $P_0 = 3.08$。代價是其他情況都比持股少賺 P_0——就像付保費買保險。

- **掩護性買權防小跌與盤整**：股價在 $S_0 + C_0 = 98.48$ 以下時，都比持股多 $C_0 = 3.48$，所以小跌能被抵掉、不動也有收入；但大跌照賠。代價是放棄大漲——漲過 98.48 後比持股少賺，漲愈多差愈多。它其實比較像把「大漲那段獲利」以 C_0 的價格賣掉，換取收入。

§6.9. Covered Positions: Spreads

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Definition 6.9.1. A **spread** consists of options of the same type and on the same underlying asset, but with different strike prices or expiration dates.

Remark 6.9.2. 價差（spread）：同一標的、同一類型（全是買權或全是賣權），但履約價或到期日不同的選擇權組合。一買一賣互相抵掉一部分成本與風險，主要用來押股價的方向。

Throughout this section we use X_L, X_M, and X_H to denote strike prices with

$$X_L < X_M < X_H.$$

All options in a spread are calls, or all are puts; mixing types produces a *combination* instead, which is Section 6.10.

§6.9.i. Bull Spreads

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Definition 6.9.3. A **bull call spread** consists of a long X_L call and a short X_H call with the same expiration date.

Remark 6.9.4. 多頭買權價差（bull call spread）：買進低履約價 X_L 的買權、賣出高履約價 X_H 的買權（到期日相同）。賣出的買權幫忙付掉一部分成本，但也把股價超過 X_H 的獲利讓出去，所以賺賠都有上限——押的是「股價會漲，但漲不多」。

Since $C_L > C_H$ (a lower strike is worth more), the position costs money to enter. Subordinated debt is a bull call spread on the firm's value (Section 10.1.iv). Its accounting is completely determined by (6.1):

- The initial investment is $C_L - C_H$.
- The payoff is nonnegative; explicitly it is $\max(0, S_T - X_L) - \max(0, S_T - X_H)$.
- The maximum payoff is $X_H - X_L$, achieved when both calls are exercised at expiration, i.e. when $S_T \geq X_H$.
- The maximum profit is therefore $(X_H - X_L) - (C_L - C_H)$.
- The maximum loss is $C_L - C_H$, incurred when neither is exercised at expiration, i.e. when $S_T \leq X_L$.

Both the gain and the loss are bounded: the short call pays for part of the long call and, in exchange, sells away every gain above X_H. The investor is betting that the stock rises, but only so far.

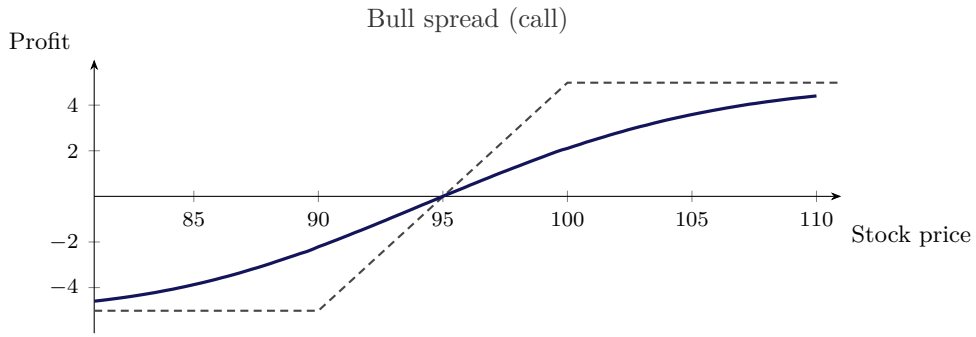


Figure 6.7.: Profit of a bull call spread with $X_L = 90$, $X_H = 100$, entered when $S = 95$ one month before expiration for a net debit of $C_L - C_H = 5.02$. Dashed: profit at expiration, a ramp clipped at both ends. Solid: profit at the moment of entry as the stock price varies. The maximum profit 4.98 and the maximum loss 5.02 are the two dashed plateaus.

Remark 6.9.5. 同樣看漲、賺賠都有上限的價差，也可以只用賣權做：賣出 X_H 賣權、買進 X_L 賣權（到期日相同），稱為多頭賣權價差（bull put spread，Section 6.9.iii 會再提到）。用和上面同樣的方式記帳：

- 期初收到 $P_H - P_L > 0$（履約價愈高的賣權愈值錢）。
- 到期 payoff 是 $\max(0, X_L - S_T) - \max(0, X_H - S_T) \le 0$：$S_T \ge X_H$ 時兩個都不履約，payoff $= 0$；$X_L < S_T < X_H$ 時只有賣出的 X_H 賣權被履約，payoff $= S_T - X_H$；$S_T \le X_L$ 時兩個都履約，payoff $= -(X_H - X_L)$。
- 最大利潤是 $P_H - P_L$，發生在 $S_T \ge X_H$：兩個賣權都作廢，權利金全部落袋。
- 最大虧損是 $(X_H - X_L) - (P_H - P_L)$，發生在 $S_T \le X_L$。
- 損益兩平點在 $S_T = X_H - (P_H - P_L)$。

和多頭買權價差對照：那邊是先付錢、payoff ≥ 0；這邊是先收錢、payoff ≤ 0。但扣掉（或加回）期初的錢之後，利潤都是「X_L 以下平、中間斜率 1、X_H 以上平」的斜坡，所以同樣是看漲、同樣賺賠都有上限。

Figure 6.8 用和 Figure 6.7 相同的設定：$X_L = 90$、$X_H = 100$，建倉時 $S = 95$、離到期一個月。此時 $P_H \approx 6.14$、$P_L \approx 1.20$，淨收入約 4.94；於是最大利潤 4.94、最大虧損 $10 - 4.94 = 5.06$、損益兩平點 95.06。這和多頭買權價差的 4.98、5.02、95.02 幾乎一樣，只差約 0.04：一個是期初付錢、到期可能收錢，另一個是期初收錢、到期可能付錢，兩者差的就是這一個月的利息。

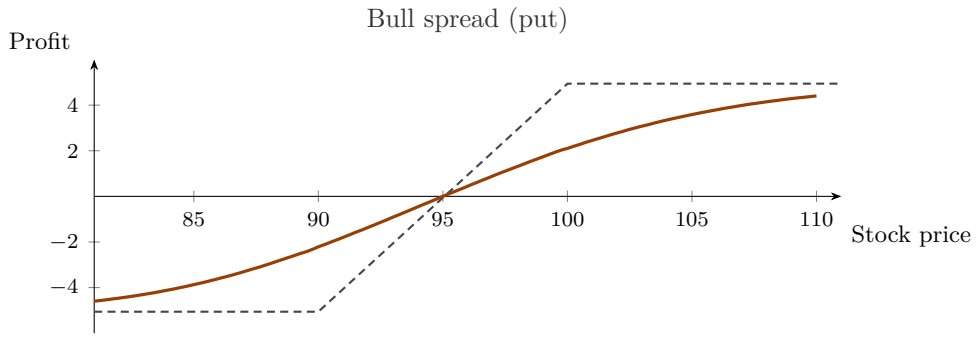


Figure 6.8.: Profit of a bull put spread with $X_L = 90$, $X_H = 100$, entered when $S = 95$ one month before expiration for a net credit of $P_H - P_L = 4.94$. Dashed: profit at expiration. Solid: profit at the moment of entry as the stock price varies. The shape is that of Figure 6.7; the plateaus sit at 4.94 and −5.06 instead of 4.98 and −5.02.

§6.9.ii. The Binary Call: A Model-Free Price

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Now take the bull call spread to a limit. Buy $(X_H - X_L)^{-1}$ units of it, so that the ramp in Figure 6.7 always rises by exactly 1 over the interval $[X_L, X_H]$, and let $X_H - X_L \to 0$. The ramp becomes vertical, and a (Heaviside) step function emerges as the payoff:

$$\text{payoff} \longrightarrow \begin{cases} 1, & S_T > X_L, \\ 0, & S_T < X_L. \end{cases}$$

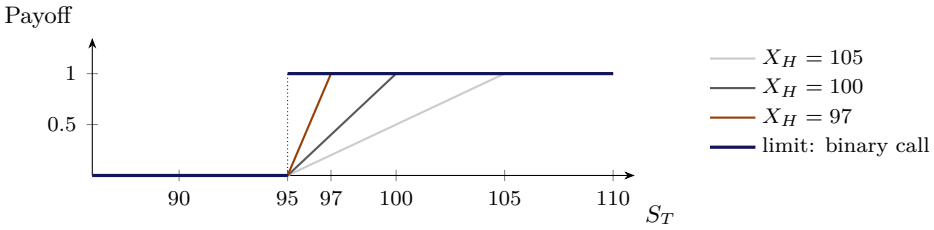


Figure 6.9.: $(X_H - X_L)^{-1}$ units of the bull call spread, $X_L = 95$ fixed: every ramp climbs to 1 and steepens into the binary call's step.

Remark 6.9.6. 為什麼要買 $(X_H - X_L)^{-1}$ 份？一份多頭買權價差的 payoff 是一段斜坡：X_L 以下是 0，X_H 以上是 $X_H - X_L$，中間斜率 1。斜坡的高度等於寬度 $X_H - X_L$。如果只是讓 X_H 靠近 X_L，斜坡會變窄、也同時變矮，最後縮成 0，什麼都沒得到。所以改買 $(X_H - X_L)^{-1}$ 份，把高度固定拉回 1；這時中間段的斜率變成 $(X_H - X_L)^{-1}$。

取極限。 Figure 6.9 固定 $X_L = 95$，讓 X_H 從 105、100 到 97 往下靠近：每條斜坡都從 0 爬到 1，但寬度愈來愈窄、愈來愈陡。寬度趨近 0 時，斜坡變成垂直，payoff 只剩兩種值：S_T 在 X_L 以上拿 1，以下拿 0——這就是下面要定義的二元買權。

Definition 6.9.7. The security with this payoff is the **binary** (or **digital**) **call**: it pays \$1 if the stock finishes above the strike, and nothing otherwise.

Remark 6.9.8. 二元買權／數位買權（binary call / digital call）：到期時股價高於履約價就付 \$1，否則付 0，報酬是一個階梯函數。它是「$(X_H - X_L)^{-1}$ 份多頭價差、兩個履約價無限靠近」的極限，所以價格就是買權價格對履約價的導數 $-\partial C/\partial X$。

Its price is the limit of the prices of the spreads that converge to it. Writing $C(X)$ for the price of a standard call struck at X, that limit is a difference quotient:

$$\frac{C(X_L) - C(X_H)}{X_H - X_L} \xrightarrow[X_H \to X_L]{} -\frac{\partial C}{\partial X}.$$

So the binary call costs

$$-\frac{\partial C}{\partial X} \tag{6.4}$$

today, where C is the price of the standard call. Under the Black–Scholes model this evaluates to $e^{-r\tau}N(x - \sigma\sqrt{\tau})$ (Example 8.14.23).

(6.4) is model independent.
We assumed no stochastic process, no volatility, no interest rate — only that standard calls of every strike trade at some price $C(X)$, and that a portfolio's price is the sum of its parts. Given the observed call prices, the binary call's price is *forced*.
This is the first appearance of a theme that organises the whole course: some facts follow from no-arbitrage alone, and only the remaining facts need a model. （這是貫穿整門課的主題第一次出現：有些結論只靠「無套利」——市場上不存在零成本、零風險卻穩賺的機會，下一章會正式介紹——就能推出來；只有剩下推不出來的部分，才需要假設股價怎麼變動的模型。像這裡的二元買權價格，只用到其他買權的報價，完全不需要模型。）

§6.9.iii. Bear Spreads

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The same directional bet can be built out of puts, and reversed.

- Writing an X_H put and buying an X_L put with identical expiration date creates the **bull put spread**. This one is entered for a net *credit*, since $P_H > P_L$, but its profit profile has the same shape as Figure 6.7.
- A **bear spread** amounts to selling a bull spread. It profits from declining stock prices.

Remark 6.9.9.
- **多頭賣權價差**（bull put spread）：賣出高履約價 X_H 的賣權、買進低履約價 X_L 的賣權。因為 $P_H > P_L$，建倉時是先收錢，但獲利形狀與多頭買權價差相同，一樣押股價上漲。
- **空頭價差**（bear spread）：把多頭價差反過來做（賣出一個多頭價差），股價下跌時獲利。

Note 6.9.10. A bull put spread collects cash up front and, most of the time, expires worthless — which is precisely what makes it dangerous. Its maximum loss, $X_H - X_L$ less the credit, is many times the credit received, so a single move can wipe out a long string of profitable months.[7]

Remark 6.9.11. 回顧 Section 6.9.i 對多頭賣權價差的記帳：期初收 $P_H - P_L$；股價只要到期時在 X_H 以上，兩個賣權都作廢，這筆錢全部落袋（最大利潤）；跌到 X_L 以下則賠最多，$(X_H - X_L) - (P_H - P_L)$。

Figure 6.8 的兩個履約價夾著現價（$90 < 95 < 100$），賺賠上限差不多（4.94 對 5.06），還看不出危險。但實務上賣這種價差的人，通常把兩個履約價都放在現價下方很遠，好讓「股價不要大跌」就能賺。例如同樣 $S = 95$、一個月到期，改用 $X_L = 80$、$X_H = 85$：

[7] See https://www.businesstoday.com.tw/article/category/80392/post/201803070 for a sad example in Taiwan on February 6, 2018.

- 兩個賣權都很價外、很便宜，只收到 $P_H - P_L \approx 0.27$；
- 股價一個月內跌破 85 的可能性不高，大約九成的情況兩個賣權都作廢、穩穩賺 0.27——這就是 Note 6.9.10 說的「多數時候到期作廢」；
- 但一旦股價跌破 80，要賠 $(85-80) - 0.27 \approx 4.73$，約是每次收入的 17 倍——這就是「最大虧損是收入的好幾倍」。

所以這個策略的損益是「經常小賺、偶爾大賠」：連續十幾個月每月賺 0.27，一次大跌就賠 4.73，把之前的獲利一次吐光。看起來穩定的月收入，其實是在承擔一個機率小但代價很大的風險，這正是 Note 6.9.10 說它危險的原因；註腳中 2018 年 2 月 6 日台股的例子，就是市場突然大跌時這類部位集體爆虧。

§6.9.iv. Butterfly Spreads

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Three calls or three puts with different strike prices and the same expiration date create a **butterfly spread**. In the call version the spread is

$$\text{long one } X_L \text{ call}, \qquad \text{long one } X_H \text{ call}, \qquad \text{short two } X_M \text{ calls}.$$

Equivalently — and this is the useful way to see it — a butterfly is

$$\underbrace{\big[\text{long } X_L \text{ call} + \text{short } X_M \text{ call}\big]}_{\text{bull call spread } (X_L, X_M)} + \underbrace{\big[\text{short } X_M \text{ call} + \text{long } X_H \text{ call}\big]}_{\text{short a bull call spread } (X_M, X_H)},$$

that is, long a bull call spread with strikes X_L and X_M and short a bull call spread with strikes X_M and X_H. The first spread's ramp goes up over $[X_L, X_M]$ and the second's comes down over $[X_M, X_H]$, so the payoffs stack into a tent.

A butterfly spread has a positive payoff at expiration only if the asset price falls between X_L and X_H. It is therefore a bet on the stock *not moving*, as opposed to the directional bets above.

Remark 6.9.12. **蝶式價差**（butterfly spread）：用三個履約價 $X_L < X_M < X_H$ 的同類選擇權組成；買權版是買一個 X_L、買一個 X_H、賣兩個 X_M。到期報酬像一個帳篷，只有股價落在 X_L 與 X_H 之間才為正、在 X_M 最高，所以它押的不是方向，而是「股價不太會動」。

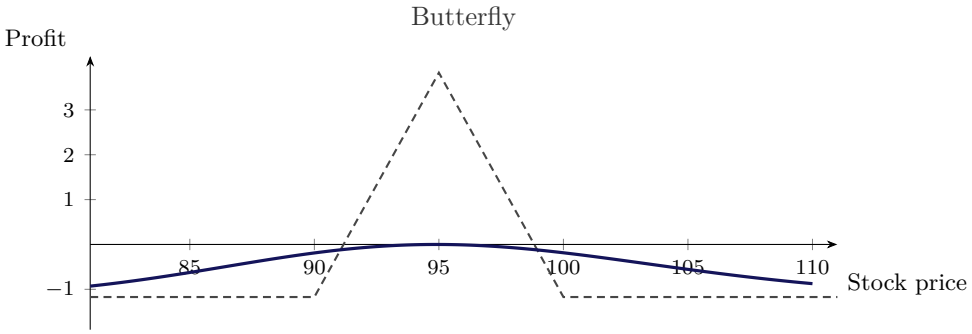


Figure 6.10.: Profit of a butterfly spread with $X_L = 90$, $X_M = 95$, $X_H = 100$, entered when $S = 95$ one month before expiration at a cost of 1.17. Dashed: the tent-shaped profit at expiration, positive only between X_L and X_H. Solid: profit at entry, which is nearly flat — with a month of time value still in the options, the market has not yet decided the bet.

Remark 6.9.13. 用和多頭買權價差同樣的方式記帳（買權版：買一個 X_L、買一個 X_H、賣兩個 X_M，X_M 在正中間）。

- 期初成本是 $C_L + C_H - 2C_M$。它一定是正的：這個部位到期時的 payoff 從不為負、有時為正，不可能免費拿到。
- 到期 payoff 分四段：

$S_T \le X_L$	$X_L < S_T \le X_M$	$X_M < S_T < X_H$	$S_T \ge X_H$
0	$S_T - X_L$	$X_H - S_T$	0

  中間兩段就是兩個多頭價差的斜坡一上一下：X_L 買權先開始賺；過了 X_M，賣出的兩個 X_M 買權開始賠，而且賠得比 X_L 買權賺得快，payoff 往下掉；到 X_H 時 X_H 買權接手，四者剛好抵銷成 0。
- 最大 payoff 是 $X_M - X_L$，發生在 $S_T = X_M$（帳篷頂點）。
- 最大利潤 $= (X_M - X_L) - (C_L + C_H - 2C_M)$，同樣發生在 $S_T = X_M$。
- 最大虧損 = 期初成本 $C_L + C_H - 2C_M$，發生在 $S_T \le X_L$ 或 $S_T \ge X_H$（股價動太多，帳篷外面）。
- 兩個損益兩平點：X_L + 成本 與 X_H − 成本。

對照 Figure 6.10：$X_L = 90$、$X_M = 95$、$X_H = 100$，成本 1.17，所以最大利潤 $5 - 1.17 = 3.83$（股價剛好停在 95）、最大虧損 1.17（股價跑出 $[90, 100]$）、股價在 91.17 到 98.83 之間才賺錢。賠有上限、賺也有上限，押的是「股價到期時停在 X_M 附近」。

§6.9.v. State Contingent Claims and State Prices

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The binary call came from squeezing a bull spread; squeezing a butterfly gives something sharper still.

想做什麼。我們想要一個「只有到期股價剛好是某個值才付錢」的證券；它的價格就代表市場對「股價落在那裡」這件事願意付多少。蝶式價差已經很接近了：它只在 X_L 到 X_H 之間付錢、在 X_M 付最多。把三個履約價往 X_M 擠，付錢的範圍就愈來愈集中在 X_M 這一點。問題是：擠的時候要買幾份？

帳篷的大小。取 $X_M = (X_H + X_L)/2$ 在正中間，記 $h = X_M - X_L = X_H - X_M$。一份蝶式價差的到期 payoff 畫在 S_T 橫軸上是一個三角形（Figure 6.10 虛線的形狀）：底從 X_L 到 X_H，寬 $2h$；頂點在 X_M，高 h；所以三角形面積是 $\frac{1}{2} \cdot 2h \cdot h = h^2$。買 k 份，高度和面積都乘上 k。

為什麼要固定面積，而不是高度。

- 固定高度 1（像二元買權那樣買 h^{-1} 份）行不通：$h \to 0$ 時帳篷縮成一根寬度 0 的細線，只有股價剛好等於 X_M 才拿得到錢。股價可以是任何實數，剛好停在某個精確數字的機率是 0，所以這東西的價格趨近 0，什麼也得不到。
- 固定面積 1 才行：帳篷變窄的同時變高，兩者剛好抵銷。帳篷很窄時，平均能拿到的錢 ≈（股價落進帳篷的機率）×（帳篷的平均高度）≈（X_M 附近每單位股價的機率）×（面積）。面積固定為 1，拿到的錢就只反映「股價落在 X_M 附近的機率有多集中」，極限是一個有限、不為零的數。

所以買 $h^{-2} = (X_M - X_L)^{-2}$ 份：面積是 $h^{-2} \cdot h^2 = 1$，不管寬度多少都一樣（高度則變成 $1/h$）。$h \to 0$ 時帳篷無限高、無限窄，極限就近似一個「只在 $S = X_M$ 這個狀態付 \$1」的證券——這裡的 \$1 精確地說是「面積 1」，也就是按每單位股價的機率來計的 \$1。[8]

[8] See Exercise 7.4.5 of the textbook.

Definition 6.9.14. A security that pays \$1 in one state of the world and nothing in any other is a **state contingent claim**, or an **Arrow security** in honour of Kenneth Arrow (1921–2017).[9] Its price is called a **state price**.[10]

Remark 6.9.15.
- **狀態或有求償權**（state contingent claim），又稱 **Arrow 證券**（Arrow security）：只在某一個特定狀態（例如 $S_T = X$）付 \$1、其他狀態都付 0 的證券。它是最基本的積木，任何報酬都能拆成它們的組合。
- **狀態價格**（state price）：這種證券今天的價格。由蝶式價差取極限可知，它等於 $\partial^2 C/\partial X^2$。

Its price follows from the same difference-quotient argument that produced (6.4), applied twice — for the butterfly is precisely a second difference of call prices in the strike. Writing $h = X_M - X_L$ and $X = X_M$, the cost of h^{-2} butterflies is

$$\frac{C(X-h) - 2\,C(X) + C(X+h)}{h^2} \xrightarrow[h\to 0]{} \frac{\partial^2 C}{\partial X^2}, \tag{6.5}$$

so the state price at X is $\partial^2 C/\partial X^2$.[11] Like (6.4), (6.5) is model independent.

State prices make pricing *linear*. Any payoff whatsoever can be chopped into state contingent claims — one for each terminal stock price — and priced by adding up the pieces, so pricing becomes integration against (6.5). Every valuation formula in the remainder of this course is an instance of that one idea.

§6.9.vi. The State Price Density

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(6.5) is a *price*, and a price is a present value. Undo the discounting — take the future value of $\partial^2 C/\partial X^2$ over the life of the option — and what is left is no longer a price at all.

Theorem 6.9.16 (Breeden & Litzenberger, 1978)
The future value of $\partial^2 C/\partial X^2$ is the probability density of the stock price $S_T = X$ at the option's maturity.

The reasoning is short once state prices are in hand. A claim that pays \$1 when S_T lands in a narrow band around X is worth the discounted value of what it is expected to pay, and what it is expected to pay is the probability of landing there. Hence

$$\underbrace{\frac{\partial^2 C}{\partial X^2}}_{\text{state price}} = e^{-r\tau}\,\underbrace{f(X)}_{\text{density}}, \qquad \text{i.e.} \qquad f(X) = e^{r\tau}\,\frac{\partial^2 C}{\partial X^2},$$

where τ is the time to maturity. Since (6.5) needed no model, neither does this: **the formula is model independent**, and a full set of quoted call prices determines the whole density.

Remark 6.9.17. 這段是直觀推理，不是嚴格證明：
(i) 由 Section 6.9.v，面積為 1、以 X 為中心的窄帳篷，今天價格是 $\partial^2 C/\partial X^2$。
(ii) 它到期平均拿到 ≈（X 附近的機率密度 $f(X)$）×（面積 1）$= f(X)$。

[9] Co-winner of the 1972 Nobel Prize in Economic Sciences.
[10] As a function of the state it is the *state-price density* (Aït-Sahalia & Lo, 1998).
[11] One can also use the put; see Exercise 9.3.6 of the textbook.

(iii) 價格 = 平均拿到的錢折現 τ 時間，即 $e^{-r\tau}f(X)$；兩邊乘 $e^{r\tau}$（換成到期時的價值，即 "future value"）就得到 $f(X)$。
注意 (iii) 的「價格 = 期望值折現」一般不成立，所以這個 f 不是真實的機率分布，見下面的 note。

Note 6.9.18. The f recovered this way is not the density under which the stock "really" evolves. It is the density the market *prices with* — the **risk-neutral** density, which folds the market's risk preferences into the probabilities. The distinction is made precise in Section 8.6, where the risk-neutral probability is constructed explicitly; for now the point is only that a density is there and that it is observable.

Remark 6.9.19. **風險中立**（risk-neutral）：從報價反推出的密度 f 不是股價「真實」的機率分布，而是市場拿來定價的分布——投資人的風險偏好已經被揉進機率裡。在這個分布下，所有資產的期望報酬都等於無風險利率，因此用無風險利率折現期望報酬就得到價格。

The practical reading of Theorem 6.9.16 closes the loop with Figure 6.10. A butterfly spread centred at X is, up to scale, a bet on $S_T \approx X$, and Theorem 6.9.16 says what the market charges for that bet. So:

buy a butterfly spread if you believe the probability of $S_T \approx X$ is higher than the probability the market is quoting.

An options market is not merely a collection of bets — it is a *published forecast.* Every strike quoted is one more point of the distribution, and Theorem 6.9.16 turns the quotes into the density with nothing but arithmetic. Trading an option is therefore trading a disagreement with that forecast, and the butterfly is the instrument for disagreeing about one particular outcome.

§6.10. Covered Positions: Combinations

Slides p. 215

Spreads bet on *direction* — where the stock goes. Combinations bet on *magnitude* — how far it goes, in either direction. Getting that distinction requires mixing the two option types, which is exactly what the definition permits.

Definition 6.10.1. A **combination** consists of options of different types on the same underlying asset. These options must be either all bought or all written.

Remark 6.10.2. **組合**（combination）：同一標的、但同時包含買權與賣權（不同類型）的部位，而且必須全部買進或全部賣出。價差押的是方向，組合押的是股價變動的幅度。

Note 6.10.3. The "all bought or all written" clause is what separates a combination from the hedges of Section 6.8 and the spreads of Section 6.9. Buying a call and writing a put is not a combination; by (6.1) and (6.2) its payoff is $S_T - X$, a straight line with no kink at all — the position is a synthetic forward, not an option strategy. Keeping all the legs on the same side is what preserves the V-shape.

§6.10.i. Straddles

Slides pp. 215–216

Definition 6.10.4. A **straddle** is a long call and a long put with the same strike price and expiration date.

Remark 6.10.5. **跨式**（straddle）：同時買進履約價、到期日都相同的一個買權與一個賣權。到期報酬 $|S_T - X|$ 呈 V 字形：不論漲跌，只要動得夠多就賺，最多賠掉兩份權利金。

By (6.1) and (6.2) the payoff is

$$\max(0, S_T - X) + \max(0, X - S_T) \;=\; |S_T - X|, \tag{6.6}$$

a V with its vertex at the strike. The holder is paid for movement and is indifferent to its direction; the loss is capped at the two premiums $C + P$, incurred only if the stock finishes exactly at X.

Since it profits from high volatility, a person who buys a straddle is said to be **long volatility**. Selling a straddle benefits from low volatility — the writer collects both premiums and keeps them if the stock stays put. A straddle is delta-neutral when $X = Se^{(r+\sigma^2/2)\tau}$, by (9.4).

Remark 6.10.6. **做多波動率**（long volatility）：部位價值隨波動度上升而增加。買進跨式賺的是「股價大幅變動」而不在乎方向，所以是做多波動率；反過來賣出跨式就是做空波動率，希望股價不動、把兩份權利金收進口袋。

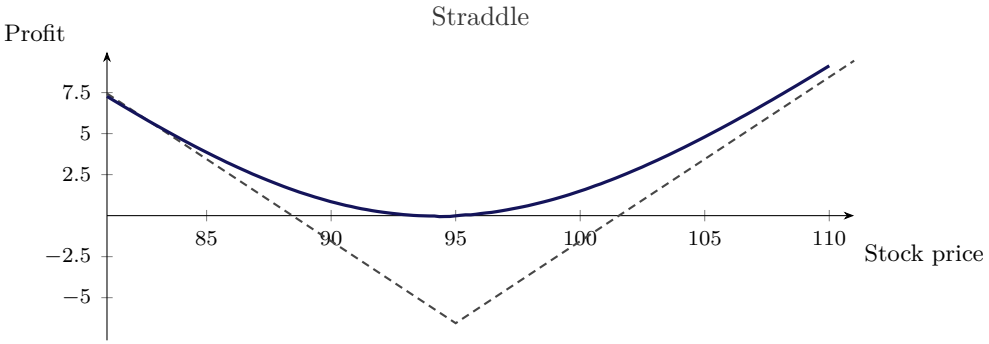


Figure 6.11.: Profit of a straddle with $X = 95$, bought when $S = 95$ one month before expiration for $C + P = 6.56$. Dashed: the V-shaped profit at expiration, (6.6) less the two premiums, with break-evens at $X \pm (C+P)$. Solid: profit at entry. The solid curve rises above the dashed one at both ends and sits above its vertex — a straddle held rather than exercised loses only time value, which is why the position is a bet on volatility and not on price.

§6.10.ii. Strangles

Slides pp. 217–218

Definition 6.10.7. A **strangle** is identical to a straddle except that the call's strike price is higher than the put's.

Remark 6.10.8. **勒式**（strangle）：和跨式一樣同時買進買權與賣權，但買權的履約價 X_H 高於賣權的履約價 X_L。兩個都是價外，所以比跨式便宜；代價是 $[X_L, X_H]$ 之間報酬為 0，股價要動得更遠才開始賺。

Writing X_L for the put's strike and X_H for the call's, the payoff is

$$\max(0, X_L - S_T) + \max(0, S_T - X_H), \tag{6.7}$$

which is zero on the whole interval $[X_L, X_H]$ and rises with slope 1 outside it. The V of the straddle has been pulled apart into a flat-bottomed bucket.

Both options are now out of the money, so the strangle is cheaper than the straddle — but the stock has to move further before it pays anything. It is the same bet on volatility, bought at a discount and with a wider dead zone.

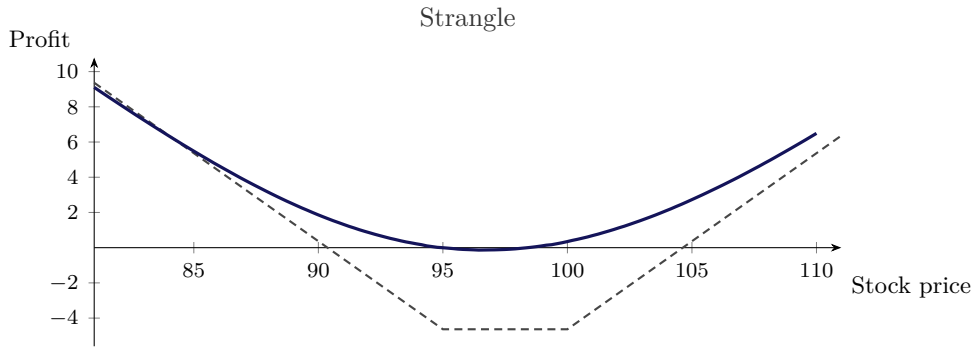


Figure 6.12.: Profit of a strangle with put strike $X_L = 95$ and call strike $X_H = 100$, bought when $S = 95$ one month before expiration for $P_L + C_H = 4.63$ — appreciably less than the straddle's 6.56. Dashed: profit at expiration, flat across $[X_L, X_H]$. Solid: profit at entry. Compared with Figure 6.11 the maximum loss is smaller but the break-evens are further apart.

Read Figure 6.10, Figure 6.11 and Figure 6.12 side by side and the whole section collapses into one picture.
The straddle and the strangle are *long* volatility: they pay when the stock leaves a neighbourhood of the strike. The butterfly is *short* volatility: it pays when the stock stays inside one. Their profit diagrams are each other turned upside down, and Theorem 6.9.16 says what both cost — the market's own probability of the stock staying put.
Direction is what spreads trade; magnitude is what combinations trade. Everything else in this section is bookkeeping around those two.

7. Arbitrage in Option Pricing

Slides pp. 219–220

All general laws are
attended with inconveniences,
when applied to particular cases.
— David Hume (1711–1776)

The problem with QE is
it works in practice,
but it doesn't work in theory.
— Ben Bernanke[1] (2014)

Chapter 6 described options without pricing any of them, and twice stopped just short of a price: the binary call cost $-\partial C/\partial X$ and the state contingent claim cost $\partial^2 C/\partial X^2$, both *relative* to quoted call prices, both obtained without a model. This chapter is that argument taken seriously.

Its single tool is the observation that a portfolio costing nothing and never losing money cannot exist. From that one sentence we will get the present-value formula we have been assuming since Chapter 2, a relation between every call and every put with the same terms, sharp bounds on both, and a complete answer to when an American call should be exercised early. None of it requires knowing how the stock price moves — which is exactly why these results survive every model the later chapters will propose and discard.

§7.1. Arbitrage

Slides p. 221

The **no-arbitrage principle** says there is no free lunch. It supplies the argument for option pricing.

Definition 7.1.1. A **riskless arbitrage opportunity** is one that, without any initial investment, generates nonnegative returns under all circumstances and positive returns under some.

Remark 7.1.2.
- **無套利原則**（no-arbitrage principle）：天下沒有白吃的午餐——市場上不存在零成本、零風險卻穩賺的機會。這是本章所有定價論證的唯一工具。
- **無風險套利機會**（riskless arbitrage opportunity）：不需任何初始投入，所有情況下報酬 ≥ 0、且某些情況下 > 0 的交易。

In an efficient market such opportunities do not exist — at least not for long.[2]

Note 7.1.3. Read the definition carefully, because the two quantifiers do all the work. *Nonnegative under all circumstances* is what makes the profit riskless; *positive under some* is what makes it a profit. Drop the first and you have an ordinary speculation; drop the second and you have a portfolio that does nothing.

[1] Co-winner of the 2022 Nobel Prize in Economic Sciences.
[2] Heakal (2013): "In the real world of investments, however, there are obvious arguments against the EMH [efficient market hypothesis]. There are investors who have beaten the market—Warren Buffett."

Note also what is *not* claimed: not that arbitrage never happens, only that it is competed away fast enough that prices may be computed as if it did not. That is an assumption about markets, not a theorem, and it is the one assumption every result below rests on.

Remark 7.1.4. 效率市場（efficient market）指價格會很快反映所有資訊的市場：一出現套利機會，大家就會搶著做，價格立刻被推回去，機會跟著消失。所以這裡並不是說套利從不發生，而是說它消失得夠快，算價格時可以當作它不存在——這是對市場的假設，不是定理，而本章所有結果都建立在它上面。

§7.2. The Portfolio Dominance Principle

Slides pp. 222–223

The no-arbitrage principle is easier to apply in the following comparative form.

Proposition 7.2.1 (Portfolio dominance principle)

Consider two portfolios A and B. Suppose A's payoff is at least as good as B's under all circumstances, and better under some. Then A should be more valuable than B.

Proof. If not, buy A and short B. The initial cash flow is nonnegative, every future cash flow is nonnegative, and at least one is positive — a riskless arbitrage opportunity. ■

Remark 7.2.2. 這裡的 payoff（以及下面兩個 corollary 的 return）指部位建立之後拿到的所有現金流——只在 T 付一次就是 T 時的價值，分好幾期付就是每一期的錢——不扣期初成本，因為期初成本正是要比較的價格。

Two corollaries do almost all the work in this chapter.

Corollary 7.2.3

A portfolio yielding a zero return in every possible scenario must have a zero PV.[a]

[a] Lyft, Inc. (2019): "We have incurred net losses each year since our inception and we may not be able to achieve or maintain profitability in the future."

Proof. Short the portfolio if its PV is positive; buy it if its PV is negative. In both cases a free lunch is created, since the portfolio itself never pays or costs anything afterwards. ■

Corollary 7.2.4

Two portfolios that yield the same return at time T must have the same price before time T.[a]

[a] Aristotle: "those who are equal should have everything alike."

Proof. Their difference is a portfolio with zero return in every scenario; apply Corollary 7.2.3. ■

Corollary 7.2.4 is the engine of everything that follows, and it deserves its own sentence: identical cash flows must carry identical prices.
Each proof below therefore has the same three steps — build a portfolio, show its payoff matches something whose price is known, conclude the prices agree.
The ingenuity is never in the argument; it is in choosing the portfolio.

§7.3. The PV Formula Justified

Slides pp. 224–226

We can now pay a debt outstanding since Chapter 2. Equations (2.6) and (4.5) were asserted, with a promise that they would be derived rather than assumed. Here is the derivation, and it uses nothing but Section 7.1.

Theorem 7.3.1

For a certain cash flow $C_1, C_2, \ldots, C_n$,

$$P = \sum_{i=1}^{n} C_i \, d(i).$$

Proof. Write P for the right-hand side and P^* for the security's market price, and suppose first that $P^* < P$.

Short the n zero-coupon bonds that match the security's n cash flows[3] — that is, short C_i face value of the i-period zero for each i. Since the i-period zero sells for $d(i)$ per dollar of face value, the proceeds are exactly $\sum_i C_i d(i) = P$ dollars.

Then use P^* of those proceeds to buy the security (Figure 7.1). At every future date i the security pays C_i, which offsets exactly the obligation C_i owed on the zeros; the net future cash flow is zero at every date. A riskless profit of $P - P^*$ dollars has therefore been realised *now*.

If $P^* > P$, just reverse the trades: buy the zeros and short the security. Hence $P^* = P$. ∎

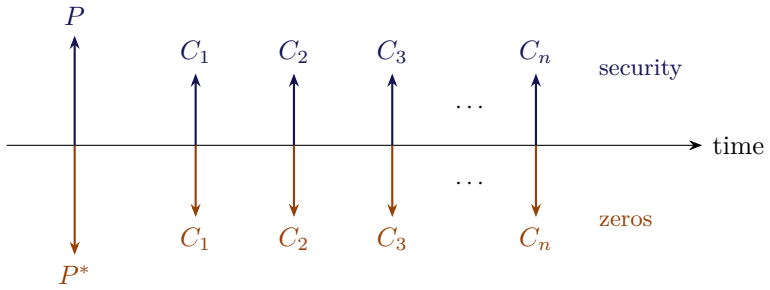


Figure 7.1.: The arbitrage of Theorem 7.3.1 when $P^* < P$. Above the axis are receipts, below it payments. Shorting the matching zeros raises P today against the obligations $C_1, \ldots, C_n$; buying the security for P^* supplies exactly those C_i. Every future column cancels, leaving $P - P^* > 0$ in hand today.

Note 7.3.2. The argument is worth one more look, because it is the template for the whole chapter. The security and the matching bundle of zeros have *identical* cash flows at every future date, so by Corollary 7.2.4 they must cost the same today — and the bundle's cost is $\sum_i C_i d(i)$ by construction. The short sale is only the mechanism that converts a price discrepancy into cash; it is Corollary 7.2.4 that does the reasoning.

Remark 7.3.3. （翻譯）這個論證值得再看一次，因為整章都照這個模板走。這張證券和那組對應的零息債券，在未來每個時點的現金流都完全相同，所以由 Corollary 7.2.4，兩者今天的價格必須一樣；而那組零息債券的價格照構造就是 $\sum_i C_i d(i)$。放空只是把「價格不一致」變成實際現金的操作手段；真正在做推理的是 Corollary 7.2.4。

[3] A key assumption: that a zero maturing at each date exists and can be shorted. Compare Section 4.13.ii, where a missing zero is manufactured instead.

Here is a second example, on the instruments of Chapter 6.

Theorem 7.3.4

A put or a call must have a nonnegative value.

Proof. Suppose otherwise, so the option has a negative price. Buy it: that is a *positive* cash flow now. By (6.1) and (6.2) the position ends up with a nonnegative amount at expiration, whatever the stock does. So an arbitrage profit is realised now. ■

§7.4. Relative Option Prices

Slides pp. 228–229

Everything from here to the end of the chapter is a *relative* pricing result: it ties the price of one instrument to the price of another, without ever saying what either is worth in absolute terms. That is the source of their power — **these relations hold regardless of the model for stock prices**.

Notation . We assume throughout, among other things, that

- there are no transaction costs[a] or margin requirements;
- borrowing and lending are available at the riskless interest rate;
- interest rates are nonnegative; and
- there are no arbitrage opportunities.

Let the current time be time zero, and write τ for the time to expiration. Then $\mathrm{PV}(x)$ stands for the PV of x dollars at expiration, so that by (4.5)

$$\mathrm{PV}(x) \;=\; x\,d(\tau). \tag{7.1}$$

[a] Schwab cut the fees of online trades of stocks and ETFs to zero on October 7, 2019.

Note 7.4.1. The nonnegative-interest-rate assumption is doing more than it looks: it is what gives $d(\tau) \le 1$, hence $\mathrm{PV}(X) \le X$. That single inequality is the reason a call is worth more than its intrinsic value and a put can be worth less — see Lemma 7.6.1 and Lemma 7.6.4.

Remark 7.4.2. 這一節只是在為後面幾節鋪設定：

- **相對定價**：後面的結果都不是算出「買權值多少」，而是「買權和賣權（或股票、債券）的價格之間必須滿足什麼關係」，例如下一節的 $C = P + S - \mathrm{PV}(X)$。因為只用到無套利、不用假設股價怎麼動，所以不管股價模型是什麼都成立。
- **假設**：沒有交易成本與保證金、可以用無風險利率自由借錢和存錢、利率不為負、沒有套利機會。
- **記號**：$\mathrm{PV}(x) = x\,d(\tau)$ 是「到期時的 x 元，換算成今天值多少」，$d(\tau)$ 就是第二章的折現因子。第三個假設「利率不為負」能推出來 $\mathrm{PV}(X) \le X$（$d(\tau) \le 1$，再乘上正的 X）：晚拿到的錢不會比現在拿到的值錢，這個不等式後面會一直用到。

§7.5. Put-Call Parity

Slides pp. 230–232

Theorem 7.5.1 (Put-call parity)

For European options on a non-dividend-paying stock, with common strike price X and

time to expiration τ,[a]

$$C \;=\; P + S - \mathrm{PV}(X). \tag{7.2}$$

[a] Castelli (1877).

Proof. Consider the portfolio consisting of

- one short European call,
- one long European put,
- one share of stock, and
- a loan of $\mathrm{PV}(X)$.

All options carry the same strike price X and time to expiration τ. The initial cash flow is therefore

$$C - P - S + \mathrm{PV}(X).$$

Now look at expiration, when the loan has grown to exactly X.

If $S_\tau \le X$, the put is worth $X - S_\tau$ and the call expires worthless, so the net future cash flow is

$$0 + (X - S_\tau) + S_\tau - X \;=\; 0.$$

If instead $S_\tau > X$, the call is worth $S_\tau - X$ — a liability, since we are short — and the put expires worthless, so the net future cash flow is again

$$-(S_\tau - X) + 0 + S_\tau - X \;=\; 0.$$

The net future cash flow is zero in either case. By Corollary 7.2.3 the initial investment must be nil as well, so $C - P - S + \mathrm{PV}(X) = 0$, which is (7.2). ∎

§7.5.i. Consequences

Slides p. 233

There is only one kind of European option.
The other can be replicated from it in combination with stock and riskless lending or borrowing, so any statement about calls is a statement about puts in disguise. Combinations such as this create synthetic securities — the same move that built the synthetic bonds of Section 4.13.ii.

Remark 7.5.2. **合成證券**（synthetic securities）：用其他證券組合出與某證券現金流完全相同的部位，也就是把它「複製」出來。由買賣權平價，任一種歐式選擇權都能用另一種選擇權加上股票與借貸合成。

例：用買權合成賣權。把 (7.2) 改寫成 $P = C - S + \mathrm{PV}(X)$，也就是「買一個買權、放空一股、存 $\mathrm{PV}(X)$ 元（到期變成 X）」。到期時：

- $S_\tau \ge X$：$(S_\tau - X) - S_\tau + X = 0$；
- $S_\tau < X$：$0 - S_\tau + X = X - S_\tau$。

和賣權的 payoff $\max(0, X - S_\tau)$ 一模一樣，所以這個組合就是一個「合成賣權」，價格也必須等於 P。而且 (7.2) 中的 τ 是剩餘時間，公式在到期前任何時點 t 都成立：$P_t = C_t - S_t + \mathrm{PV}_t(X)$；到期時 $\tau = 0$、$\mathrm{PV}(X) = X$，就退化成上面驗證的 payoff。正因為對任意時點都對，這個組合在每個時刻的價值都等於賣權，什麼時候平倉都一樣，才能真正當成賣權來用。所以只要市場上有買權，就不需要另外的賣權——這就是「只有一種歐式選擇權」的意思。

Rearranging (7.2) reads off the other synthetics:

- $S = C - P + \mathrm{PV}(X)$: a stock is equivalent to a portfolio containing a long call, a short put, and lending $\mathrm{PV}(X)$.

- $C - P = S - \mathrm{PV}(X)$: a long call and a short put amount to a long position in stock together with borrowing the PV of the strike price — that is, buying stock on margin.

Remark 7.5.3. **融資**（financing）：泛指「取得資金」，最常見的方式就是借錢。投資上的**融資買進**（buying on margin）是指買股票時自己只出一部分錢，其餘向券商借，也就是「借錢買股票」：股價漲時報酬被放大，跌時虧損也被放大，這就是槓桿（leverage）。

第二個式子正是這個意思：$S - \mathrm{PV}(X)$ 代表買一股、但其中 $\mathrm{PV}(X)$ 是借來的，自己只出 $S - \mathrm{PV}(X)$；到期時要還 X。所以「long call + short put」就等於用融資的方式買進股票。

Note 7.5.4. The second identity is the one promised in Section 6.10: buying a call and writing a put is not a combination but a synthetic forward, and (7.2) is why its payoff $S_\tau - X$ carries no kink and costs $S - \mathrm{PV}(X)$ today.

Remark 7.5.5. 這是在兌現 Section 6.10 那個 note 的伏筆：「買一個買權、賣一個賣權」為什麼不算組合。到期時股價高於 X，你會履約買權；低於 X，對方會對你履約賣權——不論哪種，你都是以 X 買進一股，payoff 恆為 $S_\tau - X$，是一條沒有折角的直線，已經沒有「可以不履約」的選擇性了。這種「約定到期時一定以 X 買進」的合約叫**遠期合約**（forward），所以這個部位是合成遠期；它今天的價格由第二個式子給出，是 $S - \mathrm{PV}(X)$。

§7.6. Bounds from Parity

Slides pp. 234–235

Parity immediately sharpens what Section 6.5 could only assert by picture.

> **Lemma 7.6.1**
>
> An American call, or a European call on a non-dividend-paying stock, is never worth less than its intrinsic value.

Proof. An American call cannot be worth less than its intrinsic value, since it may be exercised at once for exactly that amount.[4]

For European options, rearrange (7.2) as

$$C \;=\; (S - X) + \big(X - \mathrm{PV}(X)\big) + P \;\geq\; S - X,$$

because $X - \mathrm{PV}(X) \geq 0$ when interest rates are nonnegative and $P \geq 0$ by Theorem 7.3.4. Recalling $C \geq 0$, also from Theorem 7.3.4, it follows that

$$C \;\geq\; \max(S - X,\, 0),$$

the intrinsic value. ∎

Note 7.6.2. 以美式買權為例，看兩種情況為何一邊違反無套利、一邊不會（設 $S > X$）。

- $C < S - X$ **違反無套利**。今天買進買權（$-C$）、立刻履約（$-X$，拿到一股）、把股票賣掉（$+S$），淨現金 $S - X - C > 0$。三步都在今天完成、之後沒有任何現金流，是穩賺的套利，所以這種價格不可能存在。
- $C > S - X$ **不違反**。賣出買權今天收到 C，但部位沒有結清：持有人日後可在股價 S_t 很高時履約，你要付出 $S_t - X$，可能大於 C，所以不是套利。

差別在於：前者今天就能把部位完全結清，後者會留下未來的義務。多出來的 $C - (S - X)$ 正是時間價值，是賣方承擔這個義務的補償。

[4] See Lemma 8.3.1 of the textbook.

Such calls are said to have a nonnegative **time value**, defined as $C - \max(S - X, 0)$.

Remark 7.6.3. **時間價值**（time value）：這裡正式定義為 $C - \max(S - X, 0)$，即價格減去內含價值。上面的引理說明，美式買權及不配息股票的歐式買權，時間價值一定非負。

Puts are different, and the difference is visible in Figure 6.4.

> **Lemma 7.6.4**
> For European puts, $P \geq \max\big(\mathrm{PV}(X) - S,\, 0\big)$.

Proof. Rearrange (7.2) as $P = C - S + \mathrm{PV}(X)$ and use $C \geq 0$; combine with $P \geq 0$.[5] ■

Note 7.6.5. The bound is $\mathrm{PV}(X) - S$, *not* $X - S$, and since $\mathrm{PV}(X) \leq X$ the two differ by the interest on the strike. So a European put on a non-dividend-paying stock may be worth less than its intrinsic value $X - S$, and this is exactly the dip at the left edge of the right-hand panel of Figure 6.4.

直觀原因：股價很低（深度價內）時，賣權幾乎一定會被履約，等於「到期時交出一股、收到 X」。但歐式賣權只能等到期才拿到這 X，它今天只值 $\mathrm{PV}(X)$；交出的那一股今天值 S。所以整個部位大約值 $\mathrm{PV}(X) - S$，比「現在就履約」的 $X - S$ 少了 X 這段時間的利息。美式賣權可以現在就履約、立刻拿到 $X - S$，沒有這個損失——這也是第一個跡象：提前履約賣權可能是划算的。

§7.7. Early Exercise of American Calls

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Proposition 6.3.3 said an American option is worth at least as much as its European twin but left the gap uncomputed. For calls on a non-dividend-paying stock, parity closes it entirely: **European calls and American calls are identical when the underlying stock pays no dividends.**

> **Theorem 7.7.1** (Merton, 1973)
> An American call on a non-dividend-paying stock should not be exercised before expiration.

Proof. Write C_A and C_E for the American and the otherwise identical European call.

Step 1: a sharper bound for the European call. Rearrange (7.2) as $C_E = P + S - \mathrm{PV}(X)$; since $P \geq 0$ by Theorem 7.3.4, $C_E \geq S - \mathrm{PV}(X)$, and together with $C_E \geq 0$,

$$C_E \;\geq\; \max\big(S - \mathrm{PV}(X),\, 0\big).$$

This sharpens Lemma 7.6.1, whose proof threw away the nonnegative term $X - \mathrm{PV}(X)$.[6]

Step 2: pass to the American call. By Proposition 6.3.3, $C_A \geq C_E$.

Step 3: compare holding with exercising. Exercising now realises $S - X$. Interest rates are nonnegative, so $\mathrm{PV}(X) \leq X$, and chaining the steps gives

$$C_A \;\geq\; C_E \;\geq\; \max\big(S - \mathrm{PV}(X),\, 0\big) \;\geq\; S - X.$$

The left end is what the call is worth if held (or sold); the right end is what exercising yields. Holding is thus worth at least as much as exercising, at every moment before expiration. ■

[5] See Lemma 8.3.2 of the textbook.
[6] By Exercise 8.3.2 of the textbook.

Remark 7.7.2. The theorem does *not* mean American calls should be kept until maturity. What it does imply is that when early exercise is being considered, a better alternative is to *sell* the call — the market will pay at least the intrinsic value the holder was about to settle for, and generally more.

Early exercise may become optimal for American calls on a *dividend-paying* stock, however — and on a foreign currency (Section 10.3). Recall from Section 6.6 that options are assumed unprotected, and that the stock price declines as the stock goes ex-dividend: the holder who does not exercise watches that decline without being compensated for it.

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Surprisingly, even then an American call should be exercised only at a few dates.

Theorem 7.7.3 (Merton, 1973)

An American call will only be exercised at expiration or just before an ex-dividend date.[a]

[a] See Theorem 8.4.2 of the textbook.

In contrast, it might be optimal to exercise an American *put* even if the underlying stock does not pay dividends.

The asymmetry between calls and puts is now complete, and it all traces to the sign of $X - \mathrm{PV}(X)$.
A call holder pays the strike; waiting earns interest on it, so waiting is (weakly) optimal, and Theorem 7.7.3 reduces the American call to a European one priced at finitely many candidate dates. A put holder *receives* the strike; waiting forgoes that interest, so exercising early can pay.
This is why the American put has no closed-form price and the American call effectively does — and why so much of the numerical machinery in the remaining chapters exists for puts (Section 8.17).

Remark 7.7.4. 買權與賣權的差別全來自「履約價 X 是付出去還是收進來」：買權履約要付 X，晚點付可以多賺利息，所以等待不吃虧，美式買權（不配息時）等同歐式，頂多在除息前幾個時點要檢查；賣權履約是收 X，晚點收就少賺利息，所以提前履約可能划算。也因此美式買權的價格幾乎可以直接套歐式公式，美式賣權卻沒有公式，只能用後面章節的數值方法去算。

§7.8. A General Result

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One last theorem says how far the replication arguments of this chapter reach.[7]

Theorem 7.8.1 (Ross, 1976; Cox & Rubinstein, 1985)

Any piecewise linear payoff function can be replicated using a portfolio of calls and puts.

Remark 7.8.2. **設定**。設到期 payoff $g(S_T)$ 是一條折線 (Figure 7.2)：在 $S_T = 0$ 的值是 $g(0)$；折角在 $K_1 < K_2 < \cdots < K_m$；最左段斜率 s_0，過了 K_j 之後斜率變成 s_j。
從左到右一段一段拼。

[7] See Exercise 8.3.6 of the textbook.

(i) 最左段：它是直線 $g(0)+s_0S_T$，用「到期付 $g(0)$ 的零息債券 + 持有 s_0 股」就能做出來（負的就是借錢或放空）。

(ii) 過 K_1：上面那條直線會照斜率 s_0 繼續延伸（圖中虛線），但目標的斜率要變成 s_1，差了 s_1-s_0。加上 s_1-s_0 個履約價 K_1 的買權即可：買權 $\max(0,S_T-K_1)$ 在 K_1 以左是 0，不會動到已經對好的左段；在 K_1 以右斜率是 1，乘上 s_1-s_0 剛好把斜率補成 s_1。

(iii) 過 $K_2,\dots,K_m$：同理，在每個 K_j 加 s_j-s_{j-1} 個履約價 K_j 的買權（斜率變大就買、變小就賣）。

全部加起來就是

$$g(S_T) \;=\; \underbrace{g(0)}_{\text{債券}} \;+\; \underbrace{s_0\,S_T}_{\text{股票}} \;+\; \sum_{j=1}^{m} \underbrace{(s_j-s_{j-1})\max(0,\,S_T-K_j)}_{K_j\text{ 的買權}}.$$

例：跨式。$|S_T-X|$ 只在 X 折一次：$g(0)=X$，斜率由 -1 變 $+1$，差 $+2$，所以 $|S_T-X| = X-S_T+2\max(0,S_T-X)$：債券 X、放空一股、買兩個 X 買權。由 (7.2)，一個賣權等於「一個買權、放空一股、到期收 X 的債券」，所以這和「一個買權加一個賣權」是同一個部位——複製方式並不唯一。

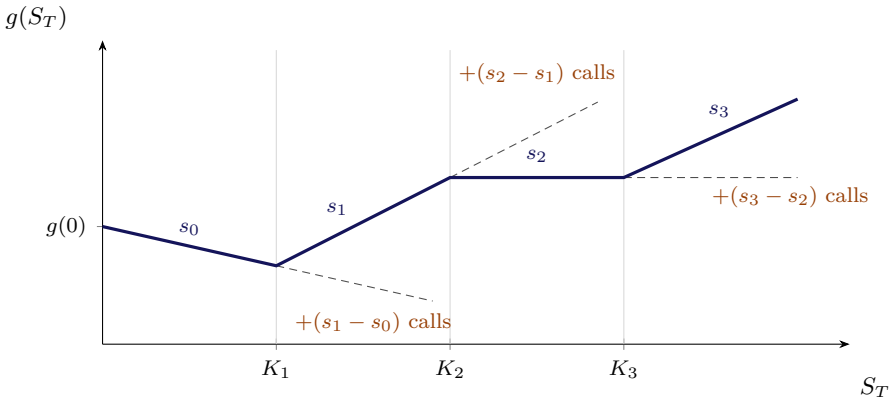


Figure 7.2.: A piecewise linear payoff with kinks K_1,K_2,K_3 and slopes $s_0,\dots,s_3$. The bond and the stock produce the first segment; at each kink the dashed line is where the previous segment would have gone, and the calls struck there (labelled in red) bend it onto the next slope.

Corollary 7.8.3

Any sufficiently well-behaved payoff function can be approximated by a portfolio of calls and puts.

Theorem 7.8.4 (Bakshi & Madan, 2000)

Any payoff function with bounded expectation can be replicated by a continuum of out-of-the-money European calls and puts.

Remark 7.8.5. *A continuum of options* 指不是買有限幾個選擇權，而是對連續一整段的每個履約價 K 都買一點點（數量 $w(K)\,dK$），把加總換成積分。平滑的 payoff 每一點都在微微彎曲，可看成「到處都是無限小的折角」；照 Theorem 7.8.1 的做法，履約價 K 的數量就是斜率

在那裡的變化率 $f''(K)$。取參考點 S_0，可寫成

$$f(S_T) = \underbrace{f(S_0) + f'(S_0)(S_T - S_0)}_{\text{債券與股票}} + \int_0^{S_0} f''(K)\,(K - S_T)^+\,dK + \int_{S_0}^{\infty} f''(K)\,(S_T - K)^+\,dK,$$

S_0 以下用賣權、以上用買權，兩邊都是價外選擇權，這就是定理中的 out-of-the-money。

Calls and puts are a *basis*.
Theorem 7.8.1 is visible in every picture of Chapter 6: each kink in a payoff diagram is one option struck at that kink, and the spreads and combinations were nothing but small linear combinations of the basis. Corollary 7.8.3 says a smooth payoff is the limit of such diagrams, and Theorem 7.8.4 makes the limit exact.
Put this next to Theorem 6.9.16 and the two halves of the story meet. Options span the payoffs; their prices reveal the density; and pricing an arbitrary claim is integrating one against the other. Everything after this chapter is the labour of computing that integral for a particular model of the stock price.

8. Option Pricing Models

Slides pp. 240–241

Black insisted that anything one could do
with a mouse could be done better
with macro redefinitions
of particular keys on the keyboard.
— Emanuel Derman (2004), *My Life as a Quant*

So we would bring in smart folks.
They didn't know anything about finance.[1]
— James Simons[2] (2015, May 13, 33:27)

Chapter 7 extracted everything no-arbitrage alone can give: parity, bounds, and the early-exercise rule for calls. It cannot give a number. This chapter supplies the missing ingredient — a model of how the stock price moves — and gets one.

§8.1. The Setting

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- The no-arbitrage principle is insufficient to pin down the exact option value.
- We need a model of the probabilistic behaviour of stock prices.
- 障礙在於折現率（discount rate）。有了 S_T 的模型（例如對數常態 lognormal，密度 f），買權的期望報酬（expected payoff）其實已經算得出來：

$$E\big[\max(0, S_T - X)\big] \;=\; \int_X^{\infty} y\, f(y)\, dy \;-\; X\, \Pr[S_T > X],$$

  第一項正是 (5.28) 取 $a = X$，這也是當初要推導那個部分期望值（partial expectation）的原因。但這是到期時的值，要變成今天的價格必須折現；而報酬有風險，投資人會要求額外的風險溢酬（risk premium），所以不能用一般的無風險利率（riskless rate），要用風險調整後的折現率（risk-adjusted discount rate）= 無風險利率 + 風險溢酬——偏偏模型本身沒告訴我們這個溢酬是多少，1973 年以前的定價嘗試都卡在這裡。

- The breakthrough came in 1973, when Black (1938–1995) and Scholes, with help from Merton, published their celebrated option pricing model,[3] known as the **Black–Scholes option pricing model**.

Remark 8.1.1. **Black–Scholes 選擇權定價模型**（Black–Scholes option pricing model）：1973 年由 Black 與 Scholes（Merton 協助）提出，第一次在給定股價模型下算出選擇權的確切價格。關鍵在於不需要風險調整後的折現率：選擇權可以用股票和債券複製，價格就是複製的成本。

[1] https://www.youtube.com/watch?v=QNznD9hMEh0
[2] James Harris Simons (1938–2024) was the founder of Renaissance Technologies. Its Medallion Fund had a 66.1% average gross annual return rate in 1988–2018.
[3] The results were obtained as early as June 1969. Merton and Scholes were winners of the 1997 Nobel Prize in Economic Sciences.

Note 8.1.2. The third bullet is the crux, and it is worth dwelling on before the machinery starts.

Discounting an uncertain payoff normally requires a rate that compensates for its risk, and that rate depends on investors' preferences — which nobody can observe. Worse, an option's risk changes continuously as the stock moves, so even a correct rate today would be wrong tomorrow. Every attempt at option pricing before 1973 foundered here.

The resolution, which Section 8.6 reaches, is that the question never has to be answered: the option can be *replicated* by stock and bonds, so its price is the cost of the replicating portfolio, and preferences drop out entirely.



Fischer Black (1938–1995)



Myron Scholes (1941–)



Robert C. Merton (1944–)

Figure 8.1.: The authors of the 1973 breakthrough. Black had died by 1997 and so could not share the Nobel Prize awarded to Scholes and Merton.

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Notation . The symbols C, P, X and S keep the meanings fixed in Section 6.5. Two are new:

- $\hat{r} > 0$: the continuously compounded riskless rate *per period*;
- $R \triangleq e^{\hat{r}}$: the **gross return** over one period.

A dollar lent today is thus worth R dollars one period later. We start from the discrete-time binomial model.

Remark 8.1.3. **總報酬** (gross return)：$R = e^{\hat{r}}$，今天存入 \$1 無風險資金，一期後變成 R 元。因為包含本金所以叫「總」報酬，淨報酬是 $R - 1$。

§8.2. The Binomial Option Pricing Model

Slides pp. 247–248

Definition 8.2.1. In the **binomial option pricing model** (BOPM), time is discrete and measured in periods. If the current stock price is S, one period later it is either

$$Su \quad \text{with probability } q, \qquad \text{or} \qquad Sd \quad \text{with probability } 1-q,$$

where $0 < q < 1$ and $d < u$.

Remark 8.2.2. **二項式選擇權定價模型** (binomial option pricing model, BOPM)：把時間切成離散的期，每一期股價只有兩種可能——乘上 u（上漲）或乘上 d（下跌）。模型雖簡單，卻已足以用複製的方法算出選擇權價格。

In fact $d \le R \le u$ must hold to rule out arbitrage.[4]

Intuition . 回憶 R 是總報酬（gross return）：今天借（或存）\$1，一期後要還（或拿回）R 元。

- **若 $R < d$**：今天借 S 元買一股，淨支出 0。一期後股票值 Su 或 Sd，要還 RS，淨得 $Su - RS$ 或 $Sd - RS$；因為 $u > d > R$，兩者都 > 0。不花錢、一定賺，就是套利（arbitrage）。直觀上，股票連「跌」的時候都比存銀行好，沒人會想持有債券。
- **若 $R > u$**：反過來做——放空一股拿到 S 元存起來。一期後拿回 RS，買回股票還掉要 Su 或 Sd，淨得 $RS - Su > 0$ 或 $RS - Sd > 0$，同樣是套利。
- **邊界 $R = d$ 或 $R = u$**：例如 $R = d$ 時，上面的交易在下跌時淨得 0、上漲時 > 0，仍符合「不虧且有機會賺」的套利定義，所以要用嚴格的 $d < R < u$（即註腳說的充分條件）。

所以 $d < R < u$ 不是為了方便而加的假設，而是確保這個模型本身不違反 Section 7.1 無套利原則（no-arbitrage principle）的條件：無風險利率必須落在股票的「最差」與「最好」報酬之間。

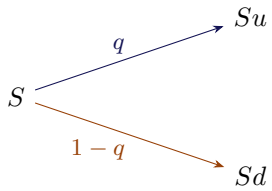


Figure 8.2.: One period of the binomial model. Only the two *multipliers* u and d matter; the current price S scales out.

Six pieces of information suffice to determine the option value on arbitrage considerations alone:

$$S, \quad u, \quad d, \quad X, \quad \hat{r}, \quad \text{and the number of periods to expiration.}$$

Conspicuously absent is q — the probability that the stock goes up. Section 8.4 explains why.

§8.3. A Call over a Single Period

Slides pp. 249–253

Take the expiration date to be only one period from now, and write C_u and C_d for the call price at time 1 after an up and a down move. Since time 1 *is* expiration, (6.1) gives them at once:

$$\begin{aligned} C_u &= \max(0,\, Su - X), \\ C_d &= \max(0,\, Sd - X). \end{aligned} \tag{8.1}$$

[4] See Exercise 9.2.1 of the textbook. The sufficient condition is the strict one, $d < R < u$ (Björk, 2009), which we shall assume.

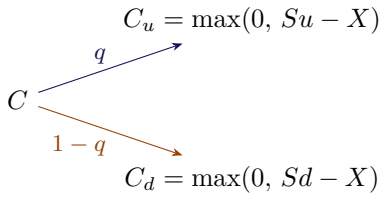


Figure 8.3.: The call written on Figure 8.2. The two terminal values are known; the problem is the value C at the root.

The trick is to build something with the same two terminal values out of instruments whose prices we already know.

Definition 8.3.1. Set up a portfolio of h shares of stock and B dollars in riskless bonds. It costs $hS + B$ today, and h is called the **hedge ratio** or **delta**.

Remark 8.3.2. **避險比率**（hedge ratio），又稱 **Delta**：複製選擇權所需持有的股數 h。由 (8.3)，$h = \frac{C_u - C_d}{Su - Sd}$，就是「選擇權價值變動 / 股價變動」，也就是選擇權對股價的敏感度。

The value of this portfolio at time one is

$$\begin{cases} hSu + RB, & \text{up move,} \\ hSd + RB, & \text{down move,} \end{cases}$$

so choosing h and B to replicate the payoff of the call means solving

$$\begin{aligned} hSu + RB &= C_u, \\ hSd + RB &= C_d. \end{aligned} \tag{8.2}$$

Two linear equations in two unknowns, with a nonsingular matrix because $u \neq d$:

$$h = \frac{C_u - C_d}{Su - Sd} \geq 0, \qquad B = \frac{uC_d - dC_u}{(u - d)\,R}. \tag{8.3}$$

Theorem 8.3.3

By the no-arbitrage principle, a European call expiring one period from now must cost the same as the equivalent portfolio:[a]

$$C = hS + B. \tag{8.4}$$

[a] Also called the **replicating portfolio**, since it replicates the option.

Proof. By construction the portfolio and the call have identical payoffs in both states at time 1, so Corollary 7.2.4 forces identical prices at time 0. ■

Remark 8.3.4. **複製投資組合**（replicating portfolio）：在每一個狀態下報酬都與選擇權完全相同的「股票 + 債券」組合。既然現金流一模一樣，由無套利，選擇權價格就必須等於這個組合今天的成本 $hS + B$。

Note 8.3.5. Read off the signs. $h \geq 0$: the replicating portfolio is *long* stock, as one would expect of a call. And since $uC_d - dC_u < 0$ we have $B < 0$, so the equivalent portfolio is a **levered long position in stocks** — buy h shares, borrow $|B|$ to help pay for them.

This also explains an option's reputation for leverage: it packages exactly that trade, and its percentage swings are correspondingly larger than the stock's.

Remark 8.3.6. 照組合的定義，「持有 B 元的無風險債券」在 $B<0$ 時代表賣出 $|B|$ 元的無風險債券，也就是今天借入 $|B|$ 元、一期後連本帶利還 $R|B|$ 元。所以這是**槓桿多頭部位**（levered long position），也就是 Section 7.5 說的融資買進：買股票（long），但錢有一部分是借來的。自己只出 $C=hS-|B|$，卻承擔 h 股的漲跌，所以同樣的股價變動，換算成自己出的錢，百分比漲跌會被放大——這就是**槓桿**（leverage），也是買權漲跌幅通常比股票大的原因。

§8.3.i. American Calls and Puts over One Period

Slides pp. 254–255

As in Section 8.3, every option in this subsection expires *one period from now*; $hS+B$ is the value of the replicating portfolio, i.e. what the option is worth if it is held to expiration.

For an American call we have to consider immediate exercise, so

$$C \;=\; \max\bigl(hS+B,\; S-X\bigr).$$

When $hS+B\ge S-X$ the call should not be exercised immediately; when $hS+B<S-X$ it should. For non-dividend-paying stocks the first case always holds, by Theorem 7.7.1, so once again $C=hS+B$.

Puts are priced the same way. Replicating the put's terminal values P_u and P_d with h shares and B dollars of bonds means solving, as in (8.2),

$$\begin{aligned} hSu+RB &= P_u,\\ hSd+RB &= P_d. \end{aligned}$$

Subtracting the second equation from the first eliminates B: $h(Su-Sd)=P_u-P_d$. Multiplying the first by d, the second by u and subtracting eliminates h: $(u-d)RB=uP_d-dP_u$. So the put's delta is

$$h \;=\; \frac{P_u-P_d}{Su-Sd} \;\le\; 0, \qquad P_u=\max(0,\,X-Su),\quad P_d=\max(0,\,X-Sd),$$

and

$$B \;=\; \frac{uP_d-dP_u}{(u-d)\,R} \;\ge\; 0,$$

since $P_d\ge P_u\ge 0$ and $u>d$ give $uP_d\ge uP_u\ge dP_u$. The European put is worth $hS+B$, while the American put is worth

$$\max\bigl(hS+B,\; X-S\bigr).$$

Note 8.3.7. The max cannot be dropped this time: early exercise *can* be optimal with American puts, exactly as Section 7.7 warned. Note also that $h\le 0$ and $B\ge 0$ — replicating a put means *shorting* $|h|$ shares and *lending* B dollars (buying the riskless bond, which pays RB one period later), the mirror image of the call's borrowing.

§8.4. Where Did the Risk Go?

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Look again at (8.3) and (8.4). Surprisingly, the option value is independent of q.[5] Hence it is also independent of the stock's expected value,

$$qSu+(1-q)\,Sd.$$

[5] More precisely, not *directly* dependent on q. Thanks to a lively class discussion on March 16, 2011.

The option value depends instead on the *sizes* of the price changes, u and d, which the investors must agree upon. Given those, the set of possible stock prices is the same whatever q is.

Two investors who violently disagree about whether the stock will rise must nevertheless quote the *same* option price, provided they agree on how far it can move.
The reason is (8.2): the replicating portfolio matches the call in *every* state, so how likely the states are never enters. This is what dissolves the risk-adjusted discount rate problem of Section 8.1 — there is no expectation to discount, only a portfolio to price.

§8.5. Pseudo Probability

Slides pp. 257–258

The formula $C = hS + B$ is correct but opaque. Substituting (8.3) and rearranging gives

$$hS + B \;=\; \frac{\dfrac{R-d}{u-d}\,C_u + \dfrac{u-R}{u-d}\,C_d}{R},$$

whose two coefficients are nonnegative and sum to 1. Naming the first of them,

$$p \;\triangleq\; \frac{R-d}{u-d}, \tag{8.5}$$

turns the expression into

$$hS + B \;=\; \frac{p\,C_u + (1-p)\,C_d}{R}. \tag{8.6}$$

Since $d < R < u$ we have $0 < p < 1$, so p may be interpreted as a probability.

Note 8.5.1. The interpretation is optional. Read without it, the numerator of (8.6) simply *interpolates*: the line through the points (Su, C_u) and (Sd, C_d), evaluated at SR, has height

$$\frac{R-d}{u-d}\,C_u + \frac{u-R}{u-d}\,C_d.$$

So (8.6) says the option is worth the linear interpolation of its payoffs at the stock's *forward* price, discounted back. Nothing probabilistic has happened yet — which is precisely why the next step is legitimate.

§8.6. The Risk-Neutral Probability

Slides p. 259

The quantity p earns its name from one calculation. Under p, the expected rate of return of the stock is exactly the riskless rate:

$$p\,Su + (1-p)\,Sd \;=\; RS. \tag{8.7}$$

Substituting (8.5) verifies it in one line.

Now, the expected rates of return of *all* securities must equal the riskless rate when investors are risk-neutral — a risk-neutral investor demands no compensation for bearing risk. Equation (8.7) says the stock behaves that way under p, and for this reason p is called the **risk-neutral probability**.

Remark 8.6.1. **風險中立機率**（risk-neutral probability）：(8.5) 的 $p = \frac{R-d}{u-d}$。(8.7) 說在這個機率下，股票的期望報酬恰好等於無風險利率，就像身處一個投資人不要求風險補償的世界。它不是股價真正上漲的機率 q，只是讓 (8.6) 可以讀成「用 p 算期望值，再用無風險利率 R 折現」的計算工具。

Read (8.6) with p as a probability: the value of an option is the expectation of its discounted future payoff *in a risk-neutral economy*. So the rate used for discounting is the riskless rate — and the obstacle raised in Section 8.1 is gone, not by being solved but by being made irrelevant.
The risk-neutral economy is a computational fiction. Real investors are not risk-neutral, and $p \neq q$ in general. What (8.6) asserts is only that the *answer* is the same as if they were, because the replicating portfolio does not care either way.
This is also the f of Theorem 6.9.16: the density the market prices with, now constructed rather than merely observed.

§8.7. The Multi-Period Model

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Consider a call with two periods remaining before expiration. Under the binomial model the stock can take on three possible prices at time two: Suu, Sud and Sdd. There are four *paths* — uu, ud, du, dd — but the tree **recombines**, since $ud = du$, so there are only three terminal prices.

At any node the next two stock prices depend only on the current price, not on the prices of earlier times; the model is **Markovian**.

Remark 8.7.1.
- **重合**（recombine）：兩期之後，「先漲再跌」的股價是 Sud，「先跌再漲」是 Sdu，兩者相等。所以這兩條路走到同一個點，樹在那裡合起來，畫出來就像 Figure 8.4：四條路徑，卻只有三個終點。期數愈多省愈多——n 期只有 $n+1$ 個終點，不用算 2^n 個。
- **馬可夫性**（Markovian）：股價下一步會變成什麼，只看它現在是多少，不管之前是怎麼走過來的。例如走到 Sud 的那個點，不論是先漲還是先跌到達，接下來的可能股價都一樣。

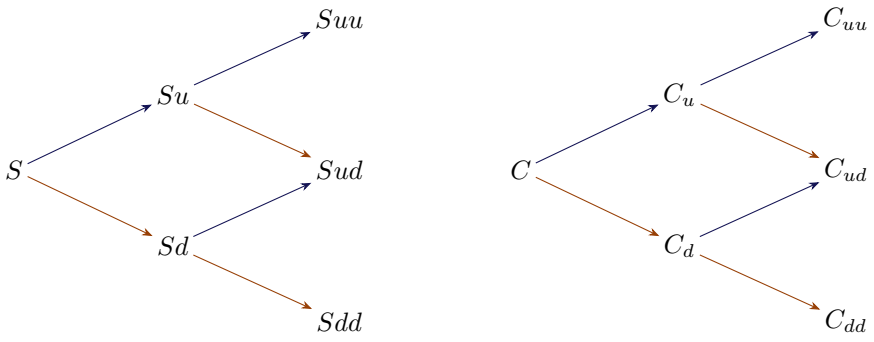


Figure 8.4.: Two periods. Four paths but three terminal states, because the tree recombines. The call tree on the right has exactly the same shape; only the numbers in the nodes differ.

At expiration the call's values are read off (6.1) as before:

$$C_{uu} = \max(0,\, Suu - X), \qquad C_{ud} = \max(0,\, Sud - X), \qquad C_{dd} = \max(0,\, Sdd - X).$$

The call values at time 1 are then obtained by applying (8.6) to each node:

$$C_u \;=\; \frac{p\,C_{uu} + (1-p)\,C_{ud}}{R}, \qquad C_d \;=\; \frac{p\,C_{ud} + (1-p)\,C_{dd}}{R}. \tag{8.8}$$

Each step is a one-period problem, so each node has its *own* replicating portfolio, given by (8.3) with that node's two successors in place of C_u, C_d:

- at the time-1 node Su (successors C_{uu}, C_{ud}): $h_u = \dfrac{C_{uu} - C_{ud}}{Suu - Sud}$, $B_u = \dfrac{uC_{ud} - dC_{uu}}{(u-d)R}$, and $C_u = h_u\,Su + B_u$;
- at the time-1 node Sd (successors C_{ud}, C_{dd}): $h_d = \dfrac{C_{ud} - C_{dd}}{Sud - Sdd}$, $B_d = \dfrac{uC_{dd} - dC_{ud}}{(u-d)R}$, and $C_d = h_d\,Sd + B_d$.

These are the portfolios held from time 1 to time 2.

We now reach the current period (time 0). Its portfolio, held from time 0 to time 1, is again (8.3) with successors C_u, C_d — which are now the time-1 *values* just computed in (8.8), not payoffs:

$$h = \frac{C_u - C_d}{Su - Sd}, \qquad B = \frac{uC_d - dC_u}{(u-d)R}, \qquad C \;=\; hS + B \;=\; \frac{p\,C_u + (1-p)\,C_d}{R}.$$

§8.7.i. Early Exercise, Again

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The tree lets us re-derive Theorem 7.7.1 by direct computation for an American call two periods from expiration. Throughout, h and B are the *time-0* portfolio just displayed, and C_u, C_d are the call's values at the two time-1 nodes.

At time 1. Each time-1 node is one period from expiration, so by Section 8.3.i the call is not exercised there even if it is American; its value is the held value, which is at least the exercise value:

$$C_u \;\geq\; Su - X, \qquad C_d \;\geq\; Sd - X.$$

At time 0. Holding is worth $hS+B$; substitute the two bounds, then use (8.7), $pu+(1-p)d = R$:

$$hS+B \;=\; \frac{p\,C_u + (1-p)\,C_d}{R} \;\geq\; \frac{p\,(Su-X) + (1-p)\,(Sd-X)}{R} \;=\; \frac{RS - X}{R} \;=\; S - \frac{X}{R} \;>\; S - X,$$

the last step because $R > 1$. The right end $S - X$ is what exercising at time 0 yields, so the call is not exercised at time 0 either, and

$$C \;=\; hS + B \;=\; \frac{p\,C_u + (1-p)\,C_d}{R}.$$

§8.8. Backward Induction

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The formula $C = \big(p\,C_u + (1-p)\,C_d\big)/R$ at the end of Section 8.7 calculates C from the two successor nodes C_u and C_d and none beyond; and the same computation happened at C_u and C_d themselves, in (8.8). This recursive procedure is called **backward induction**.[6]

[6] Ernst Zermelo (1871–1953).

Remark 8.8.1. 逆向歸納法（backward induction）：從到期日已知的報酬出發，用 (8.6) 一期一期往回推算每個節點的價值，推到根節點就是今天的價格。美式選擇權則在每個節點再與立即履約的價值取 max（如 Section 8.3.i）。

Unrolling it for two periods,

$$\begin{aligned} C &= \Big[p^2 C_{uu} + 2p(1-p)\,C_{ud} + (1-p)^2 C_{dd}\Big] \Big/ R^2 \\ &= \Big[p^2 \max\big(0, Su^2 - X\big) + 2p(1-p)\max\big(0, Sud - X\big) \\ &\qquad + (1-p)^2 \max\big(0, Sd^2 - X\big)\Big] \Big/ R^2. \end{aligned} \tag{8.9}$$

The coefficient 2 on the middle term of (8.9) is the number of paths reaching that state (ud and du), and it is the only trace the paths leave.

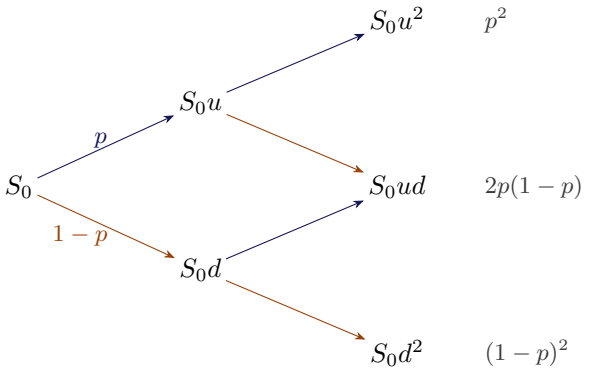


Figure 8.5.: The two-period tree with risk-neutral probabilities. The terminal weights in (8.9) are the binomial probabilities $\binom{2}{j}p^j(1-p)^{2-j}$.

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In the n-period case the same unrolling gives

$$C = \frac{\displaystyle\sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} \times \max\big(0,\, Su^j d^{n-j} - X\big)}{R^n}, \tag{8.10}$$

and similarly for the put,

$$P = \frac{\displaystyle\sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} \times \max\big(0,\, X - Su^j d^{n-j}\big)}{R^n}. \tag{8.11}$$

(8.10) 在說什麼。股價走 n 期，若其中 j 期上漲、$n-j$ 期下跌，到期價格就是 Su^jd^{n-j}，稱為 state j，此時買權的 payoff 是 $\max(0, Su^jd^{n-j}-X)$。在 risk-neutral probability p 下，走到 state j 的機率是 $\binom{n}{j}p^j(1-p)^{n-j}$（有 $\binom{n}{j}$ 條路徑，每條機率 $p^j(1-p)^{n-j}$）。所以分子就是到期 payoff 的期望值（expected payoff），再除以 R^n 折現 n 期。結論：不配息股票的買權價格 = 在 risk-neutral economy 裡，到期 payoff 的期望值折現回今天。
它不是繞過樹的捷徑。(8.10) 只是把 backward induction 在樹上一期一期往回算的結果，一次展開加總起來（(8.9) 就是 $n=2$ 的情形）。對 European option 兩者完全相同；但 American option 在每個節點都要和立即履約的 intrinsic value 取 max，這個 max 沒辦法塞進一條加總公式裡，所以只能在樹上一步步往回算。

§8.8.i. Reading the Weights as State Prices

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Define

$$p_j \triangleq \frac{\binom{n}{j}p^j(1-p)^{n-j}}{R^n}, \qquad j=0,1,\dots,n. \tag{8.12}$$

This is the *state price* of Section 6.9.vi, now for the state Su^jd^{n-j}: the price today of a claim paying \$1 there and nothing elsewhere. In general, then,

$$\text{option price} = \sum_j \big(p_j \times \text{payoff at state } j\big),$$

and (8.10) and (8.11) are two instances of it.

Note 8.8.2. The butterfly construction of Section 6.9.vi works here literally, and the discrete states simplify it.[7] Buy $(X_M - X_L)^{-1}$ units of the butterfly spread with

$$X_L = Su^{j-1}d^{n-j+1}, \qquad X_M = Su^jd^{n-j}, \qquad X_H = Su^{j+1}d^{n-j-1},$$

that is, struck at the state itself and its two neighbours. No stock price strictly between X_L and X_H is attainable except X_M, so the tent's *whole* payoff is the single value 1 at state j; the exponent -1 replaces the -2 of the continuous case in Section 6.9.v, where the tent had to be scaled by area instead. Its price is therefore p_j, and $R^np_j = \binom{n}{j}p^j(1-p)^{n-j}$ is the undiscounted state price — the risk-neutral probability.

Remark 8.8.3. (8.12) **的 p_j 是什麼**。把 (8.10) 的每一項拆成「權重 × payoff」：權重就是 p_j，也就是走到 state j（到期股價 Su^jd^{n-j}）的 risk-neutral probability 再除以 R^n 折現。

為什麼叫 state price。想像一張 Arrow security：到期時股價剛好是 state j 就付 \$1，其他 state 都付 0。照 (8.10) 同樣的算法（期望 payoff 再折現），它今天的價格是 $1 \times \binom{n}{j}p^j(1-p)^{n-j}/R^n = p_j$。所以 p_j 就是「在 state j 拿 \$1」今天的價格，和 Section 6.9.v 定義的 state price 是同一個東西。

所以任何選擇權都是一籃 Arrow securities。一個在 state j 付 payoff_j 的選擇權，等於買 payoff_j 張 state j 的 Arrow security，每個 j 都買。價格相加，就是

$$\text{option price} = \sum_j p_j \times \text{payoff}_j;$$

(8.10) 與 (8.11) 只是把買權、賣權的 payoff 代進去。

上面 note 在做什麼。它說明這張 Arrow security 真的可以用買權做出來：取以 state j 和左右相鄰兩個 state 為履約價的 butterfly spread。因為股價只會落在樹的節點上，在 X_L 與 X_H 之間只可能出現 X_M 這一個價格，所以帳篷只在 state j 有正的 payoff，其他 state 都是 0；再把份數調到使這個 payoff 剛好是 1。離散的情況下只要固定高度為 1，不必像 Section 6.9.v 的連續情況那樣固定面積——那裡是因為股價剛好落在某一點的機率為 0。最後 R^np_j 把價格換回到期時的價值，得到的正是 risk-neutral probability，對應 Theorem 6.9.16「state price 的 future value 是機率（密度）」。（嚴格說，乘法樹的三個履約價間距不相等，$X_M - X_L \neq X_H - X_M$，要讓帳篷兩端都歸零，兩側買權的份數必須不同：買 $(X_M - X_L)^{-1}$ 個 X_L、$(X_H - X_M)^{-1}$ 個 X_H，賣兩者之和個 X_M。）

[7] Bahra (1997).

§8.9. The Risk-Neutral Pricing Methodology

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Nothing in the derivation of (8.10) used the payoff $\max(0, S - X)$ beyond the fact that it was a function of the terminal stock price — (8.11) is the same sum with the put's payoff swapped in. So the conclusion generalises at once.

Every derivative can be priced as if the economy were risk-neutral. For a European-style derivative with terminal payoff function D, its value is

$$e^{-\hat{r}n}\, E^{\pi}[\, D\,], \tag{8.13}$$

where E^{π} means the expectation is taken under the risk-neutral probability.

The "equivalence" between arbitrage freedom in a model and the existence of a risk-neutral probability is called the **(first) fundamental theorem of asset pricing**.[8]

Remark 8.9.1. **資產定價第一基本定理**（first fundamental theorem of asset pricing）：一個模型沒有套利機會，若且唯若存在風險中立機率。換句話說，「無套利」與「可以用風險中立期望值定價」是同一件事的兩種說法。

一般情況的證明很難（要用泛函分析），這裡沒有證；但在二項式模型裡它其實已經證完了：

- 無套利 $\Leftrightarrow d < R < u$：Section 8.2 的 intuition 證過，$R \le d$ 或 $R \ge u$ 都能做出套利。
- 存在 risk-neutral probability $\Leftrightarrow d < R < u$：(8.7) 的唯一解是 (8.5) 的 $p = \frac{R-d}{u-d}$，它落在 $(0,1)$（真的是一個機率）恰好等價於 $d < R < u$。

兩邊等價於同一個條件，所以定理成立；多期模型則是在每個節點各用一次。



Philip H. Dybvig (1955–)



Stephen Ross (1944–2017)

Figure 8.6.: Dybvig, co-winner of the 2022 Nobel Prize in Economic Sciences, and Ross — whose 1976 replication theorem already appeared as Theorem 7.8.1.

Note 8.9.2. The theorem closes the circle this course opened with. Chapter 7 assumed no arbitrage and extracted relations between prices; Section 8.6 built a probability π out of a particular model. The fundamental theorem says these are two descriptions of the same thing: a model admits no arbitrage *exactly* when such a π exists. Pricing a derivative and checking a model for consistency are therefore one problem.

§8.10. Self-Financing

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One loose end remains. The delta h of (8.3) is computed node by node, and it changes as the

[8] Dybvig & Ross (1987).

stock moves, so the maintenance of an equivalent portfolio is *dynamic*: the replicating position must be rebalanced every period.

That sounds expensive, and it raises the obvious worry — if rebalancing requires fresh money, the option's price is not really $hS + B$. It does not.

Proposition 8.10.1

The replicating strategy is **self-financing**: there is neither injection nor withdrawal of funds throughout.[a]

[a] Except at the beginning, of course, when the option premium is paid before the replication starts.

Remark 8.10.2. **自我融資**（self-financing）：「融資」就是取得資金（見 Section 7.5 的融資買進），「自我」融資是指所需資金全由組合本身提供，中途不必再投入、也不會提出任何錢。

為什麼要擔心。在 Section 8.7，每個節點的複製組合都不一樣：time 0 持有 (h, B)，到了 time 1 的節點 Su 要改成 (h_u, B_u)，節點 Sd 要改成 (h_d, B_d)。每期都得調整部位（rebalance），若調整時要額外補錢，選擇權的價格就不只是一開始的 $hS + B$ 了。

為什麼不用補錢（以兩期為例，上漲的情況）：

(i) time 0 花 $C = hS + B$ 建立 (h, B)。

(ii) 到 time 1 股價變成 Su，這個舊組合值 $hSu + RB$；由 (8.2)，它正好等於 C_u。

(iii) 在節點 Su 要建的新組合 (h_u, B_u)，成本是 $h_uSu + B_u$；由 (8.4) 用在這個節點，它也等於 C_u。

(iv) 所以把舊組合賣掉的錢，剛好夠買新組合，不多也不少。下跌的情況同理，兩邊都是 C_d。

一路做到到期，組合的價值就是選擇權的 payoff；整個過程唯一付出的錢是一開始的 C。這就是下一段原文和 Proposition 8.10.1 的意思。

The reason is already in (8.2). At the end of the current period the portfolio is worth $hSu + RB$ or $hSd + RB$, which *is* C_u or C_d — and C_u and C_d are, by (8.4) applied one period later, exactly the cost of the next portfolio. The portfolio's value at the end of each period is therefore precisely the amount needed to set up the next one. Changes in value are due entirely to capital gains. Section 8.12.i runs the strategy through a three-period example.

Rebalancing does not depend on predicting future stock prices. The delta at each node is a function of where the stock *is*, not of where it is going; the strategy is told what to do by the tree, and it always has exactly enough money to do it. That is the whole content of "the option is worth its replicating portfolio", and it is what makes (8.13) a hedging recipe rather than merely a formula.

§8.11. The Binomial Option Pricing Formula

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Lecture 4
October 1, 2026

Section 8.10 showed that the tree is a hedging recipe. This lecture turns it into a *formula* and then into *algorithms*, takes its limit to reach the Black–Scholes formula, and finally asks how sensitive the answer is to each of its inputs.

The weights in (8.10) are binomial probabilities, so name them.

Definition 8.11.1. Denote the **binomial distribution** with parameters n and p by

$$b(j; n, p) \triangleq \binom{n}{j} p^j (1-p)^{n-j} = \frac{n!}{j!\,(n-j)!}\, p^j (1-p)^{n-j},$$

where $n! = 1 \times 2 \times \cdots \times n$ and, by convention, $0! = 1$.

If you flip a coin n times with p being the probability of getting heads, then $b(j;n,p)$ is the probability of getting j heads. In the tree, a "head" is an up move and p is the risk-neutral probability (8.5).

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The stock prices at time n are $Su^n, Su^{n-1}d, \ldots, Sd^n$, increasing in the number j of up moves. So the call finishes in the money exactly when j is large enough.

Definition 8.11.2. Let a be the minimum number of upward price moves for the call to finish in the money, i.e., the smallest nonnegative integer j such that $Su^j d^{n-j} \geq X$. Taking logarithms, $j\ln(u/d) \geq \ln\bigl(X/(Sd^n)\bigr)$, so

$$a \;=\; \max\left(0,\; \left\lceil \frac{\ln\bigl(X/(Sd^n)\bigr)}{\ln(u/d)} \right\rceil\right). \tag{8.14}$$

Every term of (8.10) with $j < a$ is zero, and for $j \geq a$ the max can be dropped. Hence

$$\begin{aligned} C \;&=\; \frac{\sum_{j=a}^{n} \binom{n}{j} p^j (1-p)^{n-j}\bigl(Su^j d^{n-j} - X\bigr)}{R^n} && (8.15)\\ &=\; S\sum_{j=a}^{n}\binom{n}{j}\frac{(pu)^j\bigl[(1-p)\,d\bigr]^{n-j}}{R^n} \;-\; \frac{X}{R^n}\sum_{j=a}^{n}\binom{n}{j}p^j(1-p)^{n-j} \\ &=\; S\sum_{j=a}^{n} b\bigl(j;n,pu/R\bigr) \;-\; Xe^{-\hat{r}n}\sum_{j=a}^{n} b(j;n,p). && (8.16) \end{aligned}$$

Note 8.11.3. 第二行只是把 S 提出來，並把 R^n 平均分給每一期：

$$p^j(1-p)^{n-j}\cdot\frac{Su^j d^{n-j}}{R^n} \;=\; S\Bigl(\frac{pu}{R}\Bigr)^j\Bigl(\frac{(1-p)\,d}{R}\Bigr)^{n-j}.$$

第三行要把這寫成 $b(j;n,pu/R)$，前提是「上漲」機率 pu/R 和「下跌」機率 $(1-p)d/R$ 加起來剛好是 1，而這正是 (8.7) 兩邊同除以 RS：

$$\frac{pu}{R} + \frac{(1-p)\,d}{R} \;=\; \frac{pu+(1-p)\,d}{R} \;=\; 1.$$

(另外 $X/R^n = Xe^{-\hat{r}n}$ 是因為 $R = e^{\hat{r}}$。) 所以 (8.16) 可以讀成

$$C \;=\; S\times \Pr_{pu/R}[\,j \geq a\,] \;-\; \mathrm{PV}(X)\times \Pr_{p}[\,j \geq a\,],$$

兩項都是「至少上漲 a 次」(即到期 in the money) 的機率，差別只在擲哪一枚硬幣：第二項用 risk-neutral probability p，第一項用 pu/R，它把上漲的權重乘上 $u/R > 1$，比 p 更偏向上漲。Theorem 8.14.19 也是完全相同的形狀。

Remark 8.11.4. **二項式選擇權定價公式** (binomial option pricing formula)：(8.16) 把 (8.10) 中為零的項 ($j < a$，到期價外) 丟掉後，拆成兩個「至少 a 次上漲」的二項尾端機率。第二項用的是風險中立機率 p，第一項用的是另一個機率 pu/R。

§8.12. Numerical Examples

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Example 8.12.1. A non-dividend-paying stock is selling for \$160, with $u = 1.5$ and $d = 0.5$. The interest rate is $r = 18.232\%$ per period, so $R = e^{0.18232} = 1.2$ and

$$p = \frac{R-d}{u-d} = \frac{1.2-0.5}{1.5-0.5} = 0.7.$$

Consider a European call on this stock with $X = 150$ and $n = 3$. The call value is \$85.069 by backward induction (Figure 8.7), or, as the PV of the expected payoff at expiration,

$$\frac{390 \times 0.343 + 30 \times 0.441 + 0 \times 0.189 + 0 \times 0.027}{(1.2)^3} = 85.069.$$

Here $a = 2$, and only the top two terminal states contribute.

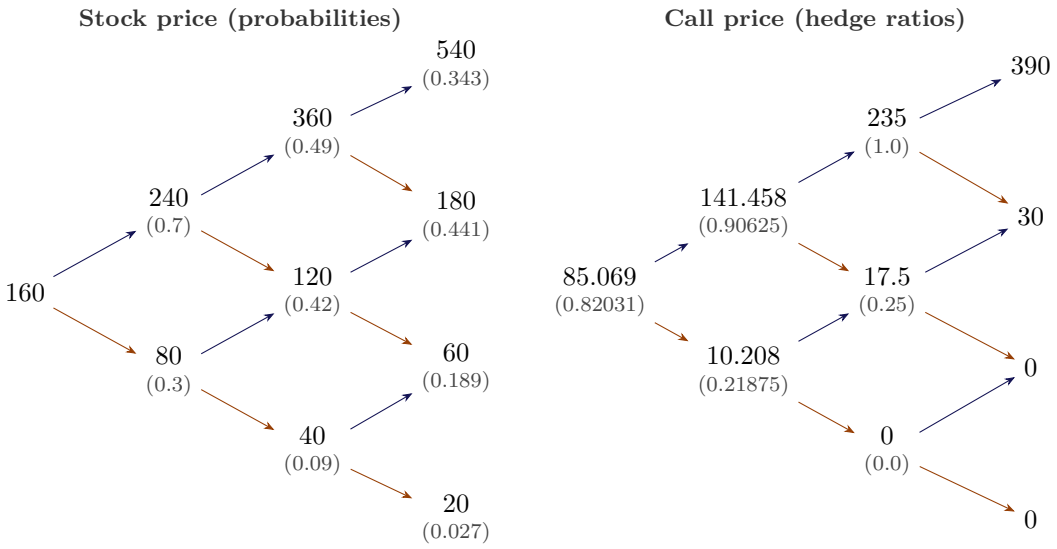


Figure 8.7.: Example 8.12.1. Left: the binomial process for the stock, with the risk-neutral probability of reaching each node. Right: the call values by backward induction, with the delta (8.3) at each node.

§8.12.i. Mispricing Leads to Arbitrage Profits

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Suppose the option is selling for \$90 instead of \$85.069. We now run Proposition 8.10.1 with real numbers.

Remark 8.12.2. 整個例子在做什麼。我們賣出一個 call（short call），同時建立並維持它的 replicating portfolio：h 股股票 + B 元債券。這裡 $B < 0$，代表借錢，所以文中的 debt 就是 $-B$。每一期只用到三條規則：

(a) **要持有幾股：**在目前節點用 (8.3) 算 delta $h = \dfrac{C_u - C_d}{Su - Sd}$，其中 Su, Sd 和 C_u, C_d 是 Figure 8.7 中下一期兩個節點的股價和 call 價值。

(b) **debt 會長利息：**借 D 元，一期後要還 $RD = 1.2D$（就是 (8.2) 中的 RB）。

(c) **rebalance：**delta 從 h_{old} 變成 h_{new}，就以目前股價買進（或賣出）$h_{\text{new}} - h_{\text{old}}$ 股；買股的錢全部用借的，賣股的錢全部拿去還債。這就是 Proposition 8.10.1。

驗算方式：每個節點上 portfolio 的價值 $hS - \text{debt} = hS + B$ 都應等於該節點的 call 價值 C ((8.4))。

Time 0. Sell the call for \$90. Invest \$85.069 in the replicating portfolio with 0.82031 shares of stock as required by the delta, borrowing $0.82031 \times 160 - 85.069 = 46.1806$ dollars to do so. The fund that remains,

$$90 - 85.069 \;=\; 4.931 \text{ dollars},$$

is the arbitrage profit, as we will see.

Remark 8.12.3. Time 0 的數字（節點 $S = 160$）。下一期的股價是 $Su = 240$、$Sd = 80$，call 價值是 $C_u = 141.458$、$C_d = 10.208$，所以由 (a)

$$h \;=\; \frac{141.458 - 10.208}{240 - 80} \;=\; \frac{131.25}{160} \;=\; 0.82031.$$

買這些股票要 $0.82031 \times 160 = 131.25$ 元，但 portfolio 只值 $C = 85.069$，差額 $131.25 - 85.069 = 46.1806$ 就是借款，即 $B = C - hS = -46.1806$。用 (8.3) 直接算也一樣：$B = \dfrac{uC_d - dC_u}{(u-d)R} = \dfrac{1.5 \times 10.208 - 0.5 \times 141.458}{1 \times 1.2} = -46.18$。

Time 1. Suppose the stock price moves to \$240. The new delta is 0.90625, so buy $0.90625 - 0.82031 = 0.08594$ more shares at the cost of $0.08594 \times 240 = 20.6256$ dollars, financed by borrowing. Debt now totals

$$20.6256 + 46.1806 \times 1.2 \;=\; 76.04232 \text{ dollars}.$$

The portfolio has a value of $0.90625 \times 240 - 76.04232 = 141.45768$, which matches the corresponding call value 141.458 in Figure 8.7. The trading strategy is self-financing: the new shares were paid for by new debt, and nothing else changed hands.

Remark 8.12.4. Time 1 的數字（節點 $S = 240$）。

- **new delta：**下一期股價是 360 和 120，call 價值是 235 和 17.5，由 (a) $h = \dfrac{235 - 17.5}{360 - 120} = \dfrac{217.5}{240} = 0.90625$。
- **舊 debt 長利息：**由 (b)，$46.1806 \times 1.2 = 55.4167$。
- **rebalance：**由 (c) 加買 $0.90625 - 0.82031 = 0.08594$ 股，花 $0.08594 \times 240 = 20.6256$ 元，全部借來，所以 debt $= 20.6256 + 55.4167 = 76.04232$。
- **驗算：**$hS - \text{debt} = 0.90625 \times 240 - 76.04232 \approx 141.458 = C_u$。

其實在 rebalance 之前，舊組合就已經值 $0.82031 \times 240 - 55.4167 \approx 141.458$：舊組合的價值剛好等於新組合的成本，所以不用補錢。

Time 2. Suppose the stock price plunges to \$120. The new delta is 0.25, so sell $0.90625 - 0.25 = 0.65625$ shares, generating an income of $0.65625 \times 120 = 78.75$ dollars. Use it to reduce the debt to

$$76.04232 \times 1.2 - 78.75 \;=\; 12.5 \text{ dollars}.$$

Remark 8.12.5. Time 2 的數字（節點 $S = 120$，從 240 下跌而來）。

- **new delta：**下一期股價是 180 和 60，call 價值是 30 和 0，由 (a) $h = \dfrac{30 - 0}{180 - 60} = 0.25$。
- **舊 debt 長利息：**由 (b)，$76.04232 \times 1.2 = 91.25$。
- **rebalance：**由 (c) 賣掉 $0.90625 - 0.25 = 0.65625$ 股，得 $0.65625 \times 120 = 78.75$ 元，全部拿去還債，debt 剩 $91.25 - 78.75 = 12.5$。
- **驗算：**$0.25 \times 120 - 12.5 = 17.5$，等於該節點的 call 價值。

Time 3, rising price. The stock price moves to \$180 and the call we wrote finishes in the money. Close out the short call by buying it back or by buying a share for delivery, at a loss of $180 - 150 = 30$ dollars. Financing this loss with borrowing brings the total debt to $12.5 \times 1.2 + 30 = 45$ dollars, which is repaid exactly by selling the 0.25 shares for $0.25 \times 180 = 45$ dollars.

Time 3, declining price. The stock price moves to \$60 and the call expires worthless. Selling the 0.25 shares brings in $0.25 \times 60 = 15$ dollars, which repays exactly the debt of $12.5 \times 1.2 = 15$ dollars.

Remark 8.12.6. Time 3 的數字。到期不再 rebalance，只要結清。由 (b)，debt 變成 $12.5 \times 1.2 = 15$，手上還有 0.25 股。

- **上漲到** 180：short call 要賠 $180-150 = 30$，總共欠 $15+30 = 45$；賣股得 $0.25 \times 180 = 45$，剛好還清。
- **下跌到** 60：call 到期 worthless，只欠 15；賣股得 $0.25 \times 60 = 15$，剛好還清。

這正是 (8.2)：$hS + RB$ 在兩種情況下都等於 call 的 payoff（$45 - 15 = 30$ 和 $15 - 15 = 0$），所以 portfolio 剛好付得起 short call 的損失。

Either way the books close at zero, so the \$4.931 pocketed at time 0 was never at risk. At no point did we need to know whether the stock would go up — each trade was dictated by the delta at the node we were standing on.

Note 8.12.7. 投影片只沿著一條路徑 $160 \to 240 \to 120 \to \{180, 60\}$ 走，但結論其實不依賴這條路徑。由 (8.2)，每個節點上 portfolio 的價值都等於 Figure 8.7 中的 call 價值；所以不論股價走哪條路徑，到期時的債務都剛好由「股票價值減去 call 的 payoff」償還。唯一的誤差來自 delta 在小數第五位的四捨五入，所以 time 1 才會出現 141.45768 對 141.45833 的差異。

§8.12.ii. Applications besides Arbitrage

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The same numbers serve two other purposes.[9]

- **Replicate** an option using stocks and bonds: set up a portfolio to replicate the call with \$85.069.
- **Hedge** the options we issued: use \$85.069 to set up a portfolio that replicates the call, so as to counterbalance its values exactly.[10]

No prediction is involved! Without the hedge, the writer may end up forking out \$390 in the worst case (the top terminal node of Figure 8.7).[11]

Remark 8.12.8. **複製**（replicate）與**避險**（hedge）是同一件事的兩面：買進複製組合 = 合成一個買權；賣出買權再買進複製組合 = 把賣出的買權完全對沖掉。兩者都只依照每個節點的 delta 調整部位，不需要預測股價。

Note 8.12.9. Hedge 的意思不是「財產數字完全不動」，而是讓整體部位的價值不受股價走勢影響，也就是把 risk 消掉；剩下的錢仍照 riskless rate 長利息。以 Section 8.12.i 為例：部位是 short call + replicating portfolio，由 (8.4)，每個節點上 portfolio 的價值都等於 call 的價值，所以淨值恆為 0；一開始拿到的 4.931 不論股價怎麼走都留得住，存到到

[9] Thanks to a lively class discussion on March 16, 2011.
[10] Hedging and replication are mirror images.
[11] Thanks to a lively class discussion on March 16, 2016.

期就是確定的 4.931×1.2^3。另外，hedge 不是建好就放著不管：每個節點的 delta h 都不同（$0.82031 \to 0.90625 \to 0.25$），每期都得 rebalance 才能維持；連續時間下則要隨時調整（Section 9.2）。

§8.13. Binomial Tree Algorithms for European Options

Slides pp. 287–288

The BOPM implies the **binomial tree algorithm** that applies backward induction. There are $\sum_{j=0}^{n}(j+1) \sim n^2/2$ nodes and each costs $O(1)$ work, so the total running time is $O(n^2)$. Storing every node needs $O(n^2)$ memory, but backward induction only ever looks one column ahead, so a single array of length $n+1$ suffices — provided its entries are updated in the right order.

Algorithm 8.1: Binomial tree algorithm for European calls, $O(n)$ space

Input: $S, u, d, X, n, \hat{r} \quad (u > e^{\hat{r}} > d)$

$R := e^{\hat{r}}$;
$p := (R-d)/(u-d)$;
for $i = 0, 1, \ldots, n$ **do**
  $C[i] := \max\bigl(0,\, Su^{n-i}d^i - X\bigr)$; `// i = number of down moves`
for $j = n-1$ **to** 0 **do**
  for $i = 0, 1, \ldots, j$ **do**
    $C[i] := \bigl(p \times C[i] + (1-p) \times C[i+1]\bigr)/R$;
return $C[0]$;

Note 8.13.1. 內層迴圈的 i 必須遞增。新的 $C[i]$ 要讀舊的 $C[i]$ 和 $C[i+1]$；往上跑時，讀 $C[i+1]$ 的那一刻它還沒被覆寫。若 i 遞減，就會讀到已經屬於時間 j 的新值 — 這就是投影片提醒的「proper updating」。

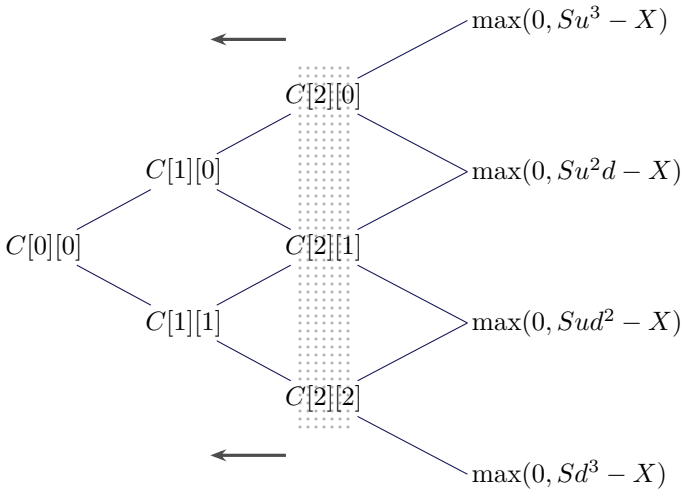


Figure 8.8.: Backward induction sweeps a strip (dotted) from the terminal payoffs towards the root. $C[j][i]$ is the call value at time j after i down moves; Algorithm 8.1 keeps only the current strip.

To find the hedge ratio, apply (8.3) to the two nodes at time 1. To price European puts, simply replace the payoff by $\max\bigl(0,\, X - Su^{n-i}d^i\bigr)$.

§8.13.i. Further Time Improvement for Calls

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If $C[j+1][i]$ and $C[j+1][i+1]$ are both zero, so is $C[j][i]$. A node with i down moves at time j can still collect at most $n-i$ up moves in total, so it is worthless whenever $n-i<a$. Letting the inner loop run only up to $\min(n-a,\, j)$ skips the whole triangle of zeros below the strike (Figure 8.9), and the algorithm runs in $O\big(n(n-a)\big)$ time with an array of size $n-a+1$.

Remark 8.13.2. Algorithm 8.1 的改進分三步：

(1) **零會往回傳**。backward induction 的公式是 $C[j][i]=\big(pC[j+1][i]+(1-p)C[j+1][i+1]\big)/R$，下一期兩個節點都是 0，這個節點就是 0。

(2) **哪些節點一定是** 0。從節點 (j,i)（時間 j、已跌 i 次）出發，最好的情況是剩下的 $n-j$ 步全部上漲，到期時總共上漲 $n-i$ 次。由 (8.14)，要至少上漲 a 次才會 in the money；所以若 $n-i<a$，即 $i>n-a$，連最好的情況都 out of the money，這個節點的 call 價值必為 0。注意這個條件和 j 無關。

(3) **省下的工作**。所以每一欄只需要算 $i=0,1,\ldots,\min(n-a,\, j)$；array 只要 $n-a+1$ 格，每欄最多 $n-a+1$ 個節點、共 n 欄，時間是 $O\big(n(n-a)\big)$。call 越 out of the money（a 越大），省得越多。

例如 Example 8.12.1 中 $n=3$、$a=2$，所以 $n-a=1$：只有 $i=0,1$ 的節點需要計算，而 $i=2,3$ 的節點（股價 40、60、20）在 Figure 8.7 中確實都是 0。

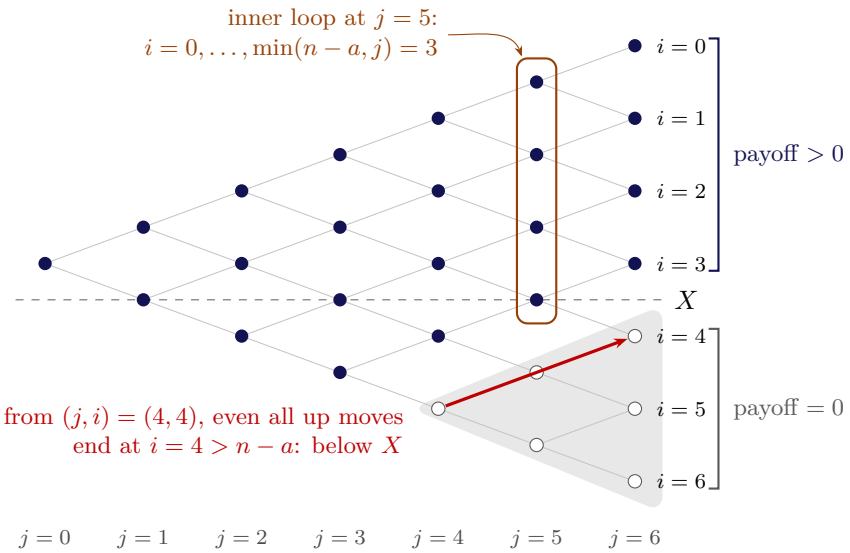


Figure 8.9.: An $n=6$ tree with $a=3$, so $n-a=3$. Filled nodes are computed; hollow nodes ($i>n-a$, shaded) are always 0 because even all remaining up moves end below the strike, so the inner loop never visits them. With $ud=1$, nodes at the same height have identical stock prices.

§8.13.ii. The Optimal Algorithm

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We can reduce the running time to $O(n)$ and the memory requirement to $O(1)$.[12] Note that

$$b(j;n,p) \;=\; \frac{p\,(n-j+1)}{(1-p)\,j}\;b(j-1;n,p),$$

[12] "We have not room enough, in our mental field, to keep the dead there as well as the living." Marcel Proust (1871–1922), *In Search of Lost Time*, Vol. 3: *The Guermantes Way*.

which follows from $\binom{n}{j} = \binom{n}{j-1}\frac{n-j+1}{j}$. So the following program computes $b(j;n,p)$ in $b[\,j\,]$:

Algorithm 8.2: Computing $b(j;n,p)$ for $j = a, a+1, \ldots, n$

```
b[a] := C(n,a) p^a (1-p)^{n-a};
for j = a+1, a+2, ..., n do
    b[j] := b[j-1] × p × (n-j+1) / ((1-p) × j);
```

It runs in $O(n)$ steps. Alternatively, $b(j;n,p)$ can be computed by the regularized incomplete beta function in $O(1)$ time.

With the $b(j;n,p)$ available, the risk-neutral valuation formula (8.15) is trivial to compute:

$$C \;=\; \frac{1}{R^n}\sum_{j=a}^{n} b(j;n,p)\big(Su^j d^{n-j} - X\big).$$

But we only need a single variable to store the $b(j;n,p)$'s as they are being sequentially computed: each one is used once, in the jth term of the sum, and then only to produce the next.

Note 8.13.3. 當 n 很大時，$b(j;n,p)$ 和 $Su^j d^{n-j}$ 的值會非常極端（遠在 $n = 10^4$ 之前 $b(a;n,p)$ 就已經 underflow），所以實作上應該改存它們的對數。[13]

This linear-time algorithm computes the discounted expected value of $\max(S_n - X, 0)$. This forward-induction approach cannot be applied to American options because of early exercise. So binomial tree algorithms for American options usually run in $O(n^2)$ time.

§8.13.iii. The Bushy Tree

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動機。Section 8.8 的遞迴式最直接的寫法就是 recursion：在 Su 和 Sd 各用剩下的 $n-1$ 期定價，再合起來。

Algorithm 8.3: Naive recursion: PRICE(S,k) is the call value with k periods left

```
if k = 0 then
    return max(S - X, 0);
else
    return (p × PRICE(Su, k-1) + (1-p) × PRICE(Sd, k-1)) / R;
```

為什麼是 $O(2^n)$。每次呼叫都會再呼叫自己兩次，所以往下一層呼叫次數就翻倍，到期時共有 2^n 個 leaves。問題在於它追蹤的是 *path* 而不是股價：Su^2d 可以由 uud、udu、duu 三條 path 到達，就被當成三個不同的 node 各算一次，tree 永遠不會 recombine（Figure 8.10）。所以 running time 是 $O(2^n)$（若每個 leaf 都用 n 次乘法重算股價則是 $O(n2^n)$），實務上不可行。

怎麼修。把算過的值用 (j,i)（時間、下跌次數）存成表（memoization），每個不同的 node 只算一次，就回到 $O(n^2)$。這其實就是 top-down 版的 backward induction；Algorithm 8.1 的 bottom-up 寫法還能把 memory 從 $O(n^2)$ 降到 $O(n)$。

那為什麼還要講 bushy tree。存表的前提是 option 的價值只取決於「現在的股價」。若 payoff 取決於整條 path（例如平均股價，見 Section 10.4），Sud 和 Sdu 股價相同但平均不同，就不能共用同一格，這時真的得面對 2^n 條 path。

[13] 課本 Programming Assignment 9.2.14。

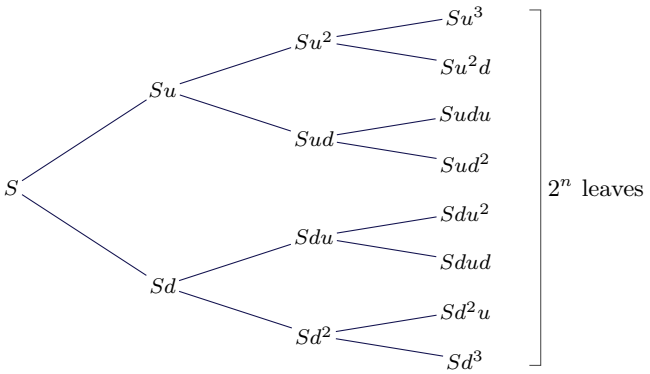


Figure 8.10.: Bushy tree：追蹤的是 path 而不是股價。像 Su^2d、$Sudu$、Sdu^2 這些相同的股價，會以不同的 nodes 出現。

同一個公式有三種演算法：brute-force recursion 是 $O(2^n)$，在 recombining tree 上做 backward induction 是 $O(n^2)$，直接對 terminal distribution 求和（Algorithm 8.2）是 $O(n)$。第一種什麼都沒利用，第二種利用了 $ud = du$，第三種利用了 European payoff 只取決於到期時的 state — 這也正是它不能用在 American options 的原因。

§8.14. Toward the Black–Scholes Formula

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The binomial model seems to suffer from two unrealistic assumptions:

(i) the stock price takes on only two values in a period;

(ii) trading occurs at discrete points in time.

As n increases, however, the stock price ranges over ever larger numbers of possible values, and trading takes place nearly continuously.[14] What is needed is to calibrate the BOPM's parameters u, d and R so that it converges to the continuous-time model. We now skim through the proof.

Remark 8.14.1 (路線圖)**.** 目標是從 binomial tree 推出 Black–Scholes 公式，只有四步：

(1) **讓 tree 像真的股票**（Section 8.14.i）：選 u, d, q，讓 tree 的 log return 平均和變異數對上真實股價的 $\mu\tau$、$\sigma^2\tau$，得到 CRR model。

(2) $n \to \infty$（Section 8.14.iv）：$\ln(S_\tau/S)$ 是 n 個小步的和，由 CLT 變成 normal，所以 S_τ 是 lognormal。

(3) **換成 risk-neutral 機率**（Lemma 8.14.15）：定價要用 p 而不是 q；換過去之後平均從 $\mu\tau$ 變成 $(r - \sigma^2/2)\tau$，變異數不變。

(4) **代入定價公式**（Section 8.14.vi）：$C = e^{-r\tau}E^\pi[\max(S_\tau - X, 0)]$，把第 3 步的 lognormal 代進去算，就是 Black–Scholes 公式。

簡報只給每一步的結論。標成「補充」的 proof 和 remark 是筆記另外補的細節，第一次讀可以跳過。每一小節開頭的 ▷ 標示它屬於第幾步。

Notation .
- τ: the time to expiration of the option, measured in years;
- r: the continuously compounded *annual* riskless rate;
- n: the number of periods during the option's life, so each period is $\Delta t \triangleq \tau/n$ years long.

The period-based quantities must then be $\hat{r} = r\tau/n = r\Delta t$ and $R = e^{\hat{r}}$.

[14] Continuous-time trading may create arbitrage opportunities in practice (Budish, Cramton, & Shim, 2015)!

§8.14.i. Matching the Mean and Variance

Slides pp. 296–300

▷ **路線圖第 1 步：讓 tree 的 log return 平均、變異數對上真實股價**

Under the binomial model, $\ln u$ and $\ln d$ are the stock's two possible continuously compounded rates of return per period.

Remark 8.14.2. 由 (2.4)，continuously compounded rate of return x 的意思是 $S_{\text{end}} = S\,e^{x}$，也就是 $x = \ln(S_{\text{end}}/S)$。一期內股價只會變成 Su 或 Sd，所以這一期的報酬率只有兩種可能：$\ln(Su/S) = \ln u$ 或 $\ln(Sd/S) = \ln d$（例如 $u = 1.1$ 時是 $\ln 1.1 \approx 9.53\%$，而不是 simple return 的 10%）。用對數的好處是多期報酬可以直接相加，這正是下面 (8.17) 的寫法：j 次上漲、$n-j$ 次下跌，總報酬就是 $j \ln u + (n-j)\ln d$。

After j up moves out of n,

$$\ln \frac{S_\tau}{S} \;=\; \ln \frac{S u^{j} d^{n-j}}{S} \;=\; j \ln(u/d) + n \ln d, \tag{8.17}$$

where $j \sim \text{Bin}(n, q)$, with q the *true* up probability of Section 8.2, so $E[\,j\,] = nq$ and $\text{Var}[\,j\,] = nq(1-q)$.

Definition 8.14.3. Let

$$\hat{\mu} \;\triangleq\; \frac{1}{n} E\left[\ln \frac{S_\tau}{S}\right], \qquad \hat{\sigma}^2 \;\triangleq\; \frac{1}{n} \text{Var}\left[\ln \frac{S_\tau}{S}\right]$$

denote the expected value and the variance of the continuously compounded rate of return *per period* of the BOPM.

From (8.17) it is not hard to show that

$$\hat{\mu} \;=\; q \ln(u/d) + \ln d, \qquad \hat{\sigma}^2 \;=\; q(1-q)\ln^2(u/d).$$

Indeed, each period contributes an independent Bernoulli variable equal to $\ln u$ or $\ln d$.

Remark 8.14.4. $\hat{\mu}$ **和** $\hat{\sigma}^2$ **怎麼算**。

(1) **從** (8.17) **直接算**。$\ln(S_\tau/S) = j\ln(u/d) + n\ln d$ 是 j 的線性函數，$\ln(u/d)$ 和 $n\ln d$ 是常數，所以由 $E[aj+b] = aE[j]+b$、$\text{Var}[aj+b] = a^2\,\text{Var}[j]$，

$$E\left[\ln \frac{S_\tau}{S}\right] = nq\ln(u/d) + n\ln d, \qquad \text{Var}\left[\ln \frac{S_\tau}{S}\right] = nq(1-q)\ln^2(u/d),$$

再除以 n 就是 $\hat{\mu}$ 和 $\hat{\sigma}^2$。

(2) **從每一期看**。第 k 期的 log return y_k 以機率 q 等於 $\ln u$、以機率 $1-q$ 等於 $\ln d$，而 $\ln(S_\tau/S) = \sum_{k=1}^{n} y_k$。一個只取兩值 a, b（機率 $q, 1-q$）的變數有

$$E[y] = qa + (1-q)b, \qquad \text{Var}[y] = qa^2 + (1-q)b^2 - \big(qa + (1-q)b\big)^2 = q(1-q)(a-b)^2,$$

代 $a = \ln u$、$b = \ln d$ 得 $E[y_k] = q\ln(u/d) + \ln d = \hat{\mu}$、$\text{Var}[y_k] = q(1-q)\ln^2(u/d) = \hat{\sigma}^2$。所以 $\hat{\mu}$、$\hat{\sigma}^2$ 就是一期的平均和變異數，這也是定義裡除以 n 的原因。

Assume the stock's true continuously compounded rate of return over τ years satisfies

$$E\left[\ln \frac{S_\tau}{S}\right] \;=\; \mu\tau, \qquad \text{Var}\left[\ln \frac{S_\tau}{S}\right] \;=\; \sigma^2\tau.$$

Call σ the stock's (annualized) **volatility**.

Remark 8.14.5. **波動率**（volatility）σ。這裡先不管 tree，假設真實股價的 τ 年 log return $\ln(S_\tau/S)$ 平均是 $\mu\tau$、變異數是 $\sigma^2\tau$。

(1) **為什麼和 τ 成正比**。把 τ 年切成 τ 個一年（設 τ 為整數，一般情況切成更小段同理），第 k 年的 log return 記為 x_k，假設 x_k i.i.d.、$E[x_k]=\mu$、$\mathrm{Var}[x_k]=\sigma^2$。log return 可以相加，所以 $\ln(S_\tau/S)=\sum_{k=1}^{\tau}x_k$，於是

$$E\left[\ln\frac{S_\tau}{S}\right]=\sum_{k=1}^{\tau}E[x_k]=\mu\tau,\qquad \mathrm{Var}\left[\ln\frac{S_\tau}{S}\right]=\sum_{k=1}^{\tau}\mathrm{Var}[x_k]=\sigma^2\tau,$$

第二式用到獨立變數的變異數可以相加。所以 μ 和 σ^2 分別是一年的平均和變異數，這就是「annualized」的意思。

(2) **σ 是什麼**。$\sigma=\sqrt{\mathrm{Var}[x_k]}$ 是一年 log return 的標準差，用來衡量股價每年波動多大。

下面 (8.18)–(8.19) 就是要求 BOPM 的 n 期 log return 的平均 $n\hat\mu$ 和變異數 $n\hat\sigma^2$ 對上這裡的 $\mu\tau$ 和 $\sigma^2\tau$。

The BOPM converges to that distribution only if

$$n\hat\mu \;=\; n\big[\,q\ln(u/d)+\ln d\,\big] \;\to\; \mu\tau, \tag{8.18}$$

$$n\hat\sigma^2 \;=\; nq(1-q)\ln^2(u/d) \;\to\; \sigma^2\tau. \tag{8.19}$$

These are two conditions on three unknowns u,d,q, so we need one more. Impose

$$ud \;=\; 1.$$

It makes nodes at the same horizontal level of the tree have identical prices (Figure 8.9); other choices are possible.[15]

Remark 8.14.6. (1) (8.18)–(8.19) **在說什麼**。由 $\hat\mu,\hat\sigma^2$ 的定義，tree 的 n 期 log return $\ln(S_\tau/S)$ 平均是 $n\hat\mu$、變異數是 $n\hat\sigma^2$。要讓 tree 逼近真實股價，至少這兩個數要對上真實的 $\mu\tau$ 和 $\sigma^2\tau$。只要求 $n\to\infty$ 時對上（$\to$）而不要求相等，因為 tree 本來就只是近似。

(2) **為什麼要多一個條件**。兩個式子只能決定兩個未知數，而 u,d,q 有三個，所以解不唯一，要再自己加一個條件。

(3) **為什麼選 $ud=1$**。$ud=1$ 等價於 $\ln d=-\ln u$，也就是在 log 座標下上漲和下跌走一樣長。於是 $Sud=S$：漲一次再跌一次會回到原價，股價只由「上漲次數 − 下跌次數」決定，畫在圖上同一高度的 node 股價相同。

Proposition 8.14.7

The requirements are satisfied by

$$u \;=\; e^{\sigma\sqrt{\Delta t}},\qquad d \;=\; e^{-\sigma\sqrt{\Delta t}},\qquad q \;=\; \frac12+\frac12\frac{\mu}{\sigma}\sqrt{\Delta t}, \tag{8.20}$$

for which

$$n\hat\mu \;=\; \mu\tau,\qquad n\hat\sigma^2 \;=\; \left[1-\left(\frac{\mu}{\sigma}\right)^2\Delta t\right]\sigma^2\tau \;\to\; \sigma^2\tau.$$

Proof. Now $\ln(u/d)=2\sigma\sqrt{\Delta t}$ and $\ln d=-\sigma\sqrt{\Delta t}$, so

$$n\hat\mu \;=\; n\sigma\sqrt{\Delta t}\,(2q-1) \;=\; n\sigma\sqrt{\Delta t}\cdot\frac{\mu}{\sigma}\sqrt{\Delta t} \;=\; n\mu\Delta t \;=\; \mu\tau.$$

For the variance, $q(1-q)=\frac14-\frac14(2q-1)^2=\frac14\big[1-(\mu/\sigma)^2\Delta t\big]$, hence

$$n\hat\sigma^2 \;=\; n\cdot\tfrac14\big[1-(\mu/\sigma)^2\Delta t\big]\cdot 4\sigma^2\Delta t \;=\; \big[1-(\mu/\sigma)^2\Delta t\big]\sigma^2\tau. \qquad\blacksquare$$

[15] Exact solutions for u,d,q are feasible if (8.18)–(8.19) are replaced by equations: 3 equations for 3 variables (Chance, 2008). Exercise 9.3.1 of the textbook instead imposes $q=1/2$.

Remark 8.14.8 (補充). (8.20) **是怎麼想出來的**。由 $ud = 1$，令 $\ln u = \alpha$、$\ln d = -\alpha$，則 $\ln(u/d) = 2\alpha$，(8.18)–(8.19) 變成

$$n\hat{\mu} = n\alpha(2q-1) \to \mu\tau, \qquad n\hat{\sigma}^2 = 4nq(1-q)\,\alpha^2 \to \sigma^2\tau.$$

(1) 第一式要求 $2q - 1 = \mu\tau/(n\alpha)$。每步很小（$\alpha \to 0$）時，$q$ 會很接近 $\frac{1}{2}$，所以第二式裡 $4q(1-q) \approx 1$。
(2) 於是第二式變成 $n\alpha^2 \approx \sigma^2\tau$，即 $\alpha = \sigma\sqrt{\tau/n} = \sigma\sqrt{\Delta t}$，這就是 $u = e^{\sigma\sqrt{\Delta t}}$、$d = e^{-\sigma\sqrt{\Delta t}}$。
(3) 代回第一式，用 $\tau/n = \Delta t$：

$$2q - 1 = \frac{\mu\tau}{n\sigma\sqrt{\Delta t}} = \frac{\mu\Delta t}{\sigma\sqrt{\Delta t}} = \frac{\mu}{\sigma}\sqrt{\Delta t},$$

即 $q = \frac{1}{2} + \frac{1}{2}\frac{\mu}{\sigma}\sqrt{\Delta t}$。

證明裡的恆等式。$q(1-q) = \frac{1}{4} - \left(q - \frac{1}{2}\right)^2 = \frac{1}{4} - \frac{1}{4}(2q-1)^2$（把右邊展開即可驗證）。代入 $2q - 1 = \frac{\mu}{\sigma}\sqrt{\Delta t}$ 後，變異數比 $\sigma^2\tau$ 多乘了 $1 - (\mu/\sigma)^2\Delta t$；$\Delta t \to 0$ 時這個因子 $\to 1$，所以是 $\to$ 而不是 $=$。

With this choice, even if σ is not calibrated correctly, the mean is still matched exactly. Note also that u and d are related to volatility exclusively: both are independent of r and μ. All the information about the drift sits in the probability; that is the same division of labour as in Section 8.4, where only u and d entered the price.

Remark 8.14.9. (1) **為什麼 σ 估錯，平均仍然對**。證明中 $n\hat{\mu} = n\sigma\sqrt{\Delta t}\,(2q-1)$，而 $2q - 1 = \frac{\mu}{\sigma}\sqrt{\Delta t}$ 裡的 $\frac{1}{\sigma}$ 剛好和前面的 σ 消掉，得到 $n\mu\Delta t = \mu\tau$。所以代入任何 σ，平均都恰好是 $\mu\tau$；σ 估錯只會影響變異數。
(2) **分工**。u, d 只含 σ，決定每一步走多遠；μ 只出現在 q，決定往上走的機率比 $\frac{1}{2}$ 大多少。

§8.14.ii. The CRR Model

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▷ **路線圖第 1 步：（續）CRR 就是第 1 步解出來的** u, d, q

The choices (8.20) result in the **CRR binomial model**.[16] Black (1992): "This method is probably used more than the original formula in practical situations." OptionMetrics's (2015) IvyDB, for instance, uses the CRR model.

Remark 8.14.10. **CRR 二項式模型** (Cox–Ross–Rubinstein binomial model)：就是 (8.20) 的選法。取 log 後

$$\ln(Su) = \ln S + \sigma\sqrt{\Delta t}, \qquad \ln(Sd) = \ln S - \sigma\sqrt{\Delta t},$$

所以每一期 $\ln S$ 固定往上或往下走 $\sigma\sqrt{\Delta t}$，在 $(t, \ln S)$ 平面上是等距的格子（Figure 8.11）。

The CRR model is best seen in logarithmic terms:

$$\ln S \longrightarrow \begin{cases} \ln S + \sigma\sqrt{\Delta t}, & \text{up move,} \\ \ln S - \sigma\sqrt{\Delta t}, & \text{down move.} \end{cases}$$

In the $(t, \ln S)$ plane the tree is a regular lattice (Figure 8.11).

[16] Cox, Ross, & Rubinstein (1979).

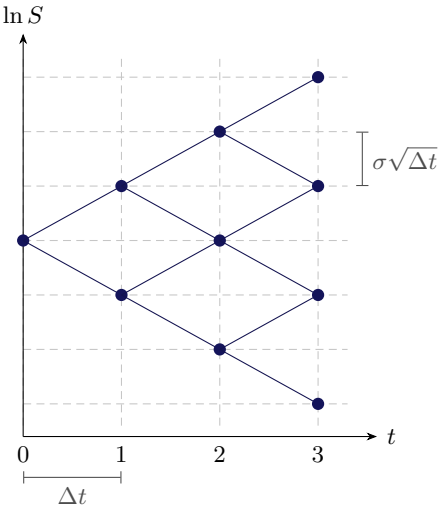


Figure 8.11.: The CRR tree in logarithmic coordinates: every move is $\pm\sigma\sqrt{\Delta t}$, and since $ud = 1$ the tree is symmetric about the starting level.

§8.14.iii. When the Probabilities Misbehave

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▷ **路線圖第 1 步：(附帶條件) CRR 的 risk-neutral p 要落在 $[0,1]$**

The no-arbitrage inequalities $d < R < u$ may not hold under (8.20) or (8.5). If this happens, the risk-neutral probabilities lie outside $[0,1]$.[17] With $r \geq 0$ the inequality $d < R$ is automatic, and $R < u$ means $e^{\sigma\sqrt{\Delta t}} > e^{r\Delta t}$, i.e., $\sigma > r\sqrt{\tau/n}$, i.e.,

$$n > \frac{r^2}{\sigma^2}\tau. \tag{8.21}$$

So the problem goes away if n is large enough.

Remark 8.14.11. (8.21) **的推導**。

(1) 由 (8.5)，$p = \frac{R-d}{u-d}$，分母 $u-d>0$。所以 $0<p<1 \Leftrightarrow 0<R-d<u-d \Leftrightarrow d<R<u$。

(2) $d<R$ **自動成立**。$d = e^{-\sigma\sqrt{\Delta t}} < 1$，而 $r \geq 0$ 時 $R = e^{r\Delta t} \geq 1$。

(3) $R<u$。$e^{r\Delta t} < e^{\sigma\sqrt{\Delta t}} \Leftrightarrow r\Delta t < \sigma\sqrt{\Delta t} \Leftrightarrow r\sqrt{\Delta t} < \sigma$。兩邊平方並代 $\Delta t = \tau/n$：

$$\frac{r^2\tau}{n} < \sigma^2 \Leftrightarrow n > \frac{r^2}{\sigma^2}\tau.$$

直觀上：每期利息 $r\Delta t$ 和 Δt 成正比，股價每步的幅度 $\sigma\sqrt{\Delta t}$ 卻和 $\sqrt{\Delta t}$ 成正比；Δt 夠小時 $\sqrt{\Delta t}$ 遠大於 Δt，所以 n 夠大就一定有 $R<u$。

Other solutions can be found in the textbook[18] or will be presented later.

Example 8.14.12. With $r = 10\%$, $\sigma = 5\%$ and $\tau = 1$, (8.21) requires $n > 4$: a two-period CRR tree for a low-volatility stock has $p > 1$. For typical equity volatilities the bound is far below 1 and never binds.

[17] Many papers and programs forget to check this condition!
[18] See Exercise 9.3.1 of the textbook.

§8.14.iv. The Limiting Distribution

Slides pp. 304–305

▷ 路線圖第 2 步：$n \to \infty$，由 CLT 得到 lognormal

By (8.17), $\ln(S_\tau/S)$ is a sum of n independent Bernoulli variables. The central limit theorem (Theorem 5.8.6) therefore says $\ln(S_\tau/S)$ converges to $N(\mu\tau, \sigma^2\tau)$.[19] So $\ln S_\tau$ approaches $N(\mu\tau + \ln S, \sigma^2\tau)$.

Remark 8.14.13. (1) **CLT 怎麼用**。$\ln(S_\tau/S) = \sum_{k=1}^{n} y_k$，$y_k$ i.i.d.，每個只取 $\ln u$ 或 $\ln d$。Theorem 5.8.6 說很多 i.i.d. 變數的和近似 normal，而 normal 由平均和變異數決定：和的平均是 $n\hat{\mu}$、變異數是 $n\hat{\sigma}^2$，由 Proposition 8.14.7 它們 $\to \mu\tau$ 和 $\sigma^2\tau$。所以 $\ln(S_\tau/S) \to N(\mu\tau, \sigma^2\tau)$。

(2) $\ln S_\tau$。$\ln S_\tau = \ln S + \ln(S_\tau/S)$，$\ln S$ 是常數；加常數只平移平均、不改變異數。

(3) **lognormal**。由 Section 5.10 的定義，$\ln Y$ 是 normal 的 Y 叫 lognormal，所以 S_τ 在極限下是 lognormal。

> **S_τ has a lognormal distribution in the limit. This is the distribution Section 5.10 prepared for; the binomial tree has now *derived* it, rather than assumed it.**

▷ 路線圖第 3 步：換成 risk-neutral 機率

Which μ? For pricing we need the tree under the risk-neutral probability, not under q.

Remark 8.14.14. 下面的 Lemma 8.14.15 要做什麼。

(1) **前面得到的**。Proposition 8.14.7 和 CLT 說：在真實機率 q 之下，$\ln(S_\tau/S) \to N(\mu\tau, \sigma^2\tau)$。

(2) **定價需要的**。由 (8.13)，option 價值 $= e^{-r\tau}E^\pi[\text{payoff}]$，期望值要在 risk-neutral 機率 $p = \frac{R-d}{u-d}$ 之下算，而不是 q。所以要問：同一棵 tree，改用 p 之後，$\ln(S_\tau/S)$ 的極限分布是什麼？

(3) **lemma 的答案**。u, d 不變，只是 q 換成 p。proof 會算出 p 和 (8.20) 的 q 長得一樣，只是 μ 被換成 $r - \sigma^2/2$。所以 Proposition 8.14.7 的結論把 μ 換掉即可：變異數仍是 $\sigma^2\tau$，平均變成 $(r - \sigma^2/2)\tau$。

(4) **用在哪**。Theorem 8.14.19 的證明就是用這個分布去算 $E^\pi[\max(S_\tau - X, 0)]$。

真實的期望報酬 μ 在 (3) 中被換掉了，這就是 Black–Scholes 公式裡沒有 μ 的原因。

Lemma 8.14.15

In a risk-neutral economy, the continuously compounded rate of return satisfies $\ln(S_\tau/S) \to N\big((r - \sigma^2/2)\,\tau,\ \sigma^2\tau\big)$ in distribution.

Proof（補充；簡報只說「令 $q = p$、$n \to \infty$，則 $\mu = r - \sigma^2/2$」）. 想法：同一棵 CRR tree，把 q 換成 $p = \frac{e^{r\Delta t} - d}{u - d}$，證明 p 和 (8.20) 的 q 形狀相同，只是 μ 換成 $r - \sigma^2/2$。用 $e^y = 1 + y + \frac{y^2}{2} + O(y^3)$，$y$ 分別代 $r\Delta t$ 和 $\pm\sigma\sqrt{\Delta t}$：

$$e^{r\Delta t} = 1 + r\Delta t + O(\Delta t^2), \qquad u, d = 1 \pm \sigma\sqrt{\Delta t} + \frac{\sigma^2\Delta t}{2} + O(\Delta t^{3/2}).$$

(1) **分子** $e^{r\Delta t} - d$：常數 1 消掉，剩 $\sigma\sqrt{\Delta t} + \big(r - \frac{\sigma^2}{2}\big)\Delta t + O(\Delta t^{3/2})$。

(2) **分母** $u - d$：1 和 $\frac{\sigma^2\Delta t}{2}$ 都消掉，剩 $2\sigma\sqrt{\Delta t} + O(\Delta t^{3/2})$。

[19] As our probabilities depend on n, this argument is only heuristic; the rigorous version uses the Lyapunov condition, $\big[(1-q)^2 + q^2\big]/\big[nq(1-q)\big] \to 0$, which holds here. See Uspensky (1937).

(3) **相除**（分子分母同除以 $2\sigma\sqrt{\Delta t}$）：

$$p = \frac{1}{2} + \frac{1}{2}\frac{r - \sigma^2/2}{\sigma}\sqrt{\Delta t} + O(\Delta t).$$

(4) **對照** (8.20)。$q = \frac{1}{2} + \frac{1}{2}\frac{\mu}{\sigma}\sqrt{\Delta t}$，所以 p 就是把 μ 換成 $r - \sigma^2/2$ 的 q。[20]
於是 Proposition 8.14.7 和 CLT（第 1、2 步）照搬，只是 μ 換成 $r - \sigma^2/2$。剩下的 $O(\Delta t)$ 不影響極限：它讓平均 $n\hat{\mu} = n \cdot 2\sigma\sqrt{\Delta t}\,(p - \frac{1}{2})$ 只差 $n\sqrt{\Delta t} \cdot O(\Delta t) = \tau \cdot O(\sqrt{\Delta t}) \to 0$。 ■

Remark 8.14.16 (第 3 步的直觀)**. 為什麼是 $r - \sigma^2/2$ 而不是 r。**
(1) risk-neutral 世界裡，股票的期望價格以 riskless rate 成長：$E^{\pi}[S_\tau] = Se^{r\tau}$（(8.7) 的連續版）。
(2) 若 $\ln(S_\tau/S) \sim N(m, \sigma^2\tau)$，由 (5.25)，$E[S_\tau/S] = e^{m+\sigma^2\tau/2}$。
(3) 兩者相等：$m + \sigma^2\tau/2 = r\tau$，所以 $m = (r - \sigma^2/2)\tau$。
也就是說，log return 的平均要比 $r\tau$ 少 $\sigma^2\tau/2$，取 exp 再平均之後才會剛好是 $e^{r\tau}$。這和 Jensen's inequality（Theorem 5.10.3）一致：ln 是 concave，所以 $E[\ln Y] \le \ln E[Y]$。

§8.14.v. Two Meanings of "Rate of Return"

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▷ **路線圖第 3 步：(續) 第 3 步的平均 $r - \sigma^2/2$ 怎麼解讀**

The expected stock price at expiration in a risk-neutral economy is $Se^{r\tau}$, by Lemma 8.14.15 and (5.25):

$$E\big[\,S_\tau\,\big] \;=\; S\exp\Big[\Big(r - \frac{\sigma^2}{2}\Big)\tau + \frac{\sigma^2\tau}{2}\Big] \;=\; Se^{r\tau}.$$

So the stock's expected annual rate of return is the riskless rate r — *if* the rate of return means

$$\frac{\ln E\big[\,S_\tau/S\,\big]}{\tau},$$

the (weighted) arithmetic average rate of return, annualized. If it means instead

$$\frac{E\big[\ln(S_\tau/S)\big]}{\tau},$$

the (weighted) geometric average rate of return, annualized, then Lemma 8.14.15 gives $r - \sigma^2/2$.

Remark 8.14.17. 兩個「rate of return」只差在 ln 和 E 的順序：
(1) $\frac{1}{\tau}\ln E[S_\tau/S]$：先平均再取 ln。上式 $E[S_\tau/S] = e^{r\tau}$，所以得到 r。$E[S_\tau/S]$ 是 gross return 的（機率加權）算術平均，所以叫 arithmetic average。
(2) $\frac{1}{\tau}E[\ln(S_\tau/S)]$：先取 ln 再平均。由 Lemma 8.14.15，$E[\ln(S_\tau/S)] = (r - \sigma^2/2)\tau$，所以得到 $r - \sigma^2/2$。$e^{E[\ln(S_\tau/S)]}$ 是 gross return 的（機率加權）幾何平均，所以叫 geometric average。

Note 8.14.18. 這兩個敘述同時成立，因為它們回答的是不同的問題。第一個講的是各情境下 *wealth* 的平均，第二個講的是單一情境中典型（median）的 *growth*。Jensen's inequality（Theorem 5.10.3）讓第二個小於第一個，對 lognormal 而言剛好差 $\sigma^2/2$。

[20] More precisely, $p = \frac{1}{2} + \frac{\mu}{2\sigma}(\Delta t)^{0.5} + \frac{\sigma^4+4\sigma^2\mu+6\mu^2}{24\sigma}(\Delta t)^{1.5} + O\big(\Delta t^{2.5}\big)$ with $\mu = r - \sigma^2/2$, consistent with (8.20). See Lemma 9.3.3 of the textbook.

§8.14.vi. The Black–Scholes Formula

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▷ **路線圖第 4 步：代入定價公式，得到 Black–Scholes 公式**

Theorem 8.14.19 (The Black–Scholes Formula, 1973)

$$C = SN(x) - Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \tag{8.22}$$
$$P = Xe^{-r\tau}N\big(-x+\sigma\sqrt{\tau}\big) - SN(-x), \tag{8.23}$$

where

$$x \triangleq \frac{\ln(S/X) + \big(r+\sigma^2/2\big)\tau}{\sigma\sqrt{\tau}}. \tag{8.24}$$

On a United flight from San Francisco to Tokyo on March 7, 2010, a real-estate manager mentioned this formula to the lecturer!

Proof（補充；簡報沒有證明）. (1) **定價公式**。由 (8.13)，European option 的價值 = 折現因子 × risk-neutral 下 payoff 的期望。n 期的折現因子是 $e^{-\hat{r}n} = e^{-r\tau}$，所以 $C = e^{-r\tau}E^{\pi}\big[\max(S_\tau - X, 0)\big]$。

(2) $\ln S_\tau$ **的分布（第 3 步）**。由 Lemma 8.14.15 再加上常數 $\ln S$：$\ln S_\tau \sim N(m, s^2)$，

$$m = \ln S + \Big(r - \frac{\sigma^2}{2}\Big)\tau, \qquad s = \sigma\sqrt{\tau}.$$

(3) **期望值**。Corollary 5.10.11 已經算好 $\ln Y \sim N(\mu,\sigma^2)$ 時的 $E[\max(Y-a,0)] = e^{\mu+\sigma^2/2}N(d_1) - aN(d_2)$。取 $Y = S_\tau$、$a = X$，參數 (μ,σ) 換成 (m,s)，逐項代入：

- $e^{m+s^2/2} = \exp\big[\ln S + (r - \frac{\sigma^2}{2})\tau + \frac{\sigma^2\tau}{2}\big] = Se^{r\tau}$；
- $d_2 = \frac{m - \ln X}{s} = \frac{\ln(S/X) + (r-\sigma^2/2)\tau}{\sigma\sqrt{\tau}}$，分子比 (8.24) 少 $\sigma^2\tau$，所以 $d_2 = x - \sigma\sqrt{\tau}$；
- $d_1 = d_2 + s = x$。

所以 $E^{\pi}\big[\max(S_\tau - X, 0)\big] = Se^{r\tau}N(x) - XN(x-\sigma\sqrt{\tau})$，乘上 $e^{-r\tau}$ 就是 (8.22)。

(4) **put**。put–call parity (Theorem 7.5.1)：$P = C - S + Xe^{-r\tau} = Xe^{-r\tau}\big[1 - N(x-\sigma\sqrt{\tau})\big] - S\big[1 - N(x)\big]$，再用 $1 - N(z) = N(-z)$ 得 (8.23)。

■

Note 8.14.20. 上面的證明是先對分布取極限再定價；課本則反過來，直接對 (8.16) 取極限。[21] 那裡同樣用 central limit theorem 證明兩個 binomial 尾端和會收斂：

$$\sum_{j=a}^{n} b\big(j; n, pu/R\big) \;\to\; N(x), \qquad \sum_{j=a}^{n} b(j; n, p) \;\to\; N\big(x - \sigma\sqrt{\tau}\big).$$

不論哪種做法，得到的公式都是 (8.16) 之後提到的那個形狀。

Remark 8.14.21. **Black–Scholes 公式**：$C = SN(x) - Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$。只需要五個輸入 S, X, σ, τ, r；股票的期望報酬 μ 不出現，正如 Section 8.4 中 q 不出現一樣。

§8.14.vii. Interpreting $N(x)$ and $N(x-\sigma\sqrt{\tau})$

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▷ **路線圖第 4 步：（續）公式裡兩個 $N(\cdot)$ 的意思**

Compare (8.22) with (8.16) for the meaning of x.

[21] 課本 Theorem 9.3.4。

- $N(x-\sigma\sqrt{\tau})$ is the limit of $\sum_{j\geq a} b(j;n,p)$: the risk-neutral probability that the call finishes in the money, $\text{Prob}^{\pi}[\,S_\tau > X\,]$ (cf. Proposition 5.10.10).
- $N(x)$ is the limit of $\sum_{j\geq a} b(j;n,pu/R)$: the probability of the same event under the tilted measure pu/R, which weights each path by how much the stock grew along it.[22]

Remark 8.14.22. (1) **對照** (8.16)：$C = S\sum_{j\geq a} b(j;n,pu/R) - Xe^{-\hat{r}n}\sum_{j\geq a} b(j;n,p)$ 和 (8.22) 逐項對應：S 對 S，$Xe^{-\hat{r}n}$ 對 $Xe^{-r\tau}$。所以兩個尾端和分別收斂到 $N(x)$ 和 $N(x-\sigma\sqrt{\tau})$。

(2) $N(x-\sigma\sqrt{\tau})$ **是價內機率**。$\sum_{j\geq a} b(j;n,p)$ 是在 p 之下至少上漲 a 次的機率，也就是 $\text{Prob}^{\pi}[S_\tau > X]$。直接算也一樣：由 Proposition 5.10.10，$\text{Prob}^{\pi}[S_\tau > X] = N\big(\frac{m-\ln X}{s}\big)$，而 Theorem 8.14.19 的證明已算出 $\frac{m-\ln X}{s} = x - \sigma\sqrt{\tau}$。

(3) $N(x)$：是同一個事件 $S_\tau > X$ 在機率 pu/R 下的機率。pu/R 把每次上漲的權重乘上 $u/R > 1$，等於依股價漲了多少替每條 path 加權，所以 $N(x) > N(x-\sigma\sqrt{\tau})$。

Example 8.14.23. The binary call of Section 6.9.ii pays \$1 exactly when $S_\tau > X$, so under Black–Scholes its price is

$$e^{-r\tau}\,\text{Prob}^{\pi}[\,S_\tau > X\,] \;=\; e^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big).$$

The model-free formula (6.4) agrees: differentiating (8.22) in X, the terms containing N' cancel by (9.1) below, and $-\partial C/\partial X = e^{-r\tau}N(x-\sigma\sqrt{\tau})$.[23]

Remark 8.14.24 (補充). $-\partial C/\partial X$ **的計算**。對 (8.22) 的 X 微分；x 也含 X，所以要用 product rule 和 chain rule：

$$\frac{\partial C}{\partial X} = SN'(x)\frac{\partial x}{\partial X} - e^{-r\tau}N(x-\sigma\sqrt{\tau}) - Xe^{-r\tau}N'(x-\sigma\sqrt{\tau})\frac{\partial(x-\sigma\sqrt{\tau})}{\partial X}.$$

$\sigma\sqrt{\tau}$ 不含 X，所以 $\frac{\partial(x-\sigma\sqrt{\tau})}{\partial X} = \frac{\partial x}{\partial X}$；再由 (9.1)，$SN'(x) = Xe^{-r\tau}N'(x-\sigma\sqrt{\tau})$，第一項和第三項抵銷，剩下 $\frac{\partial C}{\partial X} = -e^{-r\tau}N(x-\sigma\sqrt{\tau})$。

§8.15. BOPM and Black–Scholes Model

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The Black–Scholes formula needs 5 parameters: S, X, σ, τ and r. Binomial tree algorithms take 6 inputs: S, X, u, d, $\hat{r}$ and n. The connections are

$$u \;=\; e^{\sigma\sqrt{\tau/n}}, \qquad d \;=\; e^{-\sigma\sqrt{\tau/n}}, \qquad \hat{r} \;=\; r\tau/n,$$

which is the CRR model (8.20) again.

Remark 8.15.1. 對應關係。每期長度 $\Delta t = \tau/n$ 年，代入 (8.20) 的 $u = e^{\sigma\sqrt{\Delta t}}$、$d = e^{-\sigma\sqrt{\Delta t}}$；每期的連續複利利率是年利率乘上一期的年數，$\hat{r} = r\Delta t = r\tau/n$。

Example 8.15.2. Consider a 3-month option when the interest rate is 8% per annum and the volatility is 30% per annum, so $\tau = 0.25$, $r = 0.08$ and $\sigma = 0.3$. A binomial tree with $n = 5$ should use $u = e^{0.3\sqrt{0.25/5}} = 1.0694$, $d = e^{-0.3\sqrt{0.25/5}} = 0.9351$ and $\hat{r} = 0.004$.[24]

[22] See Exercise 13.2.12 of the textbook for an interpretation of the probability associated with $N(x)$ and $N(-x)$.
[23] Exercise 9.3.5 of the textbook.
[24] Example 9.3.5 of the textbook.

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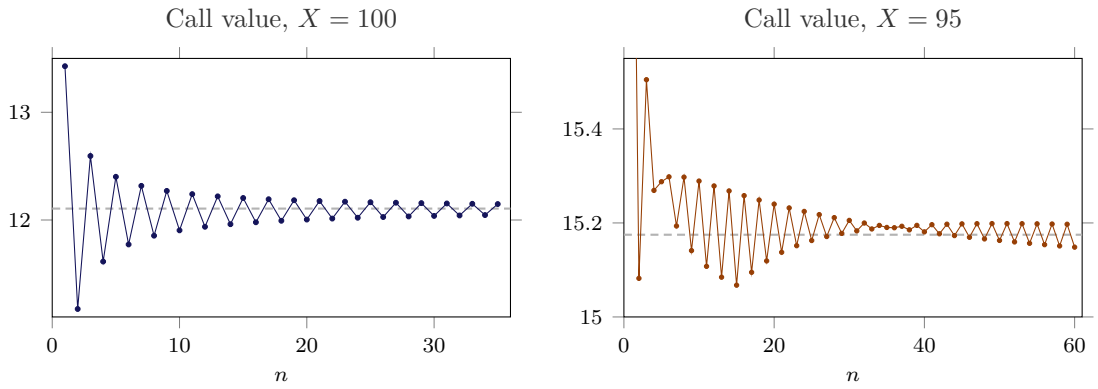


Figure 8.12.: CRR binomial call values against the number of periods n, with $S = 100$, $r = 8\%$, $\sigma = 0.2$, $\tau = 1$. Dashed: the Black–Scholes values 12.1058 (left) and 15.1749 (right). The value at $n = 1$ on the right (16.46) is off the scale.

The binomial tree algorithms converge reasonably fast: the error is $O(1/n)$.[25] Oscillations are inherent, however, and they can be dealt with by judicious choices of u and d.[26]

Remark 8.15.3. 圖在畫什麼。固定 $S = 100$、$r = 8\%$、$\sigma = 0.2$、$\tau = 1$，對每個 n 用上面的 $u, d, \hat{r}$ 建 n 期 CRR tree，由 (8.10) 算出

$$C_n = e^{-r\tau} \sum_{j=0}^{n} b(j; n, p) \max\bigl(S u^j d^{n-j} - X, 0\bigr),$$

把 C_n 對 n 畫出來；虛線是 Black–Scholes 值 C_{BS}。Section 8.14 證明了 $C_n \to C_{\mathrm{BS}}$，這張圖看的是怎麼收斂。「error is $O(1/n)$」指存在常數 K 使 $|C_n - C_{\mathrm{BS}}| \le K/n$，即 n 加倍時誤差上界減半。

為什麼會振盪。到期股價取 log 是 $\ln S + (2j - n)\sigma\sqrt{\Delta t}$，間距 $2\sigma\sqrt{\Delta t}$ 的等距格點；payoff $\max(S_\tau - X, 0)$ 只在 $\ln X$ 有一個 kink，而 tree 在 kink 附近近似得最差。所以誤差主要由「$\ln X$ 落在相鄰兩格點之間的哪個位置」決定，而這個位置隨 n 改變。左圖 $X = S$：n 偶數時 $j = n/2$ 的格點恰好在 $\ln X$ 上；n 奇數時 $2j - n$ 是奇數，$\ln X$ 落在兩格點正中間。每一步在這兩種情況間切換，誤差正負交替，形成鋸齒。右圖 $X = 95$：$\ln(X/S) = \ln 0.95$ 在格子中的相對位置隨 n 緩慢、不規則地移動，所以是比較長的波。

怎麼修。選 u, d 讓 $\ln X$ 永遠落在格點上（課本 Exercise 9.3.8：把 lattice 平移 $\frac{1}{n}\ln(X/S)$），相對位置就固定，振盪消失。同樣的 kink 問題也出現在 Section 9.7.v；barrier option 更嚴重 (Figure 10.6)。

§8.16. Implied Volatility

Slides pp. 313–314

Of the five inputs, volatility is the sole parameter not directly observable. The Black–Scholes formula can be used to compute the market's opinion of it: solve for σ given the option price, S, X, τ and r, with the numerical methods of Section 2.8. The result is the **implied volatility**.

Remark 8.16.1. 隱含波動率（implied volatility）：把市場上的選擇權價格代回 Black–Scholes 公式反解出來的 σ，可以看成市場對未來波動的「共識」。跟它相對的是**歷史波動率**（historical volatility），由過去股價的對數報酬估計。

[25] F. Diener & M. Diener (2004); L. Chang & Palmer (2007).

[26] See Exercise 9.3.8 of the textbook, which shifts the lattice by $(1/n)\ln(X/S)$ so that the strike always falls on a node.

Note 8.16.2. call 價格隨 σ 嚴格遞增（vega > 0，Section 9.5），從 $\sigma \to 0$ 的 $\max(S - Xe^{-r\tau}, 0)$ 一路升到 $\sigma \to \infty$ 的 S。所以由中間值定理，市場價格只要落在這兩者之間，就恰好有一個 σ 對應到它。實際求解用 Newton–Raphson，以 vega 作為 f'。投影片上的兩個問題：

- *American options* 呢？沒有公式可以反解，每一步都要用 tree 定價。[27]
- 為什麼 τ 很小時很難？ $\tau \to 0$ 時 vega $\to 0$，價格幾乎不隨 σ 變動，價格的一點誤差就會讓解出的 σ 差很多。

Implied volatility is

> *the wrong number to put in the wrong formula to get the right price of plain-vanilla options.*[28]

Think of it as an alternative to quoting option prices. It is often preferred to historical (statistical) volatility in practice. Is using the historical volatility like driving a car with your eyes on the rearview mirror?[29] Note that volatility is meaningful only if seen through a model![30]

§8.16.i. Problems; the Smile

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Options written on the same underlying asset usually do not yield the same implied volatility. A typical pattern is a **smile** in relation to the strike price:[31] the implied volatility is lowest for at-the-money options and becomes higher the further the option is in- or out-of-the-money. This is common for foreign exchange options.

Other patterns have also been observed. For stock options, low-strike options tend to have higher implied volatilities (a **skew**).

- One explanation is the high demand for insurance provided by out-of-the-money puts.
- Another is that volatility rises when the stock falls,[32] making in-the-money calls more likely to become in the money again.

Remark 8.16.3. **波動率微笑**（volatility smile）：同一標的、同一到期日、不同 strike 反解出的 implied vol 對 X 作圖。Black–Scholes 若正確應為水平線；外匯常見 smile，股票常見 **skew**（低 strike 的 implied vol 較高），即低 strike 的 option 比單一 σ 的 Black–Scholes 價格貴。根本原因是真實 log return 的左尾比 normal 厚（Figure 1.4），簡報給兩個來源：

- **保險需求：**大家買低 strike 的 OTM put 防大跌，價格被推高；由 put–call parity，同 strike 的 call 反解出相同的 implied vol。
- **leverage effect：**債務固定時，公司價值 V 的漲跌全由股票 E 吸收，$\sigma_E \approx \sigma_V V/E$（Section 10.1）。股價跌、$E$ 變小，V/E 變大，股票波動就變大。

TXO（台指選擇權，標的 TAIEX，2015/9/25 收 8132）：簡報的 3D 圖是當天各 strike（7000–9000）、各到期日（0–300 天）的 call 反解出的 implied vol，紅線是 at the money。固定到期日看，strike 越低 implied vol 越高，就是 skew。

[27] 課本 Exercise 9.4.2。
[28] Rebonato (2004).
[29] E.g., 1:16:23 of https://www.youtube.com/watch?v=8TJQhQ2GZ0Y.
[30] Alexander (2001).
[31] Alexander (2001).
[32] This is called the **leverage effect** (Black, 1992).

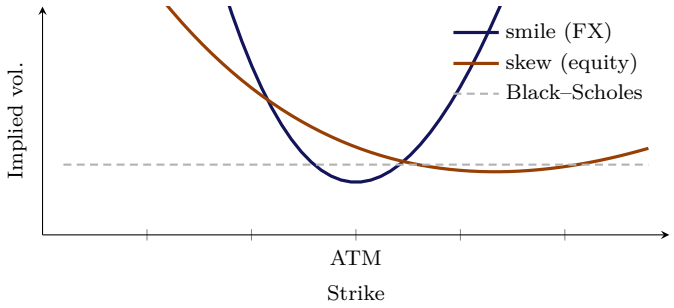


Figure 8.13.: Schematic shapes of implied volatility against strike. The slides show real data for TXO calls on September 25, 2015 (TAIEX closed at 8132), where implied volatility ranged from about 14% to 24% across strikes and maturities.

To address this issue, volatilities are often combined to produce a composite implied volatility. This practice is not sound theoretically. The existence of different implied volatilities for options on the same underlying asset shows the Black–Scholes model is not literally true.

Remark 8.16.4. **composite implied volatility**：把各 strike 的 implied vol 加權平均成一個數（常用 vega 加權，讓 at the money 的權重大），當成這支股票的 σ。理論上站不住：smile 本身就說明 Black–Scholes 不對，平均後代回公式也無法同時重現所有 option 的市場價格。比較好的做法是保留整個 implied vol 曲面，依 strike、到期日各取各的值。

A single stock has a single volatility in the Black–Scholes model, so the smile is a measurement of the model's error, not of the stock. By Theorem 6.9.16, a non-flat smile is the market telling us its risk-neutral density of S_τ is *not* log-normal — typically with fatter tails, exactly as Figure 1.4 showed for realized returns.

§8.17. Binomial Tree Algorithms for American Puts

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Early exercise has to be considered. The binomial tree algorithm starts with the terminal payoffs $\max\bigl(0,\, X - Su^j d^{n-j}\bigr)$ and applies backward induction. At each intermediate node, compare the payoff if exercised and the continuation value, and keep the larger one.

Remark 8.17.1. Algorithm 8.4（下面）和 Algorithm 8.1 只差內層那一行。時間 j、跌了 i 次的 node 股價是 $Su^{j-i}d^i$：

- 立刻履約拿到 $X - Su^{j-i}d^i$；
- 不履約則拿到 continuation value $\bigl(p\,P[\,i\,] + (1-p)\,P[\,i+1\,]\bigr)/R$，和 European 一樣的 backward induction。

持有者會選比較大的那個，所以取 max。

Algorithm 8.4: Binomial tree algorithm for American puts on a non-dividend-paying stock

Input: $S, u, d, X, n, \hat{r} \quad (u > e^{\hat{r}} > d)$

$R := e^{\hat{r}}$;
$p := (R-d)/(u-d)$;
for $i = 0, 1, \ldots, n$ **do**
  $P[i] := \max\big(0,\ X - Su^{n-i}d^i\big)$;
for $j = n-1$ **to** 0 **do**
  for $i = 0, 1, \ldots, j$ **do**
    $P[i] := \max\Big(\big(p \times P[i] + (1-p) \times P[i+1]\big)/R,\ X - Su^{j-i}d^i\Big)$;
return $P[0]$;

Example 8.17.2. Take the tree of Example 8.12.1 ($S = 160$, $u = 1.5$, $d = 0.5$, $R = 1.2$, $p = 0.7$) with an American put struck at $X = 130$.[33] At time 2, the node $S = 120$ has continuation value $(0.7 \times 0 + 0.3 \times 70)/1.2 = 17.5$, greater than the intrinsic value $130 - 120 = 10$: the put is in the money but should *not* be exercised. The node $S = 40$ has continuation value $(0.7 \times 70 + 0.3 \times 110)/1.2 = 68.33$, lower than the intrinsic value $130 - 40 = 90$: exercise. Continuing, the node $S = 80$ at time 1 is also exercised (50 against a continuation value of 32.71), and the American put is worth

$$\frac{0.7 \times 4.375 + 0.3 \times 50}{1.2} = 15.05,$$

against 9.375 for the European put.

Remark 8.17.3. 數字從哪來。terminal payoff 是 $\max(0, 130 - S_3)$，$S_3 = 540, 180, 60, 20$ 時為 $0, 0, 70, 110$。

- **time 2：**$S = 360$ 的 continuation value 為 0；$S = 120$ 取 17.5，$S = 40$ 取 90（如上）。
- **time 1：**$S = 240$ 的 continuation value 是 $(0.7 \times 0 + 0.3 \times 17.5)/1.2 = 4.375$，履約價值 $130 - 240 < 0$，不履約。$S = 80$ 是 $(0.7 \times 17.5 + 0.3 \times 90)/1.2 = 32.71 < 50$，履約。注意這裡用的是 $S = 40$ 履約後的 90。
- **European：**不能提前履約，$S = 40$ 要用 continuation value 68.33。$S = 80$ 為 $(0.7 \times 17.5 + 0.3 \times 68.33)/1.2 = 27.29$，root 為 $(0.7 \times 4.375 + 0.3 \times 27.29)/1.2 = 9.375$。

§8.17.i. Bermudan Options

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Some American options can be exercised only at discrete time points instead of continuously. They are called **Bermudan options**. Their pricing algorithm is identical to that for American options, but early exercise is considered for only those nodes when early exercise is permitted.

Remark 8.17.4. 百慕達式選擇權（Bermudan option）：只能在事先約定的幾個日期提前履約，介於歐式（只能到期履約）與美式（隨時可履約）之間——就像百慕達在地理上介於歐洲與美國之間。演算法上只在允許履約的那幾欄取 max。

§8.18. Time-Dependent Parameters

§8.18.i. Time-Dependent Volatility

Slides pp. 321–323

Suppose the (instantaneous) volatility can change over time but is otherwise predictable: $\sigma(t)$

[33] The numerical example in Section 9.5 of the textbook.

instead of σ.[34] Variances of independent increments add, so in the limit the variance of $\ln(S_\tau/S)$ is

$$\int_0^\tau \sigma^2(t)\,dt$$

rather than $\sigma^2\tau$. The annualized volatility to be used in the Black–Scholes formula should now be

$$\sqrt{\frac{\int_0^\tau \sigma^2(t)\,dt}{\tau}},$$

the root-mean-square of $\sigma(t)$ — *not* its average.

For the binomial model, u and d now depend on time:

$$u \;=\; e^{\sigma(t)\sqrt{\tau/n}}, \qquad d \;=\; e^{-\sigma(t)\sqrt{\tau/n}}.$$

But then the tree no longer combines: an up move at time t_1 followed by a down move at t_2 multiplies the price by $e^{(\sigma(t_1)-\sigma(t_2))\sqrt{\Delta t}}$, and the reverse order by its reciprocal.[35] One way out is to let the period lengths vary so that $\sigma(t_i)\sqrt{\Delta t_i}$ stays constant; then every period moves $\ln S$ by the same $\pm$ step and the lattice is regular again.

Remark 8.18.1. (1) **變異數為什麼是積分**。把 $[0,\tau]$ 切成長度 Δt 的小段，第 i 段的 log return 變異數是 $\sigma^2(t_i)\Delta t$（和固定 σ 時 τ 年變異數是 $\sigma^2\tau$ 同理）。各段獨立，變異數相加：$\sum_i \sigma^2(t_i)\Delta t \to \int_0^\tau \sigma^2(t)\,dt$（Riemann sum）。

(2) **等效的** σ。Black–Scholes 只認得「τ 年的變異數 $= \sigma^2\tau$」，所以令 $\bar\sigma^2\tau = \int_0^\tau \sigma^2(t)\,dt$，解出 $\bar\sigma = \sqrt{\frac{1}{\tau}\int_0^\tau \sigma^2(t)\,dt}$。被平均的是 σ^2 而不是 σ，所以是 root-mean-square。

(3) **tree 為什麼不 combine**。先在 t_1 漲、再在 t_2 跌，股價乘上 $e^{\sigma(t_1)\sqrt{\Delta t}}\cdot e^{-\sigma(t_2)\sqrt{\Delta t}}$；順序反過來則乘上 $e^{-\sigma(t_1)\sqrt{\Delta t}}\cdot e^{\sigma(t_2)\sqrt{\Delta t}}$。兩者只在 $\sigma(t_1)=\sigma(t_2)$ 時相等。

(4) **修正**。讓每期長度 Δt_i 不同，使 $\sigma(t_i)\sqrt{\Delta t_i}$ 都等於同一個常數 c；每步都是 $\ln S \pm c$，(3) 的兩個乘數就相等了。

The CBOE S&P 500 Volatility Index (**VIX**) tracks the 30-day implied volatility of S&P 500 index options; its history over 1990–2016 (plotted on the slides) shows how far from constant volatility actually is, with spikes above 80 during the 2008 crisis.

§8.18.ii. Time-Dependent Short Rates

Slides p. 324

Suppose the short rate (i.e., the one-period spot rate or forward rate) changes over time but predictably. The annual riskless rate r in the Black–Scholes formula should be the spot rate with a time to maturity equal to τ. In other words,

$$r \;=\; \frac{\sum_{i=0}^{n-1} r_i}{\tau},$$

where r_i is the continuously compounded short rate measured in periods for period i, i.e., the one-period forward rate.

Remark 8.18.2. 為什麼是平均。第 i 期的折現因子是 e^{-r_i}，整段的折現因子是 $\prod_i e^{-r_i} = e^{-\sum_i r_i}$。Black–Scholes 用的是 $e^{-r\tau}$，令兩者相等得 $r\tau = \sum_i r_i$。r_i 以「期」為單位，$\sum_i r_i$ 是整段的總利息，除以年數 τ 才是年利率。

[34] Merton (1973).
[35] Amin (1991); C. I. Chen (R98922127) (2011).

This is (4.19) again: the zero-coupon bond maturing at expiration is worth $\exp\left(-\sum_i r_i\right)$ per dollar, so it carries all the interest-rate information the European option needs.

Note 8.18.3. binomial tree 會不會因此不再 combine？不會。short rate 只透過 $R_i = e^{r_i}$ 進入 $p_i = (R_i - d)/(u - d)$；由 u 和 d 建出的價格 lattice 完全不受影響。backward induction 只要在第 i 期改用 p_i 和 R_i 即可 — 前提是每一期都滿足 $d < R_i < u$。

§8.18.iii. Trading Days and Calendar Days

Slides pp. 325–327

Interest accrues based on the calendar day, but σ is usually calculated based on trading days only: stock prices seem to have lower volatilities when the exchange is closed.[36] How to harmonize these two different times in the Black–Scholes formula and binomial tree algorithms?[37]

Think of σ as measuring the annualized volatility of stock price one year from now, and suppose a year has m (say 253) trading days. Variance accumulates only on trading days, so the variance up to expiration is $\sigma^2 \times$ (trading days to expiration)$/m$. Writing this in calendar time, as $\sigma_{\text{eff}}^2 \times$ (calendar days to expiration)$/365$, we can replace σ in the Black–Scholes formula with[38]

$$\sigma_{\text{eff}} = \sigma\sqrt{\frac{365}{m} \times \frac{\text{number of trading days to expiration}}{\text{number of calendar days to expiration}}}.$$

Remark 8.18.4. 為什麼要 σ_{eff}。 Black–Scholes 只有一個 τ，卻用在兩處：折現 $e^{-r\tau}$ 和變異數 $\sigma^2\tau$。利息每個日曆日都算，波動卻只發生在交易日。τ 取日曆時間 $T_{\text{cal}}/365$ 折現才對，但這樣 $\sigma^2\tau$ 會把休市日也算進變異數。所以 τ 不動、改 σ：找 σ_{eff} 使公式裡的變異數等於真正的變異數，

$$\sigma_{\text{eff}}^2 \cdot \frac{T_{\text{cal}}}{365} = \sigma^2 \cdot \frac{T_{\text{tr}}}{m},$$

右邊是每個交易日 σ^2/m、共 T_{tr} 天（$T_{\text{tr}}, T_{\text{cal}}$ 為到期前的交易日數、日曆日數），解出來就是上式。

This works only for European options. How about binomial tree algorithms?[39] One natural answer: let each period be one trading day, with the stock moving $\pm\sigma/\sqrt{m}$ in logarithm, but discount each period over its calendar length — three days of interest for the period spanning a weekend. Since only p and R change from period to period, the tree still combines, by the previous subsection.

Remark 8.18.5. tree 的做法。 tree 可以把兩種時間分開處理，不需要 σ_{eff}。每一期是一個交易日：波動用交易時間 $\Delta t = 1/m$，所以 $u = e^{\sigma/\sqrt{m}}$、$d = 1/u$ 每期相同；利息用那一期的日曆天數，例如週五到週一那期 $R = e^{3r/365}$，其他期 $R = e^{r/365}$。$p = (R - d)/(u - d)$ 因此隨期數改變，但 u, d 不變，tree 照樣 recombine（上一小節的 time-dependent short rate）。這樣也能處理 American option，而 σ_{eff} 只能用在 European 的 Black–Scholes 公式。

§8.19. Options on a Stock That Pays Dividends

Slides p. 328

Early exercise must be considered: by Theorem 7.7.3, an American call may now be exercised just before an ex-dividend date. The **proportional dividend payout model**, in which the

[36] Fama (1965); K. French (1980); K. French & Roll (1986).
[37] Recall Section 2.10 about dating issues.
[38] D. French (1984).
[39] Contributed by Mr. Lu, Zheng-Liang (D00922011) in 2015.

dividend amount is a constant proportion of the prevailing stock price, is tractable: the price goes from S to $Su(1-\delta)$ or $Sd(1-\delta)$ in a period containing an ex-dividend date, and the tree still combines.[40] In general, however, the corporate dividend policy is a complex issue.

§8.19.i. Known Dividends

Slides pp. 329–330

Constant dividends introduce complications. Use D to denote the amount of the dividend, and suppose an ex-dividend date falls in the first period. At the end of that period, the possible stock prices are $Su - D$ and $Sd - D$. Follow the stock price one more period: the number of possible stock prices is not three but four,

$$(Su - D)\,u, \quad (Su - D)\,d, \quad (Sd - D)\,u, \quad (Sd - D)\,d.$$

The binomial tree no longer combines, because $(Su - D)\,d \neq (Sd - D)\,u$ — the timing of the dividend now matters.

Remark 8.19.1. 展開得 $(Su-D)\,d = Sud - Dd$、$(Sd-D)\,u = Sud - Du$，兩者差 $D(u-d) \neq 0$。股利是固定金額：先扣再乘，D 會被之後的 u 或 d 放大或縮小，扣的時間點不同結果就不同。

With m ex-dividend dates the number of terminal nodes is at least 2^m.

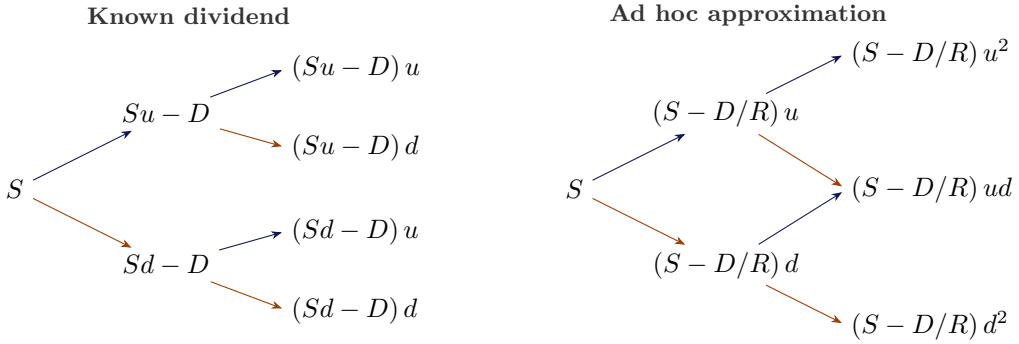


Figure 8.14.: Left: a known dividend D paid in the first period splits the tree into four terminal prices. Right: the ad hoc approximation of Section 8.19.ii, after the PV of the dividend has been added back at the root (by time 1 the dividend has been paid, so nothing is added back there). It recombines, but with a narrower range.

§8.19.ii. An Ad-Hoc Approximation

Slides pp. 331–337

Use the Black–Scholes formula with the stock price reduced by the PV of the dividends.[41] This essentially decomposes the stock price into a riskless one paying known dividends and a risky one:

- the riskless component at any time is the PV of future dividends during the life of the option;
- σ is the volatility of the process followed by the risky component.

The stock price, between two adjacent ex-dividend dates, follows the same lognormal distribution.

On a tree this becomes three steps:

[40] See Sections 9.6.1–9.6.2 of the textbook.
[41] Roll (1977); Heath & Jarrow (1988).

(1) start with the current stock price minus the PV of future dividends before expiration;
(2) develop the binomial tree for the new stock price as if there were no dividends;
(3) add back to each stock price on the tree the PV of all future dividends before expiration.

American option prices can be computed as before on this tree of stock prices, checking early exercise at each node.

The trees are different (Figure 8.14), and so are the stock prices at maturity: $(Su-D)u$, $(Su-D)d$, $(Sd-D)u$, $(Sd-D)d$ against $(S-D/R)u^2$, $(S-D/R)ud$, $(S-D/R)d^2$.[42] Note that, as $d < R < u$,

$$\begin{aligned}(Su-D)\,u-(S-D/R)\,u^2 &= Du\Big(\frac{u}{R}-1\Big) > 0,\\ (Sd-D)\,d-(S-D/R)\,d^2 &= Dd\Big(\frac{d}{R}-1\Big) < 0.\end{aligned}$$

So the ad hoc approximation has a smaller dynamic range. This explains why in practice the volatility is usually increased when using it.

Remark 8.19.2. 這一小節在講什麼。上一小節的問題是：固定金額的股利 D 讓 tree 不再 recombine。這裡用近似繞過它：

(1) **拆股價**。$S = (S - D/R) + D/R$。D/R 是確定會拿到的股利的現值，沒有風險；只有 $S - D/R$ 會隨機變動，假設它像一支不發股利的股票（lognormal）。
(2) **只對隨機部分建 tree**。它不發股利，所以是普通的 recombining tree；每個 node 再加回「之後還沒發的股利的現值」，就是那個 node 的股價（time 1 時股利已發，不用加）。European option 直接用 Black–Scholes 並把 S 換成 $S - D/R$；American option 在這棵 tree 上照常檢查 early exercise。
(3) **代價**。和真實的 tree 比，到期股價的範圍比較窄（上面的不等式：最高價較低、最低價較高），因為整段隨機成長只作用在較小的 $S - D/R$ 上。所以實務上會把 σ 調大一點補回來。

A general approach. A new tree structure handles known dividends with no approximation assumptions, together with a mathematical proof that the tree can always be constructed.[43] Its actual performance is quadratic except in pathological cases. Other approaches include adjusting σ and approximating the known dividend with a dividend yield.[44]

Remark 8.19.3. 一般做法。Dai & Lyuu 設計了一種新的 tree 結構，直接處理固定金額的股利，不做 Section 8.19.ii 那種近似，並證明這種 tree 一定建得出來（分支機率都落在 $[0,1]$）。除了極端的參數外，running time 是 $O(n^2)$，而不是不 recombine 時的至少 2^m。其他做法包括調整 σ，以及把固定股利 D 換成等價的 dividend yield（按比例發放，tree 就會 recombine）；後者對 American option 效果好，對 European option 則不然。細節在簡報 pp. 792 以後。

§8.19.iii. Continuous Dividend Yields

Slides pp. 338–339

In the **continuous-payout model**, dividends are paid continuously. This approximates a broad-based stock market portfolio, in which some company pays a dividend nearly every day. The payment of a continuous dividend yield at rate q reduces the growth rate of the stock price by q: a stock that grows from S to S_τ with a continuous dividend yield of q would have grown from S to $S_\tau e^{q\tau}$ without the dividends.

【中】continuous-payout model 假設股利連續發放，很接近大盤指數（幾乎每天都有公司配息）。以比率 q 連續配息會讓股價成長率少 q：有配息時股價從 S 漲到 S_τ，不配息的話會漲到 $S_\tau e^{q\tau}$。

[42] Contributed by Mr. Yang, Jui-Chung (D97723002) on March 18, 2009.
[43] T. Dai (B82506025, R86526008, D8852600) & Lyuu (2004). Also Arealy & Rodrigues (2013).
[44] Geske & Shastri (1985). It works well for American options but not European ones (T. Dai, 2009).

So a European option has the same value as one on a stock with price $Se^{-q\tau}$ that pays no dividends — in pricing European options, only the distribution of S_τ matters. The Black–Scholes formulas thus hold with S replaced by $Se^{-q\tau}$:[45]

$$C = Se^{-q\tau}N(x) - Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \tag{8.25}$$
$$P = Xe^{-r\tau}N\big(-x+\sigma\sqrt{\tau}\big) - Se^{-q\tau}N(-x), \tag{8.26}$$

where now

$$x \triangleq \frac{\ln(S/X) + \big(r - q + \sigma^2/2\big)\tau}{\sigma\sqrt{\tau}}. \tag{8.27}$$

Formulas (8.25) and (8.26) remain valid as long as the dividend yield is predictable, with q the average annualized yield over the option's life. A foreign currency is the prime example, with q the foreign interest rate (Section 10.3).

【中】所以 European option 的價值等於標的是「價格 $Se^{-q\tau}$、不配息的股票」時的價值，因為 European option 只看 S_τ 的分布；於是 Black–Scholes 公式把 S 換成 $Se^{-q\tau}$ 即可，得到 (8.25)–(8.27)。只要殖利率可預測就適用，q 取存續期間的平均年化殖利率；最典型的例子是外幣，q 是外國利率。

Remark 8.19.4. 連續股利殖利率（continuous dividend yield）q。

(1) **股價少成長** q。股利以每年 q 的比例從股價中發出去，股票的 total return（股價 + 股利）不變，所以股價本身的成長率少 q。若把股利都拿去買回股票，股數以速率 q 成長，τ 年後 1 股變成 $e^{q\tau}$ 股。

(2) **為什麼換成** $Se^{-q\tau}$。今天買 $e^{-q\tau}$ 股並把股利再投資，到期剛好有 1 股，價值 S_τ。所以「到期拿到 S_τ」今天值 $Se^{-q\tau}$，等同一支價格 $Se^{-q\tau}$、不發股利的股票。European option 只看 S_τ，所以把公式裡的 S 換成 $Se^{-q\tau}$ 即可。

(3) x **的變化**。(8.24) 裡的 $\ln(S/X)$ 變成 $\ln(Se^{-q\tau}/X) = \ln(S/X) - q\tau$，併入後面就是 (8.27) 的 $(r - q + \sigma^2/2)\tau$。

Slides pp. 340–341

To run binomial tree algorithms there are two equivalent-in-spirit options, both resting on the fact that the stock price grows at an expected rate of $r - q$ in a risk-neutral economy.

(1) Replace u with $ue^{-q\Delta t}$ and d with $de^{-q\Delta t}$, where $\Delta t = \tau/n$; other than this, the algorithm stays the same — in particular, p should use the *original* u and d![46]

(2) Keep u and d unchanged, but pick the risk-neutral probability as

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}. \tag{8.28}$$

Either way $E^\pi\big[S_{t+\Delta t}\big] = Se^{(r-q)\Delta t}$: in (1), $p\,ue^{-q\Delta t} + (1-p)\,de^{-q\Delta t} = Re^{-q\Delta t}$ by (8.7); in (2) it is the definition of p.

【中】tree 上有兩種做法，根據都是 risk-neutral 下股價期望成長率為 $r-q$：(1) u, d 換成 $ue^{-q\Delta t}$、$de^{-q\Delta t}$，其他不變，但 p 仍用原本的 u, d 算；(2) u, d 不變，p 改用 (8.28)。兩者都讓 $E^\pi[S_{t+\Delta t}] = Se^{(r-q)\Delta t}$。

Remark 8.19.5. 為什麼是 $r - q$。risk-neutral 下 total return 要是 r；股利拿走 q，股價本身只剩 $r - q$。(1) 把每期的股價乘上 $e^{-q\Delta t}$，也就是扣掉這一期發的股利；(2) 直接選 p 使股價的期望成長率是 $r - q$，即 (8.7) 中的 R 換成 $e^{(r-q)\Delta t}$。

Note 8.19.6. 為什麼 (1) 裡的 p 必須用原本的 u, d？持有股票的人拿到的是股價加上 dividend，在 p 之下必須賺到 R 的是這個 total return。total return 是 u 或 d；ex-dividend 價格則是 $ue^{-q\Delta t}$ 或 $de^{-q\Delta t}$。若用縮小後的乘數算 p，就等於讓 ex-dividend 價格本身賺到 R，也就是投資人除了 riskless rate 之外還白拿了 dividend。

[45] Merton (1973).
[46] Contributed by Ms. Wang, Chuan-Ju (F95922018) on May 2, 2007.

§8.20. Distribution of Logarithmic Returns of TAIEX

Slide p. 342

The slide plots the histogram of the daily log returns (in %) of TAIEX from January 3, 2003 to July 13, 2018 against a fitted normal density. The histogram has a higher central peak and heavier tails than the normal, the same picture as Figure 1.4 for the S&P 500. So the lognormal limit of Lemma 8.14.15 is an idealization, and the smile of Section 8.16 is one of its symptoms.

【中】簡報畫了 TAIEX 從 2003/1/3 到 2018/7/13 每日 log return (%) 的 histogram，並與配適的 normal density 比較：中間的峰比 normal 高、兩邊的尾巴比 normal 厚，和 S&P 500 一樣。所以 Lemma 8.14.15 的 lognormal 只是理想化，smile 就是它不完全成立的徵兆。

§8.21. Exercise Boundaries of American Options

Slides pp. 343–344

In the continuous-time model, an American put should be exercised as soon as the stock falls to a critical price $S^*(t)$, the **exercise boundary**.[47]

- The exercise boundary is a nondecreasing function of t for American puts (Figure 8.15).
- The exercise boundary is a nonincreasing function of t for American calls.
- The exercise boundary may be approximated by multipiece exponential functions.[48]

Section 10.2.iii asks whether this makes American options barrier options.

【中】連續時間模型中，American put 應在股價一跌到臨界價格 $S^*(t)$ 就履約，$S^*(t)$ 叫 exercise boundary。put 的 boundary 隨 t 遞增（Figure 8.15），call 的隨 t 遞減；boundary 可用分段指數函數近似。之後 Section 10.2.iii 會問：這樣 American option 算不算 barrier option？

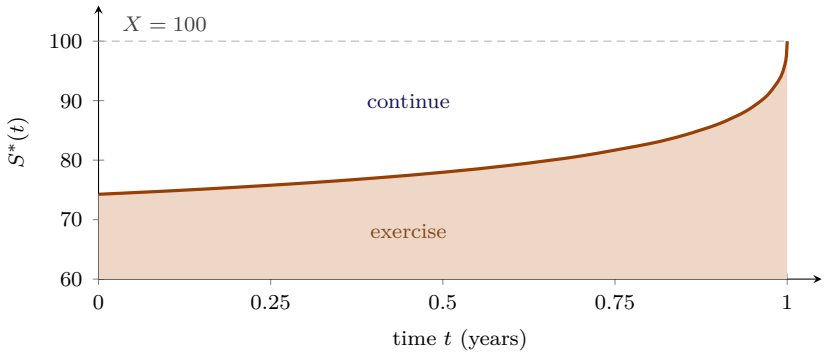


Figure 8.15.: Exercise boundary of an American put with $X = 100$, $r = 8\%$, $\sigma = 30\%$, expiring at $t = 1$, computed by Algorithm 8.4 with 600 periods and bisection on S. Below the curve, immediate exercise is optimal. The boundary rises to X at expiration.

【中】圖：$X = 100$、$r = 8\%$、$\sigma = 30\%$、一年到期的 American put，用 600 期的 Algorithm 8.4 加上對 S 二分搜尋求出 boundary。曲線下方立刻履約最好；到期時 boundary 升到 X。

Intuition . Why does the put's boundary rise towards X? Waiting keeps the chance that the stock recovers, and that chance is worth less the less time remains; exercising early earns interest on X. As expiration approaches the first effect dies, so exercise becomes optimal at prices ever closer to the strike. For an American call the roles of the two effects are taken by dividends and interest on X, which reverses the direction.

[47] See Section 9.7 of the textbook for the tree analog.
[48] Ju (1998).

Remark 8.21.1. 和到期時間的關係（以 put 為例）。在時間 t、股價 $S < X$ 時，比較「現在履約」和「繼續持有」（用 European put P 近似持有的價值）。由 put–call parity $P = C + Xe^{-r\tau} - S$，

$$\underbrace{(X - S)}_{\text{現在履約}} - \underbrace{P}_{\text{繼續持有}} = \underbrace{X\big(1 - e^{-r\tau}\big)}_{\text{早拿 } X \text{ 多賺的利息}} - \underbrace{C(S,\tau)}_{\text{放棄的 call}}.$$

也就是說，持有 put 等於「到期才拿 $X - S_\tau$」再加一個 call（股價若反彈到 X 以上，持有者不會虧）。現在履約賺到利息、放棄這個 call；利息比 C 大就該履約。

- **為什麼要股價夠低**。S 越低，股價反彈過 X 的機會越小，C 越小，越該履約。boundary $S^*(t)$ 就是兩者相等的股價。
- **為什麼和 τ 有關**。τ 長時股價有很多時間反彈，C 不小，要 S 跌很深才值得履約，所以 boundary 遠低於 X。$\tau \to 0$ 時，$S < X$ 的 call 價值以約 $e^{-c/\tau}$ 的速度消失，比利息 $\approx rX\tau$ 快得多，所以 S 只要略低於 X 就該履約，boundary 升到 X。這就是 put 的 $S^*(t)$ 遞增。

call（有股利時）是鏡像：現在履約賺到之後的股利，代價是提早付 X（損失利息）並放棄一個 put（股價跌破 X 時的保護）。τ 長時要 S 遠高於 X 才履約，到期時降到 X，所以遞減。

9. Sensitivity Analysis of Options

Slides pp. 345–346

Cleopatra's nose, had it been shorter,
the whole face of the world
would have been changed.
— Blaise Pascal (1623–1662)

Chapter 8 produced a number. This chapter asks how that number moves when each input moves — which is what a trader holding the option actually needs to know, since it is the movements that have to be hedged.
【中】上一章算出一個價格；這一章問每個輸入變動時價格怎麼變。這才是持有 option 的 trader 真正需要的，因為要避險的就是這些變動。

Remark 9.0.1 (讀法)**.** 簡報只給每個 Greek 的公式和圖。這些公式都是對 (8.22) 做偏微分，而且全部用同一個技巧（Lemma 9.1.3 和 Remark 9.1.4）。標成「補充」的 lemma、proof、remark 是筆記補的推導，第一次讀可以只記公式和直觀。

§9.1. Sensitivity Measures ("The Greeks")

Slide p. 347

How the value of a security changes relative to changes in a given parameter is key to hedging. Duration (Section 3.3.i), for instance, is the bond's sensitivity to its yield.
【中】證券價值對參數變動的反應是避險的關鍵；例如 duration 就是債券對 yield 的敏感度。

Remark 9.1.1. **希臘字母**（the Greeks）：選擇權價格對各個參數的偏導數——對股價是 Delta Δ、對時間是 Theta Θ、對股價的二階導數是 Gamma Γ、對波動率是 Vega Λ、對利率是 Rho ρ。它們之於選擇權，就像存續期間與凸性之於債券。

Notation . Throughout, x is defined by (8.24),

$$x \;=\; \frac{\ln(S/X) + \big(r + \sigma^2/2\big)\tau}{\sigma\sqrt{\tau}},$$

N is the standard normal distribution function (5.14), and its derivative

$$N'(y) \;=\; \frac{e^{-y^2/2}}{\sqrt{2\pi}} \;>\; 0$$

is the density function of the standard normal distribution.

Remark 9.1.2. x **的意義**。本章的 x 都是 (8.24)；$N(z) = \text{Prob}[Z \le z]$ 是 standard normal 的 CDF（(5.14)），N' 是它的導數，也就是 density。x 本身不好解讀，看 $x - \sigma\sqrt{\tau}$ 比較清楚：由 Lemma 8.14.15，risk-neutral 下 $\ln S_\tau \sim N\big(\ln S + (r - \sigma^2/2)\tau,\ \sigma^2\tau\big)$，所以

$$x - \sigma\sqrt{\tau} = \frac{\ln S + (r - \sigma^2/2)\tau - \ln X}{\sigma\sqrt{\tau}} = \frac{E^\pi[\ln S_\tau] - \ln X}{\text{sd}(\ln S_\tau)},$$

是「到期 log 股價的平均比 $\ln X$ 高幾個標準差」，一個 z-score：分子是現在離 strike 多遠 $(\ln(S/X))$ 加上到期前的預期漂移，分母是到期前 log 股價的標準差。它越大越可能到價內，而 $N(x-\sigma\sqrt{\tau}) = \text{Prob}^{\pi}[S_\tau > X]$（Section 8.14.vii）。$x$ 只是把平均再往上移 $\sigma^2\tau$（改用 pu/R 那個機率），$N(x)$ 就是 call 的 delta。$x \gg 0$ 是深度價內，$x \ll 0$ 是深度價外。

Every closed-form Greek below follows from one identity, which is worth isolating.
【中】下面所有 Greek 的公式都來自同一個恆等式，先把它獨立出來。

Lemma 9.1.3 (補充)

$$SN'(x) \;=\; Xe^{-r\tau}N'\big(x-\sigma\sqrt{\tau}\big). \tag{9.1}$$

Proof（補充）. (1) **相除**。$N'(y) = e^{-y^2/2}/\sqrt{2\pi}$，$\sqrt{2\pi}$ 消掉，指數相減：

$$\frac{N'(x-\sigma\sqrt{\tau})}{N'(x)} = \exp\Big(-\frac{(x-\sigma\sqrt{\tau})^2}{2} + \frac{x^2}{2}\Big) = \exp\Big(x\sigma\sqrt{\tau} - \frac{\sigma^2\tau}{2}\Big).$$

(2) **化簡指數**。(8.24) 兩邊乘 $\sigma\sqrt{\tau}$：$x\sigma\sqrt{\tau} = \ln(S/X)+(r+\sigma^2/2)\tau$，減去 $\sigma^2\tau/2$ 得 $\ln(S/X)+r\tau$，所以比值是 $\frac{Se^{r\tau}}{X}$。
(3) 兩邊乘上 $Xe^{-r\tau}N'(x)$ 即得 (9.1)。

■

Remark 9.1.4 (補充：Greeks 的通用算法)**.** 對 (8.22) 的任一參數 $\theta \in \{S, \tau, \sigma, r\}$ 微分，分成兩部分：
(1) **顯式部分：**把 $N(\cdot)$ 當常數，只對 S 和 $Xe^{-r\tau}$ 微分。
(2) **經過 $N(\cdot)$ 的部分：**chain rule 產生

$$SN'(x)\,\frac{\partial x}{\partial\theta} - Xe^{-r\tau}N'\big(x-\sigma\sqrt{\tau}\big)\Big(\frac{\partial x}{\partial\theta} - \frac{\partial(\sigma\sqrt{\tau})}{\partial\theta}\Big).$$

由 (9.1)，$Xe^{-r\tau}N'(x-\sigma\sqrt{\tau}) = SN'(x)$，所以兩個 $\frac{\partial x}{\partial\theta}$ 項抵銷，只剩 $SN'(x)\frac{\partial(\sigma\sqrt{\tau})}{\partial\theta}$。所以 $\frac{\partial C}{\partial\theta} = (1) + SN'(x)\frac{\partial(\sigma\sqrt{\tau})}{\partial\theta}$，完全不用算 $\frac{\partial x}{\partial\theta}$。

§9.2. Delta

Slide p. 348

Definition 9.2.1.

$$\Delta \;\triangleq\; \frac{\partial f}{\partial S},$$

where f is the price of the derivative and S is the price of the underlying asset.

【中】Delta 是 derivative 價格 f 對標的價格 S 的偏導數。

The delta of a portfolio of derivatives on the same underlying asset is the sum of their individual deltas, by elementary calculus. The delta used in the BOPM, (8.3), is the discrete analog. The delta of a long stock is 1.
【中】同一標的上的 derivatives 組成的 portfolio，delta 是各自 delta 的和（微分是線性的）。BOPM 的 h (8.3) 是 delta 的離散版本。一股股票的 delta 是 1。

Slides pp. 349–350

Proposition 9.2.2

The delta of a European call on a non-dividend-paying stock equals

$$\frac{\partial C}{\partial S} = N(x) > 0. \tag{9.2}$$

The delta of a European put equals

$$\frac{\partial P}{\partial S} = N(x) - 1 = -N(-x) < 0. \tag{9.3}$$

【中】不配息股票的 European call 的 delta 是 $N(x) > 0$，put 的是 $N(x) - 1 = -N(-x) < 0$。

Proof（補充；簡報只給公式）. 用 Remark 9.1.4，$\theta = S$：(1) 顯式部分只有 $SN(x)$ 前面的 S，微分得 $N(x)$；(2) $\sigma\sqrt{\tau}$ 不含 S，那一項是 0。所以 $\partial C/\partial S = N(x)$。put：$P = C - S + Xe^{-r\tau}$ 對 S 微分得 $N(x) - 1 = -N(-x)$。 ■

So the deltas of a call and an otherwise identical put cancel each other when $N(x) = 1/2$, i.e., when $x = 0$, i.e., when

$$X = Se^{(r+\sigma^2/2)\tau}. \tag{9.4}$$

The straddle (Section 6.10) $C + P$ then has zero delta!

【中】call 和同條件 put 的 delta 在 $N(x) = \frac{1}{2}$，即 $x = 0$、$X = Se^{(r+\sigma^2/2)\tau}$ 時互相抵銷，此時 straddle $C + P$ 的 delta 為 0。

Remark 9.2.3. (9.4) **的推導**。straddle 的 delta 是 $N(x) + \big(N(x) - 1\big) = 2N(x) - 1$，為 0 時 $N(x) = \frac{1}{2}$。N 是 standard normal 的 CDF，$N(0) = \frac{1}{2}$ 且嚴格遞增，所以 $x = 0$。由 (8.24)，$x = 0 \Leftrightarrow \ln(S/X) = -(r + \sigma^2/2)\tau \Leftrightarrow X = Se^{(r+\sigma^2/2)\tau}$。

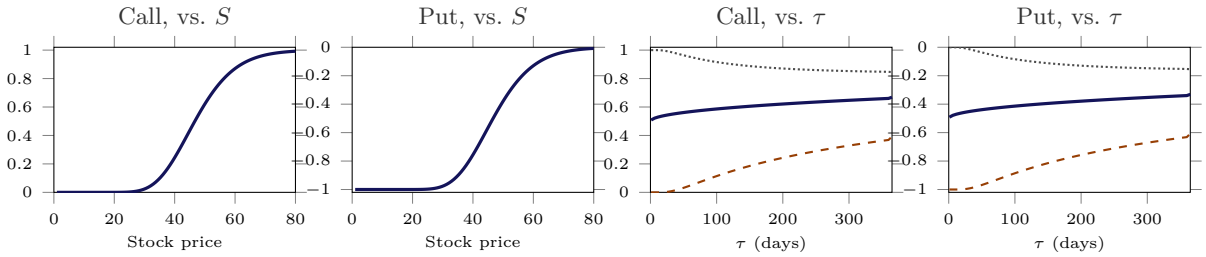


Figure 9.1.: Call and put deltas, $X = 50$, $\sigma = 30\%$, $r = 8\%$. Left two: $\tau = 201$ days. Right two: $S = 60$ (dotted), 50 (solid), 40 (dashed). Each put curve is the call curve shifted down by 1, by (9.3).

【中】圖：$X = 50$、$\sigma = 30\%$、$r = 8\%$ 的 call 和 put delta。左兩張：$\tau = 201$ 天，對股價作圖；右兩張：對剩餘天數 τ 作圖，$S = 60$（點線）、50（實線，at the money）、40（虛線）。$S = 60$ 對 call 是 in the money、對 put 是 out of the money，$S = 40$ 相反。由 (9.3)，每一條 put 曲線都是它上方的 call 曲線往下移 1——左右兩張都成立，因為 $\partial P/\partial S = \partial C/\partial S - 1$ 對每一組 (S, τ) 都成立。

Intuition . $N(x)$ is the hedge ratio of Section 8.3 in the limit, and it reads like a probability of exercise. Deep in the money the call is the stock minus a bond, so $\Delta \to 1$; deep out of the money it is nothing, so $\Delta \to 0$. Near expiration there is no time left to change sides, so the curves on the right fan out to 1, 1/2 and 0.

【中】$N(x)$ 是 Section 8.3 的 hedge ratio 取極限，也可讀成「履約的機率」。深度價內的 call 就像股票減債券，$\Delta \to 1$；深度價外幾乎沒價值，$\Delta \to 0$。接近到期時沒時間翻盤，所以右圖三條線分別趨向 1、$\frac{1}{2}$、0。

§9.2.i. Dividends, Straddles and Strangles

Slides pp. 351–353

Suppose the stock pays a continuous dividend yield of q, and define x by (8.27). Then

$$\frac{\partial C}{\partial S} = e^{-q\tau}N(x) > 0, \qquad \frac{\partial P}{\partial S} = -e^{-q\tau}N(-x) < 0,$$

since (8.25) is (8.22) evaluated at $Se^{-q\tau}$.

【中】有連續股利 q 時（x 用 (8.27)），call 和 put 的 delta 都多乘 $e^{-q\tau}$，因為 (8.25) 就是把 (8.22) 的 S 換成 $Se^{-q\tau}$。

Remark 9.2.4 (補充)**.** 記 $C_0(S')$ 為不發股利的 (8.22)。(8.25) 就是 $C_0(Se^{-q\tau})$（(8.27) 的 x 也正是 (8.24) 代 $S' = Se^{-q\tau}$）。chain rule：

$$\frac{\partial C}{\partial S} = C_0'(Se^{-q\tau}) \cdot \frac{\partial (Se^{-q\tau})}{\partial S} = N(x)\, e^{-q\tau}.$$

put 同理。

Now consider an X_1-strike call and an X_2-strike put, $X_1 \geq X_2$, otherwise identical. Let

$$x_i \triangleq \frac{\ln(S/X_i) + \big(r - q + \sigma^2/2\big)\tau}{\sigma\sqrt{\tau}}. \tag{9.5}$$

Their deltas $e^{-q\tau}N(x_1)$ and $-e^{-q\tau}N(-x_2)$ sum to zero when $x_1 = -x_2$, i.e., when $\ln(S/X_1) + \ln(S/X_2) = -(2r - 2q + \sigma^2)\tau$:

$$\frac{S}{X_1} = \frac{X_2}{S}\, e^{-(2r-2q+\sigma^2)\,\tau}. \tag{9.6}$$

The strangle (Section 6.10) $C + P$ then has zero delta! Demanding $X_1 = X_2 = X$ gives back a straddle, and (9.6) becomes $X = Se^{(r-q+\sigma^2/2)\tau}$, generalizing (9.4).

【中】strike X_1 的 call 和 strike X_2 的 put（$X_1 \geq X_2$），各自用 (9.5) 定義 x_i。兩者 delta 相加為 0 的條件是 $x_1 = -x_2$，整理得 (9.6)，此時 strangle $C + P$ 的 delta 為 0。$X_1 = X_2 = X$ 就是 straddle，條件變成 $X = Se^{(r-q+\sigma^2/2)\tau}$，推廣了 (9.4)。

Remark 9.2.5 (補充)**.** (9.6) **的推導**。

(1) $e^{-q\tau}N(x_1) - e^{-q\tau}N(-x_2) = 0 \Leftrightarrow N(x_1) = N(-x_2) \Leftrightarrow x_1 = -x_2$（$N$ 嚴格遞增）。
(2) $x_1 + x_2 = 0$ 乘上 $\sigma\sqrt{\tau}$，由 (9.5)：$\ln(S/X_1) + \ln(S/X_2) + 2(r - q + \sigma^2/2)\tau = 0$。
(3) 移項並取 exp：$\ln \frac{S}{X_1} = -\ln \frac{S}{X_2} - (2r - 2q + \sigma^2)\tau$，即 $\frac{S}{X_1} = \frac{X_2}{S} e^{-(2r-2q+\sigma^2)\tau}$。
(4) $X_1 = X_2 = X$ 時：$(S/X)^2 = e^{-(2r-2q+\sigma^2)\tau}$，開根號得 $X = Se^{(r-q+\sigma^2/2)\tau}$。

§9.2.ii. Delta Neutrality

Slide p. 354

A position with a total delta equal to 0 is **delta-neutral**. A delta-neutral portfolio is immune to small price changes in the underlying asset, so creating one serves hedging purposes.

- A portfolio consisting of a call and $-\Delta$ shares of stock is delta-neutral.
- Short Δ shares of stock to hedge a long call; long Δ shares of stock to hedge a short call.
- In general, hedge a position in a security with delta Δ_1 by shorting Δ_1/Δ_2 units of a security with delta Δ_2.

【中】總 delta 為 0 的部位叫 delta-neutral，對標的價格的小變動免疫，所以可用來避險：call 加上 $-\Delta$ 股股票就是 delta-neutral；long call 就 short Δ 股，short call 就 long Δ 股；一般而言，持有 delta 為 Δ_1 的證券，就 short Δ_1/Δ_2 單位 delta 為 Δ_2 的證券。

Remark 9.2.6. **Delta 中立**（delta-neutral）：整個部位的 delta 加總為 0。股價變動 dS 時部位價值變動約 (總 delta) $\cdot\, dS$，所以總 delta 為 0 時價值幾乎不變。

- **避險比例**。持有 1 單位 delta 為 Δ_1 的證券，再放空 h 單位 delta 為 Δ_2 的證券，總 delta 是 $\Delta_1 - h\Delta_2$；令它為 0 得 $h = \Delta_1/\Delta_2$。
- **特例**。用股票避險時 $\Delta_2 = 1$，所以 $h = \Delta_1$：long call 就 short Δ 股。

Note 9.2.7. 關鍵在「small」：delta neutrality 是一階的敘述，只在瞬間成立。S 變動時 delta 也會跟著變 — 變化率就是 gamma（Section 9.4）— 所以 hedge 必須 rebalance，這就是 Section 8.12.i 的連續時間版本。

§9.3. Theta (Time Decay)

Slide p. 355

Definition 9.3.1. **Theta** is the rate of change of a security's value with respect to time,

$$\Theta \triangleq -\frac{\partial f}{\partial \tau} = \frac{\partial f}{\partial t}.$$

The minus sign is because τ is time *to expiration*: as calendar time t passes, τ shrinks.

【中】Theta 是證券價值隨時間的變化率。有負號是因為 τ 是距到期的時間：日曆時間 t 往前走，τ 就變小。

Remark 9.3.2. Theta 是偏導數：股價 S（以及 σ, r）固定不變時，價值隨時間的變化率。它不是「價值平均會怎麼變」：S 本身也會動，而 backward induction 的定義 $C = \frac{pC_u + (1-p)C_d}{R}$ 說的是 $E^{\pi}[C_{\text{下一期}}] = RC$，平均以 riskless rate 成長。所以下面 call 的 $\Theta < 0$ 並不表示該提前履約（Theorem 7.7.1）。

Proposition 9.3.3

For a European call on a non-dividend-paying stock,

$$\Theta = -\frac{SN'(x)\,\sigma}{2\sqrt{\tau}} - rXe^{-r\tau}N\big(x - \sigma\sqrt{\tau}\big) < 0,$$

and for a European put,

$$\Theta = -\frac{SN'(x)\,\sigma}{2\sqrt{\tau}} + rXe^{-r\tau}N\big(-x + \sigma\sqrt{\tau}\big).$$

【中】不配息股票的 European call 的 theta 恆負；put 的多了正項 $rXe^{-r\tau}N(-x+\sigma\sqrt{\tau})$，可正可負。

Proof（補充；簡報只給公式）. 用 Remark 9.1.4，$\theta = \tau$：
(1) **顯式部分：** $\frac{\partial}{\partial\tau}\big(-Xe^{-r\tau}\big) = rXe^{-r\tau}$，乘上 $N(x - \sigma\sqrt{\tau})$。
(2) $\sigma\sqrt{\tau}$ **部分：** $\frac{\partial(\sigma\sqrt{\tau})}{\partial\tau} = \frac{\sigma}{2\sqrt{\tau}}$，乘上 $SN'(x)$。
所以 $\frac{\partial C}{\partial\tau} = \frac{SN'(x)\sigma}{2\sqrt{\tau}} + rXe^{-r\tau}N(x-\sigma\sqrt{\tau}) > 0$，$\Theta = -\frac{\partial C}{\partial\tau} < 0$。put：parity 給 $\frac{\partial P}{\partial\tau} = \frac{\partial C}{\partial\tau} - rXe^{-r\tau}$，再用 $N(x-\sigma\sqrt{\tau}) - 1 = -N(-x+\sigma\sqrt{\tau})$。$S \to 0$ 時 $N'(x) \to 0$、$N(-x+\sigma\sqrt{\tau}) \to 1$，所以 $\Theta_{\text{put}} \to rXe^{-r\tau} > 0$。 ■

The call loses value with the passage of time. The put's theta can be negative or positive: deep in the money the second term wins, which is the European put worth less than its intrinsic value in Figure 6.4 — waiting only delays receipt of X.

【中】call 的價值隨時間流失。put 的 theta 可正可負：深度價內時第二項較大、theta 為正，對應 Figure 6.4 中 European put 低於 intrinsic value 的情況——再等只會延後拿到 X。

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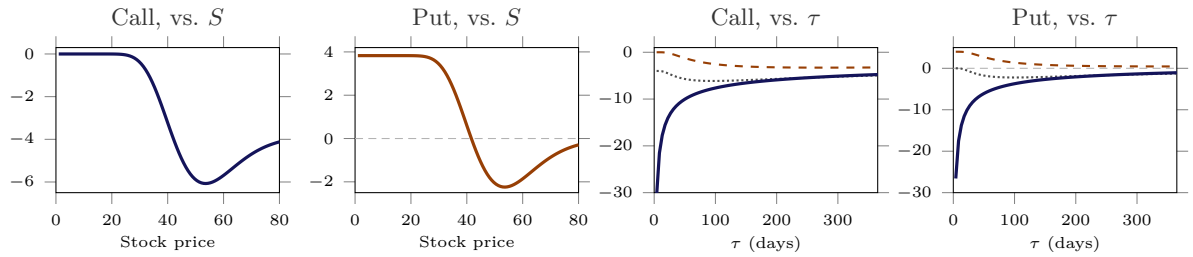


Figure 9.2.: Theta (per year), $X = 50$, $\sigma = 30\%$, $r = 8\%$. Left two: $\tau = 201$ days. Right two: $S = 60$ (dotted), 50 (solid), 40 (dashed). The call's theta is always negative; the put's turns positive deep in the money, tending to $rXe^{-r\tau}$. At the money the theta plunges as $\tau \to 0$.

【中】圖：$X = 50$、$\sigma = 30\%$、$r = 8\%$ 的年化 theta。左兩張：$\tau = 201$ 天，call 和 put 對股價作圖；右兩張：對剩餘天數 τ 作圖，$S = 60$（點線）、50（實線，at the money）、40（虛線）。call 的 theta 恆負、在價平附近最負；put 的在深度價內（$S = 40$）變正，趨近 $rXe^{-r\tau}$。價平的 theta 在 $\tau \to 0$ 時急遽變負：快到期時價平 option 的時間價值流失得最快。

Suppose the stock pays a continuous dividend yield of q, with x as in (8.27). Then each formula gains an extra term:

$$\Theta_{\text{call}} = -\frac{SN'(x)\,\sigma\, e^{-q\tau}}{2\sqrt{\tau}} - rXe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big) + qSe^{-q\tau}N(x),$$
$$\Theta_{\text{put}} = -\frac{SN'(x)\,\sigma\, e^{-q\tau}}{2\sqrt{\tau}} + rXe^{-r\tau}N\big(-x+\sigma\sqrt{\tau}\big) - qSe^{-q\tau}N(-x).$$

The extra term is the derivative of $Se^{-q\tau}$ with respect to $-\tau$: a dividend-paying call gains from the passage of time insofar as fewer dividends remain to be paid out.[1]

【中】有連續股利 q 時兩個公式各多一項，即 $Se^{-q\tau}$ 對 $-\tau$ 的導數：時間過去，剩下要發的股利變少，對 call 有利。

Remark 9.3.4 (補充). **多出來的項**。$C = Se^{-q\tau}N(x) - Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$ 的顯式部分多了

$$\frac{\partial}{\partial\tau}\big(Se^{-q\tau}\big) = -qSe^{-q\tau},$$

乘上 $N(x)$；$\Theta = -\frac{\partial C}{\partial\tau}$ 就多出 $+qSe^{-q\tau}N(x)$。第一項的 $e^{-q\tau}$ 則是因為 identity 中的 S 換成了 $Se^{-q\tau}$（見 footnote）。

§9.4. Gamma

Slide p. 358

Definition 9.4.1. Gamma is the rate of change of the delta with respect to the price of the underlying asset,

$$\Gamma \triangleq \frac{\partial^2 f}{\partial S^2}.$$

[1] The slides write the first term without $e^{-q\tau}$; with x from (8.27) the identity (9.1) becomes $Se^{-q\tau}N'(x) = Xe^{-r\tau}N'(x-\sigma\sqrt{\tau})$, and the factor $e^{-q\tau}$ belongs there. Cf. the vega with dividends in Section 9.5.

It measures how sensitive delta is to changes in the price of the underlying asset. In practice, a portfolio with a high gamma needs be rebalanced more often to maintain delta neutrality. Roughly, delta $\sim$ duration and gamma $\sim$ convexity (Section 3.10).

【中】Gamma 是 delta 對標的價格的變化率（f 對 S 的二階導數），衡量 delta 對股價有多敏感。gamma 高的 portfolio 要更頻繁 rebalance 才能維持 delta neutral。大致上 delta 對應 duration、gamma 對應 convexity。

Proposition 9.4.2

The gamma of a European call or put on a non-dividend-paying stock is

$$\Gamma \;=\; \frac{N'(x)}{S\sigma\sqrt{\tau}} \;>\; 0.$$

【中】不配息股票的 European call 和 put 有相同的 gamma $N'(x)/(S\sigma\sqrt{\tau}) > 0$。

Proof（補充；簡報只給公式）. 由 (8.24)，$x = \frac{\ln S - \ln X + (r+\sigma^2/2)\tau}{\sigma\sqrt{\tau}}$，只有 $\ln S$ 含 S，所以 $\frac{\partial x}{\partial S} = \frac{1}{S\sigma\sqrt{\tau}}$。對 (9.2) 微分：$\frac{\partial N(x)}{\partial S} = N'(x)\frac{\partial x}{\partial S} = \frac{N'(x)}{S\sigma\sqrt{\tau}}$。put 的 delta 只差常數 -1，gamma 相同。 ■

Remark 9.4.3. Gamma 恆正，表示歐式選擇權的價值是 S 的凸函數（與 Figure 6.4 一致）。持有 gamma 為正的部位並做 delta 避險，股價大漲大跌都會賺一點（凸性），代價則是 theta 每天流失的時間價值。

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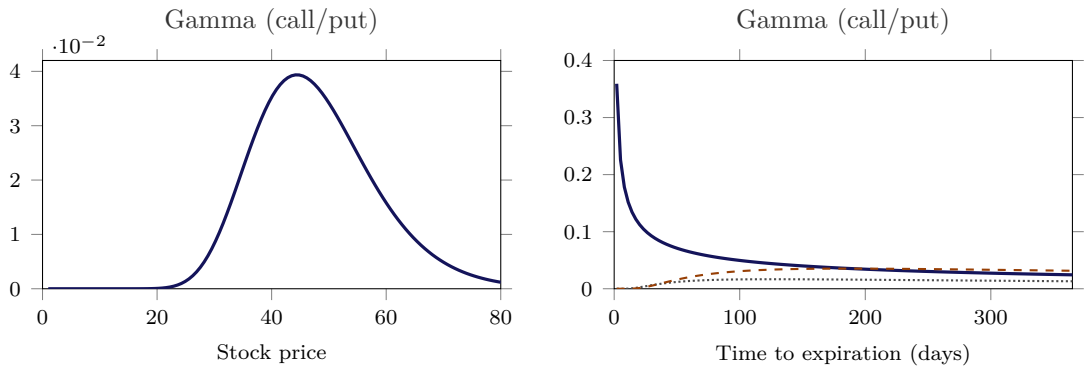


Figure 9.3.: Gamma with $X = 50$, $\sigma = 30\%$, $r = 8\%$; right: $S = 60$ (dotted), 50 (solid), 40 (dashed). The at-the-money gamma grows like $1/\sqrt{\tau}$ as $\tau \to 0$; the others die out.

【中】圖：$X = 50$、$\sigma = 30\%$、$r = 8\%$ 的 gamma；右圖 $S = 60, 50, 40$ 對剩餘天數作圖。價平的 gamma 在 $\tau \to 0$ 時像 $1/\sqrt{\tau}$ 一樣變大，其他的趨近 0。

Gamma is maximized when the option is nearly at the money, namely at

$$S \;=\; Xe^{-(r+3\sigma^2/2)\,\tau}.$$

Indeed $\partial \ln\Gamma/\partial S = -x/(S\sigma\sqrt{\tau}) - 1/S$, which vanishes at $x = -\sigma\sqrt{\tau}$; substituting into (8.24) gives the formula. As the at-the-money option approaches expiration, its gamma tends to rise, like $1/\sqrt{\tau}$; the gammas of other options, however, tend to zero, because $N'(x) \to 0$ faster as $|x| \to \infty$.

【中】Gamma 在接近價平、$S = Xe^{-(r+3\sigma^2/2)\tau}$ 時最大（令 $\partial\ln\Gamma/\partial S = 0$ 得 $x = -\sigma\sqrt{\tau}$）。價平 option 接近到期時 gamma 變大，其他的趨近 0，因為 $|x| \to \infty$ 時 $N'(x)$ 衰減得更快。

Remark 9.4.4 (補充). (1) **最大值**。$\ln\Gamma = \ln N'(x) - \ln S - \ln(\sigma\sqrt{\tau})$，而 $\ln N'(x) = -\frac{x^2}{2} - \ln\sqrt{2\pi}$。對 S 微分，用 $\frac{\partial x}{\partial S} = \frac{1}{S\sigma\sqrt{\tau}}$：

$$\frac{\partial \ln\Gamma}{\partial S} = -x\cdot\frac{1}{S\sigma\sqrt{\tau}} - \frac{1}{S} = 0 \Leftrightarrow x = -\sigma\sqrt{\tau}.$$

代入 (8.24)：$\ln(S/X) + (r+\sigma^2/2)\tau = -\sigma^2\tau$，即 $S = Xe^{-(r+3\sigma^2/2)\tau}$。

(2) **at the money**，$\tau\to 0$。$S = X$ 時 $x = \frac{(r+\sigma^2/2)\sqrt{\tau}}{\sigma} \to 0$，$N'(x)\to\frac{1}{\sqrt{2\pi}}$，所以 $\Gamma \approx \frac{1}{S\sigma\sqrt{2\pi\tau}}$，像 $1/\sqrt{\tau}$ 一樣變大。

(3) **其他情況**，$\tau\to 0$。$S\neq X$ 時 $x\approx\frac{\ln(S/X)}{\sigma\sqrt{\tau}}$，$|x|\to\infty$；$N'(x) = e^{-x^2/2}/\sqrt{2\pi}$ 以 $e^{-c/\tau}$ 的速度趨近 0，比分母的 $\sqrt{\tau}$ 快得多，所以 $\Gamma\to 0$。

Intuition . Near expiration an at-the-money option's delta is about to jump from 0 to 1 across the strike (the right panel of Figure 9.1); a jump is an infinite slope, so the delta's slope — gamma — explodes there. This is the hedger's nightmare: at-the-money options close to expiry need constant rebalancing.

【中】接近到期時，價平 option 的 delta 會在 strike 兩側從 0 跳到 1（Figure 9.1 右圖）；跳躍就是斜率無限大，所以 delta 的斜率 gamma 在那裡爆掉。這是避險者的惡夢：快到期的價平 option 要不停 rebalance。

§9.5. Vega

Slide p. 361

Definition 9.5.1. **Vega** (also lambda, kappa, sigma, zeta) is the rate of change of a security's value with respect to the volatility of the underlying asset,

$$\Lambda \triangleq \frac{\partial f}{\partial\sigma}.$$

Vega is not Greek. Alexander (2001): "This is a term that was invented by Americans, and intended to sound like a Greek letter."

Volatility often changes over time (Section 8.18), and a security with a high vega is very sensitive to changes in, or estimation error of, the volatility.

【中】Vega 是證券價值對標的波動率的變化率（vega 不是希臘字母，是美國人發明、聽起來像希臘字母的詞）。波動率常隨時間改變，vega 高的證券對波動率的變動或估計誤差非常敏感。

Proposition 9.5.2

The vega of a European call or put on a non-dividend-paying stock is

$$\Lambda = S\sqrt{\tau}\,N'(x) > 0.$$

【中】不配息股票的 European call 和 put 有相同的 vega $S\sqrt{\tau}N'(x) > 0$。

Proof（補充；簡報只給公式）. 用 Remark 9.1.4，$\theta = \sigma$：(1) S 和 $Xe^{-r\tau}$ 都不含 σ，顯式部分是 0；(2) $\frac{\partial(\sigma\sqrt{\tau})}{\partial\sigma} = \sqrt{\tau}$，乘上 $SN'(x)$。所以 $\Lambda = S\sqrt{\tau}N'(x)$。put：$P = C - S + Xe^{-r\tau}$ 中只有 C 含 σ。 ■

So higher volatility raises the option value, which incidentally guarantees the uniqueness of the implied volatility in Section 8.16.

【中】所以波動率越高 option 越值錢；這也保證 Section 8.16 的 implied volatility 唯一。

Slides pp. 362–364

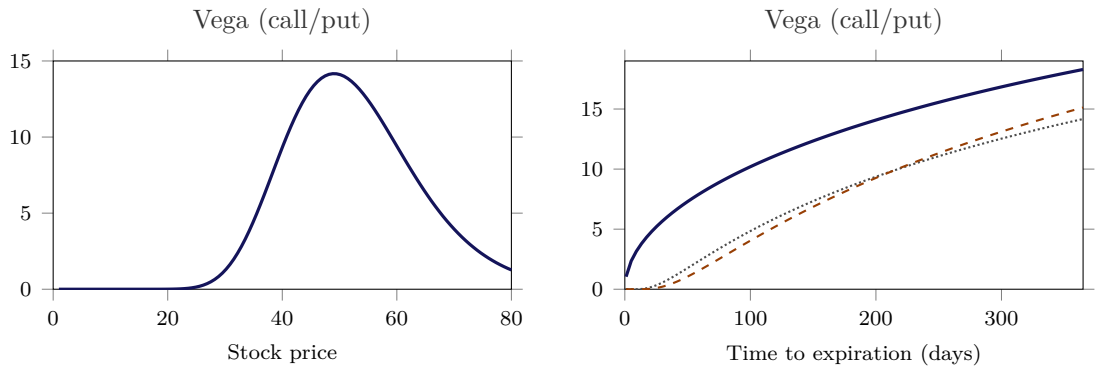


Figure 9.4.: Vega with $X = 50$, $\sigma = 30\%$, $r = 8\%$; right: $S = 60$ (dotted), 50 (solid), 40 (dashed). Every curve vanishes as $\tau \to 0$.

【中】圖：$X = 50$、$\sigma = 30\%$、$r = 8\%$ 的 vega；右圖 $S = 60, 50, 40$ 對剩餘天數作圖，每條都在 $\tau \to 0$ 時趨近 0。

Note that[2]

$$\Lambda \;=\; \tau\sigma S^2\,\Gamma, \tag{9.7}$$

as the two formulas show at a glance: $\tau\sigma S^2 \cdot \frac{N'(x)}{S\sigma\sqrt{\tau}} = S\sqrt{\tau}\,N'(x)$. If the stock pays a continuous dividend yield of q, then

$$\Lambda \;=\; Se^{-q\tau}\sqrt{\tau}\,N'(x),$$

where x is defined in (8.27).

【中】注意 $\Lambda = \tau\sigma S^2\Gamma$，比較兩個公式即可看出。有連續股利 q 時，vega 變成 $Se^{-q\tau}\sqrt{\tau}N'(x)$。

Vega is maximized when $x = 0$, i.e., when $S = Xe^{-(r-q+\sigma^2/2)\tau}$, and declines very fast as S moves away from that peak.

【中】Vega 在 $x = 0$，即 $S = Xe^{-(r-q+\sigma^2/2)\tau}$ 時最大，離開峰值後下降得很快（「最大」的精確意思見下面的 note）。

Note 9.5.3. 「Maximized」需要加個條件。固定 S 時，$x = 0$ 使 $N'(x)$ 最大，因此也使 vega 在不同 *strike* X 之間最大。若固定 strike、把 vega 看成 S 的函數，多出來的因子 S 會讓峰值移動：$\ln\Lambda = \ln S - \frac{x^2}{2} + (\text{不含} S \text{ 的項})$，用 $\frac{\partial x}{\partial S} = \frac{1}{S\sigma\sqrt{\tau}}$，

$$\frac{\partial \ln\Lambda}{\partial S} = \frac{1}{S} - \frac{x}{S\sigma\sqrt{\tau}} = \frac{1}{S}\Big(1 - \frac{x}{\sigma\sqrt{\tau}}\Big),$$

在 $x = \sigma\sqrt{\tau}$ 時為零。代入 (8.27)：$\ln(S/X) + (r - q + \sigma^2/2)\tau = \sigma^2\tau$，即 $S = Xe^{-(r-q-\sigma^2/2)\tau}$ — 這正是 Figure 9.4 左圖的峰值位置。

Now consider a portfolio consisting of an X_1-strike call C and a *short* X_2-strike put P, $X_1 \geq X_2$. Their vegas are $Se^{-q\tau}\sqrt{\tau}N'(x_i)$ with x_i as in (9.5), and they cancel out when $x_1 = -x_2$, since N' is even. This leads to (9.6) — the same condition that gave zero delta for the strangle $C + P$.

【中】long strike X_1 的 call、short strike X_2 的 put（$X_1 \geq X_2$）：因為 N' 是偶函數，兩者 vega 在 $x_1 = -x_2$ 時抵銷，得到的正是 (9.6)，和 strangle delta 為 0 的條件相同。

[2] Reiss & Wystup (2001).

Remark 9.5.4 (補充). long call 的 vega 是 $\Lambda(x_1)$，short put 的是 $-\Lambda(x_2)$，總和 $Se^{-q\tau}\sqrt{\tau}\big(N'(x_1)-N'(x_2)\big)=0 \Leftrightarrow N'(x_1)=N'(x_2)$。$N'$ 是偶函數，所以 $x_1=\pm x_2$；取 $x_1=-x_2$，就和 (9.6) 推導的第一步相同。

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Note that $\tau \to 0$ implies $\Lambda \to 0$, which answers the question in Section 8.16. The Black–Scholes formula also implies

$$C \to S, \qquad P \to Xe^{-r\tau}, \qquad \text{as } \sigma \to \infty,$$

since $x \to \infty$ and $x - \sigma\sqrt{\tau} \to -\infty$. These are the *upper* ends of the ranges allowed by no-arbitrage — recall $C \geq \max(S - Xe^{-r\tau}, 0)$ (the proof of Theorem 7.7.1) and $P \geq \max(Xe^{-r\tau} - S, 0)$ (Lemma 7.6.4), the lower ends, which are the limits as $\sigma \to 0$. These boundary conditions are handy for some numerical methods.

【中】$\tau \to 0$ 時 $\Lambda \to 0$，這回答了 Section 8.16 的問題。由 Black–Scholes 公式，$\sigma \to \infty$ 時 $C \to S$、$P \to Xe^{-r\tau}$，是 no-arbitrage 允許範圍的上界；下界 $\max(S - Xe^{-r\tau}, 0)$、$\max(Xe^{-r\tau} - S, 0)$ 則是 $\sigma \to 0$ 的極限。這些邊界條件對某些數值方法很有用。

Remark 9.5.5 (補充). (1) $\tau \to 0$ **時** $\Lambda \to 0$。$N'(x) \leq \frac{1}{\sqrt{2\pi}}$，所以 $\Lambda = S\sqrt{\tau}N'(x) \leq \frac{S\sqrt{\tau}}{\sqrt{2\pi}} \to 0$。

(2) $\sigma \to \infty$。$x = \frac{\ln(S/X)+r\tau}{\sigma\sqrt{\tau}} + \frac{\sigma\sqrt{\tau}}{2} \to \infty$，$x - \sigma\sqrt{\tau} = \frac{\ln(S/X)+r\tau}{\sigma\sqrt{\tau}} - \frac{\sigma\sqrt{\tau}}{2} \to -\infty$。代入 (8.22)：$C \to S \cdot 1 - Xe^{-r\tau} \cdot 0 = S$；代入 (8.23)：$P \to Xe^{-r\tau} \cdot 1 - S \cdot 0 = Xe^{-r\tau}$。

§9.6. Rho

Slides pp. 366–367

Definition 9.6.1. **Rho** is the rate of change in a security's value with respect to interest rates,

$$\rho \triangleq \frac{\partial f}{\partial r}.$$

Proposition 9.6.2

The rho of a European call and of a European put on a non-dividend-paying stock are, respectively,

$$X\tau e^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big) > 0, \qquad -X\tau e^{-r\tau}N\big(-x+\sigma\sqrt{\tau}\big) < 0.$$

Proof（補充；簡報只給公式）. 用 Remark 9.1.4，$\theta = r$：(1) 顯式部分 $\frac{\partial}{\partial r}\big(-Xe^{-r\tau}\big) = X\tau e^{-r\tau}$，乘上 $N(x-\sigma\sqrt{\tau})$；(2) $\sigma\sqrt{\tau}$ 不含 r，是 0。put：$\frac{\partial}{\partial r}(Xe^{-r\tau}) = -X\tau e^{-r\tau}$，所以 $\frac{\partial P}{\partial r} = X\tau e^{-r\tau}\big(N(x-\sigma\sqrt{\tau})-1\big) = -X\tau e^{-r\tau}N(-x+\sigma\sqrt{\tau})$。 ■

A call holder will pay X later, a put holder will receive it later; raising r shrinks $\mathrm{PV}(X)$, which helps the first and hurts the second.

【中】Rho 是證券價值對利率的變化率。call 的 rho 是 $X\tau e^{-r\tau}N(x-\sigma\sqrt{\tau}) > 0$，put 的是 $-X\tau e^{-r\tau}N(-x+\sigma\sqrt{\tau}) < 0$。直觀：call 持有者之後要付 X，put 持有者之後會收 X；利率上升讓 $\mathrm{PV}(X)$ 變小，對前者有利、對後者不利。

Remark 9.6.3. rho 正負的直覺。rho 是偏導數：S, σ, τ 都固定，只改 r，所以股價漲跌的風險在比較的兩邊完全一樣。把這件事寫清楚：由 Lemma 8.14.15，risk-neutral 下 $S_\tau = S\,e^{(r-\sigma^2/2)\tau+\sigma\sqrt{\tau}Z}$，$Z \sim N(0,1)$。令 $M \triangleq e^{\sigma\sqrt{\tau}Z-\sigma^2\tau/2}$（和 r 無關），則 $e^{-r\tau}S_\tau = SM$，於是

$$C = e^{-r\tau}E^\pi\big[\max(S_\tau - X, 0)\big] = E\big[\max(SM - Xe^{-r\tau}, 0)\big], \qquad P = E\big[\max(Xe^{-r\tau} - SM, 0)\big].$$

股價風險全部在 M 裡，r 只出現在 $Xe^{-r\tau}$。r 上升使 $Xe^{-r\tau}$ 變小：在每一個情境 Z 下，call 的被期望值變大、put 的變小，所以 $\rho_{\text{call}} > 0$、$\rho_{\text{put}} < 0$。直觀上，call 持有者到期才付 X、put 持有者到期才收 X；r 越高，晚付越划算、晚收越吃虧。

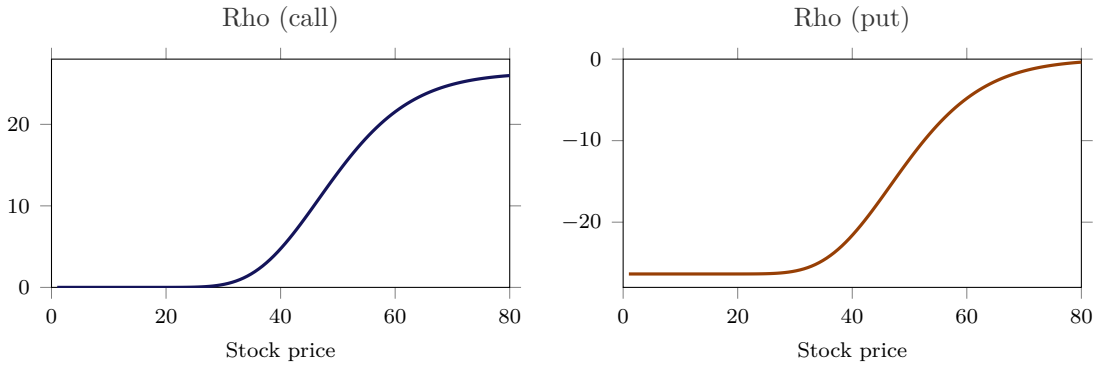


Figure 9.5.: Rho with $X = 50$, $\sigma = 30\%$, $r = 8\%$, $\tau = 201$ days. Deep in the money the call's rho tends to $X\tau e^{-r\tau} \approx 26$, the rate sensitivity of the bond $Xe^{-r\tau}$ it effectively shorts.

【中】圖：$X = 50$、$\sigma = 30\%$、$r = 8\%$、$\tau = 201$ 天的 rho。深度價內 call 的 rho 趨近 $X\tau e^{-r\tau} \approx 26$，就是它等於 short 的那張債券 $Xe^{-r\tau}$ 的利率敏感度。

§9.7. Numerical Greeks

Slide p. 368

Closed forms exist only for the simplest options; for an American put, say, the Greeks must be computed numerically. Take delta as an example. A standard method computes the finite difference

$$\frac{f(S+\Delta S) - f(S-\Delta S)}{2\Delta S},$$

which roughly doubles the computation time of evaluating the derivative itself (two extra pricings).

【中】只有最簡單的 option 有 closed form，例如 American put 的 Greeks 就得數值計算。以 delta 為例，標準做法是上式的中央差分，要多定價兩次，計算時間約為兩倍。

§9.7.i. An Alternative Numerical Delta

Slide p. 369

Use intermediate results of the binomial tree algorithm instead.[3] When the algorithm reaches the end of the first period, f_u and f_d are computed; they are the derivative values at stock prices Su and Sd. Delta is approximated by

$$\frac{f_u - f_d}{Su - Sd}, \tag{9.8}$$

[3] Pelsser & Vorst (1994).

at essentially zero extra cost — stop Algorithm 8.4 one step early and return $\big(P[0]-P[1]\big)/(Su-Sd)$. This is just (8.3).

【中】改用 tree 的中間結果：演算法算到第一期末時，已有股價 Su、Sd 處的價值 f_u、f_d，用 (9.8) 近似 delta，幾乎不用額外成本（Algorithm 8.4 提早一步停下即可）。這就是 (8.3) 的 h。

§9.7.ii. Extended Binomial Tree

Slide p. 370

Strictly speaking, (9.8) is the delta at time Δt, not now. An improved method starts the tree two periods *before* now (Figure 9.6), so that the present time is the third column, with nodes Su/d, S, Sd/u. Delta is then

$$\frac{f_{u/d}-f_{d/u}}{Su/d-Sd/u},$$

and gamma is

$$\frac{\dfrac{f_{u/d}-f}{Su/d-S}-\dfrac{f-f_{d/u}}{S-Sd/u}}{(Su/d-Sd/u)/2},$$

both at the present time.

【中】嚴格來說 (9.8) 是時間 Δt 的 delta，不是現在的。改進法讓 tree 從現在往前兩期開始 (Figure 9.6)，現在就是第三欄，有 Su/d、S、Sd/u 三個 node，可用上面兩式算出現在的 delta 和 gamma。

Remark 9.7.1. (1) **現在那一欄**。tree 從現在往前兩期的 $S/(ud)$ 開始長，兩期後（也就是現在）的三個 node 是 $\frac{S}{ud}u^2 = Su/d$、$\frac{S}{ud}ud = S$、$\frac{S}{ud}d^2 = Sd/u$。所以現在就有三個不同股價的 option 價值可用。

(2) **delta**。用 S 兩側的兩個 node 做差分（中央差分）。

(3) **gamma**。gamma 是「delta 的變化率」。$\frac{f_{u/d}-f}{Su/d-S}$ 是 S 和 Su/d 之間的 delta（約在兩者中點），$\frac{f-f_{d/u}}{S-Sd/u}$ 是 Sd/u 和 S 之間的 delta。兩個中點的距離是

$$\frac{Su/d+S}{2}-\frac{S+Sd/u}{2}=\frac{Su/d-Sd/u}{2},$$

所以 delta 的差除以這個距離就是 gamma。

(9.9)(下一小節)是同樣的公式，只是用第二期的 Suu, Sud, Sdd，中點距離是 $(Suu-Sdd)/2$。

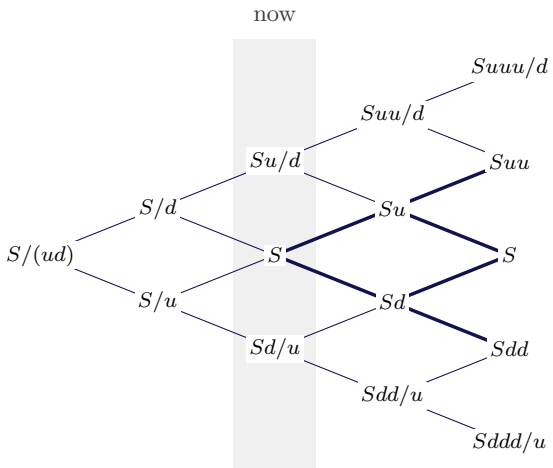


Figure 9.6.: The extended binomial tree: the original tree (bold) with time extended two periods before the present. With $ud = 1$, $S/(ud) = S$, and the present column holds Su^2, S, Sd^2.

【中】圖：原本的 tree（粗線）往前延伸兩期。$ud = 1$ 時 $S/(ud) = S$，現在那一欄是 Su^2、S、Sd^2。

§9.7.iii. Numerical Gamma

Slide p. 371

On the ordinary tree, gamma is read off the second period. At the stock price $(Suu+Sud)/2$, delta is approximately $(f_{uu} - f_{ud})/(Suu - Sud)$; at $(Sud + Sdd)/2$, it is approximately $(f_{ud} - f_{dd})/(Sud - Sdd)$. Gamma is the rate of change in deltas between these two prices:

$$\frac{\dfrac{f_{uu} - f_{ud}}{Suu - Sud} - \dfrac{f_{ud} - f_{dd}}{Sud - Sdd}}{(Suu - Sdd)/2}. \tag{9.9}$$

Remark 9.7.2. 為什麼這樣寫就是二次微分。gamma 是 $\frac{\partial^2 f}{\partial S^2} = \frac{\partial}{\partial S}\left(\frac{\partial f}{\partial S}\right)$，也就是「delta 的變化率」。tree 上沒有 f 的公式，只有第二期三個 node 的值 f_{uu}, f_{ud}, f_{dd}（股價 $Suu > Sud > Sdd$），所以分兩步做差分：

(1) **先算兩個 delta（一次差分）**。相鄰兩點連線的斜率近似它們中點的 f'：

$$\Delta_{\text{上}} = \frac{f_{uu} - f_{ud}}{Suu - Sud} \approx f'\Big(\frac{Suu + Sud}{2}\Big), \qquad \Delta_{\text{下}} = \frac{f_{ud} - f_{dd}}{Sud - Sdd} \approx f'\Big(\frac{Sud + Sdd}{2}\Big).$$

(2) **再對 delta 差分一次**。兩個中點相距 $\frac{Suu+Sud}{2} - \frac{Sud+Sdd}{2} = \frac{Suu-Sdd}{2}$，所以

$$\Gamma \approx \frac{\Delta_{\text{上}} - \Delta_{\text{下}}}{(Suu - Sdd)/2},$$

就是 (9.9)。

和熟悉的公式一樣。若三點等距、間距 h（$Suu - Sud = Sud - Sdd = h$），分子是 $\frac{f_{uu} - 2f_{ud} + f_{dd}}{h}$、分母是 h，得到 $\frac{f_{uu} - 2f_{ud} + f_{dd}}{h^2}$，正是 Section 9.7.v 的二次差分。tree 上三點不等距（間隔是乘上 u/d），所以要用上面的一般形式。注意它估的是第二期、股價 Sud（$ud = 1$ 時就是 S）附近的 gamma，不是現在的；要現在的就用 Figure 9.6 的 extended tree。

§9.7.iv. Alternative Numerical Delta and Gamma

Slide p. 372

Let $\epsilon \triangleq \ln u$ and think in terms of $\ln S$, where the CRR lattice is evenly spaced. Since $\partial f/\partial S = \frac{1}{S}\,\partial f/\partial \ln S$ and $\partial^2 f/\partial S^2 = \frac{1}{S^2}\big(\partial^2 f/\partial(\ln S)^2 - \partial f/\partial \ln S\big)$,

$$\left(\frac{f_u - f_d}{2\epsilon}\right)\frac{1}{S}$$

approximates the numerical delta, and

$$\left(\frac{f_{uu} - 2f_{ud} + f_{dd}}{\epsilon^2} - \frac{f_{uu} - f_{dd}}{2\epsilon}\right)\frac{1}{S^2}$$

approximates the numerical gamma.[4]

【中】令 $\epsilon = \ln u$，改在 $\ln S$ 座標下看，CRR lattice 在這裡是等距的。用 $\partial f/\partial S$、$\partial^2 f/\partial S^2$ 和 $\ln S$ 導數的關係，就得到上面兩個近似式（在 tree 上，gamma 式中的 ϵ 要換成 2ϵ）。

Remark 9.7.3 (補充)**. 兩個 chain rule 公式**。令 $y = \ln S$，即 $S = e^y$，$\frac{dy}{dS} = \frac{1}{S}$。記 $f_y = \partial f/\partial y$。

[4] See p. 692 of the slides, where the grid has spacing ϵ in $\ln S$. On the binomial tree, f_{uu}, f_{ud}, f_{dd} are 2ϵ apart, so ϵ in the gamma formula should then read 2ϵ.

(1) $\frac{\partial f}{\partial S} = f_y \cdot \frac{dy}{dS} = \frac{1}{S} f_y$。
(2) 再對 S 微分一次，用 product rule：$\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S^2} f_y + \frac{1}{S} \cdot f_{yy} \cdot \frac{1}{S} = \frac{1}{S^2}\left(f_{yy} - f_y\right)$。

差分。在 y 座標下 grid 是等距的，所以用一般的中央差分：$f_y \approx \frac{f(y+h)-f(y-h)}{2h}$、$f_{yy} \approx \frac{f(y+h)-2f(y)+f(y-h)}{h^2}$，$h$ 是 grid 間距。delta 用 Su, Sd（$y \pm \epsilon$，$h = \epsilon$）；gamma 的公式是寫給間距 ϵ 的 grid，在 tree 上 Suu, Sud, Sdd 間距 2ϵ，所以要把 ϵ 換成 2ϵ（見 footnote）。

§9.7.v. Finite Difference Fails for Numerical Gamma

Slide p. 373

Numerical differentiation gives

$$\frac{f(S+\Delta S) - 2f(S) + f(S-\Delta S)}{(\Delta S)^2}.$$

It does not work, and in general calculating gamma is a hard problem numerically — recall the floating-point disaster of Figure 3.5. But here the failure is structural, not only floating-point.

Remark 9.7.4. 這一節在做什麼。American put 這類沒有公式的 option，最直接的數值 gamma 是：用 tree 分別算出起始價 $S - \Delta S$、S、$S + \Delta S$ 時的價值，代入上式（二次差分；把 Taylor 展開 $f(S \pm \Delta S) = f \pm f'\Delta S + \frac{1}{2} f''(\Delta S)^2 + \cdots$ 兩式相加即得）。這一節說：ΔS 取小時這個做法會失敗，原因不只是浮點誤差，而是 tree 算出的價值函數本身的形狀（下圖）。

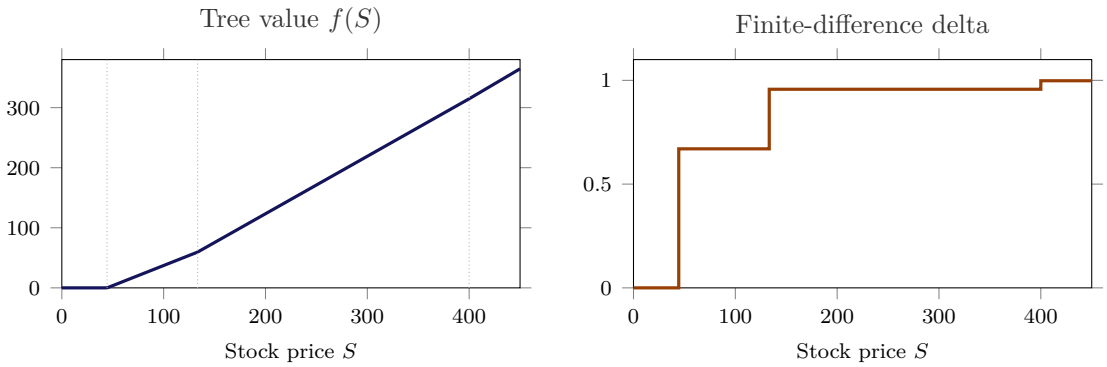


Figure 9.7.: The three-period call of Example 8.12.1 as a function of the initial price S (with u, d, R, X fixed). It is piecewise linear with kinks at $S = X/(u^j d^{3-j})$ (dotted), so a small-step finite-difference delta is a ladder, and the finite-difference gamma is 0 almost everywhere and huge at the kinks.

【中】圖：Example 8.12.1 的三期 call，固定 u, d, R, X、把價值當成起始價 S 的函數。左：價值 $f(S)$，點線是 kink；右：小步長差分算出的 delta，是階梯狀。

Remark 9.7.5. 為什麼會失敗。

(1) **tree 算出的價值是折線**。n, u, d, X 固定時，由 (8.10)，$f(S) = e^{-\hat{r}n} \sum_j b(j; n, p) \max(Su^j d^{n-j} - X, 0)$。每一項在 $S < X/(u^j d^{n-j})$ 時是 0、之後是直線，所以 f 是一段一段直線接起來的折線，kink 在 $S = X/(u^j d^{n-j})$（左圖點線）。[5]
(2) **折線的二次差分**。直線的二次差分是 0。所以 $[S - \Delta S, S + \Delta S]$ 裡沒有 kink 時，算出的 gamma 就是 0；有 kink 時，斜率在那裡跳一下（右圖的階梯），二次差分大約是 $\frac{\text{斜率跳的量}}{\Delta S}$，$\Delta S$ 越小越大。
(3) **結果**。算出來不是 0 就是很大，和真正的 gamma 無關；把 ΔS 調小只會更糟。

[5] 課本 Section 10.2.1。

But why did the binomial tree version (9.9) work? Because its step is not a free parameter: the nodes Suu, Sud, Sdd are spaced by the factor u/d, which is exactly the spacing of the kinks. Each of its two difference quotients spans one full linear piece of the tree's own value function, so the second difference sees every kink with the same weight instead of catching one by accident.[6]

【中】那 (9.9) 為什麼可行？它不是自己挑 ΔS，而是直接用同一棵 tree 第二期的三個 node。這三個股價的間隔（乘上 u/d）剛好和 kink 的間隔一樣，所以兩個斜率各自跨過一整段折線，量到的是 delta 在一整格上的變化，而不是看運氣有沒有碰到 kink。

Differentiate a numerical method on its own grid. A finite difference with a step finer than the method's resolution measures the method's artefacts, not the derivative.

【中】要在數值方法自己的 grid 上做微分。步長比方法的解析度還細的差分，量到的是方法的人工痕跡，而不是導數。

§9.7.vi. Other Numerical Greeks

Slide p. 374

The theta can be computed on the tree as

$$\frac{f_{ud} - f}{2(\tau/n)},$$

since $Sud = S$: the same stock price two periods later.

Remark 9.7.6. theta 是「股價不變時，價值隨時間的變化率」。f 是現在、股價 S 的價值；f_{ud} 是兩期後、股價 $Sud = S$ 的價值。兩者股價相同、時間差 $2\Delta t = 2\tau/n$，所以相減再除以時間差就是 $\partial f/\partial t$ 的差分近似。

In fact, the theta of a European option can be derived from delta and gamma, by an equation we shall meet later in the course. The vega of a European option can be derived from gamma, by (9.7). For rho, there seems no alternative but to run the binomial tree algorithm twice.

【中】theta 可在 tree 上用 $\frac{f_{ud}-f}{2(\tau/n)}$ 算，因為 $Sud = S$ 是兩期後的同一個股價。其實 European option 的 theta 可由 delta 和 gamma 推出（之後的課會看到那條方程式），vega 可由 gamma 用 (9.7) 推出；rho 似乎只能把 tree 跑兩次。

[6] Exercise 10.2.1 of the textbook.

10. Extensions of Options Theory

Slides pp. 375–376

Lecture 5
October 8, 2026

As I never learnt mathematics,
so I have had to think.
— Joan Robinson (1903–1983)

Chapters 8 and 9 priced one security, the plain European option on a stock, and asked how its price moves. This chapter shows how far the same machinery reaches. Three directions:

(i) **a different underlying**: the total value of a firm, which turns its stock and bonds into options, and a foreign currency, which is a stock paying a dividend yield;

(ii) **a different payoff that still depends only on a few states**: barrier options, still easy on a tree;

(iii) **a payoff that depends on the whole path**: Asian options, where the tree stops recombining and the algorithmic questions of Section 8.13 come back with a vengeance.

【中】前兩章只處理以股票為標的的 European option。這一章看同一套方法能延伸多遠：(i) 換標的——公司總價值（股票和債券都變成 option）、外幣（等於有 dividend yield 的股票）；(ii) 換成仍只看少數 state 的 payoff——barrier option，在 tree 上仍然容易；(iii) 看整條 path 的 payoff——Asian option，tree 不再 recombine，Section 8.13 的演算法問題又回來了。

§10.1. Pricing Corporate Securities

Slide p. 377

Interpret the underlying asset as the firm's total value.[1] Then the option pricing methodology can be applied to price corporate securities.[2] The result is Merton's (1974) **structural model**. Its assumptions are:

- A firm can finance payouts by the sale of assets.
- If a promised payment to an obligation other than stock is missed, the claim holders take ownership of the firm and the stockholders get nothing.

【中】把標的資產解讀成公司總價值，就能用 option 定價方法替公司發行的證券定價，這是 Merton (1974) 的 structural model。假設：公司可以賣資產來付款；若沒付到對債權人的承諾款項，債權人接管公司，股東什麼都拿不到。

Remark 10.1.1. **結構式模型**（structural model）：把公司總資產價值 V 當成標的資產，股票與債券都是以 V 為標的的衍生性商品。

- **違約由 V 決定（endogenous）**。違約就是事件 $\{V^* < X\}$：到期時資產不夠還債。一旦給定 V 的模型（例如 V 是 lognormal），違約機率就是 $\Pr[V^* < X]$，從模型算出來，不是另外輸入的參數。
- **對照：外生（exogenous）違約**。另一類模型（reduced-form model）直接假設「每單位時間以機率 λ 違約」，λ 由市場資料校準，和 V 的高低沒有關係。

所以 structural model 能回答「公司為什麼違約」：V 越接近 X、波動 σ 越大，違約機率就越高。

[1] More realistic models posit firm value = asset value + tax benefits − bankruptcy costs (Leland & Toft, 1996).
[2] Black & Scholes (1973); Merton (1974).

§10.1.i. Risky Zero-Coupon Bonds and Stock

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Consider XYZ.com, with a simple capital structure:

- n shares of its own common stock, each worth S;
- zero-coupon bonds with an aggregate par value of X.

What is the value of the bonds, B? What is the value of the XYZ.com stock, S?

【中】XYZ.com 的資本結構：n 股普通股（每股 S）和總面額 X 的 zero-coupon bonds。問題是債券值多少（B）、股票值多少（S）。

On the bonds' maturity date, suppose the total value of the firm V^* is less than the bondholders' claim X. Then the firm declares bankruptcy, and the stock becomes worthless. If $V^* > X$, then the bondholders obtain X and the stockholders $V^* - X$:

	$V^* \le X$	$V^* > X$
Bonds	V^*	X
Stock	0	$V^* - X$

【中】到期日若公司總價值 $V^* < X$，公司破產、股票歸零，債權人拿走 V^*；若 $V^* > X$，債權人拿 X、股東拿 $V^* - X$。

The stock has the same payoff as a call! It is a call on the total value of the firm with a strike price of X and an expiration date equal to the bonds'. This call provides the limited liability for the stockholders.
The bonds pay $\min(V^*, X)$: they are a covered call on the total value of the firm (Section 6.8).

【中】股票的 payoff 和 call 一樣：它是以公司總價值為標的、履約價 X、到期日同債券的 call，這個 call 給了股東 limited liability。債券拿 $\min(V^*, X)$，等於以公司總價值為標的的 covered call。

Remark 10.1.2. **有限責任**（limited liability）：股東最多賠掉投資的錢，不必用個人財產償還公司債務。用選擇權的語言，就是買權的報酬 $\max(V^* - X, 0)$ 不會是負的。

Let V stand for the total value of the firm and C for a call on V with strike X. Thus

$$\begin{aligned} nS &= C \qquad \text{(market capitalization of XYZ.com)}, \\ B &= V - C. \end{aligned}$$

Knowing C amounts to knowing how the value of the firm is divided between stockholders and bondholders. Whatever the value of C, the total value of the stock and bonds at maturity remains V^*, so the relative size of debt and equity is irrelevant to the firm's current value V.

【中】令 V 為公司總價值、C 為以 V 為標的、履約價 X 的 call，則市值 $nS = C$、$B = V - C$。知道 C 就知道公司價值怎麼分給股東和債權人。不論 C 多少，到期時股票加債券總是 V^*，所以負債與股權的比例不影響公司現在的價值 V。

Remark 10.1.3. $nS = C$ **怎麼來**。

(1) **到期 payoff 相同**。全部股票（n 股）到期拿 $\max(V^* - X, 0)$（上表），而以 V 為標的、strike X 的 call 到期也付 $\max(V^* - X, 0)$。
(2) **今天價值相同**。到期 payoff 在每個 state 都一樣，由 Corollary 7.2.4，今天價值也一樣：全部股票的價值 nS 等於 call 的價值 C。
(3) $B = V - C$。股票加債券到期總是拿 V^*，和持有整個公司相同，所以今天 $nS + B = V$，即 $B = V - nS = V - C$。

Note 10.1.4. 最後這句話就是 Modigliani–Miller irrelevance proposition，這裡只用一行 portfolio 論證就得到：股票和債券合起來在每個 state 都支付 V^*，所以由 Corollary 7.2.4，不論 payoff 怎麼切分，兩者合起來的價值都是 V。

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From Theorem 8.14.19 and put–call parity,

$$nS = VN(x) - Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \tag{10.1}$$
$$B = VN(-x) + Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \tag{10.2}$$

where now σ is the volatility of the firm value and

$$x \triangleq \frac{\ln(V/X) + \big(r+\sigma^2/2\big)\tau}{\sigma\sqrt{\tau}}.$$

The second line is $V - C$ with $1 - N(x) = N(-x)$.

【中】由 Black–Scholes 和 put–call parity 得 (10.1)–(10.2)；這裡 σ 是公司價值的波動率，x 是把 S 換成 V。

Remark 10.1.5. (1) (10.1)。$nS = C$，而 C 是標的為 V、strike 為 X 的 call，所以就是 (8.22) 把 S 換成 V（x 也一樣換）。

(2) (10.2)，**兩條路**。

(a) 直接用 $B = V - C$：$B = V\big(1-N(x)\big) + Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$，再用 $1-N(x) = N(-x)$。

(b) 用 put–call parity（Theorem 7.5.1，S 換成 V）：$V - C = Xe^{-r\tau} - P$，代入 (8.23)：$B = Xe^{-r\tau}\big[1 - N(-x+\sigma\sqrt{\tau})\big] + VN(-x)$，再用 $1 - N(-z) = N(z)$。

兩者等價，因為 $B = V - C$ 和 parity 是同一個式子。

The continuously compounded yield to maturity of the firm's bond is $\ln(X/B)/\tau$.

【中】公司債的連續複利 yield 是 $\ln(X/B)/\tau$。

Remark 10.1.6. $\ln(X/B)/\tau$ **怎麼來**。zero-coupon bond 今天價格 B、到期付 X，連續複利 yield y 滿足 $Be^{y\tau} = X$，取對數得 $y = \ln(X/B)/\tau$。riskless bond 價格 $Xe^{-r\tau}$ 代入得 $y = r$；公司債 $B = Xe^{-r\tau} - P$ 較便宜，所以 $y > r$。例：Merck 例子中 $\tau \approx 123/365$，$y = \ln(30{,}000/29{,}250)/\tau \approx 7.5\%$。

Definition 10.1.7. The **credit spread** or **default premium** is the yield difference between risky and riskless bonds, $\ln(X/B)/\tau - r$.

【中】credit spread（default premium）是公司債的 yield 減去 riskless rate。

Proposition 10.1.8

$$\frac{\ln(X/B)}{\tau} - r = -\frac{1}{\tau}\ln\left[N(-z) + \frac{1}{\omega}N\big(z-\sigma\sqrt{\tau}\big)\right], \tag{10.3}$$

where

$$\omega \triangleq Xe^{-r\tau}/V, \qquad z \triangleq \frac{\ln\omega}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau} = -x + \sigma\sqrt{\tau}.$$

【中】credit spread 可以寫成只含 $\omega = Xe^{-r\tau}/V$、σ、τ 的 (10.3)。

Proof（補充；簡報只給結論）. (1) **改寫左邊**。$\frac{\ln(X/B)}{\tau} - r = \frac{1}{\tau}\big(\ln X - \ln B - r\tau\big) = -\frac{1}{\tau}\ln\frac{Be^{r\tau}}{X}$。

(2) $Be^{r\tau}/X$。(10.2) 兩邊除以 $Xe^{-r\tau}$：$\frac{Be^{r\tau}}{X} = \frac{V}{Xe^{-r\tau}}N(-x) + N(x-\sigma\sqrt{\tau}) = \frac{1}{\omega}N(-x) + N(x-\sigma\sqrt{\tau})$。

(3) **換成** $z \circ z = -x + \sigma\sqrt{\tau}$ 代表 $x - \sigma\sqrt{\tau} = -z$、$-x = z - \sigma\sqrt{\tau}$，所以 (2) 是 $N(-z) + \frac{1}{\omega}N(z - \sigma\sqrt{\tau})$。

(4) z **的另一個寫法**。$-x + \sigma\sqrt{\tau}$ 的分子是 $-\ln(V/X) - (r + \sigma^2/2)\tau + \sigma^2\tau = \ln\frac{Xe^{-r\tau}}{V} + \frac{\sigma^2\tau}{2} = \ln\omega + \frac{\sigma^2\tau}{2}$，再除以 $\sigma\sqrt{\tau}$。

r 只透過 $\omega = Xe^{-r\tau}/V$ 出現，所以 spread 只由 ω, σ, τ 決定。 ■

Note that ω is the debt-to-total-value ratio (debt valued at the riskless rate). So the spread depends only on the capital structure ω, the operating risk σ, and the debt maturity τ — not on r separately. It is the model-based counterpart of the static spread of Section 4.8.

【中】ω 是負債（用 riskless rate 折現）占公司總價值的比例。所以 spread 只取決於資本結構 ω、營運風險 σ、負債期限 τ，不另外取決於 r；它是 Section 4.8 的 static spread 在模型裡的對應。

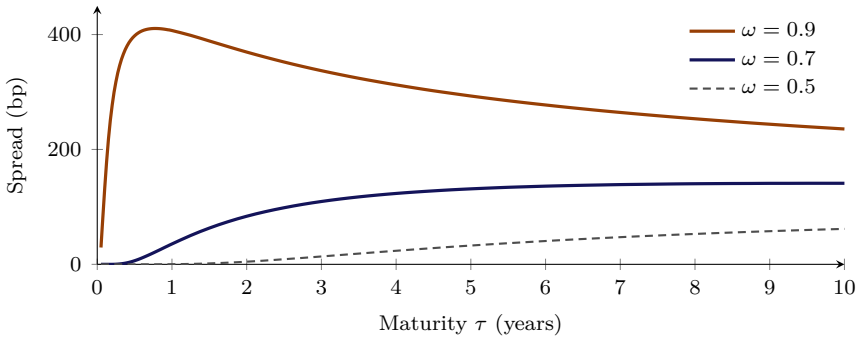


Figure 10.1.: The term structure of credit spreads (10.3) with $\sigma = 20\%$. Low leverage gives a spread rising from zero; high leverage gives a hump. (For $\omega \geq 1$ the spread explodes as $\tau \to 0$: the firm is already insolvent in PV terms.)

【中】圖：$\sigma = 20\%$ 下 credit spread 對期限的曲線。槓桿低時 spread 從 0 開始上升；槓桿高時先升後降。$\omega \geq 1$ 時公司以現值計已資不抵債，$\tau \to 0$ 時 spread 爆掉。

Remark 10.1.9. 槓桿 (leverage)：用借來的錢（債）撐起資產，Figure 10.1 用 $\omega = Xe^{-r\tau}/V$ 衡量，ω 越大槓桿越高。叫「槓桿」是因為它像槓桿一樣放大股東的報酬：股權 $nS = V - B$，V 的漲跌（債權人最多拿 X）大部分落在較小的 nS 上。例：$V = 100$、$B \approx 90$、$nS \approx 10$，V 漲 5% 時 nS 約漲 50%；反之 V 跌一點，股權就大幅縮水、離違約 $\{V^* < X\}$ 也更近，所以 ω 越大 spread 越高。

Intuition . Write $s(\tau)$ for the spread (10.3). Since $\ln(V/X) = -\ln\omega - r\tau$, Lemma 8.14.15 (with V for S) gives, under π,

$$\ln(V^*/X) \;\sim\; N\big(-\ln\omega - \sigma^2\tau/2,\ \sigma^2\tau\big),$$

and default $\{V^* < X\}$ means this falls below 0. So the firm starts a distance $-\ln\omega > 0$ above default, and the noise has size $\sigma\sqrt{\tau}$.

- **Rising.** While $\sigma\sqrt{\tau} \ll -\ln\omega$, default is almost impossible and $s \approx 0$ (a known weakness: observed short spreads are not zero). As $\sigma\sqrt{\tau}$ approaches $-\ln\omega$, i.e. $\tau \approx (\ln\omega/\sigma)^2$ (≈ 0.3, 3, 12 years for $\omega = 0.9, 0.7, 0.5$), default becomes likely and s climbs.
- **Falling.** $s\tau = -\ln(Be^{r\tau}/X)$ is the total discount for default over $[0, \tau]$, and s is its per-year average. Paths that survive the early dangerous period mostly sit well above X, so each extra year adds less than the average so far, and the average falls.

> So every curve is a hump; within 10 years only $\omega = 0.9$ shows its peak ($\omega = 0.7$ peaks near 10 years, $\omega = 0.5$ near 44).

【中】記 spread 為 $s(\tau)$。由 $\ln(V/X) = -\ln\omega - r\tau$ 和 Lemma 8.14.15，risk-neutral 下 $\ln(V^*/X) \sim N(-\ln\omega - \sigma^2\tau/2, \sigma^2\tau)$，違約就是它掉到 0 以下：公司一開始離違約 $-\ln\omega$，雜訊大小 $\sigma\sqrt{\tau}$。**上升**：$\sigma\sqrt{\tau} \ll -\ln\omega$ 時幾乎不會違約，$s \approx 0$（模型的缺點：實際短期 spread 不是 0）；$\tau \approx (\ln\omega/\sigma)^2$（$\omega = 0.9, 0.7, 0.5$ 分別約 0.3、3、12 年）時雜訊追上距離，違約變得可能，s 上升。**下降**：$s\tau = -\ln(Be^{r\tau}/X)$ 是 $[0,\tau]$ 上的總違約折價，s 是每年平均；撐過前期危險期的 path 大多已遠高於 X，之後每多一年增加的折價小於目前平均，所以平均下降。因此每條曲線都是先升後降，只是 10 年內只看得到 $\omega = 0.9$ 的峰（$\omega = 0.7$ 約在 10 年、$\omega = 0.5$ 約在 44 年）。

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In general, suppose the firm has a dividend yield at rate q and the bankruptcy costs are a constant proportion α of the remaining firm value. Then (10.1)–(10.2) become, respectively,[3]

$$\begin{aligned} nS &= Ve^{-q\tau}N(x) - Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \\ B &= (1-\alpha)\,Ve^{-q\tau}N(-x) + Xe^{-r\tau}N\big(x-\sigma\sqrt{\tau}\big), \end{aligned}$$

with $x \triangleq \big[\ln(V/X)+(r-q+\sigma^2/2)\tau\big]/(\sigma\sqrt{\tau})$ as in (8.27). The stock is unaffected by bankruptcy costs — it gets nothing in default anyway — while the bondholders recover only $(1-\alpha)V^*$.

【中】一般情況：公司有 dividend yield q、破產成本是剩餘價值的固定比例 α，股票和債券就變成上面兩式。股票不受破產成本影響（違約時本來就拿不到），債權人則只拿回 $(1-\alpha)V^*$。

Remark 10.1.10. **破產成本**（bankruptcy costs）：公司違約後走破產程序本身耗掉的價值，不歸股東也不歸債權人。包括直接成本（律師、會計師、法院費用）和間接成本（資產被迫低價拍賣（fire sale）、客戶和員工流失）。這裡假設它是剩餘價值的固定比例：違約 $\{V^* \le X\}$ 時 αV^* 被耗掉，債權人只拿到 $(1-\alpha)V^*$；$\alpha = 0$ 就是前面 (10.2) 的情形。

Remark 10.1.11 (補充). B **怎麼來：仍是「總價值 $-C$」，再扣破產成本**。到期時 V^* 分給三方：股東拿 $\max(V^*-X,0)$、破產成本吃掉 $\alpha V^*\mathbf{1}_{\{V^*\le X\}}$、債權人拿剩下的。現值是 risk-neutral 期望值再折現（(8.13)），對 payoff 是線性的，所以

$$B = \mathrm{PV}(V^*) - C - \alpha\,\mathrm{PV}\big(V^*\mathbf{1}_{\{V^*\le X\}}\big).$$

三個現值逐一來：

(a) $\mathrm{PV}(V^*) = Ve^{-q\tau}$。Section 8.19.iii remark (2)：以 q 配出去的部分在到期前就離開公司。和 $B = V - C$ 相比，這是第一個調整。

(b) $C = Ve^{-q\tau}N(x) - Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$。(8.25) 把 S 換成 V。

(c) $\mathrm{PV}\big(V^*\mathbf{1}_{\{V^*\le X\}}\big) = Ve^{-q\tau}N(-x)$。這是「只在違約時拿到 V^*」的現值，沒有現成公式，要從 (a)(b) 湊：

  (i) 用 $\mathbf{1}_{\{V^*\le X\}} = 1 - \mathbf{1}_{\{V^*>X\}}$：它等於 $\mathrm{PV}(V^*) - \mathrm{PV}\big(V^*\mathbf{1}_{\{V^*>X\}}\big)$。

  (ii) 求 $\mathrm{PV}\big(V^*\mathbf{1}_{\{V^*>X\}}\big)$：把 call 的 payoff 拆開，$\max(V^*-X,0) = V^*\mathbf{1}_{\{V^*>X\}} - X\mathbf{1}_{\{V^*>X\}}$，取現值得 $C = \mathrm{PV}\big(V^*\mathbf{1}_{\{V^*>X\}}\big) - Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$，其中 $\mathrm{PV}(\mathbf{1}_{\{V^*>X\}}) = e^{-r\tau}N(x-\sigma\sqrt{\tau})$ 是 binary call(Example 8.14.23，S 換成 $Ve^{-q\tau}$)。和 (b) 對照，$\mathrm{PV}\big(V^*\mathbf{1}_{\{V^*>X\}}\big) = Ve^{-q\tau}N(x)$，就是 (b) 的第一項。

  (iii) 相減：$Ve^{-q\tau} - Ve^{-q\tau}N(x) = Ve^{-q\tau}N(-x)$。

代入：$B = Ve^{-q\tau} - \big[Ve^{-q\tau}N(x) - Xe^{-r\tau}N(x-\sigma\sqrt{\tau})\big] - \alpha Ve^{-q\tau}N(-x) = (1-\alpha)Ve^{-q\tau}N(-x) + Xe^{-r\tau}N(x-\sigma\sqrt{\tau})$；$\alpha = q = 0$ 時回到 (10.2)。$-C$ 那一項不必拆；拆 C 只是為了得到 (c)，因為破產成本看的是違約時的 V^*，C 本身不直接給這個量。另外 $nS + B = Ve^{-q\tau}\big(1-\alpha N(-x)\big)$：破產成本讓證券總值變小，Modigliani–Miller 不再成立。

[3] Collin-Dufresne & Goldstein (2000).

§10.1.ii. A Numerical Example

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Notation . Merck 是真實的公司（股票和選擇權都有交易）；XYZ.com 是虛構的公司，唯一的資產是 1,000 股 Merck。

- n：XYZ.com 發行的股數；S：XYZ.com 每股股價。
- nS：XYZ.com 的市值（只算股票）。
- B：XYZ.com 所有債券的總價值。
- X：債券到期時承諾支付的總額（總面額）。
- $V = nS + B$：公司總價值，等於資產（1,000 股 Merck）的價值。
- C：以 V 為標的、strike 為 X 的 call，$nS = C$。

XYZ.com's assets consist of 1,000 shares of Merck as of March 20, 1995, when Merck's market value per share is \$44.5. It issues 1,000 shares of XYZ.com common stock and 30 zero-coupon bonds maturing on July 21, 1995, each promising to pay \$1,000 at maturity. So

$$n = 1{,}000, \qquad V = 44.5 \times n = 44{,}500, \qquad X = 30 \times 1{,}000 = 30{,}000.$$

As the Merck calls are being traded, we do not need formulas to price them (Table 10.1).

【中】XYZ.com 的資產是 1995/3/20 的 1000 股 Merck（每股 44.5），發行 1000 股普通股和 30 張 1995/7/21 到期、各付 1000 的 zero-coupon bond：$n = 1000$、$V = 44{,}500$、$X = 30{,}000$。Merck 的 call 有在交易，所以直接用報價，不需要公式。

			Call		Put	
Option	Strike	Exp.	Vol.	Last	Vol.	Last
Merck	30	Jul	328	$15\frac{1}{4}$	...	...
$44\frac{1}{2}$	35	Jul	150	$9\frac{1}{2}$	10	$\frac{1}{16}$
$44\frac{1}{2}$	40	Apr	887	$4\frac{3}{4}$	136	$\frac{1}{16}$
$44\frac{1}{2}$	40	Jul	220	$5\frac{1}{2}$	297	$\frac{1}{4}$
$44\frac{1}{2}$	40	Oct	58	6	10	$\frac{1}{2}$
$44\frac{1}{2}$	45	Apr	3050	$\frac{7}{8}$	100	$1\frac{1}{8}$
$44\frac{1}{2}$	45	May	462	$1\frac{3}{8}$	50	$1\frac{3}{8}$
$44\frac{1}{2}$	45	Jul	883	$1\frac{15}{16}$	147	$1\frac{3}{4}$
$44\frac{1}{2}$	45	Oct	367	$2\frac{3}{4}$	188	$2\frac{1}{16}$

Table 10.1.: Merck option quotations, March 20, 1995 (textbook Fig. 11.1). 第一欄是 Merck 當天股價 $44\frac{1}{2}$；Vol. 是成交量（volume，當天成交幾口，不是 volatility）；Last 是當天最後成交價（每股）；「...」表示當天沒有成交。1995 年美股以 $\frac{1}{16}$ 美元報價。

The Merck option relevant for pricing is the July call with a strike price of $X/n = 30$ dollars, selling for \$15.25. So XYZ.com's stock is worth $15.25 \times n = 15{,}250$ dollars, and the entire bond issue is worth

$$B \;=\; 44{,}500 - 15{,}250 \;=\; 29{,}250$$

dollars, or \$975 per bond.

【中】要用的是 strike $X/n = 30$ 的 July call，價格 15.25，所以股票值 15,250，債券總值 $44{,}500 - 15{,}250 = 29{,}250$，每張 975。

The XYZ.com bonds are equivalent to a default-free zero-coupon bond with \$$X$ par value plus n written European puts on Merck at a strike price of \$30, by put–call parity (Theorem 7.5.1).

The difference between B and the price of the default-free bond is the value of these puts.

【中】由 put–call parity，XYZ.com 的債券等於面額 X 的無違約 zero-coupon bond 加上賣出 n 個 strike 30 的 Merck European put；B 和無違約債券價格的差就是這些 put 的價值。

Remark 10.1.12. 這個例子在做什麼。目標是算出 XYZ.com 的股票 nS 和債券 B 值多少。由 $nS = C$，只要知道 call C 的價格；而 XYZ.com 的資產全是 Merck 股票，所以 C 可以直接用市場上 Merck option 的成交價，不必用 Black–Scholes（也就不用估 σ）。

(1) **股票 $= n$ 個 Merck call**。記 M^* 為 7/21 一股 Merck 的價格，公司持有 $n = 1{,}000$ 股 Merck，所以 $V^* = nM^*$，股東 payoff $\max(V^* - X, 0) = n\max(M^* - X/n, 0)$：$n$ 個 strike X/n、到期日和債券相同（July）的 Merck call。代入上面的 $n = 1{,}000$、$X = 30{,}000$，strike $X/n = 30$。由 Corollary 7.2.4，$nS = n \times 15.25 = 15{,}250$（Table 10.1 第一列的 Last）。

(2) **債券**。$B = V - nS = 44{,}500 - 15{,}250 = 29{,}250$，30 張平分，每張 975。

(3) **債券 = riskless bond − puts（第二段英文）**。英文說：債券等於「面額 X 的無違約 zero-coupon bond」加上「賣出（written）n 個 strike 30 的 Merck put」。看到期 payoff 就知道：

$$\underbrace{\min(V^*, X)}_{\text{債券}} = \underbrace{X}_{\text{無違約債券}} - \underbrace{\max(X - V^*, 0)}_{\text{put}}.$$

$V^* \geq X$ 時兩邊都是 X；$V^* < X$ 時右邊是 $X - (X - V^*) = V^*$。也就是債權人先「保證拿 X」，但賣了一個 put 給股東：公司不夠還時，債權人要吞下差額 $X - V^*$。同 (1)，$\max(X - V^*, 0) = n\max(30 - M^*, 0)$，是 n 個 strike 30 的 July Merck put。取現值：$B = Xe^{-r\tau} - n\,P_{\text{Merck}}$（和 put–call parity $V - C = Xe^{-r\tau} - P$ 是同一件事）。所以英文第二句「B 和無違約債券價格 $Xe^{-r\tau}$ 的差是這些 put 的價值」就是 $Xe^{-r\tau} - B = n\,P_{\text{Merck}}$：違約風險讓債券少值的部分。

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Table 10.2 shows the total market values of the XYZ.com stock and bonds under various debt amounts X. For example, if the promised payment is \$45,000, the relevant option is the July call with a strike price of $45{,}000/n = 45$ dollars; since it sells for $\$1\frac{15}{16}$, the stock is worth $(1 + 15/16) \times n = 1{,}937.5$ dollars.

Promised payment to bondholders X	Current market value of bonds B	Current market value of stock nS	Current total value of firm V
30,000	29,250.0	15,250.0	44,500
35,000	35,000.0	9,500.0	44,500
40,000	39,000.0	5,500.0	44,500
45,000	42,562.5	1,937.5	44,500

Table 10.2.: Distribution of corporate value under alternative capital structures, from the July calls in Table 10.1. The last column never changes.

The market value of the stock decreases as the debt-to-equity ratio increases.

【中】Table 10.2 列出不同負債 X 下股票和債券的市值；例如承諾支付 45,000 時用 strike 45 的 July call $(1\frac{15}{16})$，股票值 1,937.5。負債比例越高股票越不值錢；公司總價值那一欄不變。

Remark 10.1.13. Table 10.2 每一列都用同一個方法：$nS = n\times$(strike X/n 的 July call 價格)，$B = V - nS$，$V = 44{,}500$ 固定。

Note 10.1.14. $X = 35{,}000$ 那一列看起來很怪：$B = 35{,}000$ 表示債券以 par 出售，也就是四個月的 yield 為零。這只是報價造成的：July 35 call 最後成交價是 $9\frac{1}{2}$，剛好等於 intrinsic value $44.5 - 35$，所以隱含的 put 價值為零，利息也被忽略了。1995 年那些過時或四捨五入的報價，本來就不會精確到每一分錢都符合 no-arbitrage。

§10.1.iii. Conflicts between Stockholders and Bondholders

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There are conflicts between stockholders and bondholders. An option's terms cannot be changed after issuance, but a firm can change its capital structure. There lies one key difference between options and corporate securities: parameters such as volatility,[4] dividend, and strike price are under partial control of the stockholders or boards.

【中】股東和債權人之間有利益衝突。option 發行後條款不能改，公司卻可以改資本結構——這是 option 和公司證券的關鍵差別：volatility（asset substitution problem）、股利、strike 等參數部分由股東或董事會控制。

Issuing more debt. Suppose XYZ.com issues 15 more bonds with the same terms to buy back stock. The total debt is now $X = 45{,}000$ dollars, and Table 10.2 says the total market value of the bonds should be \$42,562.5. The new bondholders pay

$$42{,}562.5 \times (15/45) \;=\; 14{,}187.5$$

dollars, and the remaining stock is worth \$1,937.5. The stockholders therefore gain

$$14{,}187.5 + 1{,}937.5 - 15{,}250 \;=\; 875$$

dollars, and the original bondholders lose an equal amount,

$$29{,}250 - \frac{30}{45} \times 42{,}562.5 \;=\; 875.$$

This is called **claim dilution**.[5]

【中】發新債：再發 15 張同條件債券買回股票，總負債 45,000，由表所有債券值 42,562.5。新債權人付 $42{,}562.5 \times 15/45 = 14{,}187.5$，剩下股票值 1,937.5；股東賺 875，原債權人賠 875，這叫 claim dilution。

Remark 10.1.15. 數字從哪來。

(1) 發新債後總面額 45,000，由 Table 10.2，所有債券合起來值 42,562.5。新舊債券條件相同，依面額分：新的 15 張占 15/45，舊的 30 張占 30/45。

(2) **股東**。原本有 15,250 的股票；之後拿到新債的錢 14,187.5（拿去買回股票，錢回到股東手上），加上剩下的股票 1,937.5。差額 875。

(3) **原債權人**。原本 29,250，之後只值 $\frac{30}{45} \times 42{,}562.5 = 28{,}375$，少了 875。

V 不變，所以股東賺的就是原債權人賠的。

Remark 10.1.16. 債權稀釋（claim dilution）：新債的錢拿去買回股票，沒有留在公司，所以資產仍是同樣的 1,000 股 Merck，要分的人卻從 30 張變 45 張。每張舊債券的到期 payoff：

	$V^* < 30{,}000$	$30{,}000 \le V^* < 45{,}000$	$V^* \ge 45{,}000$
發新債前	$V^*/30$	1,000	1,000
發新債後	$V^*/45$	$V^*/45 < 1{,}000$	1,000

每個 state 都不會更多、有些 state 更少（違約門檻從 30,000 升到 45,000，違約時又要和新債權人平分），所以每張價值從 975 降到 $42{,}562.5/45 \approx 945.8$，30 張共少 875。$V$ 不變，這 875 就移轉給了股東。

[4] This is called the **asset substitution problem** (Myers, 1977). 這裡的 σ 是公司資產 V 的 volatility，由公司投資什麼決定（例如 XYZ.com 把 Merck 換成更高風險的資產），所以股東可以調整。股票是以 V 為標的的 call，vega > 0（Section 9.5），提高 σ 會讓 C 變大、$B = V - C$ 變小：價值從債權人移到股東。

[5] Fama & M. H. Miller (1972).

Paying a dividend. Suppose instead the stockholders sell $(1/3) \times n$ Merck shares to fund a \$14,833.3 cash dividend.[6] The stockholders now have \$14,833.3 in cash plus a call on $(2/3) \times n$ Merck shares, and the strike price remains $X = 30{,}000$. This is equivalent to owning 2/3 of a call on n Merck shares with a strike price of \$45,000, since $\max\left(\frac{2}{3}V^* - 30{,}000, 0\right) = \frac{2}{3}\max(V^* - 45{,}000, 0)$. As n such calls are worth \$1,937.5 (Table 10.2), the XYZ.com stock is worth $(2/3) \times 1{,}937.5 = 1{,}291.67$ dollars, and the bonds are worth

$$(2/3) \times n \times 44.5 - 1{,}291.67 \;=\; 28{,}375$$

dollars. Hence the stockholders gain $14{,}833.3 + 1{,}291.67 - 15{,}250 \approx 875$ dollars, and the bondholders watch their value drop from \$29,250 to \$28,375, a loss of \$875.

【中】發股利：改為賣掉 $\frac{1}{3}n$ 股 Merck 發 14,833.3 現金股利。股東有現金加上以 $\frac{2}{3}n$ 股 Merck 為標的、履約價 30,000 的 call，等於 $\frac{2}{3}$ 個履約價 45,000 的 call，值 1,291.67；債券值 28,375。股東同樣賺約 875、債權人賠 875。

Remark 10.1.17. (1) **股利金額**。賣掉 $\frac{1}{3}n$ 股 Merck，每股 44.5：$\frac{1}{3}\times 1{,}000 \times 44.5 = 14{,}833.3$。
(2) **等價的 call**。公司剩下 $\frac{2}{3}n$ 股 Merck，到期價值 $\frac{2}{3}V^*$，債務仍是 $30{,}000 = \frac{2}{3} \times 45{,}000$，所以股票 payoff 是 $\max\left(\frac{2}{3}V^* - \frac{2}{3} \times 45{,}000, 0\right) = \frac{2}{3}\max(V^* - 45{,}000, 0)$。
(3) **債券**。公司現在的總價值是 $\frac{2}{3} \times 44{,}500$，扣掉股票 1,291.67 就是債券 28,375。

Two different actions, the same \$875 transfer — because both leave the bondholders with the same claim: 2/3 of the old firm against the old promise. The same logic explains why bondholders loathe risky investments: the stock is a call, its vega is positive (Section 9.5), and raising σ moves value from the bonds to the stock with V unchanged. Covenants in bond indentures exist to limit exactly these moves.

【中】兩種做法轉移的都是 875，因為債權人最後面對同樣的處境：只剩原公司的 $\frac{2}{3}$，承諾的金額卻不變。同理，債權人討厭高風險投資：股票是 call、vega 為正，提高 σ 會在 V 不變下把價值從債券移到股票。債券契約的 covenants 就是要限制這些動作。

§10.1.iv. Further Topics

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Other examples:[7]

- **Stock as a compound call** when the company issues coupon bonds.
- **Subordinated debts as bull call spreads.**
- **Warrants as calls.**
- **Callable bonds as American calls with 2 strike prices.**
- **Convertible bonds.**
- **Bonds with safety covenants as barrier options.**

Securities issued by firms with a complex capital structure must be solved by numerical methods such as trees.[8]

【中】其他例子：發 coupon bond 時，股票是 compound call；次順位債是 bull call spread；warrant 是 call；可贖回債券讓股票成為有兩個履約價的 American call；可轉債；有安全條款的債券是 barrier option。資本結構複雜時，只能用 tree 等數值方法求解。

[6] Merck 股票是 XYZ.com 這家公司的資產，不是股東個人的；但股東透過董事會控制公司決策，可以讓公司賣掉資產、把錢當股利發給自己。這正是 Section 10.1 開頭的假設「公司可以賣資產來付款」。
[7] Cox & Rubinstein (1985); Geske (1977). See Section 11.1 of the textbook.
[8] T. Dai (B82506025, R86526008, D8852600), Lyuu, & C. Wang (F95922018) (2010).

Remark 10.1.18. 各項的意思（V 為公司總價值）：

(1) **Compound call**。債券分期付 coupon，每期股東決定付不付：付了才能繼續擁有公司，不付就違約。每期都像一個 call，疊起來就是以 option 為標的的 option（compound option）。
(2) **次順位債（subordinated debt）**。違約時要等 senior debt 拿完，junior debt 才拿剩下的，最多拿到自己的面額。這種「從 X_s 開始拿、拿到 $X_s + X_j$ 為止」的 payoff 就是 bull call spread（Section 6.9.i）。
(3) **認股權證（warrant）**。公司自己發行的 call，履約時公司發新股給持有人，會稀釋原股東。
(4) **可贖回債（callable bond）**。公司可在到期前用約定價格 X_c 把債券買回。股東因此多了提前「付錢拿回公司」的權利，股票變成有兩個履約價的 American call。
(5) **可轉債（convertible bond）**。持有人可以把債券換成公司的新股，等於普通債券再加上一個以公司股權為標的的 call。
(6) **安全條款（safety covenant）**。約定 V 一跌到某水準 H，公司就提前交給債權人，股東拿 0。所以股票是碰到 H 就作廢的 call，即 barrier option（Section 10.2）。

§10.1.v. Distance to Default

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Let μ be the total value V's rate of expected return.[9] The firm defaults at maturity exactly when $V^* < X$. Under the real-world dynamics, $\ln(V^*/V) \sim N\big((\mu - \sigma^2/2)\tau,\ \sigma^2\tau\big)$ (Section 8.14 with $\mu - \sigma^2/2$ for the log drift), so by Proposition 5.10.10 the probability of default τ years from now equals

$$N(-\mathrm{DTD}), \qquad \text{where} \qquad \mathrm{DTD} \triangleq \frac{\ln(V/X) + (\mu - \sigma^2/2)\,\tau}{\sigma\sqrt{\tau}}.$$

V/X is called the **leverage ratio**.[10]

【中】令 μ 為公司價值 V 的期望報酬率。到期違約恰好是 $V^* < X$；真實世界下 $\ln(V^*/V) \sim N((\mu - \sigma^2/2)\tau, \sigma^2\tau)$，所以 τ 年後違約機率是 $N(-\mathrm{DTD})$。V/X 叫 leverage ratio。

Remark 10.1.19. **違約距離**（distance to default, DTD）。

(1) **違約機率的推導**。$\ln V^* \sim N(m, s^2)$，$m = \ln V + (\mu - \sigma^2/2)\tau$、$s = \sigma\sqrt{\tau}$。由 Proposition 5.10.10，$\mathrm{Prob}[V^* > X] = N\big(\frac{m - \ln X}{s}\big) = N(\mathrm{DTD})$，所以 $\mathrm{Prob}[V^* < X] = 1 - N(\mathrm{DTD}) = N(-\mathrm{DTD})$。
(2) **DTD 的意思**。$\mathrm{DTD} = \frac{m - \ln X}{s}$：$\ln V^*$ 的平均 m 離違約門檻 $\ln X$ 有幾個標準差 s。DTD 越大越安全。
(3) **和 (10.1) 的關係**。$N(x - \sigma\sqrt{\tau})$ 是 risk-neutral 下不違約的機率；DTD 只是把其中的 r 換成真實的 μ。

§10.2. Barrier Options

Slides pp. 399–401

Barrier options have a payoff that depends on whether the underlying asset's price reaches a certain price level H throughout its life.[11]

【中】barrier option 的 payoff 取決於標的價格在存續期間有沒有碰到某個水準 H。

[9] This parameter is generally hard to estimate. Campbell, Hilscher, and Szilagyi (2008) use $r + 0.06$ for simplicity.
[10] Merton (1974). Useful in forecasting defaults (Bharath & Shumway, 2008).
[11] A former MBA student in finance told the lecturer on March 26, 2004, that she did not understand why barrier options were covered until she started working in a bank. She was working for Lehman Brothers in Hong Kong as of April, 2006.

Definition 10.2.1.
- A **knock-out** (KO) option is an ordinary European option which ceases to exist if the barrier H is reached by the price of its underlying asset. A knock-out call is sometimes called a **down-and-out** option if $H < S$; a knock-out put is sometimes called an **up-and-out** option when $H > S$.
- A **knock-in** (KI) option comes into existence if a certain barrier is reached. A **down-and-in** option is a knock-in call that comes into existence only when the barrier is reached and $H < S$; an **up-and-in** option is a knock-in put that comes into existence only when the barrier is reached and $H > S$.

【中】knock-out（KO）：一般的 European option，標的一碰到 H 就作廢；$H < S$ 的 knock-out call 叫 down-and-out，$H > S$ 的 knock-out put 叫 up-and-out。knock-in（KI）：碰到 H 才生效；$H < S$ 的叫 down-and-in call，$H > S$ 的叫 up-and-in put。

Remark 10.2.2. **障礙選擇權**（barrier option）：報酬取決於存續期間內股價是否碰過障礙 H。**出局**（knock-out）型一碰就作廢；**入局**（knock-in）型要碰到才生效。down/up 指障礙在目前股價的下方或上方。

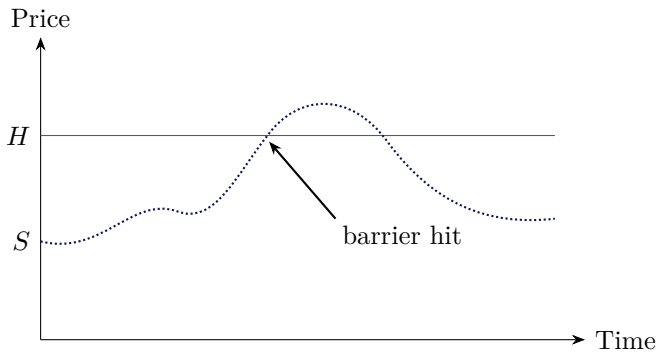


Figure 10.2.: A price path that touches the barrier H: a knock-out option on this path is dead from that moment on, and a knock-in option comes alive — even though the path ends below H.

【中】圖：一條碰到 H 的路徑：knock-out option 從那一刻起作廢，knock-in option 則生效——即使最後停在 H 下方。

The two kinds are tied by a parity relation.

Proposition 10.2.3 (In–out parity)

A European call is equivalent to a portfolio of a European down-and-out call and a European down-and-in call with an identical barrier and strike. Likewise for puts with up-and-out and up-and-in.

Proof. On every path exactly one of the two barrier options is alive at expiration, and the live one pays $\max(S_\tau - X, 0)$. So the portfolio's payoff equals the call's path by path; apply Corollary 7.2.4. ■

The argument never used that H is constant, so the parity holds for any barrier.[12] Formulas exist for all the possible barrier options mentioned above.[13]

【中】兩種 barrier option 由 in–out parity 連起來：同 barrier、同 strike 的 down-and-out call 加 down-and-in call 等於普通 European call（put 的 up-and-out 加 up-and-in 同理），因為每條路徑上恰好一個活著，活著的付 $\max(S_\tau - X, 0)$。證明沒用到 H 是常數，所以任何 barrier 都成立。上面所有 barrier option 都有公式。

[12] Comment 11.2.1 of the textbook.
[13] Haug (2006).

§10.2.i. Merton's Formulas

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Theorem 10.2.4 (Merton, 1973)

Assume $X \geq H$. The value of a European down-and-in call on a stock paying a dividend yield of q is

$$Se^{-q\tau}\left(\frac{H}{S}\right)^{2\lambda} N(x) - Xe^{-r\tau}\left(\frac{H}{S}\right)^{2\lambda-2} N\left(x - \sigma\sqrt{\tau}\right), \tag{10.4}$$

where

$$x \triangleq \frac{\ln\left(H^2/(SX)\right) + \left(r - q + \sigma^2/2\right)\tau}{\sigma\sqrt{\tau}}, \qquad \lambda \triangleq \frac{r - q + \sigma^2/2}{\sigma^2}.$$

See Exercise 17.1.6 of the textbook for a proof. A European down-and-out call can then be priced via the in–out parity: (8.25) minus (10.4).

【中】Merton 的公式（$X \geq H$）：有 dividend yield q 的 European down-and-in call 值 (10.4)，證明見課本 Exercise 17.1.6。down-and-out call 由 in–out parity 得到：(8.25) 減 (10.4)。

Theorem 10.2.5 (Merton, 1973)

Assume $X \leq H$. The value of a European up-and-in put is

$$Xe^{-r\tau}\left(\frac{H}{S}\right)^{2\lambda-2} N\left(-x + \sigma\sqrt{\tau}\right) - Se^{-q\tau}\left(\frac{H}{S}\right)^{2\lambda} N(-x).$$

Again, a European up-and-out put can be priced via the in–out parity.

【中】$X \leq H$ 時 up-and-in put 也有類似公式；up-and-out put 一樣用 in–out parity。

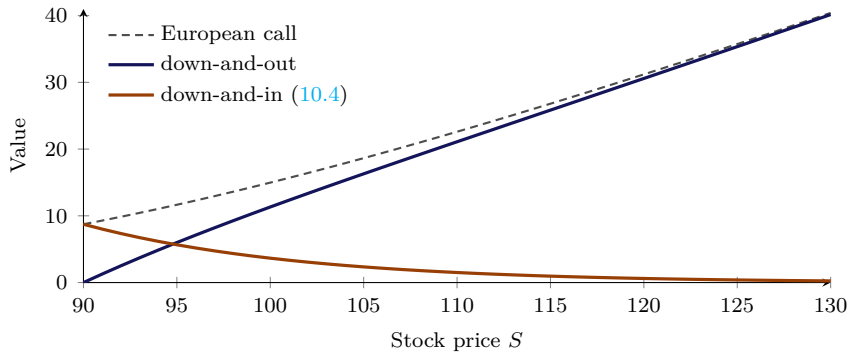


Figure 10.3.: Down-and-in and down-and-out calls with $X = 100$, $H = 90$, $r = 10\%$, $\sigma = 25\%$, $\tau = 1$, $q = 0$, as functions of $S \geq H$. At $S = H$ the down-and-in call *is* the European call and the down-and-out call is worthless; the two always add up to the European call. At $S = 95$ the down-and-in call is worth 5.6605.

【中】圖：$X = 100$、$H = 90$、$r = 10\%$、$\sigma = 25\%$、$\tau = 1$、$q = 0$ 的 down-and-in 和 down-and-out call 對 $S \geq H$ 作圖。$S = H$ 時 down-and-in 就是普通 call、down-and-out 為 0；兩者相加永遠是普通 call。$S = 95$ 時 down-and-in 值 5.6605。

Note 10.2.6. 兩個 sanity check。當 $S = H$ 時，$(H/S)^{\cdot} = 1$，且 $\frac{H^2}{SX} = \frac{H}{X} = \frac{S}{X}$，所以 x 變成 (8.27)，(10.4) 化簡為 (8.25)：barrier 一開始就被碰到，down-and-in call 就是普通的 call。當 $S \to \infty$ 時，因子 $(H/S)^{2\lambda}$ 讓價值趨近零：barrier 太遠，碰不到。

§10.2.ii. Popularity

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Knock-out options were issued in the U.S. in 1967.[14] Knock-in puts are the most popular barrier options, knock-out puts are the second most popular, and knock-out calls are the most popular among barrier call options.[15]

【中】knock-out option 1967 年在美國發行。knock-in put 最受歡迎、knock-out put 第二；call 之中最受歡迎的是 knock-out call。

Intuition . Barrier options are cheaper than their vanilla counterparts (by in–out parity each piece is worth less than the whole), and the buyer gives up only scenarios they consider unlikely or irrelevant. A knock-in put, for instance, is crash insurance that only switches on once the market has already fallen through H.

【中】barrier option 比普通 option 便宜（由 in–out parity，每一部分都小於整體），買方只放棄自己認為不太可能或不重要的情境。例如 knock-in put 是市場跌破 H 後才啟動的崩盤保險。

§10.2.iii. Are American Options Barrier Options?

Slides pp. 405–406

American options look like barrier options with the exercise boundary (Section 8.21) as the barrier and the payoff as the rebate.[16] One salient difference is that the exercise boundary must be found by backward induction; it cannot be specified arbitrarily. But the barrier is fixed by a contract,[17] and the barrier option remains European-style, without the right to early exercise.[18]

【中】American option 看起來像 barrier option：barrier 是 exercise boundary、rebate 是 payoff。差別在於 exercise boundary 要由 backward induction 求出，不能任意指定；barrier 則由合約固定，而且 barrier option 仍是 European 式，不能提前履約。

They become equal if the barrier happens to be identical to the exercise boundary and the rebate equals the intrinsic value.[19] One can also have American barrier options; one then needs to specify whether the option can be exercised early if the stock price has not touched the barrier.[20]

【中】若 barrier 恰好等於 exercise boundary、rebate 等於 intrinsic value，兩者就相同。也有 American barrier option，這時要規定還沒碰到 barrier 時能不能提前履約。

§10.2.iv. Interesting Observations

Slides pp. 407–408

Assume $H < X$ and write $C(S)$ for the Merton formula (8.25) as a function of the stock price.

- Replace S in (8.25) with H^2/S. Then (10.4) becomes $C(H^2/S)$ when $r - q = \sigma^2/2$, and $(S/H)\,C(H^2/S)$ when $r - q = 0$.

[14] Cox & Rubinstein (1985).
[15] Bennett (2014).
[16] Contributed by Mr. Yang, Jui-Chung (D97723002) on March 25, 2009.
[17] Cox & Rubinstein (1985).
[18] Contributed by Ms. Chen, Sin-Huei (Amber) (P00922005) on March 31, 2021.
[19] Contributed by Mr. Chuang, Chin-Yu (D12922014) and Mr. Hsiao, Huan-Wen (B90902081, R94922010, D13922016) on October 2, 2025.
[20] Contributed by Mr. Lu, Yu-Ming (R06723032, D08922008) on March 31, 2021.

- Conversely, replace S in (10.4) with H^2/S. It becomes $C(S)$ when $r-q=\sigma^2/2$, and $(H/S)\,C(S)$ when $r-q=0$.[21]

【中】設 $H<X$，$C(S)$ 為 (8.25)。(1) 把 S 換成 H^2/S：$r-q=\sigma^2/2$ 時 (10.4) 等於 $C(H^2/S)$，$r=q$ 時等於 $\frac{S}{H}C(H^2/S)$。(2) 反過來把 (10.4) 的 S 換成 H^2/S，分別變成 $C(S)$ 和 $\frac{H}{S}C(S)$。

Verification（補充；簡報只給結論）. 記 (10.4) 為 $\mathrm{DI}(S)$。

(1) $r-q=\sigma^2/2$。$\lambda=\frac{\sigma^2/2+\sigma^2/2}{\sigma^2}=1$，所以 $(H/S)^{2\lambda}=(H/S)^2$、$(H/S)^{2\lambda-2}=1$：

$$\mathrm{DI}(S)=\frac{H^2}{S}e^{-q\tau}N(x)-Xe^{-r\tau}N(x-\sigma\sqrt{\tau}).$$

x 的分子 $\ln\frac{H^2}{SX}=\ln\frac{H^2/S}{X}$，正是 (8.27) 把 S 換成 H^2/S。所以 $\mathrm{DI}(S)=C(H^2/S)$。

(2) $r=q$。$\lambda=\frac{1}{2}$，$(H/S)^{2\lambda}=\frac{H}{S}$、$(H/S)^{2\lambda-2}=\frac{S}{H}$：

$$\begin{aligned}\mathrm{DI}(S)&=He^{-q\tau}N(x)-\frac{S}{H}Xe^{-r\tau}N(x-\sigma\sqrt{\tau})\\&=\frac{S}{H}\Big[\frac{H^2}{S}e^{-q\tau}N(x)-Xe^{-r\tau}N(x-\sigma\sqrt{\tau})\Big]=\frac{S}{H}\,C(H^2/S).\end{aligned}$$

(3) **第二個 bullet**。$S\mapsto H^2/S$ 做兩次會回到 S。把 (1) 的 S 換成 H^2/S 得 $\mathrm{DI}(H^2/S)=C(S)$；(2) 同理得 $\mathrm{DI}(H^2/S)=\frac{H^2/S}{H}C(S)=\frac{H}{S}\,C(S)$。

■

Why? Take $r-q=\sigma^2/2$, which by Lemma 8.14.15 (with $r-q$ in place of r) makes $\ln S_t$ *driftless* under the risk-neutral measure. A driftless random walk is symmetric, so we may reflect any path about $\ln H$ from the first moment it touches the barrier — the **reflection principle**. Since $X>H$, a path starting at $S>H$ that finishes above X *after having touched* H corresponds one-to-one to a path finishing above X that started at the mirror image of S, namely $\ln\big(H^2/S\big)=2\ln H-\ln S$ (which lies below H, so it automatically crosses the barrier). Hence the down-and-in call at S is the plain call at H^2/S. With drift, the reflected paths carry different probabilities, and the powers of H/S in (10.4) are the correction.

【中】為什麼？取 $r-q=\sigma^2/2$，risk-neutral 下 $\ln S_t$ 沒有 drift，沒有 drift 的 random walk 是對稱的，所以可以把路徑從第一次碰到 barrier 起對 $\ln H$ 鏡射（reflection principle）。從 S 出發、碰過 H、最後高於 X 的路徑，和從鏡像點 $\ln\big(H^2/S\big)$ 出發、最後高於 X 的路徑一一對應，所以 S 處的 down-and-in call 等於 H^2/S 處的普通 call。有 drift 時鏡射後的機率不同，(10.4) 中 H/S 的次方就是修正。

Remark 10.2.7 (補充)**. reflection principle 的對應**。

(1) 一條從 $\ln S$ 出發、碰過 $\ln H$、最後落在 $\ln X$ 之上的 path：把它第一次碰到 $\ln H$ 之前的那一段對 $\ln H$ 做鏡射，之後不動，就得到一條從 $2\ln H-\ln S=\ln\big(H^2/S\big)$ 出發、最後也落在 $\ln X$ 之上的 path。

(2) 反過來，從 $\ln\big(H^2/S\big)<\ln H$ 出發、最後在 $\ln X>\ln H$ 之上的 path 一定穿過 $\ln H$，鏡射回去就得到 (1) 的 path。所以兩種 path 一一對應。

(3) 沒有 drift 時，每一步往上、往下的機率相同，鏡射後的 path 機率不變。所以「從 S 出發、碰過 H、最後 $>X$」的機率和 payoff，等於「從 H^2/S 出發、最後 $>X$」的，也就是 $\mathrm{DI}(S)=C(H^2/S)$。

[21] Contributed by Mr. Chou, Ming-Hsin (R02723073) on April 24, 2014.

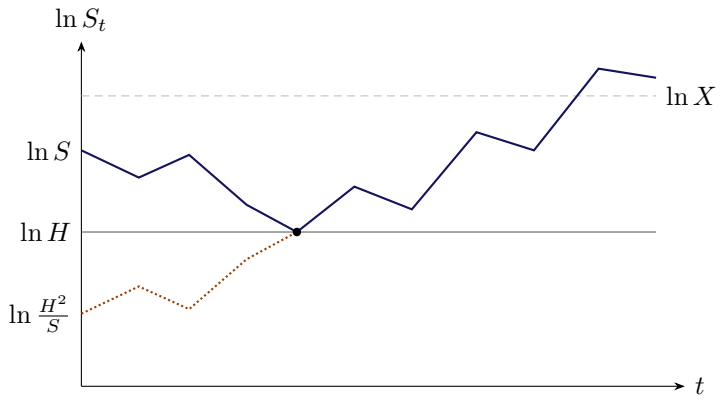


Figure 10.4.: The reflection principle. Every path from $\ln S$ that touches $\ln H$ and ends above $\ln X$ (solid) pairs with the path from $\ln\left(H^2/S\right)$ that mirrors it up to the first touch (dotted) and coincides with it afterwards.

【中】圖：從 $\ln S$ 出發、碰到 $\ln H$、最後高於 $\ln X$ 的路徑（實線），對應一條從 $\ln\left(H^2/S\right)$ 出發的路徑：碰到 $\ln H$ 前是鏡像（點線），之後重合。

This equality can be used to replicate barrier options with European options.[22]

【中】這個等式可以用來以 European option 複製 barrier option。

§10.2.v. Binomial Tree Algorithms

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Barrier options can be priced by binomial tree algorithms. For the down-and-out option, backward induction runs as usual, except that any node whose stock price is at or below the barrier has its option value set to 0 (or to the rebate, if there is one). Pricing down-and-in options is subtler: whether a node's option is alive depends on whether the path to it touched H, which the node does not know. The in–out parity sidesteps this — price the down-and-out call and subtract it from the European call.

【中】barrier option 可以用 tree 定價。down-and-out：照常 backward induction，只是股價在 barrier 或以下的 node 設為 0（有 rebate 就設成 rebate）。down-and-in 較麻煩：node 是否已生效取決於路徑有沒有碰過 H，node 本身不知道；用 in–out parity 繞過，先算 down-and-out，再用普通 call 減掉。

Example 10.2.8. $S = 8$, $X = 6$, $H = 4$, $R = 1.25$, $u = 2$ and $d = 0.5$, so $p = (1.25-0.5)/1.5 = 0.5$ and backward induction is $C = (0.5 \times C_u + 0.5 \times C_d)/1.25$ (Figure 10.5). At the node $S = 4$, the plain backward-induction value $(4.0 + 0)/2.5 = 1.6$ is replaced by 0 because the barrier is touched. So the down-and-out call is worth $12.48/2.5 = 4.992$. Without the barrier the root would be $(12.48 + 1.6)/2.5 = 5.632$, so by in–out parity the down-and-in call is worth $5.632 - 4.992 = 0.64$.

【中】例子：$S = 8$、$X = 6$、$H = 4$、$R = 1.25$、$u = 2$、$d = 0.5$，$p = 0.5$。$S = 4$ 的 node 碰到 barrier，1.6 改成 0，所以 down-and-out call 值 4.992；沒有 barrier 時是 5.632，所以 down-and-in call 值 0.64。

Remark 10.2.9. 因為 $p = 1 - p = 0.5$，backward induction $(0.5C_u + 0.5C_d)/1.25 = (C_u + C_d)/2.5$，所以每個 node 都是「兩個子節點的值相加再除以 2.5」。root 的兩個子節點是 $S = 16$（值 12.48）和 $S = 4$；down-and-out 時 $S = 4$ 碰到 barrier 設為 0，沒有 barrier 時則是 1.6。

[22] Derman, Ergener, & Kani (1995).

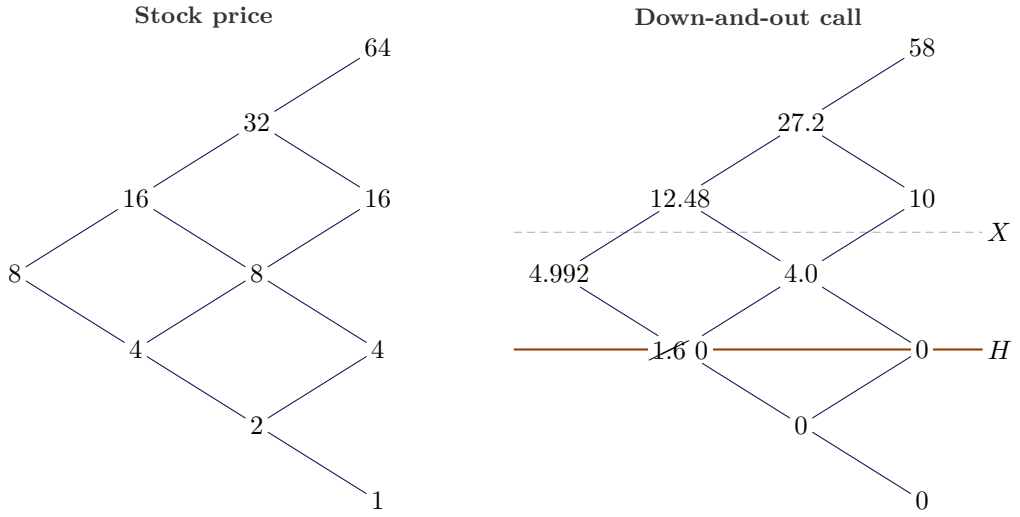


Figure 10.5.: Example 10.2.8. Nodes on or below $H = 4$ are set to zero. ($X = 6$ falls between the stock prices 4 and 8; the dashed line is only schematic.)

【中】圖：$H = 4$ 以下（含）的 node 設為 0。

Algorithm 10.1: Binomial tree algorithm for down-and-out calls on a non-dividend-paying stock

Input: $S, u, d, X, H, n, \hat{r}$ $\quad (H < X,\ H < S,\ ud = 1)$

$R := e^{\hat{r}}$;
$p := (R - d)/(u - d)$;
$h := \lfloor \ln(H/S)/\ln u \rfloor$; `// effective barrier` $Su^h \le H$
for $i = 0, 1, \ldots, n$ **do**
  $C[i] := \max\big(0,\ Su^{n-i}d^i - X\big)$;
if $n - h$ *is even and* $0 \le (n-h)/2 \le n$ **then**
  $C[(n-h)/2] := 0$; `// a hit`
for $j = n - 1$ **to** 0 **do**
  for $i = 0, 1, \ldots, j$ **do**
    $C[i] := \big(p \times C[i] + (1-p) \times C[i+1]\big)/R$;
  if $j - h$ *is even and* $0 \le (j-h)/2 \le j$ **then**
    $C[(j-h)/2] := 0$; `// a hit`
return $C[0]$;

【中】演算法：$ud = 1$，$h = \lfloor \ln(H/S)/\ln u \rfloor$ 是 effective barrier $Su^h \le H$ 的層數；每一欄做完 backward induction，就把剛好在 barrier 那層的 node 設成 0。

Note 10.2.10. h **是什麼**。因為 $ud = 1$，tree 上所有股價都是 Su^k（k 為整數）。effective barrier 是 $\le H$ 的最高那一層：$Su^k \le H \Leftrightarrow k \le \ln(H/S)/\ln u$，取最大的整數 k 就是 $h = \lfloor \ln(H/S)/\ln u \rfloor$。**哪個 node 是 barrier**。時間 j 時經過 i 次下跌的節點價格為 $Su^{(j-i)-i} = Su^{j-2i}$，它等於 Su^h 恰好在 $j - 2i = h$，即 $i = (j-h)/2$。i 必須是 0 到 j 之間的整數，所以演算法要檢查 $j - h$ 是偶數且 $0 \le (j-h)/2 \le j$。只要把那個節點設為零就夠了：它下面的每個節點都只能經由 barrier 那一層到達，而那裡的值已經是 0。若有 rebate K，就把那個節點設成 K 而不是 0。[23]

[23]課本 Figure 11.4。

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But convergence is erratic, because H is not at a price level on the tree.[24] The barrier H is moved lower (or higher, if you so choose) to the nearest node price, and this **effective barrier** changes as n increases. In fact, the binomial tree is only $O(1/\sqrt{n})$ convergent.[25] Solutions will be presented later.

【中】但收斂不規則，因為 H 通常不在 tree 的價格層上：H 被移到最近的 node 價格（effective barrier），而它會隨 n 改變。事實上 binomial tree 只有 $O(1/\sqrt{n})$ 的收斂速度，解法之後再談。

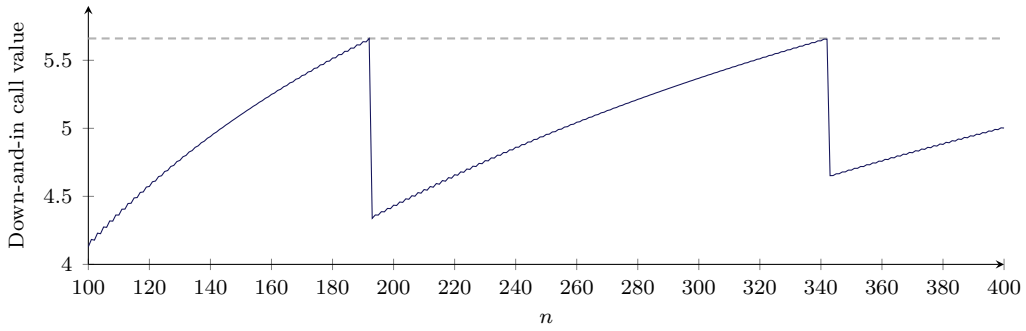


Figure 10.6.: CRR binomial values of the down-and-in call of Figure 10.3 ($S = 95$, $X = 100$, $H = 90$, $r = 10\%$, $\sigma = 25\%$, $\tau = 1$), computed by in–out parity, for $n = 100, \ldots, 400$. Dashed: the analytical value 5.6605. Each jump down happens when the effective barrier moves to the next lattice level; in between the value creeps up with an odd–even wobble. Cf. Lyuu (1998).

【中】圖：Figure 10.3 的 down-and-in call（$S = 95$）用 CRR tree 加 in–out parity 計算，$n = 100$ 到 400；虛線是解析值 5.6605。effective barrier 每移到下一層，數值就往下跳；之間慢慢上升並有奇偶擺動。

Intuition . The effective barrier Su^h sits below H by a random-looking fraction of one step $\sigma\sqrt{\Delta t}$. A barrier misplaced by $O(\sqrt{\Delta t}) = O(1/\sqrt{n})$ shifts the knock-in probability by the same order, and that is the error. As n grows the lattice level just below H creeps up towards H, the value rises, until a new level appears below H and the effective barrier drops back — the saw-tooth. The cure is to choose n (or shift the lattice) so that H falls exactly on a node.

【中】effective barrier Su^h 比 H 低了一步 $\sigma\sqrt{\Delta t}$ 中看似隨機的一部分；barrier 放錯 $O(1/\sqrt{n})$，knock-in 機率就差同樣的量級，這就是誤差。n 增加時 H 下方那層慢慢靠近 H、數值上升，直到 H 下方出現新的一層、effective barrier 又掉下去——形成鋸齒。解法是選 n（或平移 lattice）讓 H 落在 node 上。

§10.2.vi. Other Types of Barrier Options

Slide p. 413

Partial barrier options (the barrier is active during only an initial period), forward-starting barrier options (only the latter part), window barrier options, rolling barrier options,[26] and moving barrier options.[27]

【中】其他種類：partial barrier（只在開始一段時間有效）、forward-starting barrier（只在後段有效）、window barrier、rolling barrier（barrier 是階梯函數）、moving barrier。

[24] Boyle & Lau (1994).
[25] Tavella & Randall (2000); J. Lin (R95221010) (2008); J. Lin (R95221010) & Palmer (2013).
[26] The barrier is a step function. There is a pricing formula as a multiple integral (H. Lee, G. Lee, & Song, 2022). There is also an $O(n \ln n)$-time tree algorithm by Y. Lu (R06723032, D08922008) and Lyuu (2021, 2023).
[27] Armstrong (2001); Carr & A. Chou (1997); Davydov & Linetsky (2001/2002); Haug (1998).

§10.2.vii. Daily Monitoring

Slides pp. 414–416

Many barrier options monitor the barrier only for daily closing prices. If so, only nodes at the end of a day need to check for the barrier condition. We can even remove intraday nodes to create a **multinomial tree**: a node is then followed by $d+1$ nodes if each day is partitioned into d periods (Figure 10.7), with branching probabilities $b(j;d,p)$.

【中】很多 barrier option 只用每天收盤價檢查 barrier，所以只有每天結束的 node 需要檢查。甚至可以拿掉盤中 node 變成 multinomial tree：一天切成 d 期時，每個 node 接 $d+1$ 個 node，分支機率是 $b(j;d,p)$。

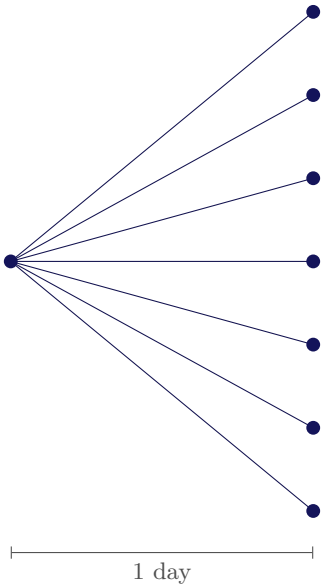


Figure 10.7.: A heptanomial tree: 6 periods per day, intraday nodes removed. The seven successors are the end-of-day prices Su^{6-2j}, $j=0,\dots,6$.

【中】圖：heptanomial tree：一天 6 期、拿掉盤中 node，七個後繼是當天收盤價 Su^{6-2j}，$j=0,\dots,6$。

Does this save time or space?[28] Not asymptotically. Over D days with d periods each, the binomial tree visits $\sim (Dd)^2/2$ nodes at two operations each, about D^2d^2 operations; the multinomial tree visits only the $\sim D^2d/2$ end-of-day nodes but at $d+1$ operations each, again of order $D^2d^2/2$. Space is a single column of $O(Dd)$ nodes either way.

【中】這樣能省時間或空間嗎？漸近上不能：binomial tree 約 $(Dd)^2/2$ 個 node、每個 2 次運算；multinomial tree 只有約 $D^2d/2$ 個收盤 node、但每個 $d+1$ 次運算，都是 D^2d^2 的量級。空間都是一欄 $O(Dd)$。

Remark 10.2.11. (1) **multinomial 的分支機率**。一天有 d 期，一天後股價由這 d 期中上漲幾次決定：上漲 j 次（$j=0,\dots,d$）共 $d+1$ 種結果，機率是 $b(j;d,p)$（Section 8.11 的 binomial distribution）。

(2) **binomial tree 的工作量**。總共 Dd 期，node 數 $\sum_{k=0}^{Dd}(k+1)\sim (Dd)^2/2$，每個 node 合併 2 個子節點。

(3) **multinomial tree 的工作量**。只留每天結束時的 node：第 k 天結束時有 $kd+1$ 個，總共 $\sum_{k=0}^{D}(kd+1)\sim dD^2/2$ 個；但每個 node 要合併 $d+1$ 個子節點，總共 $\sim dD^2/2\times d = D^2d^2/2$。

兩者同為 $O(D^2d^2)$，所以沒有省下。

[28] Contributed by Ms. Chen, Tzu-Chun (R94922003) and others on April 12, 2006.

Discrete versus continuous monitoring. Discrete barriers are more expensive for knock-out options than continuous ones, and less expensive for knock-in options — a path can cross H between two closes without being seen. Discrete barriers are far less popular than continuous ones for individual stocks; they are equally popular for indices.[29]

【中】discrete 與 continuous monitoring：knock-out 的 discrete barrier 比 continuous 貴、knock-in 則比較便宜，因為路徑可能在兩次收盤之間穿過 H 而沒被看到。個股上 discrete barrier 遠不如 continuous 普及，指數上則一樣普及。

§10.3. Foreign Currencies

Slides pp. 417–419

Data! data! data!
— Arthur Conan Doyle (1892), *The Adventures of Sherlock Holmes*

Notation .
- S: the spot exchange rate in domestic/foreign terms, i.e., the number of domestic currencies per unit of foreign currency;[a]
- σ: the volatility of the exchange rate;
- r: the domestic interest rate;
- r_f: the foreign interest rate. (The slides write $\hat{r}$; we keep $\hat{r}$ for the per-period rate of Section 8.1.)

[a] The market convention is the opposite: A/B $= x$ means one unit of currency A (the reference or base currency) is equal to x units of currency B (the counter-value currency).

【中】S 是匯率（一單位外幣值多少本國貨幣），σ 是匯率波動率，r 是本國利率，r_f 是外國利率。

A foreign currency is analogous to a stock paying a known dividend yield: holding foreign currency means holding foreign bonds, which pay a "continuous dividend yield" equal to r_f in the foreign currency.

【中】外幣就像有已知 dividend yield 的股票：持有外幣等於持有外國債券，會以 r_f 的「連續股利」生出外幣。

Remark 10.3.1. 站在本國投資人的角度，持有一單位外幣（存在外國銀行）會以利率 r_f 連續生出更多外幣，就像股票以殖利率 q 連續配息。所以 Section 8.19.iii 的所有結果，把 q 換成 r_f 就能直接用。

Slides pp. 420–429, 435–437

The slides then plot daily and minutely Euro–USD and GBP–USD rates (2005–2017), the distribution of GBP–USD between the collapse of Lehman Brothers and the Brexit referendum (September 15, 2008–June 23, 2016), and the daily JPY–USD rate, with their log returns. As for stocks (Figure 1.4), the log returns have sharper peaks and heavier tails than the normal — most visibly at the minute scale — and the volatility clusters in time.

【中】簡報畫了 2005–2017 年 Euro–USD、GBP–USD 的日資料和分鐘資料、雷曼倒閉到脫歐公投之間 GBP–USD 的分布、JPY–USD 的日匯率和 log return。和股票一樣，log return 的峰比 normal 尖、尾巴比 normal 厚（分鐘資料最明顯），而且波動在時間上會聚集。

[29] Bennett (2014).

§10.3.i. Foreign Exchange Options

Slide p. 430

In 2000 the total notional volume of foreign exchange options was US$13 trillion:[30]

- 38.5% were vanilla calls and puts with a maturity less than one month;
- 52.5% were vanilla calls and puts with a maturity between one and 18 months;
- 4% were barrier options (Section 10.2);
- 1.5% were vanilla calls and puts with a maturity more than 18 months;
- 1% were binary options (Section 6.9.ii);
- 0.7% were Asian options (Section 10.4).

【中】2000 年外匯選擇權名目總額 13 兆美元：一個月內到期的普通 call/put 占 38.5%，一到十八個月 52.5%，barrier 4%，十八個月以上 1.5%，binary 1%，Asian 0.7%。

Slides pp. 431–433

Foreign exchange options are settled via delivery of the underlying currency, and a primary use of them is to hedge currency risk.

【中】外匯選擇權以交割標的貨幣結算，主要用途是規避匯率風險。

Example 10.3.2. A U.S. company expects to receive 100 million Japanese yen in March 2000, to be exchanged for U.S. dollars. The contract size for the Japanese yen option is JPY6,250,000, so the company purchases

$$\frac{100{,}000{,}000}{6{,}250{,}000} = 16$$

puts on the Japanese yen with a strike of $.0088/JPY1 and an exercise month in March 2000. This put is in the money if the JPY–USD exchange rate drops below 0.0088, and it gives the company the right to sell 100,000,000 Japanese yen for $100{,}000{,}000 \times .0088 = 880{,}000$ U.S. dollars.

【中】例子：美國公司預計 2000 年 3 月收到 1 億日圓並換成美元。日圓選擇權每口 625 萬日圓，所以買 16 口 strike 0.0088 美元/日圓、3 月到期的 put。JPY–USD 跌破 0.0088 時 put 在價內，公司有權把 1 億日圓以 880,000 美元賣出。

Note that these puts are equivalent to the right to *buy* 880,000 U.S. dollars with 100,000,000 Japanese yen. From this angle, they become calls — on the dollar, with strike $1/0.0088 \approx 113.6$ yen per dollar. A put on one currency is a call on the other.

【中】這些 put 等於用 1 億日圓買 880,000 美元的權利，換個角度就是美元的 call，strike $1/0.0088 \approx 113.6$ 日圓/美元：一種貨幣的 put 就是另一種貨幣的 call。

Remark 10.3.3. 「用 100M JPY 換 880K USD 的權利」從美元的角度看，就是「付 100M JPY 買 880K USD 的權利」。strike 改用「每美元幾日圓」表示是 $100{,}000{,}000/880{,}000 = 1/0.0088 \approx 113.6$。日圓貶值（JPY–USD < 0.0088）等價於美元升值（USD–JPY > 113.6），所以 yen put 在價內時，dollar call 也在價內。

§10.3.ii. The Black–Scholes Model for Forex Options

Slide p. 434

Assume the exchange rate S is lognormally distributed. The formulas (8.25)–(8.26) apply with the dividend yield equal to r_f:

$$C = Se^{-r_f\tau}N(x) - Xe^{-r\tau}N\big(x - \sigma\sqrt{\tau}\big), \tag{10.5}$$

$$P = Xe^{-r\tau}N\big(-x + \sigma\sqrt{\tau}\big) - Se^{-r_f\tau}N(-x), \tag{10.6}$$

[30] Lipton (2002).

where

$$x \triangleq \frac{\ln(S/X) + \big(r - r_f + \sigma^2/2\big)\tau}{\sigma\sqrt{\tau}}.$$

These are the **Garman–Kohlhagen** formulas. On a tree, use the risk-neutral probability (8.28) with $q = r_f$.

【中】假設匯率是 lognormal，(8.25)–(8.26) 把 dividend yield 換成 r_f 即適用，得到 Garman–Kohlhagen 公式；在 tree 上用 (8.28)，$q = r_f$。

Note 10.3.4. dividend 的類比也把 early exercise 的結論一起帶過來。現在 no-arbitrage 下界是 $C \geq \max\big(Se^{-r_f\tau} - Xe^{-r\tau}, 0\big)$：到期拿到的外幣 S_τ 今天值 $Se^{-r_f\tau}$（Section 8.19.iii），要付的 X 今天值 $Xe^{-r\tau}$。當 $r_f > r$ 時它會低於 intrinsic value $S - X$：兩者相減

$$(S - X) - \big(Se^{-r_f\tau} - Xe^{-r\tau}\big) = S\big(1 - e^{-r_f\tau}\big) - X\big(1 - e^{-r\tau}\big),$$

$S \geq X$ 時第一項 $\geq X(1 - e^{-r_f\tau}) > X(1 - e^{-r\tau})$，所以差為正。所以和 Theorem 7.7.1 不同，American forex call 可能會在到期前 early exercise。[31]

§10.4. Path-Dependent Derivatives

Slides pp. 438–439

Bar the roads! Bar the paths!
Wert thou to flee from here, wert thou
to find all the roads of the world,
the way thou seekst
the path to that thou'dst find not[.]
— Richard Wagner (1813–1883), *Parsifal*

Let $S_0, S_1, \ldots, S_n$ denote the prices of the underlying asset over the life of the option; S_0 is the known price at time zero and S_n is the price at expiration. The standard European call has a terminal value $\max(S_n - X, 0)$ depending only on the last price, so its value depends only on the underlying asset's terminal price regardless of how it gets there.[32]

【中】令 $S_0, \ldots, S_n$ 為存續期間的標的價格，S_0 是現在、S_n 是到期。普通 European call 的 payoff $\max(S_n - X, 0)$ 只看最後的價格，不管怎麼走到那裡。

Slides pp. 440–441

Some derivatives are **path-dependent**: their terminal payoff depends explicitly on the path.

【中】有些 derivative 是 path-dependent：payoff 明確取決於整條路徑。

Definition 10.4.1. The (arithmetic) **average-rate call** and **average-rate put** have terminal values

$$\max\left(\frac{1}{n+1}\sum_{i=0}^{n} S_i - X,\ 0\right), \qquad \max\left(X - \frac{1}{n+1}\sum_{i=0}^{n} S_i,\ 0\right).$$

【中】算術 average-rate call / put 的 payoff 是「平均價格 $-X$」/「$X-$ 平均價格」的正部分。

Average-rate options are also called **Asian options**. They are very popular:[33] they are useful hedging tools for firms that will make a stream of purchases over a time period, because the

[31] 課本 Theorem 11.5.1。
[32] Called **simple claims** (Björk, 2009).
[33] As of the late 1990s, the outstanding volume was in the range of 5–10 billion U.S. dollars (Nielsen & Sandmann, 2003).

costs are likely to be linked to the average price. They are mostly European for the same reason. The averaging clause is also common in convertible bonds and structured notes.

【中】average-rate option 又叫 Asian option，很受歡迎：對一段時間內持續採購的公司是好用的避險工具，因為成本跟平均價格有關；也因此多半是 European。平均條款在可轉債和結構型商品中也很常見。

Remark 10.4.2. **亞式選擇權**（Asian option）：報酬取決於存續期間股價的平均而非到期價。平均比單一價格難操縱、波動也小，因此比歐式選擇權便宜，適合持續採購原料的公司避險。名稱據說來自最早用在亞洲交易量較小的股票上。

Slides pp. 442–443

Definition 10.4.3.

- A **lookback call** on the minimum pays $S_n - \min_{0\le i\le n} S_i$; a **lookback put** on the maximum pays $\max_{0\le i\le n} S_i - S_n$.
- The **fixed-strike lookback option** (also called forward lookback option) pays

$$\max\Big(\max_{0\le i\le n} S_i - X,\, 0\Big)\ \text{(call)}, \qquad \max\Big(X - \min_{0\le i\le n} S_i,\, 0\Big)\ \text{(put)}.$$

- Lookback calls and puts on the average (instead of a constant X) are called **average-strike options**.

【中】lookback call（對最小值）付 $S_n - \min S_i$，lookback put（對最大值）付 $\max S_i - S_n$。fixed-strike lookback 付 $\max(\max S_i - X, 0)$（call）或 $\max(X - \min S_i, 0)$（put）。strike 換成平均價的 lookback 叫 average-strike option。

The lookback call lets the holder buy at the lowest price of the period, the lookback put sell at the highest: "selling at the high and buying at the low." The CRR tree converges uniformly for path-dependent options.[34]

【中】lookback call 讓持有者以期間最低價買入、lookback put 以最高價賣出：「買在最低、賣在最高」。CRR tree 對 path-dependent option 一致收斂。

§10.4.i. Average-Rate Options Are Hard

Slides pp. 444–447

Average-rate options are notoriously hard to price. The binomial tree for the averages does not combine: the paths $(S_0, S_0u, S_0u^2, S_0u^2d)$ and $(S_0, S_0d, S_0du, S_0du^2)$ end at the same price but with different averages, hence different payoffs (Figure 10.8).

【中】average-rate option 出了名地難定價：平均價的 tree 不會 recombine，兩條終點同價的路徑平均不同，payoff 也不同（Figure 10.8）。

Remark 10.4.4. 兩條 path 的終點都是 S_0u^2d，但平均分別是 $\frac{S_0}{4}(1+u+u^2+u^2d)$ 和 $\frac{S_0}{4}(1+d+du+du^2)$。$u > d$ 時逐項比較，前者每一項都 $\ge$ 後者，所以平均比較大：先漲的 path 中途價格比較高。

[34] Jiang & M. Dai (2004).

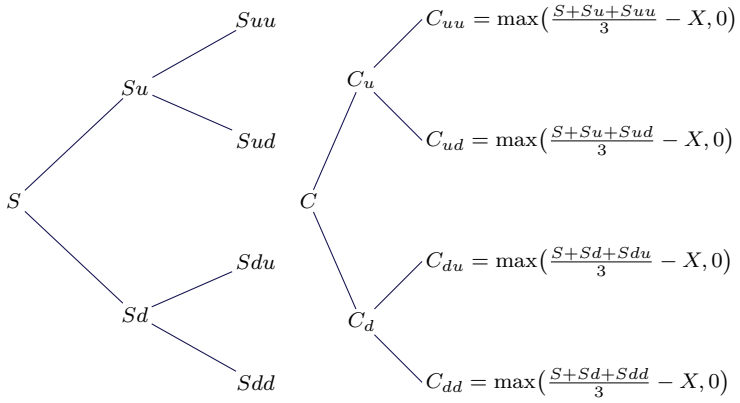


Figure 10.8.: Two periods of an average-rate call. The stock tree on the left recombines ($Sud = Sdu$); the pricing tree on the right does not, since $C_{ud} \neq C_{du}$ in general. With $C_u = \left[pC_{uu} + (1-p)C_{ud}\right]/e^{\hat{r}}$ and similarly for C_d and C, the pricing tree grows exponentially.

【中】圖：兩期的 average-rate call。左邊股價 tree 會 recombine（$Sud = Sdu$），右邊定價 tree 不會，因為一般 $C_{ud} \neq C_{du}$；定價 tree 呈指數成長。

A naive algorithm enumerates the 2^n paths for an n-period binomial tree and then averages the payoffs, but the complexity is exponential.[35] The Monte Carlo method and approximation algorithms are some of the alternatives left.

【中】天真的演算法列舉 2^n 條路徑再平均 payoff，是指數時間；剩下的選擇有 Monte Carlo 和近似演算法。

States and their transitions. The tuple (i, S, P) captures the state for the Asian option — a "sufficient statistic," if you will — where i is the time, S the prevailing stock price, and P the running sum.[36] For the binomial model the state transition is

$$(i, S, P) \longrightarrow \begin{cases} (i+1,\ Su,\ P+Su), & \text{for the up move,} \\ (i+1,\ Sd,\ P+Sd), & \text{for the down move.} \end{cases}$$

The number of states is exponential: (i, S) takes $O(n^2)$ values, but the running sum P distinguishes paths.

【中】(i, S, P) 足以描述 Asian option 的狀態（時間、股價、累積和）。上漲變成 $(i+1, Su, P+Su)$，下跌變成 $(i+1, Sd, P+Sd)$。(i, S) 只有 $O(n^2)$ 種，但累積和 P 會區分不同路徑，所以狀態數是指數的。

Remark 10.4.5. 關鍵在於「狀態」要包含哪些資訊才足以決定未來報酬。歐式選擇權只需要 (i, S)，所以二項樹會重合；亞式選擇權還需要目前的累積和 P，而不同路徑到同一個 (i, S) 幾乎都有不同的 P，樹就不再重合。

§10.4.ii. Geometric Averages

Slides pp. 448–449

Not all path-dependent derivatives are hard to price; barrier options are easy, for example. When averaging is done geometrically, the option payoffs are

$$\max\Big((S_0 S_1 \cdots S_n)^{1/(n+1)} - X,\ 0\Big), \qquad \max\Big(X - (S_0 S_1 \cdots S_n)^{1/(n+1)},\ 0\Big).$$

[35] T. Dai (B82506025, R86526008, D8852600) & Lyuu (2007) reduce it to $2^{O(\sqrt{n})}$.

[36] When the average is a moving average, a different technique is needed (C. Kao (R89723057) & Lyuu, 2003).

Theorem 10.4.6 (Angus, 1999)

The limiting analytical solutions are the Black–Scholes formulas

$$\begin{aligned} C &= Se^{-q_a\tau}N(x) - Xe^{-r\tau}N\big(x-\sigma_a\sqrt{\tau}\big), \\ P &= Xe^{-r\tau}N\big(-x+\sigma_a\sqrt{\tau}\big) - Se^{-q_a\tau}N(-x), \end{aligned} \tag{10.7}$$

with the volatility set to $\sigma_a \triangleq \sigma/\sqrt{3}$, the dividend yield set to $q_a \triangleq (r+q+\sigma^2/6)/2$, and

$$x \triangleq \frac{\ln(S/X) + \big(r - q_a + \sigma_a^2/2\big)\tau}{\sigma_a\sqrt{\tau}}.$$

【中】不是所有 path-dependent derivative 都難（barrier 就很容易）。用幾何平均時，payoff 是上面兩式；極限下的解就是 Black–Scholes 公式，只要把波動率換成 $\sigma_a = \sigma/\sqrt{3}$、dividend yield 換成 $q_a = (r+q+\sigma^2/6)/2$。

Proof（補充；簡報只給結論）. 想法：$\ln G$ 是 normal，算出它的平均和變異數，再找一支「假想股票」讓它的 $\ln S_\tau$ 有相同的分布，就能直接套 Black–Scholes。

(1) **寫成 normal 的和**。$\ln S_i = \ln S_0 + \sum_{k=1}^{i}\varepsilon_k$，在 risk-neutral 下（極限時）$\varepsilon_k$ i.i.d. $N(m\Delta t,\ \sigma^2\Delta t)$，$m \triangleq r - q - \sigma^2/2$（Lemma 8.14.15）。

(2) **交換求和順序**。ε_k 出現在所有 $i \ge k$ 的 $\ln S_i$ 裡，共 $n-k+1$ 次，所以

$$\ln G \triangleq \frac{1}{n+1}\sum_{i=0}^{n}\ln S_i = \ln S_0 + \sum_{k=1}^{n}\frac{n-k+1}{n+1}\varepsilon_k,$$

是獨立 normal 的加權和，所以也是 normal（Corollary 5.6.6）。

(3) **平均**。$\sum_{k=1}^{n}(n-k+1) = \frac{n(n+1)}{2}$，所以 $E[\ln G] - \ln S_0 = m\Delta t\cdot\frac{n}{2} = \frac{m\tau}{2}$。

(4) **變異數**。$\sum_{k=1}^{n}(n-k+1)^2 = \frac{n(n+1)(2n+1)}{6}$，所以 $\mathrm{Var}[\ln G] = \sigma^2\tau\cdot\frac{2n+1}{6(n+1)} \to \frac{\sigma^2\tau}{3}$。

(5) **對到 Black–Scholes**。dividend yield q_a、volatility σ_a 的股票有 $\ln S_\tau \sim N\big(\ln S + (r - q_a - \sigma_a^2/2)\tau,\ \sigma_a^2\tau\big)$。令它和 $\ln G$ 同分布：$\sigma_a^2 = \sigma^2/3$，且 $r - q_a - \frac{\sigma_a^2}{6} = \frac{m}{2}$，解得 $q_a = \frac{r+q}{2} + \frac{\sigma^2}{12} = \frac{r+q+\sigma^2/6}{2}$。call 價值只由 G 的分布決定，所以 (8.25) 代入 q_a, σ_a 就是答案。 ■

See the textbook for a proof based on the binomial model. Note that $\sigma_a < \sigma$: averaging cancels part of the noise, which is why average-rate options are cheaper than their vanilla counterparts.

【中】二項模型下的證明見課本。注意 $\sigma_a < \sigma$：平均會抵銷一部分雜訊，所以 average-rate option 比普通 option 便宜。

Note 10.4.7. geometric average 之所以好算，正是因為 $\ln G$ 是 log return 的線性函數，而 normal 的和仍是 normal。arithmetic average $\frac{1}{n+1}\sum S_i$ 是一堆 *lognormal* 的和，它不是 log-normal，也沒有 closed-form 的分布。實務上，average-rate options 幾乎都使用 arithmetic average。

§10.4.iii. An Approximate Formula for Asian Calls

Slide p. 450

For the continuously sampled arithmetic average there is an accurate approximation:[37]

$$C = e^{-r\tau}\left[\frac{S}{\tau}\int_0^\tau e^{\mu t+\sigma^2 t/2}\,N\left(\frac{-\gamma+(\sigma t/\tau)(\tau-t/2)}{\sqrt{\tau/3}}\right)dt \;-\; XN\left(\frac{-\gamma}{\sqrt{\tau/3}}\right)\right],$$

[37] Rogers & Shi (1995); Thompson (1999); K. Chen (R92723061) (2005); K. Chen (R92723061) & Lyuu (2006).

where $\mu \triangleq r - \sigma^2/2$ and γ is the unique value that satisfies

$$\frac{S}{\tau}\int_0^\tau e^{3\gamma\sigma t(\tau-t/2)/\tau^2+\mu t+\sigma^2[t-(3t^2/\tau^3)(\tau-t/2)^2]/2}\,dt \;=\; X.$$

【中】對連續取樣的算術平均，有一個準確的近似公式（上面的 C），其中 $\mu = r - \sigma^2/2$，γ 是讓第二個積分等於 X 的唯一值。

Note 10.4.8. 結構還是 Theorem 8.14.19 熟悉的「$S\times$ 機率 $- X\times$ 機率」。想法是對平均值最依賴的一個 normal 變數 Z 取條件（Z 基本上就是 Brownian path 的積分，變異數為 $\tau/3$）；γ 是讓「給定 Z 時平均值的條件期望」等於 X 的那個 Z 值，而對 payoff 中只由 Z 決定的部分，這個公式是精確的。

§10.5. Approximation Algorithm for Asian Options

Slides pp. 451–453

The following algorithm, due to Hull and White, is based on the BOPM. Consider a node at time j with the underlying asset price equal to $S_0u^{j-i}d^i$, and name it $N(j,i)$.

【中】以下 Hull 和 White 的演算法建立在 BOPM 上。把時間 j、股價 $S_0u^{j-i}d^i$ 的 node 記為 $N(j,i)$。

Proposition 10.5.1

The running sum $\sum_{m=0}^{j} S_m$ at node $N(j,i)$ has a maximum value of

$$S_0\big(1+u+u^2+\cdots+u^{j-i}+u^{j-i}d+\cdots+u^{j-i}d^i\big) \;=\; S_0\,\frac{1-u^{j-i+1}}{1-u} + S_0u^{j-i}d\,\frac{1-d^i}{1-d},$$

and a minimum value of

$$S_0\big(1+d+d^2+\cdots+d^i+d^iu+\cdots+d^iu^{j-i}\big) \;=\; S_0\,\frac{1-d^{i+1}}{1-d} + S_0d^iu\,\frac{1-u^{j-i}}{1-u}.$$

【中】$N(j,i)$ 的累積和最大是「先全部上漲、再全部下跌」的路徑，最小是「先全部下跌、再上漲」的路徑，兩者都是等比級數的和。

Proof（補充；簡報只給公式）. (1) **交換一對 move**。到 $N(j,i)$ 的 path 都是 $j-i$ 次 up、i 次 down，只是順序不同。path 中某處價格是 P，接著「down, up」得到 Pd、Pdu；換成「up, down」得到 Pu、Pud。第二個價格不變，第一個從 Pd 變成 Pu，變大。所以 up 越往前，和越大：全部 up 在前最大，全部 down 在前最小。

(2) **等比級數**。先 $j-i$ 次 up：$S_0(1+u+\cdots+u^{j-i}) = S_0\frac{1-u^{j-i+1}}{1-u}$；再 i 次 down：$S_0u^{j-i}(d+\cdots+d^i) = S_0u^{j-i}d\,\frac{1-d^i}{1-d}$。最小值把 u,d 和 $j-i,i$ 對調。

■

Divide these values by $j+1$ and call them $A_{\max}(j,i)$ and $A_{\min}(j,i)$; they are running averages (Figure 10.9).

【中】兩者除以 $j+1$ 就是 running average 的最大、最小值 $A_{\max}(j,i)$、$A_{\min}(j,i)$。

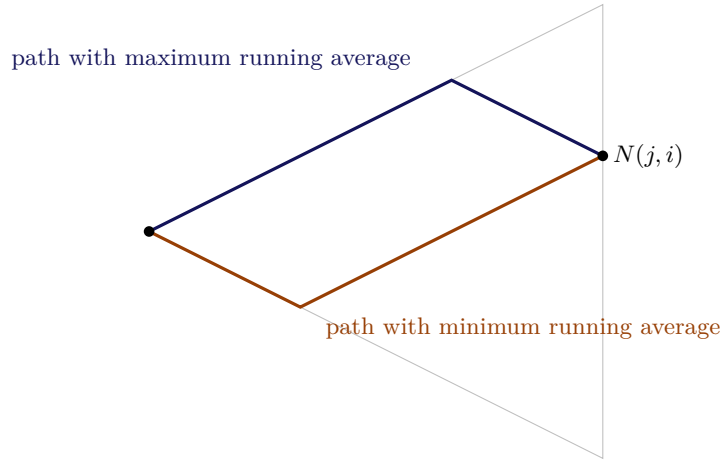


Figure 10.9.: All up moves first gives the largest running sum to $N(j,i)$, all down moves first the smallest; every other path lies in between.

【中】圖：先全部上漲得到最大的累積和，先全部下跌得到最小的；其他路徑都介於兩者之間。

Slides pp. 454–455

The number of paths to $N(j,i)$ is far too large: $\binom{j}{i}$, and for example $\binom{j}{j/2} \sim 2^j\sqrt{2/(\pi j)}$. Moreover, the distribution of the running averages at any given time seems to be bimodal.[38]

【中】到 $N(j,i)$ 的路徑數 $\binom{j}{i}$ 太多，例如 $\binom{j}{j/2} \sim 2^j\sqrt{2/(\pi j)}$；而且某個時間點的 running average 分布似乎是雙峰的。

§10.5.i. Bucketing

Slides pp. 456–458

But all averages must lie between $A_{\min}(j,i)$ and $A_{\max}(j,i)$. Pick $k+1$ equally spaced values in this range and treat them as the true and only running averages:

$$A_m(j,i) \triangleq \left(\frac{k-m}{k}\right)A_{\min}(j,i) + \left(\frac{m}{k}\right)A_{\max}(j,i), \qquad m = 0,1,\ldots,k.$$

Such **bucketing** or **binning** introduces errors, but it works reasonably well in practice.[39] A better alternative picks values whose logarithms are equally spaced;[40] still other alternatives are possible, considering the bimodal distribution of averages.

【中】所有平均都在 $A_{\min}$ 和 $A_{\max}$ 之間。在這範圍取 $k+1$ 個等距值，當成唯一可能的 running average。這種 bucketing（binning）有誤差，但實務上效果不錯。更好的做法是取對數等距的值；考慮到平均的分布是雙峰的，也可以有其他選法。

Remark 10.5.2. 分桶（bucketing）：每個節點不再追蹤所有可能的平均值（指數多個），而是只保留 $A_{\min}$ 到 $A_{\max}$ 之間等距的 $k+1$ 個代表值，每個代表值各存一個選擇權價格。狀態數從指數降到 $O(kn^2)$，代價是近似誤差。A_m 的公式可以改寫成 $A_m = A_{\min} + m\cdot\frac{A_{\max}-A_{\min}}{k}$：從 $A_{\min}$ 開始，每格寬 $\frac{A_{\max}-A_{\min}}{k}$，$m=0$ 是 $A_{\min}$、$m=k$ 是 $A_{\max}$。

[38] Contributed by Mr. Liu, Jun (R99944027) on April 15, 2014. The slide's plot uses $n=24$, $u=5/4$ and $d=4/5$.
[39] J. Hull & White (1993); Ritchken, Sankarasubramanian, & Vijh (1993).
[40] Called **log-linear interpolation**.

§10.5.ii. Backward Induction with Interpolation

Slides pp. 459–463

Backward induction calculates the option values at each node for the $k+1$ running averages. Suppose the current node is $N(j,i)$ and the running average is a, and consider the next node $N(j+1,i)$, after an up move. As the asset price there is $S_0u^{j+1-i}d^i$, we seek the option value corresponding to the new running average

$$A_u \triangleq \frac{(j+1)\,a + S_0u^{j+1-i}d^i}{j+2}.$$

But A_u is not likely to be one of the $k+1$ running averages at $N(j+1,i)$! Find the two running averages that bracket it,

$$A_\ell(j+1,i) \le A_u < A_{\ell+1}(j+1,i).$$

In "most" cases, the fastest way to nail ℓ is via

$$\ell = \left\lfloor \frac{A_u - A_{\min}(j+1,i)}{\big[A_{\max}(j+1,i) - A_{\min}(j+1,i)\big]/k} \right\rfloor.$$

But watch out for

- the rare case where $A_u = A_\ell(j+1,i)$ for some ℓ (floating-point rounding may put it on either side);
- the case where $A_u = A_{\max}(j+1,i)$, which gives $\ell = k$ and no $A_{\ell+1}$;
- the degenerate case where $A_0(j+1,i) = \cdots = A_k(j+1,i)$, which happens to the two extreme paths ($i=0$ or $i=j+1$), where the formula divides by zero.

【中】backward induction 要算每個 node 上 $k+1$ 個 running average 的 option 值。在 $N(j,i)$、running average 為 a 時，上漲到 $N(j+1,i)$ 後新的 running average 是 A_u；它通常不是那裡的 $k+1$ 個值之一，所以找夾住它的 $A_\ell \le A_u < A_{\ell+1}$，$\ell$ 大多可用 floor 公式直接算。要注意三種情況：A_u 剛好等於某個 A_ℓ（浮點誤差可能讓它落到任一邊）；$A_u = A_{\max}$ 時 $\ell = k$、沒有 $A_{\ell+1}$；最邊緣的兩條路徑（$i=0$ 或 $i=j+1$）所有 A_ℓ 都相同，公式會除以零。

Remark 10.5.3. (1) A_u。running average a 是 $S_0,\ldots,S_j$ 共 $j+1$ 個價格的平均，所以總和是 $(j+1)a$；加上新價格 $S_0u^{j+1-i}d^i$ 後共 $j+2$ 個，再除以 $j+2$。

(2) ℓ。由 bucketing 的改寫，$A_\ell = A_{\min} + \ell w$，$w = \frac{A_{\max}-A_{\min}}{k}$ 是格寬。$A_\ell \le A_u < A_{\ell+1}$ 等價於 $\ell \le \frac{A_u - A_{\min}}{w} < \ell+1$，所以取 floor。

(3) **三個例外**。(i) 浮點誤差可能讓 A_u 剛好在格線上時算錯格；(ii) $A_u = A_{\max}$ 時 $\ell = k$，沒有第 $k+1$ 格；(iii) 最上、最下的 node 只有一條 path，$A_{\max} = A_{\min}$，$w = 0$。

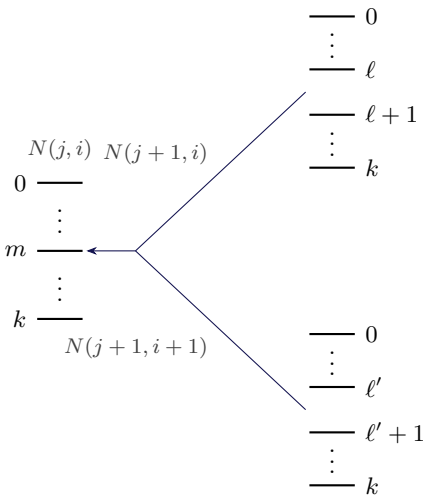


Figure 10.10.: Backward induction for arithmetic average-rate options. The $k+1$ option values at $N(j,i)$ are stored in an array; each depends on two values at $N(j+1,i)$ (buckets ℓ, $\ell+1$) and two at $N(j+1,i+1)$ (buckets ℓ', $\ell'+1$).

【中】圖：$N(j,i)$ 的 $k+1$ 個 option 值存在一個 array；每個值取決於 $N(j+1,i)$ 的兩格（$\ell, \ell+1$）和 $N(j+1,i+1)$ 的兩格（$\ell', \ell'+1$）。

Express A_u as a linearly interpolated value of the two running averages,

$$A_u \;=\; x\,A_\ell(j+1,i) + (1-x)\,A_{\ell+1}(j+1,i), \qquad 0 < x \le 1.$$

Let $C_\ell(t,s)$ denote the option value at node $N(t,s)$ with running average $A_\ell(t,s)$. Obtain the approximate option value given the running average A_u via

$$C_u \;\triangleq\; x\,C_\ell(j+1,i) + (1-x)\,C_{\ell+1}(j+1,i).$$

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This interpolation introduces the second source of error; alternatives to linear interpolation exist.

【中】把 A_u 寫成兩格的線性內插 $xA_\ell + (1-x)A_{\ell+1}$，再用同樣的權重把兩格的 option 值內插成 C_u。這是第二個誤差來源；也有線性內插以外的做法。

Remark 10.5.4. x **怎麼算**。由 $A_u = xA_\ell + (1-x)A_{\ell+1}$ 解出 $x = \frac{A_{\ell+1}-A_u}{A_{\ell+1}-A_\ell}$。$A_u$ 越靠近 A_ℓ，x 越接近 1。C_u 用同樣的權重 $x, 1-x$ 把兩格的 option 價值加權，也就是假設 option 價值在兩格之間是 running average 的線性函數。

The same steps are repeated for the down node $N(j+1,i+1)$ to obtain another approximate option value C_d. Finally obtain the option value as

$$\big[\,pC_u + (1-p)\,C_d\,\big]e^{-r\Delta t}.$$

【中】對下跌的 node $N(j+1,i+1)$ 做同樣的事得到 C_d，最後 option 值是 $[pC_u + (1-p)C_d]e^{-r\Delta t}$。

Algorithm 10.2: Approximation algorithm for European arithmetic average-rate calls

Input: $S_0, u, d, X, n, \hat{r}, k$

$p := (e^{\hat{r}} - d)/(u - d)$;
for $i = 0, 1, \ldots, n$ **do** // terminal price $S_0 u^{n-i} d^i$
  for $m = 0, 1, \ldots, k$ **do**
    $C[\,i\,][\,m\,] := \max\big(0,\ A_m(n, i) - X\big)$;
for $j = n - 1$ **to** 0 **do**
  for $i = 0, 1, \ldots, j$ **do**
    for $m = 0, 1, \ldots, k$ **do**
      $a := A_m(j, i)$;
      $A_u := \big((j + 1)\, a + S_0 u^{j+1-i} d^i\big)/(j + 2)$;
      find ℓ, x with $A_u = x A_\ell(j + 1, i) + (1 - x) A_{\ell+1}(j + 1, i)$;
      $C_u := x\, C[\,i\,][\,\ell\,] + (1 - x)\, C[\,i\,][\,\ell + 1\,]$;
      $A_d := \big((j + 1)\, a + S_0 u^{j-i} d^{i+1}\big)/(j + 2)$;
      find ℓ', x' likewise at $N(j + 1, i + 1)$;
      $C_d := x'\, C[\,i + 1\,][\,\ell'\,] + (1 - x')\, C[\,i + 1\,][\,\ell' + 1\,]$;
      $D[\,m\,] := \big(p C_u + (1 - p)\, C_d\big) e^{-\hat{r}}$;
    copy $D[\,0..k\,]$ to $C[\,i\,][\,0..k\,]$;
return $C[\,0\,][\,0\,]$;

【中】完整演算法：先設好到期時每個 node、每個 bucket 的 payoff，再一欄一欄往回做上面的內插和 backward induction。

Note 10.5.5. 需要先複製到 D，是因為下一個 i 計算它的 C_d 時還會讀 $C[\,i\,][\,\cdot\,]$ — 和 Algorithm 8.1 一樣是「proper updating」的問題。若是 American option，把對 $D[\,m\,]$ 的指定改成 $\max\big(a - X,\ (pC_u + (1 - p)C_d)e^{-\hat{r}}\big)$。

The running time is $O(kn^2)$: there are $O(n^2)$ nodes, and each node has $O(k)$ buckets. No interpolation is needed for the calculations at time step $n - 1$:[41] the option values there are simply (for calls) $C_u = \max(A_u - X, 0)$ and $C_d = \max(A_d - X, 0)$, which saves $O(nk)$ calculations and one layer of error.

【中】running time 是 $O(kn^2)$：$O(n^2)$ 個 node、每個 $O(k)$ 個 bucket。時間 $n - 1$ 那一步不用內插：call 的值直接是 $C_u = \max(A_u - X, 0)$、$C_d = \max(A_d - X, 0)$，省下 $O(nk)$ 的計算和一層誤差。

Arithmetic average-rate options were assumed to be newly issued, with no historical average to deal with. This problem can be easily addressed:[42] if the running sum of the first $m + 1$ prices is P, the payoff is

$$\max\Big(\frac{P + \sum_{i=m+1}^{n} S_i}{n + 1} - X,\ 0\Big) \;=\; \frac{n - m}{n + 1} \max\Big(\frac{1}{n - m} \sum_{i=m+1}^{n} S_i - X',\ 0\Big),$$

where $X' \triangleq \big[(n + 1)X - P\big]/(n - m)$ — a scaled, newly issued option on the remaining prices with an adjusted strike (when $X' \le 0$, the option is sure to finish in the money and is priced as a forward).

【中】前面假設 option 是新發行的、沒有歷史平均。若前 $m + 1$ 個價格的累積和是 P，payoff 可改寫成「只看剩下價格、strike 調整為 X' 的新發行 option」再乘上 $\frac{n-m}{n+1}$（$X' \le 0$ 時一定到價內，當作 forward 定價）。

[41] Contributed by Mr. Chen, Shih-Hang (R02723031) on April 9, 2014.
[42] See Exercise 11.7.4 of the textbook.

Remark 10.5.6. 等式的驗證。右邊括號內乘上 $\frac{n-m}{n+1}$：

$$\frac{n-m}{n+1}\cdot\frac{1}{n-m}\sum_{i>m}S_i=\frac{\sum_{i>m}S_i}{n+1},\qquad \frac{n-m}{n+1}X'=\frac{(n+1)X-P}{n+1}=X-\frac{P}{n+1},$$

兩者相減就是左邊括號內的 $\frac{P+\sum_{i>m}S_i}{n+1}-X$。再用 $c>0$ 時 $\max(cY,0)=c\max(Y,0)$。

How about the Greeks?[43] Delta and gamma from the first two periods (Section 9.7) are available as before, but each f_u and f_d now carries bucketing and interpolation errors of its own, which the difference quotients amplify.

【中】Greeks 呢？前兩期的 delta、gamma 照樣可用（Section 9.7），但每個 f_u、f_d 都帶有 bucketing 和內插誤差，差商會把誤差放大。

§10.5.iii. A Numerical Example

Slides pp. 467–473

Consider a European arithmetic average-rate call with strike price 50, and assume zero interest rate in order to dispense with discounting. The tree has $u=1.069$, $d=0.936$, $p=0.483$ and $S_0=50$. Node A is 2 periods from the root (price 50, after one up and one down move); its successors B and C are at expiration, with prices 53.447 and 46.775. Each node picks $k=3$, i.e., 4 equally spaced running averages (Figure 10.11). The minimum running average at node A is 48.925, the maximum 51.149.

【中】例子：strike 50 的 European 算術 average-rate call，利率設為 0 免折現；$u=1.069$、$d=0.936$、$p=0.483$、$S_0=50$。node A 在兩期後（一漲一跌，股價 50），後繼 B、C 在到期，股價 53.447、46.775。每個 node 取 $k=3$（4 個等距 running average）；A 的最小、最大 running average 是 48.925、51.149。

Remark 10.5.7. (1) **A 的** $A_{\min},A_{\max}$。A 是一漲一跌，到 A 的兩條 path 中先跌的平均最小：$\frac{S_0+S_0d+S_0du}{3}$；先漲的最大：$\frac{S_0+S_0u+S_0ud}{3}$。

(2) **四個 bucket**。格寬 $\frac{51.149-48.925}{3}$，所以四個值是 48.925、49.666、50.408、51.149（Figure 10.11）。

(3) **B、C 的 option 值**。B、C 在到期，option 值就是 payoff $\max(\text{average}-50,0)$。

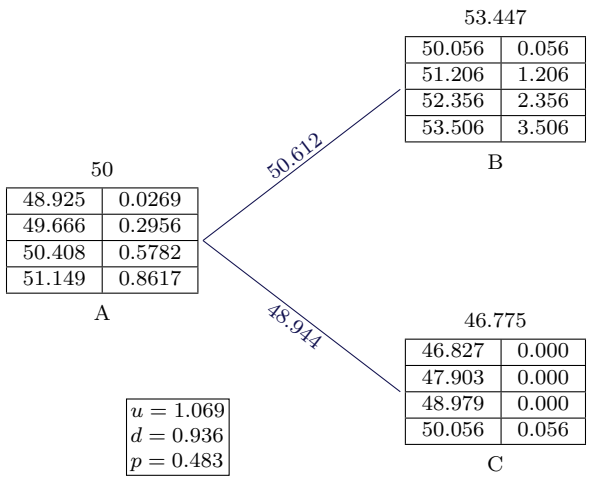


Figure 10.11.: Running averages (left columns) and option values at node A and its two successors. The edges show the new averages reached from A's second bucket, 49.666.

[43] Thanks to lively class discussions on March 31, 2004, and April 9, 2014.

【中】圖：node A 和兩個後繼的 running average（左欄）與 option 值；連線是從 A 的第二格 49.666 出發到達的新平均。

Consider the up move from node A with running average 49.666. Because the stock price at node B is 53.447, the new running average is

$$\frac{3 \times 49.666 + 53.447}{4} \approx 50.612.$$

With 50.612 lying between 50.056 and 51.206 at node B, we solve $50.612 = x \times 50.056 + (1 - x) \times 51.206$ to obtain $x \approx 0.517$. The option values corresponding to running averages 50.056 and 51.206 at node B are 0.056 and 1.206, so their contribution is weighted linearly as

$$x \times 0.056 + (1 - x) \times 1.206 \approx 0.611.$$

Now consider the down move. Because the stock price at node C is 46.775, the new running average is $(3 \times 49.666 + 46.775)/4 \approx 48.944$, lying between 47.903 and 48.979, so $x \approx 0.033$. The option values there are both 0.0, so the contribution is 0.0. Finally, the option value corresponding to running average 49.666 at node A equals

$$p \times 0.611 + (1 - p) \times 0.0 \approx 0.2956,$$

where $p = 0.483$. The remaining three option values at node A can be computed similarly.

【中】從 A 的 49.666 往上：B 的新平均 $(3 \times 49.666 + 53.447)/4 \approx 50.612$，介於 50.056 和 51.206，$x \approx 0.517$，內插得 0.611。往下：C 的新平均 48.944，介於 47.903 和 48.979，兩格的值都是 0。所以 A 這一格的值是 $0.483 \times 0.611 \approx 0.2956$；其他三格同理。

Remark 10.5.8. (1) $x \circ x = \frac{51.206-50.612}{51.206-50.056} \approx 0.517$，套上面 interpolation 的公式。
(2) **沒有折現**。利率為 0，所以 backward induction 只是 $pC_u + (1-p)C_d$，不用乘 $e^{-r\Delta t}$。

Note 10.5.9. A 在時間 $n-1$，所以由上面的 shortcut 不需要 interpolation：精確計算 $C_u = \max(50.612 - 50, 0) = 0.612$，價值為 0.2955 — 在這裡幾乎一樣。第三個 bucket（50.408）差得比較多：interpolation 給出 0.578，精確計算是 0.564。原因是 option 價值對 running average 是 *convex* 的（到期時是 $\max(\cdot - X, 0)$），所以 linear interpolation 用的弦會落在函數上方。這正是下一個 remark 背後的機制。

§10.5.iv. Remarks on Asian Option Pricing

Slides pp. 474–478

Asian option pricing is an active research area.

- The above algorithm overestimates the "true" value, and to guarantee convergence, k needs to grow with n at least.[44]
- There is a convergent approximation algorithm that does away with interpolation, with a running time of $2^{O(\sqrt{n})}$.[45]
- There is an $O(kn^2)$-time algorithm with an error bound of $O(Xn/k)$ from the naive $O(2^n)$-time binomial tree algorithm in the case of European Asian options.[46] k can be varied for trade-off between time and accuracy: picking $k = O(n^2)$ gives an error of $O(1/n)$ and a running time of $O(n^4)$.

[44] T. Dai (B82506025, R86526008, D8852600), G. Huang (F83506075), & Lyuu (2002). With $k = 50000$ the slide's plot of the value against $n = 50, \ldots, 140$ still drifts between about 0.325 and 0.35.
[45] T. Dai (B82506025, R86526008, D8852600) & Lyuu (2002, 2004).
[46] Aingworth, Motwani (1962–2009), & Oldham (2000).

- Another approximation algorithm reduces the error to $O(X\sqrt{n}/k)$ by varying the number of buckets per node.[47] If we pick $k = O(n)$, the error is $O(n^{-0.5})$; if we pick $k = O(n^{1.5})$, the error is $O(1/n)$ and the running time is $O(n^{3.5})$.
- Under "reasonable assumptions," an $O(n^2)$-time algorithm with an error bound of $O(1/n)$ exists.[48] The basic idea is a nonuniform allocation of running averages k_{ij} instead of a uniform k: few buckets where the range $A_{\max} - A_{\min}$ is narrow (near the edges of the tree), many where it is wide. It strikes a tight balance between error and complexity.

【中】Asian option 定價是活躍的研究領域：上面的演算法會高估真值，要保證收斂 k 至少要隨 n 成長；有不需內插、$2^{O(\sqrt{n})}$ 時間的收斂近似演算法；有 $O(kn^2)$ 時間、誤差 $O(Xn/k)$ 的演算法，取 $k = O(n^2)$ 得 $O(1/n)$ 誤差、$O(n^4)$ 時間；讓每個 node 的 bucket 數不同可把誤差降到 $O(X\sqrt{n}/k)$；在合理假設下有 $O(n^2)$ 時間、$O(1/n)$ 誤差的演算法，關鍵是 bucket 數 k_{ij} 不均勻：$A_{\max} - A_{\min}$ 窄處（tree 邊緣）少放、寬處多放。

> **Intuition .** Why can a nonuniform allocation help so much? The bucketing error at a node is proportional to the bucket width $(A_{\max} - A_{\min})/k$. That range is zero on the two edges of the tree and widest in the middle, so spending the same k everywhere wastes buckets where they are not needed.

【中】為什麼不均勻分配幫助這麼大？每個 node 的 bucketing 誤差和格寬 $(A_{\max} - A_{\min})/k$ 成正比；這個範圍在 tree 兩側邊緣是 0、中間最寬，所以每處都放同樣的 k 會把 bucket 浪費在不需要的地方。

§10.5.v. A Grand Comparison

Slides pp. 479–481

The slides compare several methods against exact values over a grid of strikes, volatilities and rates.[49] AA2 and AA3 are Zhang's analytical approximations; Hsu–Lyuu is the $O(n^2)$ tree; Chen–Lyuu is the formula above. A selection:

X	σ	r	Exact	AA2	AA3	Hsu–Lyuu	Chen–Lyuu
95	0.05	0.05	7.1777275	7.1777244	7.1777279	7.178812	7.177726
100			2.7161745	2.7161755	2.7161744	2.715613	2.716168
105			0.3372614	0.3372601	0.3372614	0.338863	0.337231
90	0.10	0.05	11.9510927	11.9509331	11.9510871	11.951610	11.951076
100			3.6413864	3.6414032	3.6413875	3.642325	3.641344
110			0.3312030	0.3312563	0.3311968	0.331348	0.331074
90	0.30	0.05	13.9538233	13.9555691	13.9540973	13.953937	13.952421
100			7.9456288	7.9459286	7.9458549	7.945918	7.944357
110			4.0717942	4.0702869	4.0720881	4.071945	4.070115
90	0.30	0.15	16.5129113	16.5133090	16.5128376	16.512963	16.512024
100			10.2098305	10.2110681	10.2101058	10.210039	10.208724
110			5.7301225	5.7296982	5.7303567	5.730357	5.728161

Table 10.3.: Arithmetic average-rate call values from the slides' comparison (rows for $\sigma = 0.05, 0.10, 0.30$; the slides also cover $\sigma = 0.20$ and $r = 0.09, 0.15$ throughout).

【中】簡報在一組 strike、波動率、利率上比較幾種方法和精確值：AA2、AA3 是 Zhang 的解析近似，Hsu–Lyuu 是 $O(n^2)$ 的 tree，Chen–Lyuu 是上面的近似公式。表中是部分結果。

[47] T. Dai (B82506025, R86526008, D8852600), G. Huang (F83506075), & Lyuu (2002).
[48] Hsu (R7526001, D89922012) & Lyuu (2004).
[49] Hsu (R7526001, D89922012) & Lyuu (2004); J. E. Zhang (2001, 2003); K. Chen (R92723061) & Lyuu (2006).

All four methods agree to about three or four significant digits across the board; the analytical approximations lose digits as σ grows, the tree is weakest at small σ. A problem whose naive solution is exponential is, in practice, solved — but every method trades accuracy, time and guarantees differently, and only some of them come with error bounds.

【中】四種方法大致都準到三、四位有效數字；解析近似在 σ 大時失準，tree 在 σ 小時最弱。天真解法是指數時間的問題，實務上已經解決——但每種方法在準確度、時間和保證之間的取捨不同，只有部分方法有誤差上界。

11. Forwards, Futures, Futures Options, Swaps

Lecture 6
October 15, 2026

Slides pp. 480–481

Summon the nations to come to the trial.
Which of their gods can predict the future?
— Isaiah 43:9

The sure fun of the evening
outweighed the uncertain treasure[.]
— Mark Twain (1835–1910), *The Adventures of Tom Sawyer*

Slides pp. 482–483

Notation .
- r: the riskless interest rate; continuous compounding is assumed.
- t: the current time; T: the maturity date; $\tau \triangleq T - t$: the remaining time to maturity (years).
- S: the spot price; S_T: the spot price at maturity.
- X: the delivery price.
- F: the forward or futures price for a newly written contract.
- f: the value of the contract.
- A price with a subscript t usually refers to the price at time t.

【中】r 是無風險利率（連續複利），t 是現在、T 是到期日、$\tau = T - t$ 是剩餘期間；S 是現貨價、S_T 是到期時的現貨價；X 是交割價（delivery price）；F 是新簽合約的 forward/futures price；f 是合約的價值；下標 t 通常代表時間 t 的價格。

Remark 11.0.1. X 和 F 要分清楚：X 是寫在合約上、簽約後不變的交割價；F 是現在新簽一份合約會用的交割價，隨市場變動。合約剛簽時 $X = F$。

§11.1. Forward Contracts

Slides pp. 484–485

Long **forward contracts** are for the purchase of the underlying asset for a certain delivery price on a specific time.[1] The underlying can be foreign currencies, bonds, corn, etc. They are ideal for hedging purposes. For example, a farmer enters into a forward contract with a food processor to deliver 100,000 bushels of corn for \$2.5 per bushel on September 27, 1995.[2] The farmer is assured of a buyer at an acceptable price, and the processor knows the cost of corn in advance.

【中】long forward contract 是約定在特定時間、以特定交割價買入標的資產的合約，標的可以是外幣、債券、玉米等，很適合避險。例：農夫和食品加工廠簽約，1995/9/27 以每 bushel 2.5 元交 10 萬 bushels 玉米（農夫是 short 方）。農夫確保有買家、價格可接受；加工廠事先知道成本。

[1] They have a 4,000-year history (Marthinsen, 2018).
[2] The farmer assumes a short position.

A forward agreement limits both risk and rewards. If the spot price of corn rises on the delivery date, the farmer will miss the opportunity of extra profits; if the price declines, the processor will be paying more than it would. Either side has an incentive to default. Other problems: the food processor may go bankrupt, the farmer can go bust, the farmer might not be able to harvest 100,000 bushels of corn because of bad weather, the cost of growing corn may skyrocket, etc.

【中】forward 同時限制了風險和報酬：交割日玉米漲價，農夫少賺；跌價，加工廠多付。所以雙方都有違約的誘因。其他問題：加工廠可能倒閉、農夫可能破產、天候不佳收成不到 10 萬 bushels、種植成本暴漲等。

§11.1.i. Spot and Forward Exchange Rates

Slides pp. 486–488

Notation .
- S: the spot exchange rate; F: the forward exchange rate one year from now (both in domestic/foreign terms, as in Section 10.3);
- r_f: the annual interest rate of the foreign currency; r: the annual interest rate of the local currency.

Arbitrage opportunities will arise unless these four numbers satisfy an equation, due to Keynes (1923).[3]

Theorem 11.1.1 (Interest rate parity)

$$\frac{F}{S} = e^{r-r_f}. \tag{11.1}$$

Proof. A holder of the local currency can do either of:
(1) lend the money in the domestic market to receive e^r one year from now;
(2) convert local currency for foreign currency, lend for 1 year in the foreign market, and convert foreign currency into local currency at the fixed forward exchange rate F, by selling forward foreign currency now.

No money changes hand in entering into a forward contract, so one unit of local currency becomes Fe^{r_f}/S one year from now in the second case. If $Fe^{r_f}/S > e^r$, an arbitrage profit can result from borrowing money in the domestic market and lending it in the foreign market. If $Fe^{r_f}/S < e^r$, an arbitrage profit can result from borrowing money in the foreign market and lending it in the domestic market. ■

【中】記 S 為即期匯率、F 為一年期遠期匯率（都是「本國貨幣/外幣」），r_f、r 分別是外國、本國年利率。四個數必須滿足 interest rate parity (11.1)，否則有套利。證明：一單位本國貨幣可以 (1) 存在本國，一年後得 e^r；(2) 換成外幣（$1/S$）、存在外國（變成 e^{r_f}/S）、並用 forward 鎖定以 F 換回，一年後得 Fe^{r_f}/S。簽 forward 不用付錢，所以兩者必須相等；較大的那邊借另一邊的錢來投資就是套利。

Remark 11.1.2. (2) 的 Fe^{r_f}/S 怎麼來。1 單位本國貨幣換成 $1/S$ 單位外幣；在外國存一年變成 e^{r_f}/S 單位外幣；今天先賣出這麼多外幣的 forward，一年後每單位外幣換回 F 單位本國貨幣，共 Fe^{r_f}/S。整個過程只有今天的 1 和一年後的 Fe^{r_f}/S 兩筆現金流，沒有任何風險，所以必須和 (1) 的 e^r 相等。(11.1) 也是下面 (11.6) 取 $q = r_f$、$\tau = 1$ 的特例：外幣就是 dividend yield 為 r_f 的資產（Section 10.3）。

[3] John Maynard Keynes (1883–1946) was one of the greatest economists in history. The parity broke down in late 2008 (Mancini-Griffoli & Ranaldo, 2013).

§11.1.ii. Forward Price

Slides pp. 489–491

The payoff of a forward contract at maturity is

$$S_T - X.$$

Contrast that with the call's payoff $\max(S_T - X, 0)$ (6.1). Forward contracts do not involve any initial cash flow (Figure 11.1).

Definition 11.1.3. The **forward price** F is the delivery price X which makes the forward contract zero valued; that is, $f = 0$ when $X = F$.

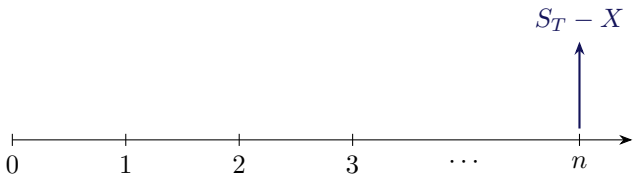


Figure 11.1.: Cash flow of a long forward contract: nothing at time 0, a single cash flow $S_T - X$ (of either sign) at maturity.

The delivery price cannot change because it is written in the contract. But the forward price may change after the contract comes into existence. So although the value of a forward contract, f, is 0 at the outset, it will fluctuate thereafter: this value is enhanced when the spot price climbs and depressed when the spot price declines. The forward price also varies with the maturity of the contract.

【中】forward 到期的 payoff 是 $S_T - X$（對比 call 的 $\max(S_T - X, 0)$），期初沒有現金流。forward price F 定義為讓合約價值 $f = 0$ 的交割價。交割價寫在合約上不會變，但 forward price 會變，所以合約價值一開始是 0，之後會隨現貨價漲而漲、跌而跌；forward price 也隨合約期限而不同。

§11.1.iii. Underlying Pays No Income

Slides pp. 492–493

Lemma 11.1.4

For a forward contract on an underlying asset providing no income,

$$F = Se^{r\tau}. \tag{11.2}$$

Proof. If $F > Se^{r\tau}$: borrow S dollars for τ years, buy the underlying asset, and short the forward contract with delivery price F. At maturity, deliver the asset for F and use $Se^{r\tau}$ to repay the loan, leaving an arbitrage profit of $F - Se^{r\tau} > 0$. If $F < Se^{r\tau}$, do the opposite. ■

【中】標的不配息時，forward price 是 $F = Se^{r\tau}$ (11.2)。若 $F > Se^{r\tau}$：借 S 買資產、同時 short forward，到期以 F 交出資產、還 $Se^{r\tau}$，淨賺 $F - Se^{r\tau} > 0$。反之則反向操作（放空資產、把錢存起來、long forward）。

Remark 11.1.5. 「**do the opposite**」**具體是什麼**。若 $F < Se^{r\tau}$：今天放空資產得 S、存起來，並 long forward（交割價 F）。到期存款變成 $Se^{r\tau}$，付 F 收到資產、還給借券的人，淨賺 $Se^{r\tau} - F > 0$。兩個方向都是今天零成本、到期確定正報酬，違反 Corollary 7.2.3。

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Example 11.1.6 (Zero-coupon bonds)**.** Let r be the annualized 3-month riskless interest rate and S the spot price of the 6-month zero-coupon bond. A new 3-month forward contract on that 6-month zero-coupon bond should command a delivery price of $Se^{r/4}$. So if $r = 6\%$ and $S = 970.87$, then the delivery price is

$$970.87 \times e^{0.06/4} \;=\; 985.54.$$

Example 11.1.7 (Options)**.** Suppose S is the spot price of the European call that expires at some time later than T. A τ-year forward contract on that call commands a delivery price of $Se^{r\tau}$. So it equals the future value of the Black–Scholes formula (Theorem 8.14.19).

【中】例：r 為 3 個月期年化利率、S 為 6 個月期 zero-coupon bond 的現價，3 個月期的 forward 交割價是 $Se^{r/4}$；$r = 6\%$、$S = 970.87$ 時是 985.54。例：標的是比 T 晚到期的 European call，τ 年期 forward 的交割價是 $Se^{r\tau}$，也就是 Black–Scholes 價格的終值。

Remark 11.1.8. 兩個例子都直接套 (11.2)，因為 zero-coupon bond 和 call 在 forward 期間都不發任何現金。第一個例子 $\tau = 3/12 = 1/4$，所以 $e^{r\tau} = e^{0.06/4}$。

Slides pp. 496–498

Proposition 11.1.9

The value of a forward contract on an underlying asset paying no income is

$$f \;=\; S - Xe^{-r\tau}. \tag{11.3}$$

Proof. Consider a portfolio consisting of one long forward contract, cash amount $Xe^{-r\tau}$, and one short position in the underlying asset. The cash will grow to X at maturity, which can be used to take delivery of the forward contract. The delivered asset will then close out the short position. Since the value of the portfolio is zero at maturity, its PV must be zero (Corollary 7.2.3): $f + Xe^{-r\tau} - S = 0$. ∎

So a forward contract can be replicated by a long position in the underlying and a loan of $Xe^{-r\tau}$ dollars. Lemma 11.1.4 is recovered by setting $f = 0$ in (11.3): then $X = Se^{r\tau}$, the forward price.

【中】不配息時 forward 合約的價值是 $f = S - Xe^{-r\tau}$ (11.3)：持有 long forward、現金 $Xe^{-r\tau}$、放空一單位資產，到期現金變 X 用來交割、拿到的資產拿去回補空頭，組合到期價值為 0，所以現值為 0。因此 forward = 買資產 + 借 $Xe^{-r\tau}$。在 (11.3) 令 $f = 0$ 就得到 $X = Se^{r\tau}$，即 Lemma 11.1.4。

Note 11.1.10. 和 put–call parity (7.2) 比較：$C - P = S - \mathrm{PV}(X)$，右邊正是 (11.3)。因為 long call + short put 到期付 $\max(S_T - X, 0) - \max(X - S_T, 0) = S_T - X$，就是一個交割價 X 的 forward。

§11.1.iv. Underlying Pays Predictable Income

Slides pp. 499–501

Lemma 11.1.11

For a forward contract on an underlying asset providing a predictable income with a PV

of I,

$$F = (S - I)\, e^{r\tau}. \tag{11.4}$$

Proof. If $F > (S-I)\, e^{r\tau}$, borrow S dollars for τ years, buy the underlying asset, and short the forward contract with delivery price F. Use the income to repay part of the loan. At maturity, the asset is delivered for F, and $(S-I)\, e^{r\tau}$ is used to repay the remaining loan. That leaves an arbitrage profit of $F - (S-I)\, e^{r\tau} > 0$. If $F < (S-I)\, e^{r\tau}$, reverse the above. ■

【中】標的有可預期收入、現值為 I 時，$F = (S-I)e^{r\tau}$。若 F 較大：借 S 買資產、short forward，收到的收入拿去還一部分借款；到期交出資產拿 F，剩下的借款是 $(S-I)e^{r\tau}$，淨賺 $F - (S-I)e^{r\tau} > 0$。

Remark 11.1.12. 為什麼剩下的借款是 $(S-I)e^{r\tau}$。把借的 S 拆成兩筆：I 和 $S-I$。收入在各時點收到，現值合計 I，所以這些收入剛好還清第一筆 I 的本息；第二筆 $S-I$ 到期本息為 $(S-I)e^{r\tau}$。

Example 11.1.13. Consider a 10-month forward contract on a \$50 stock. The stock pays a dividend of \$1 every 3 months. The forward price is

$$\left(50 - e^{-r_3/4} - e^{-r_6/2} - e^{-3\times r_9/4}\right) e^{r_{10}\times(10/12)},$$

where r_i is the annualized i-month interest rate.

【中】例：股價 50、每 3 個月配息 1 元，10 個月期 forward price 如上，r_i 是 i 個月期的年化利率。

Remark 11.1.14. 10 個月內配息發生在第 $3, 6, 9$ 個月，各 \$1，所以 $I = e^{-r_3\cdot 3/12} + e^{-r_6\cdot 6/12} + e^{-r_9\cdot 9/12}$；每筆用自己期限的利率折現。再代入 (11.4)，$\tau = 10/12$。

§11.1.v. Underlying Pays a Continuous Dividend Yield

Slide p. 502

Proposition 11.1.15

If the underlying asset pays a continuous dividend yield of q, the value of a forward contract at any time prior to T is

$$f = Se^{-q\tau} - Xe^{-r\tau}. \tag{11.5}$$

Consequently the forward price is

$$F = Se^{(r-q)\tau}. \tag{11.6}$$

Proof（補充；簡報指向課本 *p. 160*）. 買 $e^{-q\tau}$ 單位資產（成本 $Se^{-q\tau}$），收到的股利全部再買資產，則持有量以速率 q 成長，到期剛好是 $e^{-q\tau}e^{q\tau} = 1$ 單位，價值 S_T；再借 $Xe^{-r\tau}$，到期還 X。這個組合到期付 $S_T - X$，和 forward 相同，由 Corollary 7.2.4，$f = Se^{-q\tau} - Xe^{-r\tau}$。令 $f = 0$ 解 X 得 (11.6)。 ■

【中】標的有連續 dividend yield q 時，合約價值 $f = Se^{-q\tau} - Xe^{-r\tau}$ (11.5)，forward price $F = Se^{(r-q)\tau}$ (11.6)。

§11.2. Futures Contracts

Slides pp. 503–504

Futures contracts differ from forward contracts in the following ways:

- They are traded on a central exchange.
- There is a clearinghouse, so credit risk is minimized.
- Futures contracts are standardized instruments.
- Gains and losses are **marked to market** daily: adjusted at the end of each trading day based on the settlement price.

【中】futures 和 forward 的差別：在集中交易所交易；有結算所（clearinghouse），信用風險降到最低；是標準化商品；每天 marked to market，依結算價在每個交易日結束時調整損益。

The **size** of a futures contract is the amount of the underlying asset to be delivered under the contract: 5,000 bushels for the corn futures on the Chicago Board of Trade (CBOT); one million U.S. dollars for the Eurodollar futures on the Chicago Mercantile Exchange (CME).[4] A position can be closed out (or offset) by entering into a reversing trade to the original one. Most futures contracts are closed out in this way rather than have the underlying asset delivered. Forward contracts, by contrast, are meant for delivery.

【中】futures 的合約規模是一口要交割的標的數量，例如 CBOT 玉米期貨 5,000 bushels、CME 歐洲美元期貨一百萬美元。部位可以用反向交易平倉（offset），大部分 futures 都是這樣了結而不實際交割；forward 則是為了交割。

§11.2.i. Daily Settlements

Slides pp. 505–506

Price changes in the futures contract are settled daily. Hence the spot price rather than the initial futures price is paid on the delivery date. Marking to market nullifies any financial incentive for not making delivery.

Example 11.2.1. A farmer enters into a forward contract to sell 100,000 bushels of corn at \$2.00 per bushel in November. Suppose the price of corn rises to \$2.5 by November. The farmer has incentive to sell his harvest in the spot market for \$2.5. With marking to market, the farmer has transferred \$0.5 per bushel from his futures account to that of the clearinghouse by November (Figure 11.2). When the farmer makes delivery, he is paid the spot price, \$2.5 per bushel. The farmer has little incentive to default. The net price remains $\$2.5 - 0.5 = 2$ per bushel, the original delivery price.

【中】futures 每天結算價差，所以交割日付的是當時的現貨價而不是原本的 futures price。marking to market 消除了不交割的財務誘因。例：農夫約定 11 月以每 bushel 2 元賣玉米，11 月漲到 2.5 元，用 forward 的話農夫有誘因違約去現貨市場賣 2.5。用 futures 時，到 11 月農夫的保證金帳戶已經陸續付給結算所每 bushel 0.5；交割時拿到現貨價 2.5，淨價仍是 $2.5 - 0.5 = 2$，等於原本的交割價，違約沒有好處。

§11.2.ii. Daily Cash Flows

Slides pp. 507–508

Let F_i denote the futures price at the end of day i. The contract's cash flow on day i is $F_i - F_{i-1}$. The net cash flow over the life of the contract is

$$(F_1 - F_0) + (F_2 - F_1) + \cdots + (F_n - F_{n-1}) \;=\; F_n - F_0 \;=\; S_T - F_0. \tag{11.7}$$

[4] CME and CBOT merged on July 12, 2007.

A futures contract has the same accumulated payoff $S_T - F_0$ as a forward contract. The actual payoff may vary because of the reinvestment of daily cash flows and how $S_T - F_0$ is distributed.

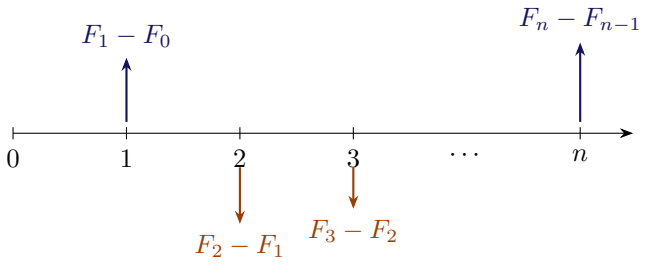


Figure 11.2.: Daily cash flows of a long futures position: one cash flow per day, up or down, summing to $S_T - F_0$ by (11.7). Compare the single cash flow of Figure 11.1.

【中】令 F_i 為第 i 天收盤的 futures price，第 i 天的現金流是 $F_i - F_{i-1}$，全部加起來 telescoping 成 $F_n - F_0 = S_T - F_0$ (11.7)，和 forward 的累積 payoff 一樣。實際 payoff 會因每天現金流的再投資、以及 $S_T - F_0$ 怎麼分配在各天而不同。

Remark 11.2.2. (11.7) 最後一步 $F_n = S_T$：第 n 天是交割日，交割日的結算價由現貨價決定（下一小節），所以 $F_n = S_T$。

§11.2.iii. Delivery and Hedging

Slide p. 509

Futures price is the delivery price that makes the futures contract zero-valued. Delivery ties the futures price to the spot price: on the delivery date, the settlement price of the futures contract is determined by the spot price. Hence, when the delivery period is reached, the futures price should be very close to the spot price.[5] Changes in futures prices usually track those in spot price, making hedging possible.

【中】futures price 是讓 futures 合約價值為 0 的交割價。交割把 futures price 和現貨價綁在一起：交割日的結算價由現貨價決定，所以接近交割期時 futures price 應該很接近現貨價。futures price 的變動通常跟著現貨價走，因此可以用來避險。

§11.2.iv. Forward and Futures Prices

Slides pp. 510–511

Surprisingly, futures price equals forward price if interest rates are nonstochastic![6] This result "justifies" treating a futures contract as if it were a forward contract, ignoring its marking-to-market feature. The West Texas Intermediate (WTI) futures price was negative on April 20, 2020![7]

【中】驚人的是，只要利率不是隨機的，futures price 就等於 forward price。這「合理化」了把 futures 當成 forward、忽略 marking to market 的做法。WTI 原油期貨在 2020/4/20 曾經是負價格。

Proposition 11.2.3 (Cox, Ingersoll, & Ross, 1981)

[5] But since early 2006, futures for corn, wheat, and soybeans occasionally expired at a price much higher than that day's spot price (Henriques, 2008).
[6] Cox, Ingersoll, & Ross (1981); see p. 164 of the textbook for proof.
[7] April 21 was the last trading day for oil delivery in May to Cushing, Oklahoma.

If the daily interest rate is a known constant δ, then $F_0 = G$, where F_0 is the futures price and G the forward price, both for delivery on day n.

Proof（補充）. 比較兩個策略在第 n 天的價值。

(1) **futures 策略**。今天存 F_0 元；在第 i 天開始時持有 $e^{i\delta}$ 口 long futures（δ 已知，所以口數事先決定）。第 i 天的損益 $(F_i - F_{i-1})e^{i\delta}$ 以利率 δ 存到第 n 天，變成 $(F_i - F_{i-1})e^{n\delta}$。由 (11.7)，合計 $(S_T - F_0)e^{n\delta}$，加上存款 $F_0e^{n\delta}$，總共 $S_Te^{n\delta}$。期初成本 F_0（futures 本身不用錢）。

(2) **forward 策略**。今天存 G 元、long $e^{n\delta}$ 口交割價 G 的 forward。第 n 天得 $(S_T - G)e^{n\delta} + Ge^{n\delta} = S_Te^{n\delta}$。期初成本 G。

兩個策略在每個 state 的 payoff 都是 $S_Te^{n\delta}$，由 Corollary 7.2.4，$F_0 = G$。 ■

When interest rates are stochastic, forward and futures prices are no longer theoretically identical. Suppose interest rates are uncertain and futures prices move in the same direction as interest rates. Then futures prices will exceed forward prices. For short-term contracts, the differences tend to be small. Unless stated otherwise, assume forward and futures prices are identical.

【中】利率是隨機時兩者理論上不再相等。若 futures price 和利率同方向變動，futures price 會高於 forward price。短期合約差異通常很小。除非特別說明，以下假設兩者相等。

Intuition. Proposition 11.2.3 的證明在 (1) 需要事先知道口數 $e^{i\delta}$，也就是利率必須是確定的。若 futures price 和利率正相關，long futures 賺錢時利率也高、賺的錢以高利率再投資，賠錢時利率低、借錢的成本也低，所以 long futures 比 long forward 有利；要讓兩者期初價值都是 0，futures 的交割價就必須更高。

§11.3. Futures Options and Forward Options

Slides pp. 512–513

The underlying of a **futures option** is a futures contract. Upon exercise, the option holder takes a position in the futures contract with a futures price equal to the option's strike price: a call holder acquires a long futures position, and a put holder acquires a short futures position. The futures contract is then marked to market, and the futures positions of the two parties are at the prevailing futures price (thus zero-valued).

It works as if the call holder received a futures contract plus cash equivalent to the prevailing futures price F_t minus the strike price X, i.e., $F_t - X$; this futures contract has zero value. It works as if the put holder sold a futures contract for $X - F_t$ dollars.

【中】futures option 的標的是 futures 合約。履約時持有人取得一個 futures price 等於 strike 的 futures 部位：call 拿到 long、put 拿到 short。接著馬上 marked to market，雙方部位變成以當時的 futures price 計價（價值為 0）。所以效果等於 call 持有人拿到一份價值 0 的 futures 加上現金 $F_t - X$；put 持有人則是拿到 $X - F_t$。

Slides pp. 514–515

For a **forward option**, what is delivered is now a forward contract with a delivery price equal to the option's strike price. Exercising a call forward option results in a long position in a forward contract; exercising a put forward option results in a short position in a forward contract. Exercising a forward option incurs no immediate cash flows as there is no marking to market.

【中】forward option 履約後拿到的是交割價等於 strike 的 forward 合約：call 拿到 long、put 拿到 short。因為沒有 marking to market，履約當下沒有現金流。

Example 11.3.1. Consider an American call with strike \$100 and an expiration date in September. The underlying asset is a forward contract with a delivery date in December. Suppose the forward price in July is \$110. Upon exercise, the call holder receives a forward contract with a delivery price of \$100. If an offsetting position is then taken in the forward market,[8] a \$10 profit in December will be assured. A call on the futures would realize the \$10 profit in July.

【中】例：strike 100、9 月到期的 American call，標的是 12 月交割的 forward。7 月時 forward price 為 110，履約得到交割價 100 的 forward；再 short 一個交割價 110 的 forward 對沖，12 月確定賺 10。若是以 futures 為標的的 call，7 月就實現這 10 元。

§11.3.i. Some Pricing Relations

Slides pp. 516–517

Let T be the delivery date, the current time be 0, and the option (on the futures or forward contract) have expiration date t ($t \le T$). Assume a constant, positive interest rate. Although forward price equals futures price, a forward option does not have the same value as a futures option. The payoffs of calls at time t are, respectively,

$$\text{futures option} = \max(F_t - X, 0), \tag{11.8}$$

$$\text{forward option} = \max(F_t - X, 0)\, e^{-r(T-t)}. \tag{11.9}$$

【中】令 T 為交割日、現在為 0、option 在 $t \le T$ 到期，利率為正的常數。雖然 forward price 等於 futures price，forward option 和 futures option 的價值卻不同：在時間 t 的 payoff 分別是 (11.8) 和 (11.9)。

Remark 11.3.2. (11.9) 的 $e^{-r(T-t)}$。在 t 履約 forward call 拿到交割價 X 的 long forward，此時新的 forward price 是 F_t。由 (11.3) 加上 (11.2) ($S_t = F_t e^{-r(T-t)}$)，它在 t 的價值是

$$S_t - X e^{-r(T-t)} = (F_t - X)\, e^{-r(T-t)}:$$

同樣的 $F_t - X$，但要到 T 才拿到，所以要折現。futures call 則在 t 就因 marking to market 拿到現金 $F_t - X$。只在 $F_t > X$ 時履約，所以取 $\max(\cdot, 0)$。

A European futures option is worth the same as a European option on the underlying asset if $T = t$, because the futures price equals the spot price at maturity. This conclusion is model independent.

【中】若 $T = t$，European futures option 和直接以標的資產為標的的 European option 價值相同，因為到期時 futures price 等於現貨價；這個結論不依賴任何模型。

§11.3.ii. Put–Call Parity

Slide p. 518

The put–call parity is slightly different from the one in (7.2).

Theorem 11.3.3 (1) For European options on futures contracts, $C = P - (X - F)\, e^{-rt}$.
(2) For European options on forward contracts, $C = P - (X - F)\, e^{-rT}$.

【中】futures option 和 forward option 的 put–call parity 分別是 (1)、(2)，和 (7.2) 稍有不同。證明見課本 Theorem 12.4.4。

[8] The counterparty will pay you \$110 for the underlying asset.

Proof（補充；簡報指向課本 *Theorem 12.4.4*）. (1) long call + short put 在 t 付 $\max(F_t - X, 0) - \max(X - F_t, 0) = F_t - X$。$F_t$ 在 t 的現金，今天價值多少？由 (11.2)，$F_t = S_t e^{r(T-t)}$；今天買 $e^{r(T-t)}$ 單位資產（成本 $Se^{r(T-t)} = Fe^{-rt}$），在 t 賣掉就得到 F_t。所以 $C - P = Fe^{-rt} - Xe^{-rt}$（Corollary 7.2.4）。

(2) 由 (11.9)，long call + short put 在 t 付 $(F_t - X)e^{-r(T-t)}$，是 (1) 的 payoff 乘上常數 $e^{-r(T-t)}$，所以今天的價值是 $(F - X)e^{-rt}e^{-r(T-t)} = (F - X)e^{-rT}$。

■

§11.3.iii. Early Exercise

Slide p. 519

The early exercise feature is not valuable for forward options.

Theorem 11.3.4

American forward options should not be exercised before expiration as long as the probability of their ending up out of the money is positive.

Early exercise may be optimal for American futures options even if the underlying asset generates no payouts.

Theorem 11.3.5

American futures options may be exercised optimally before expiration.

【中】forward option 提前履約沒有價值：只要到期時價外的機率為正，American forward option 就不該提前履約。但 American futures option 即使標的不配息，提前履約也可能是最佳的。證明見課本 Theorem 12.4.5。

Proof of Theorem 11.3.4（補充）. 比較兩個部位在到期日 t 的價值。

(1) **在 $s < t$ 提前履約**：拿到交割價 X 的 long forward。到 t 時，由上面 (11.9) 的 remark，它值 $(F_t - X)e^{-r(T-t)}$。

(2) **持有到 t 再決定**：值 $\max(F_t - X, 0)\, e^{-r(T-t)} \geq (F_t - X)e^{-r(T-t)}$，且在 $F_t < X$（價外）時嚴格大於。

兩個部位在 s 之後都沒有其他現金流，(2) 在每個 state 都不比 (1) 差、且以正機率嚴格較好，由 Proposition 7.2.1，持有的價值嚴格大於履約，所以不應提前履約。 ■

Remark 11.3.6. 關鍵差別：forward option 履約只是把「$F_t - X$」換成一個到 T 才付的 forward，沒有提早拿到任何錢，卻放棄了「價外時不履約」的權利。futures option 履約則因 marking to market 立刻拿到現金 $F_t - X$，可以先賺利息，深度價內時這個利息可能超過放棄的權利價值。下面 Black's model 會把這件事說清楚。

§11.3.iv. Black's Model

Slides pp. 520–521

Formulas for European futures options:[9]

$$C = Fe^{-rt}N(x) - Xe^{-rt}N\big(x - \sigma\sqrt{t}\big), \tag{11.10}$$

$$P = Xe^{-rt}N\big(-x + \sigma\sqrt{t}\big) - Fe^{-rt}N(-x), \tag{11.11}$$

where

$$x \triangleq \frac{\ln(F/X) + (\sigma^2/2)\, t}{\sigma\sqrt{t}}.$$

[9] Black (1976). It is also called the Black-76 model.

The above formulas are related to those for options on a stock paying a continuous dividend yield: they are (8.25)–(8.26) with q set to r and S replaced by F.

【中】European futures option 的 Black 公式 (11.10)–(11.11)；它們就是有連續 dividend yield 的 (8.25)–(8.26) 把 q 換成 r、S 換成 F。

Remark 11.3.7. (1) **為什麼** $q = r$。risk-neutral 下 futures 部位不用錢，所以 futures price 的期望成長率是 0（Section 11.3.v 會在 tree 上驗證）；而有 dividend yield q 的股價期望成長率是 $r - q$（Section 8.19.iii）。令 $r - q = 0$ 即 $q = r$。

(2) **代入檢查**。(8.25) 的 $Se^{-q\tau}$ 變成 Fe^{-rt}；(8.27) 的分子 $\ln(S/X) + (r - q + \sigma^2/2)\tau$ 變成 $\ln(F/X) + (\sigma^2/2)t$。

This observation incidentally proves Theorem 11.3.5: by (1) of the remark, an American futures option is an American option on an asset paying a dividend yield $r > 0$, and such options may be exercised early. Volatility σ is that of the stock price;[10] it is the same as the volatility of the futures price.[11] For European forward options, just multiply the above formulas by $e^{-r(T-t)}$: forward options differ from futures options by a factor of $e^{-r(T-t)}$, by (11.8)–(11.9).

【中】這個對應順便證明了 Theorem 11.3.5：American futures option 就是以 dividend yield 為 $r > 0$ 的資產為標的的 American option，而這種 option 可能提前履約。σ 是股價的波動率，也等於 futures price 的波動率。European forward option 只要把公式乘上 $e^{-r(T-t)}$。

Remark 11.3.8. 兩個 σ 為什麼相同。由 (11.2)，$F_s = S_s e^{r(T-s)}$，所以 $\ln F_s = \ln S_s + r(T-s)$：兩者的 log 只差一個確定的函數，log return 的變異數相同。

§11.3.v. Binomial Model for Forward and Futures Options

Slides pp. 522–524

In a risk-neutral economy, futures price behaves like a stock paying a continuous dividend yield of r. Let the futures price at time 0 be F. By (8.7), the expected value of S at time Δt in a risk-neutral economy is $Se^{r\Delta t}$. So by (11.2), the expected futures price at time Δt is

$$Se^{r\Delta t}\, e^{r(T-\Delta t)} \;=\; Se^{rT} \;=\; F.$$

The above observation continues to hold even if S pays a dividend yield q![12] The expected value of S at time Δt is now $Se^{(r-q)\Delta t}$ (Section 8.19.iii), so by (11.6) the expected futures price at time Δt is

$$Se^{(r-q)\Delta t}\, e^{(r-q)(T-\Delta t)} \;=\; Se^{(r-q)T} \;=\; F.$$

【中】risk-neutral 下 futures price 的行為就像 dividend yield 為 r 的股票：Δt 後股價期望值 $Se^{r\Delta t}$，代入 (11.2) 得 futures price 期望值 $Se^{r\Delta t}e^{r(T-\Delta t)} = Se^{rT} = F$，即期望值不變。股票有 dividend yield q 時也一樣：$Se^{(r-q)\Delta t}e^{(r-q)(T-\Delta t)} = F$。

Now, under the BOPM, the risk-neutral probability for the futures price is

$$p_f \;\triangleq\; \frac{1-d}{u-d} \tag{11.12}$$

by (8.28) with $q = r$. The futures price moves from F to Fu with probability p_f and to Fd with probability $1 - p_f$; the original u and d are used here. The binomial tree algorithm for forward options is identical except that (11.9) is the payoff.

【中】在 BOPM 中，futures price 的 risk-neutral 機率是 $p_f = (1-d)/(u-d)$ (11.12)，就是 (8.28) 取 $q = r$。futures price 以機率 p_f 從 F 走到 Fu、以 $1 - p_f$ 走到 Fd，用的是原本的 u, d。forward option 的 tree 演算法相同，只是 payoff 換成 (11.9)。

Remark 11.3.9. (8.28) 是 $p = \frac{e^{(r-q)\Delta t} - d}{u-d}$，$q = r$ 時分子變成 $e^0 - d = 1 - d$。等價地，p_f 是唯一使 $p_f Fu + (1-p_f)Fd = F$（期望值不變）的機率：$p_f u + (1-p_f)d = 1$ 解出 $p_f = \frac{1-d}{u-d}$。

[10] Contributed by Mr. Lu, Yu-Ming (R06723032, D08922008) on April 7, 2021.
[11] Contributed by Mr. Chen, Tung-Li (D09922014) on April 7, 2023.
[12] Contributed by Mr. Liu, Yi-Wei (R02723084) on April 16, 2014.

§11.3.vi. Spot and Futures Prices under BOPM

Slides pp. 525–526

The futures price is related to the spot price via $F = Se^{rT}$ if the underlying asset pays no dividends (Lemma 11.1.4). Recall the futures price F moves to Fu with probability p_f per period. So the stock price moves from $S = Fe^{-rT}$ to

$$Fue^{-r(T-\Delta t)} \;=\; Sue^{r\Delta t}$$

with probability p_f per period. Similarly, the stock price moves from $S = Fe^{-rT}$ to $Sde^{r\Delta t}$ with probability $1 - p_f$ per period. Note that

$$S\big(ue^{r\Delta t}\big)\big(de^{r\Delta t}\big) \;=\; Se^{2r\Delta t} \;\neq\; S.$$

This binomial model for S differs from the CRR tree. This model may not be suitable for pricing barrier options (why?).

【中】不配息時 $F = Se^{rT}$。futures price 每期以機率 p_f 乘 u，所以股價從 $S = Fe^{-rT}$ 變成 $Fue^{-r(T-\Delta t)} = Sue^{r\Delta t}$；以 $1 - p_f$ 變成 $Sde^{r\Delta t}$。注意先漲後跌回到 $Se^{2r\Delta t} \neq S$，和 CRR tree 不同。這個模型可能不適合 barrier option（為什麼？）。

Remark 11.3.10. 為什麼不適合 barrier option。CRR tree 中 $ud = 1$，第 i 期的股價是 Su^{2j-i}，各期的 node 落在同一組價格層 $\{Su^k\}$ 上。這裡第 i 期的股價是

$$S\,u^j d^{i-j} e^{ri\Delta t} \;=\; Su^{2j-i}\, e^{ri\Delta t},$$

整組價格層每期往上平移 $e^{r\Delta t}$。所以就算 H 在某一期剛好落在 node 上，下一期就不在了；Section 10.2.v 的收斂問題（effective barrier 跟著 n 跳動）在每一期都會發生，無法靠選 n 讓 H 落在 node 上來解決。

§11.3.vii. Negative Probabilities Revisited

Slide p. 527

As $0 < p_f < 1$, we have $0 < 1 - p_f < 1$ as well. The problem of negative risk-neutral probabilities is solved:

- Build the tree for the futures price F of the futures contract expiring at the same time as the option.
- Let the stock pay a continuous dividend yield of q.
- By (11.6), recover S from F at each node via $S = Fe^{-(r-q)(T-t)}$.

【中】因為 $0 < p_f < 1$，risk-neutral 機率為負的問題解決了：對和 option 同時到期的 futures 建 tree，股票有 dividend yield q 時，在每個 node 用 (11.6) 從 F 換回 $S = Fe^{-(r-q)(T-t)}$。

Remark 11.3.11. (1) **原本的問題**。(8.28) 的 $p = \frac{e^{(r-q)\Delta t} - d}{u - d}$ 只有在 $d < e^{(r-q)\Delta t} < u$ 時才落在 $(0, 1)$。CRR 下 $u = e^{\sigma\sqrt{\Delta t}}$，條件是 $|r - q|\Delta t < \sigma\sqrt{\Delta t}$；$|r - q|$ 大、σ 小或 Δt 大時就會不成立，p 變成負的或大於 1。

(2) **為什麼 p_f 沒有這個問題**。p_f 的分子把 $e^{(r-q)\Delta t}$ 換成 1，而 $u > 1 > d$ 永遠成立（如 $u = e^{\sigma\sqrt{\Delta t}}$、$d = 1/u$），所以 $0 < p_f < 1$ 不論 $r, q, \sigma, \Delta t$。

(3) **代價**。股價 node 不再是 Su^k，而是像上一小節那樣每期平移 $e^{(r-q)\Delta t}$。

§11.4. Swaps

Slide p. 528

Swaps are agreements between two counterparties to exchange cash flows in the future according to some predetermined formula. There are two basic types of swaps: interest rate and

currency. An **interest rate swap** occurs when two parties exchange interest payments periodically. **Currency swaps** are agreements to deliver one currency against another (our focus here). There are theories about why swaps exist.[13]

【中】swap 是雙方約定未來依某個公式交換現金流的合約，基本兩類：interest rate swap（定期交換利息）和 currency swap（以一種貨幣交換另一種，這裡的重點）。

§11.4.i. Currency Swaps

Slides pp. 529–531

A currency swap involves two parties to exchange cash flows in different currencies. Consider the following fixed rates available to party A and party B in U.S. dollars and Japanese yen:

	Dollars	Yen
A	$D_A\%$	$Y_A\%$
B	$D_B\%$	$Y_B\%$

Suppose A wants to take out a fixed-rate loan in yen, and B wants to take out a fixed-rate loan in dollars. A straightforward scenario is for A to borrow yen at $Y_A\%$ and B to borrow dollars at $D_B\%$. But suppose A is relatively more competitive in the dollar market than the yen market, i.e.,

$$Y_B - D_B \ < \ Y_A - D_A \qquad \text{or} \qquad Y_B - Y_A \ < \ D_B - D_A.$$

Consider this alternative arrangement: A borrows dollars, B borrows yen, and they enter into a currency swap with a bank (the swap dealer) as the intermediary.

【中】A、B 在美元和日圓市場的固定借款利率如表。A 想借日圓、B 想借美元，最直接是 A 以 $Y_A\%$ 借日圓、B 以 $D_B\%$ 借美元。但若 A 在美元市場相對較有競爭力（$Y_B - Y_A < D_B - D_A$），可以改成 A 借美元、B 借日圓，再透過銀行（swap dealer）做 currency swap。

The counterparties exchange principal at the beginning and the end of the life of the swap. This act transforms A's loan into a yen loan and B's yen loan into a dollar loan. The total interest rate is originally $(Y_A + D_B)\%$. The new arrangement has a smaller total rate of $(D_A + Y_B)\%$. So the total gain is $\big((D_B - D_A) - (Y_B - Y_A)\big)\%$. Transactions will happen only if the gain is distributed so that the cost to each party is less than the original.

【中】雙方在期初和期末交換本金，把 A 的美元借款轉成日圓借款、B 的日圓借款轉成美元借款。原本總利率 $(Y_A + D_B)\%$，新安排是 $(D_A + Y_B)\%$，總共省下 $\big((D_B - D_A) - (Y_B - Y_A)\big)\%$。只有在利益分配讓每一方的成本都比原本低時，交易才會發生。

Remark 11.4.1. 總利益怎麼來。 $(Y_A + D_B) - (D_A + Y_B) = (D_B - D_A) - (Y_B - Y_A)$，由假設 $Y_B - Y_A < D_B - D_A$ 是正的。這就是比較利益（comparative advantage）：B 在兩個市場可能都比 A 貴，但只要兩個市場的利差不同，各自去利差較小的市場借錢再交換就有利可圖。

§11.4.ii. An Example

Slides pp. 532–535

A and B face the following borrowing rates:

	Dollars	Yen
A	9%	10%
B	12%	11%

[13] Thanks to a lively discussion on April 16, 2014.

A wants to borrow yen, and B wants to borrow dollars. A can borrow yen directly at 10%; B can borrow dollars directly at 12%. The rate differential in dollars (3%) is different from that in yen (1%). So a currency swap with a total saving of $3-1=2\%$ is possible. A is relatively more competitive in the dollar market; B is relatively more competitive in the yen market.

Figure 11.3 shows an arrangement which is beneficial to all parties involved. A effectively borrows yen at 9.5% (lower than 10%), and B borrows dollars at 11.5% (lower than 12%). The gain is 0.5% for A, 0.5% for B, and, if we treat dollars and yen identically, 1% for the bank.

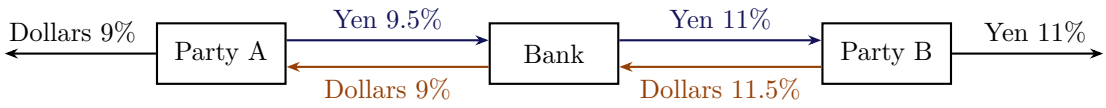


Figure 11.3.: The currency swap of Section 11.4.ii. Arrows show interest payments; the outer arrows go to the original lenders.

【中】例：A 的美元/日圓借款利率 9%/10%，B 是 12%/11%。美元利差 3%、日圓利差 1%，所以 swap 最多可省 $3-1=2\%$；A 在美元市場相對有競爭力，B 在日圓市場。Figure 11.3 的安排讓 A 實際以 9.5% 借日圓（比 10% 低）、B 以 11.5% 借美元（比 12% 低）；A、B 各省 0.5%，若把美元和日圓視為相同，銀行賺 1%。

Remark 11.4.2. 逐方核對 Figure 11.3。

(1) **A**。付原貸款人美元 9%，從銀行收美元 9%，付銀行日圓 9.5%：美元互抵，淨付日圓 9.5%。
(2) **B**。付原貸款人日圓 11%，從銀行收日圓 11%，付銀行美元 11.5%：淨付美元 11.5%。
(3) **銀行**。日圓收 9.5%、付 11%，淨付 1.5%；美元收 11.5%、付 9%，淨收 2.5%。視兩種貨幣相同時淨賺 $2.5-1.5=1\%$。

三方合計 $0.5+0.5+1=2\%$，正好是總利益。銀行承擔了匯率風險（日圓部位淨付、美元部位淨收）。

§11.4.iii. As a Package of Cash Market Instruments

Slides pp. 536–538

Assume no default risk. Take B in Figure 11.3 as an example. The swap is equivalent to a long position in a yen bond paying 11% annual interest and a short position in a dollar bond paying 11.5% annual interest. The pricing formula is

$$SP_Y - P_D,$$

where P_D is the dollar bond's value in dollars, P_Y is the yen bond's value in yen, and S is the \$/yen spot exchange rate. The value of a currency swap depends on the term structures of interest rates in the currencies involved and on the spot exchange rate. It has zero value when $SP_Y = P_D$.

【中】假設沒有違約風險。以 B 為例，swap 等於 long 一個年息 11% 的日圓債券、short 一個年息 11.5% 的美元債券，價值是 $SP_Y - P_D$（P_D 是美元債券的美元價值、P_Y 是日圓債券的日圓價值、S 是美元/日圓即期匯率）。所以 swap 的價值取決於兩國的利率期限結構和即期匯率；$SP_Y = P_D$ 時價值為 0。

Remark 11.4.3. B 在 swap 中收日圓 11%（期末收回日圓本金），等於持有日圓債券；付美元 11.5%（期末付美元本金），等於發行（short）美元債券。日圓債券的價值 P_Y 以日圓計，乘上 S（每日圓幾美元）換成美元，才能和 P_D 相減。

Example 11.4.4. Take a 3-year swap in Figure 11.3 with principal amounts of US\$1 million and 100 million yen. The payments are made once a year. The spot exchange rate is 90 yen/\$

and the term structures are flat in both nations — 8% in the U.S. and 9% in Japan. For B, the value of the swap is (in millions of USD)

$$\frac{1}{90}\times\left(11\,e^{-0.09}+11\,e^{-0.09\times2}+111\,e^{-0.09\times3}\right) - \left(0.115\,e^{-0.08}+0.115\,e^{-0.08\times2}+1.115\,e^{-0.08\times3}\right) \;=\; 0.074.$$

【中】例：3 年期 swap，本金 100 萬美元和 1 億日圓，每年付息一次，即期匯率 90 日圓/美元，兩國利率期限結構都是平的（美國 8%、日本 9%）。對 B 來說 swap 值 0.074 百萬美元。

Remark 11.4.5. (1) **日圓那一項**。日圓本金 100（百萬），年息 11% 是每年 11；第 3 年再加本金，共 111。以日圓利率 9% 折現得 $P_Y \approx 103.98$ 百萬日圓，乘 $S = 1/90$ 得 1.1553 百萬美元。
(2) **美元那一項**。本金 1（百萬），年息 11.5% 是每年 0.115，第 3 年共 1.115。以 8% 折現得 $P_D \approx 1.0812$。
(3) 相減 $1.1553 - 1.0812 \approx 0.074$。

§11.4.iv. As a Package of Forward Contracts

Slides pp. 539–541

From (11.5), the forward contract maturing i years from now has a dollar value of

$$f_i \;\triangleq\; (SY_i)\,e^{-qi} - D_i\,e^{-ri}, \tag{11.13}$$

where

- Y_i is the yen inflow at year i;
- S is the \$/yen spot exchange rate;[14]
- q is the yen interest rate;
- D_i is the dollar outflow at year i;
- r is the dollar interest rate.

For simplicity, flat term structures were assumed; generalization is straightforward.

【中】由 (11.5)，第 i 年到期的 forward 的美元價值是 $f_i = (SY_i)e^{-qi} - D_i e^{-ri}$ (11.13)：Y_i 是第 i 年收到的日圓、S 是美元/日圓即期匯率、q 是日圓利率、D_i 是第 i 年付出的美元、r 是美元利率。這裡假設期限結構是平的，推廣很直接。

Remark 11.4.6. (11.13) **怎麼套** (11.5)。第 i 年「收 Y_i 日圓、付 D_i 美元」就是一個以 D_i 美元買 Y_i 日圓的 forward。標的是 Y_i 單位日圓，現值（美元）SY_i；日圓以利率 q 生息，就像 dividend yield q（Section 10.3）。所以 (11.5) 中 $S \to SY_i$、$X \to D_i$、$\tau \to i$。用的是即期匯率 S，因為 e^{-qi} 已經處理了時間差。

Example 11.4.7. Take the swap in Example 11.4.4. Every year, B receives 11 million yen and pays 0.115 million dollars. In addition, at the end of the third year, B receives 100 million yen and pays 1 million dollars. Each of these transactions represents a forward contract: $Y_1 = Y_2 = 11$, $Y_3 = 111$, $S = 1/90$, $D_1 = D_2 = 0.115$, $D_3 = 1.115$, $q = 0.09$, and $r = 0.08$. Plug in these numbers to get $f_1 + f_2 + f_3 = 0.074$ million dollars as before.

【中】例：同 Example 11.4.4，B 每年收 11 百萬日圓、付 0.115 百萬美元，第 3 年另外收 100 百萬日圓、付 1 百萬美元。每一筆都是一個 forward，代入得 $f_1 + f_2 + f_3 = 0.074$，和前面相同。

Remark 11.4.8. $f_1 \approx 0.0055$、$f_2 \approx 0.0041$、$f_3 \approx 0.0644$。兩種算法相同並非巧合：$\sum_i f_i = S\sum_i Y_i e^{-qi} - \sum_i D_i e^{-ri} = SP_Y - P_D$，只是把債券的現金流逐年拆開。

[14] This is not the forward exchange rate. Contributed by Mr. Chen, Tzu-Chun (R12922037) and Mr. Wang, Wei-Li (R12922116) on April 19, 2024.

12. Stochastic Processes and Brownian Motion

Slides pp. 542–543

Of all the intellectual hurdles which the human mind has confronted and has overcome in the last fifteen hundred years, the one which seems to me to have been the most amazing in character and the most stupendous in the scope of its consequences is the one relating to the problem of motion.
— Herbert Butterfield (1900–1979)

§12.1. Stochastic Processes

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Definition 12.1.1. A **stochastic process** $X = \{X(t)\}$ is a time series of random variables. $X(t)$ (or X_t) is a random variable for each time t and is usually called the **state** of the process at time t. A realization of X is called a **sample path**.

- If the times t form a countable set, X is called a **discrete-time stochastic process** or a time series. In this case, subscripts rather than parentheses are usually employed, as in $X = \{X_n\}$.
- If the times form a continuum, X is called a **continuous-time stochastic process**.

【中】stochastic process $X = \{X(t)\}$ 是一串隨機變數；每個時間 t 的 $X(t)$ 叫做時間 t 的 state，X 的一個實現叫 sample path。時間是可數集合時叫 discrete-time（寫成 $\{X_n\}$），是連續區間時叫 continuous-time。

§12.2. Random Walks

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The binomial model is a random walk in disguise. Consider a particle on the integer line, $0, \pm 1, \pm 2, \ldots$. In each time step, it can make one move to the right with probability p or one move to the left with probability $1 - p$. This random walk is **symmetric** when $p = 1/2$. Connection with the BOPM: the particle's position denotes the number of up moves minus that of down moves up to that time.

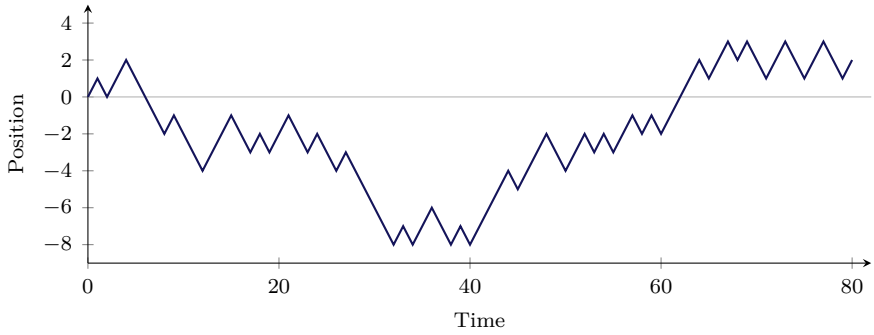


Figure 12.1.: A sample path of the symmetric random walk over 80 steps.

【中】binomial model 其實就是 random walk：粒子在整數線上，每步以機率 p 往右、$1-p$ 往左一格；$p=1/2$ 時叫 symmetric。和 BOPM 的關係：粒子的位置就是到目前為止上漲次數減下跌次數。

Remark 12.2.1. 和 BOPM 的對應。 n 步中往右 j 次時位置是 $j-(n-j)=2j-n$；CRR tree（$d=1/u$）中上漲 j 次的股價是 $Su^j d^{n-j}=Su^{2j-n}$。所以 $\ln(S_n/S)=(2j-n)\ln u$：log 股價就是步長 $\ln u$ 的 random walk，$j\sim \mathrm{Bin}(n,p)$。

§12.2.i. Random Walk with Drift

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Definition 12.2.2. A **random walk with drift** is

$$X_n \;=\; \mu + X_{n-1} + \xi_n,$$

where ξ_n are independent and identically distributed with zero mean. The **drift** μ is the expected change per period.

Note that this process is continuous in space.

【中】random walk with drift：$X_n=\mu+X_{n-1}+\xi_n$，ξ_n 是 i.i.d. 平均為 0 的隨機變數；drift μ 是每期的期望變化量。注意這個過程在空間上是連續的（ξ_n 可以取任意實數值）。

§12.3. Martingales

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Definition 12.3.1. $\{X(t), t\ge 0\}$ is a **martingale**[1] if $E[\,|X(t)|\,]<\infty$ for $t\ge 0$ and

$$E\big[\,X(t)\mid X(u),\,0\le u\le s\,\big] \;=\; X(s), \qquad s\le t. \tag{12.1}$$

In the discrete-time setting, a martingale means

$$E[\,X_{n+1}\mid X_1, X_2, \ldots, X_n\,] \;=\; X_n. \tag{12.2}$$

X_n can be interpreted as a gambler's fortune after the nth gamble. Identity (12.2) then says the expected fortune after the $(n+1)$st gamble equals the fortune after the nth gamble regardless of what may have occurred before. A martingale is therefore a notion of fair games.

【中】martingale：$E[|X(t)|]<\infty$ 且給定到 s 為止的整段歷史，$X(t)$ 的條件期望值等於 $X(s)$ (12.1)；離散版本是 (12.2)。把 X_n 想成賭徒第 n 次賭完的財富，(12.2) 說不論之前發生什麼，下一把之後的期望財富等於現在的財富，所以 martingale 是「公平賭局」的概念。

[1] The origin of the name is somewhat obscure.

Proposition 12.3.2

If $\{X_n\}$ is a martingale, then

$$E[\,X_n\,] \;=\; E[\,X_1\,] \tag{12.3}$$

for all n. Similarly, $E[\,X(t)\,] = E[\,X(0)\,]$ in the continuous-time case.

Proof. Apply the law of iterated conditional expectations (5.9) to both sides of (12.2): $E[X_{n+1}] = E\big[E[X_{n+1} \mid X_1, \ldots, X_n]\big] = E[X_n]$, and induct on n. ∎

【中】對 (12.2) 兩邊取期望值並用 tower law (5.9)，得 $E[X_{n+1}] = E[X_n]$，所以 martingale 的期望值永遠不變 (12.3)。

§12.3.i. Still a Martingale?

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Suppose we replace (12.2) with

$$E[\,X_{n+1} \mid X_n\,] \;=\; X_n.$$

It also says past history cannot affect the future. But is it equivalent to the original definition (12.2)?[2] Well, no.[3] The slides consider this random walk with drift:

$$X_i \;=\; \begin{cases} X_{i-1} + \xi_i, & \text{if } i \text{ is even,} \\ X_{i-2}, & \text{otherwise,} \end{cases}$$

where ξ_n are random variables with zero mean, and claim that

$$E[\,X_i \mid X_{i-1}\,] \;=\; \begin{cases} X_{i-1}, & \text{if } i \text{ is even,} \\ X_{i-1}, & \text{otherwise,} \end{cases}$$

so it is a martingale by the "new" definition. But

$$E[\,X_i \mid \ldots, X_{i-2}, X_{i-1}\,] \;=\; \begin{cases} X_{i-1}, & \text{if } i \text{ is even,} \\ X_{i-2}, & \text{otherwise,} \end{cases}$$

so it is not a martingale by the original definition.

【中】若把 (12.2) 改成只對前一期取條件 $E[X_{n+1} \mid X_n] = X_n$，它看起來也說「過去不影響未來」，但和原定義不等價。簡報的例子：i 為偶數時 $X_i = X_{i-1} + \xi_i$，奇數時 $X_i = X_{i-2}$；簡報說它在新定義下是 martingale，但給定整段歷史時，奇數 i 的條件期望是 $X_{i-2} \neq X_{i-1}$，所以不是原定義的 martingale。

Note 12.3.3. 簡報這個例子的「新定義」那一半其實不成立（只要 ξ 不是 0 且變異數有限）。奇數 i 時 $X_i = X_{i-2} = X_{i-1} - \xi_{i-1}$，所以

$$E[\,X_i \mid X_{i-1}\,] \;=\; X_{i-1} - E[\,\xi_{i-1} \mid X_{i-1}\,],$$

要等於 X_{i-1} 必須 $E[\xi_{i-1} \mid X_{i-2} + \xi_{i-1}] = 0$。下面的 lemma 說這只有在 $\xi_{i-1} = 0$ 時才可能。

[2] Contributed by Mr. Hsieh, Chicheng (M9007304) on April 13, 2005.
[3] Contributed by Mr. Zhang, Ann-Sheng (B89201033) on April 13, 2005.

Lemma 12.3.4

Let Y, Z have finite second moments. If $E[Z \mid Y] = Y$ and $E[Y \mid Z] = Z$, then $Y = Z$ with probability one.

Proof. 由 Proposition 5.4.6(2)（已知的 Y 可以提出來；Y 不必有界，因為由 Lemma 5.2.3，YZ 可積）與 tower law (5.9)，$E[YZ] = E\big[Y\,E[Z \mid Y]\big] = E[Y^2]$；對稱地 $E[YZ] = E[Z^2]$。所以 $E[(Z-Y)^2] = E[Z^2] - 2E[YZ] + E[Y^2] = 0$，由 Lemma 5.1.4，$Z - Y = 0$ with probability one。 ■

套在簡報的例子：$Y = X_{i-2}$、$Z = X_{i-1} = X_{i-2} + \xi_{i-1}$。偶數步的條件 $E[\xi_{i-1} \mid X_{i-2}] = 0$ 給 $E[Z \mid Y] = Y$；若奇數步也滿足新定義就是 $E[Y \mid Z] = Z$，於是 $\xi_{i-1} = 0$，整個過程是常數。一個真正的反例：

Example 12.3.5 (補充). $X_1 \in \{-1, 1\}$、$\xi_2 \in \{-1, 1\}$ 獨立、各機率 $1/2$，$X_2 = X_1 + \xi_2 \in \{-2, 0, 2\}$，令

$$X_3 = \begin{cases} X_1, & X_2 = 0, \\ X_2, & X_2 = \pm 2. \end{cases}$$

(1) $E[X_2 \mid X_1] = X_1 + E[\xi_2] = X_1$。
(2) $X_2 = \pm 2$ 時 $X_3 = X_2$；$X_2 = 0$ 時 $(X_1, \xi_2) = (1, -1)$ 或 $(-1, 1)$ 等機率，$E[X_3 \mid X_2 = 0] = E[X_1 \mid X_2 = 0] = 0 = X_2$。所以 $E[X_3 \mid X_2] = X_2$，新定義成立。
(3) 但 $E[X_3 \mid X_1, X_2 = 0] = X_1 = \pm 1 \neq 0 = X_2$，原定義 (12.2) 不成立。

新定義只看 X_2，看不出 $X_2 = 0$ 是怎麼來的；完整歷史知道 X_1，就能預測 X_3 的方向。

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Example 12.3.6. Consider the stochastic process $\{Z_n \triangleq \sum_{i=1}^{n} X_i,\ n \geq 1\}$, where X_i are independent random variables with zero mean. This process is a martingale because

$$\begin{aligned} E[\,Z_{n+1} \mid Z_1, Z_2, \ldots, Z_n\,] &= E[\,Z_n + X_{n+1} \mid Z_1, Z_2, \ldots, Z_n\,] \\ &= E[\,Z_n \mid Z_1, Z_2, \ldots, Z_n\,] + E[\,X_{n+1} \mid Z_1, Z_2, \ldots, Z_n\,] \\ &= Z_n + E[\,X_{n+1}\,] = Z_n. \end{aligned}$$

【中】例：$Z_n = \sum_{i=1}^{n} X_i$，X_i 獨立且平均為 0，則 $\{Z_n\}$ 是 martingale。

Remark 12.3.7. 最後一行用了兩件事：Z_n 是已知的，條件期望就是它自己；X_{n+1} 和 $X_1, \ldots, X_n$ 獨立，而 $Z_1, \ldots, Z_n$ 只是 $X_1, \ldots, X_n$ 的函數，所以條件期望等於無條件期望 $E[X_{n+1}] = 0$（Proposition 5.4.6）。

§12.3.ii. Probability Measure

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A **probability measure** assigns probabilities to states of the world.[4] A martingale is defined with respect to a probability measure, under which the expectation is taken. Second, a martingale is defined with respect to an **information set**. In the characterizations (12.1)–(12.2), the information set contains the current and past values of X by default, but it need not be so.

【中】probability measure 指定各個 state 的機率。martingale 是相對於某個 probability measure（期望值在它之下取）、以及某組 information set 來定義的。(12.1)–(12.2) 預設 information set 是 X 現在和過去的值，但不一定要如此。

[4] Only certain sets such as the Borel sets receive probabilities (Feller, 1972).

Definition 12.3.8. A stochastic process $\{X(t), t \geq 0\}$ is a martingale with respect to information sets $\{I_t\}$ if, for all $t \geq 0$, $E[\,|X(t)|\,] < \infty$ and

$$E[\,X(u) \mid I_t\,] \;=\; X(t)$$

for all $u > t$. The discrete-time version: for all $n > 0$, $E[\,X_{n+1} \mid I_n\,] = X_n$, given the information sets $\{I_n\}$.

The above implies

$$E[\,X_{n+m} \mid I_n\,] \;=\; X_n$$

for any $m > 0$ by (5.10). A typical I_n is the price information up to time n. Then the above identity says the FVs of X will not deviate systematically from today's value given the price history.

【中】相對於 information sets $\{I_t\}$ 的 martingale：$E[X(u) \mid I_t] = X(t)$，$u > t$；離散版 $E[X_{n+1} \mid I_n] = X_n$。由 (5.10) 可推出 $E[X_{n+m} \mid I_n] = X_n$。$I_n$ 通常是到時間 n 為止的價格資訊，這時上式說：給定價格歷史，X 的未來值不會系統性地偏離今天的值。

Remark 12.3.9. $E[X_{n+m} \mid I_n] = X_n$ **怎麼來**。對 m 歸納：$I_n \subseteq I_{n+m-1}$（資訊只會累積），由 (5.10)，

$$E[\,X_{n+m} \mid I_n\,] \;=\; E\big[\,E[\,X_{n+m} \mid I_{n+m-1}\,] \mid I_n\,\big] \;=\; E[\,X_{n+m-1} \mid I_n\,] \;=\; X_n,$$

中間用了一步的 martingale 性質，最後一步是歸納假設。

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Example 12.3.10. Consider the stochastic process $\{Z_n - n\mu,\ n \geq 1\}$, where $Z_n \triangleq \sum_{i=1}^{n} X_i$ and $X_1, X_2, \ldots$ are independent random variables with mean μ. Now,

$$\begin{aligned} E[\,Z_{n+1} - (n+1)\mu \mid X_1, X_2, \ldots, X_n\,] &= E[\,Z_{n+1} \mid X_1, X_2, \ldots, X_n\,] - (n+1)\mu \\ &= E[\,Z_n + X_{n+1} \mid X_1, X_2, \ldots, X_n\,] - (n+1)\mu \\ &= Z_n + \mu - (n+1)\mu \\ &= Z_n - n\mu. \end{aligned}$$

Define $I_n \triangleq \{X_1, X_2, \ldots, X_n\}$. Then $\{Z_n - n\mu,\ n \geq 1\}$ is a martingale with respect to $\{I_n\}$.

【中】例：X_i 獨立、平均 μ，$Z_n = \sum_{i=1}^{n} X_i$。扣掉 drift 後 $\{Z_n - n\mu\}$ 是相對於 $I_n = \{X_1, \ldots, X_n\}$ 的 martingale。注意 information set 是 X_i 而不是過程本身的值。

§12.4. Martingale Pricing

Slides pp. 560–561

Stock prices and zero-coupon bond prices are expected to rise, while call prices are expected to fall (Section 9.3). They are not martingales. Why is then martingale useful? Recall a martingale is defined with respect to some information set and some probability measure. By modifying the probability measure, we can convert a price process into a martingale.

【中】股價和 zero-coupon bond 價格預期會上漲，call 價格預期會下跌 (theta)，它們都不是 martingale。那 martingale 有什麼用？martingale 是相對於某個 information set 和 probability measure 定義的；換掉 probability measure，就能把價格過程變成 martingale。

The price of a European option is the expected discounted payoff in a risk-neutral economy (8.13). This principle can be generalized using the concept of martingale. Recall the recursive valuation of European option via (8.6):

$$C \;=\; \big[\,pC_u + (1-p)\,C_d\,\big]/R,$$

where p is the risk-neutral probability and \$1 grows to \$$R$ in a period.

【中】European option 的價格是 risk-neutral 下 payoff 折現的期望值 (8.13)，這可以用 martingale 推廣。回憶一期的遞迴 $C = [pC_u + (1-p)C_d]/R$。

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Let $C(i)$ denote the value of the option at time i. Consider the discount process

$$\{\, C(i)/R^i,\ i = 0, 1, \ldots, n \,\}.$$

Then,

$$E\left[\, \frac{C(i+1)}{R^{i+1}} \,\middle|\, S(i) \,\right] = \frac{E[\, C(i+1) \mid S(i)\,]}{R^{i+1}} = \frac{pC_u + (1-p)\, C_d}{R^{i+1}} = \frac{C(i)}{R^i}.$$

It is easy to generalize it to

$$E\left[\, \frac{C(k)}{R^k} \,\middle|\, S(i) \,\right] = \frac{C(i)}{R^i}, \qquad i \le k.$$

【中】令 $C(i)$ 為 option 在時間 i 的價值，考慮折現過程 $C(i)/R^i$。在 risk-neutral 機率下，給定 $S(i)$ 時下一期折現值的期望等於這一期的折現值，再推廣到任意 $k \ge i$。

Remark 12.4.1. (1) **第二個等號**。給定 $S(i)$，$C(i+1)$ 以機率 p 是 C_u、以 $1-p$ 是 C_d（C_u, C_d 是 $S(i)$ 往上、往下那兩個 node 的值）。
(2) **第三個等號**。由 (8.6)，$pC_u + (1-p)C_d = RC(i)$，除以 R^{i+1} 得 $C(i)/R^i$。
(3) **推廣到** k。和上面 $E[X_{n+m} \mid I_n] = X_n$ 的 remark 相同，用 (5.10) 一步一步往回推。
要注意期望值是在 risk-neutral 機率 p 下取的：真實機率下 $C(i)/R^i$ 不是 martingale。

This simplified formulation assumes:[5]
(1) The model is Markovian: the distribution of the future is determined by the present (time i) and not the past.
(2) The payoff depends only on the terminal price of the underlying asset (Asian options do not qualify); recall they are called simple claims (Section 10.4).

【中】這個簡化版本假設 (1) 模型是 Markovian：未來的分布只由現在決定，與過去無關；(2) payoff 只取決於標的的最終價格（simple claim），Asian option 不符合。

In general, the discount process is a martingale in that[6]

$$E_i^\pi\left[\, \frac{C(k)}{R^k} \,\right] = \frac{C(i)}{R^i}, \qquad i \le k, \tag{12.4}$$

where E_i^π is taken under the risk-neutral probability conditional on the price information up to time i. This risk-neutral probability is also called the EMM, or the **equivalent martingale (probability) measure**.[7]

【中】一般情形下，折現過程在 risk-neutral 機率下、相對於到時間 i 為止的價格資訊是 martingale (12.4)（這個版本 Asian option 也適用）。這個 risk-neutral 機率又叫 EMM（equivalent martingale measure）；兩個 probability measure equivalent 指它們給正機率的 state 集合相同。

Remark 12.4.2. 為什麼一般版本 Asian option 也可以。簡化版只以 $S(i)$ 為條件，但 Asian option 在時間 i 的價值還取決於到目前為止的平均價，光知道 $S(i)$ 不夠。E_i^π 以完整的價格歷史為條件，平均價也包含在內，所以 $C(i)$ 是這份資訊的函數，遞迴照樣成立。

[5] Contributed by Mr. Wang, Liang-Kai (Ph.D. student, ECE, University of Wisconsin-Madison) and Mr. Hsiao, Huan-Wen (B90902081) on May 3, 2006.
[6] In this general formulation, Asian options do qualify.
[7] Two probability measures are said to be equivalent if they assign nonzero probabilities to the same set of states.

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Equation (12.4) holds for all assets, not just options. When interest rates are stochastic, the equation becomes

$$\frac{C(i)}{M(i)} = E_i^{\pi}\left[\frac{C(k)}{M(k)}\right], \qquad i \le k, \tag{12.5}$$

where $M(j)$ is the balance in the money market account at time j using the rollover strategy with an initial investment of \$1. It is called the **bank account process**. It says the discount process is a martingale under π. If interest rates are stochastic, then $M(j)$ is a random variable: $M(0) = 1$, and $M(j)$ is known at time $j-1$ because the interest rate for the next period has been revealed then. Identity (12.5) is the general formulation of risk-neutral valuation.

【中】(12.4) 對所有資產都成立。利率是隨機時改寫成 (12.5)，$M(j)$ 是期初投入 \$1、每期滾存在貨幣市場帳戶的餘額（bank account process）。$M(j)$ 是隨機變數，$M(0)=1$，且在 $j-1$ 時就已知（下一期利率已揭露）。(12.5) 是 risk-neutral valuation 的一般形式。

Remark 12.4.3. (1) $M(j)$。記 r_ℓ 為第 ℓ 期（從 ℓ 到 $\ell+1$）的利率，在時間 ℓ 揭露，則 $M(j) = \prod_{\ell=0}^{j-1}(1+r_\ell)$（連續複利則是 $e^{\sum r_\ell}$）。最後一個因子 $1+r_{j-1}$ 在 $j-1$ 已知，所以 $M(j)$ 在 $j-1$ 已知。

(2) **和 (12.4) 的關係**。利率固定時 $M(j) = R^j$，(12.5) 就是 (12.4)。

(3) **定價怎麼用**。取 $i=0$、$k=n$：$C(0) = E^{\pi}\left[C(n)/M(n)\right]$，就是把到期 payoff 沿著實際走過的利率路徑折現，再取 risk-neutral 期望。$M(n)$ 隨機時不能拆成 $E^{\pi}[C(n)]/M(n)$。

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Theorem 12.4.4

A discrete-time model is arbitrage-free if and only if there exists an equivalent probability measure (the risk-neutral probability measure) such that the discount process is a martingale.

【中】離散時間模型沒有套利，若且唯若存在一個 equivalent probability measure（risk-neutral 機率），使折現過程是 martingale。

Note 12.4.5. 這是 **fundamental theorem of asset pricing**（Harrison & Pliska, 1981）。在一期 BOPM 中就是 Section 8.5 的結論：$p = \frac{R-d}{u-d} \in (0,1)$ 若且唯若 $d < R < u$，而 $R \le d$ 或 $R \ge u$ 時分別借錢買股票、放空股票存錢就是套利。「equivalent」要求每個真實可能發生的 state 在 π 下機率也大於 0，即 $p \ne 0, 1$。

§12.4.i. Futures Price under the BOPM

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Futures prices form a martingale under the risk-neutral probability p_f (11.12). The expected futures price in the next period is

$$p_f F u + (1-p_f) F d = F\left(\frac{1-d}{u-d}u + \frac{u-1}{u-d}d\right) = F.$$

This can be generalized to

$$F_i = E_i^{\pi}[F_k], \qquad i \le k,$$

where F_i is the futures price at time i. This equation holds under stochastic interest rates, too.[8]

【中】futures price 在 p_f 下是 martingale：下一期期望值 $p_f F u + (1-p_f) F d = F$，推廣到 $F_i = E_i^{\pi}[F_k]$，利率隨機時也成立。

[8] See Exercise 13.2.11 of the textbook.

Remark 12.4.6. 括號內 $\frac{(1-d)u+(u-1)d}{u-d} = \frac{u-ud+ud-d}{u-d} = 1$。注意 futures price 本身（不折現）就是 martingale，因為持有 futures 部位不用花錢，沒有需要賺到 R 的本金。

Futures prices do not form a martingale under the risk-neutral probability $p = (R-d)/(u-d)$ (8.5). The expected futures price in the next period equals

$$Fu\frac{R-d}{u-d} + Fd\frac{u-R}{u-d} = F\frac{uR-ud}{u-d} + F\frac{ud-Rd}{u-d} = FR.$$

But they do if u and d are changed to u/R and d/R, respectively, justified by (11.2).
【中】在股票的 risk-neutral 機率 $p = (R-d)/(u-d)$ 下，若讓 futures price 也乘 u、d，期望值是 FR，不是 martingale。但 futures price 的乘數其實應該是 u/R、d/R（由 (11.2)），這時就是 martingale。

Remark 12.4.7. **為什麼是** u/R、d/R。股價從 S 漲到 Su 時，由 (11.2)，futures price 從 $F = Se^{rT}$ 變成 $Sue^{r(T-\Delta t)} = Fu/R$（$R = e^{r\Delta t}$）：剩餘期限少了一期，少乘一個 R。於是期望值 $pFu/R + (1-p)Fd/R = FR/R = F$。這和 (11.12) 是同一件事的兩種寫法：futures price 用乘數 u, d 配 p_f，或股價用乘數 u, d、futures price 用 $u/R, d/R$ 配 p。

§12.4.ii. Martingale Pricing and Numeraire

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The martingale pricing formula (12.5) uses the money market account as **numeraire**:[9] it expresses the price of any asset relative to the money market account.[10] The money market account is not the only choice for numeraire. Suppose asset S's value is positive at all time, and choose S as numeraire. Martingale pricing says there exists a risk-neutral probability π under which the relative price of any asset C is a martingale:

$$\frac{C(i)}{S(i)} = E_i^{\pi}\left[\frac{C(k)}{S(k)}\right], \qquad i \le k,$$

where $S(j)$ denotes the price of S at time j. So the "discount" process remains a martingale.[11]
【中】(12.5) 以貨幣市場帳戶為 numeraire（計價單位），把每個資產的價格表示成相對於它的比值。numeraire 不必是貨幣市場帳戶：任何價格永遠為正的資產 S 都可以，此時存在某個 risk-neutral 機率 π，使任何資產相對於 S 的價格 C/S 是 martingale。

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Example 12.4.8. Take the binomial model with two assets. In a period, asset one's price can go from S to S_1 or S_2; asset two's price can go from P to P_1 or P_2. Both assets must move up or down at the same time. Assume

$$\frac{S_1}{P_1} < \frac{S}{P} < \frac{S_2}{P_2} \tag{12.6}$$

to rule out arbitrage opportunities.

For any derivative security, let C_1 be its price at time one if asset one's price moves to S_1, and C_2 if it moves to S_2. Replicate the derivative by solving

$$\begin{aligned} \alpha S_1 + \beta P_1 &= C_1, \\ \alpha S_2 + \beta P_2 &= C_2, \end{aligned}$$

[9] John Law (1671–1729): "Money to be qualified for exchanging goods and for payments need not be certain in its value."
[10] Leon Walras (1834–1910).
[11] This result is related to Girsanov's theorem (1960).

using α units of asset one and β units of asset two. By (12.6), α and β have unique solutions:

$$\alpha = \frac{P_2C_1 - P_1C_2}{P_2S_1 - P_1S_2} \quad \text{and} \quad \beta = \frac{S_2C_1 - S_1C_2}{S_2P_1 - S_1P_2}.$$

The derivative costs

$$C = \alpha S + \beta P = \frac{P_2S - PS_2}{P_2S_1 - P_1S_2} C_1 + \frac{PS_1 - P_1S}{P_2S_1 - P_1S_2} C_2.$$

It is easy to verify that

$$\frac{C}{P} = p\frac{C_1}{P_1} + (1-p)\frac{C_2}{P_2} \quad \text{with} \quad p \triangleq \frac{(S/P) - (S_2/P_2)}{(S_1/P_1) - (S_2/P_2)}. \tag{12.7}$$

By (12.6), $0 < p < 1$. C's price using asset two as numeraire (i.e., C/P) is a martingale under the risk-neutral probability p. The expected returns of the two assets are irrelevant.

【中】例：兩個資產的一期 binomial model，資產一從 S 變成 S_1 或 S_2，資產二從 P 變成 P_1 或 P_2，同漲同跌；(12.6) 排除套利。用 α 單位資產一、β 單位資產二複製衍生性商品，解出 α, β 並得到價格 C。可以驗證 C/P 是 C_1/P_1、C_2/P_2 以 p (12.7) 加權的平均，且 $0 < p < 1$：以資產二為 numeraire，C/P 在 p 下是 martingale。兩個資產的期望報酬無關緊要。

Remark 12.4.9. (1) **為什麼解唯一**。係數行列式 $S_1P_2 - S_2P_1 = P_1P_2\left(\frac{S_1}{P_1} - \frac{S_2}{P_2}\right)$，由 (12.6) 是負的、不為 0，用 Cramer's rule 得 α, β。

(2) **驗證** (12.7)。把 C 除以 P，C_1 的係數是 $\frac{P_2S - PS_2}{P(P_2S_1 - P_1S_2)}$；分子分母同除 P_2 再乘 P_1，等於 $\frac{1}{P_1} \cdot \frac{S/P - S_2/P_2}{S_1/P_1 - S_2/P_2} = \frac{p}{P_1}$。$C_2$ 的係數同理是 $\frac{1-p}{P_2}$。

(3) $0 < p < 1$。p 的分子 $S/P - S_2/P_2 < 0$、分母 $S_1/P_1 - S_2/P_2 < 0$，且分子絕對值比分母小（因 $S/P > S_1/P_1$），所以 $0 < p < 1$。

(4) (12.6) **為什麼排除套利**。以 P 為單位，資產一的相對價格從 S/P 變成 S_1/P_1 或 S_2/P_2；若 S/P 不夾在中間，例如 $S/P \le S_1/P_1 < S_2/P_2$，那麼賣 S/P 單位資產二、買 1 單位資產一，期初成本 0，期末 $S_j - \frac{S}{P}P_j = P_j\left(\frac{S_j}{P_j} - \frac{S}{P}\right) \ge 0$ 且不全為 0，是套利。

In the BOPM, S is the stock and P is the bond. Furthermore, p assumes the bond is the numeraire. In the binomial option pricing formula (8.16), $S\sum b(j; n, pu/R)$ uses stock as the numeraire: its probability measure pu/R differs from p, and its price at maturity is simply S. $SN(x)$ for the call and $SN(-x)$ for the put in the Black–Scholes formulas (Theorem 8.14.19) use stock as the numeraire as well.[12]

【中】在 BOPM 中 S 是股票、P 是債券，p 是以債券為 numeraire 的機率。binomial 公式 (8.16) 中 $S\sum b(j; n, pu/R)$ 是以股票為 numeraire：機率是 pu/R 而不是 p，而 numeraire 本身到期的「價格」就是 S。Black–Scholes 公式中 call 的 $SN(x)$ 和 put 的 $SN(-x)$ 也是以股票為 numeraire。

Remark 12.4.10. 驗證 pu/R 就是以股票為 numeraire 的機率。在 (12.7) 中交換兩個資產的角色：資產二換成股票（$P \to S$、$P_1 \to Su$、$P_2 \to Sd$），資產一換成債券（$S \to 1$、$S_1, S_2 \to R$）。則

$$p' = \frac{1/S - R/(Sd)}{R/(Su) - R/(Sd)} = \frac{ud - Ru}{Rd - Ru} = \frac{u(R-d)}{R(u-d)} = \frac{pu}{R}.$$

所以 (8.16) 的第一項 $S\sum_{j \ge a} b(j; n, pu/R)$ 是「股票在 $S_n \ge X$ 時的價值」：以股票為單位，這個 claim 到期值 $\mathbf{1}_{\{S_n \ge X\}}$，今天值 $\text{Prob}'[S_n \ge X]$，換回金額再乘 S。

[12] See Exercise 13.2.12 of the textbook.

§12.5. Brownian Motion

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Definition 12.5.1. **Brownian motion** is a stochastic process $\{X(t), t \geq 0\}$ with the following properties.[13]

(1) $X(0) = 0$, unless stated otherwise.
(2) For any $0 \leq t_0 < t_1 < \cdots < t_n$, the random variables $X(t_k) - X(t_{k-1})$ for $1 \leq k \leq n$ are independent.[14]
(3) For $0 \leq s < t$, $X(t) - X(s) \sim N\big(\mu(t-s),\ \sigma^2(t-s)\big)$, where μ and $\sigma \neq 0$ are real numbers.

The existence and uniqueness of such a process is guaranteed by Wiener's theorem.[15] This process will be called a (μ, σ) **Brownian motion** with drift μ and variance σ^2. Although Brownian motion is a continuous function of t with probability one, it is almost nowhere differentiable. The $(0, 1)$ Brownian motion is called the **Wiener process**. If condition (3) is replaced by "$X(t) - X(s)$ depends only on $t - s$," we have the more general **Lévy process**.[16]

【中】Brownian motion：(1) $X(0) = 0$；(2) 不重疊區間的增量互相獨立；(3) $X(t) - X(s) \sim N(\mu(t-s), \sigma^2(t-s))$，$\sigma \neq 0$。Wiener 定理保證這種過程存在且唯一，叫做 drift μ、variance σ^2 的 (μ, σ) Brownian motion。它以機率 1 是 t 的連續函數，但幾乎處處不可微。$(0, 1)$ Brownian motion 叫 Wiener process。若 (3) 只要求增量的分布只取決於 $t - s$，就是更一般的 Lévy process。

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Example 12.5.2. If $\{X(t), t \geq 0\}$ is the Wiener process, then $X(t) - X(s) \sim N(0, t - s)$. A (μ, σ) Brownian motion $Y = \{Y(t), t \geq 0\}$ can be expressed in terms of the Wiener process:

$$Y(t) \ = \ \mu t + \sigma X(t). \tag{12.8}$$

Note that $Y(t + s) - Y(t) \sim N(\mu s, \sigma^2 s)$.

【中】例：Wiener process 的增量 $X(t) - X(s) \sim N(0, t - s)$；任何 (μ, σ) Brownian motion 都可以寫成 $Y(t) = \mu t + \sigma X(t)$ (12.8)，其增量 $Y(t + s) - Y(t) \sim N(\mu s, \sigma^2 s)$。

Remark 12.5.3. $Y(t + s) - Y(t) = \mu s + \sigma\big(X(t + s) - X(t)\big)$，而 $X(t + s) - X(t) \sim N(0, s)$；常數平移 μs、乘上 σ 後仍是 normal（Corollary 5.6.6），平均 μs、變異數 $\sigma^2 s$。獨立增量也直接繼承自 X。

§12.5.i. Brownian Motion as Limit of Random Walk

Slides pp. 580–583

> **Proposition 12.5.4**
> A (μ, σ) Brownian motion is the limiting case of random walk.

A particle moves Δx to the right with probability p after Δt time, and Δx to the left with probability $1 - p$. Define

$$X_i \triangleq \begin{cases} +1, & \text{if the } i\text{th move is to the right,} \\ -1, & \text{if the } i\text{th move is to the left.} \end{cases}$$

[13] Robert Brown (1773–1858).
[14] So $X(t) - X(s)$ is independent of $X(r)$ for $r \leq s < t$.
[15] Norbert Wiener (1894–1964). He received his Ph.D. from Harvard in 1912.
[16] Paul Lévy (1886–1971).

X_i are independent with $\text{Prob}[X_i = 1] = p = 1 - \text{Prob}[X_i = -1]$. Recall

$$E[\,X_i\,] \;=\; 2p - 1, \qquad \text{Var}[\,X_i\,] \;=\; 1 - (2p-1)^2.$$

Assume $n \triangleq t/\Delta t$ is an integer. The particle's position at time t is

$$Y(t) \;\triangleq\; \Delta x\,(X_1 + X_2 + \cdots + X_n).$$

Therefore,

$$\begin{aligned} E[\,Y(t)\,] &= n\,(\Delta x)\,(2p-1), \\ \text{Var}[\,Y(t)\,] &= n\,(\Delta x)^2\big[\,1 - (2p-1)^2\,\big]. \end{aligned}$$

With $\Delta x \triangleq \sigma\sqrt{\Delta t}$ and $p \triangleq \big[\,1 + (\mu/\sigma)\sqrt{\Delta t}\,\big]/2$ — identical to (8.20) —

$$\begin{aligned} E[\,Y(t)\,] &= n\sigma\sqrt{\Delta t}\,(\mu/\sigma)\sqrt{\Delta t} \;=\; \mu t, \\ \text{Var}[\,Y(t)\,] &= n\sigma^2\Delta t\,\big[\,1 - (\mu/\sigma)^2\Delta t\,\big] \;\to\; \sigma^2 t, \end{aligned}$$

as $\Delta t \to 0$. Thus, $\{Y(t), t \geq 0\}$ converges to a (μ, σ) Brownian motion by the central limit theorem (Theorem 5.8.6).

【中】粒子每 Δt 時間以機率 p 往右、$1-p$ 往左移 Δx；$X_i = \pm 1$ 記錄第 i 步方向，彼此獨立。時間 t 的位置 $Y(t) = \Delta x(X_1 + \cdots + X_n)$，$n = t/\Delta t$。取 $\Delta x = \sigma\sqrt{\Delta t}$、$p = [1 + (\mu/\sigma)\sqrt{\Delta t}]/2$（和 (8.20) 相同），則 $E[Y(t)] = \mu t$、$\text{Var}[Y(t)] \to \sigma^2 t$，由 central limit theorem，Y 收斂到 (μ, σ) Brownian motion。

Remark 12.5.5. (1) X_i **的動差**。$E[X_i] = p - (1-p) = 2p - 1$；$X_i^2 = 1$，所以 $\text{Var}[X_i] = 1 - (2p-1)^2$。
(2) $Y(t)$ **的動差**。期望值線性；X_i 獨立，變異數相加（(5.6)），再乘上 $(\Delta x)^2$。
(3) **代入**。$2p - 1 = (\mu/\sigma)\sqrt{\Delta t}$，$n\Delta t = t$：$E[Y(t)] = n \cdot \sigma\sqrt{\Delta t} \cdot (\mu/\sigma)\sqrt{\Delta t} = \mu n \Delta t = \mu t$；$\text{Var}[Y(t)] = n\sigma^2\Delta t\,[1 - (\mu/\sigma)^2\Delta t] = \sigma^2 t - \mu^2 t\,\Delta t \to \sigma^2 t$。
(4) **為什麼要用 CLT**。(3) 只對上了平均和變異數；$Y(t)$ 是 $n \to \infty$ 個獨立項的和，由 Theorem 5.8.6（嚴格說是 p 隨 n 變的 triangular array 版本），分布趨近 normal，所以 $Y(t) \to N(\mu t, \sigma^2 t)$。不同時間區間用到的是不同的 X_i，所以增量獨立。

Brownian motion with zero drift is the limiting case of symmetric random walk by choosing $\mu = 0$. Similarity to the BOPM: the p is identical to the probability in (8.20) and $\Delta x = \ln u$. Note that

$$\text{Var}[\,Y(t + \Delta t) - Y(t)\,] \;=\; \text{Var}[\,\Delta x\, X_{n+1}\,] \;=\; (\Delta x)^2 \times \text{Var}[\,X_{n+1}\,] \;\to\; \sigma^2 \Delta t.$$

【中】取 $\mu = 0$，零 drift 的 Brownian motion 就是 symmetric random walk 的極限。和 BOPM 的相似處：p 就是 (8.20) 的機率，$\Delta x = \ln u$。另外，一步的增量變異數 $(\Delta x)^2\,\text{Var}[X_{n+1}] \approx \sigma^2 \Delta t$。

Remark 12.5.6. $\Delta x = \ln u$。CRR 的 $u = e^{\sigma\sqrt{\Delta t}}$，所以 $\ln u = \sigma\sqrt{\Delta t} = \Delta x$：$Y(t)$ 就是 CRR tree 上的 $\ln(S_t/S)$（Section 12.2 的 remark），Proposition 12.5.4 說 log 股價收斂到 Brownian motion，和 Proposition 8.14.7 是同一件事。最後一式說每一步的變異數是 $\sigma^2\Delta t$，也就是步長 Δx 的平方；位移只有 $\sqrt{\Delta t}$ 的量級，這就是 Brownian motion 連續但不可微的來源：$|\Delta Y|/\Delta t \approx \sigma/\sqrt{\Delta t} \to \infty$。

§12.6. Geometric Brownian Motion

Slides pp. 584–586

Definition 12.6.1. Let $X \triangleq \{X(t), t \geq 0\}$ be a Brownian motion process. The process

$$\left\{ Y(t) \triangleq e^{X(t)},\ t \geq 0 \right\}$$

is called **geometric Brownian motion**.

Suppose further that X is a (μ, σ) Brownian motion. By assumption, $Y(0) = e^0 = 1$. $X(t) \sim N(\mu t, \sigma^2 t)$ with moment generating function

$$E\left[e^{sX(t)} \right] = E\left[Y(t)^s \right] = e^{\mu t s + (\sigma^2 t s^2/2)}$$

from (5.17). In particular, by (5.25),

$$\begin{aligned} E[Y(t)] &= e^{\mu t + (\sigma^2 t/2)}, \\ \operatorname{Var}[Y(t)] &= E\left[Y(t)^2\right] - E[Y(t)]^2 = e^{2\mu t + \sigma^2 t}\left(e^{\sigma^2 t} - 1\right). \end{aligned}$$

【中】X 是 Brownian motion 時，$Y(t) = e^{X(t)}$ 叫 geometric Brownian motion。若 X 是 (μ, σ) Brownian motion，則 $Y(0) = 1$，$X(t) \sim N(\mu t, \sigma^2 t)$，由 MGF (5.17) 得 $E[Y(t)^s] = e^{\mu t s + \sigma^2 t s^2/2}$，所以平均和變異數如上（即 (5.25)，把 μ, σ^2 換成 $\mu t, \sigma^2 t$）。

Remark 12.6.2. 取 $s = 1$ 得 $E[Y(t)]$；取 $s = 2$ 得 $E[Y(t)^2] = e^{2\mu t + 2\sigma^2 t}$。相減：$e^{2\mu t + 2\sigma^2 t} - e^{2\mu t + \sigma^2 t} = e^{2\mu t + \sigma^2 t}(e^{\sigma^2 t} - 1)$。注意 $E[Y(t)]$ 的成長率是 $\mu + \sigma^2/2$ 而不是 μ：$X(t)$ 的平均是 μt，但 e^x 是凸函數，由 Jensen（Theorem 5.10.3）$E[e^X] > e^{E[X]}$。

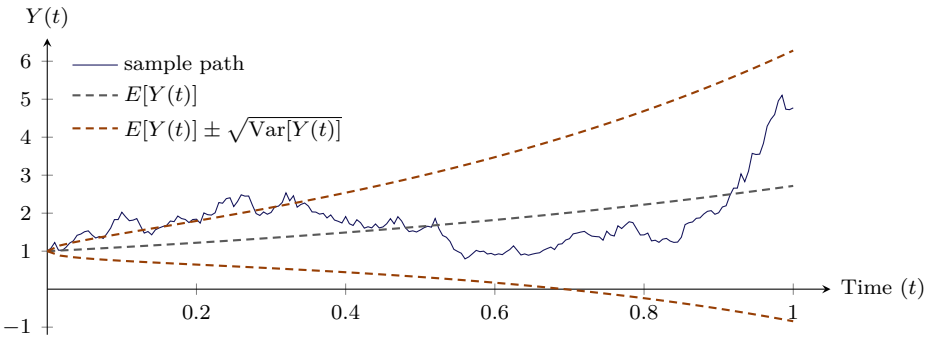


Figure 12.2.: A sample path of geometric Brownian motion with $\mu = 0.5$, $\sigma = 1$, so $E[Y(t)] = e^t$ and $\sqrt{\operatorname{Var}[Y(t)]} = e^t\sqrt{e^t - 1}$. The lower band goes negative although $Y(t) > 0$ always: one standard deviation is a poor description of a skewed distribution.

【中】圖：$\mu = 0.5$、$\sigma = 1$ 的 geometric Brownian motion 樣本路徑，虛線是 $E[Y(t)] = e^t$ 和加減一個標準差。下方那條會變成負的，但 $Y(t)$ 永遠為正——對偏斜的分布，「平均 ± 一個標準差」描述得很差。

§12.6.i. An Argument for Long-Term Investment

Slide p. 587

Suppose the stock follows the geometric Brownian motion[17]

$$S(t) = S(0)\, e^{N(\mu t, \sigma^2 t)} = S(0)\, e^{tN(\mu, \sigma^2/t)}, \qquad t \geq 0,$$

where $\mu > 0$. The annual rate of return has a normal distribution:

$$N\left(\mu,\ \frac{\sigma^2}{t}\right).$$

[17] Contributed by Dr. King, Gow-Hsing on April 9, 2015. See http://www.cb.idv.tw/phpbb3/viewtopic.php?f=7&t=1025.

The larger the t, the likelier the return is positive. The smaller the t, the likelier the return is negative.

【中】設股價服從 geometric Brownian motion $S(t) = S(0)e^{N(\mu t, \sigma^2 t)}$，$\mu > 0$。年化報酬率 $\sim N(\mu, \sigma^2/t)$：持有越久，報酬為正的機率越高；越短則越可能是負的。

Remark 12.6.3. (1) $e^{N(\mu t, \sigma^2 t)} = e^{tN(\mu, \sigma^2/t)}$。若 $Z \sim N(\mu, \sigma^2/t)$，則 $tZ \sim N(\mu t, t^2 \cdot \sigma^2/t) = N(\mu t, \sigma^2 t)$。

(2) **年化報酬率**。連續複利的年化報酬率 $\frac{1}{t} \ln \frac{S(t)}{S(0)} = Z \sim N(\mu, \sigma^2/t)$。

(3) **報酬為正的機率**。$\text{Prob}[Z > 0] = N\left(\mu/\sqrt{\sigma^2/t}\right) = N\left(\mu\sqrt{t}/\sigma\right)$，$\mu > 0$ 時隨 t 遞增、$t \to \infty$ 時趨近 1；$t \to 0$ 時趨近 $N(0) = 1/2$。

Note 12.6.4. 這個論證說的是年化報酬率的不確定性 $\sigma/\sqrt{t}$ 會縮小；總 log 報酬 $\ln(S(t)/S(0))$ 的標準差 $\sigma\sqrt{t}$ 仍然隨 t 增加。所以長期持有讓「虧錢」的機率變小，但不代表財富的絕對波動變小。